Pursuing Abstraction: Problems and Ideas on Mathematics and Physics

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ABSTRACT

This text cover problems and ideas on Physics, Mathematics and on the relation between them, corresponding to the works [119, 125, 124, 117, 115, 122, 123, 121, 118, 120, 112, 116]. The covered topics are vastly different, going the problem of determining the solutions of Tolman-Oppenheimer-Volkoff equations (which describe the hydrodynamic equilibrium of relativistic stars), to the existence and multiplicity problems of objects with given regularity on sufficiently regular manifolds.

In between, we also consider questions concerning the possible geometries of gravity and the formalization/existence of emergence between effective theories and of extensions of gauge theories. We also study an extension of Feynman rules from Feynman graphs to arbitrary hypergraphs.

Although distinct, the problems are connected by searching or using (or pursuing!) abstracting principles. During each chapter we tried to emphasizes how each of them can be understood internal to a given approach to the philosophy of Physics and Mathematics. In truth, we intend each problem as a small step towards the Axiomatization Program of Physics.

RESUMO

Este texto cobre problemas e ideias em Física, Matemática e na relação entre ambas, correspondendo aos trabalhos [119, 125, 124, 117, 115, 122, 123, 121, 118, 120, 112, 116]. Os tópicos abordados são bastantes distintos, versando desde o problema de determinação de soluções das equações de Tolman-Oppenheimer-Volkoff (que descrevem o equilíbrio hidrodinâmico de estrelas relativísticas), até os problemas de existência e multiplicidade de objetos com regularidade prescrita em variedades suficientemente regulares.

No intermédio, consideramos, também, questões ligadas às possíveis geometrias da gravidade e à formalização/existência de emergência entre teorias efetivas e de extensões de teorias de gauge. Estudamos, ainda, uma extensão das regras de Feynman de grafos de Feynman para hipergrafos arbitrários.

Embora distintos, os problemas se conectam por buscarem ou usarem (ou perseguirem!) princípios de abstração. Ao longo de cada capítulo tentamos enfatizar como todos eles podem ser entendidos dentro de abordagens para a filosofia da Física e da Matemática. Na verdade, temos a intenção que cada problema seja um pequeno passo na direção do Programa de Axiomatização da Física.

Contents

1

INTRODUCTION

1.1 STRUCTURE OF THE THESIS

At first glance, this text may look like a patchwork quilt. And it really is. The pieces of cloth correspond to papers from astrophysics to geometric regularity theory and graph reconstruction conjectures, passing through emergent gravity, extensions of Yang-Mills theories, and Feynman diagrams. What connects this zoo of topics (what sews the pieces of cloth into a quilt) is a common underlying aspect: *the fundamental role of abstraction processes*.

In this sense, besides this introduction, the thesis is structured by topic chapters:

1. *Axiomatization of Physics*, with a review on a modern approach to the Axiomatization Program of Physics through higher category theory [119].

2. *Mathematical Astrophysics*, containing two articles on the axiomatization of stellar astrophysics [125, 124].

3. *Geometry of Gravity*, with two works providing obstructions to the possible geometries that can be used to model gravity [117, 115].

4. *Emergence and Extensions*, where in two papers we propose a formalization to the notion of emergence between classical field theories and study its existence problem [121, 118], and in two other works we propose and begin the study of an abstract notion of extensions of gauge theories [122, 123].

5. *Geometric Regularity*, containing two works starting a program aiming to provide a formalization to the notion of "regular object" and to study the existence and multiplicity problems of arbitrarily regular geometric objects on sufficiently regular manifolds [120, 112].

6. *Feynman Functors*, with a work about a categorification of Feynman rules and the extension of them to structured hypergraphs, which plays a rule between reconstruction conjectures on hypergraphs and decomposition principles on the corresponding analytic expressions [116].

- In each chapter we include:
  - a *Synthesis* section, where we briefly present the main features of the works in that chapter;
  - a *Questions and Speculations* section, in which we expose some questions related to the works in that chapter, which arose during the development of the thesis,

but which at the present moment are not sufficiently explored to provide papers or preprints.

- An appendix containing a list of all those questions, providing a quick tour between future directions.

Remark 1.1. These additional questions will be studied in the context of Science 2.0 through Open Notebook Research. The discussions and developments will be made directly in the Math-Phys-Cat Wiki[1].

1.2 THE PROBLEMS

This thesis is entitled *Pursuing Abstraction: Problems and Ideas on Mathematics and Physics*. Above we commented on the sewing role of abstraction. In the following, we roughly describe those "problems and ideas".

Our objective here is to give the reader a panoramic view of the scope of the text. For a more detailed discussion of our conclusions about that questions and possible future deployments, see the corresponding topic chapters.

1. *Can one provide an axiomatization for the whole of Physics?*

   Hilbert's Program for the axiomatization of Mathematics can be viewed as an extension of Hilbert's second problem. Similarly, one could ask for an Axiomatization Program of Physics as an extension of Hilbert's sixth problem [25].

   But, what is Physics? Is it the study of physical systems? If so, what are they? Which kind of language should be used to axiomatize Physics?

2. *What is, really, a stellar system? Can they be classified?*

   Relativistic astrophysics was firstly developed in gravitational backgrounds given by General Relativity. With the advent of extensions and modifications of General Relativity, analogous versions of stellar systems have been considered and used as a tool to search for measurable aspects of those new gravity models [32, 148, 33, 34, 39].

   Thus, Astrophysics is, a priori, a branch of Physics that depends on the choice of some model for gravity. But, what are its foundational aspects which are the same independently of the background model? E.g, can one define what is a stellar system independently of a specific gravitational model? What can be said about the space of those stellar systems? Can it be parameterized? It is possible to give a bound to its "dimension"? In which sense?

3. *What are the geometries in which gravitational theories can be realized in a nontrivial way?*

   General Relativity is defined for Lorentzian geometry and most of the mathematical framework makes sense for semi-riemannian geometry [52, 87]. When regarded as an Einstein-Hilbert-Palatini (EHP) theory, it is about Cartan geometry modeled on the model geometry $\mathrm{Iso}(3,1)/O(3,1) \simeq \mathbb{R}^{3+1}$. More precisely, it is a funcional of a specific shape defined on the space of Cartan connections for that Cartan geometry.

   Can the EHP functional be defined for other kinds of Cartan geometry? That is, for Cartan connections modeled on model geometries other than $\mathrm{Iso}(3,1)/O(3,1)$? For

[1]This is an open-source wiki project in progress since 2021, designed and developed by the author. The core is based on Pmwiki software and is available at GitHub.

which geometries the corresponding functional is nontrivial? How could we interpret the triviality of the EHP functional in some geometry?

4. *Can one consider classical field theories whose "algebra of infinitesimal internal symmetries" is not a Lie algebra, but rather some more general algebraic object? What is their physical meaning?*

   Physical symmetries are normally described by a finite-dimensional Lie group G, which in turn is generated by its corresponding finite-dimensional Lie algebra $\mathfrak{g}$. In gauge theories, the Lie algebra $\mathfrak{g}$ plays a special role, since it appears as a defining part of the fields and on the action functional. Indeed, the fields are principal connections on G-bundles and, therefore, certain $\mathfrak{g}$-valued 1-forms. The Lagrangian, in turn, is typically an equivariant pairing defined by the Killing form of $\mathfrak{g}$ and evaluated in those 1-forms [50, 71, 109]. This also makes important to work with compact semi-simple Lie algebras, to ensure a negative-definite Killing form (see here).

   Let A be an arbitrary algebra with polynomial identities (not necessarily Lie algebras) endowed with arbitrary pairings (not necessarily induced by a Killing form). Can one consider functionals that are analogous (in shape) of those defining classical gauge theories, but now defined on certain A-valued 1-forms? For what kind of polynomial identities does this provide a nontrivial functional? What are the "connections" in this case?

5. *What is an emergence phenomena between physical theories? Under which conditions does a physical theory emerge from another?*

   It has been observed that some aspects of a physical theory can be rediscovered from a totally different theory. One says that the first *emerges* from the latter, which is related to the notion of *duality* between two theories [163]. E.g, in field theory it was observed that gravity emerges from the noncommutativity of spacetime [137, 12] as a consequence of a duality between string theory and gauge theories in a noncommutative background [147].

   But, what is, in formal terms, an emergence phenomenon between field theories? Is this relation symmetric? Can one establish broad criteria to determine if a theory may emerge from another without explicitly providing an emergence phenomenon between them? Is it possible to recover facts such as gravity emerging from noncommutativity as consequences of such general criteria?

6. *What it means to take an extension of a physical theory? Under which conditions a physical theory can be extended? Does it have some maximal extension?*

   Physical theories are normally viewed as effective theories. This means that they are supposed to be limits of more fundamental theories. In terms of field theories, this means that $S_0[\varphi] = \lim_\lambda S_\lambda[\varphi]$. Also, two theories are normally *coupled* to describe an entire physical phenomenon. In the case of field theories, the (minimal) coupling is obtained by summing the functionals of two independent theories and then adding a mixing term, i.e., $S[\varphi, A] = S_0[\varphi] + S_1[A] + S_{\text{int}}[\varphi, A]$.

   One can think of both examples as a newest theory *extending* a well-studied theory. Indeed, in the effective case, the fundamental theory $\sum_\lambda S_\lambda[\varphi]$ *extends* the limiting theory S_0; in the coupling case, regarding one of the theories as a background theory (fixing A), the total functional $S[\varphi] = S_0[\varphi] + S_1'[\varphi]$, where $S_1'[\varphi] = S_{\text{int}}[\varphi; A] + S_1[A]$, *extends* the "free" theory $S_0[\varphi]$.

But, what is, in formal terms, an extension of a field theory? Clearly, this should be related to the emergence relation. What is this relation? If S emerges from S' then is S' an extension of S? What can be said about the "space of extensions" of a given theory? Does it have any "universal" element?

7. *What is a regular geometric object in a regular manifold? When do they exist?*

   Global analysis and geometric analysis deal with geometric objects on differentiable manifolds. These are usually (nonlinear) tensors or (pseudo) differential operators on induced bundles [99, 157]. The existence of such objects normally depends on the regularity required, which in turn is bounded by the regularity of the manifold. Thus, to determine the existence of a geometric object with a given regularity one needs to have a minimally regular atlas.

   What can be said about the reciprocal? More precisely, how to define "geometric objects" and how to get existence theorems like "*given a regularity ε, then there exists a regularity δ such that if a manifold is δ-regular, then it admits ε-regular geometric objects*"? Given $k > 0$ are there nontrivial conditions on $\varepsilon(k)$ ensuring that every C^k-manifold admits an atlas $\varepsilon(k)$-regular?

8. *Can Feynman rules be defined for graphs other than Feynman graphs? Are they unique in some sense?*

   Feynman rules assign to each Feynman graph G the formal integration of a distribution (or products of them). Feynman graphs are flavors of graphs endowed with further structure: coloring, decomposition of edges, etc. Distributions are functionals in some function space. Thus, Feynman rules assign certain structured graphs to elements in a certain topological vector space.

   Are Feynman rules part of some functor $Z : \mathbf{Feyn} \to \mathbf{Analy}$ between the category of Feynman graphs and the category of these analytic expressions? Can they be extended to other kinds of structured (hyper)graphs? Can analytic expression be abstractly defined in other abstract monoidal categories (not arising from topological vector spaces)? What does it mean these "abstract Feynman rules"? Are they unique in some sense?

9. *What is the role of these extended Feynman rules in the underlying (hyper)graph category?*

   In (hyper)graph categories there are the so-called *reconstruction conjectures* which consider the problem of determining if the isomorphism type of graphs (in a certain category) can be determined looking at the collection of vertex-deleted subgraphs [73, 22, 21]. Thus, reconstruction conjectures are about the validity of a specific "decomposition principle" which involves the disjoint union $\sqcup_v G - v$ of vertex-deleted subgraphs.

   Are the abstract Feynman rules monoidal functors relatively to the disjoint union? What is the meaning of the corresponding decomposition principle of abstract analytic expressions? Can one use this bridge between reconstruction conjectures and decompositions of analytic expressions to study (maybe disprove) some of these conjectures?

As one can see, problems 1, 2, 5, 6 and 7 are about providing formal definitions of phenomena arising in Physics and in Mathematics. Therefore, they are about *formalization* and about *axiomatization*. In the case of problems 7, 8 one begins with certain already formalized situations and one tries to extend the models (i.e., the domain of discourse of the semantics) in which that axiomatics is evaluated. Therefore, they are about *generalization*. Finally, in problem 9 one considers an example and asks if it can be characterized in terms of categorical structures and then generalized to other models. In other words, one asks if the semantics is categorical and asks about generations of it. This can be viewed as a flavor of *categorification*.

But, formalization, axiomatization, generalization, and categorification are all manifestations of abstraction processes, which is the common essence of all these motivating problems.

Remark 1.2. One should also note that problems 1, 2, 3, 4, 5 and 6, have a more philosophical appealing, since they are about the foundations of Physics. Thus, to emphasize even more the intrinsic relation between those pieces of the patchwork quilt, in chapters 2, 3, 4 and 5, where these problems are considered we followed a discussion in a more philosophical level, trying to show how our works fits into this broad perspective[2].

1.3 LIST OF WORKS

The following is the list of "refined" works, in the order that they will appear in the next chapters. For discussion of workings in progress, see the section "Questions and Speculation" of each topic chapter.

1. Y.X. Martins, R.J. Biezuner, *Higher Category Theory and Hilbert's Sixth Problem*, Aug 2020, hal-02909681
2. Y.X. Martins, D.S.P. Teixeira, L.F.A. Campos, R.J. Biezuner, *Constraints between equations of state and mass-radius relations in general clusters of stellar systems*, **Physical Review D**, v. 99, p. 023007, Jan 2019. doi:10.1103/PhysRevD.99.023007
3. Y.X. Martins, L.F.A. Campos, D.S.P. Teixeira, R.J. Biezuner, *Existence and classification of pseudo-asymptotic solutions for Tolman-Oppenheimer-Volkoff systems*, **Annals of Physics**, v. 409, p. 167929, Aug 2019. doi:10.1016/j.aop.2019.167929
4. Y.X. Martins, R.J. Biezuner, *Topological and geometric obstructions on Einstein-Hilbert-Palatini theories*, **J. Geom. Phys.**, v. 142, p. 229, Aug 2019. doi:10.1016/j.geomphys.2019.-04.012
5. Y.X. Martins, R.J. Biezuner, *Geometric Obstructions on Gravity*, Dec 2019, arXiv:1912.11198
6. Y.X. Martins, R.J. Biezuner, *An Axiomatization Proposal and a Global Existence Theorem for Strong Emergence Between Parameterized Lagrangian Field Theories*, Sep 2020, arXiv:2009.09858
7. Y.X. Martins, L.F. A. Campos, R.J. Biezuner, *On Extensions of Yang-Mills-Type Theories, Their Spaces and Their Categories*, Jul 2020, arXiv:2007.01660
8. Y.X. Martins, L.F. A. Campos, R.J. Biezuner, *On Maximal, Universal and Complete Extensions of Yang-Mills-Type Theories*, Jul 2020, arXiv:2007.08651
9. Y.X. Martins, R.J. Biezuner, *Existence of $B^k_{\alpha,\beta}$-Structures on C^k-Manifolds*, **São Paulo J. Math. Sci.**, 2021. doi:10.1007/s40863-021-00238-z

[2] Since we are not philosophers, certain lack of philosophical formality can be observed. We apologize the experts on the field, but our aim was just to motivate non-experts on thinking about the problems in this more general perspective where the relations between them are more evident.

10. Y.X. Martins, R.J. Biezuner, *Geometric Regularity Results on $B^k_{\alpha,\beta}$-Manifolds, I: Affine Connections*, **Methods Funct. Anal. Topology**, Vol. 27, no. 3, 2021. doi:10.31392/MFAT-npu263.2021.05

11. Y.X. Martins, R.J. Biezuner, *Graph Reconstruction, Functorial Feynman Rules and Superposition Principles*, Mar 2019, arXiv:1903.06284

1.4 TECHNICAL ISSUES

We close this prologue with some technical issues.

- The papers in the topic chapters are preliminary versions of the published papers or recompiled versions with minor style modifications of those in ArXiv and HAL.
- Along with the papers there two page numbering. The lower one corresponds to the page inside the paper, while the upper one is the page number of the thesis.
- The papers have a different latex style than the thesis. This was done to make it easier to distinguish between the contents.

2

AXIOMATIZATION OF PHYSICS

2.1 SYNTHESIS

From a very broad and vague point of view, Philosophy is the study of the most fundamental questions, such as existence, reality, the nature and meaning of knowledge, characterization of good and bad conducts, nature and extension of reasoning, and so on [160]. There are some major areas, as Metaphysics, Epistemology and Logic, which deals, respectively, with existence and reality, knowledge and understanding, and truth, justification, and argumentation.

Notice that "existence", "reality", "understanding" and "truth" seems to be qualities of something, which motivates us to think of different branches of Philosophy as the study of different classes of *objects* (or *things* or *entities*) and *relations* between them, and whether these objects have certain *qualities* and/or satisfy certain *properties* (or *predicables*). This emphasizes that Philosophy is the study of other areas of study, which define such objects and their properties. Thus, e.g., one could consider existence and reality questions for the class of *physical objects*, which therefore would be ontological questions in Physics.

This naturally suggests using the language of systems to discuss Philosophy. For instance, in p.3 of [30] Bunge says:

> *Both systemics[1] experts and ontologists are interested in the properties common to all systems irrespective of their particular constitution, and both are intrigued by the peculiarities of extremely general theories [...]*

and then he argues that they differ only by the methods employed and the specific problems that they are interested in. Thus, from now on, we will refer to an area of study as being characterized by a class of *systems*, which are themselves determined by objects (possibly endowed with properties) and by relations between them.

Example 2.1.

1. Physics is assumed to be characterized by *physical systems*, consisting of *physical objects* and relations between them;

2. Similarly, Mathematics concerned with *mathematical systems*, determined by *mathematical objects* and relations between them.

[1] "Systemics" is the name coined by Bunge to describe the collection of theories and their philosophical basis that focus on a general theory of systems - see p.1 of [30].

For each class of systems (that is, for each area of study), ontological, epistemological, and logical concerns should then be considered, a priori in an independent way. For instance, the fundamental questions about existence, reality, or what is the nature of knowledge, should a priori be considered for physical systems and mathematical systems independently, leading us to talk about *ontological/epistemological/logical theories of Physics* and *ontological/epistemological/logical theories of Mathematics*.

But, independently of the class of systems $\mathcal{O}$ in which Ontology is considered, it defines three corresponding classes $\mathcal{O}_{real}$ of *real objects*, $\mathcal{O}_{abs}$ of *abstract objects* and $\mathcal{O}_{mind}$ of *mental objects*. Part of what defines an ontological theory is to say what these classes are and what is the relation between them. Some kinds of relations are:

1. *realism*, for which $\mathcal{O}_{mind} \subset \mathcal{O}_{real}$, i.e., that mental entities are real;
2. *idealism*, for which $\mathcal{O}_{real} \subset \mathcal{O}_{mind}$, i.e., that real objects are mental entities;
3. *anti-realism*, for which $\mathcal{O}_{real} = \mathcal{O}_{mind}$, meaning that an object is real iff it is mental;
4. *platonism*, such that $\mathcal{O}_{abs} \subset \mathcal{O}_{real}$ and $\mathcal{O}_{abs} \cap \mathcal{O}_{mind} = \varnothing$, so that abstract objects are real and non-mental;
5. *nominalism*, stating that $\mathcal{O}_{abs} \cap \mathcal{O}_{real} = \varnothing$, i.e., that abstract objects are not real and not necessarily mental.

Concerning Epistemology, there are also some concerns which are independent of the area of study $\mathcal{O}$. Indeed, it assigns to $\mathcal{O}$ classes $\mathrm{Belief}(\mathcal{O})$ of *beliefs*, subclasses $\mathrm{Just}(\mathcal{O}) \subset \mathrm{Belief}(\mathcal{O})$ of *justified beliefs* and $\mathrm{True}(\mathcal{O}) \subset \mathrm{Belief}(\mathcal{O})$ of *true beliefs*, as well as a class $\mathrm{Know}(\mathcal{O}) \subset \mathrm{Just}(\mathcal{O}) \cap \mathrm{True}(\mathcal{O})$ of *knowledge*. An epistemological theory should then explain what is the nature of these classes and how they are related. For example:

1. *skepticism*, for which $\mathrm{Know}(\mathcal{O}) = \varnothing$, i.e., that there is no knowledge (or, if it exists, it is inaccessible). There other forms which negate even justified beliefs;
2. *empiricism*, for which the class of $\mathrm{Know}(\mathcal{O})$ is given by a subclass $\mathrm{Exp}(\mathcal{O}) \subset \mathrm{Just}(\mathcal{O})$ of justified beliefs which are said to be *empirically justified beliefs*.

Similarly, Logic also deals with general questions which are independent of $\mathcal{O}$. It assigns classes $\mathrm{Prop}(\mathcal{O})$ of *propositions* and a corresponding class $\mathrm{Logic}(\mathcal{O}) \subset \mathrm{Prop}(\mathcal{O})$ of *logically true propositions*. Among other things, a logical theory should define what propositions and locally true propositions are, and provide a characterization of locally true propositions. To us, the most important class of logical theories are the following:

- *formal logic*, which is defined by an external language L. That language, in turn, is given by a collection $\mathrm{Letter}(L)$ of *letters*, a collection $\mathrm{Words}(L)$ of *words* (or *well-formed formulas*), and a collection $\mathrm{Thm}(\mathcal{L}) \subset \mathrm{Words}(L)$ of *theorems*. In a formal logic we have $\mathrm{Prop}(\mathcal{O}) \simeq \mathrm{Words}(L)$ and $\mathrm{Logic}(\mathcal{O}) \simeq \mathrm{Thm}(L)$;
- *axiomatic logic*, which is a particular case of formal logic such that the underlying language L has an *axiomatic apparatus*, i.e., a subclass $\mathrm{Axiom}(L) \subset \mathrm{Words}(L)$ of words corresponding to *axioms* (or *basic hypotheses*) and a map $inf : \rho(\mathrm{Axiom}(L)) \rightarrow \mathrm{Words}(L)$, called *inference rules*, that take a collection of axioms and assigns a word. In this case, the theorems are *defined* as the image of "inf", i.e., are the words arising from inference. Therefore, an axiomatic logic is one whose propositions are words of a formal logic L and whose logically true propositions are the words arising from axioms in L via inference rules.

The ontological, epistemological and logical theories are normally considered at the same time, so that one could consider a "coupled" philosophical background, where an ontological approach is based on an epistemological/logical theory, and so on. Some examples are the following.

Example 2.2 (epistemology depending on ontology/logic background)**.**

1. *epistemic idealism*, stating that, fixed an ontological theory (and therefore the class $\mathcal{O}_{mind}$ of mental entities), $\mathrm{Know}(\mathcal{O})$ actually depends only on $\mathcal{O}_{mind}$, i.e., that the knowledge is derived from mental states.
2. *rationalism*, for which $\mathrm{Know}(\mathcal{O})$ is given by $\mathrm{Logic}(\mathcal{O})$, i.e., whose knowledge is given precisely by the locally true propositions.

Example 2.3 (ontology depending on epistemology/logic)**.**

1. *epistepic ontology*, in which the class O_{real} of real objects is defined as the class of objects such that $\mathrm{Belief}(O_{real}) \simeq \mathrm{Know}(\mathcal{O})$, i.e., the class of objects restricted to which the beliefs are all the knowledge.
2. *empiricist ontology*, a particular case of epistemic ontology for the empiricism theory of epistemology. Therefore, the class O_{real} of real objects is given by those whose beliefs are empirically justified.
3. *logicist ontology*, where the class O_{real} of real objects is given by those whose propositions are locally true, i.e., such that $\mathrm{Prop}(O_{real}) \simeq \mathrm{Logic}(O_{real})$;
4. *formal ontology*, a special case of logicist ontology for the case of a formal logic, so that the class O_{real} of real objects coincides with the class of those objects restricted to which each proposition can be inferred to a theorem.

Remark 2.4. The previous relations expressing a dependence between ontology, epistemology and logic for a given class of systems $\mathcal{O}$ can be understood as arrows completing the diagram below in a commutative way. In it, all arrows are natural inclusions, and we are assuming that each object $x \in \mathcal{O}$ induces some trivial belief and some trivial proposition in any epistemological/logical theory.

$$
\begin{tikzcd}
 & \mathcal{O}_{real} \arrow[dr, hook] & \mathcal{O}_{mind} \arrow[d, hook] & \mathcal{O}_{abs} \arrow[dl, hook'] \\
 & & \mathcal{O} \arrow[dl, hook'] \arrow[dr, hook] & \\
 & \mathrm{Belief}(\mathcal{O}) & & \mathrm{Prop}(\mathcal{O}) \\
\mathrm{Just}(\mathcal{O}) \arrow[ur, hook] & \mathrm{Know}(\mathcal{O}) \arrow[l, hook'] \arrow[r, hook] \arrow[u, hook] & \mathrm{True}(\mathcal{O}) \arrow[ul, hook'] & \mathrm{Logic}(\mathcal{O}) \arrow[u, hook]
\end{tikzcd}
\tag{2.1}
$$

Above we discussed aspects of the broad areas of Philosophy and how they connect to define the philosophical foundations of an area of study $\mathcal{O}$. Although different classes of systems are a priori independent, having their own fundamental aspects, most of them are supposed to be related in some way. In this case, its philosophical foundations should also be connected.

For instance, Mathematics is in some broad sense related to Physics. The way how this relation occurs will depend directly on the fixed ontological, epistemological and logical backgrounds on both. Therefore, the existence of a relation between Physics and Mathematics satisfying specific properties will depend on compatibility between their philosophical foundations. This is precisely the subject of this chapter.

2.1.1 Mathematics

As said above, Mathematics is the study of *mathematical objects* $\mathcal{M}$, its properties and relations. Defining "mathematical object" is the concern of any approach to the Philosophy of Mathematics, which is determined by ontological, epistemological and logical theories. A common assumption, however, is that[2] [1, 150, 104]:

- *mathematical objects are abstract objects relative to the ontological theory in question.*

This produces some consequences in terms of the broad ontological theories discussed in the last section. For instance:

Example 2.5. A platonistic view of Mathematics is such that mathematical objects are real and non-mental, while a nominalistic perspective asserts that mathematical objects are non-real and not necessarily mental.

Two epistemological/logical approaches to Mathematics are the following (see also [80])[3]:

Example 2.6.

- *logicism*, which asserts that mathematical objects and mathematical knowledge has a logical nature. For instance, if we take a rationalistic perspective of Mathematics, then mathematical knowledge $\mathrm{Know}(\mathcal{M})$ is given by $\mathrm{Logic}(\mathcal{M})$ for some logical theory L_1. On other hand, in a logicist ontological theory, the real objects are determined by the property $\mathrm{Prop}(M_{real}) \simeq \mathrm{Logic}(M_{real})$ for some logical theory L_2. If, in addition we assume a platonic position, then $M_{real} \simeq \mathcal{M}$, so that if $L_1 = L_2$, we have a logicistic theory. Summarizing, a philosophical foundation to Mathematics which is platonist, rationalistic and ontologically formalistic relative to the same Logic, is logicist;
- *formalism*, which asserts that the mathematical knowledge has an axiomatic nature. This means that there exists an axiomatic logic L such that $\mathrm{Know}(\mathcal{M})$ is given by $\mathrm{Thm}(L)$, i.e., that the knowledge in Mathematics can be fully described as theorems for some axiomatic logic L. If such axiomatic logic exists, we say that Mathematics was *axiomatized* in L or that is is possible to provide an *axiomatization* of Mathematics in L.

One should contrast these approaches:

1. *role of ontology*: while logicism states that mathematical objects are logical entities, formalism does not. Thus, logicism is also an ontological theory, while formalism is epistemological in nature, although the latter can be considered together with ontological hypotheses, such as Platonism, nominalism or some formal ontology [1, 60].

[2] We observe, however, that there are approaches in which mathematical objects are not abstract [79].

[3] A third approach should also be commented: *intuitionism*, claiming that Mathematics is mental by nature. This thesis is not compatible with platonism and, if mental entities are not real, it is compatible with nominalism. Mathematical knowledge is also supposed to be obtained by certain "abstract" mental processes, making mathematical truth independent of any external language and possibly changing in time [82].

2. *role of axioms*: formalism is not worried whether axioms $\mathrm{Axiom}(L) \subset \mathrm{Word}(L)$ play some role in the mathematical knowledge or if only theorems induced by them matter. On the other hand, this question lies in the core of logicism [60].

But maybe the fundamental difference is on the role of semantics, as follows. When one takes an axiomatic language L, one has its words $\mathrm{Word}(L)$, axioms $\mathrm{Axiom}(L)$ and theorems $\mathrm{Thm}(L)$ which are obtained by the inference rules. This defines the *syntax* of L. One could analyze how they behave internally to another language L'. This is about the *semantics* of L, as follows.

Define a *morphism* $f : L \to L'$ between two formal languages L as a map $f : \mathrm{Word}(L) \to \mathrm{Word}(L')$ which preserves theorems so that it induces a map $\hat{f} : \mathrm{Thm}(L) \to \mathrm{Thm}(L')$. For instance, if L and L' are axiomatic with axioms $\mathrm{Axiom}(L)$ and $\mathrm{Axiom}(L)$, if f preserves both axioms and inference rules, then it preserve theorems and therefore is a morphism $f : L \to L'$, as sketched in the diagram below. We say, in this case, that f is a morphism of axiomatic languages (and not only of formal languages).

$$
\begin{tikzcd}
\rho(\mathrm{Axiom}(L)) \arrow[r, dotted, "\rho(f)"] \arrow[d, "inf"'] & \rho(\mathrm{Axiom}(L')) \arrow[d, "inf'"] \\
\mathrm{Thm}(L) \arrow[r, squiggly, "\hat{f}"] \arrow[d, hook] & \mathrm{Thm}(L') \arrow[d, hook] \\
\mathrm{Word}(L) \arrow[r, "f"] & \mathrm{Word}(L') \\
\mathrm{Axiom}(L) \arrow[u, hook] \arrow[r, dashed, "f"'] & \mathrm{Axiom}(L') \arrow[u, hook]
\end{tikzcd}
$$

We say that a morphism $f : L \to L'$ is an *interpretation* of L in L', or that f *models* L in L' if $f : \mathrm{Word}(L) \to \mathrm{Word}(L')$ is injective. The *semantics* of a formal language L is the study of its interpretations. For instance, by the above, if a word is a theorem in L, then it is a theorem is each of its models. Let $f : L \to L'$ be an interpretation of an axiomatic language L. Two natural questions are:

1. *If t is a theorem in L' is it induced by a theorem in L?* In other words, if f is injective, is the induced map $\hat{f} : \mathrm{Thm}(L) \to \mathrm{Thm}(L')$ surjective?

2. *if $x \in Word(L)$ is a word in L such that $f(x)$ is a theorem in L', is x a theorem in L?* In other terms, if f is injective it reflects theorems?

These questions are about how the theorems of a language are affected by the theorems of its interpretations, i.e., *whether syntactic truth depends (and how it depends) on semantic truth.* Logicists take these questions into account, while formalists normally[4] do not.

2.1.2 Axiomatization of Mathematics

Around 1900 there was an intense debate about the foundations of Mathematics, specially concerning the role of axioms and what kind of axioms should be used. On one side, Hilbert was a defender of formalism, and in 1899 published his Foundations of Geometry [76, 77] providing an axiomatization for Euclidean geometry (i.e., showing that formalism

[4] A situation in which semantic truth is considered in logicism will be discussed in the next section.

makes sense as a philosophical basis to Euclidean geometry). On the other Frege, defender of logicism which made several critiques to Hilbert's works [57, 153, 48, 20].

Hilbert was actually a defender of a specific kind of formalism that assumes the existence of an axiomatization of Mathematics by a language L in a "consistent way". Indeed, we say that an axiomatic language is:

- *syntactically consistent* if it comes equipped with a map $\neg : \mathrm{Word}(L) \to \mathrm{Word}(L)$ called *negation*, such that if $x \in \mathrm{Word}(L)$ belongs to $\mathrm{Thm}(L)$, then $\neq x$ does not belongs to $\mathrm{Thm}(L)$;
- *semantically consistent* if there is an interpretation $f : L \to L'$ in an axiomatic language with negation $\neq$: $\mathrm{Word}(L')\,\mathrm{Word}(L')$ such that if $f(x) \in \mathrm{Thm}(L')$, then $\neq f(x)$ does not belongs to $\mathrm{Thm}(L')$.

In the so-called *Hilbert's Program*, it was desired to build an axiomatization of Mathematics through a syntactically consistent axiomatic language. In his Foundations of Geometry, Hilbert provided an axiomatic to Euclidean geometry through a semantically consistent logic which in some sense disagreed with the broad philosophy of formalism and this was one of the criticisms made by Frege [57, 153, 48, 20].

As part of this program, in its emblematic talk at the International Congress of Mathematicians in 1900, Hilbert proposed the consistent axiomatization of Arithmetic as the second of his 23 problems, suggesting a sketch of proof of semantics consistency. The sketch was criticized, e.g., by Poincaré [134, 54].

To avoid the criticism, Hilbert reformulated his view on formalism and proposed a new version of formalism that one could call *Hilbertian formalism*, which adds the existence of a subclass $\mathcal{M}_{fin} \subset \mathcal{M}$ of mathematical objects (called *finitary*) which are real and such that if a word $t(x) \in \mathrm{Word}(L)$ depending on $x \in \mathcal{M}$ is a theorem $t(x)$, then there is $t(y) \in \mathrm{Word}(L)$ depending on $y \in \mathcal{M}_{fin}$ which is also a theorem and such that $t(x)$ is "equivalent" to $t(y)$. Hilbert's Program was then about the existence of a finitary axiomatization to Mathematics which is proved to be syntactically consistent.

The program was deeply constrained by Gödel's incompleteness theorems, which among other things states that a strong enough axiomatic language (meaning that Peano's arithmetic can be modeled on it) cannot prove its syntactic consistency. How much Hilbert's Program was constrained is a matter of current debate, specially because Gödel's theorems work only for axiomatic languages which model Arithmetics - see here. But, *what about parts of Mathematics which do not contain Arithmetic?*

Finally, notice that a priori one could say that a language L is *relatively consistent* if there exists another language L' with negation and a morphism (not necessarily an interpretation) $f : L \to L'$ such that if $x \in \mathrm{Word}(L)$ is such that $f(x)$ belongs to $\mathrm{Thm}(L')$, then $\neq f(x)$ does not belongs to $\mathrm{Thm}(L')$. This is the same definition of being semantically consistent, except that the morphism $f : L \to L'$ is not supposed to be an interpretation. While Gödel's theorems state that Peano's arithmetic is not *syntactically consistent*, Gentzen proved that it is *relatively consistent* for a language which is not necessarily finitary [59, 93]. Thus, Hilbert's Program could be considered for the entire Mathematics if one replaces the syntactic consistency with relative consistency and forgets the need for a finitary language.

2.1.3 Physics

Let us now move to Physics. From the previous discussion, it is the study of physical systems $\mathcal{F}$, i.e., physical objects, their properties and relations. The definition of "physical object" being part of what defines an approach to the Philosophy of Physics. A widely used conception is [27]

Definition 2.7. A physical object is an object that can be located in space and time.

This definition assumes *a priori* notions of space and time, which should also be specified as part of the approach to the Philosophy of Physics. Usually, space and time are defined as properties/functions of another entity, called *spacetime* M:

Definition 2.8. Let M be a spacetime. An object is located in space/time (relatively to M) iff it can be located in M in a given space, at a given time.

In the presence of an spacetime, Definition 2.7 rewrites as:

Definition 2.9. A physical object (relatively to a spacetime M) is an object that can be located in M[5].

Let $\mathcal{F}(M)$ be the class of physical objects relatively to M. Concerning this new entity (the spacetime), many questions should be considered, which are just some issues of the Philosophy of Space and Time [152, 136]:

- *What is a spacetime? Is it a physical object in some other sense? Is spacetime equivalent to the "physical universe"? I.e, is there some correspondence $\mathcal{F}(M) \simeq M$? If it exists, is a spacetime unique? In which sense? If an object is physical relative to a spacetime is it relative to any other? In other words, if M and N are two different spacetimes, what is the relation between $\mathcal{F}(M)$ and $\mathcal{F}(N)$?*

Example 2.10. For example, if one assumes that $\mathcal{F}(M) \simeq M$, then M could not belong to $\mathcal{F}(M)$, i.e., a spacetime could not be located in itself.

Notice that in the broad sense described above, a physical object is independent of an ontological theory. The usual approaches to the Ontology of Physics are the following:

1. *spacetime realism*, for which physical objects in any spacetime M are real, and the spacetimes themselves are real, i.e., $\mathcal{F}(M) \subset \mathcal{F}_{real}$ and $M \in \mathcal{F}_{real}$ for every M;
2. *physicalism*, which asserts that everything is physical, i.e. that every object belongs to $\mathcal{F}(M)$ for some M. In particular, the real, abstract and mental entities belongs to the union $\cup_M \mathcal{F}(M)$. This is a kind of realism in the broad sense;
3. *dualism*, which states that the mental objects are not physical objects in any spacetime, i.e., $\mathcal{F}_{mind} \cap \mathcal{F}(M) = \varnothing$ for every M;
4. *anti-realists*, which believe that spacetimes and the corresponding physical objects are not real, i.e., for every M, both M and $\mathcal{F}(M)$ do not belongs to $\mathcal{F}_{real}$. In some forms, it is assumed that both are mental.

Remark 2.11. In the presence of a spacetime, the abstract objects of an ontological theory are normally supposed to be those which cannot be located in space/time. In this sense, from Definition 2.9 physical objects in M are not abstract, i.e., $\mathcal{F}_{abs} \cap \mathcal{F}(M) = \varnothing$.

[5] Some authors also assume that the object must have causal power, meaning that "changes" in it produce "changes" in other objects [97].

Remark 2.12. From the ontological theses above it is clear that being a real object is not the same thing as being a "physical object", i.e., does not necessarily exists M such that $\mathcal{F}_{real} \simeq \mathcal{F}(M)$. For instance, in dualism we have $\mathcal{F}_{mind} \cap \mathcal{F}(M) = \varnothing$ for every M, so that if one assumes that at least part of the mental entities are real, i.e., if $\mathcal{F}_{real} \cap \mathcal{F}_{mind} \neq \varnothing$, it follows that this is a piece of $\mathcal{F}_{real}$ which does not even intersects $\mathcal{F}(M)$ for every M.

On the other hand, the typical epistemological theories of Physics, involve a methodological theory relating the physical knowledge $\mathrm{Know}(\mathcal{F})$ with some kind of information on the spacetime M. Some examples are:

1. *physical empiricism*, which states that there is a class $\mathrm{Exp}(M)$ of *experiments* on M and a map $e : \mathrm{Exp}(M) \to \mathrm{Belief}(\mathcal{F})$ such that $\mathrm{Know}(\mathcal{F})$ is contained in the image of e. The corresponding empiricist ontology is fixed, meaning that an object in $\mathcal{F}$ is real iff it corresponds to some experiment;

2. *instrumentalism*, which considers a subclass $\mathrm{Ob}(\mathcal{M}) \subset \mathrm{Belief}(\mathcal{F})$ of *observable beliefs*, defined in terms of M, which contains $\mathrm{Know}(\mathcal{F})$. This perspective is compatible with physical realism if one take $\mathrm{Ob}(\mathcal{M}) = e(\mathrm{Exp}(M))$. Here, a empiricist ontology is also fixed, so that real objects are those which can be observed;

3. *scientific realism*, which also considers a class $\mathrm{Ob}(\mathcal{M})$ of observable beliefs, but unlike instrumentalism it asserts that $\mathrm{Know}(\mathcal{F})$ intersects both $\mathrm{Ob}(\mathcal{M})$ and its complement. Again, empiricist ontology is fixed, so that real objects corresponds to both observable and non-observable beliefs.

To complete a philosophical theory of Physics one needs to couple epistemological and ontological theories. Concerning this, some remarks:

Remark 2.13.

1. The three approaches are compatible with spacetime realism and physicalism. They also are compatible with dualism, but if one assumes that mental objects are real, then they should correspond to some experiment (in the case of physical empiricism), to observable beliefs (in the case of instrumentalism), or to observable or non-observable beliefs (in the case of scientific realism).

2. A priori one could consider other ontological theory rather than empiricist ontology in each of the three approaches above. Choosing the empiricist ontology could be understood as an *operationalistic hypothesis*, since we are assuming that the nature of real objects (in particular, of physical objects) is unknown until some experiment is made.

2.1.4 Axiomatization of Physics

Notice that the main philosophical theories of Mathematics (i.e., logicism and formalism) are about extending it to Logic, so that both Epistemology and Ontology become determined. In the case of Physics, on the other hand, the philosophical theories try to reduce Epistemology and Ontology to an analysis of certain entities on the spacetime M which are defined by an auxiliary methodological theory that explains what are experiments and in which sense they produce knowledge.

Since mathematical objects are supposed abstract and abstract objects are not physical, one should not expect that Mathematics reduces to Physics. However, a natural (an interesting!) question is if Physics can be, in some sense, reduced to Logic. The analogs of logicism and formalism should be:

1. *physical logicism*: the empiricist ontology intersects the logicist ontology for some logic L such that physical knowledge intersects logically true propositions. More precisely, there exists a formal logic L and a map $f : \mathcal{F}_{real}(M) \to \mathcal{F}_{real}(L)$ connecting both ontological theories which doubly factors as in the first diagram below.

2. *(weak) physical formalism*: na ontological intersection is not needed, but the epistemological theory of Physics should intersects with the epistemology of an axiomatic formal logic L. More precisely, there is a class $\mathrm{Exp}_0(L) \subset \mathrm{Exp}(M)$ of *fundamental experiments* on M and a map $f : \mathrm{Exp}(M) \to \mathrm{Word}(L)$ assigning axioms to fundamental experiments, meaning that $\hat{f}$ exists in the second diagram, and factors as in the lower diagram.

3. *physical formalism*: notice that, as described above, physical formalism presupposes an axiomatic description only of the knowledge arising from the class $\mathrm{Exp}_0(M)$ of fundamental experiments. This is why we called it *weak physical formalism*. If we want an axiomatization of the whole physical knowledge, it is necessary to add the requirement that the fundamental experiments generate all physical knowledge, i.e., that $e(\mathrm{Exp}_0(M)) = e(\mathrm{Exp}(M))$. Thus, from now on, *physical formalism* is a weak physical formalism in which fundamental experiments generate all knowledge.

$$
\begin{tikzcd}
\mathcal{F}_{real}(M) \arrow[r, "f"] \arrow[d, hook] & \mathcal{F}_{real}(L) \arrow[d, hook] \\
\mathrm{Exp}(M) \arrow[r, dashed, "\bar{f}"] \arrow[d, "e"'] & \mathrm{Word}(L) \\
\mathrm{Know}(\mathcal{F}) \arrow[r, dotted, "\hat{f}"'] & \mathrm{Thm}(L) \arrow[u, hook]
\end{tikzcd}
\qquad
\begin{tikzcd}
\mathrm{Exp}(M) \arrow[r, "f"] & \mathrm{Word}(L) \\
\mathrm{Exp}_0(M) \arrow[u] \arrow[r, dashed, "\hat{f}"'] & \mathrm{Axiom}(L) \arrow[u] \\
\rho(\mathrm{Exp}_0(M)) \arrow[r, dashed, "\rho(\hat{f})"] \arrow[d, "\rho(e)"'] & \rho(\mathrm{Axiom}(L)) \arrow[d, "inf"] \\
\rho(\mathrm{Know}(\mathcal{F})) \arrow[r, dotted, "\hat{f}"'] & \mathrm{Thm}(L)
\end{tikzcd}
\tag{2.2}
$$

Actually, while the second of the 23 Hilbert's problems was about axiomatization of arithmetics, the sixth one was about axiomatization of certain parts of Physics which were known at the time, such as Statistical Mechanics [75, 78]:

> ***6. Mathematical Treatment of the Axioms of Physics***.
>
> *The investigations on the foundations of geometry suggest the problem: To treat in the same manner, by means of axioms, those physical sciences in which already today mathematics plays an important part; in the first rank are the theory of probabilities and mechanics.*

Thus, in the same way as Hilbert's second problem extends to an axiomatization program of Mathematics (the Hilbert's Program), one can think about extending Hilbert's sixth problem to cover current Physics and any new Physics in an *Axiomatization Program of Physics*. This was discussed by Hilbert (which extended his problem to include Quantum Mechanics and General Relativity [142, 143]) and more recently by Bunge in [28, 27] and by the authors in [119].

Notice that Hilbert's Program was about the existence of a formalistic approach to Mathematics by a formal logic that is syntactically consistent. *Which properties should one* *require to be satisfied for a language in the Axiomatization Program of Physics?* Certainly, some stability under new discoveries is fundamental. In Section 4 of [26] Bunge argues that this stability can be described by a requirement of consistency for the axiomatic language, which puts the Axiomation Program of Physics in trouble with Gödel theorems in a similar way as happened with Hilbert's Program.

In dealing with this problem, in Section 3 of [26] Bunge proposes a strategy to axiomatize Physics starting with a suitable language, consisting of 16 steps. In [29] he discusses that the most difficult part of the strategy is not to build the language L, but actually to formalize experiments and to connect them to the language, i.e., to define $\mathrm{Exp}(M)$ and to build the map $f : \mathrm{Exp}(M) \to \mathrm{Word}(L)$, which strongly depends on ontological and epistemological hypotheses on Physics.

Remark 2.14 (A note on the reviewing process)**.** Among these technical difficulties, in Section 5 of the introduction, Bunge argues in [25] that there are at least "*[t]hree obstacles blocking the progress of Foundations of Physics*". Unfortunately, they are concerned about the criticism in reviewing processes which are, "*[...] with a few exceptions, far from having attained to a scientific level.*"

2.1.5 Higher Category Theory and HoTT

We saw that, according to Bunge, the harder part of the Axiomatization Program of Physics is to construct a formal theory of experiments and measurements that is compatible with a given logic. Furthermore, if we have a suitable language, then one can try to axiomatize Physics following his strategy consisting of 16 steps. The first step is, of course, to find a suitable language. *Which are the natural candidates?*

It should be abstract enough to axiomatize any kind of physical theory. Different theories can be axiomatized internally to different kinds of languages and then modeled on differential geometry, algebra, and so on [25, 119, 113] so that the language should be unifying. Categorical languages are then the first candidates. Gauge theories are local theories and therefore modeled by stacks [145, 113]. Thus, one should consider categorical logics internal to Grothendieck topos, which are intuitionistic [88]. Actually, to define gauge theories one needs extra stuff: a *cohesive structure*, which corresponds to adding modalities [145, 113]. Thus, one needs modal intuitionistic categorical logic.

On the other hand, the stability under new physical discoveries (added with a reductionism hypothesis - see Chapter 5) forces us to include the possibility of fundamental physical objects of different dimensions: they could be particles, strings, membranes or higher-dimensional objects. Gauge theories for higher-dimensional objects are higher gauge theories, which are modeled by higher stacks [145, 113]. Therefore, the natural languages are those internal to higher Grothendieck toposes with cohesive structure, that is, intuitionistic higher categorical logic, whose prototypes are Modal Homotopy Type Theories [2, 146, 4].

In [119] our objective was precisely to improve the motivation to this statement that Modal Homotopy Type Theories are the natural languages to be considered in the Axiomatization Program of Physics.

2.1.6 An Inductive Approach

Previously we commented that in Section 3 of [26] Bunge presented an informal strategy on how to provide an axiomatization of Physics starting with a suitable language. In the last section (and with more details in [119]) we argued that a natural candidate for such a "suitable language" is Modal Homotopy Type Theory.

Here we will present a more formal strategy (following the same line as Bunge) on how to work on the Axiomatization Program of Physics (APP), based on the formalization to what is Ontology, Epistemology and Logic used previously. We will see that, in the end, realizing APP depends on (and in some cases follows from) building a systematic way to extend an arbitrarily given axiomatic language.

Recall that the APP is about the existence of a formalistic approach to Physics that is compatible with the empiricist one. More precisely, if M is the spacetime, it asserts the existence of an axiomatic language L, a set $\mathrm{Exp}_0(M) \subset \mathrm{Exp}(M)$ of *fundamental experiments* generating the entire physical knowledge $\mathrm{Know}(\mathcal{F})$ and a map $f : \mathrm{Exp}(M) \to \mathrm{Word}(L)$ which maps knowledge defined by $\mathrm{Exp}_0(M)$ into theorems obtained by the axioms of L, meaning that the dotted arrows below exist.

$$
\begin{tikzcd}
\mathrm{Exp}(M) \arrow[r, "f"] & \mathrm{Word}(L) \\
\mathrm{Exp}_0(M) \arrow[u, hook] \arrow[r, dashed, "\bar{f}"'] & \mathrm{Axiom}(L) \arrow[u, hook]
\end{tikzcd}
\qquad
\begin{tikzcd}
\rho(\mathrm{Exp}_0(M)) \arrow[r, dashed, "\bar{f}"] \arrow[d, "\rho(e)"'] & \rho(\mathrm{Axiom}(L)) \arrow[d, "inf"] \\
\rho(\mathrm{Know}(\mathcal{F})) \arrow[r, dotted, "\hat{f}"'] & \mathrm{Thm}(L)
\end{tikzcd}
\tag{2.3}
$$

Recall, furthermore, that, in the same way as mathematical formalism differs from logicism by avoiding ontological requirements, physical formalism also differs from physical logicism by avoiding additional conditions relating empiricist ontology with the logicist ontology of L. Thus, we could use empiricist ontology to parameterize the construction of such physical formalism, as follows.

Note that empiricist ontology is the epistemological ontology for empiricist epistemology (i.e., physical empiricism). This means that the class $\mathcal{F}_{real} \subset \mathcal{F}$ of real physical systems is characterized by the property of being the largest subclass of $\mathcal{F}$ for which there is a dotted arrow making commutative the following diagram:

$$
\begin{tikzcd}
\mathcal{F}_{real} \arrow[r, dashed] \arrow[d, hook] & \mathrm{Exp}(M) \arrow[d, hook] & \\
\mathcal{F} \arrow[r] & \mathrm{Belief}(\mathcal{F}) & \mathrm{Know}(\mathcal{F}) \arrow[l, hook'] \arrow[ul, "e"']
\end{tikzcd}
$$

By the above, subclasses $\mathcal{F}_0 \subset \mathcal{F}_{real}$ are determined by epistemology. We are interested in the reciprocal, that is when there is a correspondence between ontological subclasses and epistemological subclasses.

Definition 2.15. An *empiricist class of Physics* (or just *empiricist class*) is a pair $\mathcal{F}\mathcal{E}_0 = (\mathcal{F}_0, \mathcal{E}_0)$ given by subclasses $\mathcal{F}_0 \subset \mathcal{F}_{real}$ and $\mathcal{E}_0 \subset \mathrm{Exp}(M)$ such that the dotted arrows that make in the diagram below commutative exist (e_0 is just the restriction of e to $\mathcal{E}_0$).

$$
\begin{tikzcd}
 & & \mathcal{E}_0 \arrow[d, hook] \arrow[ddr, dashed, bend left, "e_0"] & \\
\mathcal{F}_0 \arrow[r, hook] \arrow[urr, dashed, "\imath_0"] & \mathcal{F}_{real} \arrow[r, "\imath"] \arrow[d, hook] & \mathrm{Exp}(M) \arrow[d, hook] \arrow[dr, "e"] & \\
 & \mathcal{F} \arrow[r] & \mathrm{Belief}(\mathcal{F}) & \mathrm{Know}(\mathcal{F})
\end{tikzcd}
$$

Remark 2.16. Every subclass $\mathcal{F}_0 \subset \mathcal{F}_{real}$ can be made an empiricist class of Physics by taking $\mathcal{E}_0 = \imath(\mathcal{F}_0)$. By the commutativity of the defining diagram, it follows that if $\mathcal{E}_0'$ is any other way to turn $\mathcal{F}_0$ into an empiricist class, then $\mathcal{E}_0' \subset \imath(\mathcal{F}_0)$.

Definition 2.17. A *proposal* to APP in an empiricist class $\mathcal{F}\mathcal{E}_0 = (\mathcal{F}_0, \mathcal{E}_0)$ is given by:

1. a subclass $\mathcal{E}_0' \subset \mathcal{E}_0$ of *fundamental experiments* in $\mathcal{F}\mathcal{E}_0$;
2. an axiomatic language L_0;
3. a map $f_0 : \mathcal{E}_0 \to \mathrm{Word}(L_0)$ such that there exists the dotted arrow in the first diagram below.

We say that a proposal *realizes* the APP in $\mathcal{F}\mathcal{E}_0$ if the dotted arrows that make the second diagram commutative exist.

$$\begin{tikzcd}
\mathcal{E}_0 \arrow[r, "f_0"] & \mathrm{Word}(L_0) \\
\mathcal{E}_0' \arrow[u, hook] \arrow[r, dashed, "\bar{f}_0"'] & \mathrm{Axiom}(L_0) \arrow[u, hook]
\end{tikzcd}
\qquad
\begin{tikzcd}
\rho(\mathcal{E}_0') \arrow[r, dashed, "\bar{f}"] \arrow[d, "\rho(e_0)"'] & \rho(\mathrm{Axiom}(L_0)) \arrow[d, "inf"] \\
\rho(\mathrm{Know}(\mathcal{F})) \arrow[r, dotted, "\hat{f}_0"'] & \mathrm{Thm}(L_0)
\end{tikzcd}
\tag{2.4}$$

Definition 2.18. Given two empirist classes $(\mathcal{F}_0, \mathcal{E}_0)$ and $(\mathcal{F}_1, \mathcal{E}_1)$ we say that $\mathcal{F}_1$ is *an extension of* $\mathcal{F}_0$ if $\mathcal{F}_0 \subset \mathcal{F}_1$, $\mathcal{E}_0 \subset \mathcal{E}_1$ and if diagram below commutes.

$$\begin{tikzcd}
 & & & \mathcal{E}_0 \arrow[d, hook] \arrow[dddr, bend left, "e_0"] & \\
 & & & \mathcal{E}_1 \arrow[d, hook] \arrow[ddr, bend left, "e_1"] & \\
\mathcal{F}_0 \arrow[r, hook] \arrow[uurrr, bend left, "\imath_0"] & \mathcal{F}_1 \arrow[r, hook] \arrow[urr, bend left, "\imath_1"] & \mathcal{F}_{real} \arrow[r, "\imath"] \arrow[d, hook] & \mathrm{Exp}(M) \arrow[d, hook] \arrow[dr, "e"] & \\
 & & \mathcal{F} \arrow[r] & \mathrm{Belief}(\mathcal{F}) & \mathrm{Know}(\mathcal{F})
\end{tikzcd}$$

Definition 2.19. Let L_0 and L_1 be two axiomatic languages. We say that L_1 is an *extension of* L_0 if $\mathrm{Words}(L_0) \subset \mathrm{Words}(L_1)$, $\mathrm{Axiom}(L_0) \subset \mathrm{Axiom}(L_1)$ and if the inductive structures factors as follows:

$$\begin{tikzcd}
\rho(\mathrm{Axiom}(L_0)) \arrow[r, hook] \arrow[d, "ind_0"'] & \rho(\mathrm{Axiom}(L_1)) \arrow[d, "ind_1"] \\
\mathrm{Thm}(L_0) \arrow[r, dashed] & \mathrm{Thm}(L_1)
\end{tikzcd}$$

With these concepts on hand, we can propose an inductive approach to APP for induction made on the collection of all empiricist classes and ordered according to the extension relation.

1. *base of induction*: the base of induction consists in showing that there exists an empiricist class $(\mathcal{F}_0, \mathcal{E}_0)$ relative to which the APP can be proved. That is, to provide a proposal $(\mathcal{E}_0', L_0, f_0)$ to APP which actually realizes APP, meaning that the dotted arrows in diagram (2.4) exist.

2. *induction hypothesis*: The induction step basically consists in showing that if APP can be realized in $\mathcal{FE}_0 = (\mathcal{F}_0, \mathcal{E}_0)$ then it can be realized in any empiricist class $\mathcal{FE}_1 = (\mathcal{F}_1, \mathcal{E}_1)$ extending $\mathcal{FE}_0$. This, in turn, can be done showing that:
   (a) if $\mathcal{FE}_1$ extends $\mathcal{FE}_0$, then every approach to APP $(\mathcal{E}'_0, L_0, f_0)$ in $\mathcal{FE}_0$ extends to an approach to APP $(\mathcal{E}'_1, L_1, f_1)$ in $\mathcal{FE}_1$ for an axiomatic language L_1 extending L_0;
   (b) if the approach $(\mathcal{E}'_0, L_0, f_0)$ realizes the APP in $\mathcal{FE}_0$, then its extension realizes the APP in $\mathcal{FE}_1$.

Discussion

Assuming both the base of induction and induction hypothesis above we conclude that *APP is realized in every empiricist class extending $\mathcal{FE}_0 = (\mathcal{F}_0, \mathcal{E}_0)$ in which the base of induction was proved.* In particular, since $(\mathcal{F}_{real}, \mathrm{Exp}(M))$ clearly extends any empiricist class, it then follows that:

- *if the induction hypothesis is true and if APP is realized in some empiricist class, then it is realized in the whole Physics.*

Classical mechanics is a well-known example of an empiricist class in which APP is realized. The problem is then to prove the induction hypothesis. The first part of the induction hypothesis depends on a process that allows us to extend any given axiomatic language. Suppose we have this process. Thus, given L_0, let L_1 be its extension.

Lemma 2.20. *Let $(\mathcal{E}'_0, L_0, f_0)$ be an approach to APP in $\mathcal{FE}_0 = (\mathcal{F}_0, \mathcal{E}_0)$ and let $\mathcal{FE}_1 = (\mathcal{F}_1, \mathcal{E}_1)$ be an extension of $\mathcal{FE}_0$. Let L_1 be an extension of L_0 and suppose that f_0 has an extension f_1 as in the diagram below. Then $(\mathcal{E}'_0, L_1, f_1)$ is an approach to APP in $\mathcal{FE}_1$. Furthermore, if the former realizes APP in $\mathcal{FE}_0$, then the later realizes it in $\mathcal{FE}_1$.*

$$\begin{tikzcd}
\mathcal{F}_1 \arrow[r] & \mathcal{E}_1 \arrow[rrd, dashed, "f_1"] & & \\
\mathcal{F}_0 \arrow[u, hook] \arrow[r] & \mathcal{E}_0 \arrow[u, hook] \arrow[r, "f_0"] & \mathrm{Word}(L_0) \arrow[r, hook] & \mathrm{Word}(L_1) \\
& \mathcal{E}'_0 \arrow[u, hook] \arrow[r, dashed, "\bar{f}_0"] \arrow[rr, bend right, "\bar{f}_1"] & \mathrm{Axiom}(L_0) \arrow[u, hook] \arrow[r, hook] & \mathrm{Axiom}(L_1) \arrow[u, hook]
\end{tikzcd}$$

Proof. The first part is straightforward: just notice that from diagram above we have the diagram below, which is the condition defining a proposal in $\mathcal{FE}_1$ (see Definition 2.17 and compare with first diagram in (2.4)).

$$\begin{tikzcd}
\mathcal{E}_1 \arrow[r, "f_1"] & \mathrm{Word}(L_1) \\
\mathcal{E}'_0 \arrow[u, hook] \arrow[r, "\bar{f}_1"'] & \mathrm{Axiom}(L_1) \arrow[u, hook]
\end{tikzcd}$$

The second part follows from the commutativity of the following diagram, which in turn is induced from the fact that L_1 is an extension of L_1 and from the diagram above.

$$\begin{tikzcd}
 & \rho(\mathrm{Axiom}(L_1)) \arrow[dr, "ind_1"] & \\
\rho(\mathcal{E}'_0) \arrow[ur, bend left, "\rho(\bar{f}_1)"] \arrow[r, "\rho(\bar{f}_0)"] \arrow[d, "\rho(e_0)"'] & \rho(\mathrm{Axiom}(L_0)) \arrow[u, hook] \arrow[d, "ind_0"] & \\
\mathrm{Know}(\mathcal{F}) \arrow[r, "\hat{f}_0"'] \arrow[rr, bend right, "\hat{f}_1"] & \mathrm{Thm}(L_0) \arrow[r, "\mu"'] & \mathrm{Thm}(L_1)
\end{tikzcd}$$

□

Corollary 2.21. *Suppose that an axiomatic language L_0 admits an extension L_1 satisfying the following property: for every classes X and Y such that $X \subset Y$, any map $f : X \to \mathrm{Word}(L_1)$ extends to Y, i.e., the dotted arrow in the first diagram below exist. In this case, if $(\mathcal{E}'_0, L_0, f_0)$ realizes the APP in an empiricist class $\mathcal{FE}_0$ which is extended by $\mathcal{FE}_1$, then APP is realized in $\mathcal{FE}_1$ by a proposal whose underlying axiomatic language is L_1.*

$$\begin{tikzcd}
Y \arrow[dr, dashed, "\hat{f}"] & \\
X \arrow[u, hook] \arrow[r, "f"'] & \mathrm{Word}(L_1)
\end{tikzcd}
\qquad
\begin{tikzcd}
\mathcal{E}_1 \arrow[drr, dashed, "f_1"] & & \\
\mathcal{E}_0 \arrow[u, hook] \arrow[r, "f_0"'] & \mathrm{Word}(L_0) \arrow[r, hook] & \mathrm{Word}(L_1)
\end{tikzcd} \tag{2.5}$$

Proof. Just notice that the hypothesis on the first diagram implies the existence of the arrow in the second diagram, and then use the lemma above. □

Corollary 2.22 (roughly). *Assume that the empiricist philosophical theory of Physics can be formulated in the axiomatic theories ZFC, NBG or MK, meaning that the classes $\mathcal{F}_{real}$, $\mathrm{Exp}(M)$ and $\mathrm{Know}(\mathcal{F})$ and the maps between them, are actually sets/classes and maps in some of these axiomatic set theories. Then the existence of a systematic process that extends any arbitrarily given axiomatic language L_0 implies that APP can be realized in the entire Physics.*

Proof. In these axiomatic set theories, we have the Axiom of (Global) Choice. In ZFC (resp. VBG and MK) it implies that every surjection between sets (resp. classes) has a section and that every injection between sets (resp. classes) with nonempty domain has a retraction. Thus, under the hypothesis the inclusion $\mathcal{E}_0 \hookrightarrow \mathcal{E}_1$ in diagram (2.5) has a retraction for every nonempty $\mathcal{E}_0$ and therefore the map f_1 exists. This nonempty $\mathcal{E}_0$ exists: take, for instance, that corresponding to classical mechanics. The result then follows from the last corollary. □

The discussion reveals that, in essence, (internal to axiomatic set theory) the APP depends on the construction of a systematic way to extend axiomatic languages, which we could call *abstraction* of which *categorification* is the main example.

Classical mechanics and gauge theory, for instance, admits a description in terms of a categorical language L_0 which satisfies APP in their empiricist class. Thus, the problem would be to formalize categorification as a process of abstracting/extending languages and to apply it iteratively in L_0 to get a limiting language L realizing APP in the whole Physics. This limit, therefore, should be some higher categorical language, agreeing with the suggestion of (Modal) Homotopy Type Theory as a natural candidate.

Remark 2.23. It should be clear that the discussion above does not aim to be a formal realization of the APP. It is only an argumentation aiming to emphasize the role of abstraction processes. Actually, even if we suggested that APP can be realized in the above way, this would not be a big deal, since as already pointed out by Bunge in Section 4 of [26] and as previously commented, the really difficult part in the axiomatization of Physics is to do that for an axiomatic language satisfying consistency.

2.1.7 Quantization Approach

The discussion above starts with a realization of the APP in some empiricist class $\mathcal{F}\mathcal{E}_0 = (\mathcal{F}_0, \mathcal{E}_0)$ and then concludes that it extends to a realization of the APP in any extension of $\mathcal{F}\mathcal{E}_0$. For it, therefore, it is not a question if there exists some $\mathcal{F}\mathcal{E}_1$ which *does not* extend nor is an extension of $\mathcal{F}_0$.

Actually, it is not a problem if there are subclasses $\mathcal{F}_0, \mathcal{F}_1 \subset \mathcal{F}_{real}$ which are disjoint. Indeed, in the end, we are interested in realizing APP in $\mathcal{F}_{real}$, which contains both. A different approach, aiming to realize APP in each "maximal component" of $\mathcal{F}_{real}$ and then provide a connection between the corresponding languages, could also be considered.

For example, if we assume that Physics splits into "classical Physics" and "quantum Physics" in such a way that they define the only maximal empiricist classes $\mathcal{F}_{class}$ and $\mathcal{F}_{quant}$, then in this other approach we should provide realizations of APP in $\mathcal{F}_{class}$ (by a language L_{class}) and $\mathcal{F}_{quant}$ (by a language L_{quant}) and then show that there is a suitable morphism $q : L_{class} \to L_{quant}$ known as *quantization*.

This approach is widely considered/assumed (even indirectly) by the Physics and Mathematical-Physics literature. To be accepted by the community, a proposal must also prove Yang–Mills existence and mass gap, which is one of the Millennium Problems [86]. It was also the approach reviewed in [119]. See also Section 7.1.

2.2 QUESTIONS AND SPECULATIONS

- Recall that, as introduced by Bunge in [30] systemics is a natural language to talk about the philosophical foundations of branches of science. Roughly, specifying a system consists of objects (or terms) and how are they related each other. In [30] Bunge formalized this idea defining a system as a tuple (T, C, E, S), where T is a collection of *terms*, $C, E \subset T$ are disjoint subsets and S is a set of relations on $C \sqcup E$ - see also here.

  Notice, on the other hand, that the notion of "category" is also about objects and relations between them. This suggests that Bunge's definition of system should be related to the notion of category in such a way that the corresponding system theory is related to category theory.

  Indeed, remember that in the internal (or source/target) perspective, a category $\mathbf{C}$ is given by classes $\mathbf{C}_0$ and $\mathbf{C}_1$, functions $s, t : \mathbf{C}_1 \mathbf{C}_0$, and maps $\circ : \mathrm{pb}(s, t) \to \mathbf{C}_1$ and $id : \mathbf{C}_0 \to \mathbf{C}_1$, which satisfies associativity and neutral element properties. But a function $f : X \to Y$ between two sets defines a relation R_f on $X \sqcup Y$. Therefore, each category defines a system for which $C = \mathbf{C}_0$, $E = \mathbf{C}_1$ and S are the relations defined by the maps $s, t, \circ, id$.

  For systems in the sense of Bunge, there is an obvious notion of morphism between them, i.e., there is an obvious way to compare (or to relate) two systems of "same type". By this we mean that the class S of relations has a fixed cardinality. More precisely, for a set Σ, define a Σ-*system* as a system such that $S \simeq \Sigma$.

Given two Σ-systems $\mathcal{S} = (T, C, E, S)$ and $\mathcal{S}' = (T', C', E', S')$, define a *morphism* $f : \mathcal{S} \to \mathcal{S}'$ between them as a map $f : C \to C'$ such that $f(C) \subset C'$ and $f(E) \subset E'$, and which preserves the relation $S_\sigma \in S$ for every $x \in \Sigma$. That is, if $xR_\sigma y$, then $f(x)R'_\sigma y$.

This clearly defines a category $\mathbf{Sys}_\Sigma$ of Σ-systems. Curiously, Bunge does not introduce any notion of morphism between systems in [30].

Q1.1 *Show that every category* $\mathbf{C}$ *defines a system* $\mathcal{S}(\mathbf{C})$ *in the sense of Bunge. What kind of systems corresponds to some category? For instance, show if* $\mathbf{C}$ *and* $\mathbf{C}'$ *are two categories, then the systems* $\mathcal{S}(\mathbf{C})$ *and* $\mathcal{S}(\mathbf{C}')$ *are actually* Σ_{cat}*-systems for certain* Σ_{cat}*. Build a functor* $F : \mathbf{Cat} \to \mathbf{Sys}_{\Sigma_{cat}}$*. Has this functor left/right adjoints?*

Q1.2 *Given* Σ*, is the category* $\mathbf{Sys}_\Sigma$ *complete/cocomplete?*

- If $\mathcal{S} = (T, C, E, S)$ is a Σ-system and if $u : \Sigma \to \Sigma'$ is a map, we define a Σ'-system $u_*(\mathcal{S})$ by $(T, C, E, u(S))$. Reciprocally, if it was given a Σ'-system $\mathcal{S}' = (T', C', E', S')$, we get a Σ-system $u^*(\mathcal{S}')$ by $(T', C', E', u^{-1}(S'))$.

Q1.3 *Do extend the rules* u_* *and* u^* *defined above to functors* $u_* : \mathbf{Sys}_\Sigma \to \mathbf{Sys}_{\Sigma'}$ *and* $u^* : \mathbf{Sys}_{\Sigma'} \to \mathbf{Sys}_\Sigma$*? Are them adjoint? Let* $\mathbf{Set}$ *be category of classes and sets. Varying* u *we get functors* $(-)_* : \mathbf{Set} \to \mathbf{Cat}$ *and* $(-)^* : \mathbf{Set}^{op} \to \mathbf{Cat}$*? This means that the study of systems can be viewed as a presheaf of categories?*

- Immediately after introducing his notion of system, Bunge focuses on studying a particular class of systems that he called *concrete systems*. These are partially defined as follows. Let C be a class of "concrete objects" which has an intrinsic relation $\leq$. To each $x \in C$ define E_x as the class of all $y \in C$ such that $x \leq y$. Define $E = \cup_x E_x$ and then take additional relations on $C \sqcup E$.

Q1.4 *Define morphisms between concrete systems of a fixed type* Σ*. Show that they fit into a category* $\mathbf{CSys}_\Sigma$*. Prove that there is an inclusion functor* $\imath : \mathbf{CSys}_\Sigma \mathbf{Sys}_\Sigma$*. This functor has a "concretification" left/right adjoint* $L : \mathbf{Sys}_\Sigma \to \mathbf{CSys}_\Sigma$*? Is the subcategory* $\mathbf{CSys}_\Sigma$ *of concrete systems a reflective/co-reflective subcategory?*

Now, recall that in the standard hom-sets definition, a category $\mathbf{C}$ is given by a class $\mathrm{Ob}(\mathbf{C})$ of objects and for every two objects $X, Y \in \mathrm{Ob}(\mathbf{C})$ a class $\mathrm{Mor}_{\mathbf{C}}(X; Y)$ of morphisms between them, and so on. Take $C = \mathrm{Ob}(\mathbf{C})$ and on C consider the relation $\leq$ such that $X \leq Y$ iff there exists some morphism $f : X \to Y$. Let E_X the class of all $Y \in C$ such that $X \leq Y$ and let E the union of all E_X. By the above we then have a concrete system (C, E, S) where the set S of relations on $C \sqcup E$ is induced by compositions and identities of $\mathbf{C}$.

Q1.5 *Prove that every category* $\mathbf{C}$ *defines a concrete system* $\mathcal{C}(\mathbf{C})$ *of a specific type* $\Sigma_{cat'}$*. Show that the construction extends to a functor* $\mathcal{C} : \mathbf{Cat} \to \mathbf{CSys}_{\Sigma_{cat'}}$*. Find a map* $u : \Sigma_{cat} \to \Sigma_{cat'}$ *such that diagram below commutes. Supposing that the concretification functor* L *exists (i.e., assuming Question 2.2), prove that the entire diagram remains commutative.*

$$
\begin{tikzcd}
 & & \mathbf{Sys}_{\Sigma_{cat}} \arrow[d, shift left, "u_*"] \\
\mathbf{Cat} \arrow[r, "\mathcal{C}"'] \arrow[urr, bend left, "\mathcal{S}"] & \mathbf{CSys}_{\Sigma_{cat'}} \arrow[r, hook, shift left] & \mathbf{Sys}_{\Sigma_{cat'}} \arrow[l, dashed, shift left, "L"] \arrow[u, shift left, "u^*"]
\end{tikzcd}
$$

- In Section 2.1 we proposed a formulation for the philosophical foundations of an area of study assuming the systemic point of view. More precisely, an area of study was defined as a class of systems $\mathcal{O}$ of "same type", while giving an ontological, epistemological and logical theories for $\mathcal{O}$ is formalized by diagram (2.1).

  Following the discussion above, assume that systems of "same type" mean that they are Σ-systems for the same Σ. Thus, an area of study should be a subcategory $\mathcal{O} \subset \mathbf{Sys}_{\Sigma}$.

Q1.6 *Reformulate what are ontological/epistemological/logical theories for an area of study now regarding them internally to categories instead of to classes. Show that they are characterized by a version of diagram (2.1) in* **Cat**.

- On the other hand, if systems are categories, then their category is a category of categories, so it should be a 2-category.

Q1.7 *Assume Question 2.2. Let $\mathbf{Sys}_{\Sigma_{cat}}$ category of systems which are defined by categories. Show that $\mathbf{Sys}_{\Sigma_{cat}}$ is actually a 2-category such that the functor $F : \mathbf{Cat} \to \mathbf{Sys}_{\Sigma_{cat}}$ is a 2-functor.*

- Let $\mathcal{O}$ be a "categorical" area of study, so that a subcategory of $\mathbf{Sys}_{\Sigma_{cat}}$. Assuming question above $\mathbf{Sys}_{\Sigma_{cat}}$ is a 2-category. Thus, it is natural to suppose that $\mathcal{O}$ is a sub-2-category. This suggests considering a 2-categorical refinement of Question 2.2.

Q1.8 *Assume Questions 2.2 and 2.2. Reformulate ontological/epistemological/logical theories for categorical areas of study internally to 2-categories instead of to categories. Prove that they can be characterized by diagrams which are analogous of (2.1), but in* 2**Cat**.

- For certain systems it would be interesting to compare not only the systems themselves but also the morphisms between them, leading to a notion of "higher systems" (or "homotopical systems") whose study should be *higher systemics* or *homotopical systemics*. If 1-categories define Σ_{cat}-systems, one should expect that higher categories define higher $\Sigma_{\infty cat}$-systems for certain $\Sigma_{\infty cat}$.

Q1.9 *Define higher (or graded) system as a tuple $\mathcal{S} = (T, C, E, S)$ where $E = E_1, \ldots$ is a list of mutually disjoint classes, $C, E_i \subset T$ and S is a collection of relations on $\sqcup_i C \sqcup E_i$. Show that every ∞-category defines a higher system and that these higher systems are always of the same type $\Sigma_{\infty cat}$.*

Q1.10 *Prove that higher Σ-systems constitute a ∞-category $\infty\mathbf{Sys}_{\Sigma}$. Show that construction of Question 2.2 extends to a ∞-functor $F : \infty\mathbf{Cat} \to \infty\mathbf{Sys}_{\Sigma_{\infty cat}}$.*

Q1.11 *In analogy to Question 2.2, show that each $u : \Sigma \to \Sigma'$ induces adjoint ∞-functors u_* and u^*. Varying u build a ∞-functor $(-)^* : \mathbf{Set}^{op} \to \infty\mathbf{Cat}$ allowing us to view higher systemics as a certain ∞-presheaf of ∞-categories.*

- If we restrict to areas of study defined by higher systems, they should be given by sub-∞-categories $\mathcal{O} \subset \infty\mathbf{Sys}_{\Sigma}$.

Q1.12 *Define what is a higher ontological/epistemological/logical theory for an area of study defined by a sub-∞-category $\mathcal{O}$ of higher systems. Prove that they can be characterized by diagram (2.1), now on* $\infty\mathbf{Cat}$.

- Suppose now that we are interested in an axiomatic approach to an area of study $\mathcal{O}$. More precisely, suppose that we fixed an epistemological theory on $\mathcal{O}$. We are then looking for an axiomatic language L such that there are the dotted arrows making diagrams (5.3) commutative.

Q1.13 *Let $\mathcal{O}$ be an area of study consisting of higher systems endowed with a higher epistemological theory. Assuming Question 2.2 show that the diagram (5.3) characterizing an axiomatic approach to $\mathcal{O}$ is defined on* $\infty\mathbf{Cat}$.

Q1.14 *What kind of languages need to be used to build axiomatic approaches to areas of study defined by of higher systems?*

- The Axiomatization Program of Physics is about finding an axiomatic approach to Physics for the epistemological theory given by empiricism. In Section 2.1.6 we proposed an inductive approach to this program and in Corollary 2.22 we argued that if the construction is made in axiomatic set theories as ZFC, NBG or MK, then then the program is fulfilled if:

  1. it is realized in some empiricist class $\mathcal{F}\mathcal{E}_0 = (\mathcal{F}_0, \mathcal{E}_0)$ with $\mathcal{E}_0$ nonempty;
  2. there exists a systematic way of extending any axiomatic language.

  Looking at the proof of Corollary 2.22 we see, however, that axiomatic set theory was used only to ensure that every injection $f : X \to Y$ with $X \neq \varnothing$ has a retraction.

Q1.15 *Show that the construction of Section 2.1.6 can be reproduced internally to any category* $\mathbf{H}$.

Q1.16 *Define the class* $\mathrm{Inj}(\mathbf{H})$ *of "injective objects" of* $\mathbf{H}$ *by the property that if $f : X \to Y$ is a monomorphism and $X \in \mathrm{Inj}(\mathbf{H})$, then f is a split monomorphism. Assume Question 2.2. Extending Corollary 2.22 prove that if*

  1. *the APP can be realized internally to $\mathbf{H}$ for an empiricist class $\mathcal{F}\mathcal{E}_0 = (\mathcal{F}_0, \mathcal{E}_0)$ such that $\mathcal{E}_0 \in \mathrm{Inj}(\mathbf{H})$;*
  2. *there exists a systematic way of extending any axiomatic language,*

  then APP can be realized internally to $\mathbf{H}$ in the entire Physics.

Q1.17 *In order to contemplate higher systems, show that Question 2.2 remains true if $\mathbb{H}$ is* $\infty\mathbf{Cat}$.

Q1.18 *What are "higher injective objects" in $\infty\mathbf{Cat}$? Are they ∞-categories $\mathbf{C}$ such that every ∞-functor $F : \mathbf{C} \to \mathbf{D}$ which is a ∞-monomorphism it is actually a split ∞-monomorphism? But, what are ∞-monomorphisms and what are split ∞-monomorphisms? Show that for the right notion of higher injective objects we have a higher analogue Corollary 2.22.*

2.3 THE WORK

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Higher Category Theory and Hilbert's Sixth Problem

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Abstract

In 1900 David Hilbert published his famous list of 23 problems. The sixth of them - the axiomatization of Physics - remains partially unsolved. In this work we will give a gentle introduction and a brief review to one of the most recent and formal approaches to this problem, based on synthetic higher categorical languages. This approach, developed by Baez, Schreiber, Sati, Fiorenza, Freed, Lurie and many others, provides a formalization to the notion of classical field theory in terms of twisted differential cohomologies in cohesive $(\infty, 1)$-topos. Furthermore, following the Atiyah-Witten functorial style of topological quantum field theories, it provides a nonperturbative quantization for classical field theories whose underlying cohomology satisfies orientability and duality conditions. We follow a pedagogical and almost non-technical style trying to minimize the mathematical prerequisites. No categorical background is required.

1 Introduction

In the year of 1900, David Hilbert (a mathematician which is mostly known in Physics for his contribution to the Einstein-Hilbert action functional and for the Hilbert spaces, used in Quantum Mechanics, named in honor to him) published a list containing 23 problems (usually known as the *Hilbert's problems*) which in his opinion would strongly influence the development of 20th century mathematics [107]. The sixth problem is about the axiomatization of the whole physics and, presently, it remains partially unsolved. Our aim is to introduce basic ideas of an approach to this problem following works of Urs Schreiber, Domenico Fiorenza, Hisham Sati, Daniel Freed, John Baez and many others, mostly inspired by the seminal topos-theoretic works of William Lawvere [134, 137, 135, 138, 136] and formalized through the development of higher topos theory by André Joyal [118, 120, 119], Michael Batanin [27], Ross Street [223], Charles Rezk [183, 184], Carlos Simpson [213], Clark Barwick [24, 25, 26], Tom Leinster [140, 48], Dominic Verity [236, 235], Jacob Lurie [147, 148] and many others. For reviews on these different approaches to higher category theory, see [47, 141, 140].

The idea goes as follows. Some points are also discussed in [179, 191, 58, 58, 44]. See [202] for a technical introduction and [160] for a gentle exposition. See [128] for a discussion on other

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mathematical problems deeply influenced by Hilbert's works, [215, 178] for a more philosophical discussion and [128] for the history of the sixth problem.

There are two kinds of math: *naive* (or *intuitive*) *math* and *axiomatic* (or *rigorous*) *math*. In naive math the fundamental objects are *primitives*, while in axiomatic math they are defined in terms of more elementary structures. For instance, we have *naive set theory* and *axiomatic set theory*. In both cases (naive or axiomatic) we need a *background language* (also called *logic*) in order to develop the theory, as in the diagram below. In the context of set theory this background language is just *classical logic*.

$$\begin{tikzcd}
 & \textbf{naive math} \\
\textbf{background language} \arrow[ur, Rightarrow] \arrow[dr, Rightarrow] & \\
 & \textbf{axiomatic math}
\end{tikzcd}$$

Notice that, when a mathematician is working (for instance, when he is trying to prove some new result in his area of research), at first he does not make use of completely rigorous arguments. In fact, he first uses his intuition, making some scribbles in pieces of paper or in a blackboard and usually considers many wrong strategies before finally discovering a good sequence of arguments which can be used to prove (or disprove) the desired result. It is only at this later moment that he tries to introduce rigor in his ideas, in order for his result to be communicated and accepted by the other members of the mathematical community.

Thus we can say that in the process of producing new mathematics, *naive arguments come before rigorous ones*. More precisely, we can say that naive mathematics produces conjectures and rigorous mathematics turns these conjectures into theorems. It happens that the same conjecture generally can be proved (or disproved) in different ways, say by using different tools or by considering different models. For instance, many concepts have algebraic and geometric incarnations (e.g, Serre-Swan theorem [224, 210, 175], Stone diality [116], Gelfand duality [174, 180] and Tannaka duality [59, 145]), suggesting the existence of a duality between algebra and geometry, as in diagram (1) below.

As will be discussed, this duality between algebra and geometry, which in some generality is formalized by Isbell-type dualities [112, 113, 220, 22], extends to the context of physics, leading to dual approaches to Hilbert's sixth problem. For another incarnation of this duality, notice that when a geometric entity x has a *commutative* algebraic incarnation $A(X)$, then we can define the *noncommutative* version of X simply by dropping the commutativity requirement on $A(X)$. This is precisely the approach used in noncommutative geometry [97, 52].

$$
\begin{tikzcd}
 & \textbf{naive math} \arrow[r, Rightarrow, "\text{produces}"] \arrow[dd, Rightarrow, "\text{model}"] & \textbf{conjectures} \arrow[dd, Rightarrow, shift right, "\text{algebra}"'] \arrow[dd, Rightarrow, shift left, "\text{geometry}"] \\
\textbf{background language} \arrow[ur, Rightarrow] \arrow[dr, Rightarrow] & & \\
 & \textbf{axiomatic math} \arrow[r, Rightarrow, "\text{produces}"'] & \textbf{theorems}
\end{tikzcd}
\tag{1}
$$

When compared with mathematics, physics is a totally different discipline. Indeed, in physics we have a restriction on the *existence* of physical objects, meaning that we have a connection with *ontology*, which is given by *empiricism* [215]. More precisely, while math is a strictly logical discipline, physics is logical and ontological. This restriction on existence produces many difficulties. For instance, logical consistence is no longer sufficient in order to establish a given sentence as physically true: ontological consistence is also needed (see [178] for a comparison between philosophical aspects of math and physics). Thus, *even though a sentence is logically consistent, in order to be considered physically true it must be consistent with all possible experiments!* This fact can be expressed in terms of a commutative diagram:

$$
\begin{tikzcd}
\textbf{philosophy} \arrow[r, Rightarrow] \arrow[d, Rightarrow] & \textbf{logic} \arrow[d, Rightarrow, "\substack{\text{mathematical}\\ \text{modelling}}"] \\
\textbf{ontology} \arrow[r, Rightarrow, "\text{empiricism}"'] & \textbf{physics}
\end{tikzcd}
$$

But now, recall that logic determines the validity of mathematical arguments, which was also translated in terms of the commutative diagram (1). So, gluing the diagram of physics with the diagram of math, we have a new commutative diagram:

$$
\begin{tikzcd}
 & \textbf{conjectures} \arrow[r, Rightarrow] & \textbf{theorems} \\
 & \textbf{naive math} \arrow[u, Rightarrow] & \\
\textbf{philosophy} \arrow[r, Rightarrow] \arrow[d, Rightarrow] & \textbf{logic} \arrow[r, Rightarrow] \arrow[u, Rightarrow] \arrow[d, Rightarrow] & \textbf{axiomatic math} \arrow[uu, Rightarrow] \\
\textbf{ontology} \arrow[r, Rightarrow, "\text{empiricism}"'] & \textbf{physics} &
\end{tikzcd}
$$

One of the most important facts concerning the relation between physics and mathematics is that, in the above diagram, the arrow

$$\textbf{logic} \Longrightarrow \textbf{physics}\ .$$

has an inverse

$$\textbf{logic} \Longleftarrow \textbf{physics}\ .$$

This last arrow is called *physical insight*. The composition of physical insight with axiomatic math produces a new arrow, called *mathematical physics*.

$$\begin{tikzcd}
\textbf{logic} \arrow[r, Rightarrow] & \begin{matrix}\textbf{axiomatic} \\ \textbf{math}\end{matrix} \arrow[dl, Rightarrow, "\text{math. phys.}"] \\
\textbf{physics} \arrow[u, Rightarrow, "\text{insight}"] &
\end{tikzcd}$$

Therefore, the full relation between physics and mathematics is given by the following diagram

$$\begin{tikzcd}
 & \textbf{conjectures} \arrow[r, Rightarrow] & \textbf{theorems} \\
 & \begin{matrix}\textbf{naive} \\ \textbf{math}\end{matrix} \arrow[u, Rightarrow] & \\
\textbf{philosophy} \arrow[r, Rightarrow] \arrow[d, Rightarrow] & \textbf{logic} \arrow[u, Rightarrow] \arrow[r, Rightarrow] \arrow[d, Leftrightarrow] & \begin{matrix}\textbf{axiomatic} \\ \textbf{math}\end{matrix} \arrow[uu, Rightarrow] \arrow[dl, Rightarrow] \\
\textbf{ontology} \arrow[r, Rightarrow, "\text{empiricism}"'] & \textbf{physics} &
\end{tikzcd}$$

leading us to the following conclusion:

Conclusion: *we can use physical insight in order to do naive mathematics and, therefore, in order to produce conjectures. These conjectures can eventually be proven, producing theorems, which in turn can be used to create mathematical models for physics* (*mathematical physics*).

The axiomatization problem of Physics[1] concerns precisely in building and studying concrete realizations of the following sequence.

$$\begin{tikzcd}
\textbf{physics} \arrow[r, Leftrightarrow] \arrow[d, Rightarrow, dashed] & \textbf{logic} \arrow[r, Rightarrow] & \begin{matrix}\textbf{naive} \\ \textbf{math}\end{matrix} \arrow[dl, Rightarrow] \\
\textbf{theorems} & \textbf{conjectures} \arrow[l, Rightarrow] &
\end{tikzcd}$$

Since this sequence depends explicitly on a background language and since there is a duality between algebra and geometry, it is natural to expect that this duality induces a duality in the level of

[1]As will be discussed in next section, there exists another different (but strictly related) problem: the *unification problem of physics*.

mathematical physics, as shown in the following diagram.

$$\begin{tikzcd}
\textbf{physics} \arrow[r, Leftrightarrow] \arrow[d, Rightarrow, shift right, "\text{geometry}"'] \arrow[d, Rightarrow, shift left, "\text{algebra}"] & \textbf{logic} \arrow[r, Rightarrow] & \textbf{naive math} \arrow[dl, Rightarrow] \\
\textbf{theorems} & \textbf{conjectures} \arrow[l, Rightarrow, shift right, "\text{algebra}"'] \arrow[l, Rightarrow, shift left, "\text{geometry}"] &
\end{tikzcd}$$

This is really the case. At the quantum level it dates from the dual Heisenberg and Schrödinger approaches to quantum mechanics (QM) and it extends to the context of Quantum Field Theories (QFT) with the so-called *Functorial* (or *Topological*) QFT (FQFT or TQFT) and *Algebraic QFT* (AQFT). Both of them are based on a functorial approach capturing a locality principle. More precisely, a TQFT is a functor F assigning to each instant of time t a Hilbert space $F(t) = \mathcal{H}_t$ and to each trajectory $\Sigma : t \to t'$ an operator $F(\Sigma) = U(t';t)$ representing time evolution [7, 238]. On the other hand, an AQFT can be defined as a functor assigning to each small region $U \subseteq M$ of spacetime the corresponding local algebra $\mathcal{A}(U)$ of observables, such that causality conditions are satisfied [18, 74].

At the classical level, it begins with the Gelfand duality between the configuration space of Classical Mechanics M and its algebra of observables $C^\infty(M)$ and generalizes to a duality between an arbitrary symplectic manifold (M, ω) and its corresponding Poisson algebra of observables.

As we will see, in order to take the axiomatization problem seriously one needs to consider arbitrarily abstract languages, which are "higher" versions of the classical language, so that one has to talk about "higher algebra" and "higher geometry", which should play a dual role. In this paper we will follow only the geometric side, so that we will review an approach to Hilbert's sixth problem by means of higher geometrical language. The "higher TQFT" are called *extended TQFT*. For the higher algebraic approach see [30, 34, 32, 35, 33, 31] and [54, 170, 171]. To a comparison between both approaches, see [200, 198]. In sum, we have the following table[2].

geometry	*algebra*
Schrödinger	Heisenberg
TQFT	AQFT
phase space	alg. observables
higher Poisson geometry	higher Poisson algebra

Table 1: geometry *vs* algebra

For us, "higher" means *homotopical*. Thus, "higher geometry" is about the coupling between homotopy theory and geometry [144, 229, 146, 230, 201]. But since in modern algebraic topology homotopy theory is described by ∞-category theory [163, 147, 49], "higher geometry" is also about the ∞-categorical version of geometry.

$$\boxed{\begin{matrix}\textbf{higher}\\ \textbf{geometry}\end{matrix} = \begin{matrix}\textbf{homotopy}\\ + \\ \textbf{geometry}\end{matrix} = \begin{matrix}\infty\textbf{-categorical}\\ \textbf{geometry}\end{matrix}} \tag{2}$$

[2] For a complete version of this table, see [1].

Because a TQFT is a functor (a categorical object) assigning time evolution operators to trajectories, it is natural do regard extended TQFT's as ∞-functors (which are ∞-categorical objects) assigning trajetories and higher trajectories (which are equivalent to trajectories of arbitrarily dimensional objects) to higher operators acting on ∞-Hilbert spaces. This is the approach of Baez-Dolan-Freed [7, 238, 81], formalized and classified by Lurie [149].

On the other hand, recalling that functors are sources of mathematical invariants, from (2) we conclude that homotopical invariants should play an important role. There are basically two kinds of such invariants: *homotopy groups* and *cohomology groups*. Gauge theories are about G-bundles with connections, which are classified by a flavor of cohomology: the *nonabelian differential cohomology* [42, 211, 109, 201]. On the other hand, the D-brane charges are supposed to be represented in twisted cohomology [239, 39], so that *twisted nonabelian differential cohomology* should be the natural context to describe gauge theories and string theory. What about M-theory? In recent years, Schreiber, Fiorenza and Sati showed that if the C-field of supergravity is quantized in twisted cohomotopy cohomology (assumption known as the *Hypothesis H*), then several folks involving M-theory, including anomaly cancellations, can be proved in all needed mathematical rigor [76, 75, 193, 43, 192, 194, 77]. Thus:

Conclusion: *the ∞-categorical language, when regarded as background language, seems to produce a promising approach for working in Hilbert's sixth problem.*

The remaining of this paper is organized as follows. In Section 2 we emphasize the difference between *axiomatization* and *unification*, where the latter is introduced as a particular approach to the former. In Section 3 we discuss why the classical set-theoretic language is not enough to attack Hilbert's sixth problem, and in Section 4 we consider the categorical language as the next natural candidate. We discuss that it is a nice choice when trying to axiomatize *particle* physics, but in Section 5 we argue that it is not sufficient to the entire Hilbert's sixth problem, so that one has to consider more abstract languages. Section 6 is devoted to the discussion of an "abstractification process", which assigns to each language a more abstract language by means of *categorification* [14, 160]. Iterating the process and taking the limit one gets an infinitely abstract language: the ∞-categorical language. It is argued that this language solves the problems of Section 5 faced by categorical language. In Section 7 the role of abstract cohomology theory in the description of classical theories is introduced and in Section 8 the quantization problem is briefly considered and reviewed. Finally, some concluding remarks are presented in Section 9.

2 Axiomatization *vs* Unification

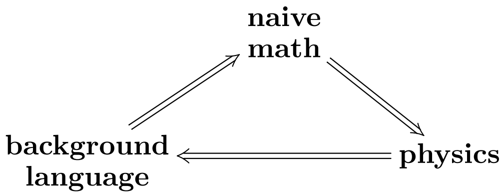
Recall that the existence of physical insight gives the "mathematical physics" arrow, as shown below. An approach to Hilbert's sixth problem can then be viewed as a way to present mathematical physics as a *surjective* arrow.

$$\begin{array}{c}\textbf{axiomatic}\\ \textbf{math}\end{array} \xRightarrow[\text{physics}]{\text{mathematical}} \textbf{physics}$$

Because axiomatic math is described by some kind of logic, the starting point is to select a proper background language. The selected background language determines directly the *naive*

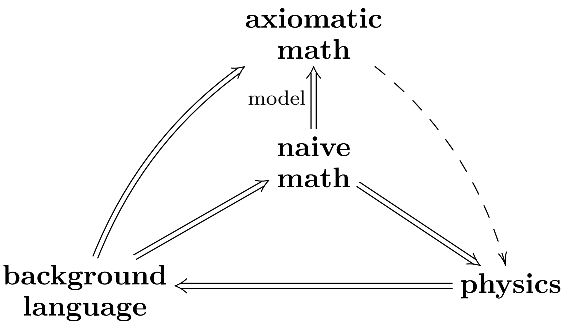
6

math, so that the next step is to analyze the following loops:

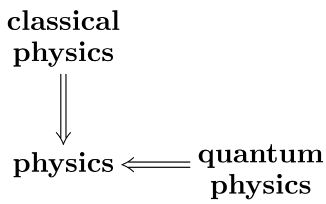


(3)

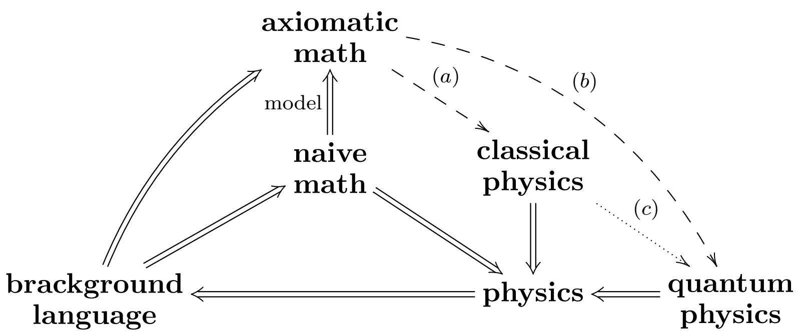
Then, once some model has been selected, we lift it to axiomatic math, as in the first diagram below. The final step is to verify if the corresponding “mathematical physics” arrow is surjective or not. In other words, we have to verify if the axiomatic concepts produced by the selected logic are general enough to model all physical phenomena.



Presently, physical theories can be divided into two classes depending on the scale of energy involved: the *classical theories* and the *quantum theories*, as shown below.



Thus, there are essentially two ways to build a surjective “mathematical physics” arrow. Either we build the arrow directly (as in the diagram above), or we first build surjective arrows (a) and (b), which respectively axiomatize classical and quantum physics, and then another arrow (c) linking these two axiomatizations, as in the diagram below.



We emphasize the difference between the two approaches: in the first one, all physical theories are described by the *same* set of axioms, while in the second classical and quantum theories are described by *different* axioms, but that are related by some process. This means that if we choose the first approach we need to *unify all physical theories*.

There are some models to this unification arrow, but they are not our focus. Just to mention one, string theory is maybe the most well-known. It is based on the assumption that the "building blocks" of nature are not particles, but rather one dimensional entities called *strings*, i.e, connected one-dimensional manifolds, which are diffeomorphic to some interval (when it has boundary or when it is not compact) or to the circle (if it is boundaryless and compact). In the first case we say that we have *open strings*, while in the second we say that we have *closed strings*.

From the mathematical viewpoint, string theory is very fruitful - see [58, 191, 241, 67] for general discussions and [62] for a brief review. This mathematical interest arises in part from the following facts. Since strings are 1-dimensional objects, their worldsheets are 2-dimensional manifolds (typically oriented). In the case of closed strings, these are Riemann surfaces, whose moduli space is finite-dimensional [60, 172]. Since the path integrals are defined on the moduli space of trajectories, it follows that for closed strings they are well-defined [61]. Not only this, they compute many geometric-topological invariants [154, 164, 69, 46], playing a fundamental role in many areas of math, such as complex algebraic geometry [154, 46, 55], complex dynamics [70, 71, 108] and symplectic topology [87, 226, 164]. See [233] for a nice pedagogical overview. In the case of open strings, they start and end in high dimensional objects, called *D-branes*, which fits into a (higher) category called *Fukaya category*, also playing a fundamental role in symplectic topology. Besides that, they appear in Mirror Symmetry, which became a highly active branch of algebraic geometry since the seminal contribution [127] of Kontsevich. See also [110, 55, 122].

On the other hand, one needs to recall that physics is not completely determined by the arrow **logic** $\Rightarrow$ **physics**, but there is also the ontological (i.e, empirical) branch, and presently there is no concrete empirical proof that strings (instead of particles) really are the most fundamental objects of nature [231, 8].

In this text, we will describe an attempt to realize the second approach. As we will see, if one starts with a sufficiently abstract (or powerful) background language, then one can effectively axiomatize *separately* a huge amount of classical and quantum theories, leaving the problem of finding a suitable quantization process, which is the really difficult part[3]. At least in the approach that will be discussed here, this difficulty comes in part from the fact that the underlying background language is itself under construction. Even so, there is a very promising process, intended to generalize geometric quantization from geometry to higher geometry, known as *motivic* or *cohomological* or *pull-push quantization* [201, 176].

3 Towards The Correct Language

As commented in the last section, independently of the approach used in order to attack Hilbert's sixth problem, the starting point is to select a proper background language. The most obvious choice is the *classical logic* used to describe set theory. This logic produces, via the arrow **logic** $\Rightarrow$ **axiomatic math**, not only set theory, but indeed all classical areas of math, such as group theory, topology and differential geometry, since each of them can be described in terms of set theory.

[3] Notice that one of the Millennium Problems is part of this quest [115].

Therefore, for this choice of background language, the different known areas of mathematics could be used in order to describe classical and quantum physical phenomena.

Since 1900, when Hilbert published his list of problems, many classical and quantum theories were formalized by means of these by now well known areas of mathematics. Indeed, quantum theories were observed to have a more algebraic and probabilistic nature, while classical theories were presumably more geometric in character.

For instance, a system in quantum mchanics (which is about quantum *particles*) can be formalized in the Schrödinger approach as a pair $(\mathcal{H}, \hat{H})$, where $\mathcal{H}$ is a complex (separable) Hilbert space and $\hat{H} : D(\hat{H}) \to \mathcal{H}$ is a self-adjoint operator defined in a (dense) subspace $D(\hat{H}) \subseteq \mathcal{H}$ [181, 103]. We say that $\hat{H}$ is the *Hamiltonian* of the system and the fundamental problem is to determine its *spectrum*, which on one hand is the set of all information about $\hat{H}$ that can be accessed experimentally and on the other hand is the spectrum (in the mathematical sense, i.e, the generalized eigenvalues) of $\hat{H}$ [227, 57]. The dynamics of the system from a instant t_0 to a instant t_1 is guided by the unitary operator $U(t_1; t_0) = e^{\frac{i}{\hbar}(t_1 - t_0)\hat{H}}$ associated to $\hat{H}$ or, equivalently, by the (time independent) Schrödinger equation $i\hbar\frac{d\psi}{dt} = \hat{H}\psi$.

On the other hand, classical theories for *particles* are given by some *action functional* $S : \mathrm{Fields}(M) \to \mathbb{R}$, defined in some "space of configurations" (or "space of fields") over a spacetime M. These configurations (or fields) generally involve paths $\gamma : I \to M$, interpreted as the trajectories of particles moving in some spacetime. If the particles are not free, i.e, if they are subjected to some interaction, we have also to consider its source as part of the configurations. The presence of an interaction can be measured by means of a force. Furthermore, it may (or not) be intrinsic to the spacetime M. For the geometrical aspects of classical field theories, see [91, 153, 104, 16] and the standard references [173, 79].

For instance, since the development of general relativity in 1916, gravity is supposed to be an intrinsic force[4], meaning that it will act on *any* particle. The presence of an intrinsic force is formalized by the assumption of a certain additional geometric structure on the spacetime M - in the case of gravity, a Lorentzian metric g on M. Other intrinsic interactions are modeled by other types of geometric structure. But not all manifolds may carry a given geometric structure, specially if integrability conditions are required [102, 221, 56]. This means that not all manifolds can be used to model the spacetime. Indeed, each geometric structure exists on a given manifold only if certain quantities, called *obstruction characteristic classes* vanish. For example, a compact manifold admits a Lorentzian metric iff its Euler characteristic $\chi(M)$ vanishes, which implies that $\mathbb{S}^4$ cannot be used to model the universe [29, 56]. On the other hand, M has a spin structure (needed to consider spinorial fields on M) iff its first two Stiefel-Whitney classes are null [133]. Furthermore, if one tries to model gravity using geometries other than the Lorentzian one the obstructions are even stronger [157, 156].

An important class of non-intrinsic interactions are given by Yang-Mills theories (or more general gauge theories). These depend on a Lie group G, called the *gauge group*, and on a G-principal bundle $P \to M$ over the spacetime M. The interaction is modelled by a connection on P, which is just a vertical equivariant $\mathfrak{g}$-valued 1-form $A : TP \to \mathfrak{g}$, where $\mathfrak{g}$ is the Lie algebra of G. We can think of A as the *potential* of the interaction, so that the *force* is just the (exterior covariant) derivative $d_A A$ of A. Here, the standard example is electromagnetism, for which G is the abelian group $U(1)$ and A is the eletromagnetic vector potential. We can think of an arbitrary Yang-Mills interaction as some

[4]There is an old but active discussion whether gravity is actually an external force (gauge theory). It is not the focus of this review. See [130, 66, 232, 114, 190, 189].

kind of "nonabelian" version of electromagnetism, leading, e.g, to the development of nonabelian Hodge theory [214, 212, 89, 111] and to the formulation of powerful topological invariants, such as Donaldson invariants [65, 64, 85] and Seiberg-Witten invariants [208, 63, 226]. Note that if one tries to extend or modify the Yang-Mills-type interactions one also finds obstructions [158, 159].

Now, let us return to focus on Hilbert's sixth problem. It requires answering for questions like these [201, 202]:

1. What is a classical theory?
2. What is a quantum theory?
3. What is quantization?

Notice that the previous discussion *does not* answer these questions. Indeed, it only reveals some examples and aspects of what classical and quantum theories should be; it does not say, axiomatically, what they really are. Furthermore, the presence of obstructions for freely extending the Lorentzian geometry of gravity and Yang-Mills theories suggests the existence of another background language in which these obstructions disappear. This leads us to the following conclusion:

Conclusion: *classical logic, viewed as a background language, is very nice in order to formulate and study properties of particular classical and quantum theory. On the other hand, a priori it does not give tools to study more deeper questions as those required by Hilbert's sixth problem.*

4 The Role of Categorical Language

The above conclusion is that, in order to attack Hilbert's sixth problem, the natural strategy is to replace classical logic by a more abstract background language. *What kind of properties should this new language have?*

Recall that by making use of classical logic we learn that classical theories are generally described by *geometric notions*, while quantum theories are described by *algebraic and probabilistic tools*. Since the quantization process is some kind of process linking classical theories to quantum theories, it should therefore connect geometric areas to algebraic/probabilistic areas. So, the idea is to search for a language which formalizes the notion of "area of mathematics" and the notion of "map between two areas".

This language actually exists: it is *categorical language*. In categorical language, an area of mathematics is determined by specifying which are the objects of interest, which are the morphisms between these objects and which are the possible ways to compose two given morphisms. This data defines a *category*. The link between two areas of mathematics described by categories $\mathbf{C}$ and $\mathbf{D}$ is formalized by the notion of *functor*. This is given by a rule $F : \mathbf{C} \to \mathbf{D}$ assigning objects into objects and mappings into mappings in such a way that compositions are preserved. See [37, 168, 150] for classical books on category theory and [186, 142, 185, 155] for more recent, introductory and pedagogical expositions. See [139] for a very elementary text. See [50, 51] for introductions devoted to physicists.

Notice that we have a category **Set**, describing set theory, whose fundamental objects are sets, whose morphisms are just maps between sets and whose composition laws are the usual compositions between functions. That all classical areas of mathematics can be described by categorical language comes from the the fact that in each of them the fundamental objects are just sets endowed with some further structure, while the morphisms are precisely the maps between the underlying sets which preserve this additional structure. For instance, linear algebra is the area of mathematics which study vector spaces and linear maps. But a vector space is just a set endowed with a linear structure, while a linear map is just a map preserving the linear structure.

Therefore, each classical area of math defines a category $\mathbf{C}$ equipped with an inclusion functor $\imath : \mathbf{C} \hookrightarrow \mathbf{Set}$ which only forgets all additional structures (in the context of linear algebra, this functor forgets the linear structure). The categories which can be included into $\mathbf{Set}$ are called *concrete*. Thus, *in order to recover classical logic from categorical language it is enough to restrict the latter to the class of concrete categories.*

The fact that categorical language is really more abstract than classical logic comes from the existence of *non-concrete* (also called *abstract*) categories. There are many examples of them. For instance, given a natural number p, we can build a category $\mathbf{Cob}_{p+1}$ whose objects are p-dimensional smooth manifolds and whose mappings $\Sigma : M \to N$ are *cobordisms* between them, i.e, $(p+1)$-manifolds Σ such that $\partial\Sigma = M \sqcup N$. The abstractness of this category comes from the fact that the morphisms are not mappings satisfying some condition, but actually higher dimensional manifolds. For a formal definition of $\mathbf{Cob}_{p+1}$ with applications to homotopical aspects of cobordism theory, see [83]. See also [222, 188, 88]. We recall that the study of cobordisms dates from the seminal work of Thom [6, 228].

Observe that for $p = 0$ the objects of $\mathbf{Cob}_1$ are just 0-manifolds: finite collection of points. The cobordisms between them are 1-manifolds having these 0-manifolds as boundaries. In other words, the morphisms are just disjoint unions of intervals, while the composition between intervals $[t_0; t_1]$ and $[t_1; t_2]$ is the interval $[t_0; t_1]$.

With categorical language on hand, let us try to attack Hilbert's sixth problem. We start by recalling that the dynamics of a system in quantum mechanics is guided by the time evolution operators $U(t_1; t_0) = e^{\frac{i}{\hbar}(t_1 - t_0)\hat{H}}$. Notice that when varying t_0 and t_1, all corresponding operators can be regarded as a unique functor $U : \mathbf{Cob}_1 \to \mathbf{Vec}_{\mathbb{C}}$, where $\mathbf{Vec}_{\mathbb{C}}$ denotes the category delimiting complex linear algebra (i.e, it is the category of complex vector spaces and linear transformations). Such a functor assigns to any instant t_0 a complex vector space $U(t) = \mathcal{H}_t$ and to any interval $[t_0; t_1]$ an operator $U(t_1; t_0) : \mathcal{H}_0 \to \mathcal{H}_1$.

At this point, the careful reader could make some remarks:

1. As commented previously, a system in quantum mechanics is defined by a pair $(\mathcal{H}, \hat{H})$, where we have a single space $\mathcal{H}$ which does not depend on time, so that for any interval $[t_0; t_1]$ the time evolution operator $U(t_1; t_0)$ is defined in the same space. On the other hand, for a functor $U : \mathbf{Cob}_1 \to \mathbf{Vec}_{\mathbb{C}}$ we have a space $\mathcal{H}_t$ for each instant of time and, therefore, for any interval the corresponding operators are defined on different spaces.

2. In quantum mechanics, the evolution is guided not by an arbitrary operator, but by a unitary operator. Furthermore, in Quantum Mechanics the space $\mathcal{H}$ for systems describing more than one (say k) particles decomposes as a tensor product $\mathcal{H}^1 \otimes ... \otimes \mathcal{H}^k$. However both conditions are not contained in the data defining a functor $U : \mathbf{Cob}_1 \to \mathbf{Vec}_{\mathbb{C}}$.

About the first remark, notice that if there is an interval $[t_0; t_1]$ connecting two different time instants t_0 and t_1, then there exists a inverse interval $[t_0; t_1]^{-1}$ such that, when composed (in $\mathbf{Cob}_1$) with the original interval, gives precisely the trivial interval. In fact, the inverse is obtained simply by flowing time in the inverse direction. In more concise terms, any morphism in the category $\mathbf{Cob}_1$ is indeed an isomorphism. But functors preserve isomorphisms, so that for any interval $[t_0;t_1]$ the corresponding spaces $\mathcal{H}_{t_0} = U(t_0)$ and $\mathcal{H}_{t_1} = U(t_1)$ are isomorphic. This means that the spaces $\mathcal{H}_t$ actually do not dependent on the time t: up to isomorphisms they are all the same.

Concerning the second remark, let us say that there are some reasons to believe that the unitarity of time evolution should not be *required* as a fundamental axiom of the true fundamental physics, but instead it should *emerge* as a consequence (or as an additional assumption) of the correct axioms. For instance, in quantum mechanics itself the unitarity of $U(t_0;t_1)$ can be viewed as a consequence of the Schrödinger equation, because its solutions involve the exponential of a Hermitian operator which by Stone's theorem is automatically unitary [227]. There are even more fundamental reasons involving the possibility of topology change in quantum gravity [9].

On the other hand, it is really true that an arbitrary functor $U : \mathbf{Cob}_1 \to \mathbf{Vec}_{\mathbb{C}}$ does not take into account the fact that in a system with more than one particle the total space of states decomposes as a tensor product of the state of spaces of each particle. In order to incorporate this condition, notice that when we say that we have a system of two fully independent particles, we are saying that time intervals corresponding to their time evolution are *disjoint*. Thus, we can interpret the time evolution of a system with many particles as a disconnected one dimension manifold, i.e, as a disconnected morphism of $\mathbf{Cob}_1$. Consequently, the required condition on the space of states can be obtained by imposing the properties

$$U(t \sqcup t') \simeq U(t) \otimes U(t') \quad \text{and} \quad U(\varnothing) \simeq \mathbb{C},$$

where the second condition only means that a system with zero particles must have a trivial space of states.

Now, notice that both categories $\mathbf{Cob}_1$ and $\mathbf{Vec}_{\mathbb{C}}$ are equipped with binary operations (respectively given by $\sqcup$ and $\otimes$), which are associative and commutative up do isomorphisms, together with distinguished objects (given by $\varnothing$ and $\mathbb{C}$), which behave as neutral elements for these operations. A category with this kind of structure is called a *symmetric monoidal category*. A functor between monoidal categories which preserves the operations and the distinguished object is called a *monoidal functor* (see [5, 150, 73] for a formal definition of monoidal categories and monoidal functors, and [209] for an interesting survey on the different flavors of monoidal structures). Therefore, these observations lead us to the following conclusion:

Conclusion. *A system in quantum mechanics is a special flavor of monoidal functor from the category $\mathbf{Cob}_1$ of 1-dimensional cobordisms to the category $\mathbf{Vec}_{\mathbb{C}}$ of complex vector spaces.*

Consequently, eyeing Hilbert's sixth problem we can use the above characterization in order to *axiomatize* a quantum theory of *particles* as being an arbitrary monoidal functor

$$U : (\mathbf{Cob}_{p+1}, \sqcup, \varnothing) \to (\mathbf{C}, \otimes, 1) \tag{4}$$

for $p = 0$ and taking values in some symmetric monoidal category. This is precisely the 1-dimensional version of the functorial topological quantum field theories (TQFT) of Atiyah-Witten [7, 238], inspired in the functorial formulation of Conformal Field Theories by Segal [207, 206, 36]. See also [9] for a comprehensive introduction. For the classification of the functors (4) in the case $p = 1$, see [2, 126, 132] and compare with the 2-dimensional conformal case [206]. For discussions on the beautiful math arising in the $p = 2$ case, see [19, 41, 23, 125, 20].

A natural question is on the viability of using the same kind of argument (and therefore the same background language) in order to get an axiomatization of the classical theories of *particles*. This really can be done, as we will outline following ideas of Chapter 1 of [201]. We start by recalling that a classical theory of particles is given by an action functional $S : \mathrm{Fields}(M) \to \mathbb{R}$, defined in some space of fields. Therefore, the first step is to axiomatize the notion of *space of fields*. In order to do this, recall that it is generally composed by smooth paths $\gamma : I \to M$, representing the trajectories of the particle into some spacetime M, and by the interacting fields, corresponding to configurations of some kind of geometric structure put in M. The canonical examples of interacting fields are metrics (describing gravitational interaction) and connections over bundles (describing gauge interactions), as discussed in Section 3.

Now we ask: *which property smooth functions, metrics and connections have in common?* The answer is: *locality*. In fact, in order to conclude that a map $\gamma : I \to M$ is smooth, it is enough to analyze it relative to an open cover $U_i \hookrightarrow M$ given by coordinate charts $\varphi_i : U_i \to \mathbb{R}^n$. Similarly, a metric g on M is totally determined by its local components g_{ij}. Finally, it can also be shown that to give a connection $A : TP \to \mathfrak{g}$ is the same as giving a family of 1-forms $A_i : U_i \to \mathfrak{g}$ fulfilling compatibility conditions at the intersections $U_i \cap U_j$. Therefore, one can say that a space of fields over a fixed spacetime M is some kind of set $\mathrm{Fields}(M)$ of structures which are local in the sense that, for any cover $U_i \hookrightarrow M$ by coordinate systems, we can reconstruct the total space $\mathrm{Fields}(M)$ from the subset of all $s_i \in \mathrm{Fields}(U_i)$ that are compatible in the intersections $U_i \cap U_j$.

Notice that we are searching for a notion of *space of fields* (without mention of any spacetime), but up to this point we got a notion of space of fields over a *fixed* spacetime. The main idea is to consider the rule $M \mapsto \mathrm{Fields}(M)$ from which follows the immediate hypothesis that it is functorial. Therefore, we could axiomatize a space of fields as a functor

$$\mathrm{Fields} : \mathbf{Diff}^{op} \to \mathbf{Set}$$

assigning to any manifold a set of local structures. This kind of functor is called a *sheaf on the site of manifolds and coordinate coverings* (or a *smooth set*, a *smooth sheaf* or even a *generalized smooth space*). See [152, 117, 123] for general discussions on sheaves on sites, and [11, 131, 201] for smooth sets and generalized smooth spaces. See also [160].

This is still not the correct notion of space of fields, however. Indeed, recall that in typical situations the set $\mathrm{Fields}(M)$ contains geometric structures. But when doing geometry we only consider the entities up to their natural equivalences (their "congruences"). This means that for a fixed space M we should take the quotient space

$$\mathrm{Fields}(M)/\mathrm{congruences}. \tag{5}$$

In order to do calculations with a quotient space we have to select an element within each equivalence class (a representative) and then prove that the results of the computations do not depend on this choice. The problem is that when we select an element we are automatically privileging it, but *there is no physically privileged element*. So, in order to be physically correct we have to work with all elements of the equivalence class *simultaneously*. This can be done by replacing the set (5) with the category whose objects are elements of $\mathrm{Fields}(M)$ and such that there is a mapping between s, s' iff they are equivalent. In this category, all mappings are obviously isomorphisms, so that it is indeed a *groupoid*. The sheaves on the site of manifolds which take values in $\mathbf{Gpd}$ are called *smooth stacks* or *smooth groupoids*, and they finally give the correct way of thinking about the space of fields for the case of *particles* [165]. See Chapter 1 of [201] for a very nice discussion and [160] for a pedagogical approach.

Therefore, in order to finish the axiomatization of the "classical theory" notion for particles we need to define what is the action functional. Given a spacetime M this should be some kind of map between the space of fields $\mathrm{Fields}(M)$ to $\mathbb{R}$. Notice that $\mathbb{R}$ is a *set*, but we promoted the space of fields to a *groupoid*. So, in order to define a map between these two entities we also need to promote $\mathbb{R}$ to a groupoid. This is done defining a groupoid whose objects are real numbers and having only trivial morphisms.

We can then finally axiomatize a classical theory of *particles* as being given by a smooth stack Fields of fields and a rule S assigning to every manifold M the action functional

$$S_M : \mathrm{Fields}(M) \to \mathbb{R}.$$

On the other hand, from the higher Yoneda embedding, every smooth groupoid X can be regarded as a smooth ∞-stack $y(X)$ [147]. Thus, $\mathbb{R}$ itself can be regarded as smooth ∞-stack $y(\mathbb{R})$ and the action functional is just a morphism $S : \mathrm{Fields} \Rightarrow y(\mathbb{R})$ between smooth ∞-stacks [201].

5 The Need of Highering

Up to this point we have seen that starting with categorical logic as the background language we can effectively axiomatize the notions of classical and quantum theories for *particles*. Indeed, a quantum theory is a symmetric monoidal functor $U : \mathbf{Cob}_1 \to \mathbf{Vec}_{\mathbb{C}}$, while a classical theory is given by a smooth stack Fields and by an action functional

$$S : \mathrm{Fields}(-) \to \mathbb{R}.$$

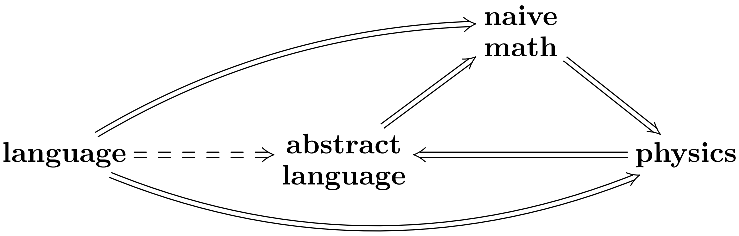
On the other hand, the nonexistence of *clues* that strings and higher dimensional objects are the correct building blocks of nature cannot be used to completely exclude them. Thus, any approach to Hilbert's sixth problems which intends to be stable under new physics should be based on a background language which allows the axiomatization of classical and quantum theories not only for *particles*, but for objects of arbitrary dimension. In this regard, categorical logic fails as background language. Indeed, we have at least the following problems (see also [201, 191, 15, 219, 160]):

- *in quantum theories*. Recall that for any p we have the category $\mathbf{Cob}_{p+1}$, so that we could immediately extend the notion of quantum theory for particles by defining a quantum theory for *p-branes* as a symmetric monoidal functor $U : \mathbf{Cob}_{p+1} \to \mathbf{Vec}_{\mathbb{C}}$, which coincides with the Atiyah-Witten p-dimensional TQFT [7, 238]. The problem is that in this case we are only replacing the assumption that particles are the correct building blocks of nature with the assumption that p-branes are the correct building blocks of nature. Indeed, in both cases we can only talk about quantum theories for a specific kind of object, while Hilbert's sixth problem requires an *absolute notion of quantum theory* (this is related with the locality principle and with the principle of general covariance [9, 200, 197, 203]);

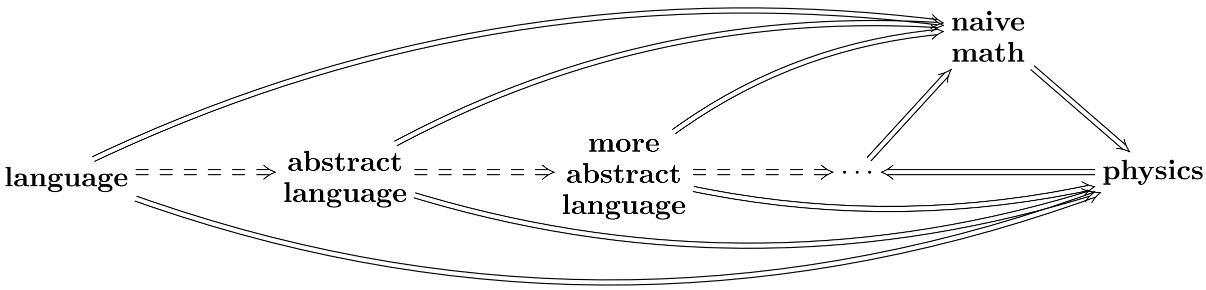
- *in classical theories*. The trajectories of particles (which are 0-dimensional objects) on a spacetime M is described by smooth paths $\varphi : I \to M$, which are smooth maps defined on a 1-dimensional manifold I. Therefore we can easily define the trajectory of p-branes (which are p-dimensional objects) on M as smooth maps $\varphi : \Sigma \to M$ defined on a $(p+1)$-dimensional manifold Σ, leading us to sigma-model theories [58]. The problem is that in the space of fields of a classical theory we have to consider not only the trajectories but also the interactions which act on the object. For particles we could consider these interactions as modeled by connections because the *notion of connection is equivalent to the notion of parallel transport along paths*. But, to the best of our knowledge, there is no obvious global notion of parallel transport along arbitrary higher dimensional manifolds[5].

Conclusion. *In order to axiomatize the notions of classical and quantum theories for higher dimensional objects we need to start with a language which is more abstract than categorical language.*

The immediate idea that comes in mind is to try building some kind of process which takes a language and returns a more abstract language in such a way that set-theoretical things are replaced by their categorical analogue. Before discussing how this can be done, recall that for a selected background language, the first step in solving Hilbert's sixth problem is to consider the loop (3) involving naive math, the language itself and physics. So, having constructed a more abstract language from a given one, we would like to consider loops for the new language as arising by extensions of the loop for the starting language, as in the diagram below.



This condition is very important, because it ensures that the physics axiomatized by the initial language is contained in the physics axiomatized by the new language. So, this means that when applying the process of "abstractification" we are getting languages that axiomatize more and more physics. Consequently, by iterating the process and taking the limit one hopes to get a background language which is abstract enough to axiomatize the whole physics. See the discussion at the introduction and section 2.3 of [160].



6 Highering

In the last sections we saw that category theory is useful to axiomatize particle physics, but not string physics and physics of higher-dimensional objects, so that we need to build some "abstrac-

[5]Actually, all attempts to higher dimensional parallel transport that we know make use, at least implicitly, of higher categorical structures [151, 205, 204, 12, 92, 217, 124].

tification process" which will be used to replace category theory by other more abstract theory. Notice that categorical language is more abstract than classical language, so that learning how to characterize the passage from classical logic to categorical logic should help knowing how to iterate the construction, getting languages more abstract than categorical language. In other words, the main approach to the "abstractification process" should be some kind of *categorification process*.

In order to get some feeling for this categorification process, notice that a set contains less information than a category. Indeed, sets consist of only one type of information: their elements. On the other hand, categories have three layers of information: objects, morphisms and compositions. Thus, we can understood the passage from set theory to category theory (and, therefore, from classical logic to categorical logic) as the *addition of information layers* (see [14] for an interesting discussion; see also Part 2 of [160]).

When iterating this process we expect to get a language describing entities containing more information than usual categories. Indeed, we expect to have not only objects, morphisms between objects and compositions between morphisms, but also morphisms between morphisms (called 2-*morphisms*) and compositions of 2-morphisms. Thus, if we call such entities 2-*categories*, adding another information layer we get 3-*categories*, and so on. Taking the limit we then get ∞-*category*. The basic example, which reveals the connection between higher category theory and homotopy theory, it the ∞-categories of topological spaces, continuous maps, homotopies, homotopies between homotopies, and so on.

Similarly, if a functor between categories maps objects into objects and morphisms into morphisms, then a k-*functor* between k-categories maps objects into objects and i-morphisms into i-morphisms for every $1 \leq i \leq k$. Taking the limit $k \to \infty$, we can define ∞-*functors*. Since the categorical language is about categories and functors (and natural transformations), then we obtain a ∞-*categorical language*, consisting of ∞-categories and ∞-functors (and higher natural transformations), which by constructions is arbitrarily abstract. See Part 3 of [160] for a more detailed intuitive discussion on this new language. We then have the following conclusion:

Conclusion. *A natural candidate for a background language sufficient to solve Hilbert's sixth problem is ∞-categorical language.*

With this conclusion in mind we need to build a math-language-physics loop (3) for the ∞-categorical language by means of extending the loop obtained using categorical language. This can be done as follows:

- *in quantum theories*: Recall that the problem with the definition of quantum theories as symmetric monoidal functors $U : \mathbf{Cob}_{p+1} \to \mathbf{Vec}_{\mathbb{C}}$ involves the fact that such functors take into account only p-branes for a *fixed* p, implying that we need to know previously what are the correct building blocks of nature. This can be avoided in the ∞-categorical context. Indeed, we can define a ∞-category $\mathbf{Cob}(\infty)$ having 0-manifolds as objects, 1-cobordisms (i.e., cobordisms between 0-manifolds) as morphisms, 2-cobordisms (i.e., cobordisms between 1-cobordisms) as 2-morphisms and so on. Notice that differently from $\mathbf{Cob}_{p+1}$ (which contains only p-manifolds and cobordisms between them), the defined ∞-category $\mathbf{Cob}(\infty)$ contains cobordisms of all orders and, therefore, describe p-branes for every p simultaneously. Thus, one can define an *absolute (or extended) quantum theory* (as required by Hilbert's sixth problem) as some kind of ∞-functor $U : \mathbf{Cob}(\infty) \to \infty\mathbf{Vec}_{\mathbb{C}}$, where $\infty\mathbf{Vec}_{\mathbb{C}}$ is some ∞-categorical version of $\mathbf{Vec}_{\mathbb{C}}$ (i.e, is some ∞-category of ∞-vector spaces). See [81, 13, 149].

- *in classical theories*: The problem with the axiomatization of classical theories via categorical language was that in the space of fields we have to consider interacting fields. For particles, these fields are modeled by connections on bundles, which are equivalent to parallel transport along paths. But, as commented, there is no canonical notion of transportation along arbitrary higher dimensional manifolds [151, 205, 204, 12, 92, 217, 124]. Another way to motivate the problem is the following: in order to define a connection A locally (i.e, in terms of data over an open covering $U_i \hookrightarrow M$) we need a family of 1-forms A_i on U_i subject to compatibility conditions on $U_i \cap U_j$. The information U_i and $U_i \cap U_j$ belong to an usual category $\check{C}(U_i)$, whose objects are elements $x_i \in U_i$ and there is a morphism $x_i \to x_j$ iff $x_i, x_j \in U_i \cap U_j$. On the other hand, if we try to define higher transportation locally we need to take into account data on U_i which is compatible not only on $U_i \cap U_j$, but also on $U_i \cap U_j \cap U_k$, on $U_i \cap U_j \cap U_k \cap U_l$, and so on. Thus, we need to consider much more information layers than can be put inside an usual category, justifying the nonexistence of connections along arbitrary higher dimensional manifolds. But they can be put inside a ∞-category $\check{C}_\infty(U_i)$, meaning that we actually have a notion of ∞-connection when we consider ∞-categorical language (see [12, 204, 195, 205]). More precisely, the initial problem is avoided if we define the space of fields not as a smooth stack, but as a *smooth ∞-stack* (or smooth *∞-groupoid*: a ∞-functor $\mathrm{Fields} : \mathbf{Diff}^{op} \to \infty\mathbf{Gpd}$ such that for any M the quantity $\mathrm{Fields}(M)$ is determined not only by $\mathrm{Fields}(U_i)$ and $\mathrm{Fields}(U_i \cap U_j)$, but also by (see [197, 203, 201, 202, 160]):

$$\mathrm{Fields}(U_i \cap U_j \cap U_k) \tag{6}$$
$$\mathrm{Fields}(U_i \cap U_j \cap U_k \cap U_l) \tag{7}$$
$$\text{and so on.} \tag{8}$$

Remark. In parts, the difficulties with this approach to Hilbert's sixth problem arise from the fact that the ∞-categorical language is still in development. Recall that a groupoid is a category such that all morphisms are invertible. When we are in a higher category we have higher morphisms, so that given a k-morphism, it can be invertible (in the classical sense) or invertible up to $k+1$-morphisms. A ∞-groupoid is a ∞-category such that each k-morphism is invertible up to $k+1$-morphism, for every $k>0$. The theory for this kind of ∞-category is well-studied and in some cases it is equivalent to the homotopy theory of topological CW-complexes, which since [10] is known as the *Homotopy Hypothesis* [182, 101] (see Section 8.4 of [160] for a discussion on its role in physics). More generally, we can define a (∞,n)-category as a ∞-category such that each k-morphism is invertible up to $k+1$-morphisms, for every $k>n$. Thus, ∞-groupoids are the same thing as $(\infty,0)$-categories. The theory of $(\infty,1)$-categories is also well-studied [118, 120, 119, 27, 140, 48, 147, 148]. But, in order to describe extended TQFT and higher prequantization we need to work on (∞,n)-categories for $n>1$ [149, 202, 197, 203, 201], whose theory is not as well developed as the theory for $n=0,1$ [24, 25, 26].

7 The Cohomological Description

In theoretical physics, the concept of "cohomology" is more well known as the cohomology groups $H^n(X^\bullet;Q)$ induced by a nilpotent operator $Q : X^\bullet \to X^\bullet$ (i.e., such that $Q^2=0$), defined in a graded vector space $X^\bullet = \oplus X^n$ (or in a graded algebra) which is typically related to the conservation of some gauge charge [218, 219, 21]. In mathematics, on the other hand, it has not only this fully algebraic meaning (which arises from homological algebra [237, 90]), but also a more topological one: the *generalized cohomology theories*. These consist of sequences of functors H^n, assigning abelian groups to topological spaces, subjected to the so-called *Eilenberg-Steenrod axioms*. Examples include the ordinary cohomology theory (which in real coefficients is equivalent to de Rham cohomology), K-theory, cobordism theory and topological modular forms [143, 4, 72]. The facts that particles are charged in ordinary cohomology and that D-brane charges can be described by K-theory show that generalized cohomology theoy is also important in physics. Thus, in view of Hilbert's sixth problem, it is natural to search for an axiomatization to cohomology which contains both flavors of cohomology as particular examples.

By Brown's representability theorem, generalized cohomologies are fully determined by sequences $\mathbb{E} = (E_n)_n$ of topological spaces known as *spectra* [143, 4, 72]. More precisely, for every $n \geq 0$ there is a topological space E_n such that for every topological space X we have $H^n(X) \simeq [X; E_n]$, where the right-hand side denotes the *set* of homotopy classes. Via Dold-Kan correspondence N, the algebraic cohomology groups $H^n(X^\bullet; Q)$ are also determined by a sequence of entities E_n such that $H^n(X^\bullet; Q) \simeq [N(X_\bullet); E_n]$, where $X_\bullet$ is the dual chain sequence of $X^\bullet$ (see Chapter 1 of [201] and Chapter 9 of [160]). The only difference is that now the representing objects are higher-categorical entities and therefore belong to a $(\infty,1)$-category instead of to a classical category, but recall that topological spaces, continuous maps, homotopies and higher homotopies also define a $(\infty,1)$-category, so that in the first case we also are in the context of ∞-categories.

Another similarity between both flavors of cohomology is the following. We say that a topological space Y is *deloopable* if there is another topological space BY whose loop space is (weak) homotopic to Y, i.e., such that $\Omega BY \simeq Y$. In a generalized cohomology theory each E_n is deloopable, with $BE_n = E_{n+1}$. In particular, $E_n = B^n E_0 = B(B...(BE_0)...)$ [162, 3]. We note that in every suitable[6] $(\infty,1)$-category we can also consider loop spaces $\mathbf{\Omega} Y$, so that we can also talk about deloopable objects [93, 147, 201]. But via Dold-Kan correspondence the spaces E_n characterizing algebraic cohomology are also deloopable, with $\mathbf{B}E_n \simeq E_{n+1}$, so that again we have $E_n = \mathbf{B}^n E_0$. This motivates us to axiomatize the notion of cohomology in any $(\infty,1)$-category as follows [40, 68, 147, 201, 160]. Given an object E_0 in an $(\infty,1)$-category, for every other object X, the 0*th cohomology of* X *with coefficients in* E_0 is given by $H^0(X;E_0) := [X;E_0]$. Analogously, if E_0 has deloopings $B^n E_0$ we define the nth cohomology as $H^n(X;E_0) := [X;E_n]$.

The amazing fact of this abstract axiomatization is that it does not only unify different flavors of cohomology, but also sets classical theories in a new powerful language [201]. More precisely, recall that, as discussed in Section 6 the space of fields is supposed to be a smooth ∞-stack $\mathrm{Fields}(-) : \mathbf{Diff}^{op} \to \infty\mathbf{Grpd}$. Via higher Yoneda embedding [147], each smooth manifold M can itself be regarded as a smooth ∞-stack $y(M)$ and $\mathrm{Fields}(M) \simeq [y(M); \mathrm{Fields}]$, so that the space of fields are actually the coefficients of a cohomology in the $(\infty,1)$-category of all smooth ∞-stacks [201]. On the other hand, $y(\mathbb{R})(M) = [y(M); y(\mathbb{R})] = H^0(M;\mathbb{R})$. Therefore, the action functional $S : \mathrm{Fields} \Rightarrow y(\mathbb{R})$ is secretly a morphism between cohomology theories. In homotopy theory, this is known as a *cohomology operation*.

Let us look at a particular example. Every Lie group G is deloopable and by the classification theorem of principal bundles, the cohomology $H^1(M;G) := [M;BG]$ classifies precisely G-bundles over M [161, 56]. The interacting fields acting on particles are not G-bundles, but G-bundles with connections, so that it is natural to ask if this classification theorem cannot be lifted to the

[6] With $(\infty,1)$-pullbacks or homotopy pullbacks.

context of bundles with connections. This is really the case if we look at the language of smooth ∞-stacks. Indeed, by the higher Yoneda embedding, we have $H^1(M;G) \simeq [y(M);\mathbf{B}G]$. On the other hand, there is another smooth ∞-stack $\mathbf{B}G_{\text{diff}}$ such that fixing a connection A in the G-bundle classified by a map $f : y(M) \to \mathbf{B}(G)$ is equivalent to giving a lifing $\hat{f} : y(M) \to \mathbf{B}\text{diff}(G)$ of f. Consequently, the abstract cohomology $H^1_{\text{diff}}(M;G) := H^0(M;\mathbf{B}_{\text{diff}}(G) = [y(M);\mathbf{B}_{\text{diff}}(G)]$, called *nonabelian differential G-cohomology* describes the space of all gauge fields [199, 42, 211, 109]. Thus, gauge theories in the classical sense are the same thing as cohomology operations on nonabelian differential cohomology.

If G is only a group, then the smooth ∞-stack $\mathbf{B}(G)$ needs not be deloopable. On the other hand, if G is a more general higher group, then $\mathbf{B}^2(G)$ exists and, in similar way, there also exists the differential refinement $\mathbf{B}^2_{\text{diff}}(G)$, leading us to define the 2th differential cohomology $H^2(M;G) := [y(M);\mathbf{B}^2_{\text{diff}}(G)]$ [199, 201]. Similarly, if G admits higher deloopings we can define higher-degree differential cohomology. Notice that if cohomology operations in 1th differential cohomology describes gauge theoris, cohomology operations in higher differential cohomology should describe higher gauge theory, i.e, classical theories whose interacting fields are not vectorial, but tensorial, as typically string theory (with the B-field) and supergravity/M-theory (with C-field). This is really the case [82, 199, 201, 225, 169, 17].

Finally, recall that cohomology theories are the natural environments in which obstruction theory can be developed [105, 161, 28]. On the other hand, there are typical anomaly cancellation conditions in (higher) gauge theories, which typically can be interpreted as obstructions to the preservation of some symmetry at the quantum level or as a charge quantization condition. Examples such as the quantization of D-brane charges in (twisted) K-theories [239, 39, 167, 84, 98, 99], the Freed-Witten anomaly as pull-push operations in (twisted) K-theory [86, 121, 45, 176], the fermionic anomaly as the trivialization of a line bundle [78, 80], the Green-Schwarz mechanism and its description in terms of liftings to twisted differential cohomology [82, 53, 196] and, more recently, the Hypothesis H, about the quantization of the C-field on (twisted) cohomotoy cohomology theory and its many consequences [76, 75, 193, 43, 192, 194, 77] seem to reveal that this cohomological approach to physics, where the meaning of obstruction is very clear, is not only formal, but very fruitful.

8 Quantization

In the previous sections we discussed that ∞-categorical language is abstract enough to provides suitable axiomatization of classical and quantum physics, making it a natural candidate to attack Hilbert's sixth problem. Thus, using this class of languages, we need to build a quantization process linking *any* classical theory to a corresponding quantum theory. Besides the problematics involving the current development of (∞, n)-categorical languages for higher n, building this full quantization process remains an open problem. Discussing complete details is beyond the scope of this paper. See [202] for a concise discussion, [160] for a guide and [201] for a complete reference. Even so, let us say a few words.

With the axiomatization of classical and quantum theories on hand we can a priori build ∞-categories Class and Quant, so that it is expected that a quantization process should be some kind of ∞-functor

$$\mathcal{Q} : \text{Class} \to \text{Quant}$$

such that it assigns to each system of Classical Mechanics a corresponding system of quantum mechanics. Since as discussed in Section 3 these disciplines can be axiomatized using 1-categorical language, one should expect that, when restricted to them, the quantization ∞-functor $\mathcal{Q}$ becomes a functor. But, by counterexamples of van Hove and others [234, 100, 95, 94, 96], this quantization 1-functor does not exist, forcing us to take into account higher layers even in the simple case.

On the other hand, there is a canonical way to map classical mechanics into quantum mechanics given by the so-called *geometric quantization* [240, 216], where the main idea was to incorporate the higher layer on it. The first step was to observe that part of the data needed to apply that quantization corresponds to an orientation in complex K-theory [166, 129]. The second one was the extension of geometric quantization from symplectic manifolds to symplectic groupoids (and therefore from set-theoretic objects to categorical-theoretic entities) [38, 106, 187]. The *higher geometric quantization* is then a program aiming to join both observations and build a quantization for higher groupoids by means of cohomological methods, which is a natural strategy in virtue of the discussion in Section 7. A very promising approach which has been more extensively developed in the last decade is the so-called *cohomological quantization*, also known as *pull-push quantization* or *motivic quantization*. See [197, 203, 177, 187, 202, 201] and the references therein.

9 Conclusion

In this paper we described and surveyed in a pedagogical way a formal approach to Hilbert's sixth problem based on works of Urs Schreiber, John Baez, Daniel Freed, Jacob Lurie and many other researchers. We discussed that the problem requires languages which are arbitrarily abstract and synthetic, so that higher categorical languages were presented as natural candidates. We saw that these higher languages seem to be abstract enough to axiomatize classical and quantum physics, leaving open the problem of building a global quantization ∞-functor. Some comments on recent cohomological quantization were made.

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3 MATHEMATICAL ASTROPHYSICS

3.1 SYNTHESIS

If Astrophysics was a discipline, by the discussion of Chapter 2, it would be the study of *astrophysical systems*, i.e., astrophysical objects, their properties are relations. But, *what are these objects?* Thinking of Astrophysics as the astronomical branch of Physics, an astrophysical system should be an astronomical system regarded as a physical system. Astronomical objects, in turn, are normally defined as real objects of "large scale". Thus, astrophysical objects should be physical objects of large scale.

Physical objects have a well-accepted conception: they are space and time located objects in a spacetime M, normally regarded as the physical universe (see Definition 2.9). But, what does it mean "of large scale"? The global aspects of the universe are studied by another discipline - Cosmology -, so that large scale should be better understood as "intermediate cosmic scale". But, again, what is it? What it means "intermediate cosmic scale?" *What is an astrophysical object?*.

In [81] Anderl emphasizes the failure (or at least the difficulty) in providing a precise scope to Astrophysics saying that "*modern Astrophysics [...] has rarely been addressed directly within the philosophy of science context.*". Some investigations have been proposed by Hacking [66, 65] trying to build an antirealistic approach to Astrophysics (in the epistemological sense)[1] , but it has been widely debated [149, 141, 138]. Anderl also says:

> *Rather than the development of autonomous mathematical theories that themselves then require specific interpretations, it is the application of existing physical theories to a vast array of cosmic phenomena and different environments that are found in the universe, which constitutes the main focus of astrophysical research.*

Indeed, the foundational mathematical aspects of Astrophysics has been poorly developed since the seminal works of Chandrasekhar [38, 37] - see [92, 72, 130, 131] for the standard modern texts on the subject.

For instance, a search for the term "mathematical astrophysics" in Google Scholar (held in December, 10, 2021) returns only 57 results, excluding citations and without any restriction in the period of publication (in contrast to the term "mathematical cosmology", with more than 1200 results). Furthermore, when restricted to results after 2017 it returned

[1]We recall that scientific realism means that true propositions built by theoretical investigations are about real objects. Thus, in scientific antirealism, the only way to investigate the real world is by means of experiments. This is some kind of (constructive) empiricism.

11 results, which either are not directly related to mathematical aspects of Astrophysics (meaning that they simply cite the term "mathematical astrophysics") or are results about very specific topics, such as plasma-astrophysics, all done by members of the same research group in the Center for Mathematical Plasma-Astrophysics at KU Leuven [105, 47][2]. The same search in arXiv returned only two results none of them about mathematical aspects of Astrophysics.

Remark 3.1. Searching in arxiv with exact matches excludes a lot of results which a priori could be about mathematical astrophysics but that do not use the term directly. For instance, the search in arxiv excluded our two works [125, 124] on the subject. However, the search in Google Scholar confirms that the topic is few explored.

The works [125, 124] in this chapter goes precisely in the opposite direction and return the focus on formal aspects of Astrophysics. Indeed, if the philosophical foundations of Astrophysics are problematic, specially concerning ontological and epistemological questions, the idea is to analyze if at least from the logical perspective the foundations of Astrophysics can be defined and studied. More precisely, the idea is to look at the known theoretical models used to describe parts of Astrophysics, to extract their essence, and propose an axiomatic approach that includes these models as instances. This could be understood as a restriction of the Axiomatization Program of Physics to "astrophysical systems". But there are two main differences:

1. *while "physical systems" are defined, "astrophysical systems" are not*. Thus, in this sense, an axiomatic definition of Astrophysics could be used as a motivation to the development of philosophical approaches to it;

2. *while in the approaches to Axiomatization Problem of Physics one should presuppose a formal ontology and a logical positivism (or even rationalism), here we are assuming just some version of scientific realism*. Indeed, we are not trying to prove that every physical system corresponds to a deductive system such that something is true iff it is a true sentence. We are only trying to propose certain formal systems whose true propositions are connected to reality.

3.1.1 Background Independence

Most of the current theoretical studies on Astrophysics can be divided more or less into two classes:

1. those doing computational results for *already existing models*, aiming to validate the model by comparison with experimental data;

2. those proposing new models, analogous to the existing ones, but defined in *alternative gravitational theories*, aiming to validate the underlying theory of gravity.

Thus, almost every research in Astrophysics stud "astrophysical systems" as something which depend *a priori* of the model to the underlying spacetime. It seems to us that these approaches are studying interpretations of an implied logic which makes sense only for certain interpretations of "spacetime" (because the astrophysical systems are defined *a priori* for them). Thus, in this perspective, the domain of discourse of Astrophysics is explicitly dependent on the interpretation of "spacetime". On the other hand, if one considers Physics as the study of physical systems, which in turn are localized at the "spacetime", we are not assuming that an interpretation of "spacetime" is fixed a priori; it becomes a variable as well.

[2] Even these works are much more related to computational than foundational aspects.

In other words, if it exists, the implied logic whose interpretations are studied in the typical research on Astrophysics is a first-order logic (with the domain of discourse fixed by a model for "spacetime"), while the logic that should be required to deal with the Axiomatization Problem of Physics is a higher-order logic (in particular, "spacetime" is a variable as well). This leads us to search for a spacetime model-independent approach to Astrophysics, which reduces to the different well-known models in different interpretations.

Notice that the first step towards such an axiomatic approach to Astrophysics is to analyze the known models and extract from them what they have in common which is independent of the model of the spacetime. In other words, it is about generalization and, therefore, about abstraction. The other part is to find a formal language that accommodates the essence of all models, so that it is about axiomatization and, therefore, about abstraction again.

Remark 3.2. Classical Astrophysics (i.e., whose background gravity is Newtonian Gravity or General Relativity) presupposes that spacetime is a four-dimensional entity. Most of the non-classical Astrophysics (i.e., whose background gravity is a different theory) do the same. There are some works that try to develop "higher dimensional Astrophysics", specially concerning the study of stars, as a source of possible experiments to test theories that use higher dimensional spacetimes, e.g., string theory [61, 19, 127, 11]. Thus, a formal definition of Astrophysics must be independent not only from the spacetime model but also from its dimension.

3.1.2 Cosmic Hierarchy

Returning again to Physics, recall the hypothesis that physical systems can be stratified through certain fundamental parameters. Similar behavior occurs in Astrophysics with the parameter being the *cosmic scale* α: for low α one has the class $\mathcal{O}_{\alpha}$ of the astronomical objects given by stars and small clusters, and when α increases one has the classes of galaxies, galaxy clusters, superclusters and filaments [??]. The fact that stars belong to clusters of stars, which in turn belong to galaxies, and so on, means that $\mathcal{O}_{\alpha} \subset \mathcal{O}_{\alpha'}$ when $\alpha \leq \alpha'$.

Of course, if the physical universe contains every physical object, it should contain $\mathcal{O}_{\alpha}$ for every α. Thus when α blows up we are becoming closer to the whole universe. But the study of large-scale physical objects should be part of the scope of Cosmology. This emphasizes, again, the difficulty in defining Astrophysics by means of its class of objects. Indeed, assume a definition of Cosmology is given. *Is it possible to distinguish Astrophysics from Cosmology by their domains?* More precisely,

- *is there a critical scale α_0 such that if $\alpha \leq \alpha_0$ one is in the domain of Astrophysics, else $\alpha_0 \leq \alpha$ one is in the domain of Cosmology?*

One should expect at least to say "Astrophysics deals with $\mathcal{O}_{\alpha}$ in the small limit of α, while Cosmology deals with $\mathcal{O}_{\alpha}$ is the large limit." But there are at least two difficulties with this approach:

1. *it presupposes an a priori knowledge of the class Γ where the cosmic scale α lives and the existence of a class of objects $\mathcal{O}_{\alpha}$ to each α.*

   If α is the cosmic parameter used to classify stars, galaxies, clusters, etc., this implies that knowing all values of α for which there exists a class $\mathcal{O}_{\alpha}$ of physical objects. Note that, if Γ has a maximal element α_0, then $\mathcal{O}_{\alpha_0}$ should be the entire universe. The Cosmological Principle states that after a certain scale the universe is homogeneous, so that Γ should have a maximum, which in this case is not the entire universe.

2. *it presupposes that it is possible to organize the classes of objects $\mathcal{O}_\alpha$ in an ascending sequence, for every value of the cosmic scale.*

   This is a hierarchical hypothesis on the entire universe. Since we are assuming a given Cosmology model, one could take it as some Hierarchical Cosmology model [43, 10, 129].

The first assumption tells us basically that if α is the cosmic scale and we want to have a class of objects for each value of α, then the Cosmological Principle must fail. The second assumption implies, more specifically, that one should fix a Hierarchical Cosmology model if, in addition, one wants a stratified structure.

> The Cosmological Principle was "statistically proved" in [??], but the obtained critical value α_0 does not agree with some observations: it has been observed non-homogeneous distribution of matter for values $\alpha > \alpha_0$ [??,??]. To explain this gap is one of the major open problems of Cosmology and Astrophysics [??,??]. Thus, a priori Hierarchical Cosmology models are not empirically excluded, but even without the Cosmological Principle they are not well-accepted, being a matter of debate [??,??]. (**peço ajuda nesse parágrafo**)

Thus, following the approach of the question above to define Astrophysics would produce a definition explicitly depending on models to Cosmology which are not well-accepted (or could change with new or more accurate experiments, that is, new Physics.). So, this leads to asking: *which aspects of the above discussion are independent of the cosmological model?*. The fact that the hierarchical structure breaks if homogeneity is observed.

Since, a priori, the hierarchical structure exists for "small" cosmic scales (stars, galaxies, etc.) and the homogeneity may exist (or not) for "large" cosmic scales, one can think in *define* Astrophysics as the study of such hierarchical structures of physical which breaks in the presence of homogeneity.

In the following, we present a very brief discussion on how this could be formalized in the presence of the Cosmological Principle.

3.1.3 A Definition Proposal

Definition 3.3. Let α be a parameter varying on some class Γ.

1. A *α-family of physical pieces* is a collection $\mathcal{O}_\alpha \subset \mathcal{O}$ of classes of physical objects, with $\alpha \in \Gamma$. In other words, it is a map $f : \Gamma \to \rho(\mathcal{O})$.
2. The physical region *covered* by a α-family is the union $\overline{f} = \cup_\alpha f(\alpha)$.
3. A subclass $\mathcal{C} \subset O$ is said to be *covered* by a α-family f if it belongs to the physical region covered by f, i.e., if $\mathcal{C} \subset \overline{f}$.

Notice that to every class $\mathcal{C}$ of physical objects one has a corresponding lattice $\rho(\mathcal{C})$ of subclasses.

Definition 3.4. Two classes of physical objects $\mathcal{C}$ and $\mathcal{C}'$ are *physically equivalent* if their lattices of subclasses are isomorphic.

Definition 3.5. Let α be a parameter varying in a class Γ. We say that a class of physical objects $\mathcal{C}$ is *α-homogeneous* (or *homogeneous relative to* α) if for every α-family f covering $\mathcal{C}$ and every $\alpha, \alpha' \in \Gamma$ the regions $\mathcal{C}_\alpha = \mathcal{C} \cap f(\alpha)$ and $\mathcal{C}_{\alpha'} = \mathcal{C} \cap f(\alpha')$ are physically equivalent.

Definition 3.6. Suppose now that Γ comes with a preorder $\leq$. Given $\alpha_0 \in \Gamma$, we say that:

1. a α-family of physical pieces f is *hierarchical up to* $\alpha_0 \in \Gamma$ if for every $\alpha, \alpha' \leq \alpha_0$ one has $f(\alpha) \subset f(\alpha')$, i.e., if f is order preserving when restricted to the interval $[\Gamma, \alpha_0]$
2. a class of physical objects $\mathcal{C}$ is *hierarchical up to* $\alpha_0 \in \Gamma$ if there exists an α-family of physical pieces f which is hierarchical up to $\alpha_0 \in \Gamma$ and such that $f(\alpha) \subset \mathcal{C}$ for every $\alpha \leq \alpha_0$.

We can now propose a definition for Astrophysics.

Definition 3.7 (Astrophysics)**.** An *astrophysical domain* consists of:

1. a parameter α, called an *astrophysical scale*, varying on a partially ordered set $(\Gamma, \leq)$
2. a value $\alpha_0 \in \Gamma$, called *homogenization scale*;
3. a class $\mathcal{C}$ of physical objects, called *astrophysical objects*,

subjected to the following conditions:

1. the class $\mathcal{C}$ is hierarchical up to α_0;
2. its complement $\mathcal{O} - \mathcal{C}$ is α-homogeneous.

Remark 3.8.

1. The definition above assumes the existence of the homogenization scale α_0, which could be understood as the Cosmological Principle. Cosmology, in this case, should correspond to the study of the complements $\mathcal{O} - \mathcal{C}$ of astrophysical domains.
2. Looking at the effective hypothesis we see that every astrophysical domain in the sense above defines an effective area of Physics which is bounded from above by the presence of homogeneity.

3.1.4 Stellar Systems

Although the draft above suggests that Astrophysics could be axiomatized, in works [125, 124] we considered the problem of formalizing not the entire Astrophysics, but only one of its branches: *Stellar Astrophysics*.

Recall that at small "cosmic scales" one knows that the universe is hierarchical: stars group into small clusters of stars, which in turn belong to galaxies and these to clusters of galaxies, and so on. Thus, in some sense, one could consider stars as the "building blocks" of Astrophysics. In an emergentistic approach to the Philosophy of Physics (see Chapter 5) this would imply, for instance, that every nontrivial phenomenon of Astrophysics was induced by the behavior at the scale of stars, i.e., that *Astrophysics is freely generated by stars*. In terms of the proposal to the definition of Astrophysics presented above, the existence of a smallest class of astrophysical objects is implemented assuming that the poset $(\Gamma, \leq)$ has a least element.

If one wants to consider planets as astrophysical objects, then, since planets and stars are observed in the same "cosmic scale', such a smallest class of astrophysical objects should not contain only stars, but also planets, allowing the generation of mixed astrophysical entities, such as planetary systems. But, *what about moons which are observed at the same cosmic scale as planets and stars, but are not planets nor stars? And what about asteroids, planetoids, small planets, dwarf planets, comets and so on?*

We could include this plethora of astrophysical objects existing in the smallest astrophysical scale α_{min} by assuming that $\mathcal{C}_{\alpha_{min}}$ has some decomposition $\mathcal{C}_{\alpha_{min}} = \cup_j \mathcal{U}_j$ in disjoint parts $\mathcal{U}_j$, with j varying over some class J. The fact is that stars, planets, moons, asteroids, etc., are not fully independent objects: some of them orbit the others. But two given objects need not be related by the ”orbit relation”. Indeed, there must be a large difference in their mass and they must be close enough[3]. In particular, being the most massive among the fundamental astrophysical objects, up to the “orbiting relation”, the fundamental entities are stars. This could be formalized saying that there is j_0 and relation R in $\mathcal{C}_{\alpha_{min}}$ whose quotient space $\mathcal{C}_{\alpha_{min}}/R$ is equivalent to $\mathcal{U}_{j_0}$.

Independently if we are taking into account fundamental entities other than stars, it is certainly because the study of stars is of greatest importance to the global understanding of Astrophysics. This leads us to questions such as “*what is a star? What can be said about the collection of them? Can it be classified?*”. In the formalization above, this corresponds to asking

- What is the nature of the elements of $\mathcal{C}_{\alpha_{min}}$? Can we put on it some additional structure? Can this structure be classified? It is possible to talk about its dimension, meaning that there is some notion of independence between the elements of $\mathcal{C}_{\alpha_{min}}$ and that there is a certain collection of independent elements which generates all the others?

From a logical point of view, the proposed definition to Astrophysics presented in Section 3.1.3 could be said to deal with syntax, in the sense that it is independent of the nature of physical objects. One can study some aspects of a formal logic from its interpretations, i.e., from its semantics. Thus, with the questions above we are going in the direction of finding an interpretation to the abstract proposed definition of Astrophysics. In order to remains focused on understanding Astrophysics independently of a model of gravity, one should search for interpretations that also are independent of a background model. This was precisely the scope of [125, 124]: *to give a formal definition of "stellar system", independent of a model for gravity, and to study the space of all of them.*

3.1.5 Main Results

What is a star? There are many ways to consider this question and it can be analyzed in parallel to the other question *how a star is formed?* For instance, one could *define* what is a star by the way it is formed: *a star* ***is*** *a physical object which is formed in a specific way*.

We note that this approach involves mixing Astrophysics with other parts of Physics, such as Nuclear Physics and Particle Physics, which are studied in physical scales other than cosmic ones. Also, if one assumes that a star is formed by more fundamental physical objects, one is adopting a reductionist approach to Stellar Astrophysics and, therefore, restricting the formulation to a fixed ontological approach.

On the other hand, one could first say what a star is (independently of an external field) and then deduce that they can be described by reductionist models. From a foundational point of view, it seems to us that the latter perspective is more agreeable.

Definition 3.9. A *star* is given by:

1. a subset $I \subset \mathbb{R}$ which is the union of non-necessarily bounded intervals;
2. two funcions $\rho, p : I \to \mathbb{R}$, corresponding to its *density* and *pressure*, respectively.

A star is said to has *class* (k, l) if ρ is a piecewise C^k and p is piecewise C^l.

[3] Of course, the precise understanding of this “orbit relation” and its properties require considering questions as “*what is n-body problem independently of a given gravitational model?*”, but this is outside our scope.

Remark 3.10. For concrete purposes, I is the closed interval, the density and pressure are differentiable at every point, and the density is assumed non-negative. But, in order to include "exotic" stars, which may have cavities and both regions with positive and density, we need to work in the broadest definition above.

Remark 3.11. Notice that the definition above is independent of the spacetime model and of its dimension.

For every k, the collection $C^k_{pw}(I)$ of piecewise C^k functions is a real vector space and, therefore, a star class (k,l) is an element of the vector space $(\rho,p) \in C^{k,l}_{pw}(I)$, where $C^{k,l}_{pw}(I)$ denotes $C^k_{pw}(I) \times C^l_{pw}(I)$. This leads us to propose a formal definition of *cluster of stars*.

Definition 3.12. A *cluster of stars of class* (k,l) or simply *cluster of class* (k,l) is just a vector subspace $\mathrm{Stellar}^{k,l}(I)$ of $C^k_{pw}(I) \times C^l_{pw}(I)$.

We notice that for every k the space $C^k_{pw}(I)$ has an extended norm (i.e., a norm possibly taking infinite values), which makes it a extended normed space, given by $\|f\| = \sum_{i \leq k} |\sup_I f^{(i)}(t)|$. On the other hand, if $(V, \|\cdot\|)$ is any extended normed space, then the balls defined by $\|\cdot\|$ generate a topology, but multiplication is typically not continuous, so that $(V, \|\cdot\|)$ is generally not a topological vector space [17]. But, there always exists a canonical topology τ which makes V not only a topological vector space, but indeed a locally convex one [18].

Thus, for every (k,l), a cluster of stars $\mathrm{Stellar}^{k,l}(I)$ inherits a canonical locally convex space structure, and therefore we are embedding Astrophysics into Functional Analysis.

In most cases, stars come with a notion of "mass" and of "radius". Here we assume a collective definition of these concepts in each cluster.

Definition 3.13. A *mass function* and *radius function* in a cluster of class (k,l) are piecewise continuous function $M, R : \mathrm{Stellar}^{k,l}(I) \to \mathbb{R}$ in the locally convex topology above.

We have two pairs of functions: (ρ, p) and (M, R). The internal structure of a star emerges in a thermodynamic picture by means of forcing a relation between ρ and p. On the other hand, what has been empirically observed are relations between M and R.

Definition 3.14. Let $\mathrm{Stellar}^{k,l}(I)$ be a cluster of class (k,l).

1. A *function of state* is a function $f : \mathrm{Stellar}^{k,l}(I) \times E \to \mathbb{R}$, where E is a Banach space;
2. A *mass-radius function* is a map $g : \mathrm{Stellar}^{k,l}(I) \times \mathrm{Stellar}^{k,l}(I) \times F \to \mathbb{R}$, where F is a locally convex space.

The equations $f(\rho, p, \epsilon) = 0$ and $g(M, R, \delta) = 0$ are called *equation of state* and *mass-radius equation*, respectively.

Now, consider the following basic (but general) questions:

- empirically it was found that a given star has a certain relation between its mass and radius. *What kind of may structures this star have?*
- trying to find tests to analyze if a model of gravity is the correct one, it was found that such model predicts the existence of stars with a certain equation of state. *For stars that satisfy what kind of mass/radius should one search?*

These questions, which are about constraints between the model and the experience, can be described by asking if the functions f and g above are independent or not. Since we have embedded Astrophysics into Functional Analysis, it can be easily studied via Implicit Function Theorem for locally convex spaces, which was the subject of [125].

Theorem 3.15. *Let* $\mathrm{Stellar}^{k,l}(I)$ *be a cluster of stellar systems of class* (k,l).

1. *Suppose given a locally monotone mass-radius function* $g(M,R,\epsilon)$.
   (a) *Any upper (resp. lower) bound on the mass and on the radius induces a bound on each monotonically locally integrable state function* $f(\rho,p,\delta)$ *depending on* ϵ *and maybe on other parameters.*
   (b) *If, in addition, the cluster comes with a continuity equation, then any upper (resp. lower) bound on the derivative of the mass and on the radius induces a bound on each monotonically locally integrable state function.*
2. *Assume given a mass-radius function* $g(M,R,\epsilon)$ *satisfying a continuity equation and a fully monotonically locally integrable function of state* $f(\rho,p,\delta)$. *If the mass function* M *is* C^2, *then any locally monotone invertible causal condition on* f *and any upper bound on the radius function* R *induce an upper bound on* f *depending only on the additional parameters of* g.

Proof. This is just the combination of Theorems 1,2,3 of [124]. □

Other problems of Stellar Astrophysics can also be formulated:

- in a given cluster $\mathrm{Stellar}^{k,l}(I)$, a property ϕ define the subset $\mathrm{Stellar}^{k,l}(I;\phi)$ of stars (ρ,p) which satisfies ϕ. It inherits a topology from $\mathrm{Stellar}^{k,l}(I)$, which coincides with the subspace topology coming from $C^{k,l}_{pw}(I)$. *For a fixed* ϕ, *what are the topological properties of this topological space? Is it connected? Compact? Convex? Discrete? Finite? And so on...*

Actually, many interesting problems can be formulated in this way, i.e., as the study of the subclasses of stars satisfying a given property. For instance, if ϕ is the property "*being in hydrodynamic equilibrium*" then $\mathrm{Stellar}^{k,l}(I;\phi)$ is the class of stars in the cluster $\mathrm{Stellar}^{k,l}(I;\phi)$ which are in hydrodynamic equilibrium. Thus, basically, the problem is *how to formalize properties* ϕ *in clusters.*

Hydrodynamic equilibrium means that the internal pressure p inside the star is balanced with the external pressure produced by gravity. Thus, to make sense of this, we need to say what gravity is, i.e., we need to specify in which background model of gravity we are working. But, we are trying to formalize things independently of such a choice. Thus, *what in the hydrodynamic equilibrium does not depend on an explicitly model of gravity?* Certainly, the gradient of pressure p' and the distribution ρ (and maybe the mass function and the radius functions, if they are being considered) should appear in an equation.

Definition 3.16. An *equilibrium equation* in a cluster $\mathrm{Stellar}^{k,l}(I)$, with $k,l\geq 1$ is a first order differential equation on it. More precisely, it is a function

$$F:\mathrm{Stellar}^{k,l}(I)\times(\mathrm{Stellar}^{k,l})'(I)\to C^0_{pw}(I)$$

of the form $F(\rho,p,\rho',p',r)$, where $(\mathrm{Stellar}^{k,l})'\subset C^{k-1,l-1}_{pw}(I)$ is the collection of derivatives (ρ',p'). A star (ρ,p) is said to be in *F-hydrodynamic equilibrium* if $F(\rho,p,\rho',p')(r)=0$ for every $r\in I$.

Different models to gravity provide different equilibrium equations. For instance, in Newtonian Gravity, we have the classical equation (with $G \equiv 1$)

$$F(\rho, p, \rho', p')(r) = p'(r) + \frac{M(r)\rho(r)}{r^2}$$

while in General Relativity we have the Tolman-Oppenheimer-Volkoff (TOV) equation (with $G \equiv 1 \equiv c$):

$$F_{TOV}(\rho, p, \rho', p')(r) = p'(r) + \frac{(\rho(r) + p(r))(M(r) + 4\pi r^3 p(r))}{r^2(1 - 2M(r)/r)}.$$

For the equilibrium equations of non-classical gravity models, such as $f(R)$-gravity, $f(R, T)$-gravity, $f(R, L)$-gravity and higher-dimensional General Relativity, see [32, 148, 33, 34, 39], respectively.

In the work [125] we considered the problem of studying the class of stars which are in hydrodynamic equilibrium relative to the TOV equation $F_{TOV}(\rho, p, p') = 0$. Let $\mathrm{TOV}^{k,l}(I)$ denote the corresponding class of stars. Since the TOV equation is nonlinear and non-autonomous, it is difficult to find analytical solutions to it. Actually, the only known analytical solution which exists for every closed interval I is with ρ constant. In [125] we showed, however, that if I is of a special type and $k, l = \infty$, then there are at least 11 analytical solutions (we actually give arguments to believe that this number is even higher, based on the fact that some of these eleven solutions may split into two independent solutions).

Theorem 3.17. *Let* $\mathrm{Stellar}^{\infty,\infty}(I)$ *a smooth cluster defined on* I*. If* I *is an union of suitable neighborhoods of* $\pm\infty$ *in the extended real line, then* $\mathrm{TOV}^{\infty,\infty}(I)$ *contains a discrete subspace with eleven points.*

Proof. This is Corollary 3 of [125]. □

In [125] we also classified the new solutions according to the sign of its density function. We realized that they typically do not have a well-defined sign, meaning that it varies between positive, negative and zero, as in Figure 1 of [125].

In terms of structure, ordinary matter is supposed to have positive density, vacuum a zero density, and negative density corresponds to some "exotic matter", such as dark matter. Thus, we showed that the model of stars above allows stars in hydrodynamic equilibrium which have cavities (regions without matter) separating regions with ordinary matter and with exotic matter.

3.2 QUESTIONS AND SPECULATIONS

Here we list some questions and speculations about what was discussed above and on possible ramifications for the works [125, 124].

- As far as the authors know, the axiomatic proposal to Astrophysics sketched in Section 3.1.3 is entirely new, so that further investigations on this axiomatics and its semantics should be interesting:

  Q3.1 *to each model* M *to gravity (General Relativity, EHP,* $f(R)$*-gravity, and so on) assigns a corresponding model* $\mathcal{C}(M)$ *to the axiomatics of Section 3.1.3.*

- in Section 3.1.4 we discussed how stellar systems should fit into this formulation to Astrophysics: they would be the minimal class of systems relatively to the cosmic scale:

Q3.2 *show that each model of gravity (General Relativity, EHP, $f(R)$-gravity, and so on) corresponds a model to Astrophysics such that this minimal class exists. Furthermore, show that the known concrete models of stellar systems belong to a corresponding model $\mathcal{C}_{\alpha_{min}}(M)$ to the axiomatics of Section 3.1.3.*

- Recall the notion of morphism between interpretations of a formal logic. We could then ask what is a morphism between models of Astrophysics. If a notion of isomorphism is provided one could also consider a classification problem of models to Astrophysics.

Q3.3 *what is a morphism between realizations of Astrophysics in the sense of Section 3.1.3? Do they constitute a category? What can be said about the classification problem of this category? Under which conditions do two models to gravity induce isomorphic models to Astrophysics? And what about the induced models restricted to the subcategories of stellar systems?*

- The stellar systems studied in General Relativity and other gravity models are usually in stable equilibrium in both hydrodynamic and evolutional senses.

Q3.4 *What is "stellar evolution" in the astrophysical context of Section 3.1.3? How to characterize the stellar systems in $\mathcal{C}_{star} \subset \mathcal{C}_{min}$ which are in evolutional equilibrium? Show that this characterization is such that in the interpretations corresponding to gravity models we recover the corresponding equations of stellar evolution.*

Time evolution is normally governed by flows. On the other hand, every star is supposed to birth from a bunch of cosmic dust. But dust clouds and the starts generalized from them are in the same cosmic scale α_{min} (recall that planets, asteroids and other classes of astrophysical systems should also exist in that scale).

This lead us to consider a distinguished subclass $\mathcal{C}_{dust} \subset \mathcal{C}_{min}$ representing "admissible" dust clouds (that are large enough to originate stars) and to assign to each $x \in \mathcal{C}_{dust}$ an *inverval of existence* $I(x) \subset \mathbb{R}$ bounded from below. On the other hand, evolution is normally described by some flow. Joining both facts, we could try to formalize stellar evolution as a rule assigning to each $x \in \mathcal{C}_{dust}$ its time evolution $E_x : I(x) \to \mathcal{C}_{star}$.

Intuitively, being in "evolutional equilibrium" means to stay in a same "stellar stage" for some considerable interval of time. This presupposes the existence of a filtration $(\mathcal{C}_{star,\lambda})_\lambda$ of $\mathcal{C}_{star}$ by subclasses $\mathcal{C}_{star,\lambda} \subset \mathcal{C}_{star}$, corresponding of different "stages of evolution", such that if $\lambda \leq \lambda'$ then of $\mathcal{C}_{star,\lambda'} \subset \mathcal{C}_{star,\lambda}$.

We could then say that a star $E_x(0) = s \in \mathcal{C}_{star}$ is in *evolutional equilibrium* relatively to a filtration $(\mathcal{C}_{star,\lambda})_\lambda$, around an instant $t_0 \in I(x)$, if there is λ and interval $I_{t_0} \subset I(X)$ around t_0 such that $E_x(t) \in \mathcal{C}_{star,\lambda})$ for every $t \in I_{t_0}$.

Q3.5 *In terms of the axiomatics of Section 3.1.3, what does it mean to say that a star is in "hydrodynamic equilibrium"? Prove that semantically the given notion contains the hydrodynamic equilibrium equations of models as General Relativity, etc. Give conditions under which this abstract notion of hydrodynamic equilibrium intersects evolutional equilibrium.*

- The stellar systems studied in the literature (including those considered in [125, 124]) presupposes that stars have spherical symmetry and that both density and pressure are radial functions. However, the Axiomatization Program of Physics presupposes a consistent axiomatic language, i.e., which is stable under new physical discoveries. Currently one knows that stars are spherically symmetric, but what about the future? Thus, it would be interesting to have models describing stars as more general entities.

Q3.6 *How to extends the classical models to stellar systems under hydrodynamic equilibrium from spherically symmetric objects to more general objects?*

- Concerning hydrodynamic equilibrium in General Relativity, in [125] we proved results on existence of *explicit* solutions for TOV equations. On the other hand, we believe that, under some conditions on pressure and density functions, TOV equations can be viewed as an elliptic differential equation on $I \times I$ or on I depending if the density is supposed to be given a priori or is a variable. Thus, one could try to use elliptic methods to determine the dimension of the solution space of TOV equations, that is, the exact number of (not necessarily explicit) linearly independent solutions.

  For instance, under periodic boundary conditions, the equations become then defined on $\mathbb{S}^1 \times \mathbb{S}^1$ and on $\mathbb{S}^1$, respectively. Since the index of elliptic operators on odd-dimensional compact manifolds is always zero, this would lead to the conclusion that *there is no nontrivial periodic solution to TOV equations*.

  This suggests to define *elliptic stellar systems* on spacetime M as submanifolds $S \subset M$ endowed with functions $\rho, p : S \to \mathbb{R}$ and an elliptic differential equation $H(p, \rho, \partial^I, \partial^j, x) = 0$ describing the *hydrodynamic equilibrium*.

Q3.7 *Verify if elliptic stellar systems can provide an attempt to Question Q3.6.*

- The models to classical Physics normally involve formulation by first principles, that is, that the physically observable objects (i.e., those which are connected with ontology) are the critical points of a functional defined on the space of all objects. Since Astrophysics is supposed to be part of classical Physics, it would be interesting to have a compatible model for them.

Q3.8 *Show that Astrophysics in the sense of Section 3.1.3 admits a model on given spacetime M as a classical theory, i.e., as the critical points of functional.*

Assuming that elliptic stellar systems is a good model (i.e., assuming Question 3.2), then one could search for variational formulations to the elliptic equation $H(p, \rho, \partial^I, \partial^j, x) = 0$. A variational formulation of Newtonian hydrodynamic equilibrium was obtained in [90].

- Concerning the explicit solutions to (extended) TOV equations that we obtained in [125], notice that the most external region of each solution has a negative density and, therefore, they could correspond to exotic matter with negative mass. Thus, by the mass-luminosity relation, the external part of the stars does not bright, which could explain why such solutions are not observed.

3.3 FIRST WORK

The following is a preliminary version of the published paper

- Y.X. Martins, L.F.A. Campos, D.S.P. Teixeira, R.J. Biezuner, *Existence and classification of pseudo-asymptotic solutions for Tolman-Oppenheimer-Volkoff systems*, **Annals of Physics**, v. 409, p. 167929, Aug 2019. doi:10.1016/j.aop.2019.167929

Constraints Between Equations of State and mass-radius relations in General Clusters of Stellar Systems

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Abstract

We prove three obstruction results on the existence of equations of state in clusters of stellar systems fulfilling mass-radius relations and some additional bound (on the mass, on the radius or a causal bound). The theorems are proved in great generality. We start with a motivating example of TOV systems and apply our results to stellar systems arising from experimental data.

1 Introduction

The structure of general relativistic and spherically symmetric isotropic stars is modeled by the Tolman-Oppenheimer-Volkoff (TOV) equations [1], described in terms of its density ρ and pressure p (setting $G = 1$ and $c = 1$):

$$p'(r) = -\frac{\Big(\rho(r) + p\Big)\Big(M(r) + 4\pi r^3 p\Big)}{r^2\big(1 - \frac{2M(r)}{r}\big)} \tag{1}$$

$$M'(r) = 4\pi r^2 \rho(r). \tag{2}$$

These equations are partially uncoupled and can be coupled through an equation of state. A typical example is the polytropic equation of state

$$f(p, \rho, k, k_0, \gamma) = p - k\rho^\gamma - k_0 = 0, \tag{3}$$

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where $k, k_0 \in \mathbb{R}$ are the *polytropic constant* and the *stiffness constant*, and $\gamma = (n+1)/n \in \mathbb{Q}$ is the *polytrope exponent*. Since $f(p, \rho) = 0$ can obviously be solved for p and ρ, it can be used to couple a TOV system. Under this coupling, we say that (p, ρ) corresponds to a *polytropic TOV system*.

Now, we recall that in thermodynamic systems where pressure and density are related by a solvable equation of state, the speed of sound within the system can be defined by $v := \sqrt{\partial p/\partial \rho}$. It folows from (3) that $v = \sqrt{k\gamma p^{\gamma-1}}$ in polytropic TOV systems. A TOV system is *causal* if the speed of sound does not exceeds the speed of light, that is, $k\gamma\rho^{\gamma-1} < 1$. In particular,

$$k\gamma\rho_c^{\gamma-1} < 1, \tag{4}$$

where $\rho_c = \rho(0)$ is the density at the origin.

Suppose a star of radius R is modeled by a TOV system, and let the *mass of the system* to be $M = M(R)$, where $M(r)$ is the function in (2). Small stars usually admit *mass-radius relations* equations of state, $g(M, R, \delta) = 0$, relating M to R and other parameters [2]. For instance, it is known [1, 2] that, in the Newtonian limit, polytropic TOV systems satisfy

$$g(M, R, k, n, \rho_c) = M - A(k, n, \rho_c)R^2, \tag{5}$$

where

$$A(k, n, \rho_c) = \Big(\frac{4\pi}{(n+1)k}\Big)^{3/2} \rho_c^{(n-3)/2n} \frac{|p'(R)|}{\rho_{\text{rel}}}$$

and $\rho_{\text{rel}} = \rho(R)/\rho_c$ is the *relative density*.

We show that there are obstructions for a Newtonian polytropic TOV system to be causal.

This follows from isolating ρ_c can be isolated in (5) and substituting in (4), leading to the following obstruction on the possible polytropic constants

$$k < \frac{n}{(n+1)}\Big(\frac{R^4}{n^3M^2}\beta\Big)^{\frac{1}{n}}, \text{ where } \beta = 64\pi^3\Big(\frac{|p'(R)|}{\rho_{\text{rel}}}\Big)^2.$$

In this article, we show that this obstruction is not particular to polytropic Newtonian TOV systems. We define precisely *clusters* of stellar systems (of which TOV systems particular examples), equations of state and mass-radius relations, in in Section 2. Then, in Section 3, we state and prove three different obstruction theorems which formalize the rough claim below. Finally, in Section 4, we apply our results to stellar systems arising from experimental data. [Roughly] Consider a small radius stellar system with a mass-radius relation $g(M, R, \epsilon)$. Each constraint on ϵ induces an obstruction on the possible equations of state depending on ϵ that can be introduced in that system. Before giving the precise statement and the proof, let us notice that this claim is reasonable. Generally, mass-radius relations are closely related to the atomic nature of the star, as stellar systems with different atomic constitution obey different relations [2, 13]. Reciprocally, constraints on the parameters of a mass-radius relation provide information about the atomic structure of the system. For instance, Newtonian polytropic stars satisfying (5), but not the Chandrasekhar limit (resp. Oppenheimer-Volkoff limit) cannot be stable white dwarfs (resp. stable neutron stars); there are also bounds on n, of course [1]. On the other hand, equations of state arise from the statistic mechanics of the atomic structure. Summarizing, mass-radius relations with constraints restrict the atomic structure and, therefore, the possible equations of state, which is precisely the content of Theorem 1.

2 Definitions

A *stellar system* is defined as a pair (p, ρ) of piecewise differentiable functions $I \to \mathbb{R}$. It is natural to define a *cluster of stellar systems* of degree (k, l) as a vector subspace $\mathrm{Stellar}^{kl}(I)$ of $C^k_{pw}(I) \times C^l_{pw}(I)$. It is important to notice that $C^k_{pw}(I)$ has a canonical generalized norm, which is a map satisfying the norm axioms but possibly taking values at infinity. In fact, if $f \in C^k_{pw}(I)$, $\|f\|_k := \sup_I |f^{(k)}(t)|$ is such a generalized norm. Like classical norms, generalized norms induce a topology which [3] can be characterized as the finest locally convex topology that makes sum and scalar multiplication continuous. Therefore, $\mathrm{Stellar}^{kl}(I)$ has a canonical locally convex space structure.

We are interested in stellar systems having well defined notions of mass and of radius. This leads us to consider clusters of stellar systems that becomes endowed with maps $M, R : \mathrm{Stellar}^{kl}(I) \to \mathbb{R}$ assigning to each system (p, ρ) its mass and its radius. In TOV systems, $M(p, \rho)$ is given by (2), and its inverse is $R(p, \rho)$. Looking at these expressions, we see that it is natural to assume M and R at least piecewise continuous. In other words, $M, R \in C^0_{pw}(\mathrm{Stellar}^{kl}(I))$. We say that a (locally convex) subspace $\mathrm{Bound}^{kl}(I) \subset \mathrm{Stellar}^{kl}(I)$ has *mass bounded from above* (resp. *radius bounded from above*) if when restricted to it the function M (resp. R) is bounded from above. Similarly, we could define subspaces with mass and radius bounded from below.

A *state function* for a cluster of stellar systems is a function $f : \mathrm{Stellar}^{kl}(I) \times E \to \mathbb{R}$, where E is a topological vector space of parameters. We say that a state function is *locally integrable at* p if the corresponding equation of state $f(p, \rho, \epsilon) = 0$ can be locally solved for p. In other words, if there is a neighborhood U of p, open sets $V \subset C^l_{pw}(I)$ and $W \subset E$, and a function $\xi : V \times W \to U$ such that $f(\xi(\rho, \epsilon), \rho, \epsilon) = 0$. additionally, if ξ is monotone in both variables we say that f is a *monotonically locally integrable (MLI)* state function.

When a locally integrable state function f is such that ξ is differentiable (in some sense to be specified below), we can define the *squared speed of sound* as $v^2 = \partial_\rho \xi$. We will be interested in situations in which both ξ and v are monotone. A state function satisfying these properties is *fully monotonically locally integrable.*

The notion of differentiability of f (and, therefore, of derivative) is linked to the nature of the space of parameters E, leading to different generalizations of the Implicit Function Theorem (IFT). These versions of the IFT imply that any function whose derivative in the direction of p satisfies mild conditions is locally integrable.

We remark that some classical generalizations of IFT, such as the IFT for Banach spaces [4] and the Nash-Moser Theorem [5] for FrÃ©chet spaces, cannot be used here, because $\mathrm{Stellar}^{kl}(I)$ isn't either Banach or FrÃ©chet. General contexts that apply here are when E is an arbitrary topological vector space and, more concretely, when E is locally convex (see [6] and [7], respectively). We will work in an intermediary context: when E is Banach, so that by embedding $\mathbb{R}$ in E we can regard f as a map $f : \mathrm{Stellar}^{kl}(I) \times E \to E$ and define the directional derivative $\partial_p f$ as usual for maps between Banach spaces, without the technicalities needed in the general situations. As proved in [8], the classical IFT holds, meaning that if $\partial_p f \neq 0$, then f is a locally integrable state function in a neighborhood of that point and ξ is differentiable (in a certain generalized sense). This IFT also gives an expression for the derivative of ξ at each point which depends on the derivative of f with respect to the other variables at points of the neighborhood where ξ is defined, and ξ is monotone if $\partial_i f \neq 0$ in those points.

A *mass-radius function* for a cluster of stellar systems is a function $g : C^0_{pw}(\mathrm{Stellar}^{kl}(I)) \times C^0_{pw}(\mathrm{Stellar}^{kl}(I)) \times F \to \mathbb{R}$, where F is a topological vector space of parameters, such that the mass-radius relation $g(M, R, \delta) = 0$ can be locally solved for some δ, in that we can locally write $\delta = \eta(M, R)$. If η is monotone, we say that g is a *locally monotone mass-radius function.* As in the previous case, if F is Banach, then a mass-radius function can be obtained by requiring $\partial_\delta g \neq 0$, and locally monotone if the other partial derivatives also do not vanish.

3 Statement and Proof

Theorem 3.1. *Let $\mathcal{C} = (\mathrm{Stellar}^{kl}(I), M, R)$ be a cluster of stellar systems of degree (k, l) endowed with a locally monotone mass-radius function $g(M, R, \epsilon)$. Then any upper (resp. lower) bound on the mass and on the radius induces a bound on each MLI state function of $\mathcal{C}$ depending on ϵ (and possibly on other parameters). If a state function is fully MLI at some p, then the bounds descend to the speed of sound near p.*

Proof. After the previous discussion, the proof becomes easy. We only work with upper bounds; the proof for lower bounds is essentially the same. By definition, the mass-radius relation $g(M, R, \epsilon) = 0$ can be solved in a neighborhood of some ϵ, that is, locally $\epsilon = \eta(M, R)$. Since M and R are bounded, say $M \leq m$ and $R \leq r$, and g is locally monotone, we see that $\eta(M, R) \leq \eta(m, R) \leq \eta(m, r)$. Therefore, ϵ is bounded too, say by ϵ_0. Now, notice that if a parameter ϵ is bounded then any monotone function $f(x, \epsilon)$ depending on that parameter is bounded by the function $h(x) = f(x, \epsilon_0)$. In particular, any MLI function of state $f(p, \rho, \epsilon)$ for the cluster $\mathcal{C}$ depending on ϵ, has an upper bound. Furthermore, if f is fully MLI at some p, then the speed of sound squared $v(\rho, \epsilon)$ near p is well defined and it also depends monotonically of ϵ, so that it also has an upper bound. □

The above theorem only applies for clusters whose function of state and mass-radius function depend on the same parameter. This hypothesis can be avoided by adding some continuity equation. Let $\mathcal{C} = (\mathrm{Stellar}^{kl}(I), M, R)$ be a cluster of stellar systems in which M is at least piecewise C^1. A *continuity equation* for $\mathcal{C}$ is an ordinary differential equation $M'(R) = F(R, \rho)$. We will only work with continuity equations which can be locally solved for ρ. The basic example is equation (2).

Theorem 3.2. *Let $\mathcal{C} = (\mathrm{Stellar}^{kl}(I), M, R)$ be a cluster of stellar systems of degree (k, l) endowed with a locally monotone mass-radius function $g(M, R, \delta)$ and a continuity equation. Then any upper (resp. lower) bound on the derivative of the mass and on the radius induces a bound on each MLI state function of $\mathcal{C}$. If a state function is fully MLI at some p, then bounds are induced on the speed of sound near p.*

Proof. The proof is similar. Because the mass-radius function $g(M, R, \delta)$ is locally monotone, it can be locally solved for M, allowing us to write $M(R, \delta)$. From the continuity equation, we have $\partial_R M(R, \delta) = F(R, \rho)$. On the other hand, the RHS can be solved for ρ as $\rho(R) = \partial_R M(R, \delta)$. It follows that any bound $\partial_R M(R, \delta) \leq M_0(\delta)$ induces a bound $\rho(R) \leq M_0(\delta)$. Consequently, if $f(p, \rho, \epsilon)$ is a MLI function of state, we have the desired bound $f(p, \rho, \epsilon) \leq M_0(\delta)$, which clearly descends to the speed of sound when f is fully MLI. □

In the last two theorems we showed that bounds on the mass-radius relations induce bounds on the functions of state, which descend to the speed of sound. On the other hand, additional conditions could be imposed *a priori* on the speed of sound, such as causal conditions. We show that these conditions lift to new bounds on the state functions. In order to do this, given a fully MLI function of state $f(p, \rho, \delta)$, let us define a *causal condition* for f near p as an upper bound

$v^2(\rho, \epsilon) \leq v_0^2(\rho, \epsilon)$ for the speed of sound near p. We assume that both v^2 and v_0^2 are *locally invertible*, meaning that we can locally invert $v^2(\rho, \epsilon)$ to write $\epsilon(v^2, \rho)$, and similarly for v_0^2. Restricting to Banach space of parameters E, this can be formally described through the IFT for Banach spaces.

Theorem 3.3. *Let $\mathcal{C} = (\mathrm{Stellar}^{kl}(I), M, R)$ be a cluster of stellar systems of degree (k, l), with M piecewise C^2, endowed with a fully MLI state function f and with a mass-radius function g fulfilling a continuity equation. Any locally monotone invertible causal condition on f and any upper bound on the radius induce an upper bound on f depending only on the additional parameters of g.*

Proof. Once more we start by writing $\rho(R) = \partial_R M(R, \delta)$. The difference is that, instead of using bounds on the right-hand side to get bounds on f, we consider the speed of sound $v^2(\rho, \epsilon)$, which becomes $v^2(R, \delta, \epsilon)$. If we have a bound for v^2 we can write $v^2(R, \delta, \epsilon) \leq v_0^2(R, \delta, \epsilon)$, translating to a bound on ϵ via the invertibility hypothesis. If $R \leq R_0$ is a bound on the radius, from the monotone property of v_0^2 we obtain $\epsilon(R, v^2\delta) \leq \epsilon_0(\delta)$, where $\epsilon_0(\delta) = \epsilon(R_0, v_0^2\delta)$. This gives the desired bound on f, due to the MLI hypothesis. $\square$

4 Applications

Here, we apply Theorem 3 to stellar systems inspired in experimental mass-radius relations. By this we mean relations found in the astrophysical literature as optimal approximations to experimental data describing zero age main sequence (ZAMS) stars and terminal age main sequence (TAMS) stars. These stars satisfy polytropic equations of state and the causal condition, so Theorem 3 can be applied.

4.1 Monomial mass-radius relations

A simple model for mass-radius relations of ZAMS and TAMS stars was developed in [11, 12] by considering monomial polynomials. The model was also applied recently to neutron stars in [13]). This means that in the cluster $\mathcal{C} = (\mathrm{Stellar}^{kl}(I), M, R)$ of ZAMS and TAMS stars we have a mass-radius relation given by a function

$$g(M, R, a, b) = M - aR^b, \tag{6}$$

where a, b are real functions on $\mathrm{Stellar}^{kl}(I)$, just as R, M. By definition, a stellar system belongs to the main sequence (MS) when the temperature at its nucleus is high enough to enable hydrogen fusion in such a way that the system becomes stable. This generally happens if the mass is at least $0.1\, M_\odot$ [2], so these systems are naturally endowed with a lower bound on the mass. On the other hand, the mass of a MS-star determines many of its properties, such as luminosity and MS lifetime. In other words, different classes of MS-stars are characterized by upper bounds $M \leq M_0$.

Additionally, stars at the beginning of the MS have smaller radius than those near MS's end. Therefore, ZAMS and TAMS have intrinsic bounds $R \leq R_0$. Furthermore, each partial derivative of g in (6) does not vanish except at $R = 0$ and $b = 0$; the point $R = 0$ is excluded by the bound $M \geq 0.1\, M_\odot$, and $b = 0$ is excluded from experimental data. Therefore, Theorem 3 applies, giving the following corollary:

Corollary 4.1. *Let $\mathcal{C} = (\mathrm{Stellar}^{kl}(I), M, R)$ be a cluster of ZAMS or TAMS stars fulfilling natural upper bounds $M \leq M_0$ and $R \leq R_0$. Then any function of state of $\mathcal{C}$ depending monotonically on a and b is bounded from above.*

Because ZAMS and TAMS stars are MS-stars, they follow Eddington's standard model, which means that polytropic state functions $f = p - K\rho^\gamma$ are good models to be chosen. In order to use Corollary 4.1 on f, we need some dependence on a and b. If the dependence is on K, i.e, if $K(a, b)$ is a monotone function, Corollary 4.1 leads to bounds on γ in terms of b, $\rho(R_0)$ and $P(R_0)$. If the dependence is on γ, we find bounds on K in terms of the same parameters. Finally, if both K and γ depend of a, b, we get bounds on b in terms of $\rho(R_0)$ and $P(R_0)$.

In Eddington's standard model, the MS-stars fulfill the continuity equation (2), allowing us to apply Theorem 3 and find bounds even when K and γ do not depend on a, b. Explicitly, Equation (2) combined with the mass-radius function ([11]) allows us to write the density as

$$\rho(R, a, b) = \frac{abR^{b-3}}{4\pi} \tag{7}$$

The speed of the sound within the star becomes

$$v(R, a, b) = k\left(\frac{ab}{4\pi}\right)^{\beta} R^{\beta-1} \text{ with } \beta = \gamma(b-3) \tag{8}$$

If we assume bounds on M' we get bounds on (7) and, therefore, on the parameters of the polytropic state equation, as well as on the speed of sound (8), as ensured by Theorem 3. For instance, if $M' \leq M_0$ and we are working only with stars of radius $R \leq R_0$ we find that γ must satisfy $M_0^{1/\gamma} \geq abR_0^{b-1}$. This can also be understood as an upper bound on the radius that a polytropic MS-star of mass M_0 fulfilling the mass-radius relation (7) may have: $R_0 \leq (\frac{M_0^{1/\gamma}}{ab})^{1/(b-1)}$. Just to illustrate, for a massive ZAMS stars, say with $M_0 \approx 120\, M_\odot$, we have [11] $\gamma = 3$, $a = 0.85$ and $b = 0.67$, so that $b - 1 < 0$, implying that R_0 very small, agreeing with massive MS-stars only staying a short time as ZAMS stars. On the other hand, $b = 1.78$ for $M_0 \approx 120\, M_\odot$ TAMS stars, meaning that in the terminal stage massive MS-stars may have a large radius.

We could also go in the direction of Theorem 3 and use causal conditions (instead of conditions on M') to get bounds. In natural units, the canonical choice of causal condition is $v < 1$. Assuming it and working in the same regime $R \leq R_0$,

$$k < \left[\left(\frac{ab}{4\pi}\right)^{\beta} R_0^{\beta-1}\right]^{-1} \text{ with } \beta = \gamma(b-3). \tag{9}$$

4.2 Rational function mass-radius relation

In Subsection 4.1, we studied monomial mass-radius relations for ZAMS and TAMS stars, found in [11, 12] following analysis of experimental data. However, for ZAMS stars with luminosity $Z = 0.02$, the monomial relation can be replaced by a rational one, as pointed in [14, 15]. So, let us consider mass-radius functions

$$g(M, R, a_i, c_i, b_i, d_i) = p(M, a_i, c_i) - Rq(M, b_i, d_i), \tag{10}$$

on a cluster of stellar systems, where $p(M) = \sum_i a_i M^{b_i}$ and $q(M) = \sum_j c_j M^{d_j}$ are polynomials and a_i, b_i, c_i, d_i are real functions on the cluster. We can apply Theorems 3, 3 and 3 to find constraints on the possible state functions. Since we are interested in MS-stars, we study the distinguished polytropic state function. Let us see how Theorems 3 and 3 works in that case. Notice that $g(M, R, a_i, c_i, b_i, d_i) = 0$ can be globally solved for R as

$$R(M, a_i, c_i, b_i, d_i) = \frac{p(M, a_i, c_i)}{q(M, b_i, d_i)}. \tag{11}$$

This function is clearly piecewise C^1 and, for each fixed $A \equiv (a_i, b_i, c_i, d_i)$, it is singular in a finite number of points. In the neighborhood of each regular point, $R(M, a_i, b_i, c_i, d_i)$ can be inverted for M, so that we have $M(R, A)$. Using the continuity equation (2), we can write $\rho(R, A, M') = M'/4\pi R^2$. Setting bounds $M' \leq M_0$ and working with $R = R_0$, we get a bound $\rho \leq M^0/4piR_0^2$ and, therefore, a bound on the polytropic function of state, as in Theorem 3. If instead of $M' \leq M_0$ we impose the canonical causal condition $v^2 = k\gamma\rho^{\gamma-1} < 1$, recalling that $\rho(R, A, M'(R, A))$, we get the constraint

$$k < \frac{1}{\gamma\rho_0^{\gamma-1}(A)}, \text{ where } \rho_0(A) = \rho(R_0, A), \tag{12}$$

exactly as in Theorem 3.

5 Concluding Remarks

We established, axiomatically and in great generality, that clusters with finite mass and finite radius can only accommodate mass-radius relations and equations of state simultaneously when certain bounds are satisfied.

Mass-radius relations can be found experimentally, while equations of state arise from theoretical modeling. In this context, we conclude that experimental data constrains *a priori* the possible theoretical models. We believe that these general results point towards an axiomatic formulation of Astrophysics, a problem pointed and extensively worked by Chandrasekhar.

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3.4 SECOND WORK

The following is a preliminary version of the published paper

- Y.X. Martins, D.S.P. Teixeira, L.F.A. Campos, R.J. Biezuner, *Constraints between equations of state and mass-radius relations in general clusters of stellar systems*, **Physical Review D**, v. 99, p. 023007, Jan 2019. doi:10.1103/PhysRevD.99.023007

Existence and Classification of Pseudo-Asymptotic Solutions for Tolman–Oppenheimer–Volkoff Systems

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Abstract

The Tolman–Oppenheimer–Volkoff (TOV) equations are a partially uncoupled system of nonlinear and non-autonomous ordinary differential equations which describe the structure of isotropic spherically symmetric static fluids. Nonlinearity makes finding explicit solutions of TOV systems very difficult and such solutions and very rare. In this paper we introduce the notion of pseudo-asymptotic TOV systems and we show that the space of such systems is at least fifteen-dimensional. We also show that if the system is defined in a suitable domain, then pseudo-asymptotic TOV systems are genuine TOV systems, ensuring the existence of new fourteen analytic solutions for TOV equations. The solutions are classified according to the nature of the matter (ordinary or exotic) and to the existence of cavities and singularities. It is shown that at least three of them are realistic, in the sense that they are formed only by ordinary matter while they contain no cavities or singularities.

1 Introduction

Since the seminal work of Chandrasekhar, the axiomatization problem of astrophysics has been neglected. In [1] the authors reintroduced the problem, showing that arbitrary clusters of stellar systems suffer fundamental constraints. In this paper we continue this work, focusing on a specific class of stellar systems: the TOV systems.

In order to state precisely the question being considered and our main results, we recall some definitions presented in [1]. A *stellar system* of degree (k, l) is a pair (p, ρ) of real functions, with p piecewise C^k-differentiable and ρ piecewise C^l-differentiable, both defined in some union of intervals $I \subset \mathbb{R}$, possibly unbounded. We usually work with systems that become endowed with an additional

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piecewise C^m-differentiable function $M : I \to \mathbb{R}$, called *mass function*. In these cases we say that we have a system of degree (k, l, m). The space of all these systems is then

$$C^k_{pw}(I) \times C^l_{pw}(I) \times C^m_{pw}(I) \equiv C^{k,l,m}_{pw}(I),$$

where $C^r_{pw}(I)$ denotes the vector space of piecewise C^r-differentiable functions on I. A vector subspace $\mathrm{Stellar}^{k,l,m}_I$ of $C^{k,l,m}_{pw}(I)$ is called a *cluster* of systems of degree (k, l, m). Let $\mathrm{Stellar}^{l,m}_I$ denote the subspace obtained from the cluster by forgetting the variable p.

We say that a system with mass function (p, ρ, M) is a *TOV system* if the Tolman–Oppenheimer–Volkoff (TOV) equation holds

$$p'(r) \quad = - \quad G\frac{\Big(\rho(r) + \frac{p(r)}{c}\Big)\Big(M(r) + 4\pi r^3 \frac{p(r)}{c^2}\Big)}{r^2\big(1 - \frac{2M(r)G}{c^2 r}\big)} \tag{1}$$

where G and c are respectively Newton's constant and the speed of light, which from now on we will normalize as $G = 1 = c$ (see [2]).

In a general cluster $\mathrm{Stellar}^{k,l,m}_I$ we define a *continuity equation* as an initial value problem $M' = F(M, \rho, r)$ in $\mathrm{Stellar}^{l,m}_I \times I$ with initial condition $M(a) = M_a$. Since the vector space of piecewise differentiable functions is locally convex but generally neither Banach nor Fréchet [1, 3], the problem of general existence of solutions for continuity equations is much more delicate [4, 5]. Therefore one generally works with continuity equations which are integrable. The classical example is

$$M' = 4\pi r^2 \rho. \tag{2}$$

Notice that if a system of degree (k, l, m) is endowed with (2), then $m = l + 1$. We define a *classical TOV system* as a TOV system equipped with the classical continuity equation (for some initial condition). Let $\mathrm{TOV}^{k,l}_I$ be the cluster of stellar systems of degree $(k, l, l+1)$ generated by (i.e, the linear span of) the classical TOV systems. We can now state the main problem:

Problem (classification of TOV clusters)**.** *Given I, k and l, determine the structure of $\mathrm{TOV}^{k,l}_I$ as a subspace of $C^{k,l,l+1}_{pw}(I)$ for some fixed locally convex topology.*

Since $(0, 0) \in \mathrm{TOV}^{k,l}_I$, this set is non-empty. Thus, the simplest thing one can ask about it is if it is nontrivial. This means asking if for every given k, l there exists at least one nonzero pair (p, ρ) for which equation (1) is satisfied when assuming (2). Before answering this question, notice that by means of isolating ρ in (2) and substituting the expression found in (1) we see that the TOV equation is of Riccati type [6, 7]:

$$p'(r) = A(r) + B(r)p(r) + C(r)p(r)^2 \tag{3}$$

where

$$A(r) \quad = \quad -\frac{M(r)M'(r)}{4\pi r^4\big(1 - 2\frac{M(r)}{r}\big)}, \tag{4}$$

$$B(r) \quad = \quad -\frac{\big(\frac{M'(r)}{r} + \frac{M(r)}{r^2}\big)}{1 - 2\frac{M(r)}{r}}, \tag{5}$$

$$C(r) \quad = \quad -\frac{4\pi r}{1 - 2\frac{M(r)}{r}}. \tag{6}$$

2

Therefore, it is a nonlinear and non autonomous equation which, added to the fact that $C^{k,l+1}_{pw}(I)$ is only a locally convex space (so that the general existence theorems of ordinary different equations do not apply), makes hard to believe that $\mathrm{TOV}^{k,l}_I$ has a nontrivial element. Even so, if ρ is a constant function, then (3) becomes integrable, showing that $\mathrm{TOV}^{k,l}_I$ is at least one-dimensional [2]. There are some results [6, 7] allowing that, under certain conditions, a solution of a TOV equation can be deformed into another solution, which may imply a higher dimension.

Remark 1. We point out that the generating theorem $(P1)$ proved in [6, 7] implies that the TOV cluster is invariant under *specific perturbations*. So, a natural question is if there exist I, k, l such that $\mathrm{TOV}^{k,l}_I$ is invariant under *arbitrary* perturbations. This is clearly false, because this would imply that any stellar system is TOV. Instead, we can ask about invariance by arbitrary *small* perturbations. Again, we assert that there are good reasons to believe that the answer is negative. Indeed, when a subset of a topological space is "invariant by small perturbations" we are saying that it is actually an open subset. Thus, assuming invariance, we are saying that $\mathrm{TOV}^{k,l}_I$ is an open subset of $C^{k,l+1}_{pw}(I)$ in our previously fixed locally convex topology. Now assume that the locally convex topology is Hausdorff and that $\mathrm{TOV}^{k,l}_I$ is finite-dimensional. Then the TOV cluster is as closed subset [8, 9]. Since topological vector spaces are contractible [8, 9], they are connected and therefore a subset which is both open and closed must be empty or the whole space. But we know that $\mathrm{TOV}^{k,l}_I$ is at least one dimensional and does not coincide with $C^{k,l+1}_{pw}(I)$. In sum, if there exists some locally convex topology in which the TOV cluster is invariant by small perturbations, then the following things cannot hold simultaneously:

- $\mathrm{TOV}^{k,l}_I$ is finite-dimensional;
- the locally convex topology is Hausdorff.

However, both conditions are largely expected to hold simultaneously, leading us to doubt the existence of a topology making $\mathrm{TOV}^{k,l}_I$ an open set[1].

Another way to get information about the TOV cluster is not to look at the TOV cluster directly, but to analyze its behavior in some regime. For instance, if in (1) we expand $1/c$ in a power series and discard all terms of higher order in c (which formally corresponds to take the limit $c \to \infty$) we get a new equation approximately describing the TOV equation. Therefore, studying the cluster $\mathrm{Newt}^{k,l}_I$ of stellar systems satisfying this new equation (called *Newtonian systems*) we are getting some information about the original TOV cluster. In fact, if we consider the subspace $\mathrm{Poly}^{k,l}_I \subset \mathrm{Newt}^{k,l}_I$ of Newtonian systems satisfying an additional equation $p = \gamma^q \rho + a$, with $\gamma, c \in \mathbb{R}$ and $q \in \mathbb{Q}$, then the Newtonian equation becomes a Lane-Emden equation, which has at least three independent solutions (besides the constant ones), showing that $\mathrm{Newt}^{k,l}_I$ is at least four-dimensional [2, 11].

In this article we analyze the structure of $\mathrm{TOV}^{k,l}_I$ in a limit other than the Newtonian one: we work in the *pseudo-asymptotic limit*. Before saying what this limit is, let us first say what it is not. We could think of defining a "genuine" asymptotic limit of TOV by taking the limit $r \to \infty$ in the TOV equation (1) in a similar way as done for getting the Newtonian limit, trying to get a new equation. In doing this we have two problems:

[1] Actually, this is not a special property of the TOV equation, but a general behavior of the solution space of elliptic differential equations [10]. The fact that TOV systems are modeled by an elliptic equation will be explored in a work in preparation.

1. differently of c (which is a parameter), r is a variable. Therefore, when taking the limit $r \to \infty$ we have to take into account the $r \to \infty$ behavior of all functions depending on r;

2. in order to get a new equation we have to fix boundary conditions for the functions, loosing part of the generality.

Thus, for us, *pseudo-asymptotic limit is not the same as a genuine asymptotic limit.* Another approach would be to work with an additional differential equation in $C^{l+1}_{pw}(I) \times I$, called a *coupling equation* and given by

$$\Lambda(M, M', M^{(2)}, ..., M^{(s)}, r) = 0, \tag{7}$$

where $M^{(i)}$ denotes the i-th derivative of M and $1 \leq s \leq l+1$ is the order of the equation. The function Λ itself is called the *coupling function* that generates equation (7). Let us consider the space $\mathrm{Ind}^{l+1}_{\Lambda}(I)$ of all C^{l+1}-differentiable mass functions M such that, if they are solutions of the coupling equation defined by Λ, then the corresponding TOV equation (3) is integrable. So, for every coupling function F we have $\mathrm{Ind}^{l+1}_{\Lambda}(I) \subset \mathrm{TOV}^{k,l}_{I}$ motivating us to define some kind of "indirect" asymptotic limit in TOV by taking the genuine asymptotic limit in (7). Again, we will have the two problems described above, but now the lack of generality is much less problematic, since we only need to consider boundary conditions for a single function (M) and for its derivatives. Even so, *the pseudo-asymptotic limit is not the same as the indirect asymptotic limit.*

Let us now explain what we really mean by pseudo-asymptotic limit. A *decomposition* for the coupling equation (7) generated by Λ is given by two other coupling functions Λ_0 and Λ_1 such that $\Lambda = \Lambda_0 + \Lambda_1$. We say that a decomposition is *nontrivial* if $\Lambda_i \neq 0$ for $i = 0, 1$. A *split decomposition* of Λ is a nontrivial decomposition such that Λ_0 generates a linear equation and Λ_1 generates a nonlinear equation. We say that a split decomposition is *maximal* when both Λ_0 and Λ_1 do not admit split decompositions. Not all coupling functions admit a maximal split decomposition, e.g, when Λ is linear. When Λ admits such a decomposition we will say that it is *maximally split.* So, let Λ be a maximally split coupling function. The *pseudo-asymptotic limit* of (3) relative to Λ is obtained by taking the genuine asymptotic limit in the nonlinear part Λ_1, added of the boundary condition $\lim_{r\to\infty} \Lambda_1 = 0$, and maintaining unchanged the linear part Λ_0. In this case, the equation replacing (1) is that generated by Λ_0. This new equation can be understood as a formal "pseudo-limit" of Λ, defined by

$$\operatorname*{psdlim}_{r\to\infty} \Lambda = \Lambda_0 + \lim_{r\to\infty} \Lambda_1.$$

For a given maximally split coupling function F, let $\mathrm{Psd}^{l+1}_{\Lambda}(I)$ denote the subspace of all mass-functions M such that

$$\lim_{r\to\infty} \Lambda_1(M(r), M'(r), ..., M^{(s)}, r) = 0,$$

i.e, which belong to the boundary conditions for Λ, and such that they are solutions of the pseudo-limit equation $\mathrm{psdlim}_{r\to\infty} \Lambda = 0$. We can now state our main result.

Theorem 1.1. *There exists I such that for every l there exists at least one maximally split coupling function such that $\mathrm{Psd}^{l+1}_{\Lambda}(I)$ is at least eleven-dimensional.*

Generally, when we take a limit in a equation, the solutions of the newer equation are not solutions of the older one. This is why we cannot use the existence of Newtonian systems to directly infer the existence of new TOV systems. But the pseudo-asymptotic limit is different, precisely because it is not a formal limit. Indeed, suppose $M \in \mathrm{Psd}^{l+1}_{\Lambda}(I)$. Then by hypothesis it satisfies Λ_0. If in addition it satisfies Λ_1 (instead of obeying only the boundary condition), then it satisfies F and, therefore, it belong to $\mathrm{TOV}^{k,l}_I$. We will show that by means of modifying I in Theorem 1.1, many of the mass functions will actually satisfy F_1, ensuring the existence of new integrable TOV systems, leading us to the following corollary:

Corollary 1. *For suitable I the space $\mathrm{TOV}^{\infty,\infty}_I$ of piecewise C^∞-differentiable TOV systems is at least eleven-dimensional.*

The proof of Theorem 1.1 and its extrapolation to TOV systems will be made in Section 2.1 and in Section 2.2, respectively, by making use of purely analytic arguments. In Section 2.3 we argue that $\mathrm{Psd}^{\infty}_{\Lambda}(I)$ must have a higher dimension and we give two strategies that can be used to verify this. In Section 3 we present a physical classification for the densities associated with mass functions in Theorem 1.1 and, therefore, for the new TOV systems. We classify them according to if they possess or not cavities/singularities and to if they are composed of ordinary or exotic matter. In the process of classifying them we prove that they generally admit a topology change phenomenon (similar to a phase transition), allowing us to improve Theorem 1.1 and Corollary 1, showing that for suitable I the space $\mathrm{TOV}^{\infty,\infty}_I$ is at least fifteen-dimensional. We finish the paper in Section 4 with a summary of the results.

2 Proof of Theorem 1.1

Before giving the proof of Theorem 1.1, let us emphasize that finding integrable TOV systems is a nontrivial problem. Notice that the simplest way to find explicit solutions of the Riccati equation is choosing coefficients satisfying one of the following conditions:

1. C identically zero. In this case the equation becomes a linear homogeneous first order ODE, which is separable.

2. A identically zero. Then the equation is of Bernoulli type and, therefore, integrable by quadrature.

3. There are constants $a, b, c \in \mathbb{R}$ such that $A(r) = a$, $B(r) = b$ and $C(r) = c$ simultaneously. In this case, the equation is separable.

Thus, recalling that TOV systems are described by a Riccati equation, we could think of applying some of these conditions in (3). It happens that the coefficients of this Riccati equation are not independent, but rather satisfy the conditions

$$\frac{4\pi r^4}{M(r)M'(r)}A(r) = \frac{r^2}{rM'(r) + M(r)}B(r) = \frac{1}{4\pi r}C(r). \tag{8}$$

Therefore, the first two possibilities are ruled out for making (3) trivial. Also, the third condition along with (6) implies that

$$M(r) = r/2 - 2\pi r^2/c \tag{9}$$

and $M'(r)$ is the corresponding linear polynomial. However, in (8) the above $M(r)$ makes the A term a quadratic polynomial, and the B term a rational function of degree 1, so the equality can never hold.

2.1 The proof

Now we will prove Theorem 1.1.

Proof. We start by noticing that in [12] it was shown that if the coefficients of any Riccati equation satisfy additional differential or integral conditions, then the nonlinearity of the starting equation can be eliminated, making it fully integrable. Each class of conditions is parametrized by functions $f : I \to \mathbb{R}$ and real constants. Because the authors of [12] worked only with smooth Riccati equations, they assumed $f : I \to \mathbb{R}$ smooth. However, it should be noticed that in the general situation we may have $f \in C^m_{pw}(I;\mathbb{R})$ or $f \in C^{m+1}_{pw}(I;\mathbb{R})$, where m is the least order of differentiability of the coefficients of the Riccati equation. Furthermore, from (4), (5) and (6) we see that $m = l$, where l is the class of M.

In the following, we will use these additional equations in order to build maximally split coupling functions. Precisely, one of the integral conditions presented in [12] is

$$A(r) = \frac{f(r) - \left[B(r) + C(r)\left[\int \frac{f(s)-B^2(s)}{2C(s)}\, ds - c_1\right]\right]^2}{4C(r)}, \tag{10}$$

under which a explicit solution for Riccati equation (3) is given by

$$p(r) = \frac{e^{\int^r (B(s)+C(s)h(s))\, ds}}{c_0 - \int^r C(s) e^{\int^s (B(\phi)+C(\phi)h(\phi))\, d\phi}\, ds} + \frac{h(r)}{2} \tag{11}$$

where c_0 is a constant of integration and

$$h(r) = \int^r \frac{f(s) - B^2(s)}{2C(s)}\, ds - c_1. \tag{12}$$

Let us fix $f(r) = B^2(r) + 2g(r)C(r)$, where $g : I \to \mathbb{R}$ is some integrable function. We will also take $c_1 = 0$. Let $h(r) = \int g(s)\, ds$. Using the coefficients of TOV equation (3), (10) becomes

$$M'(r) + \frac{M(r)M'(r)}{2\pi r^3 h(r)} - \left(\frac{2h'(r)}{h(r)} - \frac{1}{r}\right) M(r) + 2\pi r^2 h(r) + \frac{rh'(r)}{h(r)} = 0, \tag{13}$$

which is a coupling equation of order 1. Notice that it is maximally split, with maximal splitting decomposition given by

$$\Lambda_0(M, M', r) = M' - \left(\frac{2h'(r)}{h(r)} - \frac{1}{r}\right) M + 2\pi r^2 h(r) + \frac{rh'(r)}{h(r)} \tag{14}$$

$$\Lambda_1(M, M', r) = \frac{MM'}{2\pi r^3 h(r)}. \tag{15}$$

Therefore, giving M and h such that $\lim_{r\to\infty} \Lambda_1(M, M', r) = 0$, then M will automatically belong to $\mathrm{Psd}^k_{\Lambda}(I)$. We found ten such pairs. They are obtained by regarding h as solutions of the differential equation

$$F(r) = 2\pi r^3 h^2(r) + r^2 h'(r), \tag{16}$$

for different $F : I \to \mathbb{R}$, as organized in Table 1 of Appendix A. Only to illustrate the method, we will show how the first row is obtained. The other rows are direct analogues, only involving more calculations.

Recall that we are trying to find pairs (M, h) such that $\mathrm{psdlimit}_{r\to\infty} \Lambda = 0$, i.e, such that $\lim_{r\to\infty} \Lambda_1 = 0$ and such that the coupling equation induced by Λ_0 is satisfied. Suppose such a pair was found. Then, by the linearity of Λ_0, they become related (up to addition of a constant) by

$$M(r) = \frac{h^2(r)}{r}\left(\int_1^r -\frac{s^2\left(h'(s) + 2\pi s h^2(s)\right)}{h^3(s)}\,ds + c_2\right) \tag{17}$$

where c_2 is a integration constant. Taking $F = 0$ in (16) we see that it is solved for

$$h(r) = \frac{1}{\pi(r^2 + c_1/\pi)}, \tag{18}$$

so that (17) becomes

$$M(r) = \frac{c_2 h^2(r)}{r}, \tag{19}$$

whose derivative is

$$M'(r) = \frac{c_2\left(2rh'(r)h(r) - h^2(r)\right)}{r^2}.$$

Thus Λ_1 is given by the following expression, which clearly goes to zero as $r \to \infty$.

$$\Lambda_1(M, M', r) = \Lambda_1(r) = \frac{c_2^2 h^2(r)\left(2rh'(r) - h(r)\right)}{2\pi r^6} = -\frac{c_2^2\left(5\pi r^2 - c_1\right)}{2\pi r^6\left(c_1 - \pi r^2\right)^4}.$$

Defining (M, h) by (19) and (18), they will clearly satisfy the desired conditions, showing that $M \in \mathrm{Psd}^l_{\Lambda}(I)$.

Now, notice that all involved functions are actually piecewise C^∞, so that we can take l arbitrary. On the other hand, the domains of the functions in Table 1 are different, but we can restrict them to the intersection of the domains and then (since we are working with piecewise differentiable functions) extend all of them trivially to the starting I. This finishes the proof of Theorem 1.1, except by the fact that Table 1 contains *ten* linearly independent mass functions instead of the *eleven* ones stated in Theorem 1.1. The one missing is just the well known constant density solution. For completeness, let us show that it can also be directly obtained from our method. Indeed, by taking $f(r) = B^2(r)$ in (10), writing $c = 8\pi^2 c_1/3$, and using the coefficients of the TOV system, the coupling equation (13) becomes

$$M'(r) + \frac{3M(r)M'(r)}{4\pi c r^3} + \frac{M(r)}{r} + \frac{4\pi c r^2}{3} = 0,$$

which has solution $M(r) = 4\pi c r^3/3$, whose associated density is $\rho(r) = c$. □

Remark 2. In the construction, the motivation of the definition of $h(r)$ in (11) is the control of the integral term. By control, we mean that the h function is a integral of a integrable function g, freely chosen.

Remark 3. *A note on the differentiability of the pressure.* From (11) we see that $p \in C^{k'}_{pw}(I)$, where k' is the minimum between the differentiability class of h and the class of M. The latter is ∞, as obtained above, so that k' is the class of h. But looking at Table 1 we see that the class of h is ∞ in each of the cases. Thus, $p \in C^{\infty}_{pw}(I)$.

2.2 Extending

In the last section we proved that there exists a subset $I \subset \mathbb{R}$, which can be regarded as a disjoint union of intervals, such that we have a maximally split coupling function F of order 1, whose corresponding space of pseudo-asymptotic TOV systems $\mathrm{Psd}^{\infty}_{\Lambda}(I)$ is at least eleven-dimensional. As discussed in the Introduction, a pseudo-asymptotic TOV system does not need to be a TOV system. Here we will show that by means of modifying I properly, we can assume that some of the pseudo-asymptotic systems that we have obtained really define TOV systems.

Recall that if there exists a coupling function Λ such that a pseudo-asymptotic mass function M satisfies not only the linear part Λ_0 but also the nonlinear one Λ_1, then M actually defines an integrable TOV system. So, our problem is to analyze when the pseudo-asymptotic M obtained in the last section satisfies the differential equation $\Lambda_1(M, M', r) = 0$ for Λ_1 given by (15). We will give sufficient conditions on the general pseudo-asymptotic mass functions in order for this to happen. That these conditions are satisfied for our mass functions will be a consequence of their classification. The fundamental step is the following result from real analysis. We point out that in the following a *neighborhood* of a point $x \in \mathbb{R}$ is meant to be any connected subset (i.e, not necessarily an open interval) containing x.

Lemma 2.1. *Let $f : I \to \mathbb{R}$ be continuous at a point $a_0 \in I$ and such that $f(x) \to 0$ when $x \to a_0$. Assume that there exists a neighborhood $U \subset I$ of a_0 in which one of the following conditions is satisfied:*

c1) *f is non-negative and decreasing;*

c2) *f is non-positive and increasing.*

In this case, there exists a possibly smaller neighborhood U_0 of a_0 in which f is constant and equal to zero in U_0.

Proof. Because f is continuous in a_0 and $f(x) \to 0$ when $x \to a_0$, we have $f(a_0) = 0$. If f satisfies the first condition, let U_0 be some neighborhood within U and whose left extreme is a_0, i.e, some interval $[a_0, \varepsilon)$ or $[a_0, \varepsilon]$ for ε smaller enough. Since f is decreasing in U, it follows that if $x \geq y$ in U, then $f(x) \leq f(y)$. If we are in U_0, this means that for every x we have $f(x) \leq f(a_0) = 0$. But f is non-negative in U and thus in U_0. Therefore, we must have $f(x) = 0$ in U_0. For the second condition, take U_0 as some neighborhood which have a_0 as the right extreme. An analogous argument will then give the result. □

Now, recall that we can extend the real line $\mathbb{R}$ in two different ways: by adding a point at infinity ∞ or by adding both $+\infty$ and $-\infty$. In the first case we have the *extended real line* $\overline{\mathbb{R}}$, while in the second we have the *projectively extended real line* $\hat{\mathbb{R}}$. For the arithmetic construction of these objects, see [13]. Topologically, both spaces acquire natural Hausdorff compact topologies: $\hat{\mathbb{R}}$ is the one-point compactification of $\mathbb{R}$ and, therefore, is homeomorphic to the circle $\mathbb{S}^1$, while $\overline{\mathbb{R}}$ has an order topology homeomorphic to $[0,1]$ [14].

Notice that any piecewise continuous function $f : I \to \mathbb{R}$ admits an extension $\overline{f}$ to $\overline{\mathbb{R}}$, as follows. We first extend it to $\mathbb{R}$ by defining $\overline{f}(x) = 0$ when $x \notin I$ and then take $\overline{f}(\pm\infty) = \lim_{x\to\pm\infty} f(x)$. If both limits coincide, $\overline{f}(+\infty) = \overline{f}(-\infty)$ and the function has also an extension $\hat{f}$ to $\hat{\mathbb{R}}$. Furthermore, by definition, when they exist these extensions are continuous in $\pm\infty$ and ∞[2].

From the above remarks and from Lemma 2.1 we get the following corollary:

Corollary 2. *Let Λ be a maximally split coupling function, M a pseudo-asymptotic mass function for the coupling function Λ and $\overline{\Lambda_1}$ the extension to $\overline{\mathbb{R}}$ of the function*

$$r \mapsto \Lambda_1(M(r), M'(r), .., M^{(s)}(r), r).$$

If there exists a neighborhood U of $-\infty$ (resp. $+\infty$) such that $\overline{\Lambda_1}$ satisfies condition (c1) *(resp.* (c2)*) of Lemma 2.1, then there exists a neighborhood I_- (resp. I_+) of $-\infty$ (resp. $-\infty$) in which $\overline{\Lambda_1}(r) = 0$ and, therefore, in which M defines a TOV system.*

In our case of interest, the pseudo-asymptotic mass function M is that obtained in Section 2.1, whose underlying density functions are in Table 1, so that Λ_1 is given by (15), which depends on a function h, also listed in Table 1. So, what we need to do is to write down (15) explicitly for each (M, h) and analyze if there exists some neighborhood of infinity in which conditions (c1) or (c2) are satisfied. If it exists, then M defines a new TOV system when restricted to an appropriate domain. This can be done graphically. We then have:

Corollary 3. *If I is an union of suitable neighborhoods of $\pm\infty$, then the space* $\mathrm{TOV}_I^{\infty,\infty}$ *of piecewise C^∞-differentiable TOV systems is at least eleven-dimensional.*

2.3 Beyond

We proved Theorem 1.1 ensuring the existence of a coupling function Λ whose space $\mathrm{Psd}_\Lambda^\infty(I)$ of pseudo-asymptotic solutions is *at least* eleven-dimensional. In this section we show that it is at least fifteen-dimensional and discuss why it is natural to believe that it has an even higher dimension. The first assertion is due to the following reason:

1. *Existence of critical configurations exhibiting phase transitions.* Notice that the integrability conditions of [12], such as (10), (20) and (21), depends on two constants c_1 and c_2. Consequently, the pseudo-asymptotic solutions of the corresponding coupling equations also depend on such constants. Let us write $M_{1,2}$ to emphasize that M is a pseudo-asymptotic mass function depending on c_1 and c_2. Now, recall that two mass functions M, N are linearly dependent in $\mathrm{Psd}_\Lambda^l(I)$ if there exists a real number c such that $M(r) = cN(r)$ for every $r \in I$. This means that if the dependence of $M_{1,2}$ on c_1 (resp. c_2) is **not** in the form $M_{1,2} = c_1 M_2$ (resp. $M_{1,2} = c_2 M_1$), then when varying c_1 (resp. c_2) we get at least two linearly independent pseudo-asymptotic mass functions. These linearly independent mass functions can be obtained by defining a new piecewise differentiable map $\mathcal{M} : I \times \mathbb{R} \times \mathbb{R} \to \mathbb{R}$, such that

[2]This clearly does not means that $\overline{f}$ and $\hat{f}$ are continuous in the whole domain $\overline{\mathbb{R}}$ or $\hat{\mathbb{R}}$.

$\mathcal{M}(r, c_1, c_2) = M_{1,2}(r)$, and then by studying its critical points. In typical cases (as for those obtained in Section 2.1, i.e, for those whose density function is in Table 1) the function $\mathcal{M}$ is piecewise a submersion. Therefore, the pre-images $\mathcal{M}^{-1}(c)$ are submanifolds S_c of $\mathbb{R}^3$ and the linearly independent mass functions can be obtained by searching for topology changes in S_c when c vary (in a similar way we search for phase transitions in a statistical mechanics system). In the next section we will analyze the topology of the surface S_c corresponding to rows 1, 2 and 7 of Table (1), showing that they admit one, two and one topology changes, respectively. This means that Theorem 1.1 and Corollary 3 can be improved, giving the theorem below.

Theorem 2.1. *There exists a maximally split coupling function Λ such that $\mathrm{Psd}^{\infty}_{\Lambda}(I)$ is at least fifteen-dimensional. Furthermore, if I is a union of suitable neighborhoods of $\pm\infty$, then $\mathrm{TOV}^{\infty,\infty}(I)$ is at least fifteen-dimensional.*

The second assertion is suggested by the following reason:

2. *Existence of other maximally split coupling functions.* Recall that our starting point to get Λ in (13) was the integral equation (10) obtained in [12] which when satisfied induces a solution for the TOV equation. In [12], besides (10), other nine integral/differential equations playing the same role are presented. Applying to these other nine equations a strategy analogous to that used in (10) to obtain (13), allows us to obtain new coupling equations. We recall that each integrability equation in [12] becomes parametrized by certain constants and by an arbitrary function f. By making a suitable choice of f in the sixth and eighth cases of [12], we see that the induced coupling equations coincide with (13). Explicitly, the sixth case is given by

$$B(r) = \frac{f_3(r) - A(r)C(r) - C^2(r)\left[\int \frac{f_3(r)}{2C(r)}\,dr - c_7\right]^2}{2C(r)\left[\int \frac{f_3(r)}{2C(r)}\,dr - c_7\right]} \tag{20}$$

where $f_3 \in C^{k+1}_{pw}(I)$ and c_7 is a constant of integration. For the choice $f_3(s) = 2C(s)g(s)$, where $g \in C^{k+1}_{pw}(I)$, the induced coupling equation reduces to (13). Furthermore, the eighth case is

$$B(r) = \frac{1}{f_4(r)}\left[C(r)\frac{d}{dr}\left(\frac{f_4(r)}{C(r)}\right) - \frac{f_4^2(r)}{2} - 2A(r)C(r)\right] \tag{21}$$

where $f_4 \in C^{k+1}_{pw}(I)$. If we choice $f_4(r) = h(r)C(r)$ again we get (13). Therefore, in view of the methods developed in Section (2.1), equations (20) and (21) do not differ from (10). On the other hand, we could not find f which makes the coupling equation induced by each of the other seven cases equals to (13). This does not means that they will produce new pseudo-asymptotic mass functions which will eventually define (via Corollary 2) new TOV systems[3]. Even so, it suggests the possibility.

[3]Indeed, many of the induced maximally split coupling equation have a nonlinear part which is really high non-linear.

3 Classification

So far we have focused on getting new integrable TOV systems. In this section we will work to give physical meaning to discovered systems. In order to do this, we propose a simple classification of general stellar systems in which we will consider the new TOV systems. Indeed, let $(p, \rho) \in C^k_{pw}(I) \times C^l_{pw}(I)$ be a stellar system of degree (k, l). We say that it is

- *ordinary (resp. exotic)* in an open interval $J \subset I$ if the density ρ is positive (resp. negative) in each point of J. That is, $\rho(r) > 0$ (resp. $\rho(r) < 0$) for r in J;
- *without cavities* if ρ has no zeros, i.e, $\rho(r) \neq 0$ for every r in I;
- *without singularities* if its domain is an open interval $(0, R)$, where R can be ∞;
- *smooth* if it is without singularities and $\rho \in C^\infty(I)$;
- *realistic* if it is smooth and ordinary in I.

A *cavity radius* of (p, ρ) is a zero of ρ. Similarly, a *singularity* of (p, ρ) is a discontinuity point of ρ. So, (p, ρ) is without cavities (resp. without singularities) iff it has no cavity radius (resp. singularity).

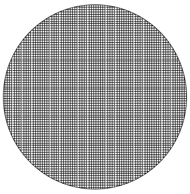
When looking at Table 1 it is difficult to believe that some of the stellar systems there described are realistic. In fact, as can be rapidly checked, when defined in their maximal domain, these systems are in fact unrealistic. But as we will see, when restricted to a small region, many of them becomes realistic. This becomes more clear looking at the figures below, which describe the classification of certain rows of Table 1. In the schematic drawings, the filling by the grid is associated to exotic matter, whereas the filling by the hexagons to ordinary matter. The dashed circles represents a singularity radius and the dot circles a cavity radius.

Notice that in order to do this classification we need to search for the zeros of the density function, which will give the radii in which there is no matter inside the star. Suppose that we found one of them, say r_o. If ρ is continuous in that radius, then it is a cavity; otherwise, it is a singularity. The fundamental difference between them is that continuity implies that ρ cannot change sign in neighborhoods of r_o. This means that a star containing only cavities is composed either of ordinary matter or of exotic matter. On the other hand, stellar systems with singularities may contain both ordinary and exotic matter.

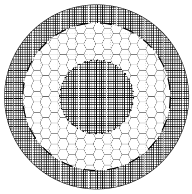
The stellar systems considered in Table 1 have densities of the form $\rho(r) = \frac{p(r)}{q(r)} \log(o(r))$, where p, q and o are polynomials with integer or fractional powers. Singularities are identified with zeros of q, while cavities are given by zeros of p and $o - 1$. The existence of real roots for a given polynomial is strongly determined by its coefficients and, in the present situation, the coefficients of p, q and o depend on two real parameters c_1 and c_2. We write p_{12}, q_{12} and o_{12} in order to emphasize this fact. We can then search for *critical configurations*, in which a small change of c_1 and c_2 produces a complete modification of the system, as in a phase transition in statistical mechanics. The critical configurations can be captured by defining new functions

$$\mathcal{P}, \mathcal{Q}, \mathcal{O} : [0, \infty) \times \mathbb{R} \times \mathbb{R} \to \mathbb{R}$$

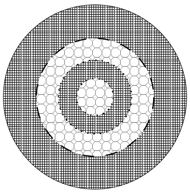
by $\mathcal{P}(r, c_1, c_2) = p_{12}(r)$, and so on, which are piecewise submersions. The solution of $\mathcal{Q}$ is then an algebraic submanifold $S_p \subset \mathbb{R}^3$, possibly with boundary, that completely determines the behavior



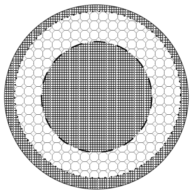
(a) Schematic Drawing of Figure 2



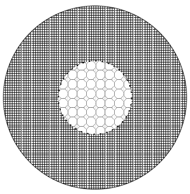
(b) Schematic Drawing of Figure 3



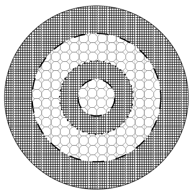
(c) Schematic Drawing of Figure 4d



(d) Schematic Drawing of Figure 4e



(e) Schematic Drawing of Figure 5a



(f) Schematic Drawing of Figure 5b

Figure 1: Classification of some new TOV systems.

of the singular set of ρ_{12} when we vary c_1 and c_2. Similarly, the disjoint union of the solutions sets of $\mathcal{P}$ and $\mathcal{O}-1$ also defines an algebraic submanifold S_{po} of $\mathbb{R}^3$ which determines the behavior of the cavities when we vary c_1 and c_2.

Notice that a point $(r, c_1, c_2) \in S_q$ is a critical singularity of ρ_{12} iff it admits topologically distinct neighborhoods. Analogously, the critical cavities are given by points in S_{po} with non-homeomorphic neighborhoods. Finally, the critical configurations of the stellar system with density ρ_{12} are the points of $S_q \sqcup S_{po}$ which are critical singularities or critical cavities. A manifold defined by the inverse image of a function is locally homeomorphic to the graph of the defining function. This means that a neighborhood for (r, c_1, c_2) is just a piece of the graph of $\mathcal{P}$, $\mathcal{Q}$ or $\mathcal{O}-1$. Since the topology of a graph changes only at an asymptote, zeros of Q with fixed (c_1, c_2) give the singularities, while zeros with fixed (r, c_2) and (r, c_1) will give the critical singularities, and similarly for critical cavities.

Having obtained singularities, cavities and critical configurations, the classification is completed by determining the kind of matter, which can be done from graphical analysis. In next sections we will apply this strategy to rows 1, 2 and 7 of Table 1. The row 1 will produce the schematic drawings (1a) and (1b), while row 2 will produce (1c) and (1d), and row 7 will give (1e) and (1f).

3.1 Row 1 of Table 1

In this case the density function in given by

$$\rho(r,c_1,c_2)=\frac{c_2\left(c_1-5\pi r^2\right)}{4\pi r^4\left(\pi r^2-c_1\right)^3}. \tag{22}$$

Notice that c_2 is a multiplicative constant, so that it will give linearly dependent solutions, leading us to fix $c_2=1$. This means that the singular set S_q is a submanifold of $\mathbb{R}^2$, while the set of cavities S_{po} is a submanifold of $\mathbb{R}^3$ of the form $S\times\mathbb{R}$, with $S\subset\mathbb{R}^2$. More explicitly, singularities are given by the solutions of the algebraic equation

$$4\pi r^4(\pi r^2-c_1)^3=0, \tag{23}$$

while cavities are determined by the solutions of

$$c_1-5\pi r^2=0 \tag{24}$$

Both equations depend explicitly of c_1. If $c_1\le 0$, the only singularity radius is the origin $r=0$ and there are no cavities. An example of this behavior is given by Figure 2, for $c_1=0$:

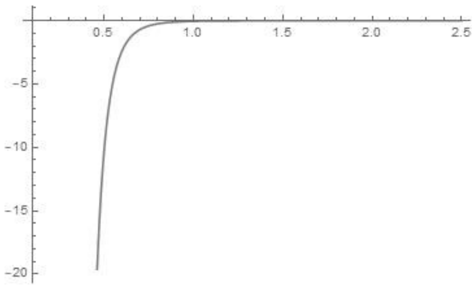


Figure 2: Plot of (22) for $c_1=0$ and $c_2=1$.

It then follows that, for our choice $c_2=1$, the stellar system is composed only by exotic matter. On the other hand, if $c_1>0$, the singularities happen at $r=0$ and at $r_0=\sqrt{c_1/\pi}$, which is the non-negative solution of $\pi r^2-c_1=0$. The single cavity is given by the single positive root $r_1=\sqrt{c_1/5\pi}$ of (24), which lies in the interval $(0,r_0)$ bounded by the singularities. Thus, the type of matter inside $(0,r_0)$ may change, but it remains the same after crossing r_0. In order to capture this change of matter we analyze the sign of the derivative of ρ_{12} at r_1. The derivative is

$$\rho_{12}'(r)=\frac{c_2\left(c_1^2-5c_1\pi r^2+10\pi^2r^4\right)}{\pi r^5\left(\pi r^2-c_1\right)^4} \tag{25}$$

and we see that $\rho_{12}'(r_1)>0$. So, in that point, the star matter stops being exotic and becomes ordinary. Moreover, we have $\lim_{r\to r_{0+}}\rho_{12}(r)=+\infty$ and $\lim_{r\to r_{0-}}\rho_{12}(r)=-\infty$. It follows that in (r_0,∞) the star is composed of exotic matter. An example of this behavior is given by Figure 3, where $c_1=7$:

Now, solving (24) and (23) we see that for each given r_0 there exists a single c_1 making r_0 a critical configuration.

13

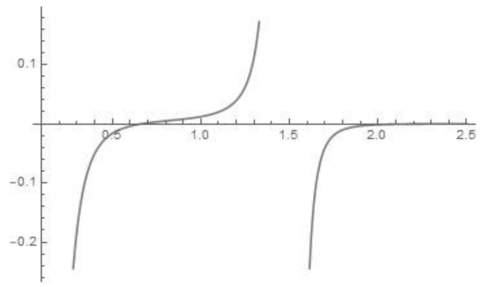


Figure 3: Plot of 22 for $c_1 = 7$ and $c_2 = 1$.

3.2 Row 2 of Table 1

We start by noticing that in the present case the density function can be written as

$$\rho_{12}(r) = -\frac{\Gamma_{12}(r)}{8\pi r^4 \left(r^2-1\right)\left(-c_1+\pi r^2+\pi \log\left(r^2-1\right)\right)^3} \tag{26}$$

where

$$\begin{aligned}\Gamma_{12}(r) = &-2c_1c_2\left(r^2-1\right)+3\pi^2\left(r^2-1\right)\left(\pi r^2-c_1\right)\\ &\times\left[\log\left(r^2-1\right)\right]^2-2\pi[c_2\left(1-5r^2\right)r^2\\ &+c_1^2\left(r^4+2r^2-1\right)]+\pi\left(r^2-1\right)\\ &\times\left[2\pi c_1\left(1-3r^2\right)+2\left(c_1^2+c_2\right)+\pi^2\left(3r^4-1\right)\right]\\ &\times\log\left(r^2-1\right)-\pi^2c_1\left(3r^6-13r^4+r^2+1\right)\\ &+\pi^3\left(r^2-1\right)\left[\log\left(r^2-1\right)\right]^3\\ &+\pi^3\left(r^8-r^6-5r^4+r^2\right).\end{aligned} \tag{27}$$

The function ρ_{12} depends non-trivially on both variables c_1 and c_2 only in Γ_{12}, so that a priori its singular set and its set of zeros are arbitrary submanifolds of $\mathbb{R}^3$ and $\mathbb{R}^2$, respectively, and therefore we may expect many critical configurations. The interval $[0,1]$ is clearly singular, while the singularities for $r > 1$ are determined by solutions of the equation

$$-c_1+\pi r^2+\pi\log\left(r^2-1\right)=0.$$

We assert that for each fixed c_1 there is precisely one solution. In other words, we assert that the singular set is diffeomorphic to $([0,1]\times\mathbb{R})\sqcup\mathbb{R}$. Indeed, let $Y : [0,\infty)\times\mathbb{R}\to\mathbb{R}$ be the piecewise differentiable function

$$Y(r,c_1)=-c_1+\pi r^2+\pi\log\left(r^2-1\right). \tag{28}$$

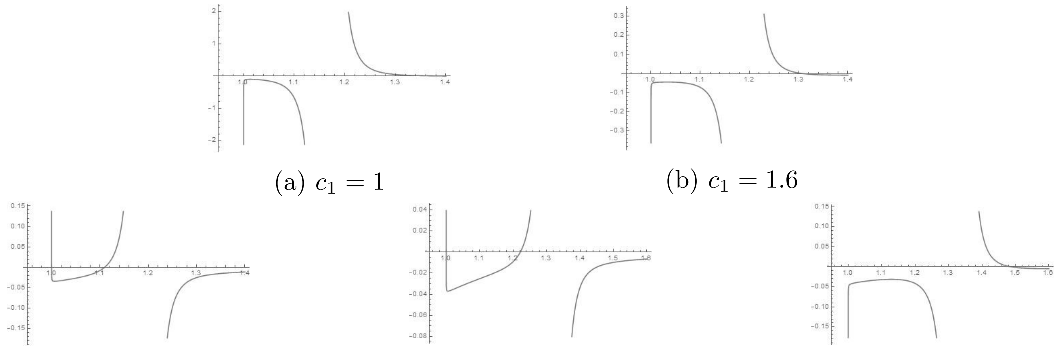
Clearly, there are r_0 and r_1 such that $Y(r_0,c_1)<0$ and $Y(r_1,c_1)>0$. But, the derivative of $Y(r,c_1)$ in the direction of r is always positive for $r>1$. If fact, if $r>1$, then

$$\frac{\partial Y}{\partial r}(r,c_1)=\frac{2r(r^2+\pi-1)}{r^2-1}>0.$$

Consequently, Y has a single zero in $r > 1$ corresponding to the unique singularity of ρ_{12} in this interval. Moreover, for any $r_0 > 1$, we can set the constant c_1 appropriately so that r_0 corresponds the singularity. In fact, just set

$$c_1 = \pi r_0^2 + \pi \log\left(r_0^2 - 1\right).$$

The cavities of ρ_{12} in (26) are given by zeros of Γ_{12}. Differently of the previous case, we cannot isolate any of the variables, so that $S_{po} \subset \mathbb{R}^3$ will have complicated topology. We assert that the manifold of critical configurations is $\mathbb{R}^2$. Indeed, if we fix c_1 and vary c_2 we do not find any change of topology, while if we fix c_2, say $c_2 = 1$, we find two critical points, approximately at $c_1 = 1,7$ and at $c_1 = 4,6$, as shown in Figure 4.



(a) $c_1 = 1$ (b) $c_1 = 1.6$

(c) $c_1 = 1.8$ (d) $c_1 = 4.5$ (e) $c_1 = 4.7$

Figure 4: Plot of (26) for different values of c_1 and constant $c_2 = 1$.

3.3 Row 7 of Table 1

Previously we studied a TOV system with discrete critical configurations and another with a continuum of critical configurations. In both cases, graphical analysis were used to identify the critical configurations. Here we will discuss a third system, whose special feature is to show that even when we have an analytic expression for cavities and singularities, a graphical analysis is fundamental to complete the classification.

The density function is now of the form

$$\rho(r, c_1, c_2) = f(r, c_1, c_2)/q(r, c_1), \tag{29}$$

where f is a linear combination of polynomials and logarithmic functions, depending on both parameters c_1 and c_2. We will focus on singularities, so that only the expression of q matters. It is the polynomial

$$q(r, c_1) = 12\pi r^2(-c_1 r + \pi r^3 + 1)^3.$$

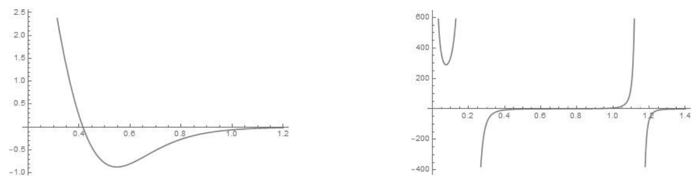
Because c_2 does not affect q we see that the singular set is a submanifold of $\mathbb{R}^2$. For each fixed c_1 the complex roots of q can be written analytically with help of some software (we used Mathematica®). They are $r_0 = 0$, with multiplicity two, and

$$r_1 = \frac{\sqrt[3]{2}\left(\sqrt{81\pi - 12c_1^3} - 9\sqrt{\pi}\right)^{2/3} + 2\sqrt[3]{3}c_1}{6^{2/3}\sqrt{\pi}\sqrt[3]{\sqrt{81\pi - 12c_1^3} - 9\sqrt{\pi}}} \tag{30}$$

$$r_2 = \frac{\sqrt[3]{2}\sqrt[6]{3}\left(-1+i\sqrt{3}\right)\left(\sqrt{81\pi-12c_1^3}-9\sqrt{\pi}\right)^{2/3}-2\left(\sqrt{3}+3i\right)c_1}{2\ 2^{2/3}3^{5/6}\sqrt{\pi}\sqrt[3]{\sqrt{81\pi-12c_1^3}-9\sqrt{\pi}}} \tag{31}$$

$$r_3 = \frac{\sqrt[3]{2}\sqrt[6]{3}\left(-1-i\sqrt{3}\right)\left(\sqrt{81\pi-12c_1^3}-9\sqrt{\pi}\right)^{2/3}-2\left(\sqrt{3}-3i\right)c_1}{2\ 2^{2/3}3^{5/6}\sqrt{\pi}\sqrt[3]{\sqrt{81\pi-12c_1^3}-9\sqrt{\pi}}}, \tag{32}$$

each of them with multiplicity three. Our stellar system then have singularities only for the values c_1 such that some of the radii above are real and non-negative. One can be misled to think that r_1 corresponds to a real zero iff $c_1 \leq \sqrt[3]{\frac{81\pi}{12}} \approx 2.77$. Although $c_1 \approx 2.77$ is clearly a critical configuration, a graphical and numerical analysis shows that r_1 is always real. In order to see this, we compare in Figure 5 the density function for $c_1 = 5 > 2.77$ and $c_1 = 1 < 2.77$.



(a) $c_1 = 1$ (b) $c_1 = 5$

Figure 5: Plot of (29) of c_1 and constant $c_2 = 1$.

Although Figure (5a) seems physically interesting, it is not: a numerical evaluation shows that $r_1(c_1) \approx -0.84$, so that despite being real, it is negative. In turn, Figure (5b) should appear strange: it contains two nonzero real roots while the analytic expressions above suggest that there is at most one non-null real root. Again, a numerical evaluation gives $r_1(c_1 = 5) \approx 1.15$, $r_2(c_1 = 5) \approx -1.35 + 1.69 \times 10^{-21} i$ and $r_3(c_1 = 5) \approx 0.21 - 4.24 \times 10^{-22} i$. So, we see that when c_1 grows, the imaginary parts of the roots r_2 and r_3 become increasingly smaller, allowing us to discard them. Thus, $c_1 \approx 2.77$ determines a critical configuration such that for $c_1 < 2.77$ there are no singularities and for $c_1 > 2.77$ there are two of them.

4 Summary

In this paper we considered the problem of classifying the stellar systems modeled by the piecewise differential TOV equation (1). We began by introducing the problem in the general context presented in [1], allowing us to formalize the problem as the determination of the structure of $\mathrm{TOV}_I^{k,l}$ as a subspace of $C_{pw}^{k,l,l+1}(I)$ relative to some locally convex topology. We showed that this subspace is generally not open if the topology is Hausdorff. We introduced another subspace $\mathrm{Psd}^{l+1}(I) \subset C_{pw}^{k,l,l+1}(I)$ of pseudo-asymptotic systems and we showed that for $l = \infty$ this space is at least fifteen-dimensional and that there are good reasons to believe that its dimension is larger. We proved that if I is some suitable neighborhood of the infinity (in the sense of the extended real line), then $\mathrm{Psd}^{l+1}(I) \subset \mathrm{TOV}_I^{k,l}$, leading us to conclude that $\mathrm{TOV}_I^{\infty,\infty}$ is at least fifteen-dimensional. So, we have new fourteen integrable TOV systems besides the well known constant density solution. We presented a method of classification of these solutions, applying it to some of them, allowing us to show that there are new TOV systems consisting only of ordinary matter and that contain no cavities or singularities.

A Table of Solutions

Function $F(r)$ (Eq. 16)	$h(r)$	$\rho(r)$
0	$\frac{1}{\pi r^2 - c_1}$	$\frac{c_2\left(c_1 - 5\pi r^2\right)}{4\pi r^4\left(\pi r^2 - c_1\right)^3}$
$6{*}h'(r)$	$6{*}\frac{1}{-c_1+\pi r^2+\pi\log\left(r^2-1\right)}$	$[-2c_1c_2\left(r^2-1\right)+3\pi^2\left(r^2-1\right)\left(\pi r^2-c_1\right)\left[\log\left(r^2-1\right)\right]^2$ $-2\pi\left(c_2\left(1-5r^2\right)r^2+c_1^2\left(r^4+2r^2-1\right)\right)+\pi\left(r^2-1\right)$ $\times\left(2\pi c_1\left(1-3r^2\right)+2\left(c_1^2+c_2\right)+\pi^2\left(3r^4-1\right)\right)\log\left(r^2-1\right)$ $-\pi^2c_1\left(3r^6-13r^4+r^2+1\right)+\pi^3\left(r^2-1\right)\left[\log\left(r^2-1\right)\right]^3$ $+\pi^3\left(r^8-r^6-5r^4+r^2\right)] : [8\pi r^4\left(r^2-1\right)$ $\times\left(-c_1+\pi r^2+\pi\log\left(r^2-1\right)\right)^3]$
$9{*}rh'(r)$	$9{*}\frac{1}{-c_1+\pi r^2+2\pi r+2\pi\log(r-1)-3\pi}$	$[60\pi^2(r-1)\left(9c_1+\pi\left(8r^3-9r^2-18r+31\right)\right)\log^2(r-1)$ $+30\pi\left(c_1^2\left(4r^4-r^3-3r^2+17r-11\right)-3c_2\left(5r^3+r^2-5r+3\right)\right)$ $+2\pi(r-1)[-30\pi c_1\left(8r^3-9r^2-18r+31\right)-90\left(c_1^2+c_2\right)$ $+\pi^2\left(24r^5+135r^4-1460r^3+390r^2+1860r-1669\right)]$ $\times\log(r-1)-\pi^2c_1[24r^6+111r^5-1595r^4+1850r^3+1470r^2$ $-3529r+1669]+90c_1c_2(r-1)+\pi^3[9r^7-115r^6-803r^5$ $+4465r^4-3365r^3-3529r^2+5375r-2037]$ $-360\pi^3(r-1)\log^3(r-1)] : [360\pi(r-1)r^4[-c_1+\pi\left(r^2+2r-3\right)$ $+2\pi\log(r-1)]^3]$
$h^2(r)$	$\frac{r}{-c_1r+\pi r^3+1}$	$\frac{-\pi r^3(10c_1r+15c_2+7)+3\left(-c_1r+5\pi r^3-1\right)\log(r)+3(c_1(c_2+3)r+c_2-1)+2\pi^2r^6}{12\pi r^2\left(-c_1r+\pi r^3+1\right)^3}$
$3{*}rh^2(r)$	$3{*}\frac{1}{-c_1+\pi r^2-\log(r)}$	$[-3\pi\left(c_1-1\right)r^4-\left(2c_1^2-3c_1+20\pi c_2+2\right)r^2$ $+\left(\left(3-4c_1\right)r^2+4c_2-3\pi r^4\right)\log(r)+4\left(c_1+2\right)c_2+\pi^2r^6$ $-2r^2\log^2(r)] : [-16\pi r^4\left(c_1-\pi r^2+\log(r)\right)^3]$
$r^2h^2(r)$	$\frac{1}{-c_1+\pi r^2-r}$	$-\frac{(8\pi c_1-15)r^5-45c_1r^4-40c_1^2r^3+300\pi c_2r^2-180c_2r-60c_1c_2+9\pi r^6}{240\pi r^4\left(c_1-\pi r^2+r\right)^3}$
$r^3h^2(r)$	$\frac{2}{-2c_1+2\pi r^2-r^2}$	$\frac{7(2\pi-1)c_1r^6-18c_1^2r^4+15(1-2\pi)c_2r^2+6c_1c_2-(1-2\pi)^2r^8}{12\pi r^4\left((2\pi-1)r^2-2c_1\right)^3}$
$3{*}\frac{h^2(r)}{r}$	$3{*}\frac{2r^2}{-2c_1r^2+2\pi r^4+1}$	$[\left(-2c_1r^2+2\pi r^4+1\right)\left(4c_1r^2+3c_2r^2+12c_1r^2\log(r)-10\pi r^4+1\right)$ $-2\left(8\pi r^3-4c_1r\right)\left(c_2r^3+4c_1r^3\log(r)-2\pi r^5+r\right)]$ $: [4\pi r^2\left(-2c_1r^2+2\pi r^4+1\right)^3]$
$r^{1/2}h^2(r)$	$\frac{\sqrt{r}}{-c_1\sqrt{r}+\pi r^{5/2}+2}$	$\frac{-7c_1^2r^{3/2}-34\pi c_1r^{7/2}-105\pi c_2r^2+42c_1r+21c_1c_2+9\pi^2r^{11/2}+126\pi r^3-84\sqrt{r}}{84\pi r^{5/2}\left(-c_1\sqrt{r}+\pi r^{5/2}+2\right)^3}$
$r^{3/4}h^2(r)$	$\frac{\sqrt[4]{r}}{-c_1\sqrt[4]{r}+\pi r^{9/4}+4}$	$[440c_1r^{7/4}-525\pi c_2r^{9/4}-118\pi c_1r^4-45c_1^2r^2+105c_1c_2\sqrt[4]{r}$ $-210c_2-1120r^{3/2}+616\pi r^{15/4}+35\pi^2r^6]$ $: [420\pi r^{7/2}\left(-c_1\sqrt[4]{r}+\pi r^{9/4}+4\right)^3]$

Table 1: Solutions of the coupling equation (13) in the pseudo-asymptotic limit.

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4 GEOMETRY OF GRAVITY

4.1 SYNTHESIS

Since General Relativity appeared in 1915 as a proposal to describe gravity, a movement of "geometrization of Physics" has been developed following the primary ideas of physicists as Einstein and Eddington, mathematicians as Weyl, Klein and Hilbert, and philosophers as Meyerson and Reichenbach, which provided different lines of thought (for interesting expositions, see [140]):

1. that geometrization is a methodological doctrine, where geometry should be the primary language to be used in order to model Physics;
2. that the fundamental laws of Physics *can* be modeled using geometry;
3. that the fundamental laws are geometric identities, so that they *must* be modeled using geometry;
4. that the entire Physics reduces to the study of geometry;
5. that Physics and geometry are equivalent in some sense.

The first one does not establish a philosophical constrain between Physics and geometry; neither from an ontological nor from an epistemological point of view[1]. It postulates just an "statute" on how to research in Physics: in the step of searching for an analogous formal model, look first at those defined using geometry. This is discussed, e.g., in [89], where the author argues that it is closely related to the unification problem of Physics.

Note that, a priori, assuming the first notion of "geometrization", a physical system could be modeled by a class of formal systems other than geometry. The possibility of a physical system that cannot be modeled by geometry is also allowed. In the second perspective to "geometrization" the latter possibility is remove, while in the third line of thought the former possibility is removed too. Here, again, there is no strong ontological and epistemological appeal, specially concerning the second line of thought: they are more about the modeling processes. These are so far the most widespread perspective.

Actually, how far we observed, when the expression "geometrization of Physics" is used without a precise meaning, it was implicitly used in some of these two perspectives. An example is Palais's notes [132], entitled "Geometrization of Physics", but whose goal, as explained in the preface,

[1] Actually, as discussed in Chapter 2, this kind of assumption comes with the hypothesis that it is possible to study some aspects of a physical system trough its models, which has an epistemological content.

> *[...] was to develop carefully the mathematical tools necessary to understand the classical [...] aspects of gauge fields and then to present the essentials [...] of the physics.*

In the abstract to Researchgate version of these notes, Palais also says: "*We give a survey of the way much of Physics was reduced to Geometry in the Twentieth Century*", which could be interpreted as an explicit assumption of fourth perspective, but a careful inspection would reveal the misunderstanding. A similar kind of misunderstanding can be observed in the works of many of the 21th-century mathematicians who worked directly with geometry arising from modeling Physics.

One could think that this misunderstanding is purposeful, since the main objective of these works is to study the mathematics arising from modeling the physical entities, so that they are directed to geometers, which normally do not take into account the philosophy of physics concerns. Another possibility, if the work in question concerns mathematical models to gravity, is that they were influenced by the folklore that Einstein was the father of the fourth line of thinking above, i.e., that Physics is nothing but a sort of geometry [103]. For instance, in [159] Wanas says that

> *[...] Einstein has used the geometrization philosophy in constructing his theory. Among other principles of this philosophy is that laws of nature are just differential identities in the chosen geometry.*

But, in the scientific medium of Philosophy of Science and History of Sciences, it is argued the opposite: *that Einstein disagreed with the idea of Physics reduced to geometry*, which is specially clear in letters exchanged by him, Reichenbach and Meyerson [103, 62, 63, 64]. For instance, referring to a letter from Einstein to Reichenbach, Giovalli writes in [62] that

> *[Einstein] explicitly shared Reichenbach's skepticism toward the rhetorics of the 'geometrization of physics' that were widespread in technical and philosophical literature.*

As another example, in his classical talk "On the Method of Theoretical Physics", Einstein emphasizes his belief that reality is not reduced to mathematics, making explicitly his proximity to the first line of though above [51, 89]:

> *Experience remains, of course, the sole criterion of the physical utility of a mathematical construction. But the creative principle resides in mathematics. In a certain sense, therefore, I hold it true that pure thought can grasp reality, as the ancients dreamed.*

Concerning the philosophical foundations of Physics, this discussion is very important: unlike the first three approaches to geometrization, the forth and the fifth ones rely on strong ontological and epistemological hypotheses. Indeed, in the case of the fourth, assuming that everything on Physics is about geometry we conclude that physical reality is determined by logical existence, which is the assumption of some kind of formal ontology. Also, if everything that can be learned about the physical world depends on the investigation in geometry, then we are in the hypothesis of formal epistemology[2]. It seems that this was the position of Weyl and Eddington about geometrization that was used in their unification theories [139]. For example, according to Kouneiher in [98], referring to his work on unification, Weyl said in [162] that (see also [144]):

> *According to this theory, everything real, that is in the world, is a manifestation of the world metric; the physical concepts are no different from the geometrical ones.*

[2]Of course, here we are thinking in "geometry" as a formal theory, which itself involves a lot of debate, e.g., between Hilbert and Fregue at the beginning of 20th-century [77, 57, 153, 48].

The fifth perspective to geometrization is even more restrictive, since it presupposes not only that Physics can be reduced to geometry, but also that geometry reduces to Physics. If Physics is the study of physical objects, which in turn are real entities, then this position assumes that geometric objects are real too. Thus either geometrical objects are at the same time abstract objects and real objects or they are not abstract objects but are physical entities. In the first case, we are in a form of mathematical realism, such as mathematical platonism and its variations, while the second could be attended assuming the thesis of physicalism. In particular, if we are on an empiricist ontology basis, then physicalism implies that geometrical objects should have existence detected by experiments. From the epistemological side, this also implies a positivistic perspective.

This more intricate relation between Physics and geometry was the point of view adopted by Hilbert and others who believed in a unified approach to science [143]. In the case of Hilbert, this belief was explicit in Hilbert's sixth problem, as discussed in Chapter 2. For a discussion on the geometrical realism in the works of Weyl, see [64, 140].

4.1.1 Geometry of Gravity

Which is the geometry of gravity? Of course, by the previous discussion, this question depends on what we mean by "geometrization of Physics", as in the five lines of tough above. The assumption of each of them presupposes its own ontological and epistemological basis, which can be realized by different theses. Independently of the relation between geometry and Physics, all the five lines of tough agree that geometry is at least a natural language to model Physics. But, at the same time, they do not deny the existence of two non-equivalent geometric systems modeling the same physical system. A similar situation occurs with axiomatic systems, which typically admit a lot of non-equivalent models.

Thus, an interesting question should be: *what kind of geometries cannot be used to model gravity?* General Relativity (GR), as formulated in terms of Lorentzian metrics, is the classical model and so far the most well-accepted from the empiricism point of view, so that if we are in an empiricist ontology, then Lorentzian geometry should belong to the class of the geometries that model gravity. Also, there are other formulations, physically equivalent to GR, such as Einstein-Hilbert-Palatini theory (EHP) and Teleparallel Gravity, which also are formulated in Lorentzian geometry, but considering different geometric structures (Cartan connections and affine connections) as the fundamental ones [35, 42, 94].

There is also a plethora of theories aiming to model gravity, but which are not physically equivalent to GR (for reviews, see [40, 74]). They contain GR at some limit, trying to regard GR as an effective theory. The interest in such alternative theories is partially motivated by the fact that, unless being the most well-accepted theory of gravity, it faces some technical difficulties in its quantization[3] and with cosmological predictions[4] [44, 158]. Since its proposal in 1915, there is also some philosophical criticism to GR [140, 23].

The alternatives to GR occurs essentially in the following forms:

1. *extending the domain of discourse*[5], which is obtained extending the domain of the action functional. This is the case of Einstein-Cartan, which include GR in the limit of symmetric affine connections, but is physically different to GR because it allows the

[3] The quantization problem, as discussed in Chapter 2, has its own mathematical and philosophical concerns.
[4] But again this have its philosophical considerations, as commented in Chapter 3.
[5] About extensions of gauge theories and their relation with emergence phenomena, see the discussion of Chapter 5.

existence of singularities arising from geodesic incompleteness (i.e., Penrose-Hawking theorems may fail) [135];

2. *extending the shape of the action functional*. The main example here is the so-called $f(R)$-gravity, whose action functional is $\int f(R_g)\omega_g$, which differs from the Einstein-Hilbert action functional $\int R_g \omega_g$ in the case of a function $f : C^{\infty}(M) \to C^{\infty}(M)$ which is not the identity. Another example is the Infinite Derivative Gravity, with action functional $\int f(R_g, g)\omega_g$, where $f(R_g, g)$ is a function depending not only on scalar curvature and its derivatives, but also on powers of the d'Alambert operator $\Box_g$ of g. Again, they are not physically equivalent to GR because they avoid Penrose-Hawking singularity theorems [49].

3. *both the processes together*. Einstein-Cartan, itself, is an example since it also has a modified action functional. However, the main examples are theories that presuppose that the correct description of gravity appears after coupling GR with some other theory. This is the case of supergravity theories, which differ physically from GR to predict the existence of such fundamental theory which should couple to GR.

Although there are a lot of well-studied alternatives to GR, we notice that all of them are about semi-Riemannian geometry, primarily about Lorentzian geometry. As far as we understand, the only exceptions are the supergravity theories, which have the part of the geometry arising from the other fundamental theory, Complex General Relativity of Esposito [45, 53] (which are based on Penrose-Rindler twistor theory [133]) and the symplectic stuff used in the canonical formalism [56, 100]. The latter, however, is much more about geometric structures on the phase space of gravity than the geometry of gravity itself.

Thus, one could affirm that, besides Lorentzian geometry, complex geometry and the geometries of supergravity are in the class of those geometries which can be used to model gravity, but there is a lack of formal results on which geometries could **not** be used. It is precisely in this gap that the works [117, 115] fit. Indeed, in these works, we developed a method that we call *Geometric Obstruction Theory*, which could be used to determine the obstructions that a geometry suffers to be used as a model to gravity, as we will discuss in the sequence.

4.1.2 Obstructions on GR Based Gravity

If we think of geometry as described by tensors, the obvious idea to build models to gravity based on geometries other than Lorentzian is to replace the Lorentzian metric g_{ij} by another tensor t_{ij}. This is the approach of the so-called *tensor theories of gravity*. For instance, this works well in the case of GR if t_{ij} is a semi-Riemannian metric with arbitrary signature, which is just a symmetric nondegenerate rank 2 tensor. We could think of lacking the nondegeneracy hypothesis, but then we would lack the possibility to raise and lower indexes, which is something widely used to define the geometric quantities of GR. There is an extension of semi-Riemannian geometry to the case of degenerate metrics (i.e., rank 2 symmetric tensors) which is called Singular Semi-Riemannian Geometry and allows the description of an extension of GR to the degenerate case [154, 155, 156]. Thus, we should include the geometry of arbitrary symmetric rank 2 tensors in the class of those which can be used to model gravity.

Recall that any rank 2 tensor t_{ij} decomposes into its symmetric and antisymmetric parts $t_{ij} = g_{ij} + \omega_{ij}$, and it is nondegenerate iff both parts are. In the nondegenerate case, therefore, t_{ij} it would have a semi-Riemannian part and an almost symplectic part, so that a version of GR based on arbitrary nondegenerate rank 2 tensors should have contributions arising from the semi-Riemannian metric g_{ij} and from the almost symplectic form ω_{ij}. However, as we discuss in the introduction of [117], a theory analogous to GR but based purely on an almost symplectic form has no physical content, since the action functional identically vanishes. This is due basically to the fact that the Ricci tensor $(Ric_t)_{ij}$ of 2-rank tensor t_{ij} is symmetric independently if t_{ij} is, so that the contraction $t^{ij}(Ric_t)_{ij}$, which is the scalar curvature R_t of t_{ij}, vanishes identically if t_{ij} is antisymmetric. Also, notice that a symplectic form may exist only on even-dimensional spacetime, which is an imposition of the model to the underlying physical objects and, therefore, presupposes some version of logical ontology and an interpretation of "geometrization of Physics" following the fourth line of though previously discussed.

As well as a semi-riemannian metric, an almost symplectic form ω also induces a volume form on the underlying $2n$-dimensional manifold by ω^n, so that a nongenerate rank 2 tensor has a total volume form $\omega_t = \omega_g + \omega^n$. The analogue of GR using general nongenerate rank 2 tensors should then have the action functional

$$\int R_t \omega_t = \int g^{ij}(Ric_t)_{ij}\omega_g + \int g^{ij}(Ric_t)_{ij}\omega^n.$$

Thus, a priori (assuming some philosophical concerns) one could also propose models to gravity based on semi-Riemannian geometry and almost symplectic geometry together. But, *what it means to take two geometries "together"*? The idea that geometry corresponds to tensors can be understood in terms of the more abstract point of view of reductions on the structural group of principal bundles. Recall that if $P \to M$ is principal G-bundle over a manifold M and $H \subset G$ is a subgroup (not necessarily closed), then G acts in the homogeneous space G/H and this induces an associated bundle $E = P \times_G G/H$ such that group structure reduction of P from G to H are in bijective correspondence with global sections of E. In the context of Klein geometry, the pair (H, G) defines a geometry that is modeled by the homogeneous space G/H. I we take G itself as a subgroup of $GL(n;\mathbb{R})$ and H_t as the group of linear transformations in $\mathbb{R}^n$ that leaves invariant a *fixed* tensor t in $\mathbb{R}^n$, then G/H is the space of *all* tensors in $\mathbb{R}^n$ which are invariant by H_t. Thus, in this case, the existence of group reduction on a G-bundle $PM$is equivalent to the existence of a globally defined tensor in M, i.e., geometry (in the sense of Klein) is about tensors.

Nondegenerate semi-riemannian metrics and almost symplectic structures correspond, respectively, to the reduction of $GL(n;\mathbb{R})$ to the orthogonal group with signature $O(n,r)$ and to the symplectic group $Sp(2n;\mathbb{R})$. They are the groups of transformations which leaves invariant the canonical nondegenerate form $g_0 = \mathrm{diag}(1,...,1,-1,...,-1)$ of signature (r,s) in $\mathbb{R}^{2n}$ and the canonical symplectic $\omega_0 = dp \wedge dp$ form in $\mathbb{R}^{2n}$, respectively. We then ask *what it means to take semi-riemannian geometry and symplectic geometry together?* By the above, this would be to consider group reduction from $GL(2n)$ to the subgroup H_t which leaves invariant a *fixed* nondegerate bilinear form t in $\mathbb{R}^{2n}$.

Above we noticed that $O(r,s)$ and $Sp(2n;\mathbb{R})$ are the subgroups that leaves invariant the *canonical* symmetric and antisymmetric forms in $\mathbb{R}^{2n}$. By g_0 being canonical we mean that each other symmetric nondegenerate 2-tensor g is conjugated with g_0 by the action of $GL(\mathbb{R}^{2n};\mathbb{R}^n)$, which in turn implies the subgroups that leaves each g invariant is isomorphic to $O(n,r)$. This is basically Sylvester's law of inertia. Similarly, ω_0 is canonical in the sense that each other ω is conjugated to ω_0, which also implies that its corresponding subgroup is isomorphic to $Sp(2n;\mathbb{R})$. This can be viewed as a version of Darboux's theorem. Thus, for the case of semi-Riemannian geometry and almost symplectic geometry alone, the corresponding geometries (in the sense of group reduction) is well-defined, because $H_g \simeq O(r,s)$ and $H_\omega \simeq Sp(2n;\mathbb{R})$ for every g, ω. But if one tries to consider them together, the corresponding group reduction is not well-defined, because there is no canonical nondegenerate 2-tensor, i.e., there is no canonical non-singular real matrix up to conjugation[6]. Thus, a priori if t and t' are two non-conjugated tensors, then H_t need not be isomorphic to $H_{t'}$ and therefore they could define non-equivalent geometries both describing the union of semi-Riemannian geometry with almost symplectic geometry.

Instead of looking for geometries that describe the *union* of semi-Riemannian geometry with almost symplectic geometry, one could consider its *intersection*. In terms of the language of group reductions, this would be to consider reductions from $GL(2n;\mathbb{R})$ to $O(r,s) \cap Sp(2n;\mathbb{R})$. The intersection is nontrivial only if $(r,s) = (2n,0)$, i.e. if we are in Riemannian signature. In this case, $O(2n) \cap Sp(2n;\mathbb{R}) \simeq U(n)$. Thus, the study of models to gravity based on the intersection of symplectic geometry with Riemannian geometry is actually about almost Kähler geometry, which we already know to be possible for the case of GR, as commented in the last section.

To summarize, we have the following conclusions:

- it is possible to formalize nontrivial analogs of GR on semi-Riemannian geometry for every signature (r,s);
- there is no physically nontrivial analog of GR based on almost symplectic geometry;
- it is possible to define nontrivial analogs of GR for arbitrary nondegenerate 2-forms and therefore on the union of semi-Riemannian geometry (of arbitrary signature) with almost symplectic geometry, but the corresponding theories are not well-defined in the sense that they depend on the choice of a nondegenerate 2-form in $\mathbb{R}^{2n}$;
- one can define a nontrivial analogue of GR in the intersection of Riemannian geometry with almost symplectic geometry, and therefore on Kähler geometry, which should be related with Complex General Relativity.

4.1.3 From GR to EHP

The discussion above on GR provides important insights on the search for a systematic method to determine the possibles geometries in which a gravity model can be realized:

1. the different analogues of GR are obtained taking an action functional with the same shape as GR functional (i.e, something with the same format as Einstein-Hilbert functional), but evaluated in other classes of tensors, which describe different geometries;
2. the impossibility to build a physically nontrivial model to gravity analogue to GR, based on a certain geometry (in the case above, in almost symplectic geometry), does *not* means that an Einstein-Hilbert functional cannot be constructed for such geometry, but actually that it identically vanishes.

The idea to formalize these facts is to think in building some kind of "synthetic GR", given by an abstract functional $S[x] = \int R_x \omega_x$, where x is an indeterminate range over all geometries t for which the corresponding expression $S[t]$ makes sense. The geometries whose corresponding theory is physically trivial are therefore that belong to the kernel of

[6]recall that Jordan normal form depends on Fundamental Theorem of Algebra, so that it works on $\mathbb{C}$ and therefore the Jordan matrix of a real matrix need not be real.

the map $t \mapsto S[t]$. Thus, the problem of describing the geometries in which GR is physically trivial consists of two steps:

1. to extract from the GR action functional a synthetic expression;
2. to determine the kernel of such expression.

This is the essence of the Geometric Obstruction Theory, as discussed in [117, 115]. We note, however, that it is hard to follow it if we start with the model to gravity given by GR. Indeed, the dependence of the tensor g_{ij} in the Einstein-Hilbert functional is both on the scalar curvature R_g and on the volume form ω_g. In turn, R_g involves products of g_{ij} and its first and second-order derivatives. Thus, the synthetic version of Einstein-Hilbert functional contains a term like $R(x, \partial x, \partial^2 x)$. The kernel of this synthetic functional would contain the solution to the partial differential equations $R(x, \partial x, \partial^2 x) = 0$. Therefore, *Geometric Obstruction Theory for GR involves much global analysis and nonlinear PDEs of different types (e.g., elliptic and hyperbolic).*

Observe that the analytical nature of Geometric Obstruction Theory for GR relies on the fact that geometry was considered in the sense of being given by tensors, which are precisely the variables in GR. But, as we discussed in the last section, geometry as tensors is a particular setup of Klein geometry (G, H) for which $G \subset GL(n; \mathbb{R})$ and H is the group of transformations that preserves a fixed tensor t in $\mathbb{R}^n$. Thus, in this sense, geometry is about Lie groups stuff, which is much more topological and analytical than algebraic. The idea is then to move the perspective to work in the infinitesimal setting of the Lie algebras instead of the Lie groups.

To motivate how this can be done, recall that the incarnation of tensor geometry in terms of Weil geometry is made by the bijection between reductions from G to H and global sections in a bundle with typical fiber G/H. Topologically, a principal G-bundle $P \to M$ is classified up to isomorphism by a homotopy class of continuous maps $f : M \to BG$, where BG is the classifying space of G. A reduction to H is then given by lifting to BH, as below. Therefore, on one side H-reductions are ways to get H-data from G-data, while on the other it is about getting G/H-data from G-data. This means that H-reductions are actually about splitting G-data into H-data and G/H-data.

$$\begin{tikzcd} & BH \arrow[d] \\ M \arrow[r, "f"'] \arrow[ur, dotted, "\hat{f}"] & BG \end{tikzcd}$$

This intuition becomes more accurate if we suppose that $H \subset G$ is a normal subgroup (so that G/H is itself a group) and such that the projection $G \to G/H$ has a section $s : G/H \to G$, as in the diagram below, then the exact sequence is right-splitting and the splitting lemma implies $G \simeq H \ltimes (G/H)$.

$$\begin{tikzcd} 0 \arrow[r] & H \arrow[r, "\imath"] & G \arrow[r, shift left, "\pi"] & G/H \arrow[l, shift left, "s"] \arrow[r] & 0 \end{tikzcd}$$

Then, a H-reduction induces a (G/H)-bundle, as in the first diagram below. If the sequence happens to be left-splitting too, i.e., if there is a retraction $r : G \to H$, then the reciprocal holds, meaning that a reduction to a (G/H)-bundle induces an H-reduction, as in the second diagram below. Actually, in this case, we have $G \simeq H \times (G/H))$, and since the classifying space functor preserves products we also have $BG \simeq BH \times B(G/H)$, so that every G globally splits into an H-bundle and a G/H-bundle.

$$
\begin{tikzcd}
BH \arrow[r, "B\iota"'] \arrow[rr, bend left, "B(\pi \circ \iota)"] & BG & B(G/H) \arrow[l, "Bs"'] \\
& M \arrow[u, "f"] \arrow[ul, dashed, "\hat{f}"] \arrow[ur, dotted] &
\end{tikzcd}
\qquad
\begin{tikzcd}
BH & BG \arrow[l, "Br"'] \arrow[r, "B\pi"'] & B(G/H) \arrow[ll, bend right, "B(r \circ s)"'] \\
& M \arrow[u, "f"] \arrow[ul, dotted] \arrow[ur, dashed, "\hat{f}"'] &
\end{tikzcd}
$$

Thus, the Lie algebra counterpart of this discussion should be about giving a $\mathfrak{g}$-data over P which decomposes into $\mathfrak{h}$-data and $(\mathfrak{g}/\mathfrak{h})$-data which are coherent in the case when an H-reduction is given. This is precisely the setup of Cartan geometry. Notice that here the first advantage of Lie algebras in comparison to Lie groups becomes evident: for an arbitrary subgroup $H \subset G$ the quotient G/H exists as a topological space but a priori there is no way to decomposes G into H and G/H, even in the category of topological spaces. The decomposition exists (and it occurs in the category of groups) under the hypothesis that H is normal and $\pi : G/H$ has a section. But, for a Lie subalgebra $\mathfrak{h} \subset \mathfrak{g}$ the quotient $\mathfrak{g}/\mathfrak{h}$ exists in the category of vector spaces in which the decomposition $\mathfrak{g} \simeq \mathfrak{h} \oplus \mathfrak{g}/\mathfrak{h}$ is always satisfied independently of additional conditions on $\mathfrak{h}$. This is because Lie groups forget to categories that are not so well-behaved, as manifolds or topological spaces, while Lie algebras forget to an abelian category.

In this new setting of geometry we work with connections on principal G-bundles $P \to M$, which are $\mathfrak{g}$-valued 1-forms $D \in \Omega^1(P;\mathfrak{g})$ satisfying additional conditions. The decomposition $\mathfrak{g} \simeq \mathfrak{h} \oplus \mathfrak{g}/\mathfrak{h}$ induces an isomorphism

$$\Omega^1(P;\mathfrak{g}) \simeq \Omega^1(P;\mathfrak{h}) \oplus \Omega^1(P;\mathfrak{g}/\mathfrak{h}),$$

so that one can write $D = \omega + e$. We say that D is a *Cartan connection* relatively to the subalgebra $\mathfrak{h}$ if e is a pointwise isomorphism.

Remark 4.1. In the comparison between Cartan Geometry and Klein Geometry, this condition corresponds to the nondegeneracy condition on tensors (for a concrete example, see Remark 4.2). Thus, as one could forget nondegeneracy and develop Singular Riemannian Geometry, here we can work in a more ample class of geometries by forgetting the requirement of e being a pointwise isomorphism.

If $\mathfrak{h}$ is an *ad*-invariant subalgebra of $\mathfrak{g}$, then the 1-form $\omega : TP \to \mathfrak{h}$ becomes equivariant and, if P admits an H-reduction, then ω is a connection in the corresponding H-bundle. Under this hypothesis, we say that we are working with *reductive Cartan geometry*.

Now, returning to discuss the method of Geometric Obstruction Theory, suppose that we found a model to gravity that is based on reductive Cartan geometry instead of tensor geometry, i.e., whose fields are reductive Cartan connections instead of tensors. The action functional should be of the form $S[\omega, e] = \int L(\omega, e)$, where $\omega : TP \to \mathfrak{h}$ and $e : TP \to \mathfrak{g}/\mathfrak{h}$ are vector-valued 1-forms and the Lagrangian density takes values into something which is integrable over M, such as the volume forms (if M is orientable) or on densities (if M is non-orientable). Let us consider some basic assumptions:

1. the Lagrangian density $L(\omega, e)$ is actually of the form $F(\mathcal{L}(\omega, e))$, where $\mathcal{L}(\omega, e)$ is some expression returning an $\mathfrak{g}$-valued n-form (instead of a volume form or a density in M) and $F : \mathcal{C}(\mathfrak{g}) \subset \Omega^1(P;\mathfrak{g}) \to \mathrm{Den}(M)$ is a map defined on a class $\mathcal{C}(\mathfrak{g}) \subset \Omega^1(P;\mathfrak{g})$ of $\mathfrak{g}$-valued n-forms containing the the expression $\mathcal{L}(\omega, e)$, and taking values into densities or volume forms;

2. the map F satisfies $F(\alpha) = 0$ iff $\alpha = 0$;
3. the expression $\mathcal{L}(\omega, e)$ does not depends explicitly on vector spaces $\mathfrak{g}$, $\mathfrak{h}$ and $\mathfrak{g}/\mathfrak{h}$, but only on 1-forms with values on them and wedge products between them;

Thus, the two steps of the Geometric Obstruction Theory can be attacked in the following way:

1. *synthetic action*: from the third hypothesis above, the expression $\mathcal{L}(\omega, e)$ makes sense if we think of ω and e not as $\mathfrak{h}$-valued and $\mathfrak{g}/h$-valued 1-forms, but actually as A_0-valued and A_1-valued 1-forms, where $A_0, A_1 \subset A$ vector subspaces of an arbitrary algebra $(A, *)$ such that $A \simeq A_0 \oplus A_1$. Therefore, if the F exists not only form a Lie algebra $\mathfrak{g}$ but for an arbitrary algebra A, then the corresponding Lagrangian density $L(\omega, e)$ can also be defined for every algebra A with a vector space decomposition $A \simeq A_0 \oplus A_1$, so that one can look $L(\omega, e)$ as formal expression $L(x, y)$ which is evaluated for different algebras with decomposition.
2. *obstructions*: the problem of determining the kernel of $(A_0, A_1) \mapsto \int L(x, y; A_0, A_1)$ is equivalent to find algebras with a decomposition for which $\mathcal{L}(\omega, e)$ vanishes identically for every ω, e.

This means that now we have an equation in the space of algebras, instead of in the space of tensors, which tends to be much handier. The basic question now is: *which models to gravity are based on some reductive Cartan geometry?* There are some of them, e.g., Teleparallelism, Einstein-Cartan, Einstein-Hilbert-Palatini (EHP) and Metric-affine gravitation theory. Am them, Teleparallel Gravity and EHP have their versions that are physically equivalent to GR, so it seems natural to study Geometric Obstruction Theory for them first. In [117, 115] we done this for EHP theory.

4.1.4 Obstructions on EHP Based Gravity

Let $\mathbb{R}^{n-1,1}$ be the space $\mathbb{R}^n$ with the canonical metric of signature $(n-1, 1)$, i.e., let $\mathbb{R}^{n-1,1}$ be the n-dimensional Minkowski space. Let $G = \mathrm{Iso}(\mathbb{R}^{n-1,1})$ be its isometry group and consider the subgroup $H = O(n-1, 1)$ given by the isometry group of the origin. Thus, $\mathrm{Iso}(\mathbb{R}^{n-1,1})$ is just the n-dimensional Poincaré group and $O(n-1, 1)$ the corresponding Lorentz group.

The Klein geometry for (H, G) has the quotient G/H as a model, which in this case is just $\mathbb{R}^n$. Given a smooth manifold M, let FM be its frame bundle. By the correspondence between tensors and reductions, the existence of a $\mathrm{Iso}(\mathbb{R}^{n-1,1})$-structure in FM with a $O(n-1, 1)$-reduction is equivalent to the existence of a Lorentzian metric in M. Thus, suppose that M admits a Lorentzian metric, which is the case if M noncompact or if it is compact and $\chi(M) = 0$, where $\chi(M)$ is the Euler characteristics.

Cartan geometry for this Klein model is about Cartan $\mathrm{Iso}(\mathbb{R}^{n-1,1})$-connections in FM relatively to $O(n-1, 1)$. Since $\mathfrak{o}(n-1, 1)$ is an invariant subalgenra, it follows that the Cartan geometry is reductive. These reductive connections are certain 1-forms $D : TFM \to \mathfrak{iso}(n-1, 1)$ which decomposes as $D = \omega + e$, where $\omega : TFM \to \mathfrak{o}(n-1, 1)$ is a $O(n-1.1)$-connection, usually called *spin connection*, and $e : TFM \to \mathbb{R}^{n-1,1}$ is known as *tetrad* or *vielbein*.

Remark 4.2. The canonical metric in $\mathbb{R}^{n-1,1}$ can be pulled back to TFM defining a symmetric 2-form which is nondegenerate (and, therefore, a Lorentzian metric) iff e is an isomorphism. This is an explicit example of what was commented in Remark 4.1. Thus, if e is not an isomorphism, we can develop the version of EHP which should be physically equivalent to Singular Semi-Riemannian Geometry.

Recall that to every G-bundle $P \to M$, the collection $\mathrm{Con}_G(P) \subset \Omega^1(P;\mathfrak{g})$ of G-connections on P is an affine space of the space $\Omega^1_{he}(P;\mathfrak{g})$ of horizontal equivariant 1-forms. Also, the curvature Ω of any connection ω is an horizontal equivariant 2-forn On the other hand, horizontal equivariant k-forms in P, with $k \geq 0$, are equivalent to k-forms in M with values into the adjoint bundle $ad(P) = P \times_{ad} \mathfrak{g}$. In other words, there is an isomorphism $\Omega^k_{he}(P;\mathfrak{g}) \simeq \Omega^k(M; ad(P))$. A Lie algebra has its Killing form $B \in \Omega^2(\mathfrak{g})$ which is a symmetric 2-form which induces a symmetric bilinear map $B : \Gamma(ad(P)) \otimes \Gamma(ad(P)) \to C^\infty(M)$. Since $\Omega^k(M; ad(P)) \simeq \Omega^k(M) \otimes \Gamma(ad(P))$, we get, in turn, pairings

$$B^{kl} : \Omega^k(M; ad(P)) \otimes \Omega^l(M; ad(P)) \to \Omega^{k+l}(M) \otimes C^\infty(M) \simeq \Omega^{k+l}(M).$$

Therefore, if $k + l = n$ we get n-forms in M, which are equivalent to densities if M is orientable. In sum, we have the following diagram.

$$\begin{tikzcd}
\Omega^k_{he}(P;\mathfrak{g}) \otimes \Omega^l_{he}(P;\mathfrak{g}) \arrow[r, "\simeq"] \arrow[rr, bend right, "F_{kl}"'] & \Omega^k(M; ad(P)) \otimes \Omega^l(M; ad(P)) \arrow[r, "B^{kl}"] & \Omega^{k+l}(M)
\end{tikzcd}$$

Definition 4.3. Let M be an orientable n-dimensional manifold admiting Lorentzian metrics (i.e., which is noncompact or such that $\chi(M) = 0$). Let $\mathrm{Cartan}(M^{n-1,1})$ be the collection of Cartan $\mathrm{Iso}(\mathbb{R}^{n-1,1})$-connections in FM relatively to $O(n-1,1)$. Let ω_0 be a fixed $O(n-1.1)$-connection in FM and $\Lambda \in \mathbb{R}$ a real number, regarded as the cosmological constant. The *Einstein-Hilbert-Palatini functional* is the map $S_{EHP} : \mathrm{Cartan}(M^{n-1,1}) \to \mathbb{R}$, given by

$$S[\omega, e] = \int_M F(\wedge^{n-2}_{\mathfrak{g}} e \wedge_{\mathfrak{g}} \Omega + \frac{\Lambda}{(n-1)!} \wedge^n_{\mathfrak{g}} e).$$

Remark 4.4. The formal proof that the phase space of S_{EHP} coincides with the phase space of the Einstein-Hilbert functional in the case $n = 4$ is due to Cattaneo and Schiavina [35].

Let

$$\alpha^n_{EHP} = \wedge^{n-2}_{\mathfrak{g}} e \wedge_{\mathfrak{g}} \Omega + \frac{\Lambda}{(n-1)!} \wedge^n_{\mathfrak{g}} e,$$

so that $S_{EHP}(e,\omega) = \int F(\alpha^n_{EHP})$. Now, notice that:

1. the functional S_{EHP} makes perfect sense in other $\mathrm{Iso}(\mathbb{R}^{n-1,1})$-bundle P with a $O(n-1,1)$-reduction;
2. it also makes sense for other Klein geometry (G, H);
3. the form α_{EHP} can be analogously defined for any given algebra $(A, *)$ with vector space decomposition $A \simeq A_0 \oplus A_1$. Indeed, in this case

$$\alpha^n_{EHP}(A_0, A_1) = \wedge^{n-2}_* e \wedge_* \Omega + \frac{\Lambda}{(n-1)!} \wedge^n_* e, \tag{4.1}$$

where $\omega : TP \to A_0$ and $e : TP \to A_1$ are 1-forms, $\Omega = d\omega + \omega \wedge_* \omega$ and $\wedge_*$ is the wedge product induced by the multiplication of A.

Therefore, by the discussion above, for every way to extend the map F from Lie algebras to arbitrary algebras, we can define $S_{EHP}(e,\omega;A_0,A_1) = \int F(\alpha^n_{EHP}(A_0,A_1))$, so that if (A_0,A_1) is such that $\alpha_{EHP}(A_0,A_1) = 0$, then there is no physically nontrivial version of EHP based on the pair (A_0,A_1). In [117, 115] we showed that these extensions of F corresponds to what we called *functorial algebra bundle systems* and we studied conditions on (A_0,A_1) such that $\alpha^n_{EHP}(A_0,A_1) = 0$.

4.1.5 Main Results

Looking at expression (4.1) we see that if (A_0,A_1) are such that $\wedge^i_* e = 0$ for some $i > 0$, then $\alpha^n_{EHP}(A_0,A_1) = 0$ if $n \leq i$. The condition $\wedge^i_* e = 0$ says precisely that e is an element of nilpotency degree at most i in the graded algebra $(\Omega(P;A),\wedge_*)$. We can then restate the above as follows: *if M is a manifold of dimension n and A is such that every $e \in \Omega^1(P;A_1)$ is nilpotent of degree $i \leq n$, then $\alpha^n_{EHP}(A_0,A_1) = 0$.* In particular, *under the nilpotency hypothesis on A, $alpha^n_{EHP}(A_0,A_1)$ is non-null (i.e., the theory S_{EHP} is physically nontrivial) only if the dimension of M satisfies some bound*. It is precisely this kind of result that we got in [117, 115].

In a first result we noticed that the functional S_{EHP} can be extended to the graded case with a general coefficient ring R and we find obstructions in this more general setting:

Theorem 4.5. *Let $P \to M$ be a graded bundle over a n-dimensional manifold and $A \simeq \oplus_m A^m$ a graded R-algebra endowed with a R-module decomposition $A_0 \oplus A_1$ such that each $A_0 \cap A^m$ is a weak (k_m,s_m)-solv submodule of A_0. Let (k,s) be $\min_m(k_m,s_m)$. If $n \geq k+s+1$, then the A-EHP action functional is homogeneous (i.e, $\Lambda = 0$). If $n \geq k+s+3$, then it is trivial.*

Proof. This is a particular case of Theorem 2.1 of [117]. □

The result above gives bounds on the dimension of M for the existence of nontrivial EHP-theories based on certain pairs (A_0,A_1). As a second result we showed that the constraints on M can be stronger, providing even a classification up to diffeomorphisms in the concrete case of Lie algebras:

Theorem 4.6. *Let M be an n-dimensional Berger manifold endowed with an H-structure. If there are natural numbers $k_1 + ... + k_r = n$ such that $\mathfrak{h} \subset \mathfrak{so}(k_1) \oplus ... \oplus \mathfrak{so}(k_r)$ properly, then the torsionless EHP functional based in the pair $(\mathfrak{h}, \oplus_i \mathfrak{so}(k_i)/\mathfrak{h})$ is nontrivial only if $n = 2,4$ and M is Kähler.*

Proof. This is a simplified version of Theorem 3.1 of [117]. □

Corollary 4.7. *If, in addition to the hypotheses of theorem above, M is four-dimensional, then the torsionless EHP functional for $(\mathfrak{h}, \oplus_i \mathfrak{so}(k_i)/\mathfrak{h})$ is nontrivial only if M is diffeomorphic to Fermat's quartic given by the solution space over $\mathbb{C}$ of $x^4 + y^4 + z^4 + w^4 = 0$.*

Proof. By definition, a Berger manifold is simply connected. But every 4-dimensional simply connected Kähler is a complex analytic K3-surface, which by Kodaira's theorem are diffeomorphic to Fermat's quartic. □

The condition $\mathfrak{h} \subset \mathfrak{so}(k_1) \oplus ... \oplus \mathfrak{so}(k_r)$ may seems stranger, but it actually describes most geometries. It does *not* describes semi-Riemannian geometry, because it is about $O(r,s) \hookrightarrow \mathrm{Iso}(\mathbb{R}^{r,s})$ and $\mathfrak{iso}(\mathbb{R}^{r,s})$ does not satisfies that property. But, we already known that EHP functional is nontrivial for semi-Riemannian geometry. Some examples of geometries satisfying that condition include:

1. *those in Berger's classification theorem*, i.e., Kähler, quaternionic-Kähler and hyper-Kähler geometries, Calabi-Yau geometry, $\mathrm{Spin}(7)$-geometry and G_2-geometry;

2. *some generalized geometries*, as Type II geometry, generalized complex geometry, generalized Calabi-Yau geometry, generalized Kähler geometry and generalized Calabi geometry;

3. *Cayley-Dickson extensions of these generalized geometries*, such as generalized quaternionic geometry and generalized octonionic geometry;

4. *some exceptional geometries*, as G_2-geometry above, but also F_4, E_6, E_7 and E_8-geometries,

and much more, as exhaustively discussed in Section 4 of [117] and in Section 5 of [115].

Remark 4.8. Almost symplectic and almost complex geometries do not satisfy the condition $\mathfrak{h} \subset \mathfrak{so}(k_1) \oplus ... \oplus \mathfrak{so}(k_r)$, so that a priori one could not use the previous result to conclude that EHP functional is physically trivial for them. But both geometries are contained as particular cases of generalized complex geometry, for which the theorem applies.

A third result, discussed along Section 4 of [117], is the following:

Theorem 4.9. *Let A be an algebra with vector space decomposition $A_0 \oplus A_1$. Suppose that EHP functional is nontrivial for the pair (A_0, A_1), over a n-dimensional manifold M. Now, let K be a (k,s)-solv algebra and consider an extension E of A by K. If $n \geq k+s+1$, then the EHP action functional for (K, A) is homogeneous. If $n \geq k+s+3$, then it is trivial.*

Proof. This is a particular case of Theorem 4.6. □

The result is telling us that, even if EHP is nontrivial for some geometry, it becomes trivial after some extension by (k,s)-solv algebras. Particular examples are abelian extensions and Lie superalgebras extensions. Thus, an interesting corollary is the following:

Corollary 4.10. *If M is a manifold of dimension $n \geq 6$, then the EHP functional for the super-Poincaré Lie superalgebra with N central charges $\mathfrak{siso}_N(n-1,1)$, with the decomposition $\mathfrak{siso}(n-1,1) \simeq \mathfrak{iso}(n-1,1) \oplus \mathbb{R}^{0|N}$, is physically trivial.*

Proof. Just note that the super point $\mathbb{R}^{0|N}$ is $(2,1)$-nil. □

4.2 QUESTIONS AND SPECULATIONS

The following are questions and speculations on the discussion above.

- Recall that the obstruction results obtained in [117, 115] was based on a method that we call *Geometric Obstruction Theory*. Although introduced for the case of Einstein-Hilbert-Palatini theories, as discussed above, the method could be applied to any (reductive) gauge theory whose algebra of infinitesimal symmetries is arbitrary (not necessarily a Lie algebra), so that a (reductive) algebra-valued gauge theory.

(Q4.1) *Define in formal terms what (reductive) algebra-valued gauge theories are. Show that the Geometric Obstruction Theory method can be extended to them.*

- As commented above, other models of gravity, such as Teleparallel Gravity and Einstein-Cartan, are examples of reductive gauge theory. Thus, one could try to extend them to reductive algebra-valued theories and use Geometric Obstruction Theory to find obstructions for them.

Q4.2 *Show that Teleparallel Gravity and Einstein-Cartan extends to reductive algebra-valued gauge theories and apply Geometric Obstruction Theory on them. Compare the corresponding obstructions with those obtained for EHP theories in [117, 115].*

Q4.3 *Yang-Mills theories, Chern-Simons theories and other non-reductive gauge theories can be extended to algebra-valued gauge theories? What are the corresponding obstructions?*

- In [117, 115] we do not find obstructions for EHP theories defined in an arbitrary algebra A with a $\mathbb{Z}_2$-vector space decomposition $A \simeq A_0 \oplus A_1$, but also for EHP theories with a $\mathbb{Z}_m$-graded vector space $A \simeq A_{01} \oplus ... \oplus A_m$. In this case, the principal bundle P is replace for a fiber bundle P with a vector bundle decomposition $TP \simeq E_0 \oplus ... \oplus E_m$. This motivates us to ask if the construction cannot be lifted to the case of super/graded-geometry.

Q4.4 *Define what is a graded algebra-valued gauge theory on a graded manifold M. Prove that the Geometric Obstruction Theory extends to this case.*

Q4.5 *Show that the typical super gauge theories admitting a superspace formalism extend to a super algebra-valued gauge theory on a supermanifold. Find the corresponding obstructions.*

- Independently if a super gauge theory admits a superspace formalism, it is obtained extending a gauge theory with gauge Lie algebra $\mathfrak{g}$ to a super Lie algebra $\hat{\mathfrak{g}}$. On the other hand, in Section 5 of [117] we argued that, in the case of EHP theories, obstructions on an algebra-valued gauge theory induce obstructions on each extension.

Q4.6 *Define what is a super-extension of an algebra-valued theory. Establish a correspondence between obstructions on a theory and obstructions on its super-extensions.*

Q4.7 *Assuming Questions Q4.3 and Q4.6, find obstructions on super Yang-Mills theories. If a super Yang-Mills theory admits a superspace formalism, compare these obstructions with those obtained from Questions Q4.4 and Q4.5.*

- Recall that supergravity theories are not just super-extensions of gravity theories. In other words, supergravity is not the internalization of gravity in super-geometry. Indeed, super gauge theories are about superconnections. Connections are locally $\mathfrak{g}$-valued 1-forms $A_i : U_i \to \mathfrak{g}$ for a Lie algebra $\mathfrak{g}$. Similarly, superconnections are locally $\hat{\mathfrak{g}}$-valued $(1|1)$-form for a super Lie algebra $\hat{\mathfrak{g}} \simeq \mathfrak{g}_0 \oplus \mathfrak{g}_1$. Equivalently, they are locally an even $\mathfrak{g}_0$-valued 1-form A_i and an odd $\mathfrak{g}_1$-valued 1-form ψ_a.

  In the case of super-extension of a gauge theory with Lie algebra $\mathfrak{g}$, the extension provides the addition of the spinorial field ψ_a. Thus, a super-extension of a gravity model is about the inclusion of the gravitino field. However, in supergravity we do not have only the addition of the gravitino but also of entities which locally are $\mathfrak{g}_0$-valued 2-forms (the B-field) or 3-forms (the C-field).

  These higher-dimensional fields are locally captured if instead of considering super-extensions we consider *higher-extensions*, i.e., extensions from Lie algebras to higher Lie algebras L_∞-algebras.

Q4.8 *Define what is a higher-algebra-valued gauge theory. Show that the Geometric Obstruction Theory can be extended to this setting. Define what is a higher-extension of an algebra-valued gauge theory. Prove that obstructions on an algebra-valued gauge theory induce obstructions on its higher-extensions.*

- The globalization of the higher-dimensional fields appearing in supergravity leads to consider a reformulation of gauge theory in terms of stacks. More precisely, recall that G-bundles on M are classified up to isomorphisms by homotopy classes of maps $f : M \to BG$. In other words, we have a bijection $\mathrm{Iso}_G(M) \simeq [M; BG]$. Both sides can be viewed as groupoids whose objects are the elements and the morphisms exist iff they are in the same equivalence class. Varying M we get then presheaves of groupoids which are actually stacks.

  On the other hand, recall that $BG = |\mathbf{B}G|$ is the geometric realization of the groupoid $\mathbf{B}G$ of G (there is a single object and morphisms are elements of G). Each groupoid induces a stack, so that $\mathbf{B}G$ can be viewed as a stack $\bar{\mathbf{B}}G$. The stack $[-; BG]$ is equivalent to the stack $[-; \bar{\mathbf{B}}G]$ where now the brackets represent the internal hom in the category of stacks.

  Take now the groupoid $\mathbf{B}_{conn}G(M)$ whose objects are $\mathfrak{g}$-valued 1-forms on M and there is a morphism $f : A \to A'$ iff there is a smooth function $g : M \to G$ such that $A' = g^1 \cdot A \cdot g + dg$. Varying M we get another stack $\mathbf{B}_{conn}G$ with an obvious projection $\mathbf{B}_{conn}G \to \bar{\mathbf{B}}G$. Notice that giving a connection to a G-bundle is the same thing that giving an extension, as below.

  Therefore, the global object which locally describes L_∞-algebra-valued forms is some kind of version of $\mathbf{B}_{conn}G$, but for L_∞-algebras instead of Lie algebras. The Lie group G is the integration of the Lie algebra $\mathfrak{g}$, so we will need to consider the objects integrating L_∞-algebras. These are the so-called ∞-Lie groups. Now, G-bundles need to be replaced by ∞-bundles.

  The fields in supergravity are these kinds of higher stacks defined on higher bundles. To apply Geometric Obstruction Theory to them we need, therefore, to provide an algebra-valued version of this stack description of (higher) gauge theory.

Q4.9 *Define what is an "algebraic context" in which stacks taking values into objects of them exist. Define the algebra-valued stack classifying an algebra-valued bundle P. Define what is a "refinement" of an algebra-valued stack. Show that the refinements of the algebra-valued stack classifying an algebra-valued bundle P correspond to geometric objects on P which locally can be written as higher-algebra-valued forms. Define algebra-valued higher gauge theory as the study of these stacks. Prove that the Geometric Obstruction Theory extends to this context.*

Q4.10 *Prove that the supergravity theories extend to algebra-valued higher gauge theories. Compute the obstructions on them.*

In working in progress we argue that an "algebraic context" can be defined in terms of enriched categories whose enrichment ambient may vary, so that on *parameterized enriched categories*. These entities are, in some sense, dual to Shulman's *enriched indexed categories* [151]. The study of stacks with valued into parameterized enriched categories is therefore about parameterized enriched topos theory.

On the other hand, recall that the set of homotopy classes of maps $[X; BG]$ are nonabelian cohomology $H^1_{na}(M; G)$, while $[M; \mathbf{B}_{conn}G]$ is the differential refinement of nonabelian cohomology: *nonabelian differential cohomology*. Thus, the process of building refinements for algebra-valued stacks (as in Question Q4.10) is about constructing a general theory of refinements of abstract cohomologies on parameterized enriched toposes.

- In the work [115] we discovered that one can study the problem of finding obstructions on an algebra-valued gauge theory from the obstructions on certain dual theories. It would be interesting to know if for some class of algebras these mathematically dual theories are physically dual.

Q4.11 *Let S be a reductive algebra-valued gauge theory. Show that the geometric/algebraic duals S^* and ${}^*S$ are well-defined. For which classes of algebras do these action functionals induce equivalent perturbative series? Similar questions for super-algebra-valued gauge theories and higher-algebra-valued gauge theories.*

- A final question. As discussed above, EHP is a reductive gauge theory for which the vielbein e is a pointwise isomorphism, i.e., EHP is Cartan geometry. This condition is equivalent to saying that the metric induced by e is nondegenerate. It was proved that EHP is physically equivalent to General Relativity [35]. On the other hand, General Relativity has an extension to degenerate metrics given by Singular Semi-Riemannian Geometry [154, 155, 156].

Q4.12 *Is EHP with e not necessarily a pointwise isomorphism physically equivalent to Singular Semi-Riemannian Geometry?*

4.3 FIRST WORK

The following is a preliminary version of the published paper

- Y.X. Martins, R.J. Biezuner, *Topological and geometric obstructions on Einstein-Hilbert-Palatini theories*, **J. Geom. Phys.**, v. 142, p. 229, Aug 2019. doi:10.1016/j.geomphys.2019.-04.012

Topological and Geometric Obstructions on Einstein-Hilbert-Palatini Theories

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Abstract

In this article we introduce A-valued Einstein-Hilbert-Palatini functional (A-EHP) over a n-manifold M, where A is an arbitrary graded algebra, as a generalization of the functional arising in the study of the first order formulation of gravity. We show that if A is weak (k,s)-solvable, then A-EHP is non-null only if $n < k+s+3$. We prove that essentially all algebras modeling classical geometries (except semi-Riemannian geometries with specific signatures) satisfy this condition for $k = 1$ and $s = 2$, including Hitchin's generalized complex geometry, Pantilie's generalized quaternionic geometries and all other generalized Cayley-Dickson geometries. We also prove that if A is concrete in some sense, then a torsionless version of A-EHP is non-null only if M is Kähler of dimension $n = 2, 4$. We present our results as obstructions to M being an Einstein manifold relative to geometries other than semi-Riemannian.

1 Introduction

The standard theory of gravity is General Relativity [32, 16], which is formulated in Lorentzian geometry: spacetime is regarded as a Lorentzian manifold (M, g) whose Lorentzian metric g is a critical point of the Einstein–Hilbert action functional

$$S_{EH}[g] = \int_M (R_g - 2\Lambda) \cdot \omega_g, \tag{1}$$

defined on the space of all possible Lorentzian metrics in M, $\Lambda \in \mathbb{R}$ is the cosmological constant and R_g and ω_g are, respectively, the scalar curvature and the volume form of g. The Euler-Lagrange equations of S_{EH} are the Einstein's equations

$$\mathrm{Ric}_g - \frac{1}{2} R_g g + \Lambda g = 0, \tag{2}$$

which are equivalent to saying that (M, g) is an Einstein manifold, i.e, $\mathrm{Ric}_g = \frac{2}{n-2}\Lambda g$. The problem of determining which manifolds admit an Einstein structure is old and a general classification of them remains an open problem [7, 28].

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The action functional S_{EH} can be considered not only in Lorentzian metrics, but also in metrics of arbitrary signature. The Euler-Lagrange equations have the same form and they are also equivalent to the Einstein manifold condition. Despite the similarity, being Einstein depends strongly on the signature (and, therefore, on the underlying geometry). For instance, if M is compact, orientable, non-parallelizable and Kähler, then it does not admit any Lorentzian metric[1], while Calabi-Yau manifolds, hyperKähler and quaternion Kähler manifolds are the basic examples of Riemannian Einstein manifolds.

Due to the difficulty of classifying semi-Riemmanian Einstein-manifolds, many generalizations and closely related concepts were introduced and studied, such as generalized Einstein manifolds [1], quasi-Einstein manifolds [10], generalized quasi-Einstein manifolds [11], super quasi-Einstein manifolds [24], and many others. Each of these generalizations concerns semi-Riemannian manifolds. Here we will consider another generalization: the problem of finding Einstein manifolds for geometries other than semi-Riemannian. Because semi-Riemannian Einstein manifolds are critical points of the Einstein-Hilbert functional, the idea is to define Einstein manifolds relative to other geometries as critical points of functionals which are analogues of S_{EH} in those geometries. The classical examples of geometries can be described by tensors t on M fulfilling additional integrability conditions. Considering connections on M compatible with t (in the sense that $\nabla t = 0$) we can take their curvature tensor and Ricci curvature. In order to write down the analogous action we need to contract the Ricci curvature with the tensor t, getting the *scalar curvature* R_t of t. If M is oriented we can define an action functional $S[t] = \int (R_t - 2\Lambda)\cdot \omega$ on the moduli space of those tensors. This makes sense only when t has rank two. For example, we can write a direct analogue of General Relativity in symplectic geometry, with action functional $S[\omega] = \int_M (\pi^{ij}\operatorname{Ric}_{ij} - 2\Lambda)\cdot \omega^n$ on a $2n$-dimensional manifold, where π^{ij} are the components of the bivector induced by ω. But the tensor Ric_{ij} is always symmetric, so that $\pi^{ij}\operatorname{Ric}_{ij} = 0$ and the action therefore is trivial [8, 9]. Also, there is no obvious analogue of S_{EH} (and, consequently, of Einstein manifolds) relative to Kähler, quaternion Kähler and hyper-Kähler geometry. This fact should not be viewed as a contradiction to the above cited fact that Calabi-Yau manifolds, quaternion Kähler and hyper-Kähler manifolds are Riemmanian Einstein manifolds nor to the applications of those geometries in classical (i.e, semi-Riemannian) General Relativity, specially in the canonical formalism [2, 14].

We recall, however, that Einstein's equations can be obtained as critical points of another functional, usually known as the *Einstein–Hilbert–Palatini* (EHP) *action functional*, which arises in the so-called *first order formulation of gravity* (also called *tetradic gravity*). Instead of a semi-Riemannian metric, this functional is defined on the space of reductive Cartan connections on the frame bundle $FM \to M$, relative to the group reduction $O(r,s) \hookrightarrow \mathbb{R}^{r,s} \rtimes O(r,s)$. These can be identified with pairs of 1-forms e and ω in FM, called *tetrad* and *spin connection*, with values in $\mathbb{R}^{n-1,1}$ and $\mathfrak{o}(n-1,1)$, respectively. It is usual to assume that e is pointwise an isomorphism. The action itself is given by

$$S_{EHP}[e,\omega] = \int_M \operatorname{tr}(\curlywedge_{\rtimes}^{n-2} e \curlywedge_{\rtimes} \Omega + \frac{\Lambda}{(n-1)!} \curlywedge_{\rtimes}^{n} e), \tag{3}$$

where $\curlywedge_{\rtimes}$ is a type of "wedge product[2]" induced by matrix multiplication in $O(r,s)$, $\curlywedge_{\rtimes}^{k}\alpha = \alpha \curlywedge_{\rtimes} ... \curlywedge_{\rtimes} \alpha$ and $\Omega = d\omega + \omega \curlywedge_{\rtimes} \omega$ is the curvature of ω. It is a well-known fact [4, 33] that in dimension

[1] In fact, Kähler condition requires $b_i(M) = 0$ for i odd, so that $\chi(M) = \sum_i b_{2i}(M)$, which is positive, since M is non-parallelizable. On the other hand, compact manifolds admit Lorentzian metrics only if $\chi(M) = 0$.

[2] In this paper we will work with many different types of wedge products, satisfying very different properties.

$n = 4$ and Lorentzian signature varying (3) in relation to e and ω we get

$$e \curlywedge_{\rtimes} \Omega + \frac{\Lambda}{(n-1)!} e \curlywedge_{\rtimes} e \curlywedge_{\rtimes} e = 0 \quad \text{and} \quad e \curlywedge_{\rtimes} \Theta = 0, \tag{4}$$

where $\Theta = d\omega + \omega \curlywedge_{\rtimes} e$ is the *torsion of* ω. Lorentz metrics on M can be identified as pullbacks of the canonical Minkowski metric on $\mathbb{R}^{r,s}$ via tetrad and the connections ω are compatible with them. Furthermore, since e is pointwise an isomorphism, the second equation in (4) is equivalent to the torsion-free condition $\Theta = 0$, so that ω is necessarily a Levi-Civita connection. Via these identifications, the first equation in (4) is just Einstein's equation. Therefore, a 4-dimensional Einstein manifold can also be regarded as a critical point (relative to e) of S_{EHP}.

The advantage of this new approach is that unlike S_{EH}, the EHP functional S_{EHP} (and, consequently, the notion of Einstein manifold) makes sense relative to any geometry. In fact, recall that tensor geometries on M are equivalent to reductions on the structural group of FM, while connections compatible with tensors are equivalent to Cartan connections for the corresponding group reductions. The difference from the previous problematic approach is that, while S_{EH} involves scalar curvature (which makes canonical sense only for particular tensorial geometries), S_{EHP} contains only the curvature form, which can be defined for arbitrary Cartan connections. Not only this, EHP theories can be defined in a purely algebraic sense: given an R-algebra A endowed with a vector decomposition $A \simeq A_0 \oplus A_1$, we can define a *reductive* A*-connection* in a bundle $P \to M$ as a pair of 1-forms $e : TP \to A_0$ and $\omega : TP \to A_1$. The corresponding A*-valued EHP theory* (A-EHP) is given by the functional[3]

$$S_{EHP}[e, \omega] = \int_M (\wedge_*^{n-2} e \wedge_* \Omega + \frac{\Lambda}{(n-1)!} \wedge_*^n e), \tag{5}$$

where $\wedge_*$ is the wedge product induced by the multiplication $* : A \otimes A \to A$ and $\Omega = d\omega + \omega \wedge_* \omega$ is the *field strength* (or *curvature* of ω). An A*-valued Einstein manifold* is then defined as a manifold M endowed with an A-connection $A = e + \omega$ fulfilling the Euler-Lagrange equations of (5). This definition can be enlarged even more to include graded geometries, where A is a graded algebra and $P \to M$ is a graded bundle over a non-graded manifold.

Since EHP theories make sense in arbitrary geometries defined by abstract algebras, we can ask if for each of those geometries the corresponding theory is nontrivial in some sense, implying the existence of A-valued Einstein manifolds. The minimal requirement is, of course, that the functional must be non-null (recall that the symplectic version of S_{EH} vanishes identically, so that it is expected that a similar situation will happen in EHP). In this article we give general obstruction results for the minimal nontriviality of A-EHP (and, therefore, for the existence of A-valued Einstein manifolds) in terms of properties of the underlying algebra. We will prove two main theorems, which are different in nature. The first of them applies to concrete and abstract EHP functionals and gives obstructions on the dimension of the base manifold, while the second applies only to concrete geometries, but gives obstructions on the dimension and on the topology of the base manifold.

Theorem A. *Let $P \to M$ be a graded bundle over a n-dimensional manifold and $A \simeq \oplus_m A^m$ a graded R-algebra endowed with a R-module decomposition $A_0 \oplus A_1$ such that each $A_0 \cap A^m$ is a*

Therefore, in order to avoid confusion, we will not follow the literature, but introduce specific symbols for each of them.

[3]In the way it is written, it only makes sense when the algebra $\Lambda(P; A)$ satisfies some associativity-like polynomial identity. We will see, nonetheless, that it is possible to define S_{EHP} for arbitrary algebras.

weak (k_m, s_m)-solv submodule of A_0. Let (k,s) be $\min_m(k_m, s_m)$. If $n \geq k+s+1$, then the A-EHP action functional is homogeneous (i.e, $\Lambda = 0$). If $n \geq k+s+3$, then it is trivial.

Theorem B. *Let M be an n-dimensional Berger manifold endowed with an H-structure. If there are natural numbers $k_1 + k_r = n$ such that $\mathfrak{h} \subset \mathfrak{so}(k_1) \oplus ... \oplus \mathfrak{so}(k_r)$ properly, then the torsionless EHP functional is nontrivial only if $n = 2, 4$ and M is Kähler. In particular, if M is compact and $H^2(M;\mathbb{C}) = 0$, then it must be a K3-surface.*

This paper is organized as follows: in Section 2 we study the solvability conditions appearing in Theorem A and this theorem is stated in a more general context and proved. In Section 3 we recall some facts concerning algebra extensions and Theorem B is stated also in a more general context and proved. In Section 4 many examples are given, where by an example we mean a specific algebraic context fulfilling the solvability hypothesis of Theorem A, so that the corresponding EHP theory cannot be realized in the underlying geometry.

2 Theorem A

Let $(A,*)$ be an R-algebra, where R is a commutative ring. Given a smooth manifold P, let $\Lambda(P;A)$ denote the $\mathbb{N}$-graded algebra of A-valued forms on P with the wedge product $\wedge_*$ induced by $*$. An *A-valued reductive connection* (or simply A-connection) on a smooth bundle $P \to M$, relative to a R-module decomposition $A \simeq A_0 \oplus A_1$, is a pair $\nabla = e + \omega$ of 1-forms $e : TP \to A_0$ and $\omega : TP \to A_1$. Therefore, the space of A-connections is given by

$$\mathrm{Conn}_A(P) = \Lambda^1(P;A_0) \oplus \Lambda^1(P;A_1).$$

The *pure field strength* (or *curvature*) and the *torsion* of an A-connection ∇ are respectively defined as the A-valued 2-forms $\Omega = d\omega + \omega \wedge_* \omega$ and $\Theta = de + \omega \wedge_* e$. If $(A,*)$ is associative, then given $\Lambda \in \mathbb{R}$ there is no ambiguity in considering the *Hilbert-Palatini form*

$$\alpha_{n,\Lambda} = \wedge_*^{n-2} e \wedge_* \Omega + \frac{\Lambda}{(n-1)!} \wedge_*^n .$$

In this case, fixed any FABS $\jmath$ and any trace transformation tr (see Appendix, p. 14) we have a natural map $\mathrm{tr} \circ \jmath : \Lambda(P;A) \to \Lambda(M;\mathbb{R})$ preserving the $\mathbb{N}$-grading, allowing us to define the *A-valued EHP action functional* in $P \to M$ as the functional (if $\Lambda = 0$ we say that this is the homogeneous A-valued EHP functional)

$$S_{n,\Lambda} : \mathrm{Conn}_A(P) \to \mathbb{R} \quad \text{such that} \quad S_{n,\Lambda}[e,\omega] = \int_M \mathrm{tr}(\jmath(\alpha_{n,\Lambda})).$$

But, if $(A,*)$ does not satisfy associative-like polynomial identities (PIs), in order to define Hilbert-Palatini forms and, consequently, A-EHP theories, we need to work with A up to those associative-like PIs. More precisely, let us consider the associativity-like polynomials

$$(x_1 \cdot ... \cdot (x_{s-2} \cdot (x_{s-1} \cdot x_s))), \quad (((x_1 \cdot x_2) \cdot (x_3 \cdot x_4)) \cdot ... \cdot x_s), \quad \text{etc.},$$

Due to the structure of the Hilbert-Palatini forms, we are interested in the more specific polynomials

$$(x \cdot ... \cdot (x \cdot (x \cdot y))), \quad \text{etc.} \quad \text{and} \quad (x \cdot ... \cdot (x \cdot (x \cdot x))), \quad \text{etc.} \tag{6}$$

4

When evaluated at $\Lambda(P;A)$, the set of these polynomials generates an ideal $\mathfrak{I}(P,A)$ and we can take the quotient $\Lambda(P;A)/\mathfrak{I}(P,A)$, whose quotient map we denote by π. Fixing a FABS invariant under the ideal $\mathfrak{I}(P,A)$ (see Appendix, p. 14) we can now consider $S_{n,\Lambda}$ for arbitrary algebras:

$$S_{n,\Lambda}[e,\omega] = \int_M \operatorname{tr}(\jmath(\pi(\alpha_{n,\Lambda}))). \tag{7}$$

Suppose now that (as an R-module) A is also $\mathfrak{m}$-graded for some abelian group $(\mathfrak{m},+)$, meaning that $A \simeq \oplus_m A^m$ with $m \in \mathfrak{m}$. Such a decomposition induces a decomposition in each A_i, with $i = 0,1$, by $A_i \simeq \oplus_m A_i^m$, where $A_i^m = A_i \cap A^m$. Suppose that $P \to M$ is also $\mathfrak{m}$-graded, in that $TP \simeq \oplus_m E^m$ for vector bundles $E^m \to M$. In this case, the previous notions can also be generalized to the graded context. Indeed, given $l \in \mathfrak{m}$ we define an *A-connection of degree l in P* as a pair $\nabla = e + \omega$ of $\mathfrak{m}$-graded A-valued 1-forms of degree l, i.e, as a sequence of 1-forms $e^m : E^m \to A_0^{m+l}$ and $\omega^m : E^m \to A_1^{m+l}$. The space of those connections is then

$$\operatorname{Conn}_A^l(P) = \oplus_m(\Lambda^1(E^m;A_0^{m+l}) \oplus \Lambda^1(E^m;A_1^{m+l})).$$

The curvature and the torsion are defined analogously, being determined by the curvature and the torsion of each connection $\nabla^m = e^m + \omega^m$. As discussed in the Appendix, the notions of FABS and trace transformation can be internalized into the category of graded principal bundles (in the sense above), allowing us to define (for each graded-FABS and each graded-trace transformations) the *$\mathfrak{m}$-graded A-EHP functional of degree l in $P \to M$* by the same expression as (7), but now with domain $\operatorname{Conn}_A^l(P)$.

We will restate Theorem A considering the following more general conditions on the $\mathfrak{m}$-graded algebra A endowed with the a R-module decomposition $A \simeq A_0 \oplus A_1$:

(G1) A_0 is indeed a subalgebra and each $A_0^j = A^i \cap A_0$ is a (k_j,s_j)-weak solvable submodule of A_0.

(G2) each A^j is a (k_j,s_j)-weak solvable submodule of A.

Before restating and proving Theorem A, let us explain what we mean by a (k,s)-weak solvable submodule of an R-algebra A. We say that an algebra $(A,*)$ is *(k,s)-nil*, with $k,s \geq 0$ if (for any smooth manifold P) every A-valued k-form in P has nilpotency degree s, i.e, if for every $\alpha \in \Lambda^k(P;A)$ we have $\wedge_*^s \alpha \neq 0$, but $\wedge_*^{s+1}\alpha = 0$. The *(k,s)-solv* algebras are those that can be decomposed as finite sums of (k,s)-nil algebras. We say that a submodule $V \subset A$ is a *(k,s)-nil submodule of A* if every V-valued k-form has nilpotency degree s when regarded as a A-valued form. Similarly, we say that V is a *(k,s)-solv submodule* if it decomposes as a sum of (k,s)-nil submodules. When $A = \oplus_m A^m$ is $\mathfrak{m}$-graded, there are other kind of submodules $V \subset A$ that can be introduced. For instance, we say that V is *graded (k,s)-solv* if each $V_m = V \cap A_m$ is a (k,s)-solv submodule. In any submodule V of a $\mathfrak{m}$-graded algebra we get a corresponding grading by $V \simeq \oplus_m V_m$. So, any V-valued form α can be written as $\alpha = \sum_m \alpha^m$. We say that V is *weak (k,s)-nilpotent* if for every k-form α, any polynomial $p_{s+1}(\alpha^{m_1},...,\alpha^{s+1})$ vanishes. Similarly, we say that V is *weak (k,s)-solvable* if it decomposes as a sum of weak (k,s)-nilpotent submodules.

Examples will be given in Section 4. Just to mention, if an algebra A is nilpotent (resp. solvable) of degree s, then it is $(1,s)$-nil (resp. $(1,s)$-solv). Similarly, degree s nilpotent (resp. solvable) subalgebras of any algebra are (k,s)-nil (resp. (k,s)-solv) submodules.

Restatement and Proof

Now we can give a more precise statement for Theorem A (in the formulation below it is more general than that present in the Introduction).

Theorem 2.1. *Let M be an n-dimensional manifold, $P \to M$ be a bundle such that TP is $\mathfrak{m}$-graded and $(A, *)$ be a $\mathfrak{m}$-graded algebra endowed with a R-module decomposition $A_0 \oplus A_1$. Given $l \geq 0$, assume that $A[-l]$ satisfies condition* (G1) *or* (G2). *Furthermore, let (k, s) be the minimum of (k_j, s_j). If $n \geq k+s+1$, then for any compatible graded FABS and any trace transformation, the corresponding graded A-EHP of degree l equals the homogeneous ones. If $n \geq k+s+3$, then A-EHP is trivial.*

We will divide the proof in two cases:

1. when only the algebra A is graded;
2. when both A and P are graded and the theory may have arbitrary degree.

Proof of first case. The proofs considering (G1) or (G2) are very similar, so we will only explain the (G1) case. Since everything in the EHP action is linear, grading-preserving and invariant under the action of $\mathfrak{I}(P; A)$, it is enough to prove that $\alpha_{n,\Lambda} = 0$ for some representative Einstein-Hilbert form in the quotient $\Lambda(P; A)/\mathfrak{I}(P; A)$. We choose

$$\alpha_{n,\Lambda} = \wedge_*^{n-2} e \wedge_* \Omega + \frac{\Lambda}{(n-1)!} \wedge_*^n e.$$

Under the hypothesis, we can locally write $e = \sum_m e^m$ with $e^m : TP \to A_0^m$. From condition (G1) each A_0^m is a weak (k_m, s_m)-solvable subspace, so that it writes as a sum $A_0^m = \oplus_i V_i^m$ of weak (k_m, s_m)-nilpotent subspaces, which means that we can write $e^m = \sum_i e_i^m$ locally and, therefore, $e = \sum_m \sum_i e_i^m$. Consequently, for every l we have $\wedge_*^l e = p_l(e_{i_1}^{m_1}, ..., e_{i_l}^{m_l})$ for some polynomial of degree l. If we now consider the minimum (k, s) (over m) of (k_m, s_m), the fact that each A_0^m is weakly (k_m, s_m)-nilpotent then implies $\wedge_*^{k+s+1} e = 0$. Consequently,

$$(\wedge_*^{k+s+1} e) \wedge_* \alpha = 0 \tag{8}$$

for any A-valued form α. In particular, for

$$\alpha = \frac{\Lambda}{(n-1)!} \wedge_*^{n-(k+s+1)} e$$

(8) is precisely the inhomogeneous part of $\alpha_{\Lambda,n}$. Therefore, for every $n \geq k+s+1$ we have $\alpha_{n,\Lambda} = \alpha_n$ and consequently $\pi(\alpha_n) = \pi(\alpha_{n,\Lambda})$, implying $S_n[e, \omega] = S_{n,\Lambda}[e, \omega]$ for any A-valued connection, and thus $S_n = S_{n,\Lambda}$. On the other hand, for

$$\alpha = (\wedge_*^{n-(k+s+1)+2} e) \wedge_* \Omega$$

we see that (8) becomes exactly α_n. Therefore, when $n \geq k+s+3$ we have $\alpha_n = 0$, implying (because $\jmath$ is linear) $\jmath(\alpha_n) = 0$ and thus $S_n = 0$. But $k+s+3 > k+s+1$, so that $S_{n,\Lambda} = 0$ too, ending this first case.

Proof of second case. Notice that to give a morphism $f : A' \to A$ of degree l between two graded algebras is the same as giving a zero degree morphism $f : A' \to A[-l]$, where $A[-l]$ is the graded algebra obtained by shifting A. Consequently, the obstructions of a degree l graded A-EHP are just the obstructions of degree zero graded $A[-l]$-EHP, so that it is enough to analyze theories of degree zero. If we assume that TP is a graded bundle, so that $TP \simeq \oplus_m E^m$, and if e has degree zero, then the only change in comparison to the previous "partially graded" context is that instead of decomposing e as a sum $\sum_m e^m$, we can now write it as a genuine direct sum $e = \oplus_m e^m$, with $e^m : E^m \to A_0 \cap A^m$, meaning that the same argumentation of the case case applies here, ending the proof. □

3 Theorem B

Until now we worked in full generality and we showed that action functionals defined on spaces of algebra valued forms suffer minimal obstructions (in terms of the properties of the algebra) which affect the possible dimensions of the base manifold. Here we will show that if we restrict ourselves to certain concrete situations the base manifold suffers not only those minimal obstructions, but also strong topological obstructions.

Recall that given algebras E, A and H, we say that E is an *extension of A by H* when they fit into a short exact sequence (as R-modules):

$$
\begin{tikzcd} 0 \arrow[r] & H \arrow[r, "\imath"] & E \arrow[r, "\pi"] & A \arrow[r] & 0 \end{tikzcd}
$$

Furthermore, the extension is called *separable* if π has a section, i.e, a R-linear map $s : A \to E$ such that $\pi \circ s = id_A$. In this case we can use s to pushforward the product of A to E, as below:

$$
\begin{tikzcd} E \otimes E \arrow[r, "\pi\otimes\pi"] \arrow[rrr, bend right, "*'"'] & A \otimes A \arrow[r, "*"] & A \arrow[r, "s"] & E \end{tikzcd} \tag{9}
$$

Concerning this new product, all that we need is the following result:

Lemma 3.1. *The algebra $(A, *)$ is (k, s)-nil iff $(E, *')$ is (k, s)-nil.*

Proof. Relative to the new product A is clearly a subalgebra of E. Therefore, A is (k, s)-nil only if $(E, *')$ is. In order to get the reciprocal, recall that any algebra A induces a corresponding graded-algebra structure in $\Lambda(P; A)$. Due to the functoriality of $\Lambda(P; -)$, we then get the commutative diagram below, where the horizontal rows are just (9) for the algebras $(E, *)$ and $(A, *')$, composed with the diagonal map. The commutativity of this diagram says just that $\wedge^2_{*'}\alpha = \wedge^2_{*} f(\alpha)$ for every $\alpha \in \Lambda(P; A)$. From the same construction we get, for each given $s \geq 2$, a commutative diagram that implies $\wedge^s_{*'}\alpha = \wedge^s_{*} f(\alpha)$. Therefore, if $(A, *)$ is (k, s)-nil it immediately follows that $(E, *')$ is (k, s)-nil too.

$$
\begin{tikzcd}
\Lambda(P;E) \arrow[r, "\Delta"] \arrow[d, shift left, "\Lambda\pi"] & \Lambda(P;E)\otimes\Lambda(P;E) \arrow[r] \arrow[d, shift left, "(\Lambda\pi)\otimes(\Lambda\pi)"] & \Lambda(P;E\otimes E) \arrow[r] \arrow[d, shift left, "\Lambda(\pi\otimes\pi)"] & \Lambda(P;E) \arrow[d, shift left, "\Lambda\pi"] \\
\Lambda(P;A) \arrow[r, "\Delta"] \arrow[u, shift left, "\Lambda s"] & \Lambda(P;A)\otimes\Lambda(P;A) \arrow[r] \arrow[u, shift left, "(\Lambda s)\otimes(\Lambda s)"] & \Lambda(P;A\otimes A) \arrow[r] \arrow[u, shift left, "\Lambda(s\otimes s)"] & \Lambda(P;A) \arrow[u, shift left, "\Lambda s"]
\end{tikzcd}
$$

□

7

In the following we will be interested in:

1. *k-algebras.* These are algebras $(E, *')$ arising as splitting extensions of real matrix algebras $(A, *)$ which are associative subalgebras of $\mathfrak{so}(k_1) \oplus ... \oplus \mathfrak{so}(k_r)$ for some $k_1, .., k_r$. We notice that $\mathfrak{so}(k)$ is $(1,2)$-nil for any k. It then follows that A (and therefore E, due to lemma above) is $(1,2)$-nil, because $A \subset \mathfrak{so}(k_1 + .. + k_r)$;

2. *Berger k-manifolds N.* These have dimension k and are simply connected, locally irreducible and locally non-symmetric;

3. *k-proper bundles $P \to M$.* These are such that there exists an immersed Berger k-manifold $N \hookrightarrow M$ whose frame bundle is a subbundle of P, i.e, such that $FN \subset \imath^* P$. The motivating examples are the frame bundle of a Berger k-manifold and, more generally, the pullback of the frame bundle by the immersion $f : N \to M$ of a Berger k-manifold.

4. *torsion-free EHP theories.* By this we mean A-EHP funcional (7) restricted to the subspace $\mathrm{Conn}_A^{\Theta=0}(P) \subset \mathrm{Conn}_A(P)$ of A-valued connections $\nabla = e + \omega$ such that $\Theta_\omega = 0$.

Restatement and Proof

Let us now restate and proof Theorem B and analyze its fundamental consequences.

Theorem 3.1. *Let $P \to M$ be a k-proper H-bundle over a n-manifold M. If $\mathfrak{h}$ is a k-algebra, then $k_1 = k$ and $k_{i>1} = 0$. Furthermore, for any FABS, the torsionless EHP theory with values into $\mathbb{R}^k \rtimes \mathfrak{h}$ is nontrivial only if one of the following conditions is satisfied*

(B1) *$k = 2, 4$ and M contains a Kähler Berger k-manifold;*

(B2) *$k = 4$ and M contains a quaternionic-Kähler Berger k-manifold.*

Proof. Since $\mathfrak{h}$ is a k-algebra, it is $(1,2)$-nil and, therefore, the splitting extension $\mathbb{R}^k \rtimes \mathfrak{h}$ is $(1,2)$-nil too. Consequently, by Theorem 8 the actual (and, in particular, the torsionless) EHP is trivial if $n \geq 6$, so that we may assume $n < 6$. Since $P \to M$ is k-proper, M contains at least one immersed Berger k-manifold $\imath : N \hookrightarrow M$ such that $FN \subset \imath^* P$. Let κ denote the inclusion of FN into $\imath^* P$. On the other hand, we also have an immersion $\imath^* P \hookrightarrow P$, which we denote by $\imath$ too. Lie algebra-valued forms can be pulled-back and the pullback preserves horizontability and equivariance. So, for any $\nabla = e + \omega$ the corresponding 1-form $(\imath \circ \kappa)^* \omega \equiv \omega|_N$ is an H-connection in FN whose holonomy is contained in H. Since H is the Lie integration of a k-algebra we have

$$H \subset SO(k_1) \times ... \times SO(k_r) \subset SO(k).$$

In particular, the holonomy of $\omega|_N$ is contained in $SO(k)$, implying that $\omega|_N$ is compatible with some Riemannian metric g in N. But, we are working with torsion-free connections, so that $\omega|_N$ is actually the Levi-Civita connection of g and, because N is irreducible, de Rham decomposition theorem implies that there exists $i \in 1, ..., r$ such that $k_i = k$ and $k_j = 0$ for $j \neq i$. Without loss of generality we can take $i = 1$. Because N is a Berger manifold, Berger's theorem [6] applies, implying that the holonomy of $\omega|_N$ is classified, giving conditions (B1) and (B2). $\square$

Corollary 3.1. *Let M be a Berger n-manifold with an H-structure, where H is such that $\mathfrak{h}$ is a n-algebra. In this case, for any FABS, the EHP theory with values into $\mathbb{R}^n \rtimes H$ is nontrivial only if M has dimension $n = 2, 4$ and admits a Kähler structure.*

Proof. The result follows from the last theorem by considering the bundle $P \to M$ as the frame bundle $FM \to M$ and from the fact that every orientable four-dimensional smooth manifold admits a quaternionic-Kähler structure [7, 27]. $\square$

This corollary shows how topologically restrictive it is to internalize *torsionless* EHP in geometries other than Lorentzian. Indeed, if the manifold M is compact and 2-dimensional, then it must be $\mathbb{S}^2$. On the other hand, in dimension $n = 4$ compact Kähler structures exist iff the Betti numbers $b_1(M)$ and $b_3(M)$ are zero, so that $\chi(M) = b_2(M) + 2$. As a consequence, if we add the (mild) condition $H^2(M; \mathbb{R}) \simeq 0$ on the hypothesis of Corollary 3.1 we conclude that M must be a K3-surface!

4 Examples

In this section we will give realizations of the obstruction theorems studied previously. Due to the closeness with Lorentzian EHP (which is where EHP theories are usually formulated), our focus is on concrete situations, meaning that we will be focused in A-EHP functionals for A a k-algebra or an splitting extension of a k-algebra. Even so, some abstract examples will also be given.

Concrete Examples

Let us start by considering A-EHP for A an $\mathbb{R}$-algebra endowed with a distinguished subalgebra $\overline{A} \subset A$. We then have two vector space decompositions $A \simeq A_0 \oplus A_1$: one for which $A_0 = \overline{A}$ and $A_1 = A/\overline{A}$ and another one for the opposite. From Theorem 2.1 and the fact that k-algebras are $(1, 2)$-nil we have the following conclusions for $n \geq 6$:

(E1) if $\overline{A}$ is an ideal and $A/\overline{A}$ is a k-algebra, then the EHP funcional relative to the opposite decomposition is trivial;

(E2) if $\overline{A}$ is a k-subalgebra, then the EHP functional relative to the first decomposition is trivial;

(E3) if A is k-algebra, then EHP is trivial in both decompositions.

Conditions (E2) and (E3) are immediately satisfied for k-groups, i.e, Lie subgroups of $SO(k_1) \times ... \times SO(k_r)$, or, equivalently, Lie subalgebras of $\mathfrak{so}(k_1) \oplus ... \oplus \mathfrak{so}(k_r)$. Indeed, if G_0 is a such subgroup, then the EHP theory for any associative algebra algebra A extending $\mathfrak{g}_0$ obviously satisfies (E2), so that if $n \geq 6$ the dual theories are trivial. On the other hand, for any subalgebra $A_0 \subset \mathfrak{g}_0$ the EHP theory with values into $\mathfrak{h}$ clearly satisfies (E3) so that in dimension $n \geq 6$ both the dual and the actual EHP theory are trivial. Some obvious examples of subgroups of k-groups $G_0 \subset SO(k)$ are given in the table below. Except for $Spin(4) \hookrightarrow O(8)$, which arises from the exceptional isomorphism $Spin(4) \simeq SU(2) \times SU(2)$, all the other inclusions appear in Berger's classification [6].

Let us focus on (E2). We can think of each element of Table 1 as included in some $GL(k; \mathbb{R})$, i.e, as a G-structure on a manifold and then as a geometry. We could get more examples by taking finite products of arbitrary elements in the table. In terms of geometry, this can be interpreted as follows. Recall that a distribution of dimension k on an n-manifold can be regarded as a G-structure for $G = GL(k) \times GL(n-k)$. Therefore, we can think of a product $O(k) \times O(n-k)$ as a distribution of Riemannian leaves on a Riemannian manifold, $U(k) \times U(n-k)$ as a hermitan distribution, and so on.

(n,k)	$(1,k)$	$(2,k)$	$(4,k)$	$(1,7)$	$(1,8)$
$H \subset O(n \cdot k)$	$SO(k)$	$U(k)$	$Sp(k)$	G_2	$Spin(4)$
	$U(k)$	$SU(k)$	$Sp(k) \cdot Sp(1)$		$Spin(7)$

Table 1: First examples of concrete geometric obstructions

Other special cases where condition (E2) applies are in the table below. In the first line, $O(k,k)$ is the so-called *Narain group* [23], i.e, the orthogonal group of a metric with signature (k,k), whose maximal compact subgroup is $O(k) \times O(k)$. Second line follows from an inclusion similar to $U(k) \hookrightarrow O(2k)$, first studied by Hitchin and Gualtieri [17, 15, 18], while the remaining lines are particular cases of the previous ones. The underlying flavors of geometry arose from the study of Type II gravity and Type II string theory [12, 20]. That condition (E2) applies for the second and third lines follows from the fact that by complexifying $U(k,k) \hookrightarrow O(2k,2k)$ we obtain $U(k,k) \hookrightarrow O(4k;\mathbb{C})$.

G	G_0	geometry
$O(k,k)$	$O(k) \times O(k)$	Type II
$O(2k,2k)$	$U(k,k)$	Generalized Complex
$O(2k,2k)$	$SU(k,k)$	Generalized Calabi-Yau
$O(2k,2k)$	$U(k) \times U(k)$	Generalized Kähler
$O(2k,2k)$	$SU(k) \times SU(k)$	Generalized Calabi

Table 2: More examples of concrete geometric obstructions

About these two tables, some remarks:

1. Table 1 contains any classical flavors of geometry, except symplectic and complex. The reason is that $\mathrm{Sp}(k;\mathbb{R})$ and $GL(k;\mathbb{C})$ are not contained in some $O(r)$. But this does not mean that condition (E2) is not satisfied by those geometries Indeed, generalized complex geometry contains both of them [15], so that they actually fulfill (E2).

2. We could create a third table with "exotic k-groups", meaning that a priori they are not related to any classical geometry, so that they describe some kind of "exotic geometry". For instance, in [25] all Lie subgroups $G_0 \subset O(k)$ satisfying
$$\frac{(k-3)(k-4)}{2} + 6 < \dim H < \frac{(k-1)(k-2)}{2}$$
were classified and in arbitrary dimension k there are fifteen families of them. Other exotic (rather canonical, in some sense) subgroups that we could add are maximal tori. Indeed, both $O(2k)$ and $U(k)$ have maximal tori, say denoted by T_O and T_U, so that the reductions $T_O \hookrightarrow O(2k)$ and $T_U \hookrightarrow U(k)$ will satisfy condition (E2). More examples of exotic subgroups to be added are the *point groups*, i.e, $H \subset \mathrm{Iso}(\mathbb{R}^k)$ fixing at least one point. Without loss of generality we can assume that this point is the origin, so that $H \subset O(k)$. Here we have the

symmetric group of any spherically symmetric object in $\mathbb{R}^k$, such as regular polyhedrons and graphs embedded on $\mathbb{S}^{k-1}$.

3. If we are interested only in condition (E2), then the tables above can be enlarged by including embeddings of $O(k_1) \times ... \times O(k_r)$ into some other larger group $\tilde{G}$. In this condition it only matters that H is a k-group. Particularly, $O(k)$ admits some exceptional embeddings (which arise from the classification of simple Lie algebras), as in the table below.

k	3	9	10	12	16
$O(k) \hookrightarrow$	G_2	F_4	E_6	E_7	E_8

Table 3: Exceptional embbedings of orthogonal groups.

4. Despite (E2) making sense sense for arbitrary real algebras A with distinguished k-subalgebra $A_0 \subset A$, the examples above were all for matrix algebras underlying Lie algebras of matrix Lie groups. But, as pointed out previously, in order to get (E2) we only need that A_0 be a k-algebras, meaning that **any** extension (not necessarily a matrix extension) will satisfy (E2). In particular, for every G_0 in the above tables $\mathbb{R}^k \rtimes \mathfrak{g}_0$ satisfies this condition. Recall that EHP theories with values in those algebras are immediate generalizations of Lorentzian EHP theory. Therefore, what we showed is that the EHP functional cannot be realized (in dimension $n \geq 6$) in essentially all flavors of geometry other then semi-Riemannian with specific signature.

5. Condition (E2) is on the Lie algebra level. A Lie group and its universal covering have the same Lie algebra. Therefore, we can double the size of our current tables by adding the universal cover (when it exists) of each group. For instance, in Table 1 the spin groups $\mathrm{Spin}(4)$ and $\mathrm{Spin}(7)$ were added due to exceptional isomorphisms on the level of Lie groups. Now, noticing that for $k > 2$ (in particular for $k = 4, 7$) $\mathrm{Spin}(k)$ is the universal cover of $SO(k)$, we can automatically add all spin groups to our table. We can also add the universal covering of the symplectic group $Sp(k; \mathbb{R})$, the metapletic group.

6. Due to the above discussion, we can use Tables 1 and 2, as well as the "table of exotic k-groups", as a source of examples for condition (E3). Indeed, we can just consider the algebra A in condition (E3) as the algebra of some H in the mentioned tables and take a fixed subalgebra $A_0 \subset \mathfrak{g}$. This produces a long list of examples, because G/G_0 is a priori an arbitrary homogeneous space subjected only to the condition of G being a k-group (here G_0 is the integrated Lie group of A_0). In geometric terms, if a geometry fulfills condition (E2), then any induced "homogeneous geometry" fulfills (E3). On the other hand, differently from what happens with condition (E2), Table 3 *cannot* be used to get more examples of condition (E3). Indeed, if $G_0 \hookrightarrow G$ is a reduction fulfilling (E3) and $G \hookrightarrow \tilde{G}$ is an embedding, then $G_0 \hookrightarrow \tilde{G}$ fulfills (E3) iff it is fulfilled by $G \hookrightarrow \tilde{G}$ (see diagram below).

$$\begin{tikzcd} H \arrow[r, hook] \arrow[rrr, bend right] & G \arrow[r, hook] & O(k_1) \times ... \times O(k_r) \arrow[r, hook] & \tilde{G} \end{tikzcd}$$

Let us now analyze condition (E1). First of all we notice that it is very restrictive, because we need to work with subalgebras $A_0 \subset A$ of k-algebras which are ideals. In terms of Lie groups this means that we need to restrict to normal subgroups $G_0 \subset G$ such that the quotient G/G_0 is a k-group. For instance, in the typical situations above, G_0 is not normal. Even so, there are two dual conditions under which $G_0 \subset G$ fulfills (E1): when G is a k-group with $G/G_0 \subset G$ and, dually, when G_0 is k-group with $G/G_0 \subset G_0$. However, in both situations we are also in (E3), so that there is nothing much new here.

Cayley-Dickson Examples

Here we will show that the list of concrete examples can be extended indefinitely, allowing us to conclude that EHP theories are trivial in an infinite number of geometries. This will be done by making use of the Cayley-Dickson construction. Let $(A, *)$ be an arbitrary $\mathbb{R}$-algebra with involution $\overline{(-)} : A \to A$. For each fixed $l > 0$ we get a sesquilinear map $s : A^l \times A^l \to A$ such that $s(x, y) = \sum_i \overline{x}_i * y_i$. Consider the subspace of all $l \times l$ matrices $M \in \mathrm{Mat}_{l\times l}(A)$ with coefficients in A which preserve s, in the sense that $s(Mx, My) = s(x, y)$. We say that these are the *unitary matrices in* A, and we denote this set by $U(k; A)$. If the involution is trivial (i.e, the identity map), then call them the *orthogonal matrices in* A, writing $O(k; A)$ to denote the corresponding space.

Recall that Cayley-Dickson takes an involutive algebra A and gives another involutive algebra $\mathrm{CD}(A)$ with weakened PIs [29]. As an $\mathbb{R}$-module, the newer algebra is given by a sum $A \oplus A$ of real and imaginary parts. We have an inclusion $A \hookrightarrow \mathrm{CD}(A)$, obtained by regarding A as the real part, which induces an inclusion into the corresponding unitary groups $U(k; A) \hookrightarrow U(k; \mathrm{CD}(A))$. This inclusion can be extended in the following way

$$U(k; A) \hookrightarrow U(k; \mathrm{CD}(A)) \hookrightarrow U(2k; A),$$

defined by setting the real part and the imaginary part as diagonal block matrices. Iterating we see that for every k and every l there exists an inclusion

$$U(k; \mathrm{CD}^l(A)) \hookrightarrow U(2^l k; A). \tag{10}$$

We assert that, at least when the starting algebra A is finite-dimensional, each $U(k; \mathrm{CD}^l(A))$ is a Lie group. Taking $A = \mathbb{R}$ in the inclusion above we will then get a corresponding sequence of matrix Lie algebras (and, therefore, a sequence of underlying associative algebras). Consequently, each of such algebras will satisfy condition (E2), meaning that, for every $k > 0$ and $l \geq 0$, the EHP theories with values into extensions of $\mathfrak{u}(k; \mathrm{CD}^l(A))$ are trivial.

$$U(k; \mathrm{CD}^l(\mathbb{R})) \longrightarrow \cdots \longrightarrow U(2^{l-2}k; \mathbb{H}) \longrightarrow U(2^{l-1}k; \mathbb{C}) \longrightarrow O(2^l k)\ .$$

The idea for proving the assertion is to reproduce the proof that the standard unitary/orthogonal groups $U(k)$ and $O(k)$ are Lie groups. The starting point is to notice that the involution of A induces an involution $(-)^\dagger$ in $\mathrm{Mat}_{k\times k}(A)$, defined by composition with the transposition map. The next step is to see that $k \times k$ matrices preserve the involution s iff they are invertible with $M^{-1} = M^\dagger$, i.e, iff $M \cdot M^\dagger = 1_k = M^\dagger \cdot M$. Since $\overline{(-)} : A \to A$ is an algebra morphism, we have the usual property $(M \cdot N)^\dagger = M^\dagger \cdot N^\dagger$, allowing us to characterize the unitary matrices as those satisfying $M \cdot M^\dagger = 1_k$. Therefore, defining the map

$$f : \mathrm{Mat}_{k\times k}(A) \to \mathrm{Mat}_{k\times k}(A) \quad \text{by} \quad f(M) = M \cdot M^\dagger,$$

in order to proof that $U(k;A)$ is a Lie group it is enough to verify that the above map is, in some sense, a submersion (which will give the smooth structure) and that the multiplication and inversion maps are smooth. When A is finite-dimensional this is immediate (although this requeriment is not necessary).

Since classical geometries (meaning geometries in the sense of Weil) are described by inclusions of Lie groups, this leads us to think of (10) as modeling some flavor of geometry, say the *Cayley-Dickson geometry in A, of order l*. With this nomenclature we have concluded that EHP theories are trivial in real Cayley-Dickson geometries of arbitrary order, generalizing the examples in Table 1. In the following we will show that Table 2 can be generalized as well. In fact we will show that there is a notion of *generalized Cayley-Dickson geometry in A* which, for $A = \mathbb{R}$, contains Hitchin's generalized complex geometry and in which EHP theories are trivial. This follows directly from the fact that for every A we have the inclusion $U(k,k;\mathrm{CD}(A)) \subset O(2k;2k,A)$, so that by iteration we get

$$U(k,k;\mathrm{CD}^l(A)) \subset O(2^l k;2^l k,A). \tag{11}$$

For $A = \mathbb{R}$ and $l = 1$ this model generalized complex geometry, leading us to say that the inclusion above models *generalized Cayley-Dickson geometry in A of degree l*. For instance, if we take $A = \mathbb{R}$ and $l = 2$ this becomes generalized quaternionic geometry, which is a poorly studied theory, beginning with the works [26, 13] . For $l = 3, 4, ...$ it should be generalized octonionic geometry, generalized sedenionic geometry, and so on. The authors are unaware of the existence of substantial works on these theories.

When tensoring inclusion (11) with $\mathrm{CD}^l(A)$ we get

$$U(k,k;\mathrm{CD}^l(A)) \subset O(2^l k;2^l k,\mathrm{CD}^l(A)) \simeq O(2^l k + 2^l k;\mathrm{CD}^l(A)),$$

so that condition (E2) above implies that, independently of the present development of generalized Cayley-Dickson geometries, EHP theory is trivial in each of them.

Abstract Examples

Here we give abstract examples of algebras in which EHP cannot be realized. By abstract we mean that they are not necessarily k-algebras. We will restrict to the cases in which only A is $\mathfrak{m}$-graded. From Theorem 2.1 we have that:

(F1) when A admits a vector space decomposition $A \simeq A_0 \oplus A_1$, where A_0 is a subalgebra such that each $A_0^m = A_0 \cap A^m$ is a weak (k_m, s_m)-solvable subspace, then A-EHP is trivial in dimension $n \geq k + s + 3$, where (k, s) is the minimum over (k_m, s_m).

The most basic examples are those for $\mathfrak{m} = 0$ and $A_1 = 0$, i.e, the non-graded setting with A itself (k, s)-solv. For instance, any Lie algebra is $(k, 1)$-nil for any even k. As a consequence, *in dimension $n \geq 4$, any Lie algebra valued EHP functional is trivial.* In particular, EHP with values in the Poincaré group $\mathfrak{iso}(n-1, 1)$ is trivial. We emphasize that this **does not** mean that the standard Lorentzian EHP theory is trivial (which would be absurd). Indeed, while Lorentzian EHP theory and $\mathfrak{iso}(n-1, 1)$-valued EHP theory take values in the same **vector space**, and their action functionals have the same shape, the **algebras** (and, therefore, their properties) used to define the corresponding wedge product are *totally different*.

If we now allow $\mathfrak{m}$ to be nontrivial, but with $A_1 = 0$, then condition (F1) is satisfied if each A^m is a weak (k_m, s_m)-solvable subespace. In particular, it remains satisfied if A is itself (k, s)-solv. It happens that not only Lie algebras are solv $(2, 1)$-nil, but also graded Lie algebras. Consequently, EHP theories are also trivial in the domain of graded Lie algebras. One can generalize even more thinking of EHP theories with values in Lie superalgebras and in graded Lie superalgebras. Indeed, recall that a $\mathfrak{m}$-*graded Lie superalgebra* $\mathfrak{g}$ is just a $\mathfrak{m}$-graded Lie algebra whose underlying PIs (i.e, skew-commutativity and Jacobi identity) hold in the graded sense. Particularly, this means that the $\mathbb{Z}_2$-grading writes $\mathfrak{g} \simeq \mathfrak{g}^0 \oplus \mathfrak{g}^1$, with $\mathfrak{g}^0$ a $\mathfrak{m}$-graded Lie algebra and, therefore, $(2, 1)$-nil. It then follows that $\mathfrak{g}$ satisfies condition (F1). Summarizing, in dimension $n \geq 6$, *the EHP functional cannot be realized in any Lie algebraic context.*

Graded Lie superalgebras are the first examples of algebras satisfying (F1) with $A_1 \neq 0$, but they are far from being the only one. Indeed, when we look at a decomposition $A \simeq A_0 \oplus A_1$, where A_0 is a subalgebra, it is inevitable to think of A as an extension of A_1 by A_0, meaning that we have an exact sequence as shown below. If A_0 is (k, s)-solv, then (F1) holds. This can be interpreted as follows: suppose that we encountered an algebra A_1 such that EHP is not trivial there. So, EHP will be trivial in any (splitting) extension of A_1 by a (k, s)-solv algebra.

$$0 \longrightarrow A_0 \longrightarrow A \longrightarrow A_1 \longrightarrow 0$$

In particular, because $(\mathbb{R}^k, +)$ is abelian and, therefore, $(1, 1)$-nil, any algebra extension by $\mathbb{R}^k$ will produce a context in which EHP is trivial. Recall the previous concrete examples, in which we considered splitting extensions of matrix algebras by $\mathbb{R}^k$. This would seem to suggest that every concrete EHP theory is trivial, which is not true. Once again the problem is in the wedge products: we have the same vector space structures, but in the concrete contexts and here the way of multiplying forms with values in those spaces are different. The main difference is that the product used there does not take the abelian structure of $\mathbb{R}^k$ into account.

Another example about extensions is as follows: the paragraph above shows that abelian extensions of nontrivial EHP are trivial, but what about super extensions? They remain trivial. Indeed, being a Lie superalgebra, the translational superalgebra $\mathbb{R}^{k|l}$ (the cartesian superspace, regarded as a Lie superalgebra) is $(2, 1)$-nil, so that any algebra extension by it is trivial. So, if we think on supergravity as a realization of gravity not in the domain of Cartan connections but in the domain of super Cartan connections on the super Poincaré Lie algebra $\mathfrak{siso}(n-1, 1)$ we find that EHP functional is not a good model, because it will be trivial if $n \geq 6$.

Finally, there are also purely abstract examples. Just to mention, in [30] the authors build "mathematically exotic" examples of nilpotent algebras, which fulfill (F1). In [5] necessary and sufficient conditions were given under which arbitrary orthogonal groups $O(q, \mathbb{K})$, where $q : V \to \mathbb{K}$ is a positive-definite quadratic form on a finite-dimension $\mathbb{K}$-space q and $\mathrm{ch}(\mathbb{K}) \neq 2$, admit an embedded maximal torus $\mathbf{T}(q; \mathbb{K})$. Independently of the quadratic space (V, q), the corresponding orthogonal group is a Lie group and, as in the real case, the algebra $\mathfrak{o}(q; \mathbb{K})$ is $(2, 1)$-nil. Therefore, under the conditions of [5], the EHP functional will be trivial in those toroidal geometries.

5 Appendix

Here we will introduce a general tool that allows us to pushforward algebra-valued forms on the total space P of a bundle $P \to M$ to the base manifold M. The first step is to build some process allowing us to replace A-valued forms in P by forms in M with coefficients in some other bundle, say E_A. It is more convenient to think of this in categorical terms. Let $\mathbf{Alg}_R$ be the category of R-algebras, $\mathbb{Z}\mathbf{Alg}_R$ be the category of $\mathbb{Z}$-graded R-algebras and, given a manifold M, let $\mathbf{Bun}_M$ and $\mathbf{Alg}_R\mathbf{Bun}_M$ denote the categories of bundles and of R-algebra bundles over M, respectively. A *functorial algebra bundle system* (FABS) consists of

1. a subcategory $\mathbf{C} \subset \mathbf{Bun}_M \times \mathbb{Z}\mathbf{Alg}_R$;
2. a functor E_- assigning to any algebra $A \in \mathbf{C}$ a corresponding algebra bundle $E_A \to M$ whose typical fiber is A;
3. a functor $S(-;-)$ that associates an algebra to each pair $(P, A) \in \mathbf{C}$;
4. natural transformations $\imath : S(-;-) \Rightarrow \Lambda(-;-)$ and $\xi : S(-;-) \Rightarrow \Lambda(M;-)$ such that $\imath$ is objectwise injective.

Now that we know how to replace A-valued forms in P with E_A-valued forms in M, let us see how to transfer the latter to classical $\mathbb{R}$-valued forms in M. This is done by taking some "trace". A *trace* compatible with a FABS is given by a functor $\mathrm{tr} : \mathbb{Z}\mathbf{Alg}_R \to \mathbb{Z}\mathbf{Alg}_{\mathbb{R}}$ together with a natural transformation τ between $\mathrm{tr} \circ \Lambda(M; E_-)$ and the constant functor at $\Lambda(M;\mathbb{R})$. All this data fits in the following diagram:

$$
\begin{tikzcd}
\mathbf{Alg}_{\mathbf{R}} \arrow[rr, dashed, "E_-"] & & \mathbf{Alg}_{\mathbf{R}}\mathbf{Bun}_M \arrow[d, "\Lambda(M;-)"] \\
\mathbf{C} \arrow[u, "\pi_1"] \arrow[rr, dashed, "{S(-,-)}"] \arrow[rr, bend right, "{\Lambda(-;-)}"'] \arrow[urr, phantom, "\Uparrow \jmath"] \arrow[rr, phantom, bend right=15, "\Downarrow \imath"] & & \mathbb{Z}\mathbf{Alg}_{\mathbf{R}}
\end{tikzcd}
$$

Such systems exist, as shown in the next example. The fundamental properties and constructions involving FABS will appear in a work under preparation [22].

Example 5.1 (*standard case*)**.** Let $\mathbf{C}$ be composed by pairs (P, A), where $P \to M$ is a G-bundle whose group G becomes endowed with a representation $\rho : G \to GL(A)$. In that case we define E_- as the rule assigning to each A the corresponding associated bundle $P \times_\rho A$. The functor S is such that $S(P, A)$ is the algebra $\Lambda_\rho(P; A)$ of *ρ-equivariant A-valued forms α in P*, i.e, of those which satisfy the equation $R_g^*\alpha = \rho(g^{-1}) \cdot \alpha$, where $R : G \times P \to P$ denotes the canonical free action characterizing P as a principal G-bundle. This algebra of ρ-equivariant forms naturally embeds into $\Lambda(P; A)$, giving $\imath$. Finally, it is a standard fact [21] that each ρ-equivariant A-valued form on P induces an $P \times_\rho A$-valued form on M, defining the transformation ξ. This is the standard approach used in the literature, so that we will refer to it as the *standard FABS*.

In order to define tghe EHP action functional we need to consider invariant FABS. Let us introduce them. Indeed, we say that a FABS on $\mathbf{C}$ compatible with a trace (tr, τ) is *invariant* under a functor $I : \mathbf{C} \to \mathbb{Z}\mathbf{Alg}_R$ if

(a) for all $(P, A) \in \mathbf{C}$ the corresponding $I(P, A)$ is an ideal of $S(P; A)$, so that we can take the quotient functor S/I and we have a natural transformation $\pi : S \Rightarrow S/I$;

(b) there exists another $J : \mathbf{C} \to \mathbb{Z}\mathbf{Alg}_{\mathbb{R}}$ such that $J(P, A)$ is an ideal of $\Lambda(M; E_A)$ and whose projection we denote by π';

(c) there exists a natural transformation $\jmath' : S/I \Rightarrow \Lambda(M; -)/J$ such that $\jmath' \circ \pi = \pi' \circ \jmath$, i.e, the diagram below commutes for every (P, A);

$$\begin{tikzcd} S(P,A) \arrow[r, "\jmath_{(P,A)}"] \arrow[d, "\pi_{(P,A)}"'] & \Lambda(M;E_A) \arrow[d, "\pi'_{(P,A)}"] \\ S(P,A)/I(P,A) \arrow[r, "\jmath'_{(P,A)}"'] & \Lambda(M;E_A)/J(P,A) \end{tikzcd} \tag{12}$$

(d) the transformation τ passes to the quotient by $\tau \circ I$.

Finally, notice that all these definitions make perfect sense in the more general category $\mathfrak{m}\mathbf{Alg}_R$ of $\mathfrak{m}$-graded R-algebras.

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4.4 SECOND WORK

The following is a recompilation of the ArXiv preprint

- Y.X. Martins, R.J. Biezuner, *Geometric Obstructions on Gravity*, Dec 2019, arXiv:1912.11198

Geometric Obstructions on Gravity

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Abstract

These are notes for a short course and some talks gave at Departament of Mathematics and at Departament of Physics of Federal University of Minas Gerais, based on the author's paper [26]. Some new information and results are also presented. Unlike the original work, here we try to give a more physical emphasis. In this sense, we present obstructions to realize gravity, modeled by the tetradic Einstein-Hilbert-Palatini (EHP) action functional, in a general geometric setting. In particular, we show that if spacetime has dimension $n \geq 4$, then the cosmological constant plays no role in any "concrete geometries" other than Lorentzian. If $n \geq 6$, then the entire EHP theory is trivial, meaning that Lorentzian geometry is (essentially) the only "concrete geometry" in which gravity (i.e, the EHP action functional) makes sense. Examples of "concrete geometries" include those locally modeled by group reductions $H \hookrightarrow O(k; A)$ for some k and some algebra A, so that Riemannian geometry, Hermitian geometry, Kähler geometry and symplectic geometry, as well as Type II geometry, Hitchin's generalized complex geometry and G_2-geometry are included. We also study EHP theory in "abstract geometries", such as graded geometry (and hence supergeometry), and we show how the obstruction results extend to this context. We construct two theories naturally associated to EHP, which we call the geometric/algebraic dual of EHP, and we analyze the effect of the obstructions in these new theories. Finally, we speculate (and provide evidence for) the existence of a "universal obstruction condition".

1 Introduction

Einstein's theory of gravity is formulated in Lorentzian geometry: spacetime is regarded as an orientable four-dimensional Lorentzian manifold (M, g) whose Lorentzian metric g is a critical point of the Einstein-Hilbert action functional

$$S_{EH}[g] = \int_M (R_g - 2\Lambda) \cdot \omega_g, \tag{1}$$

defined on the space of all possible Lorentzian metrics in M. Here, $\Lambda \in \mathbb{R}$ is a parameter (the cosmological constant), while R_g and ω_g are, respectively, the scalar curvature and the volume-form

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of g, locally written as $g^{ij}\operatorname{Ric}_{ij}$ and $\sqrt{|\det g|}\,dx^1 \wedge ... \wedge dx^4$, where Ric_{ij} are the components of the Ricci tensor. Such critical points are precisely those satisfying the vacuum Einstein's equation

$$\operatorname{Ric}_{ij} - \frac{1}{2} R_g g_{ij} + \Lambda g_{ij} = 0. \tag{2}$$

The action (1) is about tensors, so that we can say that General Relativity is generally formulated in a "tensorial approach". Electromagnetism (or, more generally, Yang-Mills theories) was also initially formulated in a "tensorial approach". Indeed, the action functional for electromagnetism is given by

$$S_{YM}[A] = -\frac{1}{4}\int_M F_{ij}\tilde{F}^{ij},$$

where F and $\tilde{F}$ are the "electromagnetic tensor" and the "dual eletromagnetic tensor", with components

$$F_{ij} = \partial_i A_j - \partial_j A_i \quad \text{and} \quad \tilde{F}^{ij} = \frac{1}{2}\epsilon^{ijkl} F_{kl}\sqrt{|\det g|},$$

respectively. Varying this action we get Maxwell's equations in their tensorial formulation:

$$\partial_k \tilde{F}^{kl} + A_k \tilde{F}^{kl} = 0. \tag{3}$$

Notice that this is actually **one half** of the actual Maxwell's equations, corresponding to the "inhomogeneous equations". In order to get the full equations we need to take into account an external set of equations, corresponding to the "homogeneous part" of Maxwell's equations:

$$\epsilon^{ijklmn}\partial_l F_{mn} = 0. \tag{4}$$

It happens that, when we move from the tensorial language to the language of differential forms and connections on bundles, we rediscover (4) as a *geometric identity* (Bianchi identity), so that (4) actually holds *a priori* and not *a posteriori*, as was suggested using tensorial language. This is not a special feature of Yang-Mills theories. In fact, General Relativity (GR) also manifests this type of behavior: alone, Einstein's equation (2) does not completely specify a system of (GR); we also need to assume that the connection in question is precisely the Levi-Civita connection of g. But, as in the case of Bianchi identity for Yang-Mills theories, this assumption can be avoided if we use the language of differential forms.

Indeed, in the so-called *first order formulation of gravity* (also known as *tetradic gravity*) we can rewrite Einstein-Hilbert action functional as the *Einstein-Hilbert-Palatini* (EHP) *action*:

$$S_{EHP}[e,\omega] = \int_M \operatorname{tr}(e \curlywedge_{\rtimes} e \curlywedge_{\rtimes} \Omega + \frac{\Lambda}{6} e \curlywedge_{\rtimes} e \curlywedge_{\rtimes} e \curlywedge_{\rtimes} e), \tag{5}$$

where e and ω are 1-forms in the frame bundle FM with values in the Lie algebras $\mathbb{R}^{3,1}$ and $\mathfrak{o}(3,1)$, called *tetrad* and *spin connection*, respectively, and $\curlywedge_{\rtimes}$ is a type of "wedge product"[1] induced by matrix multiplication in $O(3,1)$. Because M is Lorentzian, its frame bundle is structured over $O(3,1)$ and the spin connection is actually a connection in this bundle. The tetrad e is such that

[1]In this paper we will work with many different types of "wedge products", satisfying very different properties. Therefore, in order to avoid confusion, we will not follow the literature, but introduce specific symbols for each of them.

for every $a \in FM$ the corresponding map $e_a : FM_a \to \mathbb{R}^{3,1}$ is an isomorphism, so that it has the geometrical meaning of a soldering form. Furthermore, in (5) we have a 2-form Ω with values in $\mathfrak{o}(3,1)$, representing the curvature $\Omega = d\omega + \omega \curlywedge_{\rtimes} \omega$ of ω. The translational algebra and the Lorentz algebra fits into the Poincaré Lie algebra $\mathfrak{iso}(3,1) = \mathbb{R}^{3,1} \rtimes \mathfrak{o}(3,1)$. The equations of motion for the EHP action are [?]

$$d_\omega e \curlywedge_{\rtimes} e = 0 \quad \text{and} \quad e \curlywedge_{\rtimes} \Omega + \frac{\Lambda}{6} e \curlywedge_{\rtimes} e \curlywedge_{\rtimes} e = 0.$$

The second of them is just Einstein's equation (2) rewritten in the language of forms, while the first, due to the fact that e is an isomorphism, is equivalent to $d_\omega e = 0$. As the 2-form $\Theta = d_\omega e$ describes precisely the torsion of the spin connection ω, the first equation of motion implies that ω is actually the only torsion-free connection compatible with the metric: the *Levi-Civita connection.*

These examples emphasize that the language of differential forms seems to be a nice way to describe physical theories. Indeed, [18, 19] started a program that attempts to unify different physical theories by using the same type of functional on forms, the so-called *Hitchin's functional.* Furthermore, in [11] it was shown that all known "gravity theories defined by forms" can really be unified in that they are particular cases of a single *topological M-theory*. Motivated by this philosophy, in this article we will consider generalizations of classical EHP theories (5).

First of all notice that if the underlying manifold is n-dimensional spacetime (instead of four-dimensional), (5) can be immediately generalized by considering $\mathfrak{iso}(n-1,1)$-valued forms and an action functional given by

$$S_{n,\Lambda}[e,\omega] = \int_M \mathrm{tr}(e \curlywedge_{\rtimes} \ldots \curlywedge_{\rtimes} e \curlywedge_{\rtimes} \Omega + \frac{\Lambda}{(n-1)!} e \curlywedge_{\rtimes} \ldots \curlywedge_{\rtimes} e). \tag{6}$$

However, this is not the only generalization that can be considered. We notice that in the modern language of differential geometry, the data defined by the pairs (ω, e) above corresponds to *reductive Cartan connections* on the frame bundle FM with respect to the inclusion $O(n-1,1) \hookrightarrow \mathrm{Iso}(n-1,1)$ of the Lorentz group into the Poincaré group. Indeed, given a G-bundle $P \to M$ and a structural group reduction $H \hookrightarrow G$, we recall that a *Cartan connection* in P for such a reduction is a G-connection ∇ in P which projects isomorphically onto $\mathfrak{g}/\mathfrak{h}$ in each point. Explicitly, it is an (horizontal and equivariant) $\mathfrak{g}$-valued 1-form $\nabla : TP \to \mathfrak{g}$ such that in each $a \in P$ the composition below is an isomorphism of vector spaces:

$$\begin{tikzcd} TP_a \arrow[r, "\nabla_a"] \arrow[rr, bend right, "\simeq"'] & \mathfrak{g} \arrow[r, "\pi"] & \mathfrak{g}/\mathfrak{h} \end{tikzcd}$$

Due to the decomposition (as vector spaces) $\mathfrak{g} \simeq \mathfrak{g}/\mathfrak{h} \oplus \mathfrak{h}$, the existence of the isomorphism above allows us to write $\nabla_a = e_a + \omega_a$ in each point. If this varies smoothly we say that the Cartan connection is *reductive* (or *decomposable*).

Now, looking at (6) we see that the groups $O(n-1,1) \hookrightarrow \mathrm{Iso}(n-1,1)$ and the bundle FM do not appear explicitly, so that we can think of considering an analogous action for other group reductions $H \hookrightarrow G$ in other bundle P. Intuitively, this means that we are trying to realize gravity (as modeled by EHP theory) in different geometries other than Lorentzian. But, *is* (6) *always nontrivial?* In other words, *is it possible to realize gravity in any geometry?* Notice that it is natural to expect that the algebraic properties of H and G will be related to the fundamental properties of the corresponding version of (6), so that a priori there should exist some abstract algebraic conditions under which (6) is trivial. In this article we will search for such nontrivial conditions, which we call *geometric obstructions*. Thus, one can say that *we will do Geometric Obstruction Theory applied to EHP theory*. For instance, one of the results that we will prove is the following, which emphasizes that some notion of "solvability" is crucial:

Theorem A. Let M be an n-dimensional spacetime and $P \to M$ be a $\mathbb{R}^k \rtimes H$-bundle, endowed with the group reduction $H \hookrightarrow \mathbb{R}^k \rtimes H$. If $\mathfrak{h}$ is a (k,s)-solv algebra and $n \geq k+s+1$, then the cosmological constant plays no role; if $n \geq k+s+3$, then the entire EHP theory is trivial.

This type of obstruction theorem gives restrictions only on the dimension of spacetime. We also show that if we restrict to torsion-free connections, then there are nontrivial geometric obstructions that gives restrictions not only on the dimension, but also on the topology of spacetime. For instance,

Theorem B. Let M be an n-dimension Berger manifold endowed with an H-structure. If $\mathfrak{h} \subset \mathfrak{so}(n)$, then the torsionless EHP theory is nontrivial only if $n = 2, 4$ and M is Kähler. In particular, if M is compact and $H^2(M;\mathbb{C}) = 0$, then it must be a K3-surface.

The generalization of EHP from Lorentzian geometry to arbitrary geometry is not the final step. Indeed, looking at (6) again we see that it remains well defined if one forgets that e and ω together define a Cartan connection. In other words, all one needs is the fact they are 1-forms that take values in some algebra. This leads us to consider some kind of *algebra-valued EHP theories*, defined on certain algebra-valued differential forms, for which we show that Theorem A remains valid almost *ipsis litteris*. We also show that if the algebra in question is endowed with a grading, then the "solvability condition" in Theorem A can be weakened.

This paper is organized as follows: in Section 2 we review some facts concerning algebra-valued differential forms and we study the "solvability conditions" that will appear in the obstructions results. In Section 3 we introduce EHP theory into the "linear"/ "extended-linear" contexts and we prove many obstruction theorems, including Theorem A and Theorem B. In Section 4 EHP is internalized into the "abstract context", where matrix algebras are replaced by arbitrary (possibly graded) algebras. We then show how the "fundamental obstruction theorem", namely Theorem A, naturally extends to this context. We also give geometric obstructions to realize EHP into the "full graded context", i.e, into "graded geometry" and, therefore, into "supergeometry". This section ends with a conjecture about the existence of some kind of "universal geometric obstruction". Finally, in Section 5 many examples are given, where by an "example" we mean an specific algebra fulfilling the hypothesis of some obstruction result, so that EHP cannot be realized in the underlying geometry.

2 Polynomial Identities in Algebra-Valued Forms

Let A be a real vector space[2] and P be a smooth manifold. A *A-valued k-form in P* is a section of the bundle $\Lambda^k TP^* \otimes A$. In other words, it is a rule α assigning to any $a \in P$ a skew-symmetric k-linear map

$$\alpha_a : TP_a \times \ldots \times TP_a \to A.$$

[2] All definitions and almost all results of this section hold analogously for modules over any commutative ring R of characteristic zero. For some results in Subsection 2.4 we must assume that the module is free, which is guaranteed if R is a field.

The collection of such maps inherits a canonical vector space structure which we will denote by $\Lambda^k(P;A)$. Given an A-valued k-form α and a B-valued l-form β we can define an $(A\otimes B)$-valued $(k+l)$-form $\alpha\otimes\beta$, such that

$$(\alpha\otimes\beta)_a(v,w)=\frac{1}{(k+l)!}\sum_{\sigma}\operatorname{sign}(\sigma)\alpha_a(v_\sigma)\otimes\beta_a(w_\sigma),$$

where $v=(v_1,...,v_k)$ and $w=(v_{k+1},...,v_{k+l})$ and σ is a permutation of $\{1,...,k+l\}$. This operator is obviously bilinear, so that it extends to a bilinear map

$$\otimes:\Lambda^k(P;A)\times\Lambda^l(P;B)\to\Lambda^{k+l}(P;A\otimes B).$$

We are specially interested when A is an algebra, say with multiplication $*:A\otimes A\to A$. In this case we can compose the operation $\otimes$ above with $*$ in order to get an exterior product $\wedge_*$, as shown below.

$$\begin{tikzcd}\Lambda^k(P;A)\times\Lambda^l(P;A) \arrow[r, "\otimes"] \arrow[rr, bend right, "\wedge_*"'] & \Lambda^{k+l}(P;A\otimes A) \arrow[r, "*"] & \Lambda^{k+l}(P;A)\end{tikzcd} \tag{7}$$

Explicitly, following the same notations above, if α and β are A-valued k and l forms, we get a new A-valued $(k+l)$-form by defining

$$(\alpha\wedge_*\beta)_a(v,w)=\frac{1}{(k+l)!}\sum_{\sigma}\operatorname{sign}(\sigma)\alpha_a(v_\sigma)*\beta_a(w_\sigma).$$

This new product defines an $\mathbb{N}$-graded algebra structure on the total A-valued space

$$\Lambda(P;A)=\bigoplus_k\Lambda^k(P;A),$$

whose properties are deeply influenced by the properties of the initial product $*$. For instance, recall that many properties of the algebra $(A,*)$ can be characterized by its polynomial identities (PI's), i.e, by polynomials that vanish identically when evaluated in A. Just to mention a few, some of these properties are commutativity and its variations (as skew-commutativity), associativity and its variations (Jacobi-identity, alternativity, power-associativity), and so on [22, 12].

The following proposition shows that each such property is satisfied in A iff it is satisfied (in the graded-sense) in the corresponding algebra of A-valued forms.

Proposition 2.1. *Any PI of degree m in A lifts to a PI in the graded algebra of A-valued forms. Reciprocally, every PI in the graded algebra restricts to a PI in A.*

Proof. Let f be a polynomial of degree m. Given arbitrary A-valued forms $\alpha_1,...,\alpha_m$ of degrees $k_1,...,k_m$, we define a corresponding A-valued form of degree $k=k_1+...+k_m$ as

$$\hat{f}(\alpha_1,...,\alpha_m)(v_1,...,v_k)=\frac{1}{k!}\sum_{\sigma}\operatorname{sign}(\sigma)f(\alpha_1(w_{\sigma(1)}),\alpha_2(w_{\sigma(2)})...,\alpha_m(w_{\sigma(m)})),$$

where

$$w_{\sigma(1)}=(v_{\sigma(1)},...,v_{\sigma(k_1)}),\quad w_2=(v_{\sigma(k_1+1)},...,v_{\sigma(k_1+k_2)}),\quad\text{and so on.}$$

Therefore, $\hat{f}$ is a degree m polynomial in $\Lambda(P;A)$. It is clear that if f vanishes identically then $\hat{f}$ vanishes too, so that each PI in A lifts to a PI in $\Lambda(P;A)$. On the other hand, if we start with a polynomial F in $\Lambda(P;A)$, we get a polynomial $F|_A$ in A of the same degree by restricting F to constant 1-forms. Clearly, if F is a PI, then $F|_A$ is also a PI. □

It follows that the induced product $\wedge_*$ is associative, alternative, commutative, and so on, iff the same properties are satisfied by $*$. Notice that, because the algebra $\Lambda(P;A)$ is graded, its induced properties must be understood in the graded sense. For instance, by commutativity and skew-commutativity of $\wedge_*$ one means, respectively,

$$\alpha \wedge_* \beta = (-1)^{kl}\beta \wedge_* \alpha \quad \text{and} \quad \alpha \wedge_* \beta = -(-1)^{kl}\beta \wedge_* \alpha.$$

Let us analyze some useful examples.

Example 2.1 (*Cayley-Dickson forms*)**.** There is a canonical construction, called *Cayley-Dickson construction* [2], which takes an algebra $(A,*)$ endowed with an involution $\overline{(-)}: A \to A$ and returns another algebra $\mathrm{CD}(A)$. As a vector space it is just $A \oplus A$, while the algebra multiplication is given by

$$(x,y)*(z,w) = (x*z - \overline{w}*y, z*x + y*\overline{z}).$$

This new algebra inherits an involution $\overline{(x,y)} = (\overline{x}, -y)$, so that the construction can be iterated. It is useful to think of $\mathrm{CD}(A) = A \oplus A$ as being composed of a "real part" and an "imaginary part". We have a sequence of inclusions into the "real part"

$$A \longrightarrow \mathrm{CD}(A) \longrightarrow \mathrm{CD}^2(A) \longrightarrow \mathrm{CD}^3(A) \longrightarrow \cdots$$

For any manifold P, such a sequence then induce a corresponding sequence of inclusions into the algebra of forms

$$\Lambda(P;A) \longrightarrow \Lambda(P;\mathrm{CD}(A)) \longrightarrow \Lambda(P;\mathrm{CD}^2(A)) \longrightarrow \Lambda(P;\mathrm{CD}^3(A)) \longrightarrow \cdots$$

The Cayley-Dickson construction weakens any PI of the starting algebra [2, **?**] and, therefore, due to Proposition 2.1, of the corresponding algebra of forms. For instance, if we start with the commutative and associative algebra $(\mathbb{R},\cdot)$, endowed with the trivial involution, we see that $\mathrm{CD}(\mathbb{R}) = \mathbb{C}$, which remains associative and commutative. But, after an iteration we obtain $\mathrm{CD}^2(\mathbb{R}) = \mathrm{CD}(\mathbb{C}) = \mathbb{H}$, which is associative but not commutative. Another iteration gives the octonions $\mathbb{O}$ which is non-assocative, but alternative. The next is the sedenions $\mathbb{S}$ which is non-alternative.

Example 2.2 (*Lie algebra valued forms*)**.** Another interesting situation occurs when $(A,*)$ is a Lie algebra $(\mathfrak{g},[\cdot,\cdot])$. In this case, we will write $\alpha[\wedge]_{\mathfrak{g}}\beta$ or simply $\alpha[\wedge]\beta$ instead of[3] $\alpha \wedge_{[\cdot,\cdot]} \beta$. Lie algebras are not associative and in general are not commutative. This means that the corresponding algebra $\Lambda(P;\mathfrak{g})$ is not associative and not commutative. On the other hand, any Lie algebra is skew-commutative and satisfies the Jacobi identity, so that these properties lift to $\Lambda(P;\mathfrak{g})$, i.e, we have

$$\alpha[\wedge]\beta = (-1)^{kl+1}\beta[\wedge]\alpha$$

and

$$(-1)^{km}\alpha[\wedge](\beta[\wedge]\gamma) + (-1)^{kl}\beta[\wedge](\gamma[\wedge]\alpha) + (-1)^{lm}\gamma[\wedge](\alpha[\wedge]\beta) = 0$$

for any three arbitrarily given $\mathfrak{g}$-valued differential forms α, β, γ of respective degrees k, l, m.

[3] Some authors prefer the somewhat ambiguous notations $[\alpha\wedge\beta]$, $[\alpha,\beta]$ or $[\alpha;\beta]$. The reader should be very careful, because for some authors these notations are also used for the product $[\wedge]_{\mathfrak{g}}$ without the normalizing factor $(k+l)!$. Here we are adopting the notation introduced in [25].

We end with an important remark.

Remark 2.1. In the study of classical differential forms we know that if α is an **odd**-degree form, then $\alpha \wedge \alpha = 0$. This follows directly from the fact that the algebra $(\mathbb{R}, \cdot)$ is commutative. Indeed, in this case $\Lambda(P;\mathbb{R})$ is graded-commutative and so, for α of **odd**-degree k, we have

$$\alpha \wedge \alpha = (-1)^{k^2} \alpha \wedge \alpha = -\alpha \wedge \alpha,$$

implying the condition $\alpha \wedge \alpha = 0$. Notice that the same argument holds for any commutative algebra $(A, *)$. Dually, analogous arguments show that if $(A, *)$ is skew-commutative, then $\alpha \wedge_* \alpha = 0$ for any **even**-degree A-valued form. So, for instance, $\alpha [\wedge]_{\mathfrak{g}} \alpha = 0$ for any given Lie algebra $\mathfrak{g}$.

2.1 Matrix Algebras

In Example 2.2 above, we considered the algebra $\Lambda(P;\mathfrak{g})$ for an arbitrary Lie algebra $\mathfrak{g}$. We saw that, because a Lie algebra is always skew-commutative, the corresponding product $[\wedge]$ is also skew-commutative, but now in the graded sense, i.e,

$$\alpha [\wedge] \beta = (-1)^{kl+1} \beta [\wedge] \alpha$$

for every $\mathfrak{g}$-valued forms α, β. As a consequence (explored in Remark 2.1) we get $\alpha[\wedge]\alpha = 0$ for even-degree forms.

From now on, let us assume that the Lie algebra $\mathfrak{g}$ is not arbitrary, but a subalgebra of $\mathfrak{gl}(k;\mathbb{R})$, for some k. In other words, we will work with Lie algebras of $k \times k$ real matrices. We note that in this situation $\Lambda(P;\mathfrak{g})$ can be endowed with an algebra structure other than $[\wedge]$. In fact, for each k there exists an isomorphism

$$\mu : \Lambda^l(P;\mathfrak{gl}(k;\mathbb{R})) \simeq \mathrm{Mat}_{k\times k}(P;\Lambda^l(P;\mathbb{R}))$$

given by $[\mu(\alpha)]_{ij}(a) = [\alpha(a)]_{ij}$, allowing us to think of every $\mathfrak{g}$-valued l-form α as a $k \times k$ matrix $\mu(\alpha)$ of classical forms. It happens that matrix multiplication gives an algebra structure on

$$\mathrm{Mat}_{k\times k}(P;\Lambda(P;\mathbb{R})) = \bigoplus_l \mathrm{Mat}_{k\times k}(P;\Lambda^l(P;\mathbb{R})),$$

which can be pulled-back by making use of the isomorphism μ, giving a new product on the graded vector space $\Lambda(P;\mathfrak{g})$, which we will denote by the symbol "$\curlywedge$".

Because $\mathfrak{g}$ is a matrix Lie algebra, its Lie bracket is the commutator of matrices, so that we have an identity relating both products. From Proposition 2.1 it follows that we have an analogous identity between the corresponding "wedge products" $\curlywedge$ and $[\wedge]$:

$$\alpha [\wedge] \beta = \alpha \curlywedge \beta - (-1)^{kl} \beta \curlywedge \alpha. \tag{8}$$

This relation clarifies that, while the product $[\wedge]$ is skew-commutative for arbitrary Lie algebras, we **cannot** conclude the same for the product $\curlywedge$, i.e, it is not always true that $\alpha \curlywedge \beta = (-1)^{kl+1} \beta \curlywedge \alpha$. Consequently, *it is **not** true that $\alpha \curlywedge \alpha = 0$ for every even-degree $\mathfrak{g}$-valued form.* It is easy to understand why: recalling that $\curlywedge$ is induced by matrix multiplication, the correspondence between PI's on the algebra $(\mathfrak{g}, *)$ and on the algebra of $\mathfrak{g}$-forms teaches us that *$\curlywedge$ is skew-commutative exactly when the matrix multiplication is skew-commutative*, in other words, iff $\mathfrak{g}$ is a Lie algebra of skew-commutative matrices.

Useful examples are given in the following lemma.

Lemma 2.1. *The condition $\alpha \curlywedge \alpha = 0$ is satisfied for even-degree forms with values into subalgebras*

$$\mathfrak{g} \subset \mathfrak{so}(k_1) \oplus ... \oplus \mathfrak{so}(k_r).$$

Proof. By definition, $\mathfrak{so}(k)$ is the algebra of skew-symmetric matrices, which anti-commute, thus the result holds for $\mathfrak{g} = \mathfrak{so}(k)$ for every k. It is obvious that if $\overline{\mathfrak{g}}$ is some algebra of skew-commuting matrices, then every subalgebra $\mathfrak{g} \subset \overline{\mathfrak{g}}$ is also of skew-commuting matrices. Therefore, the result holds for subalgebras $\mathfrak{g} \subset \mathfrak{so}(k)$. But it is also clear that the direct sum of algebras of anti-commuting matrices remains an algebra of anti-commuting matrices, so that if $\mathfrak{g}_1$ and $\mathfrak{g}_2$ fulfill the lemma, then $\mathfrak{g}_1 \oplus \mathfrak{g}_2$ fulfills too. In particular, the lemma holds for subalgebras $\mathfrak{g} \subset \mathfrak{so}(k_1) \oplus \mathfrak{so}(k_2)$. Finite induction ends the proof. □

Remark 2.2. We could have given a proof of the last lemma without using its invariance under direct sums. In fact, assuming that it holds for subalgebras of $\mathfrak{so}(k)$, notice that for every decomposition $k = k_1 + ... + k_r$ we have a canonical inclusion

$$\mathfrak{so}(k_1) \oplus ... \oplus \mathfrak{so}(k_r) \hookrightarrow \mathfrak{so}(k).$$

While the last lemma is a useful source of examples in which the discussion above applies, let us now give a typical non-example.

Example 2.3. Let H be a linear group of $k \times k$ real matrices. We have a canonical action of H on the additive abelian group $\mathbb{R}^k$, allowing us to consider the semi-direct product $G = \mathbb{R}^k \rtimes H$, of which $H \hookrightarrow G$ can be regarded as subgroup. The same holds in the level of Lie algebras, so that $\mathfrak{g} \simeq \mathbb{R}^k \rtimes \mathfrak{h}$. As a vector space we have $\mathfrak{g}/\mathfrak{h} \simeq \mathbb{R}^k$. Now, because $\mathfrak{g}/\mathfrak{h}$ is abelian, it follows that $\alpha[\wedge]\beta = 0$ for every two $\mathfrak{g}/\mathfrak{h}$-valued forms and from (8) we then conclude that $\alpha \curlywedge \alpha = 0$ for every odd-degree form α. In particular, if $A = \omega + e$ is a reductive Cartan connection for the reduction $H \hookrightarrow \mathbb{R}^k \rtimes H$, then e is a $\mathfrak{g}/\mathfrak{h}$-valued 1-form, so that $e \curlywedge e = 0$. Consequently, the action (5) is trivial. But, for $H = O(3,1)$ this is precisely General Relativity, which is clearly nontrivial. Therefore, at some moment we made a mistake! The problem is that we applied the relation (8) to $\mathbb{R}^n$-valued forms, but the product $\curlywedge$ is **not** defined for such forms. Indeed, it can be defined only for Lie algebras arising from matricial algebras.

The example above teaches us two things:

1. The way it was defined, the product $\curlywedge$ makes sense only for matrix Lie algebras. Particularly, it **does not** makes sense for a semi-direct sum of a matrix algebra with other algebra, so that $\curlywedge$ is **not** the product appearing in the classical EHP action (5). Therefore, if we need to abstract the structures underlying EHP theory we need to show how to extend $\curlywedge$ to a new product $\curlywedge_{\rtimes}$ defined on forms taking values in semi-direct sums. This will be done in Subsection 2.2;

2. Once $\curlywedge_{\rtimes}$ is defined, the corresponding Einstein-Hilbert-Palatini action functional can be trivial. In fact, looking at (6) we see that we have terms like $e \curlywedge_{\rtimes} e \curlywedge_{\rtimes} .... \curlywedge_{\rtimes} e$, where the number of e's depends on the spacetime dimension. So, if we are in an algebraic context in which $\alpha \curlywedge_{\rtimes} \alpha = 0$ for **odd**-degree forms, then $e \curlywedge_{\rtimes} e = 0$ and, consequently, the action will vanishes in arbitrary spacetime dimensions! On the other hand, if $\alpha \curlywedge_{\rtimes} \alpha = 0$ for **even**-degree forms, then $(e \curlywedge_{\rtimes} e) \curlywedge_{\rtimes} (e \curlywedge_{\rtimes} e) = 0$ and the theory is trivial when $n \geq 6$. *This simple idea is the*

core of almost all obstruction results that will be presented here. Notice that what we need is $\curlywedge_{\rtimes}^{k}\alpha = 0$ for some k, which is a nilpotency condition. The correct nilpotency conditions that will be used in our abstract obstruction theorems will be discussed in Subsections 2.3 and 2.4.

2.2 Splitting Extensions

In this section we will see how to define the product $\curlywedge_{\rtimes}$ abstractly. We start by recalling the notion of "extension of an algebraic object". Let **Alg** be a category of algebraic objects, meaning that we have a null object $0 \in \mathbf{Alg}$ (understood as the trivial algebraic entity) and such that every morphism $f : X \to Y$ has a kernel and a cokernel, computed as the pullback/pushout below.

$$\begin{tikzcd} \ker(f) \arrow[r] \arrow[d] & 0 \arrow[d] \\ X \arrow[r, "f"'] & Y \end{tikzcd} \qquad \begin{tikzcd} \operatorname{coker}(f) & 0 \arrow[l] \\ Y \arrow[u] & X \arrow[l, "f"] \arrow[u] \end{tikzcd}$$

Example 2.4. The category **Grp** of groups and the category $\mathbf{Vec}_{\mathbb{R}}$ of real vector space are examples of models for **Alg**. A non-example is **LieAlg**, the category of Lie algebras, since a Lie algebra morphism $f : \mathfrak{h} \to \mathfrak{g}$ may not have a cokernel, which means that the vector space $\mathfrak{g}/f(\mathfrak{h})$ may not have a canonical Lie algebra structure.

If a morphism has a kernel and a cokernel we can then define its *image* as the kernel of its cokernel. Consequently, internal to **Alg** we can talk of an "exact sequence": this is just an increasing sequence of morphisms $f_i : X_i \to X_{i+1}$ such that for every i the kernel of f_{i+1} coincides with the image of f_i. An *extension* is just a short exact sequence, i.e, an exact sequence condensed in three consecutive objects. More precisely, in a short exact sequence as the one below we say that the middle term A was obtained as an *extension of H by E*.

$$0 \longrightarrow E \xrightarrow{\imath} A \xrightarrow{\jmath} H \longrightarrow 0 \tag{9}$$

Remark 2.3. From the last example we conclude that we cannot talk of extensions internal to the category of Lie algebras. This does **not** mean that we cannot define a "Lie algebra extension" following some other approach. In fact, notice that we have a forgetful functor $U : \mathbf{LieAlg} \to \mathbf{Vec}_{\mathbb{R}}$, so that we can define a *Lie algebra extension* as a sequence of maps in **LieAlg** which is exact in $\mathbf{Vec}_{\mathbb{R}}$. The same strategy allows us to enlarge the notion of extension in order to include categories that, a priori, are not models for **Alg**.

Now, a typical example of extensions.

Example 2.5. The Poincaré group is just an extension of the Lorentz group by the translational group. More generally, given a Lie group H endowed with an action $H \times \mathbb{R}^n \to \mathbb{R}^n$ we can form the semi-direct product $\mathbb{R}^n \rtimes H$, which fits into the canonical exact sequence below, where the first map is an inclusion and the second is obtained restricting to pairs $(0, h)$.

$$0 \longrightarrow \mathbb{R}^n \longrightarrow \mathbb{R}^n \rtimes H \longrightarrow H \longrightarrow 0 \tag{10}$$

The extensions in the last example are special: in them we know how to include the initial object H into its extension $\mathbb{R}^n \rtimes H$. An extension with this property is called *splitting*. More precisely, we say that an abstract extension as (9) is *splitting* if there exists a morphism $s : H \to A$ such that $\jmath \circ s = id_H$.

$$\begin{tikzcd} 0 \arrow[r] & E \arrow[r, "\imath"] & A \arrow[r, "\jmath"] & H \arrow[l, bend left, "s"] \arrow[r] & 0 \end{tikzcd} \tag{11}$$

The name comes from the fact that in some good situations, the category **Alg** have a notion of "product", say #, such that a sequence (9) is splitting iff A "splits" as $A \simeq E\#H$. For instance, if **Alg** is an abelian category, then the Splitting Lemma shows that such a product # is just the coproduct $\oplus$, i.e, the "direct sum". For general groups or Lie algebras, # is the corresponding notion of semi-direct product/sum.

Now, assume that the ambient category **Alg** is actually monoidal, meaning that we have a fixed bifunctor $\otimes : \mathbf{Alg} \times \mathbf{Alg} \to \mathbf{Alg}$ and an object $1 \in \mathbf{Alg}$ such that $\otimes$ is associative (up to natural isomorphisms) and has 1 as a neutral object (also up to natural isomorphisms). We can then talk of *monoids in* **Alg**. These are objects $X \in \mathbf{Alg}$ endowed with morphisms $* : X \otimes X \to X$ and $e : 1 \to X$ which satisfy the associavitity-type and neutral element-type diagrams.

The main point is that we can use sections morphisms to transfer (to pullback) monoid structures. Indeed, if $f : X \to Y$ is a morphism with section $s : Y \to X$, then for any monoid structure $(*, e)$ in Y we get a corresponding monoid structure $(*', e')$ in Y with $*'$ as shown below and $e' = s \circ e$. In particular, if in an splitting extension (11) H is a monoid, then we can use the section $s : H \to A$ to get a monoid structure in A.

$$\begin{tikzcd} X \otimes X \arrow[r, "f \otimes f"] \arrow[rrr, bend right, "*'"'] & Y \otimes Y \arrow[r, "*"] & Y \arrow[r, "s"] & X \end{tikzcd} \tag{12}$$

In the case when **Alg** admits a product # characterizing splitting extensions it is natural to write $*_{\#}$ and $e_{\#}$ (instead of $*'$ and e') to denote the monoid structure induced on an extension.

Example 2.6. Let us take the category $\mathbf{Vec}_{\mathbb{R}}$ endowed with the monoidal structure given by the tensor product $\otimes_{\mathbb{R}}$. Its monoid objects are just real algebras. From the last paragraphs, if A is a splitting extension of a vector space H by another vector space E and H is an algebra, say with product $*$, then A is automatically an algebra with product $*_{\oplus}$. Therefore, for a given manifold P we will have not only a wedge product $\wedge_*$ between H-valued forms, but also a product $\wedge_{*_{\oplus}}$. In the very particular case when H is a matrix algebra, recall that $\wedge_*$ is denoted by $\curlywedge$, which motivate us to denote the corresponding $\wedge_{*_{\oplus}}$ by $\curlywedge_{\oplus}$. It is exactly this kind of multiplication that appears in (5).

2.3 (k, s)-Nil Algebras

In the last subsection we showed how to build the products that will enter in the abstract definition of EHP theory. Here we will discuss the nilpotency conditions that will be imposed into these products in order to get obstruction theorems for the corresponding EHP theory.

We start by recalling that an element $v \in A$ in an algebra $(A, *)$ has *nilpotency degree* s if $v^s \neq 0$, but $v^{s+1} = 0$, where $v^i = v * ... * v$. In turn, the nilpotency degree of A is the minimum over the nilpotency degree of its elements. An algebra with non-zero nilpotency degree is called a *nil algebra*. This "nil" property can also be characterized as a PI: A is nil iff for some $s \neq 0$ the polynomial $p_1(x) = x^s$ does not vanish identically, but x^{s+1} does.

The first nontrivial examples of nil algebras are the anti-commutative algebras, which include Lie algebras, for which the nilpotency degree is $s = 1$. For such kind of objects we usually consider the more restrictive notion of *nilpotent algebra*. Indeed, we say that A is *nilpotent* of degree s if not only $p_1(x) = x^{s+1}$ vanishes, but also

$$p_{s+1}(x_1, ..., x_{s+1}) = x_1 \cdot .... \cdot (x_{s-1} \cdot (x_s \cdot x_{s+1}))$$

and any other polynomial obtained from p_{s+1} by changing the ordering of the parenthesis. In general, being nilpotent is much stronger than being nil. For associative algebras, on the other hand, such concepts coincide [**?**, **?**]. A fact more easy to digest is that an anti-commutative and associative algebra A is nilpotent iff it is an *Engel's algebra*, in the sense that the PI's

$$p_{s,1}(x, y) = p_{s+1}(x, ..., x, y) = 0$$

are satisfied for any "p_{s+1}-type" polynomial. Due to Engel's theorem, other important examples of Engel's algebras are the Lie algebras.

From Proposition 2.1, if an algebra A is nil or nilpotent, then the corresponding graded-algebra of A-valued exterior forms, with the product $\wedge_*$, have the same properties, but in the **graded** sense. Notice that $x^{s+1} = 0$ iff $x^{s+1} = -x^{s+1}$, so that the graded-nil condition becomes $x^{s+1} = (-1)^{k^{s+1}+1}x^{s+1}$, where $k = \deg x$, which is nontrivial only when k is even. The conclusion is the following: *if $(A, *)$ is nil with nilpotency degree $s > 0$, then $\Lambda^{\text{even}}(P; A)$ is graded-nil with the same degree.*

The next example sets Lemma 2.1 in terms of this new language.

Example 2.7. If A is anti-commutative, then it is nil with degree $s = 1$, which implies that even A-valued forms are also nil with degree $s = 1$, i.e, $\alpha \wedge_* \alpha = 0$ for even forms. If A and B are nil of respective degrees r and s, then the direct sum $A \oplus B$ is also nil, with degree given by $\min\{r, s\}$. Subalgebras of nil algebras are also nil algebras. Consequently,

$$A \hookrightarrow A_1 \oplus ... \oplus A_r$$

is nil when each A_i is nil. This is exactly a generalization of Lemma 2.1.

The same kind of discussion applies to the nilpotent property. The analysis is easier when A is skew-commutative and satisfies an associativity or Jacobi identity, because we can work with $p_{s,1}$ instead of p_{s+1}. Noting that $p_{s,1}$ is a PI iff $x^s \cdot y = -x^s \cdot y$ we conclude that the graded-nilpotent property is described by

$$x^s \cdot y = (-1)^{k^s l+1}(x^s \cdot y),$$

where $k = \deg x$ and $l = \deg y$. This condition is nontrivial iff $k^s l + 1$ is odd, i.e, iff $k^s l$ is even, which implies that k and l have the same parity. Summarizing: *if A is nilpotent of degree s, then $\Lambda^{\text{odd}}(P; A)$ or $\Lambda^{\text{even}}(P; A)$ is graded-nilpotent with the same degree.*

Notice that the nil property gives nontrivial conditions only on even-degree forms. On the other hand, while the nilpotency property can be used to give nontrivial conditions on even-degree or odd-degree forms, it is too strong for most purposes. This leads us to define an intermediary concept of *(k, s)-nil algebra* containing both nil and nilpotent algebras as particular examples. Indeed, given integers $k, s > 0$, we will say that an algebra $(A, *)$ is *(k, s)-nil* if in the corresponding graded-algebra $\Lambda(P; A)$ every A-valued k-form has nilpotency degree s.

2.4 (k,s)-Solv Algebras

In this subsection we will explore more examples of (k,s)-nil algebras. In the study of Lie algebras (as well as of other kinds of algebras), there is a concept of *solvable algebra* which is closely related to the concept of *nilpotent algebra*. Indeed, to any algebra A we can associate two decreasing sequences A^n and $A^{(n)}$ of ideals, respectively called the *lower central series* and the *derived series*, inductively defined as follows:

$$A^0 = A, \quad A^k = \bigoplus_{i+j=k} A^i * A^j \quad \text{and} \quad A^{(0)} = A, \quad A^{(k)} = A^{(k-1)} * A^{(k-1)},$$

where, for given subsets $X, Y \subset A$, by $X * Y$ we mean the ideal generated by all products $x * y$, with $x \in X$ and $y \in Y$. Clearly, the polynomials like p_{s+1} are PI's for A iff $A^{s+1} = 0$. So, A is nilpotent of degree s iff its lower central series stabilizes in zero after $s+1$ steps. Analogously, we say that A is *solvable of solvability degree s* if its derived series stabilizes in zero after $s+1$ steps.

We notice that $A^{(k+1)}$ can be regarded not only as an ideal of A, but indeed as an ideal of $A^{(k)}$. Furthermore, the quotient $A^{(k)}/A^{(k+1)}$ subalgebra is always commutative [**?**]. Starting with $A^{(1)}$ we get the first exact sequence (the first line) below. Because we are working over fields, the quotient is a free module and then the sequence splits, allowing us to write $A \simeq A^{(1)} \oplus A/A^{(1)}$. Inductively we then get $A \simeq A'_s \oplus ... \oplus A'_0$, where $A'_i := A^{(i)}/A^{(i+1)}$. Summarizing: *as a vector space, a solvable algebra can be decomposed into a finite sum of spaces each of them endowed with a structure of commutative algebra.*

$$\begin{tikzcd}
0 \arrow[r] & A^{(s)} \arrow[r] \arrow[d] & A^{(s-1)} \arrow[r] \arrow[d] & A/A^{(1)} \arrow[r] \arrow[d] & 0 \\
 & \vdots \arrow[d] & \vdots \arrow[d] & \vdots \arrow[d] & \\
0 \arrow[r] & A^{(2)} \arrow[r] \arrow[d] & A^{(1)} \arrow[r] \arrow[d] & A^{(1)}/A^{(2)} \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & A^{(1)} \arrow[r] & A \arrow[r] & A/A^{(1)} \arrow[r] & 0
\end{tikzcd}$$

With the remarks above in our minds, let us prove that any solvable algebra is "almost" a (k,s)-nil algebra.

Proposition 2.2. *Let $(A,*)$ be a solvable algebra and let α be an A-valued k-form in a smooth manifold P. If k is odd and α is pointwise injective, then $\alpha \wedge_* \alpha = 0$.*

Proof. Let $A \simeq A'_s \oplus ... \oplus A'_0$ be the decomposition above. Given a k-form α, assume that α_a is injective for every $a \in P$. Then, by the isomorphism theorem, α_a induces an isomorphism from its domain on its image. Notice that the subespace $\operatorname{img}(\alpha) \subset A$ can be decomposed as $\overline{A}_s \oplus ...\overline{A}_0$, where $\overline{A}_i = \operatorname{img}(\alpha) \cap A'_i$. Let V_i be the preimage $\alpha_a^{-1}(\overline{A}_i)$, so that the domain of α_a decomposes as $V'_s \oplus ... \oplus V'_1$, allowing us to write $\alpha_a = \alpha_a^s + ... + \alpha_a^0$. We assert that $\alpha \wedge_* \alpha = 0$. From the bilinearity of $\wedge_*$ it is enough to verify that $\alpha_a^i \wedge_* \alpha_a^j = 0$ for every i, j and $a \in P$. If $i \neq j$ this is immediate because α_a^i and α_a^j are nonzero in different subspaces. So, let us assume $i = j$. In this case, since A is solvable, the algebras A'_i are commutative, so that by the discussion in Section 2 we have $\beta \wedge_* \beta = 0$ for every A'_i-valued odd-degree form and, in particular, $\alpha^i_a \wedge_* \alpha^i_a = 0$. □

Up to the injectivity hypothesis, the last proposition is telling us that solvable algebras are $(k,1)$-nil algebras for every k odd. So, we can say that solvable algebras are $(k,1)$-nil "on the class of injective forms". This motivates us to define the following: given a class $\mathcal{C}^k \subset \Lambda^k(P;A)$ of A-valued k-forms, we say that A is a *$(\mathcal{C}^k, s)$-nil algebra* if any A-valued form belonging to $\mathcal{C}^k$ has nilpotency degree s.

We are now in position of generalizing the last proposition. Indeed, the structure of its proof is very instructive in the sense that it can be easily abstracted by noticing that the "solvable" hypothesis over A was used only to get a decomposition of $A \simeq A'_s \oplus ... \oplus A'_0$ in terms of commutative algebras. The commutativity, in turn, was important only to conclude $\alpha \wedge_* \alpha = 0$. Therefore, using the same kind of proof we immediately obtain an analogous result if we consider algebras A endowed with a vector space decomposition $A \simeq A_s \oplus ... \oplus A_0$ where each A_i is a (k,s)-nil algebra. We will call this kind of algebras *(k,s)-solv algebras*, because they generalize solvable algebras in the same sense as (k,s)-nil algebras generalize nil and nilpotent algebras.

Summarizing, in this new terminology we have the following result.

Proposition 2.3. *Every (k,s)-solv algebra is $(\mathcal{C}^k,s)$-nil over the class of pointwise injective forms.*

Remark 2.4. Exactly as nilpotent algebras are always solvable, (k,s)-nil algebras are (k,s)-solv. The last proposition shows that the reciprocal is almost true. On the other hand, we could have analogously defined $(\mathcal{C}^k,s)$-solv algebras and, in this case, if $\mathcal{C}^k$ is contained in the class of pointwise injective forms, the last proposition would be rephrased as stating an equivalence between the concepts of $(\mathcal{C}^k,s)$-solv algebras and $(\mathcal{C}^k,s)$-nil algebras.

Remark 2.5. In Subsection 4.2 we will work with special vector subspaces of a given algebra. Let us seize the opportunity to introduce them. Given an arbitrary algebra A, we say that a vector subspace $V \subset A$ is a *(k,s)-nil subspace of* A if every V-valued k-form has nilpotency degree s when regarded as a A-valued form. Similarly, we say that V is a *(k,s)-solv subspace* if it decomposes as a sum of (k,s)-nil subspaces. When $A = \oplus_m A^m$ is $\mathfrak{m}$-graded, there are other kind of subespaces $V \subset A$ that can be introduced. For instance, we say that V is *graded (k,s)-solv* if each $V_m = V \cap A_m$ is a (k,s)-solv subspace. In any subspace V of a $\mathfrak{m}$-graded algebra we get a corresponding grading by $V \simeq \oplus_m V_m$. So, any V-valued form α can be written as $\alpha = \sum_m \alpha^m$. We say that V is *weak (k,s)-nilpotent* if for every k-form α, any polynomial $p_{s+1}(\alpha^{m_1}, ..., \alpha^{s+1})$ vanishes. Similarly, we say that V is *weak (k,s)-solvable* if it decomposes as a sum of *weak (k,s)-nilpotent* subspaces.

2.5 Functorial Algebra Bundle System

Now, let us discuss the last ingredient before applying Geometric Obstruction Theory to EHP theories. In previous subsections, P was an arbitrary smooth manifold. Let us now assume that it is the total space of a G-bundle $\pi : P \to M$. EHP theories (which are our aim) are not about forms on the total space P, but about forms on the base manifold M. So, we need some process allowing us to replace A-valued forms in P by forms in M with coefficients in some other bundle, say E_A. More precisely, we are interested in rules assigning to every pair (P,A) a corresponding algebra bundle E_A, whose typical fiber is A, in such a way that there exists a graded-subalgebra $S(P,A) \subset \Lambda(P;A)$ and a canonical morphism $\jmath : S(P,A) \to \Lambda(M;E_A)$.

It is more convenient to think of this in categorical terms. Let $\mathbf{Alg}_{\mathbb{R}}$ be the category of real finite-dimensional[4] algebras, $\mathbb{Z}\mathbf{Alg}_{\mathbb{R}}$ be the category of $\mathbb{Z}$-graded real algebras and, given a manifold M, let $\mathbf{Bun}_M$ and $\mathbf{Alg}_{\mathbb{R}}\mathbf{Bun}_M$ denote the categories of bundles and of $\mathbb{R}$-algebra bundles over M, respectively. As in the first diagram below, we have two canonical functors, which assign to each pair (A,P) the corresponding superalgebra of A-valued forms in P, and to each algebra bundle E over M the graded algebra of E-valued forms in M. We also have the projection $(P,A)\mapsto A$. So, our problem of passing from forms in P to forms in M could be solved by searching for a functor F making commutative the second diagram below.

$$\begin{tikzcd} & \mathbf{Alg}_{\mathbb{R}}\mathbf{Bun}_M \arrow[d, "\Lambda(M;-)"] \\ \mathbf{Bun}_M \times \mathbf{Alg}_{\mathbb{R}} \arrow[r, "\Lambda(-;-)"'] & \mathbb{Z}\mathbf{Alg}_{\mathbb{R}} \end{tikzcd} \qquad \begin{tikzcd} \mathbf{Alg}_{\mathbb{R}} \arrow[r, dashed, "E_-"] & \mathbf{Alg}_{\mathbb{R}}\mathbf{Bun}_M \arrow[d, "\Lambda(M;-)"] \\ \mathbf{Bun}_M \times \mathbf{Alg}_{\mathbb{R}} \arrow[u, "\pi_1"] \arrow[r, "\Lambda(-;-)"'] & \mathbb{Z}\mathbf{Alg}_{\mathbb{R}} \end{tikzcd}$$

But this would be a stronger requirement; for instance, it would imply the equality of two functors, a fact that can always be weakened by making use of natural transformations. Therefore, we could search for functors E_- endowed with natural transformations $\jmath$, as in the diagram below. This condition would remain stronger than we need: it requires that for **any** bundle $P\to M$ and for **any** algebra A we have a canonical algebra morphism $\Lambda(P;A)\to\Lambda(M;E_A)$. We would like to include rules that are defined only for certain classes of bundles and algebras. This leads us to work in subalgebras $\mathbf{C}$ of $\mathbf{Bun}_M\times\mathbf{Alg}_{\mathbb{R}}$ as in the second diagram below.

$$\begin{tikzcd} \mathbf{Alg}_{\mathbb{R}} \arrow[r, dashed, "E_-"] & \mathbf{Alg}_{\mathbb{R}}\mathbf{Bun}_M \arrow[d, "\Lambda(M;-)"] \\ \mathbf{Bun}_M \times \mathbf{Alg}_{\mathbb{R}} \arrow[u, "\pi_1"] \arrow[r, "\Lambda(-;-)"'] \arrow[ur, Rightarrow, dashed, "\jmath"] & \mathbb{Z}\mathbf{Alg}_{\mathbb{R}} \end{tikzcd} \qquad \begin{tikzcd} \mathbf{Alg}_{\mathbb{R}} \arrow[r, dashed, "E_-"] & \mathbf{Alg}_{\mathbb{R}}\mathbf{Bun}_M \arrow[d, "\Lambda(M;-)"] \\ \mathbf{C} \arrow[u, "\pi_1"] \arrow[r, "\Lambda(-;-)"'] \arrow[ur, Rightarrow, dashed, "\jmath"] & \mathbb{Z}\mathbf{Alg}_{\mathbb{R}} \end{tikzcd}$$

But, once again, we should consider weaker conditions. Indeed, the above situation requires that for $(P,A)\in\mathbf{C}$ the algebra morphism $\Lambda(P;A)\to\Lambda(M;E_A)$ is "canonical" in the entire algebra $\Lambda(P;A)$. It may happen that Λ be "canonical" only in some subalgebra $S(P;A)\subset\Lambda(P;A)$, meaning that it is first defined in $S(P;A)$ and then trivially extended to $\Lambda(P;A)$. Therefore, the correct approach seems to be to replace Λ by another functor S endowed with an objectwise injective transformation $\imath : S\Rightarrow\Lambda$, as in the following diagram.

$$\begin{tikzcd} \mathbf{Alg}_{\mathbb{R}} \arrow[r, dashed, "E_-"] & \mathbf{Alg}_{\mathbb{R}}\mathbf{Bun}_M \arrow[d, "\Lambda(M;-)"] \\ \mathbf{C} \arrow[u, "\pi_1"] \arrow[r, dashed, bend left, "S(-,-)"{name=S}] \arrow[r, bend right, "\Lambda(-;-)"'{name=L}] & \mathbb{Z}\mathbf{Alg}_{\mathbb{R}} \arrow[from=S, to=1-2, Rightarrow, dashed, "\jmath"] \arrow[from=S, to=L, Rightarrow, dashed, "\imath"] \end{tikzcd}$$

In sum, given a manifold M, *the last diagram describes, in categorical terms, the transition between algebra-valued forms in bundles over M and algebra bundle-valued forms in M*. The input

[4] Actually, we can work in the infinite-dimensional setting, but in this case we have to take many details into account. For instance, we would need to work with topological algebras, bounded linear maps, etc.

needed to do this transition corresponds to the dotted arrows in the last diagram. We will say that they define a *functorial algebra bundle system* (FABS) *for the manifold* M. Concretely, a FABS for M consists of

1. a category $\mathbf{C}$ of pairs (P, A), where $P \to M$ is a bundle and A is a real algebra;
2. a functor E_{-} assigning to any algebra $A \in \mathbf{C}$ a corresponding algebra bundle $E_A \to M$ whose typical fiber is A;
3. a functor $S(-;-)$ that associates an algebra to each pair $(P, A) \in \mathbf{C}$;
4. natural transformations $\imath : S(-;-) \Rightarrow \Lambda(-;-)$ and $\xi : S(-;-) \Rightarrow \Lambda(M;-)$ such that $\imath$ is objectwise injective.

Such systems always exists, as showed by the next examples. The fundamental properties and constructions involving FABS will appear in a work under preparation [27].

Example 2.8 (*trivial case*)**.** Starting with any subcategory $\mathbf{C}$ of pairs (P, A) that contains only the trivial algebra, up to natural isomorphisms there exists a single E_{-}: the constant functor at the trivial algebra bundle $M \times 0 \to M$. Noting that $\Lambda(M; M \times 0) \simeq 0$ since the trivial algebra is a terminal object in $\mathbf{Alg}_{\mathbb{R}}$, independently of the choice of S and $\imath : S \Rightarrow \Lambda$ there is only one $\jmath : S \Rightarrow \Lambda(M;-)$: the trivial one.

Example 2.9 (*almost trivial case*)**.** The last situation has the defect that it can be applied only for subcategories $\mathbf{C}$ whose algebraic part is trivial. It is easy, on the other hand, to build FABS for arbitrarily given algebras. Indeed, let $\mathbf{C}$ be defined by pairs (P, A), where A is an arbitrary algebra, but P is the trivial principal bundle $M \times GL(A) \to M$. Putting E_A as the trivial algebra bundle $M \times A \to A$, we have a 1-1 correspondence between A-valued forms in P and forms in M with values in E_A. Therefore, we actually have a map $\jmath : \Lambda(P; A) \to \Lambda(M; E_A)$ leading us to take $S = \Lambda$ and $\imath = \jmath$.

Example 2.10 (*standard case*)**.** In the first example we considered $\mathbf{C}$ containing arbitrary bundles, but we paid the price of working only with trivial algebras. In the second example we were faced with a dual situation. A middle term can be obtained by working with pairs (P, A), where $P \to M$ is a G-bundle whose group G becomes endowed with a representation $\rho : G \to GL(A)$. In that case we define E_{-} as the rule assigning to each A the corresponding associated bundle $P \times_\rho A$. The functor S is such that $S(P, A)$ is the algebra $\Lambda_\rho(P; A)$ of *ρ-equivariant A-valued forms α in P*, i.e, of those satisfy the equation $R_g^* \alpha = \rho(g^{-1}) \cdot \alpha$, where here $R : G \times P \to P$ is the canonical free action characterizing P as a principal G-bundle. This algebra of ρ-equivariant forms naturally embeds into $\Lambda(P; A)$, giving $\imath$. Finally, it is a standard fact [24] that each ρ-equivariant A-valued form on P induces an $P \times_\rho A$-valued form on M, defining the transformation ξ. This is the standard approach used in the literature, so that we will refer to it as the *standard FABS*.

Example 2.11 (*canonical case*)**.** There is an even more canonical situation: that obtained from the latter by considering only associative algebras underlying matrix Lie algebras. Indeed, in this case we have also a distinguished representation $\rho : G \to GL(\mathfrak{g})$, given by the adjoint representation. The corresponding bundles $E_{\mathfrak{g}}$ correspond to what is known as the *adjoint bundles*, leading us to say that this is the *adjoint FABS*.

3 Concrete Obstructions

After the algebraic prolegomena developed in the previous section, we are ready to introduce and study EHP theories in the general setting. We will start in Subsection 3.1 by considering a very simple framework, modeled by inclusions of linear groups, where our first fundamental obstruction theorem is obtained. At each following subsection the theory will be redefined (in the direction of the full abstract context) in such a way that the essence of the obstruction theorem will remain the same.

Thus, in Subsection 3.2 we define a version of EHP for reductions $H \hookrightarrow G$ such that G is not a linear group, but a splitting extension of H. We then show that, as desired, the fundamental obstruction theorem remains valid, after minor modifications.

Independently, if one is interested in matrix EHP theories or EHP theories arising from splitting extensions, we also work with reductive Cartan connections. These are 1-forms $\nabla : TP \to \mathfrak{g}$ that decompose as $\nabla = e + \omega$, where $e : TP \to \mathfrak{g}/\mathfrak{h}$ is a pointwise isomorphism and $\omega : TP \to \mathfrak{h}$ is an H-connection. What happens if we forget the pointwise isomorphism hypothesis on e? In Subsection 3.3 we discuss the motivations to do this and we show that the fundamental obstruction theorem not only holds, but becomes stronger.

Finally, in Subsection 3.5 we show that if we restrict our attention to connections $\nabla = e + \omega$ such that ω is torsion-free, then we get, as a consequence of Berger's classification theorem, a new obstruction result which is independent of the fundamental obstruction theorem. In particular, this new result implies topological obstructions to the spacetime in which the EHP theory is being described.

3.1 Matrix Gravity

Let us start by considering a smooth G-bundle $P \to M$ over an orientable smooth manifold M, with G a linear group, and fix a group structure reduction $H \hookrightarrow G$ of P. Given an integer $k > 0$ and $\Lambda \in \mathbb{R}$, we define the *homogeneous* and the *inhomogeneous linear* (or *matrix*) *Hilbert-Palatini form* of degree k of a reductive Cartan connection $\nabla = e + \omega$ in P, relative to the group structure reduction $H \hookrightarrow G$, as

$$\alpha_k = e \curlywedge ... \curlywedge e \curlywedge \Omega \quad \text{and} \quad \alpha_{k,\Lambda} = \alpha_k + \frac{\Lambda}{(k-1)!} e \curlywedge ... \curlywedge e, \tag{13}$$

respectively. In α_k the term e appears $(k-2)$-times, while in right-hand part of $\alpha_{k,\Lambda}$ it appears k-times. Because we are working with matrix Lie algebras we have the adjoint FABS, so that these $\mathfrak{g}$-valued forms correspond to k-forms in M with values in the adjoint bundle $P \times_{\mathrm{ad}} \mathfrak{g}$, respectively denoted by $\jmath(\alpha_k)$ and $\jmath(\alpha_{k,\Lambda})$.

We define the *homogeneous* and the *inhomogeneous linear EHP theories* in P with respect to $H \hookrightarrow G$ as the classical field theories whose spaces of configurations are the spaces of reductive Cartan connections $\xi = e + \omega$ and whose action functionals are respectively given by

$$S_n[e,\omega] = \int_M \mathrm{tr}(\jmath(\alpha_n)) \quad \text{and} \quad S_{\Lambda,n}[e,\omega] = \int_M \mathrm{tr}(\jmath(\alpha_{n,\Lambda})). \tag{14}$$

Warning: The expression "linear EHP" is used here to express the fact that the EHP action is realized in a geometric background defined by linear groups. It **does not** mean that the equations of motion were linearized or something like this.

Remark 3.1. A priori, H is an arbitrary subgroup of G, which implies that $\mathfrak{h}$ is a Lie subalgebra of $\mathfrak{g}$. It would be interesting to think of $\mathfrak{g}$ as the Lie algebra extension of some $\mathfrak{r}$ by $\mathfrak{h}$. This makes sense only when $\mathfrak{h} \subset \mathfrak{g}$ is not just a subalgebra, but an ideal, which is true iff $H \subset G$ is a **normal** subgroup.

With the remark above in mind, we can state our first obstruction theorems. In them we will consider theories for reductions $H \hookrightarrow G$ fulfilling one of the following conditions:

(C1) the subgroup $H \subset G$ is normal and, as a matrix algebra, $\mathfrak{g}$ is an splitting extension by $\mathfrak{h}$ of a (k,s)-nil algebra $\mathfrak{r}$.

(C2) as a matrix algebra, $\mathfrak{g}$ is a (k,s)-nil algebra.

Theorem 3.1. *Let M be an n-dimensional spacetime and $P \to M$ be a G-bundle, endowed with a group reduction $H \hookrightarrow G$ fulfilling (C1) or (C2). If $n \geq k+s+1$, then the linear inhomogeneous EHP theory equals the homogeneous ones. If $n \geq k+s+3$, then both are trivial.*

Proof. Assume (C1). As vector spaces we can write $\mathfrak{g} \simeq \mathfrak{g}/\mathfrak{h} \oplus \mathfrak{h}$ and, because $\mathfrak{g}$ is a Lie algebra extension of $\mathfrak{r}$ by $\mathfrak{h}$, it follows that $\mathfrak{g}/\mathfrak{h} \simeq \mathfrak{r}$. How the extension splitting, $\mathfrak{r}$ is a subalgebra of $\mathfrak{g}$ and $\mathfrak{g}/\mathfrak{h} \simeq \mathfrak{r}$ is indeed a Lie algebra isomorphism, So, $\mathfrak{g}/\mathfrak{h}$ can be regarded as a (k,s)-nil algebra and, therefore, $\curlywedge^{s+1}\alpha = 0$ for every $\mathfrak{g}/\mathfrak{h}$-valued k-form in P. If $\nabla = e + \omega$ is a reductive Cartan connection in P relative to $H \hookrightarrow G$, then e is a $\mathfrak{g}/\mathfrak{h}$-valued 1-form in P and, because $\mathfrak{g}/\mathfrak{h}$ is a subalgebra of $\mathfrak{g}$, $\curlywedge^k e$ is a $\mathfrak{g}/\mathfrak{h}$-valued k-form. Consequently,

$$(\curlywedge^{k+s+1} e) \curlywedge \alpha = 0 \tag{15}$$

for any $\mathfrak{g}$-valued form α. In particular, for

$$\alpha = \frac{\Lambda}{(n-1)!} \curlywedge^{n-(k+s+1)} e$$

(15) is precisely the inhomogeneous part of $\alpha_{\Lambda,n}$. Therefore, for every $n \geq k+s+1$ we have $\alpha_{\Lambda,n} = \alpha_n$ and, consequently, $\jmath(\alpha_k) = \jmath(\alpha_{\Lambda,n})$, implying $S_n[e,\omega] = S_{n,\Lambda}[e,\omega]$ for any Cartan connection $\nabla = e + \omega$, and then $S_n = S_{n,\Lambda}$. On the other hand, for

$$\alpha = (\curlywedge^{n-(k+s+1)+2} e) \curlywedge \Omega$$

we see that (15) becomes exactly α_n. Therefore, when $n \geq k+s+3$ we have $\alpha_n = 0$, implying (because $\jmath$ is linear) $\jmath(\alpha_n) = 0$ and then $S_n = 0$. But $k+s+3 > k+s+1$, so that $S_{n,\Lambda} = 0$ too. This ends the proof when the reduction $H \hookrightarrow G$ fulfills condition (C1). If we assume (C2) instead of (C1), we can follow exactly the same arguments, but now thinking of e as a $\mathfrak{g}$-valued 1-form. □

Recall that, as remarked in Section 2, examples of (k,s)-nil algebras $\mathfrak{g}$ include nil and nilpotent algebras. In particular, from Lemma 2.1 we see that subalgebras of $\mathfrak{so}(k_1) \oplus ... \oplus \mathfrak{s}(k_r)$ are (k,s)-nil for every (k,s) such that k is even and $s \geq 1$. This leads us to the following corollary:

Corollary 3.1. *For subalgebras as above, in a spacetime of dimension $n \geq 4$, the inhomogeneous and the homogeneous linear EHP are equal. If $n \geq 6$ both are trivial.*

3.2 Gravity Arising From Extensions

Now, let us move on to a slightly more abstract situation: when G is not a linear group, but a splitting extension of a given linear group H, i.e, we will deal with group reductions $H \hookrightarrow \mathbb{R}^k \rtimes H$. First of all, notice that if $P \to M$ is a bundle structured over $H \subset GL(k;\mathbb{R})$, then we can always extend its group structure to $\mathbb{R}^k \rtimes H$. Indeed, up to isomorphisms this bundle is classified by a map $f : M \to BH$ and getting a group extension is equivalent to lifting f as shown below. It happens that this lifting actually exists, since we are working with splitting extensions (second diagram). It is in this context that we will now internalize EHP theories.

$$
\begin{tikzcd}
& \mathbb{R}^k \rtimes H \arrow[d, "\jmath"] \\
M \arrow[r, "f"'] \arrow[ur, dashed] & H
\end{tikzcd}
\qquad\qquad
\begin{tikzcd}
& \mathbb{R}^k \rtimes H \arrow[d, "\jmath"'] \\
M \arrow[r, "f"'] \arrow[ur, dashed, "s \circ f"] & H \arrow[u, bend right, "s"']
\end{tikzcd}
$$

Thus, given a linear group H, consider bundles $P \to M$ endowed with group reductions $H \hookrightarrow \mathbb{R}^k \rtimes H$. As previously, fixed $k > 0$ and $\Lambda \in \mathbb{R}$ we define *homogeneous* and *inhomogeneous extended-linear Hilbert-Palatini form* of reductive Cartan connections $\nabla = e + \omega$ in P as

$$
\alpha_k = e \curlywedge_{\rtimes} \dots \curlywedge_{\rtimes} e \curlywedge_{\rtimes} \Omega \quad \text{and} \quad \alpha_{k,\Lambda} = \alpha_k + \frac{\Lambda}{(k-1)!} e \curlywedge_{\rtimes} \dots \curlywedge_{\rtimes} e. \tag{16}
$$

Furthermore, fixed a FABS, the corresponding *extended-linear EHP theories* are defined by (14). At first sight, the only difference between the "extended-linear" and the "linear" theories is that we now consider the induced product $\curlywedge_{\rtimes}$ instead of $\curlywedge$. However, in the context of Geometric Obstruction Theory, this replacement makes a fundamental difference. For instance, the condition (C1) used to get Theorem 3.1 no longer makes sense, because $H \hookrightarrow \mathbb{R}^k \rtimes H$ generally is **not** a normal subgroup. Immediate substitutes for (C1) and (C2) are

(C1') as an associative algebra $(\mathfrak{h}, \curlywedge_{\rtimes})$ is (k,s)-nil;

(C2') as an associative algebra, $(\mathbb{R}^k \rtimes \mathfrak{h}, \curlywedge_{\rtimes})$ is (k,s)-nil.

But, differently from (C1) and (C2), these new conditions are intrinsically related. It is obvious that (C2)' implies (C1)', because $\mathfrak{h}$ is a subalgebra of $\mathbb{R}^k \rtimes \mathfrak{h}$. The reciprocal is also valid, as shown in the next lemma.

Lemma 3.1. *Let $(A,*)$ be an algebra, B a vector space and $f : A \to B$ a linear map that admits a section $s : B \to A$. In this case, if $(A,*)$ is (k,s)-nil, then $(B,*')$ is too, where $*'$ is the pulled-back multiplication (12).*

Proof. Recall that any algebra X induces a corresponding graded-algebra structure in $\Lambda(P;X)$ according to (7). Due to the functoriality of $\Lambda(P;-)$, we then get the commutative diagram below, where the horizontal rows are just (7) for the algebras $(A,*)$ and $(B,*')$, composed with the diagonal map. The commutativity of this diagram says just that $\wedge^2_{*'}\alpha = \wedge^2_{*} f(\alpha)$ for every $\alpha \in \Lambda(P;B)$. From the same construction we get, for each given $s \geq 2$, a commutative diagram that implies

$\wedge^s_{*'}\alpha = \wedge^s_* f(\alpha)$. Therefore, if $(A,*)$ is (k,s)-nil it immediately follows that $(B,*')$ is (k,s)-nil too.

$$\begin{tikzcd}
\Lambda(P;B) \arrow[r, "\Delta"] \arrow[d, shift left, "\Lambda f"] & \Lambda(P;B)\otimes\Lambda(P;B) \arrow[r] \arrow[d, shift left, "(\Lambda f)\otimes(\Lambda f)"] & \Lambda(P;B\otimes B) \arrow[r] \arrow[d, shift left, "\Lambda(f\otimes f)"] & \Lambda(P;B) \arrow[d, shift left, "\Lambda f"] \\
\Lambda(P;A) \arrow[r, "\Delta"] \arrow[u, shift left, "\Lambda s"] & \Lambda(P;A)\otimes\Lambda(P;A) \arrow[r] \arrow[u, shift left, "(\Lambda s)\otimes(\Lambda s)"] & \Lambda(P;A\otimes A) \arrow[r] \arrow[u, shift left, "\Lambda(s\otimes s)"] & \Lambda(P;A) \arrow[u, shift left, "\Lambda s"]
\end{tikzcd}$$

□

As a consequence of the lemma above, the version of Theorem (3.1) for extended-linear EHP is the following:

Theorem 3.2. *Let M be an n-dimensional spacetime and $P\to M$ be a H-bundle, where H is a linear group such that $(\mathfrak{h},\curlywedge)$ is (k,s)-nil. If $n\geq k+s+1$, then the inhomogeneous extended-linear EHP theory equals the homogeneous ones. If $n\geq k+s+3$, then both are trivial.*

Proof. From the last lemma we can assume that $(\mathfrak{g},\curlywedge_{\rtimes})$ with $\mathfrak{g}=\mathbb{R}^l\rtimes\mathfrak{h}$ is (k,s)-nil. An argument similar to that of the (C2)-case in Theorem 3.1 gives the result. □

Corollary 3.2. *In a spacetime of dimension $n\geq 4$, the cosmological constant plays no role in any extended-linear EHP theory with $\mathfrak{h}\subset\mathfrak{so}(k_1)\oplus\ldots\oplus\mathfrak{so}(k_r)$. If $n\geq 6$ the full theory is trivial.*

Remark 3.2. This simple result will be our primary source of examples of geometric obstructions. This is due to the fact that essentially all "classical" geometry satisfies the hypothesis of the last corollary, implying that EHP cannot be realized in them in higher dimensions.

Remark 3.3. For future reference, notice that any *linear EHP theory induces, in a canonical way, a corresponding extended linear EHP theory.* Indeed, if $G\subset GL(k;\mathbb{R})$ is a linear Lie group, then any subgroup $H\subset G$ inherits a canonical action on $\mathbb{R}^k$, allowing us to consider the semidirect product $\mathbb{R}^k\rtimes H$ and then H as a subgroup of it. So, if we start with a linear theory for the inclusion $H\hookrightarrow G$ we can always move to a a extended-linear theory for the inclusion $H\hookrightarrow\mathbb{R}^k\rtimes H$. We will refer to this induced theory as the *canonical extended-linear theory associated* to a given linear theory.

3.3 Gauged Gravity

In the last subsection we defined EHP theories in linear geometries $H\hookrightarrow G$ and in extended-linear geometries $H\hookrightarrow\mathbb{R}^k\rtimes H$ as direct analogues of the EHP action functional. This means that we considered theories on (reductive) Cartan connections for the given group reduction, which are pairs $\nabla=e+\omega$, where ω is an usual H-connection and e is a pointwise isomorphism.

In order to regard EHP theories as genuine (reductive) gauge theories, it is necessary to work with arbitrary (i.e, not necessarily Cartan) reductive connections. In practice, this can be obtained by just forgetting the hypothesis of pointwise isomorphism in e. This leads us to define *homogeneous* and *inhomogeneous gauge linear/extended-linear Hilbert-Palatini forms* of a reductive (not necessarily Cartan) connection $\nabla=e+\omega$ as in (13). Similarly, we can then define *homogeneous* and *inhomogeneous gauge linear/extended-linear EHP theories* as in (14), whose configuration space is this new space of arbitrary reductive connections.

A linear map may not be an isomorphism when it is not injective or/and when it is not surjective. We would like that gauge EHP theories have a nice physical interpretation, leading us to ask: *what is the physical motivation for e be injective or/and surjective?*

At least for geometries described by G-structures (which in our context means extend-linear geometries), the quotient $\mathfrak{g}/\mathfrak{h}$ can be identified with some $\mathbb{R}^l$ endowed with a tensor t such that the pair $(\mathbb{R}^l, t)$ is the "canonical model" for the underlying geometry. In such cases, the injectivity hypothesis on $e_a : TP_a \to \mathfrak{g}/\mathfrak{h}$ is important to ensure that we have a good way to pull-back this "canonical geometric model" to each fiber of TP. For instance, when $G = \mathbb{R}^l \rtimes O(n-1,1)$ and $H = O(n-1,1)$, the quotient $\mathfrak{g}/\mathfrak{h}$ is just Minkowski space endowed with its standard metric η, and the injectivity of e implies that $g = e^*\eta$ is also a Lorentzian metric TP. But, even if we forget injectivity, the tensor $g = e^*\eta$ **remains** well-defined, but now it is no longer non-degenerate. Furthermore, for some models of Quantum Gravity, this non-degeneracy is welcome (say to ensure the possibility of topology change [20, 1]), which means that *physically it is really interesting to consider reductive connections $\xi = \omega + e$ such that e is pointwise non-injective.* With this in mind, we will do gauge EHP theories precisely for this kind of reductive connections.

On the other hand, in the context of Geometric Obstruction Theory, it is natural to expect that working with gauge EHP theories such that e in non-injective will impact the theory more than usual. But, how deep will be this impact? Notice that the proofs of Theorem 3.1 and Theorem 3.2 were totally based on the fact that in each point e take values in some (k,s)-nil algebra. In general, as discussed in Subsection 2.4, being (k,s)-solv is weaker than being (k,s)-nil. But, if we are in the class of injective forms, Proposition 2.3 shows that both concepts agree. Therefore, when working with Cartan connections, no new results can be obtained if we replace "(k,s)-nil" with "(k,s)-solv". However, forgetting injectivity we may build stronger obstructions.

In order to get these stronger obstructions, let us start by recalling that until this moment we worked with the class of **reductive** connections. These are $\mathfrak{g}$-valued 1-forms ξ which, respectively to the vector space decomposition $\mathfrak{g} \simeq \mathfrak{g}/\mathfrak{h} \oplus \mathfrak{h}$, can be globally written as $\xi = e + \omega$. This notion of "reducibility" extends naturally to k-forms with values in A, endowed with some vector space decomposition $A \simeq A_s \oplus ... \oplus A_0$. Indeed, we say that an A-valued k-form α in a smooth manifold P is *reductive* (or *decomposable*) respectively to the given vector space decomposition if it can be globally written as $\alpha = \alpha_s + ... + \alpha_0$. We can immediately see that when P is parallelizable every A-valued k-form is reductive respectively to an arbitrarily given decomposition of A. It then follows that *in an arbitrary manifold algebra-valued k-forms are locally reductive.*

After this digression, we state and proof the obstruction result for gauge linear/extended-linear EHP theories. Here, once fixed a group reduction $H \hookrightarrow G$, instead of (C1) and (C2) (which are nilpotency conditions) we will consider the following analogous solvability conditions:

(S0) the group G is isomorphic to $\mathbb{R}^l \rtimes H$ and $(\mathfrak{h}, \curlywedge)$ is (k,s)-solv;

(S1) the subgroup $H \subset G$ is normal and, as a matrix algebra, $(\mathfrak{g}, \curlywedge)$ is a splitting extension by $\mathfrak{h}$ of a (k,s)-solv algebra $\mathfrak{r}$;

(S2) the full algebra $(\mathfrak{g}, \curlywedge)$ is a (k,s)-solv algebra.

Theorem 3.3. *Let M be an n-dimensional spacetime and $P \to M$ be a G-bundle, endowed with a group reduction $H \hookrightarrow G$ fulfilling* (S0), (S1) *or* (S2). *If $n \geq k+s+1$, then the inhomogeneous gauge EHP theory equals the homogeneous ones. If $n \geq k+s+3$, then both are trivial.*

Proof. By definition, a (k,s)-solv algebra A comes endowed with a decomposition $A \simeq A_s \oplus ... \oplus A_0$ by (k,s)-nil algebras A_i. Therefore, under condition (S1) if, e is an $\mathfrak{g}/\mathfrak{h}$-valued 1-form, where $\mathfrak{g}/\mathfrak{h}$ is

(k,s)-solv, the digression above tells us that it can be locally written as $e = e_s + ... + e_0$. Furthermore, we have

$$\curlywedge^r e = \curlywedge^r e_s + ... + \curlywedge^r e_0,$$

because for $i \neq j$ the forms e_i and e_j are non-zero in different spaces. Since each e_i take values in a (k,s)-nil algebra, it then follows that $\curlywedge^{k+s+1} e = 0$ locally, which implies $\curlywedge^{k+s+1} e \curlywedge \alpha = 0$ for every $\mathfrak{g}$-valued form α. The remaining part of the proof is identical to that of Theorem 3.1, starting after equation (15). Case (S2) is analogous to (S1) and (after using Lemma 3.1) case (S0) is analogous to (S2). □

3.4 Dual Gravity

Here we will see that there are two "dual theories" associated to any EHP theory and we will see how the geometric obstructions of the actual EHP theory relate to the obstructions affecting these new theories.

We start by recalling that the fields in a gauge EHP theory (being it linear or extended-linear) are reductive connections for a given reduction $H \hookrightarrow G$. As a vector space we always have $\mathfrak{g} \simeq \mathfrak{h} \oplus \mathfrak{g}/\mathfrak{h}$ and these connections are given by 1-forms $e : TP \to \mathfrak{g}/\mathfrak{h}$ and $\omega : TP \to \mathfrak{h}$ such that ω is a connection. We can then think of a EHP theory as being determined by four variables: the fields e and ω together with the spaces $\mathfrak{g}/\mathfrak{h}$ and $\mathfrak{h}$. This allows us to define two types of *"dual EHP theories"* by interchanging such variables.

More precisely, we define the *geometric dual* of a given EHP theory as that theory having the same space of configurations, but whose action functional ${}^*S_{n,\Lambda}$ is ${}^*S_{n,\Lambda}[e,\omega] = S_{n,\Lambda}[\omega, e]$. Explicitly, for any fixed FABS we have

$$ {}^*S_{n,\Lambda}[e,\omega] = \int_M \operatorname{tr}(\jmath({}^*\alpha_{n,\Lambda})), \quad \text{with} \quad {}^*\alpha_{n,\Lambda} = \curlywedge_*^{n-2}\omega \curlywedge_* E + \frac{\Lambda}{(n-1)!} \curlywedge_*^n \omega, \tag{17}$$

where $E = de + e \curlywedge_* e$ and $\curlywedge_*$ must be interpreted as $\curlywedge$ or $\curlywedge_{\rtimes}$ depending if the starting theory is linear or extended-linear.

On the other hand, we define the *algebraic dual* of a EHP theory as that theory having an action functional with the same shape, but now defined in a "dual configuration space". This is the theory whose action function $S^*_{n,\Lambda}$ is given by

$$ S^*_{n,\Lambda}[e,\omega] = \int_M \operatorname{tr}(\jmath(\alpha^*_{n,\Lambda})), \quad \text{with} \quad \alpha^*_{n,\Lambda} = \curlywedge_*^{n-2} e \curlywedge_* \Omega + \frac{\Lambda}{(n-1)!} \curlywedge_*^n e, \tag{18}$$

where e and ω take value in $\mathfrak{h}$ and $\mathfrak{g}/\mathfrak{h}$, respectively. Furthermore, as above, $\curlywedge_*$ must be interpreted as $\curlywedge$ or $\curlywedge_{\rtimes}$, depending of the case.

In very few words we can say that the *geometric dual* of a given theory makes a change at the *dynamical level*, in the sense that only the *action funcional* is changed. Dually, the *algebraic dual* of a given theory produce changes only at the *kinematic level*, meaning that the dualization refers to the *configuration space*. This can be summarized in the table below.

theories /variables	e	ω	$\mathfrak{g}/\mathfrak{h}$	$\mathfrak{h}$
EHP	e	ω	$\mathfrak{g}/\mathfrak{h}$	$\mathfrak{h}$
* EHP	ω	e	$\mathfrak{g}/\mathfrak{h}$	$\mathfrak{h}$
EHP*	e	ω	$\mathfrak{h}$	$\mathfrak{g}/\mathfrak{h}$

Table 1: Relation between EHP theories and its geometric and algebraic dualizations.

Once such dual theories are introduced, let us now see how the obstruction results for EHP theories apply to them. Let us start by focusing on the geometric dual. Our main result until now is Theorem 3.3. It is essentially determined by two facts. First, e take values in an algebra fulfilling some "solvability condition" (i.e, any one of hypotheses (S0), (S1) or (S2)); and second, the EHP action contains powers of e.

When we look at the action (17) of the geometric dual theory we see that it contains powers of ω (instead of e). Therefore, assuming that ω take values in some "solvable" algebra we will get a result analogous to Theorem 3.3. Looking at Table 1 we identify that in the geometric dual theory ω takes values in $\mathfrak{h}$ which is a subalgebra of $\mathfrak{g}$, where the product $\curlywedge$ makes sense, so that the "solvability condition" should be on $\mathfrak{h}$. Dually, the action (18) also contains powers, now of e. Table 1 shows that here e take values in $\mathfrak{h}$, so that *if* $(\mathfrak{h}, \curlywedge)$ *is "solvable", then not only the geometric dual theory is trivial, but also the algebraic dual one!* Formally, we have the following result, whose proof follows that of Theorem 3.3

Theorem 3.4. *Let M be an n-dimensional spacetime and $P \to M$ be a G-bundle, endowed with a group reduction $H \hookrightarrow G$. Assume that $(\mathfrak{h}, \curlywedge)$ is a (k,s)-solv algebra. If $n \geq k+s+3$, then both the geometric dual and the algebraic dual of gauge EHP theory are trivial.*

Corollary 3.3. *If condition* (S0) *is satisfied and $n \geq k+s+3$, then gauge EHP theory and all its duals are trivial.*

Remark 3.4. In the **linear** context, where G is a linear Lie group and $H \subset G$ is a subgroup, Theorem 3.4 above implies that if $(\mathfrak{h}, \curlywedge)$ is (k,s)-solv, then the duals of the actual EHP theory are trivial. This **does not** mean that the actual EHP is trivial. In fact, Corollary 3.3 needs condition (S0), which subsumes that we are working in the **extended-linear** context.

3.5 The Role of Torsion

Until this moment we have given conditions under which gauge EHP theories and their duals are trivial. One of this conditions is (S0), which makes sense in the extended-linear context and states that the subalgebra $(\mathfrak{h}, \curlywedge)$ is (k,s)-solv. Indeed, from Corollary 3.3, if this condition holds then EHP and their duals are all trivial for $n \geq k+s+3$. Due to Lemma 2.1, the standard examples of $(1,2)$-nil algebras are subalgebras of $\mathfrak{so}(k_1) \oplus ... \oplus \mathfrak{so}(k_r)$, so that *for such types of geometry EHP and their duals are trivial if $n \geq 6$*. Here we will see that if we work with a special class of connections, then there are even more obstructions, due essentially to Berger's classification theorem on Riemannian holonomy [4, 29].

We start by recalling that if ω is a connection on a G-principal bundle $\pi : P \to M$, each point $a \in P$ determines a Lie subgroup $\mathrm{Hol}(\omega, a) \subset G$, called the *holonomy group of ω in a*, which measures the failure of a loop in M, based in $\pi(a)$, remaining a loop after horizontal lifting [24]. In typical situations, P is the frame bundle FM of a n-manifold M, regarded as a G-structure

22

under some reduction $G \hookrightarrow GL(n;\mathbb{R})$. Such a G-structure, in turn, generally is determined as the space of frame transformations which preserve some additional tensor field t in M. In this context, the condition $\mathrm{Hol}(\omega,a) \subset G$ is equivalent to saying that there exists one such tensor field which is parallel relative to ω, i.e, such that $\nabla_\omega t = 0$, where ∇_ω is the covariant derivative on tensors induced by ω. Particularly, $SO(n)$-reductions of FM correspond to Riemannian metrics on an oriented manifold, so that if ω is a connection on FM such that $\mathrm{Hol}(\omega,a) \subset SO(n)$, then $\nabla_\omega g = 0$ for some Riemannian metric g. Consequently, *a torsion-free connection ω in FM whose holonomy is contained in $SO(n)$ is the Levi-Civita connection of some metric.*

Recall that, as pointed in Remark 2.2, for every $k_1 + ... + k_r = n$ we have a canonical inclusion

$$SO(k_1) \times ... \times SO(k_r) \hookrightarrow SO(n) \tag{19}$$

as diagonal block matrices. Given a connection ω with special Riemannian holonomy (meaning that it is contained in $SO(n)$) we can ask: *when is it indeed contained in the product subgroup above?* From de Rham decomposition theorem [?, ?] we see that if ω is torsion-free (and, therefore, the Levi-Civita connection of a metric g) and M is simply connected, this happens iff (M,g) is locally isometric[5] to a product of Riemannian k_i-manifolds (M_i, g_i), with $i = 1, ..., r$, and $\mathrm{Hol}(\omega)$ is actually the product of $\mathrm{Hol}(\omega_i) \subset SO(k_i)$, where ω_i is the Levi-Civita connection of g_i.

Assuming (M,g) simply connected and locally irreducible in the above sense, the holonomy reduction (19) does not exist. In this case, it is natural to ask for which proper subgroups $G \hookrightarrow SO(n)$ the holonomy of the Levi-Civita connection of g can be reduced. When (M,g) is not locally isometric to a symmetric space, we have a complete classification of such proper subgroups, given by *Berger's classification theorem*, as in Table 2.

$G \subset SO(n)$	$\dim(M)$	nomenclature
$U(n)$	$2n$	Kähler
$SU(n)$	$2n$	Calabi-Yau
$Sp(n)\cdot Sp(1)$	$4n$	Quaternionic-Kähler
$Sp(n)$	$4n$	Hyperkähler
G_2	7	G_2
Spin(7)	8	Spin(7)

Table 2: Berger's classification theorem for Riemannian signature.

Let us see how we can use Berger's classification theorem to get geometric obstructions similar to those given in Theorem 3.4 and in Corollary 3.3. We will need some definitions. We say that a linear group H is a *k-group* if there are nonnegative integers $k_1, ..., k_r$, with $k_1 + ... + k_r = k$, such that $(\mathfrak{h}, \curlywedge)$ is a subalgebra of $\mathfrak{so}(k_1) \oplus ... \oplus \mathfrak{so}(k_r)$. In other words, k-groups are fundamental examples of $(1,2)$-nil algebras. Furthermore, we say that a manifold N is a *Berger k-manifold* if it has dimension k and it is simply connected, locally irreducible and locally non-symmetric. A bundle $P \to M$ is called *k-proper* if there exists an immersed Berger k-manifold $N \hookrightarrow M$ whose frame bundle is a subbundle of P, i.e, such that $FN \subset i^*P$. The motivating (trivial) examples are the following:

Example 3.1. The frame bundle of a Berger n-manifold is, of course, n-proper.

[5] If (M,g) is geodesically complete, the decomposition is global.

Example 3.2. If $f : N \hookrightarrow M$ an immersion with N a Berger k-manifold, then FM is automatically k-proper, because $FN \subset f^*FM$.

Despite the notions introduced above, in the next theorem we will work with *torsionless (extended-linear) EHP theories*, i.e, extended-linear EHP theories restricted to reductive connections $\nabla = e+\omega$ such that $\Theta_\omega = d_\omega e = 0$. For the case of linear EHP theories, recall that as discussed in Remark 3.3, to any of them we have an associated canonical extended-linear theory.

Theorem 3.5. *Let $P \to M$ be a k-proper H-bundle over a n-manifold M. If H is a k-group, then $k_1 = k$ and $k_{i>1} = 0$. Furthermore, for any FABS, a torsionless extended-linear EHP theory based on H is nontrivial only if one of the following conditions is satisfied*

(B1) *$k = 2, 4$ and M contains a Kähler Berger k-manifold;*

(B2) *$k = 4$ and M contains a quaternionic-Kähler Berger k-manifold.*

Proof. From Lemma 2.1, the hypothesis on $\mathfrak{h}$ implies that it is $(1, 2)$-nil and, therefore, because we are working with extended-linear theories, condition (S0) is satisfied. Consequently, by Theorem 3.3 the actual (and, in particular, the torsionless) EHP theory is trivial if $n \geq 6$, so that we may assume $n < 6$. Since $P \to M$ is k-proper, M contains at least one immersed Berger k-manifold $\imath : N \hookrightarrow M$ such that $FN \subset \imath^*P$. Let κ denote the inclusion of FN into $\imath^*P$. On the other hand, we also have an immersion $\imath^*P \hookrightarrow P$, which we denote by $\imath$ too. Lie algebra-valued forms can be pulled-back and the pullback preserves horizontability and equivariance. So, for any $\nabla = e + \omega$ the corresponding 1-form $(\imath \circ \kappa)^*\omega \equiv \omega|_N$ is an H-connection in FN and its holonomy is contained in H. By hypothesis H is an k-group so that, via Lie integration,

$$H \subset SO(k_1) \times ... \times SO(k_r) \subset SO(k).$$

In particular, the holonomy of $\omega|_N$ is contained in $SO(k)$, implying that $\omega|_N$ is compatible with some Riemannian metric g in N. But, we are working with torsion-free connections, so that $\omega|_N$ is actually the Levi-Civita connection of g and, because N is irreducible, de Rham decomposition theorem implies that there exists $i \in 1, ..., r$ such that $k_i = k$ and $k_j = 0$ for $j \neq i$. Without loss of generality we can take $i = 1$. Because N is a Berger manifold, Berger's theorem applies, implying that the holonomy of $\omega|_N$ is classified by Table 2, giving conditions (B1) and (B2). □

Corollary 3.4. *Let M be a Berger n-manifold with an H-structure, where H is a n-group. In this case, for any FABS, a torsionless extended-linear EHP based on H is nontrivial only if M has dimension $n = 2, 4$ and admits a Kähler structure.*

Proof. The result follows from the last theorem by considering the bundle $P \to M$ as the frame bundle $FM \to M$ and from the fact that every orientable four-dimensional smooth manifold admits a quaternionic-Kähler structure [5, **?**]. □

Remark 3.5. This corollary shows how topologically restrictive it is to internalize *torsionless* extended-linear EHP in geometries other than Lorentzian. Indeed, if the spacetime M is compact and 2-dimensional, then it must be $\mathbb{S}^2$. On the other hand, in dimension $n = 4$ compact Kähler structures exist iff the Betti numbers $b_1(M)$ and $b_3(M)$ are zero, so that $\chi(M) = b_2(M)+2$. As a consequence, if we add the (mild) condition $H^2(M; \mathbb{R}) \simeq 0$ on the hypothesis of Corollary 3.4 we conclude that M must be a K3-surface!

The last theorem was obtained as a consequence of Theorem 3.3 and of Berger's classification theorem. So, this is a geometric obstruction result for *extended-linear* EHP theories. However, due to Theorem 3.4 and Corollary 3.3 we can get exactly the same result for the geometric and algebraic *duals* of *linear* EHP. The same result does not hold for the *actual linear* EHP theories, because there is no analogue of Theorem 3.3 or Corollary 3.3 for them, as emphasized in Remark 3.4.

On the other hand, Berger's theorem remains valid, allowing us to get an obstruction result for linear EHP theories independently of the previous ones. Indeed, recall that (as pointed in Remark 3.3) we can always associate a canonical (extended-linear) theory to a given linear one. Let us define a *torsionless liner EHP theory* as that obtained by restricting a linear theory to the space of connections such that the corresponding canonical theory is a torsionless extended-linear EHP theory in the sense introduced above. We then have the following:

Theorem 3.6. *Let $P \to M$ be a k-proper G-bundle over a n-manifold M endowed with a group structure reduction $H \hookrightarrow G$. If H is a k-group, then $k_1 = k$ and $k_{i>1} = 0$. Furthermore, a torsionless linear EHP theory for this reduction is nontrivial only if one of the following conditions is satisfied*

(B1') *k is even and M contains a Kähler Berger k-manifold;*

(B2') *k is divisible by 4 and M contains a quaternionic-Kähler Berger k-manifold;*

(B3') *$k = 7$ and M contains a Berger k-manifold with a G_2-structure.*

Proof. The proof follows the same lines of the last theorem, except by the first argument, which makes explicit use of Theorem 3.3. □

The above results are about different versions of torsionless *gauge* EHP theories. By this we mean that no requirement was made on e. We close this section remarking that if we work not on the *gauge* context, but on the *Cartan* context (in the sense that e is a pointwise isomorphism), then there is a physical appeal for working with torsionless connections.

Indeed, recall that by the very abstract definition, a *classical theory* is given by an *action functional* $S : \mathrm{Conf} \to \mathbb{R}$ defined on some *space of configurations*. The interesting classical theories are those in that Conf has some kind of "smooth structure" relative to which S can be regarded as a "smooth function" and, as such, has a "derivative". In such cases, there exists a distinguished subspace $\mathrm{Cut}(S) \subset \mathrm{Conf}$ constituted by the "critical points" of S. This is the *phase space* of the underlying classical theory, which contains all configurations which a priori can be observed in nature.

If S is now the action of EHP theory, we find that the phase space is determined by the pairs (e, ω) which satisfy the equations

$$\curlywedge_*^{n-2} e \curlywedge_* \Omega + \frac{\Lambda}{(n-1)!} \curlywedge_*^{n} e = 0 \quad \text{and} \quad \curlywedge_*^{n-2} e \curlywedge \Theta_\omega = 0$$

simultaneously, where (once again) $\curlywedge_*$ must be interpreted as $\curlywedge$ or $\curlywedge_{\rtimes}$ depending if we are in the linear or in the extended-linear context. The first of these equations is just an analogue of Einstein's equations. The second, in turn, if we are working with *Cartan connections*, reduces to $\Theta_\omega = 0$, i.e, to the "torsionless" condition previously used.

4 Abstract Obstructions

Until this moment we considered EHP theory for group structure reductions $H \hookrightarrow G$ other than $O(n-1,1) \hookrightarrow \mathrm{Iso}(n-1,1)$, where G could be an arbitrary linear group or some semidirect product $\mathbb{R}^k \rtimes H$. In other words, we realized gravity, as modeled by EHP theories, in other geometries than Lorentzian geometry. We then showed that for certain classes of geometries the corresponding theory is actually trivial, meaning that we have geometric obstructions.

In Subsection 4.1 we will generalize EHP theories even more by replacing the algebras $\mathfrak{h}$ and $\mathfrak{g}$, which are induced by the Lie groups G and H, by general (not necessarily satisfying PI's) algebras. We call these theories *algebra-valued EHP theories*[6]. One motivation to consider this new generalization is the following. In the previous extended-linear context, a group reduction $H \hookrightarrow \mathbb{R}^n \rtimes H$ on a manifold M is (as discussed in the beginning of Subsection 3.5) a geometry modeled by some kind of tensor t in M. A *global symmetry* of (M,t) is an automorphism in the category of H-structures, i.e, a difeo such that $f^*t = t$. If we apply this to the canonical geometric model $(\mathbb{R}^n,t)$ we see that its group of global symmetries $\mathrm{Aut}(\mathbb{R}^n,t)$ is precisely $\mathbb{R}^n \rtimes H$. Therefore, the algebra $(\mathbb{R}^n \rtimes \mathfrak{h}, \curlywedge_{\rtimes})$ takes the role of the associative algebra of *infinitesimal global symmetries of the canonical geometric model*, which is where the reductive connections for the reduction $H \hookrightarrow \mathbb{R}^n \rtimes H$ take values.

It happens that in many physical situations we have more hidden/internal/worldsheet symmetries, so that the *full* algebra of infinitesimal symmetries has a more abstract structure than just an associative algebra, leading us to consider "reductive connections with values in arbitrary algebras". Once the notion of algebra-valued EHP theoriesis introduced, we show that the previous obstruction results (Theorem 3.3), hold *ipsis litteris* in this new abstract context.

A typical situation in that the full algebra of infinitesimal symmetries is not just an associative algebra is when the underlying "canonical geometric model" is not a cartesian space $\mathbb{R}^n$ endowed with a tensor, but actually a supercartesian space $\mathbb{R}^{n|m}$ endowed with a supertensor. Indeed, in this case the algebra of infinitesimal symmetries inherits a $\mathbb{Z}_2$-graded algebra structure. This motivates us to analyze the effects of gradings in the obstruction results, which is done in Subsection 4.2.

Notice that if the pair $(\mathbb{R}^n, t_n)$ describes geometry, the pair $(\mathbb{R}^{n|m}, t_{n|m})$ describes *supergeometry*. Therefore, it would be natural to consider not only "EHP with values in superalgebras", but actually "super EHP theories". Closing the section, in Subsection 4.3 we present an approach to the notion of "reductive graded-connection", allowing us to define "graded EHP theories", and we show how to extend the obstruction results to this context.

4.1 Algebra-Valued Gravity

We start by recalling that a reductive connection in P for $H \hookrightarrow G$ is a $\mathfrak{g}$-valued 1-form ∇ which decomposes as $\omega + e$, where ω is a $\mathfrak{h}$-valued 1-form and e is a $\mathfrak{g}/\mathfrak{h}$-valued 1-form (not necessarily an isomorphism, due to previous discussion). Notice that it is completely determined by the pair (ω, e) and by the vector space decomposition $\mathfrak{g} \simeq \mathfrak{h} \oplus \mathfrak{g}/\mathfrak{h}$, a fact that was already used in Subsection 3.4. In typical cases $\mathfrak{g}$ is actually a splitting extension (the previous extended-linear context) or $H \subset G$ is a normal subgroup (previous linear context), so that both $\mathfrak{h}$ and $\mathfrak{g}/\mathfrak{h}$ acquire algebra structures.

[6] We could call such theories *algebraic EHP theories*, but this nomenclature would suggest that exactly as EHP theories are about connections, "algebraic EHP" should be about "algebraic connections". The problem is that the notion of "algebraic connection" actually exists in the literature [9, 23, 13], being applied in a more general context.

This leads us to the following generalization: given an $\mathbb{R}$-algebra $(A, *)$ endowed with a vector space decomposition $A \simeq A_0 \oplus A_1$, where A_0 and A_1 have algebra structures $*_0$ and $*_1$ (not necessarily subalgebras of A), we define an *A-connection* in a manifold P as an A-valued 1-form ∇ in P which is reductive respective to $A \simeq A_1 \oplus A_0$. In }

{kmore concrete terms, it is an A-valued 1-form ∇ which is written as $\nabla = \omega + e$, where ω and e take values in A_1 and A_0, respectively. The *curvature* of ω is the 2-form $\Omega = d\omega + \omega \wedge_1 \omega$ in A_1, where $\wedge_1$ is the product induced by $*_1$. Similarly, the *torsion* of ω is the A-valued 2-form $\Theta_\omega = de + \omega \wedge_* e$. Some related concepts are considered in [6, 7].

With these structures in hand we can generalize gauge EHP theories (about $\mathfrak{g}$-valued connections) to *algebra-valued EHP theories* (about A-connections in the above sense). Indeed, given a spacetime manifold M, this generalization is obtained following the following steps:

1. consider some analogue of the Hilbert-Palatini forms (13);
2. show that this Hilbert-Palatini form induces a corresponding form in M with values in some bundle;
3. turn this bundle-valued form into a real-valued form;
4. define the action functional as the integral over M of this real-valued form.

In order to do the second step, the immediate idea is to select a FABS, say defined on a subcategory $\mathbf{C} \subset \mathbf{Bun}_M \times \mathbf{Alg}_{\mathbb{R}}$, as in Section 2.5. However, in order to realize the first step we need to work with FABS fulfilling certain "invariance property": we say that a FABS on $\mathbf{C}$ is *invariant* by a functor $I : \mathbf{C} \to \mathbb{Z}\mathbf{Alg}_{\mathbb{R}}$ if

(a) for all $(P, A) \in \mathbf{C}$ the corresponding $I(P, A)$ is an ideal of $S(P; A)$, so that we can take the quotient functor S/I and we have a natural transformation $\pi : S \Rightarrow S/I$;

(b) there exists another functor $J : \mathbf{C} \to \mathbb{Z}\mathbf{Alg}_{\mathbb{R}}$ such that $J(P, A)$ is an ideal of $\Lambda(M; E_A)$ and whose projection we denote by π';

(c) there exists a natural transformation $\jmath' : S/I \Rightarrow \Lambda(M; -)/J$ such that $\jmath' \circ \pi = \pi' \circ \jmath$, i.e, the diagram below commutes for every (P, A).

$$
\begin{tikzcd}
S(P,A) \arrow[r, "\jmath(P,A)"] \arrow[d, "\pi_{(P,A)}"'] & \Lambda(M;E_A) \arrow[d, "\pi'_{(P,A)}"] \\
S(P,A)/I(P,A) \arrow[r, "\jmath'_{(P,A)}"'] & \Lambda(M;E_A)/J(P,A)
\end{tikzcd}
\tag{20}
$$

Returning to the first step, notice that a priori the algebras A and A_1 may not be associative, so that due to ambiguities we cannot simply replace "$\curlywedge$" by "$\wedge_*$" and "$\wedge_0$" in (13) to get an A-valued version of α_{HP}; instead, we should take into account all possibilities simultaneously. This can be done as follows. Recall the "associativity-like" polynomials

$$(x_1 \cdot ... \cdot (x_{s-2} \cdot (x_{s-1} \cdot x_s))), \quad (((x_1 \cdot x_2) \cdot (x_3 \cdot x_4)) \cdot ... \cdot x_s), \quad \text{etc.},$$

which were introduced in Subsection 2.3 in order to describe nilpotency and nil properties as PI's. Due to the structure of the Hilbert-Palatini forms, we are interested in the more specific polynomials

$$(x \cdot ... \cdot (x \cdot (x \cdot y))), \quad \text{etc.} \quad \text{and} \quad (x \cdot ... \cdot (x \cdot (x \cdot x))), \quad \text{etc.} \tag{21}$$

When evaluated at $\Lambda(P;A)$, the set of these polynomials generates an ideal $\mathfrak{I}(P,A)$ and the idea is to consider FABS which are invariant by it.

More precisely, given a bundle $P \to M$ over a manifold M and an algebra A endowed with vector space decomposition $A \simeq A_0 \oplus A_1$, we say that a FABS over M, defined in $\mathbf{C}$, is *compatible* with the given data if

(a) the pair (P,A) belongs to $\mathbf{C}$;

(b) each possibility of defining $\alpha_{n,\Lambda}$ (i.e, the evaluation of each polynomial (21) in each A-connection) belongs to $S(P,A)$;

(c) the FABS in question is invariant by $\mathfrak{I}$, as defined above.

Considering FABS satisfying (a)-(c) we realize the first two steps needed to define "algebra-valued EHP theory". In order to realize the third step, we need something like a "trace". A *trace transformation* for a FABS on $\mathbf{C}$ invariant by I is a natural transformation tr between $\Lambda(M;E_{-})/J$ and the constant functor in $\Lambda(M;\mathbb{R})$. In other words, it is a rule that assigns (in a natural way) a map of graded-algebras $\mathrm{tr}_{(P,A)} : \Lambda(M;E_A) \to \Lambda(M;\mathbb{R})$ to every pair $(P,A) \in \mathbf{C}$.

Now, we can finally define what is an EHP theory in the general algebraic setting. Given a principal bundle $P \to M$, an algebra A and a vector space decomposition $A \simeq A_0 \oplus A_1$, choose a compatible FABS endowed with a trace transformation. The corresponding *A-valued inhomogeneous*[7] *EHP theory* in P is the classical theory whose configuration space is the space of all reductive A-connections in P and whose action functional is given by (we omit the subscripts in the maps π, j' and tr in order to simplify the notation)

$$S_{n,\Lambda}[e,\omega] = \int_M \mathrm{tr}(\jmath'(\pi(\alpha_{HP}))),$$

where $\alpha_{HP} \in S(P,A)$ is any "Hilbert-Palatini"-type form, say

$$\alpha_{HP} = (e \wedge_* ...(\wedge_*(e \wedge_* (e \wedge_* \Omega))) + \frac{\Lambda}{(n-1)!}(e \wedge_* ...(\wedge_*(e \wedge_* (e \wedge_* \Omega))).$$

Notice that we recover the gauge linear EHP theories discussed previously by taking A as a matrix Lie algebra $\mathfrak{g}$ and considering the adjoint FABS endowed with the classical trace map. Therefore, at least in this restricted domain we have obstruction Theorems 3.1, 3.3 and 3.4. These results remain valid if:

1. *we replace the adjoint FABS by any other compatible FABS* (indeed, due to the linearity of the FABS, once one shows that $\alpha_{HP} = 0$ it immediately follows that the bundle-valued form $\jmath'(\pi(\alpha_{HP}))$ is also trivial independently of the FABS chosen);
2. *we replace the classical trace by any other trace transformation* (also due to the linearity of trace transformations);

[7]The corresponding A-valued *homogeneous* EHP theory is defined analogously.

3. *we replace the algebra $\mathfrak{g}$ by any other algebra A endowed with a vector space decomposition* $A \simeq A_0 \oplus A_1$ (this follows from the structure of the proofs of Theorems3.1, 3.3 and 3.4, which in essence depends only on general algebraic hypotheses on the decomposition $\mathfrak{g} \simeq \mathfrak{g}/\mathfrak{h} \oplus \mathfrak{h}$).

Summarizing: the obstruction theorems hold not only in the domain of matrix algebras, adjoint FABS and classical trace, but also for arbitrary algebras, arbitrary compatible FABS and arbitrary trace transformations. Concretely, we have the following general obstruction theorem whose proof is in essence Remarks 1-3 above.

Theorem 4.1. *Let M be an n-dimensional spacetime and $P \to M$ be a bundle and $(A,*)$ an algebra endowed with a vector space decomposition $A_0 \oplus A_1$ fulfilling one of the following conditions*

(A1) the algebra $(A_0, *_0)$ is a (k,s)-solv subalgebra[8] *$(A,*)$;*

(A2) $(A,*)$ is itself (k,s)-solv.

If $n \geq k+s+1$, then for any compatible FABS and any trace transformation, the corresponding inhomogeneous A-valued EHP theory equals the homogeneous ones. If $n \geq k+s+3$, then both theories are trivial.

4.2 Graded-Valued Gravity

Here we shall indicate how the previous discussion can be extended to the case when the background algebra A is itself graded (for more details see [27]). Given a monoid $\mathfrak{m}$, let $P \to M$ be a principal bundle and A be a $\mathfrak{m}$-graded real algebra. For any vector space decomposition $A \simeq A_0 \oplus A_1$ where A_i are algebras, we define a *connection in P with values in the graded algebra A* exactly as previously: as a A-valued 1-form ∇ in P which writes as $\nabla = e + \omega$ for $e : TP \to A_0$ and $\omega : TP \to A_1$. The only difference is that now the $\mathfrak{m}$-grading of A induces a corresponding grading in each A_i by $\oplus_m A_i^m$, where $A_i^m = A_i \cap A^m$, which means that locally we can write e and ω as

$$e = \sum\nolimits_m e^m \quad \text{and} \quad \omega = \sum_m \omega^m,$$

where $e^m : TP \to A_0^m$ and $\omega : TP \to A_1^m$ are just the projections of e and ω onto the corresponding A_i^m.

In order to extend FABS to this graded context we notice that if A is $\mathfrak{m}$-graded, then the algebra of A-valued exterior forms $\Lambda(P;A)$ is $(\mathbb{Z} \times \mathfrak{m})$-graded. On the other hand, an algebra bundle E_A whose typical fiber is A is not necessarily $\mathfrak{m}$-graded, because the pointwise decomposition may not vary continuously to allow us to globally decompose E_A as $\oplus_m E_{A_m}$. Therefore, in order to incorporate graded-algebras it is enough to work with "graded FABS" characterized by the following diagram:

[8] We can forget the subalgebra hypothesis by modifying a little bit the ideal $\mathfrak{I}(P,A)$ used to define the notion of "compatible FABS".

$$\begin{tikzcd}
\mathfrak{m}\mathbf{Alg} \arrow[rr, dashed, "E_-"] & & \mathbf{AlgBun}_M \arrow[d, "\Lambda(M;-)"] \\
& \arrow[ur, phantom, "\jmath"] & \mathbb{Z}\mathbf{Alg} \\
\mathbf{C} \arrow[uu, "\pi_1"] \arrow[rr, dashed, bend left, "S(-,-)"] \arrow[rr, bend right, "\Lambda(-;-)"'] \arrow[rr, phantom, "\Downarrow \imath"] & & (\mathbb{Z}\times\mathfrak{m})\mathbf{Alg} \arrow[u]
\end{tikzcd}$$

For a chosen bundle $P\to M$ and a $\mathfrak{m}$-graded algebra $A\simeq A_0\oplus A_1$, we define a *compatible graded FABS* exactly as in the last subsection. Notice that now the natural transformation $\imath : S\Rightarrow\Lambda$ is $(\mathbb{Z}\times\mathfrak{m})$-graded, while $\jmath : S\Rightarrow \Lambda(M;E_-)$ remains $\mathbb{Z}$-graded (due precisely to the fact that E_A may not be $\mathfrak{m}$-graded). In particular, the diagram (20) remains in $\mathbb{Z}\mathbf{AlgBun}_M$. This means that the trace transformation that we need to consider is also only $\mathbb{Z}$-graded, as previously. For every such compatible graded FABS endowed with a trace transformation tr we define the corresponding A-*valued inhomogeneous EHP theory* as the classical theory whose configuration space is the collection of all A-valued connections in P, and whose action functional is given by

$$S_{n,\Lambda}[e,\omega]=\int_M \mathrm{tr}(\jmath'(\pi(\alpha_{HP}))). \tag{22}$$

Thus, up to minor modifications everything works as in the last subsection. The crux of these "minor modifications" is that we can now locally decompose e and ω. This allows us to get a stronger version of Theorem 4.1 following the same strategy used to get Theorem 3.3 from Theorem 3.1. Indeed, consider the following conditions about the vector space decomposition $A\simeq A_0\oplus A_1$:

(G1) $(A_0,*_0)$ is a subalgebra and each $A_0^j=A^i\cap A_0$ is a (k_j,s_j)-weak solvable subspace of A_0 (recall definition in Remark 2.5).

(G2) restricted to each A^j the algebra A is (k_j,s_j)-weak solvable.

We can now prove

Theorem 4.2. *Let M be an n-dimensional spacetime, $P\to M$ be a bundle and $(A,*)$ be a $\mathfrak{m}$-graded algebra endowed with a vector space decomposition $A_0\oplus A_1$ fulfilling conditions* (G1) *or* (G2) *above. Furthermore, let (k,s) be the minimum of (k_j,s_j). If $n\geq k+s+1$, then for any compatible FABS and any trace transformation, the corresponding inhomogeneous A-valued EHP theory equals the homogeneous ones. If $n\geq k+s+3$, then both theories are trivial.*

Proof. The proofs for (G1) and (G2) are very similar, so that we will only explain the (G1) case. Once again, since everything is linear and grading-preserving, it is enough to prove that $\alpha_{HP}=0$ for some representative Einstein-Hilbert form. Particularly, we can prove for

$$\alpha_{HP}=\wedge_*^{n-2}e\wedge_*\Omega+\frac{\Lambda}{(n-1)!}\wedge_*^n e.$$

Under the hypothesis, we can locally write $e=\sum_m e^m$ with $e^m : TP\to A_0^m$. From condition (G1) each A_0^m is a weak (k_m,s_m)-solvable subspace, so that it writes as a sum $A_0^m=\oplus_i V_i^m$ of weak

(k_m, s_m)-nilpotent subspaces, which means, in particular, that we can write $e^m = \sum_i e_i^m$ locally and, therefore, $e = \sum_m \sum_i e_i^m$. Consequently, for every l we have $\wedge_*^l e = p_l(e_{i_1}^{m_1}, ..., e_{i_l}^{m_l})$ for some polynomial of degree l. If we now consider the minimum (k, s) (over m) of (k_m, s_m), the fact that each A_0^m is weakly (k_m, s_m)-nilpotent then implies $\wedge_*^{k+s+1} e = 0$. The remaining steps in the proof are identical to every other given in the previous theorems. □

4.3 Graded Gravity

The last section was about "gauge theories with values in graded algebras". We would like to work not only with graded algebras but in the "full graded context", i.e, with genuine graded gauge theories (particularly, with graded EHP theories). A reductive A-valued gauge theory is about A-valued connections, i.e, 1-forms $\nabla : TP \to A$ which are reductive respective to some decomposition $A \simeq A_0 \oplus A_1$. So, "graded gauge theories" should be about "A-valued graded connections". There are many approaches to formalize the notion of "graded connection". For instance, we have:

1. Quillen superconnections [32], which are defined as operators on graded vector bundles over a non-graded manifold;

2. connections on graded manifolds in the spirit of Kostant-Berezin-Leites, which are defined for graded principal bundles $P \to M$ over graded manifolds [?];

3. connections on principal ∞-bundles, which can be applied in the domain of any cohesive ∞-topos (particularly in the ∞-topos of formal-super-smooth manifolds), where the notion of differential cohomology can be axiomatized and a ∞-bundle with connection is defined as a cocycle of such cohomology [14, ?, ?].

Here we will not work with any of the models above. Instead, given a$\mathfrak{m}$-graded algebra A, we assume that the bundle TP is also $\mathfrak{m}$-graded (in the sense that it decomposes as a sum of vector bundles $TP \simeq \oplus_m E^m$) and we define an *A-valued graded connection in P of degree l* as a smooth 1-form $\nabla : TP \to A$ which has degree l, meaning that it decomposes as a sequence of usual vector-valued 1-forms $\nabla^m : E^m \to A^{m+l}$. Particularly, for us the *reductive graded connections* are those that can be written as $\nabla = e + \omega$, where each e and ω can themselves be decomposed as maps of degree l

$$e^m, \omega^m : E^m \to A^{m+l}, \quad \text{with} \quad e^m + \omega^m = \nabla^m.$$

With the notion of "reductive graded connections of degree l", we can define "graded EHP theories of degree l" in a natural way. Indeed, chosen a $\mathfrak{m}$-graded FABS, the corresponding *$\mathfrak{m}$-graded A-valued EHP theory of degree l* is the classical theory given by the action functional (22) restricted to the class of graded connections of degree l.

Remark 4.1. If we think of A as a graded-algebra describing the infinitesimal symmetries of the theory and if we interpret the graded structure of the bundle TP as induced by the different flavors of fundamental objects of the theory, then it is more natural to consider the connections of degree zero, because they will map each piece E^m into each corresponding subspace of infinitesimal symmetries A^m. On the other hand, in some situations (say in the BV-BRST formalism, where we have ghosts, anti-ghosts and anti-fields) we need to work with "shifted symmetries", meaning that degree $l > 0$ graded connections should also have some physical meaning.

Now, let us focus on the geometric obstructions of the A-valued EHP theory that appear in the full graded context. First of all, notice that to give a morphism $f : A' \to A$ of degree l between two graded algebras is the same as giving a zero degree morphism $f : A' \to A[-l]$, where $A[-l]$ is the graded algebra obtained shifting A. Consequently, the obstructions of a degree l graded A-valued EHP theory are just the obstructions of degree zero graded $A[-l]$-valued EHP theory, so that it is enough to analyze theories of degree zero.

The core idea of the proof of Theorem 4.2 was to use that A is graded and the hypothesis (G1) or (G2) in order to conclude that $\wedge_*^{k+s+1} e = 0$. More precisely, the grading of A was needed in order to write e as $e \simeq \sum_m e^m$, with $e^m : TP \to A_0^m$, while the "solvability hypothesis" (G1) or (G2) allowed us to decompose each e^m as $e^m = \oplus_i e_i^m$, so that $e = \oplus_{i,m} e_i^m$. The hypothesis (G1) or (G2) was then used once again to wield $\wedge_*^{k+s+1} e = 0$.

If we assume that TP is a graded bundle, so that $TP \simeq \oplus_m E^m$, and if e has degree zero, then the only change in comparison to the previous "partially graded" context is that instead of decomposing e as a sum $\sum_m e^m$, we can now write it as a genuine direct sum $e = \oplus_m e^m$, with $e^m : E^m \to A_0 \cap A^m$. Therefore, under the same hypothesis (G1) or (G2), for theories of **degree zero** we get exactly the same obstruction results. Due to the argument of the last paragraph we then have the following general obstruction result, which applies for graded theories of arbitrary degree.

Theorem 4.3. *Let M be an n-dimensional spacetime, $P \to M$ be a bundle such that TP is $\mathfrak{m}$-graded and $(A, *)$ be a $\mathfrak{m}$-graded algebra endowed with a vector space decomposition $A_0 \oplus A_1$. Given $l \geq 0$, assume that $A[-l]$ satisfies condition* (G1) *or* (G2). *Furthermore, let (k, s) be the minimum of (k_j, s_j). If $n \geq k + s + 1$, then for any compatible graded FABS and any trace transformation, the corresponding inhomogeneous A-valued graded EHP theory of degree l equals the homogeneous ones. If $n \geq k + s + 3$, then both theories are trivial.*

Assume that the $\mathfrak{m}$-graded algebra A is bounded from below, meaning that $\mathfrak{m}$ has a partial order $\leq$ and that there is $m_o \in \mathfrak{m}$ such that $A^m \simeq 0$ if $m < m_o$. Consider a $\mathfrak{m}$-graded theory of degree l taking values in that kind of algebra. In order to apply the last theorem we need to verify one of conditions (G1) or (G2). Such conditions are about $A[-l]$. This algebra has grading $A[-l]^m = A^{m+l}$. Because A is bounded, the first solvability condition falls on A^{m_o+l}, i.e, *no condition is needed between m_o and $m_o + l$.* This is, in essence, the new phenomenon obtained when we work with "full graded theories" in the sense introduced above. As a corollary:

Corollary 4.1. *In the same notation of the last theorem, assume that A is bounded from below and from above, respectively in degrees m_o and m_1. If l does not divide $m_1 - m_0$, then* (G1) *and* (G2) *are automatically satisfied and, therefore, for any graded bundle P the corresponding graded EHP theory of degree l is trivial. Otherwise, i.e, if $m_1 - m_0 = k.l$ for some k, then conditions* (G1) *and* (G2) *need to be fulfilled exactly for k terms.*

A particular consequence is the following:

Corollary 4.2. *Let $A \simeq A^0 \oplus A^1$ be a $\mathbb{Z}_2$-graded algebra, endowed with a vector space decomposition $A \simeq A_0 \oplus A_1$, where A_i is not necessarily A^i and A_0 is a subalgebra of A. If $A_0 \cap A^1$ is a weak (k, s)-solvable subspace of A_0, then any A-valued $\mathbb{Z}_2$-graded EHP theory of degree one over a n-dimensional spacetime M is trivial if $n \geq k + s + 3$.*

Examples will be explored in next section.

4.4 Some Speculation

When we look at the previous obstruction theorems, all of them (except Theorem 3.5) were conceived as abstractions of a single "fundamental obstruction theorem", namely Theorem 3.1, introduced in the most concrete situation: the *linear/matrix context*. Despite the fact that "derived obstruction theorems" hold in more abstract contexts, they are very closely related to the first one, in that they require the same kind of hypothesis on the underlying algebra: a "solvability condition". Furthermore, the more abstract the context is, the weaker the required "solvability condition" is. Indeed, Theorem 3.1 (for linear EHP theories) requires "(k,s)-nil condition", while Theorem 3.3 (for gauge EHP theories) requires "(k,s)-solv condition". Furthermore, Theorem 4.1 (for arbitrary A-valued EHP theories) requires "partially (k,s)-solv condition", in the sense that $A\simeq A_0\oplus A_1$, with A_0 (k,s)-solv, and Theorem 4.2 (for arbitrary graded-valued EHP theories) requires "locally (k,s)-solv condition", meaning that each A_0^m is weak (k,s)-solvable.

On the other hand, the same strategy used here can a piori be applied to get geometric obstructions for any classical theory (Conf,S) whose space of configurations Conf is some class of algebra-valued smooth forms. Let us call such theories *smooth forms theories*. However, recall that our strategy here was based in working first in the "linear context", where we identify a "fundamental algebra condition". Then, abstracting the context we could consider weaker conditions than the "fundamental" one. Therefore, in order to use this strategy in other results we need to find a corresponding "fundamental algebra condition". This leads us to speculate:

Conjecture (roughly)**.** *Any smooth forms theory admits a fundamental algebraic condition.*

For instance, in the way that it is stated, it is easy to verify that this conjecture is true for "polynomial smooth forms theories", i.e, for theories whose configuration space is $\mathrm{Conf}=\Lambda(P;A)$, for some algebra A endowed with a vector space decomposition $A\simeq A_1\oplus...\oplus A_k$, and whose action functional is

$$S[\alpha_1,...,\alpha_k]=\int_M \jmath(p_s(\alpha_1,...,\alpha_k)),$$

where $\jmath$ is some FABS and p_s is a polynomial of degree s in $\Lambda(P;A)$. Indeed, the desired condition is that p_s be a PI of A. Now, notice that EHP theories are "polynomial smooth forms theories of degree" in the above sense. Therefore, they have a "fundamental algebra condition" given by the vanishing of p_s. This is a *nilpotent condition*, which is much stronger than the (k,s)*-solv condition* previously established. This teaches us that the same smooth forms theory may admit two fundamental algebraic conditions, leading us to search for the optimal one:

Conjecture (roughly)**.** *Any smooth forms theory admits an optimal fundamental algebraic condition.*

We can go further and ask if there is some kind of "universal algebraic condition". More precisely, suppose a collection $\mathcal{C}$ of smooth forms theories satisfying last conjecture is given. Given two of those theories, if the optimal algebraic condition of one is contained in the optimal algebraic condition of the other, then both can be simultaneously trivialized. If this is not the case, the union of those optimal conditions will clearly trivialize them simultaneously. But, the union of optimal conditions is not necessarily the optimal one. This leads us to define the *optimal fundamental algebraic condition* of $\mathcal{C}$ as the weaker algebraic condition under which each classical theory in $\mathcal{C}$ becomes trivial, and to speculate its existence:

Conjecture (roughly)**.** *Any collection $\mathcal{C}$ of smooth forms theories admits an optimal fundamental algebraic condition.*

5 Examples

In the present section we will give realizations of the obstruction theorems studied previously. Due to the closeness with the genuine Lorentzian EHP theory, our focus is on the "concrete context", meaning that we will give many examples of "concrete geometries" which realize the obstruction theorems of Section 3. Even so, some examples for the "abstract context" of Section 4 will also be given.

5.1 Linear Examples

We start by considering the "fully linear" context of Subsection 3.1, i.e, EHP theories defined on a G-principal bundle $P \to M$ respective to a group reduction $H \hookrightarrow G$, where G is a linear group and M is a smooth n-manifold. The objects of interest are the classical reductive Cartan connections on P, i.e, pairs (e, ω) of 1-forms such that e takes values in $\mathfrak{g}/\mathfrak{h}$ and ω takes values in $\mathfrak{h}$. It follows from Corollary 3.1 and Theorem 3.4 that if $n \geq 6$ and

(E1) $H \subset G$ is normal, with $\mathfrak{g}/\mathfrak{h} \subset \mathfrak{so}(k_1) \oplus ... \oplus \mathfrak{so}(k_r)$, then the corresponding gauge EHP theory is trivial;

(E2) $\mathfrak{h} \subset \mathfrak{so}(k_1) \oplus ... \oplus \mathfrak{so}(k_r)$, then the geometric/algebraic dual EHP theories are trivial;

(E3) $\mathfrak{g} \subset \mathfrak{so}(k_1) \oplus ... \oplus \mathfrak{so}(k_r)$, then both EHP and the dual theories are trivial.

Conditions (E2) and (E3) are immediately satisfied if we identify k-groups, i.e, Lie subgroups of $SO(k_1) \times ... \times SO(k_r)$, or, equivalently, Lie subalgebras of $\mathfrak{so}(k_1) \oplus ... \oplus \mathfrak{so}(k_r)$. Indeed, if A is a such subgroup, then the EHP theory for $A \hookrightarrow G_A$, where G_A is any matrix extension of A, obviously satisfies (E2), so that if $n \geq 6$ the dual theories are trivial. On the other hand, if we consider EHP theories for reductions $H \hookrightarrow A$, then (E3) is clearly satisfied and in dimension $n \geq 6$ both the dual and the actual EHP theory are trivial.

Some obvious examples of subgroups of $A \subset SO(k)$, are given in the table below. Except for $Spin(4) \hookrightarrow O(8)$, which arises from the exceptional isomorphism $Spin(4) \simeq SU(2) \times SU(2)$, all the other inclusions already appeared in Subsection 3.5. Let us focus on condition (E2). In this case we can think of each element of Table 3 as included in some $GL(k; \mathbb{R})$, i.e, as a G-structure on a manifold and then as a geometry (recall Table 2).

(n,k)	$(1,k)$	$(2,k)$	$(4,k)$	$(1,7)$	$(1,8)$
$H \subset O(n,k)$	$SO(k)$	$U(k)$	$Sp(k)$	G_2	$Spin(4)$
	$U(k)$	$SU(k)$	$Sp(k) \cdot Sp(1)$		$Spin(7)$

Table 3: First classical examples of geometric obstructions

We could get more examples by taking finite products of arbitrary elements in the table. In terms of geometry, this can be interpreted as follows. Recall that a regular distribution of dimension k on an n-manifold can be regarded as a G-structure for $G = GL(k) \times GL(n-k)$. Therefore, we can think of a product $O(k) \times O(n-k)$ as a regular distribution of Riemannian leaves on a Riemannian manifold, $U(k) \times U(n-k)$ as a hermitan distribution, and so on.

Other special cases where condition (E2) applies are in table below. In the first line, $O(k,k)$ is the so-called *Narain group* [28], i.e, the orthogonal group of a metric with signature (k,k), whose maximal compact subgroup is $O(k) \times O(k)$. Second line follows from an inclusion similar as $U(k) \hookrightarrow O(2k)$, first studied by Hitchin and Gualtieri [16, 15, 17], while the remaining lines are particular cases of the previous ones. The underlying flavors of geometry arose from the study of Type II gravity and Type II string theory [8, 21]. That condition (E2) applies for the second and third lines follows from the fact that complexifying $U(k,k) \hookrightarrow O(2k,2k)$ we obtain $U(k,k) \hookrightarrow O(4k;\mathbb{C})$, as will be discussed in the next section.

G	H	geometry
$O(k,k)$	$O(k) \times O(k)$	Type II
$O(2k,2k)$	$U(k,k)$	Generalized Complex
$O(2k,2k)$	$SU(k,k)$	Generalized Calabi-Yau
$O(2k,2k)$	$U(k) \times U(k)$	Generalized Kähler
$O(2k,2k)$	$SU(k) \times SU(k)$	Generalized Calabi

Table 4: More examples of classical geometric obstructions

About these two tables, some remarks:

1. Table 3 contains any "classical" flavors of geometry, except symplectic geometry. The reason is that the symplectic group $\mathrm{Sp}(k;\mathbb{R})$ is not contained in some $O(r)$. But this does not mean that condition (E2) cannot be satisfied by symplectic geometry. Indeed, generalized complex geometry contains symplectic geometry [15], so that Table 4 implies that symplectic geometry fulfill condition (E2).

2. We could create a third table with "exotic k-groups", meaning that a priori they are not related to any "classical geometry", so that they describe some kind of "exotic geometry". For instance, in [30] all Lie subgroups $H \subset O(k)$ satisfying
$$\frac{(k-3)(k-4)}{2} + 6 < \dim H < \frac{(k-1)(k-2)}{2}$$
were classified and in arbitrary dimension k there are fifteen families of them. Other exotic (rather canonical, in some sense) subgroups that we could add are maximal tori. Indeed, both $O(2k)$ and $U(k)$ have maximal tori, say denoted by T_O and T_U, so that the reductions $T_O \hookrightarrow O(2k)$ and $T_U \hookrightarrow U(k)$ will satisfy condition E2. More examples of exotic subgroups to be added are the *point groups*, i.e, $H \subset \mathrm{Iso}(\mathbb{R}^k)$ fixing at least one point. Without loss of generality we can assume that this point is the origin, so that $H \subset O(k)$. Here we have the symmetric group of any spherically symmetric object in $\mathbb{R}^k$, such as regular polyhedrons and graphs embedded on $\mathbb{S}^{k-1}$.

3. If we are interested only in condition (E2), then the tables above can be enlarged by including embeddings of $O(k_1) \times \ldots \times O(k_r)$ into some other larger group $\tilde{G}$. Indeed, in this condition it only matters that H is a k-group. Particularly, $O(k)$ admits some exceptional embeddings (which arise from the classification of simple Lie algebras), as in the table below [?].

k	3	9	10	12	16
$O(k) \hookrightarrow$	G_2	F_4	E_6	E_7	E_8

Table 5: Exceptional embbedings of orthogonal groups.

Due to the above discussion, Tables 3 and 4, as well as the possible "table of exotic k-groups", are not only a source of examples for condition (E2), but also for condition (E3). Indeed, we can just consider G in condition (E3) as some "H" in the tables and take an arbitrary $H \subset G$. This produces a long list of examples, because G/H is a priori an arbitrary homogeneous space subject only to the condition that G is a k-group. In geometric terms, if a geometry fulfills condition (E2), then any induced "homogeneous geometry" fulfills (E3). On the other hand, differently from what happens with condition (E2), Table 5 *cannot* be used to get more examples of condition (E3). Indeed, if $H \hookrightarrow G$ is a reduction fulfilling (E3) and $G \hookrightarrow \tilde{G}$ is an embedding, then $H \hookrightarrow \tilde{G}$ fulfills (E3) iff it is satisfied by $G \hookrightarrow \tilde{G}$ (see diagram below).

$$\begin{tikzcd} H \arrow[r, hook] \arrow[rrr, bend right] & G \arrow[r, hook] & O(k_1) \times ... \times O(k_r) \arrow[r, hook] & \tilde{G} \end{tikzcd}$$

Let us now analyze condition (E1). First of all we notice that it is very restrictive, because we need to work with reductions $H \hookrightarrow G$ in which $H \subset G$ is normal. For instance, in the typical situations above, H is not normal. Even so, there are two dual conditions under which $H \hookrightarrow G$ fulfills (E1): when G is a k-group with $G/H \subset G$ and, dually, when H is k-group with $G/H \subset H$. In the first situation, we are just in condition (E3) for $G/H \hookrightarrow G$, while in the second we are in condition (E3) for $G/H \hookrightarrow H$. Therefore, there are not many new examples here.

Conclusion. *Assume $n \geq 6$. So, for any compatible FABS:*

1. *geometric/algebraic dual EHP is trivial in each geometry modeled by tables above;*
2. *the actual gauge EHP is trivial in each homogeneous geometry associated to the first two tables above.*

5.2 Extended-Linear Examples

In the last subsection we constructed explicit examples of geometric obstructions for linear EHP theories and their duals. Let us now consider extended-linear theories. This means that we will work with bundles $P \to M$ structured over $\mathbb{R}^k \rtimes H$, where H is a linear group and M is a n-dimensional smooth manifold. Furthermore, we will take into account reductive connections for the group reduction $H \hookrightarrow \mathbb{R}^k \rtimes H$. Our obstruction theorem is now Corollary 3.3, which implies that if $n \geq 6$ and

- $\mathfrak{h} \subset \mathfrak{so}(k_1) \oplus ... \oplus \mathfrak{so}(k_r)$, then EHP action and its duals are trivial.

Notice that this is exactly the same condition as (E2). This means that *if we are working in the extended-linear context, in dimension $n \geq 6$ EHP theory cannot be realized in any geometry of the last subsection.*

Remark 5.1. Recall that it is in the extended-linear context, with $P = FM$, that "abstract EHP theory" becomes very close to the concrete one, in that the geometries of the last subsection really describe geometries (in the most classical sense, in terms of tensors) on the manifold M. Therefore, the fact that EHP is trivial in a lot of situations is a strong manifestation that the geometry of gravity is very rigid, especially in higher dimension.

5.3 Cayley-Dickson Examples

In the last two sections we presented obstructions to the realization of linear/extended-linear (dual) EHP theories in commonly studied geometries, as well as in some "exotic" geometries. Here we would like to show that there are also obstructions for less studied (and some never studied in detail) kinds of geometry. So, for now, let $(A, *)$ be an arbitrary $\mathbb{R}$-algebra and let $\mathrm{Mat}_{k\times l}(A)$ be the $\mathbb{R}$-module of $k \times l$ matrices with coefficients in A. The multiplication of A induces a corresponding multiplication

$$\cdot : \mathrm{Mat}_{k\times l}(A) \times \mathrm{Mat}_{l\times m}(A) \to \mathrm{Mat}_{k\times l}(A).$$

In particular, for $k = 0 = m$ we see that for every l we have a bilinear map

$$b : A^l \times A^l \to A, \quad \text{given by} \quad b(x, y) = x_1 * y_1 + ... + x_l * y_l.$$

where we used the identifications

$$\mathrm{Mat}_{0\times l}(A) \simeq A^l \simeq \mathrm{Mat}_{l\times 0}(A) \quad \text{and} \quad \mathrm{Mat}_{0\times 0}(A) \simeq A.$$

Remark 5.2. The bilinear map above is symmetric iff the algebra A is commutative. Furthermore, its non-degeneracy depends on whether A has zero divisors or not, and a priori it is not possible to ask about its positive definiteness, because it takes values in A and not in $\mathbb{R}$. Thus, it is **not** an inner product in $\mathrm{Mat}_{0\times l}(A)$. But, if we choose $A = \mathbb{R}$, then it is the standard inner product of $\mathbb{R}^l$. Now, assume that A is endowed with an involution $\overline{(-)} : A \to A$. In this case, it can be combined with b to get a sesquilinear map s in A^l, as in the diagram below.

$$\begin{tikzcd} A^l \times A^l \arrow[r, "\overline{(-)}^l \times id"] \arrow[rr, bend right, "s"'] & A^l \times A^l \arrow[r, "b"] & A \end{tikzcd}$$

Explicitly,

$$s(x, y) = \overline{x}_1 * y_1 + ... + \overline{x}_l * y_l.$$

We can now consider the subspace of all $l \times l$ matrices $M \in \mathrm{Mat}_{l\times l}(A)$ with coefficients in A which preserve the sesquilinear form s, in the sense that $s(Mx, My) = s(x, y)$. We say that these are the *unitary matrices in* A, *respective to the involution* s *induced by the involution* $\overline{(-)}$, and we denote this set by $U(k; A)$. If the involution is trivial (i.e, the identity map), then call them the *orthogonal matrices in* A, writing $O(k; A)$ to denote the corresponding space.

Example 5.1. If we consider $A = \mathbb{R}, \mathbb{C}$ and the trivial involution we get, respectively, the real and the complex orthogonal groups $O(k)$ and $O(k; \mathbb{C})$. If we consider the canonical involution of $\mathbb{C}$, we get the unitary group $U(k)$, while if we consider $\mathbb{H}$ with its canonical involution we get the quaternionic unitary group $U(k; \mathbb{H})$, which already have appeared in Tables 2 and 3 with the more used notation $Sp(k)$.

The last example leads us to compare different unitary groups of involutive algebras which are related by the Cayley-Dickson construction discussed in Example 2.1. Indeed, recall that this construction takes an involutive algebra A and gives another involutive algebra $\mathrm{CD}(A)$ with weakened PI's. As an R-module, the newer algebra is given by a sum $A \oplus A$ of "real" and "imaginary" parts. We have an inclusion $A \hookrightarrow \mathrm{CD}(A)$, obtained by regarding A as the real part, which induces an inclusion into the corresponding unitary groups $U(k; A) \hookrightarrow U(k; \mathrm{CD}(A))$. This inclusion can be extended in the following way

$$U(k; A) \hookrightarrow U(k; \mathrm{CD}(A)) \hookrightarrow U(2k; A),$$

defined by setting the "real part" and the "imaginary part" as diagonal block matrices. Iterating we see that for every k and every l there exists an inclusion

$$U(k; \mathrm{CD}^l(A)) \hookrightarrow U(2^l k; A). \tag{23}$$

Example 5.2. If we start with $A = \mathbb{R}$ endowed with the trivial involution, the first inclusion is $U(k) \hookrightarrow O(2k)$, which describes complex geometry. The next iteration gives $Sp(k) \hookrightarrow O(4k)$, which is quaternionic geometry. These are precisely the first lines of Table 3, but we can continue getting octonionic geometry, sedenionic geometry, etc.

Recalling that geometry can be regarded as the inclusion of Lie groups $H \hookrightarrow G$, with the previous examples in ours minds, the idea is to think of inclusion (23) as some flavor of geometry, which we could name *Cayley-Dickson geometry in A, of order l*. But, this makes sense only if the unitary groups in (23) are Lie groups. We notice that this is the case at least when A is finite-dimensional, as it will be sketched now.

The idea is to reproduce the proof that the standard unitary/orthogonal groups $U(k)$ and $O(k)$ are Lie groups. The starting point is to notice that the involution of A induces an involution in $\mathrm{Mat}_{k\times k}(A)$, defined by the following composition, where t is the transposition map.

$$\begin{tikzcd} \mathrm{Mat}_{k\times k}(A) \arrow[r, "t"] \arrow[rr, bend right, "(-)^{\dagger}"'] & \mathrm{Mat}_{k\times k}(A) \arrow[r, "\overline{(-)}"] & \mathrm{Mat}_{k\times k}(A) \end{tikzcd}$$

We then notice that a $k \times k$ matrix M with coefficients in A is unitary respective to s iff M is invertible with $M^{-1} = M^{\dagger}$, i.e, iff $M \cdot M^{\dagger} = 1_k = M^{\dagger} \cdot M$. Since $\overline{(-)} : A \to A$ is an algebra morphism, we have the usual property $(M \cdot N)^{\dagger} = M^{\dagger} \cdot N^{\dagger}$, allowing us to characterize the unitary matrices as those satisfying $M \cdot M^{\dagger} = 1_k$. Therefore, defining the map

$$f : \mathrm{Mat}_{k\times k}(A) \to \mathrm{Mat}_{k\times k}(A) \quad \text{by} \quad f(M) = M \cdot M^{\dagger},$$

in order to proof that $U(k; A)$ is a Lie group it is enough to verify that the above map is, in some sense, a submersion (which will give the smooth structure) and that the multiplication and inversion maps are "smooth". It is at this point that we require that A to be finite-dimensional.

Now, we can return to the context of Geometric Obstruction Theory. Taking $A = \mathbb{R}$, for every k and l the discussion above gives a sequence of Lie group inclusions

$$U(k; \mathrm{CD}^l(\mathbb{R})) \longrightarrow \cdots \longrightarrow U(2^{l-3}k; \mathbb{O}) \longrightarrow U(2^{l-2}k; \mathbb{H}) \longrightarrow U(2^{l-1}k; \mathbb{C}) \longrightarrow O(2^l k)$$

which imply that (in spacetime dimension $n \geq 6$) EHP theory and its duals are trivial for each of these geometries (because they satisfy (E2)). In sum, Table 3 can be augmented to include Cayley-Dickson geometries in $\mathbb{R}$ of arbitrary order, as shown below.

G	H	geometry
$\mathbb{R}^k \rtimes H$	$O(k)$	Riemannian
$\mathbb{R}^{2k} \rtimes H$	$U(k;\mathbb{C})$	hermitean
$\mathbb{R}^{4k} \rtimes H$	$U(k;\mathbb{H})$	quaternionic
$\mathbb{R}^{8k} \rtimes H$	$U(k;\mathbb{O})$	octonionic
$\mathbb{R}^{16k} \rtimes H$	$U(k;\mathbb{S})$	sedenionic
$\vdots$	$\vdots$	$\vdots$
$\mathbb{R}^{2^l k} \rtimes H$	$U(k;\mathrm{CD}^l(\mathbb{R}))$	real CD of order l
$\vdots$	$\vdots$	$\vdots$

Table 6: Extended examples of concrete geometric obstructions

In Subsections 5.1 and 5.2 we showed that EHP theory cannot be realized in the most classical flavors of geometry. These, however, constitute a *finite* amount. As a corollary of the construction above, we now know that EHP theory is actually trivial in an *infinite* number of geometries:

Corollary 5.1. *In dimension $n \geq 6$, the EHP theory and its duals cannot be realized in an infinite number of geometries.*

Table 6 can, in turn, be extended in three different directions:

1. *by adding distinguished subgroups*. For every involutive algebra A we can define the subgroup $SU(k;A) \subset U(k;A)$ of those matrices whose determinant equals the identity $1 \in A$. When A is finite-dimensional, it will be a Lie subgroup, allowing us to include $SU(k;\mathrm{CD}^l(\mathbb{R}))$ for every $l \geq 0$ in Table 6. This means that if a geometry belongs to the "table of obstructions", then its "oriented version" belongs too;

2. *by replacing the base field*. Notice that condition $\mathfrak{h} \subset \mathfrak{so}(k_1) \oplus ... \oplus \mathfrak{so}(k_r)$ was used above only in order to have $\alpha \curlywedge \alpha = 0$ for every even-degree $\mathfrak{h}$-valued form, i.e, to ensure that $\mathfrak{h}$ is $(k,1)$-nil for each k even. In turn, the condition $\alpha \curlywedge \alpha = 0$ is satisfied exactly because $\mathfrak{so}(n)$ is an algebra of skew-symmetric matrices. But this remains valid for the Lie algebra $\mathfrak{so}(n;A)$ of $SO(n;A)$, independently of the involutive algebra. This means that we can replace $\mathbb{R}$ by any algebra A in Table 6. New interesting examples that arise from this fact are the following. For given $p,q > 0$, there is no k such that $O(p,q) \subset O(k)$, so that a priori we cannot add $O(p,q)$ to Table 6. On the other hand, after complexification, i.e, after replacing $\mathbb{R}$ by $\mathbb{C}$ we have $O(p,q) \otimes_{\mathbb{R}} \mathbb{C} \simeq O(p+q) \otimes_{\mathbb{R}} \mathbb{C}$ for every $p,q \geq 0$, so that they immediately enter in the "obstruction table". This means that we can add to Table 6 "complex semi-Riemannian geometry" and, similarly, "quaternionic semi-Hermitean geometry", and so on.

3. *by making use of Lie theory*. The condition that we need is on the **Lie algebra** level. It happens that in general there are many groups with the same algebra. Therefore, once we find a Lie group whose algebra fulfills what we need, we can automatically add to our "table

of obstructions" every other Lie group that induces the same algebra. In particular, we can double the size of our current table by adding the universal cover (when it exists) of each group. For instance, in Tables 3 and 6 the spin groups Spin(4) and Spin(7) were added due to exceptional isomorphisms on the level of **Lie groups**. Now, noticing that for $k > 2$ (in particular for $k = 4, 7$) Spin(k) is the universal cover of $SO(k)$, we can automatically add all theses spin groups to our table, meaning that the dual EHP cannot be realized in "spin geometry". Similarly, we can add the universal coverings of the (connected component at the identity of) $U(k;\mathrm{CD}^l(\mathbb{R}))$. We can also add the universal cover of the symplectic group $Sp(k;\mathbb{R})$, usually known as the *metapletic group*. Therefore, dual EHP cannot be realized in "metapletic geometry" too.

Up to this point we gave examples which extend Table 3. We notice, however, that Table 4 can also be extended. Indeed, for every A we have the inclusion $U(k,k;\mathrm{CD}(A)) \subset O(2k;2k,A)$, so that by iteration we get

$$U(k,k;\mathrm{CD}^l(A)) \subset O(2^l k; 2^l k, A). \tag{24}$$

For $A = \mathbb{R}$ and $l = 1$ this model generalized complex geometry, leading us to say that the inclusion above models *"generalized Cayley-Dickson geometry in A of degree l"*. For instance, if we take $A = \mathbb{R}$ and $l = 2$ this becomes generalized quaternionic geometry, which is a poorly studied theory, started with the works [31, 10] . For $l = 3, 4, ...$ it should be "generalized octonionic geometry", "generalized sedenionic geometry", and so on. The authors are unaware of the existence of substantial works on these theories.

When tensoring inclusion (24) with $\mathrm{CD}^l(A)$ we get

$$U(k,k;\mathrm{CD}^l(A)) \subset O(2^l k; 2^l k, \mathrm{CD}^l(A)) \simeq O(2^l k + 2^l k; \mathrm{CD}^l(A)),$$

so that condition 2 above implies that, independently of the present development of abstract generalized Cayley-Dickson geometries, EHP theory (and its duals) is trivial in each of them.

Conclusion: *In dimension $n \geq 6$ and for any FABS, extendend-linear EHP theory and its duals cannot be realized in a lot of geometries, which include:*

1. *real Cayley-Dickson geometries of all orders, their "connected components of the identity" and their universal covering geometries;*

2. *generalized Cayley-Dickson geometries of all orders;*

3. *the exceptional geometry G_2.*

5.4 Partially Abstract Examples

The last examples were considered "concrete" because the algebras underlying them are matrix algebras over the real numbers. So, in order to give non-concrete examples it is enough to work with algebras which are not matrix algebras "and/or" which have coefficient rings other than $\mathbb{R}$. Let us call the "or" examples *partially abstract examples.*

The interesting part of the partially abstract situations is that we can give abstract examples of obstructions without using the abstract theorems of Section 4. For instance, some obstructions for non-real matrix algebras where actually given in last subsection in "*by replacing the base-field*".

Indeed, there we considered situations in which the new field $\mathbb{K}$ arises as an extension of $\mathbb{R}$ and the matrix algebra $A(k;\mathbb{K})$ in consideration is indeed a scalar extension $A(k;\mathbb{K}) = A(k;\mathbb{R}) \otimes_{\mathbb{R}} \mathbb{K}$. But, since $\mathbb{R}$ is characteristic zero, it follows that every extension $\mathbb{K} \supset \mathbb{R}$ is also characteristic zero, allowing us to ask: *can we give examples when $\mathbb{K}$ has prime characteristic?*

In [3] necessary and sufficient conditions were given under which arbitrary orthogonal groups $O(q,\mathbb{K})$, where $q : V \to \mathbb{K}$ is a positive-definite quadratic form on a finite-dimension $\mathbb{K}$-space q and $\mathrm{ch}(\mathbb{K}) \neq 2$, admit an embedded maximal torus $\mathbf{T}(q;\mathbb{K})$. Independently of the quadratic space (V,q), the corresponding orthogonal group is a Lie group and, as in the real case, the algebra $\mathfrak{o}(q;\mathbb{K})$ is $(2,1)$-nil. Therefore, under the conditions of [3], *EHP theory cannot be realized in any of these "toroidal geometries"* $\mathbf{T}(q;\mathbb{K}) \hookrightarrow O(q;\mathbb{K})$. Notice that for $\mathbb{K} = \mathbb{R}$ the same result appeared in 5.1 as a "exotic linear example".

5.5 Fully Abstract Examples

Finally, we give "abstract examples" of geometries in which EHP theory cannot be realized. First we deal with algebra-valued geometry, with graded geometry considered in the sequence. This means that we work with a bundle $P \to M$ endowed with A-valued connections $\nabla : TP \to A$, where A is a graded a$\mathfrak{m}$-graded algebra $A \simeq \oplus_m A_m$. The obstruction theorems are now Theorems 4.1, 4.2 and 4.3. In summary, if

(F1) *the algebra A admits a vector space decomposition $A \simeq A_0 \oplus A_1$, where A_0 is a subalgebra such that each $A_0^m = A_0 \cap A^m$ is a weak (k_m, s_m)-solvable subspace*, then EHP theory is trivial in dimension $n \geq k + s + 1$, where $(k,s) = \min(k_m, s_m)$.

The most basic examples are those for $\mathfrak{m} = 0$ and $A_1 = 0$, i.e, the non-graded setting with A itself (k,s)-solv. As discussed in Subsections 2.3 and 2.4, there are many natural examples of (k,s)-solv algebras, e.g, any Lie algebra is $(k,1)$ for any even k. As a consequence, *in dimension $n \geq 4$, any Lie algebra valued EHP theory is trivial.* In particular, EHP theories with values in the Poincaré group $\mathfrak{iso}(n-1,1)$ are trivial. But, $\mathfrak{iso}(n-1,1)$-valued EHP theory is just classical EHP theory, thus we conclude that General Relativity does not makes sense in dimension $n \geq 4$, which is absurd. We made a similar mistake in Example 2.3: while the classical EHP theory and the $\mathfrak{iso}(n-1,1)$-valued EHP theory take values in the same **vector space**, at the same time that their action functionals have the same shape, the **algebra** (and, therefore, its properties) used to define the corresponding wedge product is *totally different.* Indeed, in the discussed cases, the exterior products $\curlywedge_{\rtimes}$ and $[\wedge]$, respectively.

If we now allow $\mathfrak{m}$ to be nontrivial, but with $A_1 = 0$, then condition (5.5) is satisfied if each A^m is a weak (k_m, s_m)-solvable subespace. In particular, it remains satisfied if A is itself (k,s)-solv. It happens that not only Lie algebras are solv $(2,1)$-nil, but also a class of graded Lie algebras. Consequently, EHP theories are also trivial in the domain of graded Lie algebras. One can generalize even more thinking in EHP theories with values in Lie superalgebras and in graded Lie superalgebras. Indeed, a $\mathfrak{m}$-*graded Lie superalgebra* $\mathfrak{g}$ is just a $\mathfrak{m}$-graded Lie algebra whose underlying PI's (i.e, skew-commutativity and Jacobi identity) hold in the graded sense. Particularly, this means that the $\mathbb{Z}_2$-grading writes $\mathfrak{g} \simeq \mathfrak{g}^0 \oplus \mathfrak{g}^1$, with $\mathfrak{g}^0$ a $\mathfrak{m}$-graded Lie algebra and, therefore, $(2,1)$-nil. It then follows that $\mathfrak{g}$ satisfies condition F1. Summarizing, *EHP theories cannot be realized in any "Lie algebraic" context.*

Graded Lie Superalgebras are the first examples of algebras satisfying (F1) with $A_1 \neq 0$, but they are far from being the only one. Indeed, when we look at a decomposition $A \simeq A_0 \oplus A_1$, where

A_0 is a subalgebra, it is inevitable to think of A as an extension of A_1 by A_0, meaning that we have an exact sequence as shown below. If A_0 is (k,s)-solv, then (F1) holds. This can be interpreted as follows: *suppose that we encountered an algebra A_1 such that EHP is not trivial there. So, EHP theory will be trivial in any (splitting) extension of A_1 by a (k,s)-solv algebra.*

$$0 \longrightarrow A_0 \longrightarrow A \longrightarrow A_1 \longrightarrow 0$$

In particular, because $(\mathbb{R}^k,+)$ is abelian and, therefore, $(1,1)$-nil, any algebra extension by $\mathbb{R}^k$ will produce a context in which EHP theory is trivial. For instance, recall the extended-linear context, which was obtained taking splitting extensions of matrix algebras by $\mathbb{R}^k$. Thus, EHP theory is trivial in every extended-linear context, so that classical EHP is trivial. This implies tha GR is trivial, showing that *we made another mistake*. Once again, *the mistake resides on the underlying "wedge products": the wedge product induced on an algebra extension by the abelian group* $(\mathbb{R}^k,+)$ *is* ***not*** *the wedge product $\curlywedge_{\rtimes}$ studied in Subsection 2.2*. Indeed, $\curlywedge_{\rtimes}$ does **not** take the abelian structure of $\mathbb{R}^k$ into account.

Another example about extensions is as follows: the paragraph above shows that abelian extensions of nontrivial EHP theories are trivial, but what about "super extensions"? They remain trivial. Indeed, being a Lie superalgebra, the translational superalgebra $\mathbb{R}^{k|l}$ (the cartesian superspace, regarded as a Lie superalgebra) is $(2,1)$-nil, so that any algebra extension by it is trivial.

Finally, let us say that we can also consider "fully exotic abstract examples" meaning geometries modeled by algebras fulfilling (F1), but that have no physical meaning. Just to mention, in [?] the authors build "mathematically exotic" examples of nilpotent algebras, which fulfill (F1). If, for any reason, one tries to model gravity as EHP with values in those algebras, one will find a trivial theory.

6 Conclusion

In this notes, based on [26], we considered EHP action functional in different contexts and we gave obstructions to realize gravity (modeled by these EHP actions) in several geometries. In particular, we showed that EHP cannot be realized in almost all "classical geometries", including Riemannian geometry, hermitean geometry, Kähler geometry, generalized complex geometry and many extensions of them, as well as some exceptional geometries, such as G_2-geometry. We also introduced the notions of geometric/algebraic duals of an EHP theory and we have show that many obstructions also affect them. A physical understanding of these "dual theories" is desirable.

In this process of finding geometric obstructions for EHP theories, we identified a "general obstruction", corresponding to a "solvability condition" on the underlying algebra, which led us to speculate on the existence of a "general obstruction" for each gauge theory and of a "universal obstruction", unifying all such "general obstructions".

The speculations and the obstruction theorems developed here are independent of the choice of a *Functional Algebra Bundle System*, a concept conceived in the present work, which codifies the data necessary to pass from A-valued forms on a bundle $P \to M$ to bundle-valued forms on M. A specific study of the category of these objects is desirable (we have some work in developing stage [27]). For instance, which kind of limits/colimits exist in such category? It has at least initial objects (corresponding to universal FABS)?

We believe that the main contribution of the present work is to clarify that *when working with classical theories defined by algebra-valued differential forms, one needs to be very careful about the* *"wedge products" used in the action funcional, given that the properties of the underlying algebra deeply affect the properties of the corresponding wedge product.*

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5

EMERGENCE AND EXTENSIONS

5.1 SYNTHESIS

As discussed in Chapter 2, it is suggestive to use system theory to talk about Philosophy, as was extensively developed in the case of Philosophy of Science by Bunge in [30]. We also saw that the philosophical foundations of an area of study $\mathcal{O}$ (viewed as a class of systems) is made, among other things, by fixing ontological, epistemological, and logical theories for it.

For instance, recall that an epistemological theory on $\mathcal{O}$ consists of a class $\mathrm{Belief}(\mathcal{O})$ of believes, subclasses $\mathrm{Just}(\mathcal{O}) \subset \mathrm{Belief}(\mathcal{O})$ of justified believes and $\mathrm{True}(\mathcal{O}) \subset \mathrm{Belief}(\mathcal{O})$ of true believes, as well as a class $\mathrm{Know}(\mathcal{O}) \subset \mathrm{Just}(\mathcal{O}) \cap \mathrm{True}(\mathcal{O})$ of knowledge, and similarly for ontological and logical theories, which can be summarized in diagram (2.1), which for simplicity we will denote by $D(\mathcal{O})$.

Notice, however, that diagram (2.1) and the stuff characterizing ontological, epistemological and logical theories, are assigned to a fixed area o study $\mathcal{O}$, which is a collection of systems. Each system $X \in \mathcal{O}$, in turn, has its own subsystems which should characterize another area of study $\mathcal{O}_x \subset \mathcal{O}$. One could think in use the same philosophical theories in $\mathcal{O}$ to build philosophical foundations for $\mathcal{O}_x$, so that we have a corresponding diagram $D(\mathcal{O}_x)$ for every $x \in \mathcal{O}$, as below.

$$
\begin{tikzcd}
 & & (\mathcal{O}_x)_{real} \arrow[dr, hook] & (\mathcal{O}_x)_{mind} \arrow[d, hook] & (\mathcal{O}_x)_{abs} \arrow[dl, hook'] & \\
 & & & \mathcal{O}_x \arrow[dll, hook'] \arrow[drr, hook] & & \\
 & \mathrm{Belief}(\mathcal{O}_x) & & & & \mathrm{Prop}(\mathcal{O}_x) \\
\mathrm{Just}(\mathcal{O}_x) \arrow[ur, hook] & \mathrm{Know}(\mathcal{O}_x) \arrow[l, hook'] \arrow[u, hook] \arrow[r, hook] & \mathrm{True}(\mathcal{O}_x) \arrow[ul, hook'] & & & \mathrm{Logic}(\mathcal{O}_x) \arrow[u, hook]
\end{tikzcd}
\tag{5.1}
$$

Varying $x \in \mathcal{O}_x$ we get a a family $(\mathcal{O}_x)_{x \in \mathcal{O}}$ of system and a corresponding family of diagrams $(D(\mathcal{O}_x))_{x \in \mathcal{O}}$, describing the induced philosophical theories in each subarea of study. Notice that the union of $(\mathcal{O}_x)$ gives back $\mathcal{O}$. The natural question is, therefore:

- *how the collection of diagrams (5.1) relates with (2.1)? In other words, can we recover the philosophical foundations of an area of study from the philosophical foundations of its subareas?*

An answer to this question can be understood as an additional hypothesis on the foundations of $\mathcal{O}$. *Reductionism* asserts an affirmative answer, while *emergentism* supports a negative answer - see [30] for a definition of emergence in systemic. Thus:

1. *ontological emergentism* corresponds to saying, for instance, that $\cup_x(\mathcal{O}_x)_{real} \neq \mathcal{O}_{real}$, i.e., that being real when belonging to the parts is not equivalent to being real when inside the whole;
2. *ontological reductionism* states that $\cup_x(\mathcal{O}_x)_{real} \simeq \mathcal{O}_{real}$, i.e., that being real reduces to being real in the parts;
3. *epistemological emergentism* presupposes that $\cup_x \operatorname{Know}(\mathcal{O}_x) \neq \operatorname{Know}(\mathcal{O})$, i.e., that the knowledge of the parts is not equivalent to the knowledge of the whole;
4. *epistemological reductionism* assumes that $\cup_x \operatorname{Know}(\mathcal{O}_x) \simeq \operatorname{Know}(\mathcal{O})$, i.e., that it is enough to have the knowledge on the parts;
5. *logical emergentism* says that $\cup_x \operatorname{Logic}(\mathcal{O}_x) \neq \operatorname{Logic}(\mathcal{O})$ a sentence being logically true when restricted to the parts does not necessarily implies that it is globally true;
6. *logical emergentism*, in which a $\cup_x \operatorname{Logic}(\mathcal{O}_x) \simeq \operatorname{Logic}(\mathcal{O})$, so that it is enough to prove locally truth in the parts.

Thus, the emergentistic hypothesis states that reality, epistemological truth and logical truth are *global properties*, while reductionism asserts that they are local.

5.1.1 Stabilization Hypothesis

Let $\mathcal{O}$ be a class of systems and for every system $x \in \mathcal{O}$, let $\mathcal{O}_x$ its class of subsystems. For every $y \in \mathcal{O}_x$ one could take the class $(\mathcal{O}_x)_y$ of subsystems of the subsystem y. In turn, given $z \in (\mathcal{O}_x)_y$ one could considers $(\mathcal{O}_x)_z$, and so on. A natural question is: *this inductive process stabilizes at some point?*. More precisely, for a given area of study $\mathcal{O}$, define its *fundamental series* $\mathcal{O}^{(k)}$, with $k \geq 0$, as follows: $\mathcal{O}^0 := \mathcal{O}$, and $\mathcal{O}^{(k)} := \{\mathcal{O}_x^{(k-1)} | x \in \mathcal{O}^{(k-1)}\}$ for every $k \geq 1$. Thus:

- *is there some k such that if $i \geq k$, then $\mathcal{O}^{(i)} \simeq \mathcal{O}^{(k)}$?*

This stabilization hypothesis is also part on the philosophical foundations of $\mathcal{O}$. Indeed, notice that if it is assumed the existence of such $k \geq 0$, then the members of $\mathcal{O}^{(k)}$ plays the role of fundamental parts or "building blocks" of $\mathcal{O}$. This could be understood as some kind of *fundamentalistic* view[1] [15].

Another question concerning subsystems is how subsystems of subsystems of a system x are related with subsystems of x. More precisely, given $x \in \mathcal{O}$ and $y \in \mathcal{O}_x$, *what is the relation between $(\mathcal{O}_x)_y$ and $\mathcal{O}_x$?* In other words, *how the higher-order terms of the fundamental series $\mathcal{O}^{(k)}$ relates with the lower-order terms $\mathcal{O}^{(l)}$, with $l < k$?*

Again, this is on the philosophical foundations of $\mathcal{O}$. A typical assumption is that each member in $(\mathcal{O}_x)_y$ can itself be regarded as member in $\mathcal{O}_x$, so that we have inclusions $\mathcal{O}^{(k)} \hookrightarrow \mathcal{O}^{(l)}$ for $l \leq k$, suggesting us to think of the fundamental series of $\mathcal{O}$ as a inverse system and the "fundamental parts", if they exists, as belonging to the inverse limit $\varprojlim_k \mathcal{O}^{(k)}$.

[1] In the sense of accepting *fundamentality* and not *fundamentalism* - the latter is used in the context of Philosophy of Religion.

Now, notice that the questions on how the philosophical foundations of $\mathcal{O}_x$ relates to the philosophical foundations of $\mathcal{O}$ and how $(\mathcal{O}_x)_y$ relates with $\mathcal{O}_x$ are different in nature. Thus, they could be considered at the same time. In particular, *one could assume an emergentistic/reductionistic approach to a fundamentalist area of study*. The conclusions in both cases are widely different:

1. *fundamentalist reductionism* asserts that there are fundamental entities and whose ontology/epistemology/logic determine the entire philosophical theory of $\mathcal{O}$;

2. *fundamentalist emergentism* argue that, although the fundamental entities exist, its ontological/epistemological/logical aspects are not enough to determine the philosophical theory of $\mathcal{O}$.

Remark 5.1. A discussion on emergence ontology under the fundamentalist hypothesis with this level of generality is in [15].

5.1.2 Effectiveness Hypothesis

Recall the above discussion on fundamentalistic philosophical theories, which relies on the *stabilization hypothesis*, stating that the fundamental series $\mathcal{O}^{(k)}$ stabilizes for some k largely enough. A way to formalize this was assuming that if $k \leq l$, then $\mathcal{O}^{(k)} \subset \mathcal{O}^{(l)}$, to that for larger k we have smaller $\mathcal{O}^{(k)}$, allowing us to take inverse limits.

Notice that the above formalization of a fundamentalistic area of study was based on the assumption that the systems in $\mathcal{O}$ can be organized in a specific way, allowing us to compare them. One could be considered areas of study organized in different ways.

Definition 5.2. Let $\mathcal{O}$ be a class of systems. An *effective structure* on $\mathcal{O}$ is given by:

1. a class Λ in which varies a parameter λ, called *fundamental scale* (or *effective scale*);

2. to each $\lambda \in \Lambda$ a subclass $\mathcal{O}_\lambda \subset \mathcal{O}$.

Definition 5.3. An *increasing effective structure* (resp. *decreasing effective structure*) is an effective structure with a relation $\leq$ in Λ, called *fundamental relation* (or *effectiveness relation*), such that $\lambda \leq \lambda'$, implies $\mathcal{O}_\lambda \subset \mathcal{O}_{\lambda'}$ (resp. $\mathcal{O}_{\lambda'} \subset \mathcal{O}_\lambda$).

Definition 5.4. An *effective approach to the foundations of* $\mathcal{O}$ is obtained by fixing an effective structure on $\mathcal{O}$. The effective approach is said to be *increasing* or *decreasing* if the underlying effective structure is increasing or decreasing.

Remark 5.5. The existence of an effective structure (and therefore the possibility to follow an effective approach) is another hypothesis to be added in a philosophical theory for $\mathcal{O}$: the *effectiveness hypothesis*.

Notice that in an increasing (resp. decreasing) effective approach to an area of study $\mathcal{O}$ we can view the family $(\mathcal{O}_\lambda)_{\lambda\in\Lambda}$ as an indirect (resp. direct) system, so that its limit (resp. colimit), if it exists, can be considered as a generalization of the fundamental system described in the last section.

On the other hand, even if this limit/colimit does not exist (or even if we does not worry about its existence), an effective structure can be used as an "organizational principle" to study $\mathcal{O}$, which in the increasing or decreasing case in an "hierarchical organizational principle". This motivates some authors to consider effective structures as an investigation method for the philosophical foundations of $\mathcal{O}$ instead of as part of the proper foundations [30].

Thinking of the effectiveness hypothesis as a generalization of the fundamentalistic hypothesis, the fundamentalistic emergentistic and fundamentalistic reductionistic approaches, as well as the ontological/epistemological/logical reductionism/emergentism, also generalizes, as follows:

1. *effective ontological reductionism (resp. emergentism)* corresponds to saying, for instance, that $\cup_\lambda(\mathcal{O}_\lambda)_{real}$ is equivalent to (resp. is different from) $\mathcal{O}_{real}$, i.e., that being real when belonging to the different fundamental scales is the same thing (resp. is not equivalent to) being real when inside the whole;

2. *effective epistemological reductionism (resp. emergentism)* assuming that $\cup_\lambda \mathrm{Know}(\mathcal{O}_\lambda)$ is not equal to (resp. is equal to) $\mathrm{Know}(\mathcal{O})$, i.e., that the knowledge of the fundamental scales is equivalent (resp. is not equivalent to) the knowledge of the whole;

3. *effective logical reductionism (resp. emergentism)*, which says that $\cup_\lambda \mathrm{Logic}(\mathcal{O}_\lambda)$ is just (resp. is not) $\mathrm{Logic}(\mathcal{O})$, i.e., that a sentence being logically true when restricted to the parts implies (resp. does not necessarily implies) that it is globally true.

5.1.3 Relations Between Effective Structures

Let $\mathcal{O}$ be a class of systems. Under the effective hypothesis, it admits an effective structure $(\mathcal{O}_\lambda)_{\lambda\in\Lambda}$, but *what about the uniqueness of such way to organize the systems in X?* More precisely, *given two different effective structures, are they related? In which sense?*

Recall that every partially ordered set (poset) defines a category. In particular, every class X defines a category: that of the poset $(\rho(X),\subseteq)$. Thus, each area of study $\mathcal{O}$ defines a category. If $(\Lambda,\leq)$ is a poset, then an increasing (resp. decreasing) effective structure with effective scale in $(\Lambda,\leq)$ is equivalent to a covariant (resp. contravariant) functor $F:(\Lambda,\leq)\to(\rho(\mathcal{O}),\subseteq)$. In these case, therefore, there is a well-defined notion of morphism between effective increasing/decreasing structures given by natural transformations. They are characterized as follows:

Proposition 5.6. *There exists (and in this case it is unique) a morphism between two increasing (resp. decreasing) effective structures $(\mathcal{O}_\lambda)_{\lambda\in\Lambda}$ and $(\mathcal{O}'_\lambda)_{\lambda\in\Lambda}$ iff for every $\lambda\in\Lambda$ we have $\mathcal{O}_\lambda\subset\mathcal{O}'_\lambda$.*

Proof. Just notice that the natural transformation diagram is equivalent to that condition. Uniqueness follows from the uniqueness of morphisms between two objects in a poset category. □

Thus, we can form the categories $\mathrm{Eff}_{inc}(\mathcal{O};\Lambda)$ and $\mathrm{Eff}_{dec}(\mathcal{O};\Lambda)$ of increasing and decreasing effective structures, which are just the functor categories $\mathrm{Func}((\Gamma,\leq);(\rho(\mathcal{O}),\subseteq))$ and $\mathrm{Func}((\Gamma,\leq)^{op};(\rho(\mathcal{O}),\subseteq))$.

Remark 5.7. Even if $(\Gamma,\leq)$ is not a poset the characterization above could be used to define morphisms between increasing/decreasing effective structures.

We would like to give a definition of morphism between arbitrary effective structures, not necessarily increasing or decreasing, but again defined on the same effective scale Γ. This can be done following an analogous discussion, as follows.

Recall that every class X defines another category $\mathbf{E}X$, now a discrete one, whose class of objects is X and having only identities as morphisms. A functor $F:\mathbf{E}\Lambda\to\rho(\mathcal{O})$ is the same thing that an effective structure, so that we have again a notion of morphism given by natural transformations. Thus, the functor category $\mathrm{Func}(\mathbf{E}\Gamma;(\rho(\mathcal{O}),\subseteq))$ describes the category $\mathrm{Eff}(\mathcal{O};\Gamma)$ and same characterization in Proposition 5.6 can can be use to characterize the morphisms.

The question now is:

- *How to deal with the case of effective structures defined on different fundamental scales? In other words, how to compare different hierarchical organizational principles based on different criteria?*

Recall that a functor $F : \mathbf{A} \to \mathbf{B}$ induces a corresponding functor $F^* : \mathrm{Func}(\mathbf{B};\mathbf{C}) \to \mathrm{Func}(\mathbf{A};\mathbf{B})$ which makes a "change of coefficients". By the above, effective structures are functors defined on $\mathbf{E}\Gamma$ or on $(\Gamma,\leq)$ (resp. $(\Gamma,\leq)^{op}$) if the effective structures are increasing (resp. decreasing). Thus, one case use functors $F : \mathbf{E}\Lambda \to \mathbf{E}\Gamma$ to first make a change of fundamental scale and then define morphisms as previously.

Definition 5.8. Let $\mathcal{O}_\Lambda = (\mathcal{O}_\lambda)_{\lambda\in\Lambda}$ and $\mathcal{O}'_\Gamma = (\mathcal{O}'_\gamma)_{\gamma\in\Gamma}$ be effective structures on $\mathcal{O}$. A *left morphism* (resp. *right morphism*) $F : \mathcal{O}_\Lambda \to \mathcal{O}_\Gamma$ between them is given by:

1. a functor $F : \mathbf{E}\Gamma \to \mathbf{E}\Lambda$ (resp. $F : \mathbf{E}\Lambda \to \mathbf{E}\Gamma$);
2. a morphism $F^*(\mathcal{O}_\Lambda) \to \mathcal{O}'_\Gamma$ (resp. $F : \mathcal{O}_\Lambda \to F^*(\mathcal{O}'_\Gamma)$).

Explicitly, it is given by a function $F : \Gamma \to \Lambda$ (resp. $F : \Lambda \to \Gamma$) such that $\mathcal{O}_{F(\gamma)} \subset \mathcal{O}'_\gamma$ (resp. $\mathcal{O}_\lambda \subset \mathcal{O}'_{F(\lambda)}$).

The definition above provides the categories $\mathrm{Eff}^{left}(O)$ and $\mathrm{Eff}^{right}(O)$ of all effective structures in an area of study $\mathcal{O}$ with left or right morphisms, respectively. Also, analogous definitions can be made for the case of increasing/decreasing ffective structures, giving the categories $\mathrm{Eff}^{left}_{inc}(O)$, $\mathrm{Eff}^{left}_{dec}(O)$, etc.

5.1.4 The Emergence Problem

By the discussion above, both reductionistic and emergentistic approaches to the philosophical foundations of an area of study $\mathcal{O}$ are a priori compatible with the effectiveness hypothesis. On the other hand, the same class of systems $\mathcal{O}$ may admit different effective structures. This motives the following question:

- *considered two effective structures on $\mathcal{O}$, each of them compatible with a different foundational theory, can they be related in some sense? In other words, suppose that $\mathcal{O}_\Lambda$ and $\mathcal{O}'_\Gamma$ are two different effective structures such that there is a morphism $F : \mathcal{O}_\Lambda \to \mathcal{O}'_\Gamma$. This constrains induces some constrain on the foundational theories compatible with each of them?*

Just to exemplify, let us was consider an epistemological theory on $\mathcal{O}$ whose knowledge is compatible with an effective structure $\mathcal{O}'_\Gamma$ in the sense that is *increasing*, i.e., if $X_\gamma \subset X'_{\gamma'}$, then $\mathrm{Know}(X_\gamma) \subset \mathrm{Know}(X'_{\gamma'})$. Consider now another effective structure $\mathcal{O}_{Lambda}$ and suppose that there is a left-morphism $F : \mathcal{O}_\Lambda \to \mathcal{O}'_\Gamma$. Thus, $\mathcal{O}_{F(\gamma)} \subset \mathcal{O}'_\gamma$, so that

$$\cup_\gamma\,(Know)(\mathcal{O}_{F(\gamma)}) \subset \cup_\gamma \mathcal{O}'_\gamma. \tag{5.2}$$

Therefore, if the effective epistemological theory is reductionistic relatively to $\mathcal{O}'_\Gamma$, it follows that the right-hand side of (5.2) is just the whole knowledge, i.e., $\mathrm{Know}(\mathcal{O})$. This means that if the same effective epistemological theory is considered relatively to $F^*(\mathcal{O}_\Lambda)$, then it would be typically emergentistic. If, in addition, $F : \Gamma \to \Lambda$ is surjective, then the left-hand side of (5.2) equals to $\cup_\lambda \mathrm{Know}(\mathcal{O}_\lambda)$, so that the epistemological theory relatively to $\mathcal{O}_\Lambda$ is emergentistic too.

The same discussion applies to ontological or logical theories. The conclusion is then the following:

- if there is a left-morphism $F : \mathcal{O}_\Lambda \to \mathcal{O}'_\Gamma$ of effective structures on $\mathcal{O}$, then a reductionistic effective theory on $\mathcal{O}'_\Gamma$ induces an effective theory on $F^*(\mathcal{O}_\Lambda)$ which is normally emergentistic. If F is surjective, the the emergentistic effective theory becomes defined on $\mathcal{O}_\Lambda$.

One of the basic problems concerning emergentistic approaches to an area of study is its existence:

- *given $\mathcal{O}$, it admits some emergentistic ontological/epistemological/logical theory which is compatible with other previously supposed principles?*

The discussion above tells us how to work on this problem under the effectiveness hypothesis. Indeed, it consists of the following steps:

1. provides an effective structure $\mathcal{O}'_\Gamma$ compatible with the supposed principles;
2. consider another effective structure $\mathcal{O}_\Lambda$ relatively to which ones want an emergentistic theory;
3. build a surjective left-morphism $F : \mathcal{O}_\Lambda \to \mathcal{O}'_\Gamma$.

The main problem is then to build surjective left-morphisms between two given effective structures. In [121, 118] we considered this problem for $\mathcal{O}$ being the part of Physics which can be modeled by Lagrangian field theory, as follows.

5.1.5 Syntactic and Semantical Emergence

Let $\mathcal{O}$ be an area of study and suppose given an axiomatic language L describing it. More precisely, suppose given an epistemological theory endowed with a distinguished class $\mathrm{Belief}_0(\mathcal{O})$ of believes and a map f, which factors trough knowledge, as below[2].

$$
\begin{tikzcd}
 & \mathrm{Belief}(\mathcal{O}) \arrow[r, "f"] & \mathrm{Word}(L) & \rho(\mathrm{Belief}_0(\mathcal{O})) \arrow[r, dashed, "\rho(\bar{f})"] & \rho(\mathrm{Axiom}(L)) \arrow[d, "ind"] \\
\mathrm{Know}(\mathcal{O}) \arrow[ur] \arrow[r, hook] & \mathrm{Belief}_0(\mathcal{O}) \arrow[u, hook] \arrow[r, dashed, "\bar{f}"'] & \mathrm{Axiom}(L) \arrow[u, hook] & \rho(\mathrm{Know}(\mathcal{O})) \arrow[u, hook] \arrow[r, dotted, "\hat{f}"'] & \mathrm{Thm}(L)
\end{tikzcd}
\tag{5.3}
$$

Internally to the logic L, the condition of being an emergentistic epistemological theory admits a formal characterization, which may or not be satisfied in different interpretations of L. Thus, in this case, one can say that the epistemological is:

1. *syntactically emergentistic* if it is emergentistic internally to the language L;
2. *semantically emergentistic* if it is emergentistic in a given interpretation of L.

And, of course, one should expect that epistemological emergentism implies synthetic emergentism for any formal logic L, while syntactic emergentism for L should imply

[2]see the similarity with diagram (2.2) required in the Axiomatization Program of Physics. The latter is actually a particular case of the former for an empiricist epistemology.

semantical emergentism for every interpretation of L. Thus, *the failure of being emergentistic of some interpretation of L can be viewed as an obstruction to be epistemologically emergentistic.*

The effective hypothesis and the discussion on effective structures could also be internalized in the language L and studied in its different interpretations, leading us to consider:

1. *syntactic effectiveness hypothesis*, which is about the existence of an effective structure for $\mathcal{O}$ internally to L;
2. *semantical effectiveness hypothesis*, asserting the existence of effective structures for $\mathcal{O}$ in a given interpretation of L.

For example, if L has categorical semantics (e.g., if L is some flavor of type theory), then in an interpretation of $\mathcal{O}$ it is regarded as a category $\mathbf{C}_{\mathcal{O}}$, so that an effective structure on this interpretation would be just a functor $F : \mathbf{E}\Gamma \to \mathbf{C}_{\mathcal{O}}$ (i.e., a family of objects $\mathcal{O}_\lambda \in \mathbf{C}_{\mathcal{O}}$, with $\lambda \in \Lambda$), which defines a category $\mathrm{Eff}(\mathbf{C}_{\mathcal{O}})$ in the same lines discussed above.

Furthermore, a semantical effective emergentistic approach would be about the impossibility to reconstruct global data about $\mathbf{C}_{\mathcal{O}}$ from data assigned to each $\mathcal{O}_\lambda$. Again from the discussion above, the existence of such an approach could be studied by analyzing the existence of certain morphisms between semantical effective structures.

In [121, 118] we studied the problem of building effective emergentistic philosophical foundations of Physics focusing on the class of systems $\mathcal{O}$ admitting a formalization as Lagrangian field theories.

5.1.6 Extension Problem

Let $\mathcal{O}$ be an area of study. Given two systems X and Y the basic question is whether one is "more fundamental" than the other. In a reductionistic perspective, if X is a subsystem of Y, then it would be more fundamental. This goes in the direction of the stabilization hypothesis and fundamentalistic approaches previously discussed.

An ampler way to make sense of this task is by considering increasing/decreasing effective structure $\mathcal{O}_\Lambda$ on $\mathcal{O}$. Indeed, if the effective structure is increasing (resp. decreasing) we can say that X is more fundamental than Y (relatively to that effective structure) if $X \in \mathcal{O}_\lambda$ and $Y \in \mathcal{O}_\gamma$, with $\lambda \neq \gamma$ and $\lambda \leq \gamma$ (resp. $\gamma \leq \lambda$). But, what happens if $\lambda = \gamma$? In other words:

- *how to compare two systems that belong to the same fundamental scale? Are they necessarily equivalent or one can be more fundamental?*

We observe that the concept of *extension* could be used to compare two systems in this case. More precise, recall that a system (in the sense of Bunge [30]) is a tuple $\mathcal{S} = (T, C, E, R)$ given by:

1. a class T defining the *terms* of $\mathcal{S}$;
2. disjoint subclasses $C, E \subset S$ of terms, i.e., such that $C \cap E = \varnothing$, where C corresponds to the *composition* and E is the *environment* of $\mathcal{S}$;
3. a collection R of relations on the union $C \cup E$, called *structural relations* between components and the environment..

Consider now an effective structure $\mathcal{O}_\Lambda$ on the collection $\mathcal{O}$ of all those systems, and writes $\mathcal{S}_\lambda$ to denote a system which belongs to $\mathcal{O}_\lambda$. Since a system is actually a tuple, even if $\lambda = \gamma$ one can compare $\mathcal{S}_\lambda$ and $\mathcal{S}'_\gamma$ in many different ways:

Definition 5.9. Let $\mathcal{S} = (T, C, E, R)$ and $\mathcal{S}' = (T', C', E', R')$ be two systems as above. We say that:

1. $\mathcal{S}'$ is a *term extension* of $\mathcal{S}$ if $T \subset T'$;
2. $\mathcal{S}'$ is a *composition extension* of $\mathcal{S}$ if $C \subset C'$ (in particular, $C \subset T'$);
3. $\mathcal{S}'$ is an *environment extension* of $\mathcal{S}$ if $E \subset E'$ (in particular, $E \subset T'$);
4. $\mathcal{S}'$ is a *structural extension* of $\mathcal{S}$ if $C \subset C'$, $E \subset E'$ (in particular, $C \cup E \subset C' \cup E' \subset T'$) and if xRy, thus $xR'y$ for every $x \in C$ and $y \in E$; $\mathcal{S}'$ is a *full extension* of $\mathcal{S}$ if it is both a term extension and a structural extension of $\mathcal{S}$.

The *extension problem* of a given system $\mathcal{S} = (T, C, E, R)$ is to classify all its extensions in the senses above. More precisely, for a fixed $\hat{T}$ with $T \subset \hat{T}$, let $\mathrm{Ext}(\mathcal{S}; \hat{T})$ be the class of all term extensions of $\mathcal{S}$ whose terms is $\hat{T}$. Similarly, defines $\mathrm{Ext}(\mathcal{S}; \hat{C})$, $\mathrm{Ext}(\mathcal{S}; \hat{E})$ and $\mathrm{Ext}(\mathcal{S}; \hat{R})$. Thus the extension problems are:

- *given a system $\mathcal{S}$ and a class $\hat{T}$ (resp. $\hat{C}$, $\hat{E}$ and $\hat{R}$), determines the corresponding class* $\mathrm{Ext}(\mathcal{S}; \hat{T})$ *(resp.* $\mathrm{Ext}(\mathcal{S}; \hat{C})$, $\mathrm{Ext}(\mathcal{S}; \hat{E})$ *and* $\mathrm{Ext}(\mathcal{S}; \hat{R})$*).*

In the presence of an effective structure $\mathcal{O}_\Lambda$ on $\mathcal{O}$ we normally restricts to extensions which are in the same fundamental scale. Thus, given $\mathcal{S} \in \mathcal{O}_\lambda$, let $\mathrm{Ext}_\lambda(\mathcal{S}; \hat{T}) \subset \mathrm{Ext}(\mathcal{S}; \hat{T}) \cap \mathcal{O}_\lambda$, and similarly for the other forms of extension. The *effective extension problem* is about classifying this space of extensions in the fixed fundamental scale:

- *given $\mathcal{S} \in \mathcal{O}$ and an effective structure $\mathcal{O}_\Lambda$ for $\mathcal{O}$, if $\mathcal{S} \in \mathcal{O}_\lambda$, determines* $\mathrm{Ext}(\mathcal{S}; \hat{T})$ *(resp.* $\mathrm{Ext}_\lambda(\mathcal{S}; \hat{C})$, $\mathrm{Ext}_\lambda(\mathcal{S}; \hat{E})$ *and* $\mathrm{Ext}_\lambda(\mathcal{S}; \hat{R})$*).*

Suppose now that $\mathcal{O}$ is described by an axiomatic language L. In this case, the notions of extension above (and therefore the corresponding extension problems) should also make sense syntactically and semantically, i.e., they should have an analogous version internally to L and for each interpretation of L. Same for effective extension problems.

In the works [122, 123] we considered the problem of extension for the part of Physics whose systems can be interpreted in some axiomatic language as gauge theories, in the sense that they correspond to some functional $S : \mathrm{Conn}_G(P) \to \mathbb{R}$ in the space of principal connections on a principal G-bundle $P \to M$ over a manifold M, supposed to model the spacetime. We also analyzed which kind of extension $\hat{S}$ of S makes sense as an effective extension and also which does not, suggesting in this case that $\hat{S}$ *emerges* from S.

5.1.7 Main Results

Let M be a smooth manifold, regarded as the spacetime, and let $P \to M$ be a fiber bundle with a subset $\mathrm{Par}(P) \subset \Gamma(P)$ of global sections, playing the role of the class of fundamental scales (or *parameters*).

Remark 5.10. In [121, 118] we assumed M compact and orientable, but we believe that with some care with the integration process these hypotheses can be avoided.

Let $E \to M$ be a $\mathbb{K}$-vector bundle over M, with $\mathbb{K} = \mathbb{R}, \mathbb{C}$. Supposed fixed in $\Gamma(E)$ a $\mathbb{R}$-bilinear nondegenerate map $\langle \cdot, \cdot \rangle$, which could be induced by some metric M, for instance.

Lagrangian systems (the class of systems which are interested) are described by functionals $S:\Gamma(E)\to\mathbb{R}$ which are typically of the form $S[\varphi]=\int_M\langle\varphi,D\varphi\rangle$, where $D\in\mathrm{Diff}(E)$ is a differential operator on E. Thus, for fixed field bundle with pairing, the class of systems of interest is in 1-1 correspondence with $\mathrm{Diff}(E)$. On the other hand, as previously discussed, the effective structure is a map from fundamental scales to the class of systems. Thus, which plays the role of effective structures is the following:

Definition 5.11. A *differential parameterized theory* is function $D:\mathrm{Par}(P)\to\mathrm{Diff}(E)$ assigning to each parameter $\varepsilon\in\mathrm{Par}(P)$ a differential operator D_ε on E.

The emergence problem for two given effective structures is about the existence of suitable morphisms between them. Therefore, in the present context, given $D:\mathrm{Par}(P)\to\mathrm{Diff}(E)$ and $D':\mathrm{Par}(P')\to\mathrm{Diff}(E)$, one would like to build suitable morphisms between them, meaning functions $F:\mathrm{Par}(P)\to\mathrm{Par}(P')$ such that $D_\varepsilon=D'_{F(\varepsilon)}$.

The class of systems described above is strict to our purposes. We want to extend it into two directions:

1. allowing operators on $\Gamma(E)$ which are more general than differential operators, e.g., pseudo-differential operators and other Fourier operators, which belong to a $\mathbb{K}$-algebra $\mathrm{Op}(E)$ extending $\mathrm{Diff}(E)$;
2. considering not the operator itself, but actually functions on the algebra of such generalized operators.

Also, instead of considering the fundamental scale (i.e., the parameters) as belonging to $\mathrm{Par}(P)\subset\Gamma(P)$, we will allow it to belong to a subset of $\Gamma(P)^\ell=\Gamma(P)\times\ldots\times\Gamma(P)$ for some $\ell\geq 1$. Thus, we will allow a multiple fundamental scale (i.e., multiple parameters) $\varepsilon(\ell)=(\varepsilon_1,\ldots,\varepsilon_\ell)$. We will write $\mathrm{Par}^\ell(P)$ to denote that it belongs to $\Gamma(P)^\ell$.

Remark 5.12. Some additional technical properties of the space of fundamental scales $\mathrm{Par}^\ell(P)$ are also needed. Indeed, it must admit a structure of $\mathbb{K}$-algebra which

1. admits square roots;
2. acts on the algebra $\mathrm{Op}(E)$ of generalized operators by an action $\cdot^\ell$ such that:
   (a) action on the identity operator I is injective, i.e., such that the map $\ell I:\mathrm{Par}^\ell(P)\to\mathrm{Op}(E)$ given by $\varepsilon(\ell)\mapsto\varepsilon(\ell)\cdot^\ell I$ is injective;
   (b) is compatible to some functional calculus, i.e., there is a subset $C_\ell(P;\mathbb{K})\subset\mathrm{Maps}(\mathrm{Par}^\ell(P);\mathbb{K})$ and a function $\psi^\ell_- : C_\ell(P;\mathbb{K})\to\mathrm{Op}(E)$ such that $\psi^\ell_f\circ[f(\varepsilon(\ell))\Psi]=\varepsilon(\ell)\cdot^\ell\Psi$.

These conditions are satisfied, for instance, if $P=M\times\mathbb{C}$ is the complex line bundle, $\ell=1$ and $\mathrm{Par}(P)\subset C^\infty(M;\mathbb{C})$ the constant functions.

The class of effective systems we will work on is then the following:

Definition 5.13. A *parameterized polynomial theory* of degree (r,l) is given by

1. a polynomial $p^l_\ell\in\mathrm{Map}(\mathrm{Par}^\ell(P);\mathbb{K})[x_1,\ldots,x_r]$ of degree l in r variables;
2. generalized operators $\Psi_1,\ldots,\Psi_r\in\mathrm{Op}(E)$,

which in turn defines a rule $p^l_\ell(\Psi_1,\ldots,\Psi_r):\mathrm{Par}^\ell(P)\to\mathrm{Op}(E)$.

We then proved the existence of nontrivial morphisms between a parameterized theory and parameterized polynomial theory in terms of properties of the defining operators

$\Psi_1, ..., \Psi_r$ and on the dependence of the coefficients of p^l_ℓ relatively to the fundamental scales $\varepsilon(\ell) = (\varepsilon_1, ..., \varepsilon_\ell)$.

Theorem 5.14 (Emergence Theorem)**.** *Let $\Psi^\ell : \mathrm{Par}^\ell(P) \to \mathrm{Op}(E)$ be a parameterized theory and let $p^l_\ell(\Psi_1, ..., \Psi_r)$ a parameterized polynomial theory of degree (r, l). Suppose that:*

1. *the rule $\Psi : \mathrm{Par}^\ell(P) \to \mathrm{Op}(E)$ is an algebra homomorphism;*
2. *the generalized operators $\Psi_1, ..., \Psi_r \in \mathrm{Op}(E)$ are right-invertible;*
3. *the coefficients of p^l_ℓ belongs to some functional calculus on $\mathrm{Par}^\ell(P)$.*

Then there is a morphism $F : \Psi^\ell \to p^l_\ell$, i.e., the effective theory described by Ψ^ℓ emerges from that described by p^l_ℓ.

Proof. This is a particular version of Theorem 3.1 in [121]. □

Let us consider now another class of systems over a (compact and orientable) smooth manifold M, again regarded as the spacetime: *gauge theories*. Instead of field theories which were defined by a functional $S : \Gamma(E) \to \mathbb{R}$ on global sections of a vector bundle, they are given by functionals $S : \mathrm{Conn}_G(P) \to \mathbb{R}$ defined on G-connections of some principal G-bundle $P \to M$.

If field theories were normally of the form $S[\varphi] = \int_M \langle \varphi, \Psi\varphi \rangle$ for some generalized operator $\Psi \in \mathrm{Op}(E)$, the main examples of gauge theories are Yang-Mills-Type theories, given by $S[A] = \int_M \langle F_A, F_A \rangle$, where $F_A = dA + A[\wedge]A$ is the curvature of A and $\langle \cdot, \cdot \rangle : \Omega^2_{he}(P; \mathfrak{g}) \otimes_{\mathbb{R}} \Omega^2_{he}(P; \mathfrak{g}) \to \Omega^n(M)$ is some nondegenerate $\mathbb{R}$-bilinear pairing from horizontal equivariant $\mathfrak{g}$-valued 2-forms on P to volume forms on M.

Remark 5.15. If M is semi-Riemannian, then the metric g on M and the killing form of $\mathfrak{g}$ induce such a pairing which is nondegenerate only if $\mathfrak{g}$ is semisimple. For this pairing, the Yang-Mills-Type theory above is what is known as Yang-Mills theory in the literature [50, 71, 109]. In Section 2 of [122] we proved that for fixed G, M and P, if G is not discrete and $\dim M \geq 2$, then the space $\mathrm{YMT}_G(P; M)$ of Yang-Mills-Type theories over this data is an infinite-dimensional real vector space and finite projective $C^\infty(M)$-modules. Otherwise, it is a trivial real vector space (and then a trivial $C^\infty(M)$-module too), so that the only Yang-Mills-Type theory is the classical Yang-Mills theory. For that, see Theorem 2.1 and Corollary 1 of [122]. We also gave upper bounds for the rank of $\mathrm{YMT}_G(P; M)$ - see Corollary 2 of [122]. However, these facts do not matter for what follows.

Since gauge theories are our systems of interest, in order to consider extension problems for them it is interesting (keeping in mind our previous discussion) to regard them as tuples $\mathcal{S} = (G, P, \mathrm{Conn}_G(P), S)$. Thus, another gauge theory $\hat{\mathcal{S}} = (\hat{G}, \hat{P}, \mathrm{Conn}_{\hat{G}}(\hat{P}), \hat{S})$ would be an extension of $\mathcal{S}$ if, in some sense, $G \subset \hat{G}$, $P \subset \hat{P}$, $\mathrm{Conn}_G(P) \subset \mathrm{Conn}_{\hat{G}}(\hat{P})$ and $S \subset \hat{S}$. But, if G is a Lie subgroup of G, i.e., if the first condition $G \subset \hat{G}$ is satisfied, then it induces an inclusion $P \hookrightarrow P'$ and in turn a map on the level of the connections, so that one could just require that S' is an extension of S relatively to that map, meaning that diagram below commutes.

$$\begin{tikzcd} \mathrm{Conn}_G(P) \arrow[r, "\imath_*"] \arrow[dr, "S"'] & \mathrm{Conn}_{\hat{G}}(\hat{P}) \arrow[d, "S'"] \\ & \mathbb{R} \end{tikzcd}$$

This definition of extension has an obvious limitation: *all the entries of the tuple defining the extended system are fully characterized by one of them*. Thus, to get a more ample notion of extension one should allow functionals $\hat{S}$ which are defined on some domain $\widehat{\mathrm{Conn}}_{\hat{G}}(\hat{P})$ ampler than $\mathrm{Conn}_{\hat{G}}(\hat{P})$. We then would have an obvious replacement for the diagram above:

$$
\begin{tikzcd}
\mathrm{Conn}_G(P) \arrow[r, "\imath_*"] \arrow[rrd, bend right=10, "S"'] & \mathrm{Conn}_{\hat{G}}(\hat{P}) \arrow[r, hook] \arrow[rd, dashed, "\hat{S}|_{\mathrm{Conn}_{\hat{G}}(\hat{P})}"] & \widehat{\mathrm{Conn}}_{\hat{G}}(\hat{P}) \arrow[d, "\hat{S}"] \\
 & & \mathbb{R}
\end{tikzcd}
$$

Thus, the condition $G \subset \hat{G}$ are still fixing $\hat{S}$, which happens because the map $\imath_*$ still connecting the domains of S and $\hat{S}$, and we are requiring direct commutativity of the induced diagram. These constraints can be avoided by adding a complement to the diagram above (as below) and then replacing commutativity with some other weaker condition. In our case, we choose additive commutativity, meaning that $\hat{S} \circ \jmath = S \circ \delta + C$.

$$
\begin{tikzcd}
 & & & C(\hat{P};\hat{\mathfrak{g}}) \arrow[llld, bend right=15, "\delta"'] \arrow[lld, "\jmath"'] \arrow[ldd, bend left=20, "C"] \\
\mathrm{Conn}_G(P) \arrow[r, "\imath_*"] \arrow[rrd, bend right=10, "S"'] & \mathrm{Conn}_{\hat{G}}(\hat{P}) \arrow[r, hook] & \widehat{\mathrm{Conn}}_{\hat{G}}(\hat{P}) \arrow[d, "\hat{S}"] & \\
 & & \mathbb{R} &
\end{tikzcd}
$$

This leads us to the following definition of extension of a gauge theory.

Definition 5.16. Let $S : \mathrm{Conn}_G(P) \to \mathbb{R}$ be a gauge theory. An *extension* of S is given by:

1. a Lie group $\hat{G}$ having G as a subgroup;
2. an *extended domain* $\widehat{\mathrm{Conn}}_{\hat{G}}(\hat{P})$ extending $\mathrm{Conn}_{\hat{G}}(\hat{P})$;
3. a *correction domain* $C(\hat{P};\hat{\mathfrak{g}})$ which is extended by $\mathrm{Conn}_{\hat{G}}(\hat{P})$ and related with $\mathrm{Conn}_G(P)$ via a map δ;
4. an *extended functional* $\hat{S}$ and a *correction functional* C, defined on the extended domain and on the correction domain, respectively.

This data is subjected to additive commutativity: $\hat{S} \circ \jmath = S \circ \delta + C$.

Remark 5.17.

1. In Section 3.1, 3.2, 3.3 of [122] we discussed how this definition can be useful to unify a plethora of examples of extensions of gauge theories considered in the literature.
2. In Section 3.4 of [122] we analyzed how this notion of extension relates with the notion of emergence of field theories introduced in [121, 118] and briefly discussed above. Indeed, we proved in Theorem 3.1 that:
   (a) every gauge-fixed Yang-Mills-Type theory $S : \mathrm{Conn}_G(P) \to \mathbb{R}$ can be viewed as a differential parameterized theory (Definition 5.11 above) for every parameter bundle F and every class of parameters $\mathrm{Par}(F) \subset \Gamma(F)$;

(b) if some other generalized parameterized theory $\Psi : \mathrm{Par}(F) \to \mathrm{Op}(E)$ is such that $\mathrm{Conn}_{\hat{G}}(\hat{P}) \subset \Gamma(E)$ for some $\hat{G}$ extending G, and if S emerges from Ψ, then Ψ_{ε} is an extension of S for every $\varepsilon \in \mathrm{Par}(F)$.

The final objective of [122, 123] was to study the extension problem for gauge theories. The main results can be summarized as follows:

Theorem 5.18. *For a given gauge theory S, let $\mathrm{Ext}(S;\hat{G})$ be the collection of all extensions of S for a fixed Lie group $\hat{G}$ containing G and let $\mathrm{ED}(S;\hat{G})$ the collection of all those possible extended domains. In this case:*

1. *for every ring R and every additive group $\mathbb{G}$ acting on $\mathbb{R}$ the fibers of the obvious projection $\pi : \mathrm{Ext}(S;\hat{G}) \to \mathrm{ED}(S;\hat{G})$ have an structure of $R[\mathbb{G}]$-module, so that it is a $R[\mathbb{G}]$-module bundle in the category of sets;*
2. *this $R[\mathbb{G}]$-module bundle is subundle of a trivial bundle $R[\mathbb{G}]$-module bundle.*

Proof. This follows from Corollary 3 of [122] and from the discussion on Section 4.3. □

Remark 5.19. In [122] we also showed that there are natural topologies to be considered on $\mathrm{Ext}(S;\hat{G})$ and on $\mathrm{ED}(S;\hat{G})$, relatively to that the projection $\pi : \mathrm{Ext}(S;\hat{G}) \to \mathrm{ED}(S;\hat{G})$. We argue that the entire bundle structure could be internalized in the category of topological spaces using this topology, but leaving the details for future works.

Another aspect of an extension problem is whether one can modify an extension to get a better one. This is related to the question of whether a subspace is dense. In [123] we proved that for every gauge theory S and every fixed $\hat{G}$, the class $\mathrm{Coh}(S;\hat{G})$ of *almost coherent extension* is "dense". By definition, they have a null correction functional, their extended functionals are injective after quotient by the extended domain, and they are isomorphic to an extension obtained by a pullback. We also proved that every other class largely enough is maximal internally to $\mathrm{Coh}(S;\hat{G})$.

Definition 5.20. The *index of coherency* of extensions of S^G relatively to $\hat{G}$, denoted by $\mathrm{ind}(S;\hat{G})$, is the minimum over the cardinality $|I|$ of all subsets $I \subset \mathrm{ED}(S;\hat{G})$ of mutually disjoint extended domains.

Theorem 5.21. *Let S be a gauge theory and fixes a Lie group $\hat{G}$ extending G.*

1. *There is a map $\mu : \mathrm{Ext}(S;\hat{G}) \to \mathrm{Coh}(S;\hat{G})$ which maps nontrivial extensions into nontrivial extensions;*
2. *if a class of extensions $\mathcal{E} \subset \mathrm{Ext}(S;\hat{G})$ is such that $|\mathcal{E}| \geq \mathrm{ind}(S;\hat{G})$, then $\mathcal{E}$ is maximal relatively to the map $\mu|_{\mathcal{E}} : \mathcal{E} \to \mathrm{Coh}(S;\hat{G})$.*

Proof. This is Theorem C (see also Theorem 5.1) of [123]. □

5.2 QUESTIONS AND SPECULATIONS

In the following, we present some questions and speculations on future developments for the topics of this chapter.

5.2.1 Broad Questions and Speculations on Emergence

- In the discussion above we presented reductionism as the possibility to recover ontology/epistemology/logic of an area of study from the ontology/epistemology/logic of its subareas.

  Normally the assignment of data along subobjects defines a presheaf $F : \mathbf{C}^{op} \to \mathbf{Set}$. The fact that a presheaf is a sheaf means that given an object $X \in \mathbf{C}$ there exists a collection X_i of subobjects of X such that the data $F(X_i)$ assigned to the subobjects satisfy compatibility conditions in the intersections $X_{ij} = X_i \cap X_j$ in such a way that the collection of all local data $F(X_i)$ determines $F(X)$.

  Sometimes compatibility along intersections is not enough and compatibility along higher intersections $X_{i,j,k...} = X_i \cap X_j \cap X_k \cap ...$ are needed. In this case we say that F is a ∞-*sheaf*.

(Q5.1) *Show that an area of study $\mathcal{O}$ is ontological reduionist (resp. epistemological reduionist or logical redutionist) iff the rule $(-)_{real}$ (resp.* Know *or* Logic*) assigning to each system $X \in \mathcal{O}$ its class $(\mathcal{O}_x)_{real}$ (resp.* Know$(\mathcal{O}_x)$ *or* Logic$(\mathcal{O}_x)$*) of real subsystems (resp. knowledge or logically true propositions) is a ∞-sheaf. Dually, show that $\mathcal{O}$ is ontological/epistemological/logical emergentist iff the corresponding rules are not ∞-sheaves.*

- If an area of study $\mathcal{O}$ is formalized as some category (as discussed in Question ??), then the rule assigning to each system $X \in \mathcal{O}$ its class of subsystems should be a Grothendieck topology relatively to which the rules $(-)_{real}$, Know or Logic above are ∞-sheaves in an ontological, epistemological or logical reductionist approach to $\mathcal{O}$.

  On the other hand, notice that if $\mathcal{O}$ has an effective structure $(\mathcal{O}_\lambda)_{\lambda \in \Lambda}$, then to each $X \in \mathcal{O}$ we assigns a collection $\mathcal{O}_{x,\Lambda} \subset \mathcal{O}_x$ of subsystems by $\mathcal{O}_{x,\Lambda} = \cup_\lambda \mathcal{O}_x \cap \mathcal{O}_\lambda$.

(Q5.2) *Prove that an effective structure in an area of study $\mathcal{O}$ induces a Grothendieck topology on it (if this is not the case, give conditions ensuring it). Furthermore, $\mathcal{O}$ is effective ontological, epistemological or logical reducionist in that effective structure iff the corresponding rules $(-)_{real}$,* Know *or* Logic *are ∞-sheaves in the induced Grothendieck topology. Dually, show that effective ontological/epistemological/logical emergentism is on the failure of those rules being ∞-sheaves.*

(Q5.3) *Show that each morphism between effective structures in Definition 5.8 induces a morphism between sites relative to the corresponding Grothendieck topologies of Question (Q5.2). Use this to give a sheaf-theoretic proof for the fact that reductionist approaches can be pulled back and to the strategy to build emergentist approaches.*

5.2.2 Questions and Speculations on Emergence Between Field Theories

- In [121, 118] we studied the emergence problem for parameterized Lagrangian field theory from a global, but particular point of view: although parameterized Lagrangian field theories are general maps $\Psi : \mathrm{Par}(P) \to \mathrm{Op}(E)$, we considered the emergence problem for particular examples of them. It would be interesting to find local, but general results.

  A *parameterized* (not necessarily Lagrangian) *field theory* with (not necessarily vectorial) field bundle E could be defined as a map $S_- : \mathrm{Par}(P) \to \mathrm{Map}(\Gamma(E); \mathbb{R})$ assigning to each fundamental parameter $\epsilon \in \Gamma(E)$ a functional $S_\epsilon : \Gamma(E) \to \mathbb{R}$. The emergence problem for two parameterized field theories S_- and S'_- is to find a map $F : \Gamma(P) \to \Gamma(P')$ such that diagram below commutes.

$$
\begin{tikzcd}
 & \mathrm{Par}(P') \arrow[d, "S'_-"] \\
\mathrm{Par}(P) \arrow[r, "S_-"'] \arrow[ur, dashed, "F"] & F(\Gamma(E);\mathbb{R})
\end{tikzcd}
$$

Suppose that $\mathrm{Par}(P)$ and $\mathrm{Par}(P')$ are Banach manifolds and that there exists a set $\mathcal{S}(M)$ of *admissible functionals*, which is also a Banach manifold, and such that S_- and S'_- actually take values in $\mathcal{S}(M)$. We say that a parameterized field theory is *of class C^k* if it is of class C^k as a function $S_- : \Gamma(P) \to \mathcal{S}(M)$ between Banach manifolds. One can then consider the emergence problem in the context of global analysis: to find a C^r-map $F : \mathrm{Par}(P) \to \mathrm{Par}(P')$ between Banach manifolds which makes diagram above commutative.

On the other hand, observe that the existence of F in the diagram above consists of an extension problem, which can also be studied from a homotopical point of view.

Q5.4 *Use of global analysis and homotopy theory to provide general, but local, emergence theorems between arbitrary parameterized field theories.*

- Let X be a topological space and A an element of a spectrum $\mathbb{A}$. The cohomology of X with coefficients on A is the set of homotopy classes $H(X;A) = [X;A]$ of maps $X \to A$. The collection of those maps can be identified with global sections of the trivial bundle $X \times A \to X$. Let $\pi_A : P_A \to X$ some other nontrivial bundle over X whose typical fiber is A. The homotopy class $\Gamma(P_A)/\simeq$ of global sections is a *twisted A-cohomology of X*.

  The cocycles of twisted cohomology are homotopy classes of maps $s : P_A \to X$ such that $\pi_A \circ s = id_X$. Consider now two different coefficients A and A' and suppose given twistings P_A and $P'_{A'}$. Given a morphism $F : A \to A'$ which extends to the bundles $F : P_A \to P'_{A'}$, it induces a map $F* : \Gamma(P_A) \to \Gamma(P'_{A'})$ between the global sections and, therefore, between the twisted cohomologies. This means that for every $s : X \to P_A$ there is $s' : X \to P'_{A'}$ such that $s = F \circ s'$.

  If we take X as a space $\mathcal{S}(E)$ of admissible functionals, as above, and $\Gamma(P_A) = \mathrm{Par}(P)$, then parameterized field theories are maps $S_- : \Gamma(P_A) \to X$ and the emergence problem is about finding maps $F : \Gamma(P_A) \to \Gamma(P'_{A'})$ such that $S_- = S'_{-'} \circ F$. By the above, maps like F can be built from spectrum morphisms $F : A \to A'$.

Q5.5 *Use of the discussion above and conditions on the homotopy of $\mathcal{S}(E)$ to find emergence relations between parameterized field theories whose fundamental parameters are cocycles of twisted generalized cohomologies of $\mathcal{S}(E)$.*

- We speculate that parameterized field theories model data analysis, so that emergence relations between parameterized field theories (in particular the emergence theorems) could be used to infer the existence of emergence between conclusions that can be obtained from the data but using different methods. This goes as follows.

  Take the base manifold M as a discrete metric space (the set of data with a notion of distance between them). Furthermore, take the field bundle E as a subbundle of the trivial line bundle $M \times \mathbb{R}$ such that for every datum $x \in M$, the corresponding fiber is some interval centered in x, describing the error of measuring x. The global sections $s : M \to E$ then correspond to the choice of a value around x, so that $\Gamma(E)$ contains all possible values around every datum.

An "index" or "measure" takes these values (i.e., data and errors) and produces a conclusion in the form of a real number. This define a functional $S : \Gamma(E) \to \mathbb{R}$. Thus, a set of indexes/measures defines a set of acceptable functionals $\mathcal{S}(E)$, i.e., a set of acceptable conclusions that can be obtained from the data and errors. For instance, if the conclusion has meaning only if the real number obtained is non-negative, then we should take $\mathcal{S}(E)$ as the collection of all non-negative functionals $S : \Gamma(E) \to \mathbb{R}$.

If Λ is a set, a function $S_{-} : \Lambda \to \mathcal{S}(E)$ selects a family of acceptable indexes/measures $S_{\lambda} in \mathcal{S}(E)$ and, therefore, a parameterized field theory. One could think of S_{-} as an "investigation method" of the data M. If S_{-} and $S'_{-'}$ are two families of acceptable indexes/measures (i.e., two parameterized field theories or two investigation methods), the condition that S_{-} emerges from $S'_{-'}$ means precisely that every conclusion from S_{-} could be obtained from $S'_{-'}$.

Q5.6 *Formalize discussion above and use of Questions Q5.4 Q5.5 to build local emergence theories between data investigation methods. What it means a Lagrangian field theory in this context? How could the emergence theorems of [121, 118] be interpreted? How this relates to topological data analysis?*

5.2.3 Questions and Speculations on Extensions of Gauge Theories

- Recall that in [122] we showed that for every gauge theory S with gauge group G and every fixed extended gauge group $\hat{G}$, the collection $\mathrm{Ext}(\hat{S};\hat{G})$ of extensions $\hat{S}$ of S with extended gauge group $\hat{G}$ is a topological $R[\mathbb{G}]$-module bundle on the space of extended domains, for every group $\mathbb{G}$ acting on $(\mathbb{R}, +)$ and every unital ring R.

  Notice, on the other hand, that we could be obtained other bundle structures by projecting on correction domains, correction functionals, etc., rather than on extended domains.

  Q5.7 *What can be said about the $R[\mathbb{G}]$-module bundle structures that can be put on* $\mathrm{Ext}(\hat{S};\hat{G})$*? How are they related? Are there bundle morphisms between them?*

- The fundamental invariant of topological vector bundles in K-theory. The study of topological module bundles and the construction of a K-theory for them has been designed for certain rings by Mallios and Abel in [106, 107, 108, 7, 8, 9]. The existence of a K-theory for module bundles on arbitrary rings is unknown by the author.

  Q5.8 *Use some kind of K-theory and characteristic classes for group-algebra bundles to study the $R[\mathbb{G}]$-module bundle structures on* $\mathrm{Ext}(\hat{S};\hat{G})$.

- In the work [123] we considered the existence problem of complete extensions, i.e., whose correction term is null (see Theorem 5.21).

  Q5.9 *Let* $\mathrm{Inc}(\hat{S},\hat{G})$ *the class of all incomplete extensions $\hat{S}$ of S with extended gauge group $\hat{G}$. What can be said about* $\mathrm{Inc}(\hat{S},\hat{G})$*? It inherits a sub-$R[\mathbb{G}]$-module bundle structure from* $\mathrm{Ext}(\hat{S};\hat{G})$*?*

  In working in progress we showed that the existence problem of incomplete extensions for a given extended functional $\hat{S}$ and fixed extended gauge group $\hat{G}$ is closely related to the existence problem of equivariant extensions, i.e., whose extended functional is invariant under the action of the group $\mathrm{Gaug}_{\hat{G}}(\hat{P})$ of extended gauge transformations. The latter problem, in turn, corresponds to the existence of solutions for a system of nonlinear partial differential equations: *the extended gauge invariance equation*.

- For a given topological group Γ, recall the notion of Γ-equivariant cohomology. While K-theory is about virtual classes of vector bundles, equivariant K-theory is about virtual classes of Γ-equivariant vector bundles. Given a Γ-equivariant bundle E, let $K_{\Gamma}(E)$ and $K(E)$ denote its corresponding Γ-equivariant K-theory and K-theory, respectively. There is a forgetful map $\imath : K_{\Gamma}(E) \to K(E)$.

  Assume Questions Q5.8, meaning that we found a K-theory for $\mathrm{Ext}(\hat{S};\hat{G})$. Assume that the notions of "equivariant module-bundle" and of "equivariant K-theory for module-bundles" are well-defined. Give conditions under which $\mathrm{Ext}(\hat{S};\hat{G})$ is actually a $\mathrm{Gaug}_{\hat{G}}(\hat{P})$-equivariant module bundle.

Q5.10 *Construct equivariant extensions from equivariant cocycles of* $\mathrm{Gaug}(\hat{G})$*-equivariant* K*-theory of* $\mathrm{Ext}(\hat{S};\hat{G})$.

- From the physical point of view, assuming that the quantization approach is the only path to the Axiomatization Program of Physics, a gauge theory is physically acceptable only if it remains to be gauge-invariant after (perturbative) quantization. A gauge theory satisfying this condition is called *anomaly-free*.

  Let $\mathrm{Free}(\hat{S},\hat{G})$ the class of extensions of anomaly-free extensions of a gauge theory. Of course, $\mathrm{Free}(\hat{S},\hat{G}) \subset \mathrm{Equiv}(\hat{S},\hat{G})$, where $\mathrm{Equiv}(\hat{S},\hat{G})$ denotes the class of equivariant extensions.

  For theories $S : \mathrm{Conf}(M) \to \mathbb{R}$ which are invariant under the action of a gauge group $\mathrm{Gaug}(M)$ and in which BV-BRST formalism applies there is a cohomological characterization of the anomaly-free condition.

Q5.11 *Use the cohomological description of the anomaly-free condition to characterize the* $\mathrm{Free}(\hat{S},\hat{G})$. *In particular, characterize whether equivariant extensions $\hat{S}$ of anomaly free gauge theories are anomaly-free.*

- Notice that the concept of extension of gauge theories introduced in [122] could be extended to the case of algebra-valued gauge theories. This suggests using Geometric Obstruction Theory to work on some questions above.

Q5.12 *Assuming Question Q4.1, let S be an algebra-valued gauge theory with an algebra of infinitesimal symmetries A. Let $\hat{A}$ an algebra which has A as a subalgebra. Define what is an extension $\hat{S}$ of S with algebra of infinitesimal symmetries given by $\hat{A}$.*

Q5.13 *What is the analog of the group of gauge transformations for algebra-valued gauge theories? Let* $\mathrm{Gaug}_A(P)$ *this group. Let* $\mathrm{Equiv}(\hat{S};\hat{A})$ *be the class of extensions of a given algebra-valued gauge theory S which are invariant under the action of* $\mathrm{Gaug}_{\hat{A}}(\hat{P})$. *Show that as above these equivariant extensions can be characterized as solutions of a system of partial differential equations, now depending on $\hat{A}$. Use Geometric Obstruction Theory to find classes of algebras in which these equations simplify to equations whose solution space is well-known, allowing us to classify* $\mathrm{Equiv}(\hat{S};\hat{A})$.

Q5.14 *Is it possible to extend the BV-BRST formalism to the case of algebra-valued gauge theories? For which class of algebras is this possible? For them, find an analogous cohomological characterization of anomaly-free theories. Use Geometric Obstruction Theory to find conditions on A for which the gauge anomaly vanishes identically, so* *that algebra-valued gauge theories with coefficients in A are anomaly-free. Similarly, find conditions on $\hat{A}$ such that extensions of anomaly-free algebra-valued gauge theories remain anomaly-free.*

5.3 FIRST WORK

The following is a recompilation of the ArXiv preprint

- Y.X. Martins, R.J. Biezuner, *An Axiomatization Proposal and a Global Existence Theorem for Strong Emergence Between Parameterized Lagrangian Field Theories*, Sep 2020, arXiv:2009.09858

An Axiomatization Proposal and a Global Existence Theorem for Strong Emergence Between Parameterized Lagrangian Field Theories

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Abstract

In this paper we propose an axiomatization for the notion of strong emergence phenomenon between field theories depending on additional parameters, which we call parameterized field theories. We present sufficient conditions ensuring the existence of such phenomena between two given Lagrangian theories. More precisely, we prove that a Lagrangian field theory depending linearly on an additional parameter emerges from every multivariate polynomial theory evaluated at differential operators which have well-defined Green functions (or, more generally, that has a right-inverse in some extended sense). As a motivating example, we show that the phenomenon of gravity emerging from noncommutativy, in the context of a real or complex scalar field theory, can be recovered from our emergence theorem. We also show that, in the same context, we could also expect the reciprocal, i.e., that noncommutativity could emerge from gravity. Some other particular cases are analyzed.

1 Introduction

The term *emergence phenomenon* has been used for years in many different contexts. In each of them, Emergence Theory is the theory which studies those kinds of phenomena. E.g, we have versions of it in Philosophy, Art, Chemistry and Biology [1, 49, 15]. The term is also often used in Physics, with different meanings (for a review on the subject, see [13, 18]. For an axiomatization approach, see [19]). This reveals that the concept of emergence phenomenon is very general and therefore difficult to formalize. Nevertheless, we have a clue of what it really is: when looking at all those instances of the phenomenon we see that each of them is about describing a system in terms of other system, possibly in different *scales*. Thus, an emergence phenomenon is about a relation between two different systems, the *emergence relation*, and a system emerges from another when it (or at least part of it) can be recovered in terms of the other system, which is presumably more fundamental, at least in some scale. The different emergence phenomena in Biology, Philosophy,

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Physics, and so on, are obtained by fixing in the above abstract definition a meaning for system, scale, etc.

Notice that, in this approach, in order to talk about emergence we need to assume that to each system of interest we have assigned a *scale*. In Mathematics, scales are better known as *parameters*. So, emergence phenomena occur between *parameterized systems*. This kind of assumption (that in order to fix a system we have to specify the scale in which we are considering it) is at the heart of the notion of effective field theory, where the scale (or parameter) is governed by Renormalization Group flows [12, 31, 26, 22]. Notice, in turn, that if a system emerges from another, then the second one should be more fundamental, at least in the scale (or parameter) in which the emergence phenomenon is observed. This also puts Emergence Theory in the framework of searching for the fundamental theory of Physics (e.g Quantum Gravity), whose systems should be the minimal systems relative to the emergence relation [13, 18]. The main problem in this setting is then the existence problem for the minimum. A very related question is the general existence problem: *given two systems, is there some emergence relation between them?*

One can work on the existence problem at different levels of depth. Indeed, since the systems is question are parameterized one can ask if there exists a correspondence between them in *some scales* or in *all scales*. Obviously, requiring a complete correspondence between them is much stronger than requiring a partial one. On the other hand, in order to attack the existence problem we also have to specify which kind of emergence relation we are looking for. Again, is it a full correspondence, in the sense that the emergent theory can be fully recovered from the fundamental one, or is it only a partial correspondence, through which only certain aspects can be recovered? Thus, we can say that we have the following four versions of the existence problem for emergence phenomena:

	weak	*weak-scale*	*weak-relation*	*strong*
relation	partial	full	partial	full
scales	some	some	all	all

Table 1: Types of Emergence

In Physics one usually works on finding weak emergence phenomena. Indeed, one typically shows that certain properties of a system can be described by some other system at some limit, corresponding to a certain regime of the parameter space. These emergence phenomena are strongly related with other kind of relation: the *physical duality*, where two different systems reveal the same physical properties. One typically builds emergence from duality. For example, AdS/CFT duality plays an important role in describing spacetime geometry (curvature) from mechanic statistical information (entanglement entropy) of dual strongly coupled systems [42, 39, 46, 8, 10, 9].

There are also some interesting examples of weak-scale emergence relations, following again from some duality. These typically occur when the action functional of two Lagrangian field theories are equal at some limit. The basic example is gravity emerging from noncommutativity following from the duality between commutative and noncommutative gauge theories established by the Seiberg-Witten maps [43]. Quickly, the idea was to consider a gauge theory $S[A]$ and modify it into two different ways:

1. by considering $S[A]$ coupled to some background field χ, i.e, $S_{\chi}[A;\chi]$;
2. by using the Seiberg-Witten map to get its noncommutative analogue $S_{\theta}[\hat{A};\theta]$.

Both new theories can be regarded as parameterized theories: the parameter (or scale) of the first one is the background field χ, while that of the second one is the noncommutative parameter $\theta^{\mu\nu}$. By construction, the noncommutative theory $S_\theta[A;\theta]$ can be expanded in a power series on the noncommutative parameter, and we can also expand the other theory $S_\chi[A;\chi]$ on the background field, i.e, one can write

$$S_\chi[A;\chi] = \sum_{i=0}^{\infty} S_i[A;\chi^i] = \lim_{n\to\infty} S_{(n)}[A;\chi] \quad \text{and} \quad S_\theta[\hat{A};\theta] = \sum_{i=0}^{\infty} S_i[A;\theta^i] = \lim_{n\to\infty} S_{(n)}[A;\theta],$$

where $S_{(n)}[A;\chi] = \sum_{i=0}^{n} S_i[A;\chi^i]$ and $S_{(n)}[A;\theta] = \sum_{i=0}^{n} S_i[A;\theta^i]$ are partial sums. One then tries to find solutions for the following question:

Question 1. *Given a gauge theory $S[A]$, is there a background version $S_\chi[A;\chi]$ of it and a number n such that for every given value $\theta^{\mu\nu}$ of the noncommutative parameter there exists a value of the background field $\chi(\theta)$, possibly depending on $\theta^{\mu\nu}$, such that for every gauge field A we have $S_{(n)}[A;\chi(\theta)] = S_{(n)}[A;\theta]$?*

Notice that if rephrased in terms of parameterized theories, the question above is precisely about the existence of a weak-scale emergence between S_χ and S_θ, at least up to order n. This can also be interpreted by saying that, in the context of the gauge theory $S[A]$, the background field χ emerges in some regime from the noncommutativity of the spacetime coordinates. Since the noncommutative parameter $\theta^{\mu\nu}$ depends on two spacetime indexes, it is suggestive to consider background fields of the same type, i.e, $\chi^{\mu\nu}$. In this case, there is a natural choice: metric tensors $g^{\mu\nu}$. Thus, in this setup, the previous question is about proving that in the given gauge context, gravity emerges from noncommutativity at least up to a perturbation of order n. This has been proved to be true for many classes of gauge theories and for many values of n [40, 52, 4, 30, 16]. On the other hand, this naturally leads to other two questions:

1. Can we find some emergence relation between gravity and noncommutativity in the nonperturbative setting? In other words, can we extend the weak-scale emergence relation above to a strong one?

2. Is it possible to generalize the construction of the cited works to other kinds of background fields? In other words, is it possible to use the same idea in order to show that different fields emerge from spacetime noncommutativity? Or, more generally, is it possible to build a version of it for some general class of field theories?

The first of these questions is about finding a strong emergence phenomena and it has a positive answer in some cases [51, 5, 44, 41]. The second one, in turn, is about finding systematic and general conditions ensuring the existence (or nonexistence) of emergence phenomena. At least to the authors knowledge, there are no such general studies, specially focused on the strong emergence between field theories. It is precisely this point that is the focus of the present work. Indeed we will:

1. based on Question 1, propose an axiomatization for the notion of *strong emergence* between field theories;

2. establish sufficient conditions ensuring that a given Lagrangian field theory emerges from each theory belonging to a certain class of theories.

We will work on the setup of *parameterized field theories*, which are given by families $S[\varphi;\varepsilon]$ of field theories depending on a *fundamental parameter* ε. In the situations described above, ε is the the noncommutative parameter $\theta^{\mu\nu}$ or the background field $\chi^{\mu\nu}$. In the cases where the emergence was explicitly obtained, it was of summary importance that the parameters $\theta^{\mu\nu}$ and $\chi^{\mu\nu}$ belongs to the *same* class of fields and that the corresponding action functions are defined on the *same* field A_μ. Thus, given two parameterized theories $S[\varphi;\varepsilon]$ and $S'[\varphi;\varepsilon']$ we always assume that they are defined on the same fields and that the parameters ε and ε' belong to the same space. Keeping Question 1 as a motivation, let us say that $S[\varphi;\varepsilon]$ *emerges* from $S'[\varphi;\varepsilon']$ if there is a map F on the space of parameters such that, for every ε and every field φ we have $S[\varphi;\varepsilon] = S'[\varphi;F(\varepsilon)]$. We call F a *strong emergence phenomenon* between S and S'.

Our main result states that for certain $S[\varphi;\varepsilon]$ and $S'[\varphi;\varepsilon']$, depending on ε and ε' in a suitable parameter space, in the sense that it has some special algebraic structure, then these strong emergence phenomena exist. The formal statement of this result will be presented in Section 3 after some technical digression. But, as a motivation, let us state a particular version of it and show how it can be used to recover an example of emergence between gravity and noncommutativity.

First of all, recall that the typical field theory has a kinetic part and an interacting part. The kinetic part is usually quadratic and therefore of the form $\mathcal{L}_{\mathrm{knt},i}(\varphi_i) = \langle \varphi_i, D_i\varphi_i\rangle_i$, with $i = 1,...,N$, where N is the number of fields, D_i are differential operators and $\langle\cdot,\cdot\rangle_i$ are pairings in the corresponding space of fields. Summing the space of fields and letting $\varphi = (\varphi_1,...,\varphi_N)$, $D = \oplus_i D_i$ and $\langle\cdot,\cdot\rangle = \oplus_i\langle\cdot,\cdot\rangle_i$, one can write the full kinetic part as $\mathcal{L}_{\mathrm{knt}}(\varphi) = \langle\varphi, D\varphi\rangle$. On the other hand, the interacting part is typically polynomial, i.e., it is of the form $\mathcal{L}_{\mathrm{int}}(\varphi) = \langle\varphi, p^l[D_1,...,D_N]\varphi\rangle$, where $l \geq 0$ is the degree of the interaction and $p^l[D_1,...,D_N] = \sum_{|\alpha|\leq l} f_\alpha \cdot D^\alpha$. Since the space of differential operators constitutes an algebra, it follows that $p^l[D_1,...,D_N]$ is a differential operator too, so that both the kinetic and the interacting parts (and therefore the sum of them, which constitute the typical lagrangians) are of the form $\mathcal{L}(\varphi) = \langle\varphi, D\varphi\rangle$.

Notice, furthermore, that the pairing $\langle\cdot,\cdot\rangle$ is typically induced by fixed geometric structures (such as metrics) in the spacetime manifold M and in the field bundle E. Thus, in the parameterized context the natural dependence on the parameter is on the differential operator, i.e., the typical parameterized Lagrangian theories are of the form $\mathcal{L}(\varphi;\varepsilon) = \langle\varphi, D_\varepsilon\varphi\rangle$. In the case of polynomial theories (e.g., those describing interations), it is more natural to assume that the dependence on D_ε is actually on the coefficient functions f_α, i.e., $p^l_\varepsilon[D_1,...,D_N] = \sum_{|\alpha|\leq l} f_\alpha(\varepsilon)D^\alpha$. These coefficient functions could be scalar functions or, more generally, parameter-valued functions if we have an action of parameters in differential operators. In this last case, the polynomial is $p^l_\varepsilon[D_1,...,D_N] = \sum_{|\alpha|\leq l} f_\alpha(\varepsilon)\cdot D^\alpha$, where the dot denotes the action of parameters in differential operators.

Emergence Theorem (rough version) *Let M be a spacetime manifold and $E \to M$ a field bundle which define a pairing $\langle\cdot,\cdot\rangle$ in E. Let $\mathcal{L}_1(\varphi;\varepsilon) = \langle\varphi, D_\varepsilon\varphi\rangle$ be an arbitrary parameterized theory and $\mathcal{L}_2(\varphi;\delta) = \langle\varphi, p^l_\delta[D_1,...,D_N]\varphi\rangle$ a polynomial theory, e.g., an interaction term. Suppose that:*

1. *the parameters φ are nonnegative real numbers or, more generally, positive semi-definite operators or tensors acting on the space of differential operators;*
2. *the dependence of D_ε on ε is of the form $D_\varepsilon\varphi = \varepsilon\cdot D\varphi$, where D is a fixed differential operator;*
3. *the coefficient $f_\alpha(\delta)$ of $p^l_\delta[D_1,...,D_N]$ are nowhere vanishing scalar or parameter-valued functions, and the differential operators $D_1,...,D_N$ have well-defined Green functions.*

Then the theory $\mathcal{L}_1(\varphi;\varepsilon)$ emerges from the theory $\mathcal{L}_2(\varphi;\varepsilon)$.

Now, looking at [40], consider again the question of emergent gravity in a four dimensional space-time. The first order perturbation of a real scalar field theory φ, coupled to a semi-Riemannian[1] gravitational field $g=\eta+h$, in an abelian gauge background, is given by equation (10) of [40][2]:

$$\begin{aligned}\mathcal{L}_{\text{grav}}(\varphi;h) &= \partial^{\mu}\varphi\partial_{\mu}\varphi - h^{\mu\nu}\partial_{\mu}\varphi\partial_{\nu}\varphi &(1)\\ &= -\langle\varphi,\Box\varphi\rangle + \langle\varphi,(h\cdot D_1)\varphi\rangle &(2)\\ &= \mathcal{L}_0(\varphi)+\mathcal{L}_1(\varphi;h). &(3)\end{aligned}$$

where in the second step we did integration by parts and used the fact that the spacetime manifold is boundaryless. Furthermore, the pairing is just $\langle\varphi,\varphi'\rangle=\varphi\varphi'$, while $D_1=\partial_{\mu}\partial_{\nu}$ and for every 2-tensor $t=t^{\mu\nu}$, we have the trace action $t\cdot D=t^{\mu\nu}\partial_{\mu}\partial_{\nu}$. In particular, $\eta\cdot D_1=\partial^{\mu}\partial_{\nu}=\Box$. On the other hand, the first order expansion in θ of the Seiberg-Witten dual of a real scalar field theory, in an abelian gauge background, is given by (equation (7) of [40]):

$$\begin{aligned}\mathcal{L}_{\text{ncom}}(\varphi;\theta) &= \partial^{\mu}\varphi\partial_{\mu}\varphi + 2\theta^{\mu\alpha}F_{\alpha\kappa}\eta^{\kappa\nu}(-\partial_{\mu}\varphi\partial_{\nu}\varphi+\frac{1}{4}\eta_{\mu\nu}\partial^{\rho}\varphi\partial_{\rho}\varphi) &(4)\\ &= \langle\varphi,(-\Box)\varphi\rangle + \langle\varphi,(f(\theta)\cdot D_2)\varphi\rangle &(5)\\ &= \langle\varphi,p^1_{\theta}[\Box,D_2]\varphi\rangle, &(6)\end{aligned}$$

where in the first step we again integrated by parts, and used the fact that $p^1_{\theta}[x,y]$ is the first order polynomial $(-1)\cdot x+f(\theta)\cdot y$, where $f(\theta)=\theta$ and, in coordinates, $D_2=2F_{\alpha\kappa}\eta^{\kappa\nu}(\partial_{\mu}\partial_{\nu}-\frac{1}{4}\eta_{\mu\nu}\Box)$.

Notice that if $h^{\mu\nu}$ is a positive-definite tensor (which is the case in *Riemannian signature*), then $\mathcal{L}_1(\varphi;h)$ in (3) satisfies the hypotheses of the emergence theorem. On the other hand, since in the Riemannian setting $\Box$ is the Laplacian and D_2 is essentially a combination of generalized Laplacians, both of them have Green functions[3]. Thus, if we forget the trivial case $\theta^{\mu\nu}=0$, then $\mathcal{L}_{\text{ncom}}(\varphi;\theta)$ also satisfies the hypotheses of the emergence theorem. Thus, as a consequence we see that *the gravitational term $\mathcal{L}_1(\varphi;h)$ emerges from the noncommutative theory $\mathcal{L}_{\text{ncom}}(\varphi;\theta)$*, as also proved in [40]. In other words, there is a function F on the space of 2-tensors, such that for every $h^{\mu\nu}$ we have $\mathcal{L}_1(\varphi;h^{\mu\nu})=\mathcal{L}_{\text{ncom}}(\varphi;F(h^{\mu\nu}))$.

Some comments.

1. In [40] the author proved *explicitly* that gravity emerges from noncommutativy. More precisely, he gave an explicit expression for the function $F(h^{\mu\nu})$. Here, however, our result is only about the existence of such function,

2. While above we had to assume Rimemannian signature, our main result (Theorem 3.1) applies equally well to the Lorentzian signature. It is different, however, of the rough version stated above.

[1] In [40] the author considered only Lorentzian spacetimes. However, we notice that in order to get the emergence phenomena the Lorentzian signature was not explicitly used, so that the same holds in the semi-Rimennanian case.

[2] In [40] an expansion of the form $g_{\mu\nu}=\eta_{\mu\nu}+h_{\mu\nu}+h\eta_{\mu\nu}$ was considered, where h is some function, but the author concluded that the emergence phenomenon exists only if $h=0$. Thus, we are assuming this necessary condition from the beginning.

[3] Actually, they are elliptic and thus have Fredholm inverses; this is enough for us.

3. In [40] analogous emergence phenomena were established in the case of complex scalar fields. Our main theorem also holds for complex fields.

4. Notice that the scalar theory in the gravitational background (3) can also be written as $\mathcal{L}_{\text{grav}}(\varphi; h) = \langle \varphi, p_h^1[\Box, D_1]\varphi \rangle$, where $p_h^1[x, y]$ is the first order polynomial $(-1) \cdot x + f(h) \cdot y$, where, again $f(h) = h$. Thus, we can also see the gravitational theory as a polynomial theory, defined by the same polynomial, but evaluated in different differential operators. On the other hand, (6) can be written as $\mathcal{L}_0(\varphi) + \mathcal{L}_2(\varphi; \theta)$, where $\mathcal{L}_2(\varphi; \theta) = \langle \varphi, (\theta \cdot D_2)\varphi \rangle$. Since $\Box$ and D_1 are generalized Laplacians, they have Green functions. Thus, if one restricts to the case of noncommutativity parameters θ which are positive definite (say, e.g., positive definite symplectic forms), then one can again use our emergence theorem to conclude the reciprocal fact: that $\mathcal{L}_2(\varphi; \theta)$ emerges from $\mathcal{L}_{\text{grav}}(\varphi; h)$, i.e., that *noncommutativity may also emerges from gravity.*

This work is organized as follows. In Section 2 we propose, based on the lines of this introduction, a formal definition for the notion of strong emergence between parameterized field theories, and we introduced the *emergence problem* which is about determining if there is some emergence phenomenon between such two given theories. We discuss why it is natural (or at least reasonable) to restrict to a certain class of parameterized field theories, defined by certain "generalized operators". In Section 3 our main theorem is stated. We begin by introducing its "syntax", leaving the precise formal statement (or "semantic") to Section 3.1. Before giving the proof, which is a bit technical and based on induction arguments, in Section 3.2 we analyze some particular cases of the main theorem trying to emphasize its real scope. In Section 4 the main result is finally proved.

Remark. Although we used the example of gravity emerging from noncommutativity as a motivating context, we would like to emphasize that this paper is not intended to focus on it or in building concrete examples. Indeed, our aim is to propose a formalization to the notion of strong emergence, at least in the context of Lagrangian field theories, and a general strategy to investigate the existence problem for such phenomena. With this remark we admit the need of additional concrete applications of the methods proposed here, which should appear in a future work. In particular, the case of gravity emerging from noncommutative in its difference incarnations, is to appear in a work in progress.

2 The Strong Emergence Problem

Recall that a *field theory* on a n-dimensional manifold M (regarded as the spacetime) is given by an action functional $S[\varphi]$, defined in some space of fields (or configuration space) $\text{Fields}(M)$, typically the space of sections of some real or complex vector bundle $E \to M$, the *field bundle*. A *parameterized field theory* consists of another bundle $P \to M$ (the *parameter bundle*), a subset $\text{Par}(P) \subset \Gamma(P)$ of global sections (the *parameters*) and a collection $S_\varepsilon[\varphi]$ of field theories, one for each parameter $\varepsilon \in \text{Par}(P)$. A more suggestive notation should be $S[\varphi; \varepsilon]$. So, e.g, for the trivial parameter bundle $P \simeq M \times \mathbb{K}$ we have $\Gamma(P) \simeq C^\infty(M; \mathbb{K})$ and in this case we say that we have *scalar parameters.* If we consider only scalar parameters which are constant functions, then a parameterized theory becomes the same thing as a 1-parameter family of field theories. Here, and throughout the paper, $\mathbb{K} = \mathbb{R}$ or $\mathbb{K} = \mathbb{C}$ depending on whether the field bundle in consideration is real or complex.

We will think of a parameter ε as some kind of "physical scale", so that for two given parameters ε and ε', we regard $S[\varphi;\varepsilon]$ and $S[\varphi;\varepsilon']$ as *the same theory* in two different physical scales. Notice that if P has rank l, then we can locally write $\varepsilon = \sum \varepsilon^i e_i$, with $i = 1,...,l$, where e_i is a local basis for $\Gamma(P)$. Thus, locally each physical scale is completely determined by l scalar parameters ε^i which are the fundamental ones. In terms of these definitions, Question 1 has a natural generalization:

Question 2. *Let $S_1[\varphi;\varepsilon]$ and $S_2[\psi;\delta]$ be two parameterized theories defined on the same spacetime M, but possibly with different field bundles E_1 and E_2, and different parameter bundles P_1 and P_2. Arbitrarily giving a field $\varphi \in \Gamma(E_1)$ and a parameter $\varepsilon \in \mathrm{Par}_1(P_1)$, can we find some field $\psi(\varphi) \in \Gamma(E_2)$ and some parameter $\delta(\varepsilon) \in \mathrm{Par}_2(P_2)$ such that $S_1[\varphi;\varepsilon] = S_2[\psi(\varphi);\delta(\varepsilon)]$? In more concise terms, are there functions $F : \mathrm{Par}_1(P_1) \to \mathrm{Par}_2(P_2)$ and $G : \Gamma(E_1) \to \Gamma(E_2)$ such that $S_1[\varphi;\varepsilon] = S_2[G(\varphi);F(\varepsilon)]$?*

We say that the theory $S_1[\varphi;\varepsilon]$ *emerges from the theory* $S_2[\psi;\delta]$ if the problem above has a positive solution, i.e, if we can fully describe S_1 in terms of S_2. Notice, however, that as stated the emergence problem is fairly general. Indeed, if P_1 and P_2 have different ranks, then, by the previous discussion, this means that the parameterized theories S_1 and S_2 have a different number of fundamental scales, so that we should not expect an emergence relation between them. This leads us to think of considering only the case in which $P_1 = P_2$. However, we could also consider the situations in which $P_1 \neq P_2$, but $P_2 = f(P_1)$ is some nice function of P_1, e.g, $P_2 = P_1 \times P_1 \times ... \times P_1$. In these cases the fundamental scales remain only those of P_1, since from them we can generate those in the product. Throughout this paper we will also work with different theories defined on the same fields, i.e, $E_1 = E_2$. This will allow us to search for emergence relations in which G is the identity map $G(\varphi) = \varphi$.

Hence, after these hypotheses, we can rewrite our main problem, whose affirmative solutions axiomatize the notion of strong emergence we are searching for:

Question 3. *Let $S_1[\varphi;\varepsilon]$ and $S_2[\varphi;\delta]$ be two parametrized theories defined on the same spacetime M, on the same field bundle E and on the parameter bundles P_1 and $P_2 = f(P_1)$, respectively. Does there exists some map $F : \mathrm{Par}_1(P_1) \to \mathrm{Par}_2(f(P_1))$ such that $S_1[\varphi;\varepsilon] = S_2[\varphi, F(\varepsilon)]$?*

Our plan is to show that the problem in Question 3 has an affirmative solution for certain class of parameterized field theories, which by the absence of a better name we call *generalized parameterized field theories*. This will be obtained from the class of *differential parameterized theories* by means of allowing dynamical operators which are not necessarily differential, but some generalized version of them.

2.1 Differential Parameterized Theories

In order to motivate the need for looking at a special class of field theories, we begin by noticing that typically the field theories are *local*, in the sense that they are defined by means of integrating some Lagrangian density $\mathcal{L}(x,\varphi,\partial\varphi,\partial^2\varphi,...)$, i.e, $S[\varphi] = \int_M \mathcal{L}(j^\infty\varphi)dx^n$, where $j^\infty\varphi = (x,\varphi,\partial\varphi,\partial^2\varphi,...)$ is the jet prolongation. On the other hand, a quick look at the standard examples of field theories shows that, when working in a spacetime without boundary, after integration by parts and using Stoke's theorem, those field theories can be stated, at least locally, in the form $\mathcal{L}(x,\varphi,\partial\varphi) = \langle\varphi, D\varphi\rangle$, where $\langle\varphi,\varphi'\rangle$ is a nondegenerate pairing on the space of fields $\Gamma(E)$ and $D : \Gamma(E) \to \Gamma(E)$ is a differential operator of degree d, which means that it can be locally written as $D\varphi(x) = \sum_{|\alpha|\leq d} a_\alpha(x)\partial^\alpha\varphi$, where $\alpha = (\alpha_1, ..., \alpha_r)$ is some mult-index, $|\alpha| = \alpha_1 + ... + \alpha_r$ is its degree and $\partial^\alpha = \partial_1^{\alpha_1} \circ ... \circ \partial_r^{\alpha_r}$, with $\partial_i^l = \partial^l/\partial^l x_i$. This is the case, e.g, of φ^3 and φ^4 scalar field theories, the standard spinorial field theories, Einstein-Hilbert-Palatini-Type theories [33, 34] as well as Yang-Mills-type theories and certain canonical extensions of them [36, 37]. More generally, recall that the first step in building the Feynman rules of a field theory is to find the (kinematic part of the) operator D and take its "propagator" .

In these examples, the pairing $\langle\varphi,\varphi'\rangle$ is typically symmetric (resp. skew-symmetric) and the operator D is formally self-adjoint (resp. formally anti-self-adjoint) relative to that pairing. Furthermore, $\langle\varphi,\varphi'\rangle$ is usually a L^2-pairing induced by a semi-Riemannian metric g on the field bundle E and/or on the spacetime M, while D is usually a generalized Laplacian or a Dirac-type operator relative to g [17]. For example, this holds for the concrete field theories (scalar, spinorial and Yang-Mills) above. The skew-symmetric case generally arises in gauge theories (BV-BRST quantization) after introducing the Faddeev-Popov ghosts/anti-ghosts and it depends on the grading introduced by the ghost number [17].

Another remark, still concerning the concrete situations above, is that if the metric g inducing the pairing $\langle\varphi,\varphi'\rangle$ is actually Riemannian (which means that the gravitational background is Euclidean), then $\langle\varphi,\varphi'\rangle$ becomes a genuine L^2-inner product and D is elliptic and extends to a bounded self-adjoint operator between Sobolev spaces [20, 23]. Working with elliptic operators is very useful, since they always admits parametrices (which in this Euclidean cases are the propagators) and for generalized Laplacians the heat kernel not only exists, but also has a well-known asymptotic behavior [50], which is very nice in the Dirac-type case [11].

From the discussion above, it is natural to focus on parameterized theories such that each functional $S[\varphi;\varepsilon]$ is local and defined by a Lagrangian density of the $\mathcal{L}(\jmath^\infty\varphi) = \langle\varphi, D_\varepsilon\varphi\rangle$, i.e, are determined by a single nondegenerate pairing $\langle\varphi,\varphi'\rangle$ in $\Gamma(E)$, fixed a priori by the nature of M and E, and by a family of differential operators $D_\varepsilon \in \mathrm{Diff}(E)$, one for each parameter $\varepsilon \in \mathrm{Par}(P)$, where $\mathrm{Diff}(E) = \bigoplus_d \mathrm{Diff}^d(E;E)$ denotes the $\mathbb{K}$-vector space of all differential operators in E (which is actually a $\mathbb{K}$-algebra which the composition operation) and $\mathrm{Diff}^d(E;E)$ is the space of those operators of degree d. Thus:

Definition 2.1. Let M be a compact and oriented manifold. A *background for doing emergence theory* (or simply *background*) over M is given by the following data:

1. a $\mathbb{K}$-vector bundle $E \to M$ (the field bundle);
2. a pairing $\langle\varphi,\varphi'\rangle$ in $\Gamma(E)$;
3. a parameter bundle $P \to M$ and a set of parameters $\mathrm{Par}(P) \subset \Gamma(P)$.

Definition 2.2. A *differential parameterized theory* in a background over M is a collection of differential operators $D_\varepsilon \in \mathrm{Diff}(E)$ with $\varepsilon \in \mathrm{Par}(P)$. The *parameterized Lagrangian density* is given by $\mathcal{L}(\varphi;\varepsilon) = \langle\varphi, D_\varepsilon\varphi\rangle$. The *parameterized action functional* is the integral of the parameterized Lagrangian in M.

2.2 Generalized Parameterized Theories

Sometimes working on the narrow class of field theories defined by differential operators is not enough. For instance, notice that in building the "propagator" of a Lagrangian field theory

$\mathcal{L}(\jmath^\infty\varphi) = \langle\varphi, D\varphi\rangle$ we are actually finding some kind of "quasi-inverse" D^{-1} for the differential operator D. For example, in the Riemannian case, where differential operators $D \in \mathrm{Diff}^d(E)$ extend to bounded operators $\hat{D} \in B(W^{d,2}(E))$ in the Sobolev space, if D is elliptic, then building the propagator is equivalent to building parametrices, which in turn can described in terms of Fredholm inverses for $\hat{D}$ [17]. On the other hand, in the Lorentzian setting (where the spacetime manifold is assumed globally hyperbolic and the typical differential operators are hyperbolic), building the propagator is about finding its advanced and retarded Green functions [7, 6].

Independently of the case, it would be very useful if the quasi-inverse D^{-1} could exist as a differential operator, i.e., if $D^{-1} \in \mathrm{Diff}(E)$. Indeed, in this case D^{-1} would also define a differential field theory. In particular, in the parameterized case, if each D_ε has a "quasi-inverse" described by a differential operator D_ε^{-1}, the collection of them define a parameterized differential theory over the same background. In turn, from the physical viewpoint, it would be very interesting if the "quasi-inverse" D^{-1} of the differential operator D was a genuine inverse for D in the algebra $\mathrm{Diff}(E)$. Indeed, in this case one could use the relations $D \circ D^{-1} = I$ and $D^{-1} \circ D = I$ in order to get global solutions for the equation of motion $D\varphi = 0$ [45].

Notice, however, that the equations $D \circ D^{-1} = I = D^{-1} \circ D$ typically has no solutions in $\mathrm{Diff}(E)$. Indeed, recall that $\mathrm{Diff}(E) = \bigoplus_d \mathrm{Diff}^d(E;E)$ is a $\mathbb{Z}_{\geq 0}$-graded algebra with composition, so that if D^{-1} was a left or right inverse for D, then $\deg(D^{-1}) = -\deg D$, which has no solution if $\deg D \neq 0$. This forces us to search for extensions of the algebra $\mathrm{Diff}(E)$ such that left and/or right inverses of differential operators may exist. The obvious idea is to consider extensions by adding a negative grading to $\mathrm{Diff}(E)$ getting the structure of a $\mathbb{Z}$-graded $\mathbb{K}$-algebra, and the natural choice is the $\mathbb{Z}$-graded algebra $\mathrm{Psd}(E)$ of pseudo-differential operators, which are defined via symbol-theoretic (i.e., microlocal analysis) approach [32, 38].

For constant coefficient operators, the right-inverse really exist in $\mathrm{Psd}(E)$. In the case of non-constant coefficients, for Riemannian spacetimes, elliptic operators satisfying specific ellipticity conditions admit right-inverse in $\mathrm{Psd}(E)$ [28, 14, 47, 48, 2, 29, 3, 25]. For globally hyperbolic Lorentzian spacetimes, at least when restricted to fields with support in the causal cones, Green hyperbolic operators (in particular normally hyperbolic operators) have both left and right inverses in $\mathrm{Psd}(E)$ [7, 6]. For flat spacetimes, more abstract examples exist [27]. On the other hand, if a differential operator is not right-invertible in $\mathrm{Psd}(E)$ it could be invertible in another extension of $\mathrm{Psd}(E)$, such as in the class of Fourier-type operators [45].

Our main result does not depend on the extension of $\mathrm{Diff}(E)$; the only thing we need is the existence of a right-inverse for a differential operator as *some* kind of operator, i.e., we only need some $\mathbb{K}$-algebra $\mathrm{Op}(E)$ such that $\mathrm{Diff}(E) \subset \mathrm{Op}(E) \subset \mathrm{End}(\Gamma(E))$ and such that the subset of differential operators $\mathrm{Op}(E)$ contains nontrivial elements.

Definition 2.3. Let M be a compact and orientable manifold. A *generalized background* over M, denoted by $\mathrm{GB}(M)$ is given by the same data in Definition 2.1 and, in addition, a $\mathbb{K}$-algebra $\mathrm{Op}(E) \subset \mathrm{End}(\Gamma(E))$ extending $\mathrm{Diff}(E)$ and such that $\mathrm{Diff}(E) \cap R\,\mathrm{Op}(E)$ contains non-multiplies of the identity, where $R\,\mathrm{Op}(E)$ is the set of right-invertible elements of $\mathrm{Op}(E)$.

Definition 2.4. A *generalized parameterized theory* (GPT) in a generalized background $\mathrm{GB}(M)$ is a collection of generalized operators $\Psi_\varepsilon \in \mathrm{Op}(E)$, with $\varepsilon \in \mathrm{Par}(P)$. The corresponding Lagrangian density action funcional are defined analogously to Definition 2.2, just replacing D_ε with Ψ_ε.

Definition 2.5. We say that a GPT over M is *right-invertible* if the generalized operators Ψ_ε are right-invertible, i.e, if they belongs to $R\,\mathrm{Op}(E)$ for every $\varepsilon \in \mathrm{Par}(P)$.

2.3 Polynomial Parameterized Theories

Until this moment we considered theories which are defined by a 1-parameter family of differential (or more general) operators. As discussed, these provide a description of *free* theories and of their parameterized version. We will show, however, that the concept in broad enough to describe interacting theories as well. In order to do this, recall that the classical examples of interacting field theories has interacting terms given by multivariate polynomials with variables corresponding to operators of different theories at interaction, as discussed at the introductory section.

But, in order to talk about polynomials we need a ring of coefficients. In our parameterized context, the natural idea is to consider coefficients depending on the parameters. This leads us to look at the ring $\mathrm{Map}(\mathrm{Par}(P);\mathbb{K})$ of scalar functions $f:\mathrm{Par}(P)\to\mathbb{K}$ on the space of parameters. For a given $l\geq 0$ and formal variables $x_1,...,x_r$, let $\mathrm{Map}_l(\mathrm{Par}(P);\mathbb{K})[x_1,...,x_r]$ be the corresponding $\mathbb{K}$-vector space of multivariate polynomials of degree l in the given variables. Thus, an element of it can be written as $p^l[x_1,...,x_r]=\sum_{|\alpha|\leq l} f_\alpha\cdot x^\alpha$, where $f_\alpha\in\mathrm{Map}(\mathrm{Par}(P);\mathbb{K})$ and α is a multi-index. If $\Psi_1,...,\Psi_r\in\mathrm{Op}(E)$ are fixed generalized operators and $p^l[x_1,...,x_r]$ is a polynomial as above, recalling that $\mathrm{Op}(E)$ is a $\mathbb{K}$-algebra, by means of replacing the formal variables x_i with the operators Ψ_i we get a family of operators $p^l[\Psi_1,...,\Psi_r](\varepsilon)=\sum_{|\alpha|\leq l} f_\alpha(\varepsilon)\Psi^\alpha$, with $\varepsilon\in\mathrm{Par}(P)$. The following definitions are then natural.

Definition 2.6. Let M be a compact and oriented manifold. A *polynomial parameterized theory* (*PPT*) *of degree* $l\geq 0$ *in* r *variables*, defined on a generalized context $\mathrm{GB}(M)$, is given by an element p^l of $\mathrm{Map}_l(\mathrm{Par}(P);\mathbb{K})[x_1,...,x_r]$ and by operators $\Psi_i\in\mathrm{Op}(E)$, with $i=1,...,r$. The parameterized Lagrangian and the parameterized action functional are

$$\mathcal{L}(j^\infty\varphi;\varepsilon)=\langle\varphi,p^l[\Psi_1,...,\Psi_r](\varepsilon)\varphi\rangle \quad \text{and} \quad S[\varphi;\varepsilon]=\ll\varphi,p^l[\Psi_1,...,\Psi_r](\varepsilon)\varphi\gg,$$

while the extended Lagrangian and the extended action functional are defined analogously by means of replacing ψ_i with $\widetilde{\Psi}_i$.

Now, to check that the definition above really captures the standard examples of interacting terms, if the interaction to be described is of $j>1$ different fields, the corresponding PPT is usually of j variables and such that $E\simeq\oplus_i E_i$, with $i=1,...,j$, and $\Psi_i=(0,...,\Phi_i,...,0)$, where $\Psi_i\in\Gamma(E)$ and $\Phi_i\in\Gamma(E_i)$. Here, E_i is the field bundle of the ith field theory[4].

Remark 2.1. Every PPT of arbitrary degree l and in arbitrary number r of variables is a GPT, on the same generalized background, with parameterized operator $\Psi_\varepsilon=p^l[\Psi_1,...,\Psi_r](\varepsilon)$, so that the concept of GPT is really broad enough to describe both free and interacting terms of a typical Lagrangian field theory.

Closing this section, notice that a priori we have two possible definitions of right-invertibility for a PPT. Since by the above every PPT is a GPT, we could say that a PPT is *right-invertible* if it is as a GPT, i.e., if for every ε the generalized operator $p^l[\Psi_1,...,\Psi_r](\varepsilon)$ is right-invertible. But we could also define a *right-invertible* PPT as such that each generalized operator Ψ_i, with $i=1,...,r$, is right-invertible. These conditions are very different. Indeed, the first one is about the invertibility of *multivariate polynomials* in noncommutative variables, while the second one is about the invertibility of the *variables* of a multivariate polynomial. Thus, the first one relies on

[4] Gauge field with gauge group G are incorporated as follows. Recall that they are principal G-connections, which can be regarded as 1-forms in M taking values in the adjoint G-bundle $E_{\mathfrak{g}}$. Thus, we just take $E_i=TM^*\otimes E_{\mathfrak{g}}$.

constraints on the polynomials p^l, while the second one is a condition only on the operators Ψ_i. Luckily, we will need only the second condition, leading us to the define:

Definition 2.7. We say that a PPT in r variables is *right-invertible* if the defining operators Ψ_i, with $i = 1, ..., r$, are right-invertible.

2.4 Higher Parameter Degree

Notice that, as in Definitions 2.2, 2.5 and 2.6, a parameterized field theory has as parameters a *single* element $\varepsilon \in \mathrm{Par}(P)$. On the other hand, as discussed at the beginning of Section 2, a physical theory may depend on many fundamental scales. This leads us to consider parameterized theories depending on a list $\varepsilon(\ell) = (\varepsilon_1, ..., \varepsilon_\ell)$ of elements of $\mathrm{Par}(P)$, i.e, such that $\varepsilon(\ell) \in \mathrm{Par}(P)^\ell$. In this case, we say that the number $\ell \geq 1$ is the *fundamental degree* of the parameterized theory. We will also use the convention that $\mathrm{Par}(P)^0$ is a singleton, whose element we denote by $\varepsilon(0)$. More precisely, we have the following definition.

Definition 2.8. Let M be a compact and oriented manifold. A *GPT of degree* $\ell \geq 0$ in a generalized background $\mathrm{GB}(M)$ is given by a collection of generalized operators $\Psi_{\varepsilon(\ell)} \in \mathrm{Op}(E)$. Particularly, a *PPT of degree* (l, ℓ) *in* r *variables* is given by an element $p^l_\ell \in \mathrm{Map}_l(\mathrm{Par}(P)^\ell; \mathbb{K})[x_1, ..., x_r]$ and by operators $\Psi_i \in \mathrm{Op}(E)$, with $i = 1, ..., r$.

3 The Emergence Theorem

After the previous digression we are now ready to state our main theorem. The syntax is the following:

Main Theorem Syntax. *Let M be a compact and oriented manifold with a fixed generalized background* $\mathrm{GB}(M)$. *Let S_1 be a GPT of degree ℓ and S_2 a PPT of degree (l, ℓ') in r variables. Suppose that:*

1. *S_1 depends on the fundamental parameters $\varepsilon(\ell)$ in a "linear" way;*
2. *the PPT theory S_2 is right-invertible and its coefficients f_α are "suitable" functions in a sense that depends on r, l and ℓ'.*

Then S_1 emerges from S_2.

In order to turn this syntax into a rigorous statement, let us described what we meant by a "linear" dependence on the parameters and by "suitable" functions. Since the term "linear" typically means "preservation of some algebraic structure", it is implicit that we will need to assume some algebraic structure on the space of parameters $\mathrm{Par}(P)^\ell$. To begin, we will require an associative and unital $\mathbb{K}$-algebra structure, whose sum and multiplication we will denote by "$+^\ell$" and "$*^\ell$", respectively, or simply by "$+$" and "$*$" when the number ℓ of fundamental parameters is implicit. Thus,

Definition 3.1. A GPT of degree ℓ in a generalized background $\mathrm{GB}(M)$ is *linear* (or *homomorphic*) if $\mathrm{Par}(P)^\ell$ has a $\mathbb{K}$-algebra structure and the rule $\varepsilon(\ell) \mapsto \Psi_{\varepsilon(\ell)}$ is a $\mathbb{K}$-algebra homomorphism.

Some properties of the emergence phenomena (as those proved in Subsection 4.1) depends only on the preservation of sum "+", scalar multiplication, or product "$*$". This motivates the following definition:

Definition 3.2. Under the same notations and hypotheses of the last definition, we say that a GPT is *additive* (resp. *multiplicative*) if the rule $\varepsilon(\ell) \mapsto \Psi_{\varepsilon(\ell)}$ is $\mathbb{K}$-linear (resp. a multiplicative monoid homomorphism). If it is only required $\Psi_{c\varepsilon(\ell)} = c\Psi_{\varepsilon(\ell)}$ we say that the GPT is *scalar invariant*.

On the other hand, notice that in a PPT the generalized operators Ψ always appear multiplied by a *number* $f(\varepsilon(\ell)) \in \mathbb{K}$, which depends on the parameters $\varepsilon(\ell)$. However, if one recalls that the parameters are interpreted as fundamental scales, it should be natural to consider parameters $\varepsilon(\ell)$ (instead of numbers $f(\varepsilon(\ell)) \in \mathbb{K}$ assigned to them) multiplying the operators Ψ. Thus, we need an action $\cdot^{\ell} : \mathrm{Par}(P)^{\ell} \times \mathrm{Op}(E) \to \mathrm{Op}(E)$. But, since by Definition 2.3 and by the above assumption both $\mathrm{Op}(E)$ and $\mathrm{Par}(P)^{\ell}$ are $\mathbb{K}$-algebras, it is natural to require some compatibility between the action $\cdot^{\ell}$ and these $\mathbb{K}$-algebra structures. More precisely, we will assume that $\cdot^{\ell}$ is $\mathbb{K}$-bilinear and that the following condition are satisfied for every $\Psi, \Psi' \in \mathrm{Op}(E)$ and every $\varepsilon(\ell) \in \mathrm{Par}(P)^{\ell}$:

$$(\varepsilon(\ell) \cdot^{\ell} \Psi) \circ \Psi' = \varepsilon(\ell) \cdot^{\ell} (\Psi \circ \Psi') \quad \text{and} \quad (\varepsilon(\ell) * \delta(\ell)) \cdot^{\ell} \Psi = \varepsilon(\ell) \cdot^{\ell} (\delta(\ell) \cdot^{\ell} \Psi). \tag{7}$$

Remark 3.1. If not only conditions (7) are satisfied, but also

$$\Psi \circ (\varepsilon(\ell) \cdot^{\ell} \Psi') = \varepsilon(\ell) \cdot^{\ell} (\Psi \circ \Psi') \tag{8}$$

is satisfied for every $\varepsilon(\ell)$ and every Ψ, Ψ', then the algebra $\mathrm{Par}(P)^{\ell}$ must be commutative. See Comment 3. On the other hand, if (7) is satisfied for every $\varepsilon(\ell)$ and every Ψ, Ψ', but (8) is satisfied for every $\varepsilon(\ell)$ and a *single* $\Psi' = \Psi$, then $\mathrm{Par}(P)^{\ell}$ need not be commutative.

Now, to make a rigorous sense of Condition 2 in the previous syntactic statement, let us clarify what we mean by a "suitable" function $f \in \mathrm{Map}(\mathrm{Par}(P)^{\ell}; \mathbb{K})$. In few words, a function f is "suitable" if it belongs to some functional calculus. For us, a *functional calculus of degree* ℓ is a subset $C_{\ell}(P; \mathbb{K})$ of $\mathrm{Map}(\mathrm{Par}(P)^{\ell}; \mathbb{K})$ endowed with a function $\Psi^{\ell}_{-} : C_{\ell}(P; \mathbb{K}) \to \mathrm{Op}(E)$ assigning to each function $f \in C_{\ell}(P; \mathbb{K})$ a generalized operator Ψ^{ℓ}_f, which is compatible with the action $\cdot^{\ell}$, in the sense that

$$\Psi^{\ell}_f \circ [f(\varepsilon(\ell))\Psi] = \varepsilon(\ell) \cdot^{\ell} \Psi. \tag{9}$$

Definition 3.3. A functional calculus is *unital* if $C_{\ell}(P; \mathbb{K})$ contains the constant function $f \equiv 1$.

We notice that, although our main emergence theorem will depend on the choice of a unital functional calculus, the existence of them is not an obstruction.

Lemma 3.1. *In every generalized background* $\mathrm{GB}(M)$*, for every* $\ell \geq 0$ *there exists a unique functional calculus of degree* ℓ *such that* $C_{\ell}(P; \mathbb{K})$ *is the set of nowhere vanishing functions* $f : \mathrm{Par}(P)^{\ell} \to \mathbb{K}$ *if* $\ell > 0$*, and the constant function* $f \equiv 1$ *if* $\ell = 0$*.*

Proof. First, assume existence for every $\ell \geq 0$. Uniqueness in the case $\ell = 0$ is obvious. In the case $\ell > 0$, from the existence hypothesis we have $\Psi^{\ell}_f \circ [f(\varepsilon(\ell))\Psi] = \varepsilon(\ell) \cdot^{\ell} \Psi$ for every f, Ψ and $\varepsilon(\ell)$, so that $\Psi^{\ell}_f \circ \Psi = (\varepsilon(\ell) \cdot^{\ell} \Psi)/f(\varepsilon(\ell))$. In particular, for $\Psi = I$, we get $\Psi^{\ell}_f = (\varepsilon(\ell) \cdot^{\ell} I)/f(\varepsilon(\ell))$, proving uniqueness. In order to prove existence, for each $\ell \geq 0$ define $\Psi^{\ell}_f = (\varepsilon(\ell) \cdot^{\ell} I)/f(\varepsilon(\ell))$, so that

$$\Psi^{\ell}_f \circ [f(\varepsilon(\ell))\Psi] = (\varepsilon(\ell) \cdot^{\ell} I) \circ \Psi = \varepsilon(\ell) \cdot^{\ell} (I \circ \Psi) = \varepsilon(\ell) \cdot^{\ell} \Psi,$$

where in the last step we used the compatibility between $\cdot^{\ell}$ and $\circ$, as described in (7). $\square$

Finally, let us state a few more necessary technical assumptions:

1. *The* $\mathbb{K}$*-algebra of fundamental parameters* $\mathrm{Par}(P)^\ell$ *is required to have square roots.* This means that the function $\varepsilon(\ell) \mapsto \varepsilon(\ell) * \varepsilon(\ell)$ is surjective, i.e, for every $\varepsilon(\ell)$ we there is another $\sqrt{\varepsilon(\ell)} \in \mathrm{Par}(P)^\ell$ such that
$$(\sqrt{\varepsilon(\ell)})^2 = \sqrt{\varepsilon(\ell)} * \sqrt{\varepsilon(\ell)} = \varepsilon(\ell).$$

2. *Right multiplication by* I *is injective.* More precisely, for every $\varepsilon(\ell), \delta(\ell) \in \mathrm{Par}(P)^\ell$, if $\varepsilon(\ell) \cdot^\ell I = \delta(\ell) \cdot^\ell I$, then $\varepsilon(\ell) = \delta(\ell)$, i.e., $\varphi_i = \delta_i$ for $i = 1, ..., \ell$.

Some comments concerning these technical conditions:

1. The square roots $\sqrt{\varepsilon(\ell)}$ in condition 1. need not be unique.

2. Condition 2 above and compatibility conditions (7) imply that the right multiplication $r^\ell_\Psi : \mathrm{Par}^\ell(P) \to \mathrm{Op}(E)$ is injective for every right-invertible operator $\Psi \in R\,\mathrm{Op}(E)$. Thus, for every such Ψ, the map r^ℓ_Ψ is actually an isomorphism $\mathrm{Par}(P)^\ell \simeq \mathrm{Par}(P)^\ell \cdot^\ell \Psi$ between its domain and its image.

3. If both compatibility conditions (7) and (8) are satisfied, then Condition 2 above implies that the $\mathbb{K}$-algebra $(\mathrm{Par}(P)^\ell, *^\ell)$ is commutative. This follows basically from the Eckmann-Hilton argument [21, 24].

Now, suppose that $\mathrm{Par}(P)^\ell$ has a $\mathbb{K}$-algebra structure. Then $\mathrm{Par}(P)^{k\ell}$ has an induced $\mathbb{K}$-algebra structure, with $k > 0$, given by componentwise sum and multiplication. If $\mathrm{Par}(P)^\ell$ has square roots, then $\mathrm{Par}(P)^{k\ell}$ has too, given by $\sqrt{\varepsilon(k\ell)} = (\sqrt{\varepsilon(\ell_1)}, ..., \sqrt{\varepsilon(\ell_k)})$, where $\varepsilon(k\ell) = (\varepsilon(\ell_1), ..., \varepsilon(\ell_k))$ and $\varepsilon(\ell_i) = (\varepsilon_{i,1}, ..., \varepsilon_{i,\ell})$. On the other hand, since $\mathrm{Par}(P)^{l\ell}$ embeds in $\mathrm{Par}(P)^{k\ell}$ as $\varepsilon(\ell) \mapsto (\varepsilon(\ell_1), ..., \varepsilon(\ell_l), 0, ..., 0)$, if $l \leq k$, it follows that every action $\cdot^{k\ell}$ of $\mathrm{Par}(P)^{k\ell}$ can be pulled back to an action $\cdot^{l\ell}$ of $\mathrm{Par}(P)^{l\ell}$.

If $C_{k\ell}(P;\mathbb{K}) \subset \mathrm{Map}(\mathrm{Par}(P)^{k\ell};\mathbb{K})$ is a set of functions, pullback by the inclusion $\imath : \mathrm{Par}(P)^{l\ell} \hookrightarrow \mathrm{Par}(P)^{k\ell}$ above defines a new set of functions $C_{l\ell;k}(P;\mathbb{K}) \subset \mathrm{Map}(\mathrm{Par}(P)^{l\ell};\mathbb{K})$. Furthermore, if $C_{k\ell}(P;\mathbb{K})$ is actually a functional calculus defined by a map $\Psi^{k\ell}_{-} : C_{k\ell}(P;\mathbb{K}) \to \mathrm{Op}(E)$, we have an induced functional calculus in $C_{l\ell;k}(P;\mathbb{K})$, defined by the function $\Psi^{l\ell;k}_{-} : C_{l\ell;k}(P;\mathbb{K}) \to \mathrm{Op}(E)$ such that $\Psi^{l\ell;k}_{\imath\circ f} = \Psi^{k\ell}_f$. Notice that if $C_{k\ell}(P;\mathbb{K})$ is unital and/or satisfies the third technical conditions above, then $C_{l\ell;k}(P;\mathbb{K})$ does.

3.1 Formal Statement

We can now rigorously state the main result of this paper. First, notice that the three technical conditions of last section are about algebraic properties of the space $\mathrm{Par}^\ell(P)$ of fundamental parameters and on its action on the algebra $\mathrm{Op}(E)$ of generalized operators. On the other hand, recall from Definition 2.3 that both $\mathrm{Par}^\ell(P)$ and $\mathrm{Op}(E)$ are part of the data defining a generalized background. Thus, those conditions are actually conditions on the underlying generalized background $\mathrm{GB}(M)$. This motivates the following definition.

Definition 3.4. Let M be a compact and oriented smooth manifold and let $\ell \geq 0$ and $k > 0$ be non-negative integers. We say that a generalized background $\mathrm{GB}(M)$ is of (ℓ, k)*-type* if:

1. the space of fundamental parameters $\mathrm{Par}^{\ell}(P)$ has an structure of $\mathbb{K}$-algebra with square roots;
2. the induced $\mathbb{K}$-algebra $\mathrm{Par}^{k\ell}(P)$ acts in $\mathrm{Op}(E)$ by an action $\cdot^{k\ell}$ which is injective at the identity operator I and compatible with a functional calculus $C_{k\ell}(P;\mathbb{K})$.

Our main theorem is then the following:

Theorem 3.1 (Emergence Theorem)**.** *Let M be a compact and oriented manifold and let* $\mathrm{GB}(M)$ *be a generalized background of (ℓ,k)-type, with $k>0$. Let S_1 be a GPT of degree ℓ and let S_2 be a PPT of degree (l,ℓ') in r variables, where $\ell'=k'\ell$, with $0<k'\leq k$, defined on* $\mathrm{GB}(M)$*. Assume:*

1. *S_1 is homomorphic;*
2. *S_2 is right-invertible and the coefficient functions $f_\alpha:\mathrm{Par}(P)^{k'\ell}\to\mathbb{K}$ belongs to the functional calculus $C_{k'\ell;k}(P;\mathbb{K})$.*

Then S_1 emerges from S_2.

Before proving the emergence theorem, let us say that the proof which will be given here can be generalized, with basically the same steps and arguments, in some directions (for more details, see [35]):

1. *the spacetime manifold M need not be compact nor orientable.* Notice that these conditions were used only to define integrals. Thus, instead, one could only assume integrability conditions on global sections (such as compact supportness) and consider Lagrangians as taking values on general densities.
2. *the space of fundamental parameters need not be a $\mathbb{K}$-algebra, but a more lax algebraic entity.* As will be clear during the proofs, we only need the "nonnegative" part of a $\mathbb{K}$-algebra. More precisely, what we really need is that the set $\mathrm{Par}(P)^{\ell}$ can be regarded as the subset of a $\mathbb{K}$-algebra A, which is closed by sum, multiplication, and scalar multiplication by $\mathbb{R}_{\geq 0}$. Notice that if P is a $\mathbb{K}$-algebra bundle, then $\mathrm{Par}(P)^{\ell}$ can always be realized as a subset of the $\mathbb{K}$-algebra $\Gamma(P)^{\ell}$.
3. the coefficient functions f_α need not be scalar functions, but actually maps $f_\alpha:\mathrm{Par}(P)^{\ell'}\to\mathrm{Par}(P)^{\ell}$, so that the scalar multiplication $f_\alpha(\varepsilon(\ell))\Psi$ is replaced by the action $\cdot^{\ell}$. The notions of functional calculus, etc., can be defined in an analogous way such that the syntax of the theorem remains the same[5].

3.2 Some Particular Cases

Let M be a compact and orientable manifold, $\mathbb{K}=\mathbb{C}$ an $E=M\times\mathbb{C}$ the complex trivial line bundle, regarded as a field bundle with space of fields given by complex scalar functions $C^{\infty}(M;\mathbb{C})$, endowed with the pairing $\langle\varphi,\psi\rangle=\varphi\overline{\psi}$. In addition, let $P=M\times\mathbb{C}$, viewed now as the parameter bundle, and take the constant functions as parameters, so that $\mathrm{Par}(P)\simeq\mathbb{C}$. This data clearly defines a background over M. Let $\mathrm{Op}(E)$ be the complex algebra $\mathrm{Psd}(M\times\mathbb{C})$ of pseudo-differential

[5]Recall from the discussion at Introduction that parameter-valued coefficient functions plays an important role in the description of emergent gravity in terms of emergence phenomena. Further explanations will appear in a work in progress.

operators. By the discussion of Section 2.2 we then have a generalized background. Notice that $\mathrm{Par}(P) \simeq \mathbb{C}$ is a $\mathbb{C}$-algebra with square roots and with their action in $\mathrm{Psd}(M \times \mathbb{C})$ via scalar multiplication, if $z \cdot I = z' \cdot I$, then clearly $z = z'$. Finally, let $C_1(P; \mathbb{C})$ be the unital functional calculus given by nowhere vanishing functions $f : \mathrm{Par}(P) \simeq \mathbb{C} \to \mathbb{C}$, as in Lemma 3.1, defining a generalized background over M of $(1,1)$-type.

As a particular case of Theorem 3.1 we then have:

Corollary 3.1. *Let M be a compact and oriented manifold and let $\mathrm{GB}(M)$ be the generalized background of $(1,1)$-type defined above. Let $D \in \mathrm{Diff}(M \times \mathbb{C})$ be any idempotent differential operator, i.e., there exists $n > 0$ such that $D^{2n} = D^n$. For given $r > 0$ and $l \geq 0$, let $p^l[x_1, .., x_r]$ be a polynomial of degree l in r variables and whose coefficients $f_\alpha : \mathbb{C} \to \mathbb{C}$ are nowhere vanishing functions. Let $D_1, ..., D_r \in \mathrm{Diff}(M \times \mathbb{C})$ other differential operators and assume one of the following conditions:*

1. *the operators D_i, with $i = 1, ..., r$ are of constant coefficient;*
2. *there exists a Riemannian metric in M such that D_i, with $i = 1, ..., r$ are strongly elliptic in the sense of any of references [28, 14, 47, 48, 2, 29, 3, 25];*
3. *there exists a Lorentzian metric in M such that M is globally hyperbolic and each D_i, with $i = 1, ..., r$, is Green hyperbolic.*

Then theory $\mathcal{L}_1(\jmath^\infty \varphi; \varepsilon) = \overline{\varphi} \varepsilon D^n \varphi$ emerges from theory

$$\mathcal{L}_2(\jmath^\infty \varphi; \delta) = \overline{\varphi} p^l[D_1, ..., D_r]\varphi = \sum_{|\alpha| \leq l} \overline{\varphi} f_\alpha(\delta) D^\alpha \varphi.$$

Proof. Since $D^{2n} = D^n \circ D^n = D^n$, the rule $\varepsilon \mapsto \varepsilon D^n$ is clearly homomorphic. On the other hand, from the discussion on Section 2.2 each of the three hypotheses above implies that D_i, for $i = 1, ..., r$, are right-invertible as objects of $\mathrm{Psd}(M \times \mathbb{C})$. Since the coefficient functions f_α are nowhere vanishing and therefore belong to the functional calculus of $\mathrm{GB}(M)$, the result follows from Theorem 3.1. □

Remark 3.2. From Comment 1, the same construction of $\mathrm{GB}(M)$ holds if M is a bounded open set of some $\mathbb{R}^N$. From Comment 3 it remains valid if $\mathrm{Par}(P) \simeq \mathbb{R}_{\geq 0}$ are the constant non-negative real functions and the functional calculus $C_1(\mathbb{R}_{\geq 0}; \mathbb{C})$ consists of the nowhere vanishing functions taking values in $\mathbb{R}_{\geq 0}$. The difference, in this case, is that both ε and $f_\alpha(\delta)$ are real numbers, so that the Lagrangians in the last corollary are real too. See [35] for further details.

Other generalizations of the last corollary, and particular cases of Theorem 3.1, are the following:

1. *We can consider other kind of fields.* In Corollary 3.2 we considered a generalized background defined on the trivial line bundle $M \times \mathbb{C}$. Notice, however, that if E is any complex bundle with an Hermitian metric at the fibers, then the space of pseudo-differential operators $\mathrm{Psd}(E)$ remains well-defined as a $\mathbb{Z}$-graded $\mathbb{C}$-algebra, so that the same thing holds equally well if instead of scalar fields one considered vector fields and tensor fields.

2. *We can consider other kind of parameters.* In Corollary 3.2 we considered $\ell = 1$ and $\mathrm{Par}(P) \simeq \mathbb{C}$ (or $\mathbb{R}_{\geq 0}$, due to the remark above). We could consider, more generally, $\mathrm{Par}(P)$ as any complex algebra with square roots endowed with an action $\cdot : \mathrm{Par}(P) \times \mathrm{Psd}(E) \to \mathrm{Psd}(E)$, where E is a complex vector bundle (due to the last remark), such that conditions (7) and Condition 2 are satisfied. Let Ψ_0 be such that Ψ_0^n is idempotent and suppose that (8) are satisfied for fixed $\Psi = \Psi' = \Psi_0^n$. Then the rule $\varepsilon \mapsto \varepsilon \cdot \Psi_0^n$ is an algebra homomorphism and Corollary 3.2 holds equally well.

   Example 3.1 (operator parameters)**.** Take $P = \mathrm{End}(E)$, so that $\Gamma(P) \simeq \mathrm{End}(\Gamma(E))$, which is an associative $\mathbb{C}$-algebra with an obvious action in $\mathrm{Psd}(E)$ by composition such that conditions (7) and Condition 2 are clearly satisfied. Let $\Psi_0 \in \mathrm{Psd}(E)$ be such that Ψ_0^n is idempotent and let $Z(\Psi_0^n)$ denote its centralizer, i.e., the subalgebra of all elements $\sigma \in \mathrm{End}(\Gamma(E))$ such that $\sigma \circ \Psi_0^n = \Psi_0^n \circ \sigma$, so that (8) is satisfied. Then take $\mathrm{Par}(P)$ as some subalgebra of $Z(\Psi_0^n)$ with square roots. As a concrete example, one can take $\mathrm{Par}(P)$ as the subalgebra of nonnegative bounded self-adjoint operators in $\Gamma(E)$ which commutes with Ψ_0^n.

3. *We can consider other kinds of parameterized operators.* In Corollary 3.2 and in the above generalizations we considered only parameterized operators of the form $\Psi_\varepsilon = \varepsilon \cdot \Psi$, which forced us to assume Ψ idempotent. Indeed, notice that the nilpotency condition was used only to ensure that $\varepsilon \mapsto \varepsilon \cdot \Psi$ is an algebra homomorphism. More generally, let $\mathrm{Par}(P)$ be some complex algebra with square roots endowed with a representation $\rho : \mathrm{Par}(P) \to \mathrm{End}_{\mathbb{C}}(\Gamma(E))$ and define the action of $\mathrm{Par}(P)$ in $\mathrm{Psd}(E)$ by $\varepsilon \cdot \Psi := \rho(\varepsilon) \circ \Psi$, so that the second condition in (7) is clearly satisfied. If the action is faithful, then Condition 2 is satisfied too. Finally, if the action is compatible with the algebra structure of $\mathrm{Psd}(E)$, i.e., if

$$\rho(\varepsilon) \circ (\Psi \circ \Psi') = (\rho(\varepsilon) \circ \Psi) \circ (\rho(\varepsilon) \circ \Psi') \tag{10}$$

   for every $\varepsilon \in \mathrm{Par}(P)$ and $\Psi, \Psi' \in \mathrm{Psd}(E)$, then the first part of (7) is also satisfied and for every fixed Ψ the rule $\varepsilon \mapsto \rho(\varepsilon) \circ \Psi$ is homomorphic, so that Corollary 3.2 holds analogously.

   Example 3.2. Recall that a ring R is *Boolean* if each element is idempotent, i.e., if $x * x = x$ for every $x \in R$. Let $\mathrm{Bol}(P)$ be a Boolean ring[6] and take $\mathrm{Par}(P) = \mathrm{Bol}(P) \otimes_{\mathbb{Z}} \mathbb{C}$. Let $\rho : \mathrm{Bol}(P) \to \mathrm{End}_{\mathbb{C}}(\Gamma(E))$ be a faithful representation of this Boolean ring and notice that (10) is immediately satisfied (recall that every Boolean ring is commutative). Tensoring with $\mathbb{C}$ we get a faithful representation of $\mathrm{Par}(P) = \mathrm{Bol}(P) \otimes_{\mathbb{Z}} \mathbb{C}$. Finally, notice that every Boolean ring has square roots, since for every x we have $x^2 = x$, i.e., $\sqrt{x} = x$.

   Example 3.3 (a more concrete case)**.** Let $P = M \times A$ be a trivial algebra bundle, so that $\Gamma(P) \simeq C^\infty(M; A)$. Let $\mathrm{Bol}(A) \subset A$ be the Boolean ring of the idempotent elements of A and take $\mathrm{Bol}(P)$ as the set of functions $f : M \to A$ such that $f(x) \in \mathrm{Bol}(A)$ for every $x \in M$. Now, let $\rho : A \to \mathrm{End}(F)$ be a faithful representation of A in the typical fiber of E. It induces a faithful representation $\rho : \Gamma(P) \to \mathrm{End}_{\mathbb{C}}(\Gamma(E))$ and, therefore, by restriction a faithful representation of $\mathrm{Par}(P)$.

[6] We suspect that the same holds, more generally, for Von Neumann regular rings, but we do not have a proof of this.

4 Proof of Theorem 3.1

In this section we prove our emergence theorem. The proof will be inductive on the number r of variables of S_2. In order to prove the base case, i.e., the emergence theorem when S_2 is a univariate polynomial, we will need to use some additivity and multiplicativity properties of the emergence phenomena. In turn, the induction step will be based on a technical lemma.

In order to better understand the whole proof, this section will be organized as follows. In Subsection 4.1 we prove the basic properties of the emergence phenomena needed to prove the base step. In Subsection 4.2 this base step is proved. In Subsection 4.4, Theorem 3.1 is finally demonstrated, with the technical lemma used for the induction step being presented before in Subsection 4.3.

4.1 Properties of Emergence Phenomena

- In the following discussion, when the degree of a GPT does not matter it will be made implicit in order to simplify the notation. In these cases we will also write ε instead of $\varepsilon(\ell)$. Thus, from now on, by saying "*let* Ψ_ε *be a GPT over* $\mathrm{GB}(M)$" we mean that it is any GPT of any degree ℓ.

Definition 4.1. Let Ψ_ε and $\Psi'_{\varepsilon'}$ be GPT over the same generalized background $\mathrm{GB}(M)$. The *sum* and the *composition* between them are the GPT over $\mathrm{GB}(M)$ given by $\Psi^{+}_{(\varepsilon,\varepsilon')} = \Psi_\varepsilon + \Psi'_{\varepsilon'}$ and $\Psi^{\circ}_{(\varepsilon,\varepsilon')} = \Psi_\varepsilon \circ \Psi'_{\varepsilon'}$. Notice that the sum and the composition between GPT of degrees ℓ and ℓ' has degree $\ell + \ell'$.

Lemma 4.1. *Let* $\Psi_{1,\varepsilon}$, $\Psi_{2,\delta}$ *and* $\Psi_{3,\kappa}$ *be three GPT over the same generalized background* $\mathrm{GB}(M)$ *with fundamental parameter algebra* $\mathrm{Par}(P)^\ell$, *where* ℓ *is the degree of the first GPT, such that:*

1. $\Psi_{1,\varepsilon}$ *is multiplicative;*
2. $\Psi_{1,\varepsilon}$ *emerges from both* $\Psi_{2,\delta}$ *and* $\Psi_{3,\kappa}$.

Then $\Psi_{1,\varepsilon}$ *emerges from the compositions* $S_{2,\delta} \circ S_{3,\kappa}$ *and* $S_{3,\kappa} \circ S_{2,\delta}$.

Proof. From the second hypothesis we conclude that $\Psi_{1,\varepsilon} = \Psi_{2,F(\varepsilon)}$ and $\Psi_{1,\varepsilon} = \Psi_{3,G(\varepsilon)}$ for certain functions F, G. Composing them and using the first hypothesis, we find

$$\Psi_{1,\varepsilon^2} = \Psi_{1,\varepsilon} \circ \Psi_{1,\varepsilon} = \Psi_{2,F(\varepsilon)} \circ \Psi_{3,G(\varepsilon)} = \Psi_{3,G(\varepsilon)} \circ \Psi_{2,F(\varepsilon)}.$$

Let $\sqrt{-} : \mathrm{Par}(P)^\ell \to \mathrm{Par}(P)^\ell$ be a function selecting to each fundamental parameter ε' a square root $\sqrt{\varepsilon'}$, which exists by hypothesis. Then, for every ε' one gets

$$\Psi_{1,\varepsilon'} = \Psi_{2,F(\sqrt{\varepsilon'})} \circ \Psi_{3,G(\sqrt{\varepsilon'})} = \Psi_{3,G(\sqrt{\varepsilon'})} \circ \Psi_{2,F(\sqrt{\varepsilon'})} = \Psi^{\circ}_{H(\varepsilon')},$$

finishing the proof. □

In a completely analogous way one proves the following.

Lemma 4.2. *Let* $\Psi_{1,\varepsilon}$, $\Psi_{2,\delta}$ *and* $\Psi_{3,\kappa}$ *be three GPT over the same generalized background* $\mathrm{GB}(M)$, *such that:*

1. $\Psi_{1,\varepsilon}$ *is scalar invariant;*
2. $\Psi_{1,\varepsilon}$ *emerges from both* $\Psi_{2,\delta}$ *and* $\Psi_{3,\kappa}$.

Then $\Psi_{1,\varepsilon}$ *also emerges from the sum* $\Psi_{2,\delta} + \Psi_{3,\kappa}$.

4.2 Base of Induction

In this subsection, using the additivity and multiplicativity properties of last section, we will prove the following lemma, which will be the base of the induction step in the proof of Theorem 3.1:

Lemma 4.3. *Let* $\mathrm{GB}(M)$ *be generalized background of* (ℓ, k)*-type. Let* $\Psi_{1,\varepsilon(\ell)}$ *be a GPT of degree* ℓ *and let* $\Psi_{2,\delta(\ell')}$ *a PPT of degree* (l, ℓ') *in* $r = 1$ *variables, defined on* $\mathrm{GB}(M)$ *and such that* $\ell' = k'\ell$*, with* $0 < k' \leq k$*. Suppose that:*

1. $\Psi_{1,\varepsilon(\ell)}$ *is homomorphic;*
2. $\Psi_{2,\delta(\ell')}$ *is right-invertible and the coefficient functions* $f_\alpha : \mathrm{Par}(P)^{k'\ell} \to \mathbb{R}$ *of the polynomial* $p^l_{\ell'}$ *defining* $\Psi_{2,\delta(\ell')}$ *belongs to the functional calculus* $C_{k'\ell;k}(P; \mathbb{K})$*.*

Then $\Psi_{1,\varepsilon(\ell)}$ *emerges from* $\Psi_{2,\delta(\ell')}$*.*

We begin with another lemma.

Lemma 4.4. *Let* $\Psi_{1,\varepsilon(\ell)}$ *be a GPT of degree* ℓ *defined on a generalized background* $\mathrm{GB}(M)$ *of* (ℓ, k)*-type, with* $k > 0$*. Then* $\Psi_{1,\varepsilon(\ell)}$ *emerges from every GPT* $\Psi_{2,\delta(\ell')}$ *over* $\mathrm{GB}(M)$*, which has degree* $\ell' = k'\ell$ *for some* $0 < k' \leq k$*, and such that* $\Psi^l_{2,\delta(\ell')} = g(\delta(\ell'))\Psi^l$*, with* $l \geq 0$*, where* Ψ *is right-invertible and* $g \in C_{k'\ell;k}(P; \mathbb{K})$*.*

Proof. Since $\Psi^0 = I$ is right-invertible, the case $l = 0$ is a particular setup of case $l = 1$. Furthermore, if $l > 1$ and Ψ is right-invertible, then $\Xi = \Psi^l$ is right-invertible too, so that the case $l > 1$ also follows from the $l = 1$ case. Thus, it is enough to work with $l = 1$. Thus, let $R_\Psi \in \mathrm{Op}(E)$ be a right-inverse for Ψ and notice that to find an emergence from $\Psi_{1,\varepsilon(\ell)}$ to $\Psi_{2,\delta(\ell')}$ is equivalent to building a function $F : \mathrm{Par}(P)^\ell \to \mathrm{Par}(P)^{\ell'}$ such that $\Psi_{1,\varepsilon(\ell)} \circ R_\Psi = g(F(\varepsilon(\ell)))I$. From (9) and from the fact that the right multiplication by right-invertible operators is injective, last condition is in turn equivalent to the existence of F such that $\Psi^{\ell'}_g \circ \Psi_{1,\varepsilon(\ell)} \circ R_\Psi = F(\varepsilon(\ell)) \cdot^{\ell'} I$, but this actually defines F via $\mathrm{Par}(P)^{\ell'} \cdot^{\ell'} I \simeq \mathrm{Par}(P)^{\ell'}$. $\square$

Sketch of proof of Lemma 4.3. Given a PPT $\Psi_{2,\delta(\ell')} = \sum_i f_i(\delta(\ell'))\Psi^i$ in the hypothesis, for each $j = 1, ..., l$ let $\Gamma_j = \sum_{i=j}^l f_i(\delta(\ell'))\Psi^{i-1}$ and notice that

$$\Psi_{2,\delta(\ell')} = (\sum_{i=1}^{l} f_i(\delta(\ell'))\Psi^{i-1}) \circ \Psi = \Gamma_1(\delta(\ell')) \circ \Psi.$$

Since Ψ is right-invertible and $\mathrm{GB}(M)$ is of (ℓ, k)-type, with $k > 0$, from Lemma 4.4 it follows that $\Psi_{1,\varepsilon(\ell)}$ emerges from $1 \cdot \Psi$[7]. Thus, if $S_{\varepsilon(\ell)}$ itself emerges from Γ_1 one can use Lemma 4.1 to conclude that it actually emerges from $\Gamma_1 \circ \Psi$. In turn, notice that $\Gamma_1 = f_1 \cdot I + \Gamma_2 \circ \Psi = \Gamma_2 \circ \Psi + f_1 \cdot I$. But, since I is right-invertible and since $f_1 \in C_{\ell';k}(P; \mathbb{K})$, from Lemma 4.4 we get that $S_{\varepsilon(\ell)}$ emerges from the theory defined by $f_1 \cdot I$, while by the same argument we see that $\Gamma_2 \circ \Psi$ emerges from $f_1 \cdot I$. Therefore, if $\Psi_{1,\varepsilon(\ell)}$ emerges from $\Gamma_2 \circ \Psi$ we will be able to use Lemma 4.2 to conclude that it emerges from Γ_1, finishing the proof. It happens that, as done for $\Gamma_1 \circ \Psi$, we see that Γ_2

[7] Here we are using explicitly that the functional calculus is unital.

emerges from Ψ and we already know that $\Psi_{1,\varepsilon(\ell)}$ emerges from Ψ. Thus, our problem is to prove that $\Psi_{1,\varepsilon(\ell)}$ emerges from Γ_2 instead of from Γ_1. A finite induction argument proves that if $\Psi_{1,\varepsilon(\ell)}$ emerges from Γ_l, then it emerges from Γ_j, for each $j=1,...,l$. Recall that $\Gamma_l = f_l \cdot \Psi^{l-1}$. Since $f_l \in C_{\ell';k}(P;\mathbb{K})$ we can use Lemma 4.4 to see that $S_{\varepsilon(\ell)}$ really emerges from Γ_l. □

4.3 Technical Lemma for Induction Step

Also as a consequence of the properties of the emergence phenomena, we can now prove the following technical lemma, which will be used in the induction step of Theorem 3.1.

Lemma 4.5. *Let $\Psi_{1,\varepsilon}$ be a GPT over a generalized background $\mathrm{GB}(M)$. Given $l \geq 1$, let Ψ_{2_j,δ_j} and Ψ_{3_s,κ_s}, with $1 \leq j,s \leq l$ be two families of GPT, also defined over $\mathrm{GB}(M)$. Assume that:*

1. *$\Psi_{1,\varepsilon}$ is homomorphic;*
2. *$\Psi_{1,\varepsilon}$ emerges from Ψ_{2_j,δ_j} and from Ψ_{3_s,κ_s} for every j,s.*

Then $\Psi_{1,\varepsilon}$ emerges from $\Psi^s_{\delta_J,\kappa_J} = \sum_{j=1}^{s} \Psi_{2_j,\delta_j} \circ \Psi_{3_j,\kappa_j}$, for every $s=1,...,l$

Proof. We proceed by induction in l. First of all, notice that from the first two hypotheses and from Lemma 4.1 we see that $\Psi_{1,\varepsilon}$ emerges from the composition $\Psi_{2_j,\delta_j} \circ \Psi_{3_j,\kappa_j}$ for every $j=1,...,l$. In particular, it emerges from $\Psi^1_{\delta_1,\kappa_1} = \Psi_{2_1,\delta_1} \circ \Psi_{3_1,\kappa_1}$, which is the base of induction. For every $m=1,...,l-1$ it also emerges from $\Psi_{2_{m+1},\delta_{m+1}} \circ \Psi_{3_{m+1},\kappa_{m+1}}$. For the induction step, suppose that $\Psi_{1,\varepsilon}$ emerges from $\Psi^m_{\delta_J,\kappa_J} = \sum_{j=1}^{m} \Psi_{2_j,\delta_j} \circ \Psi_{3_j,\kappa_j}$ for every $1 \leq m \leq l-1$ and let us show that it emerges from $\Psi^{m+1}_{\delta_J,\kappa_J}$. Notice that

$$
\begin{aligned}
\Psi^{m+1}_{\delta_J,\kappa_J} &= \sum_{j=1}^{m+1} \Psi_{2_j,\delta_j} \circ \Psi_{3_j,\kappa_j} = \sum_{j=1}^{m} (\Psi_{2_j,\delta_j} \circ \Psi_{3_j,\kappa_j}) + \Psi_{2_{m+1},\delta_{m+1}} \circ \Psi_{3_{m+1},\kappa_{m+1}} \\
&= \Psi^m_{\delta_J,\kappa_J} + (\Psi_{2_{m+1},\delta_{m+1}} \circ \Psi_{3_{m+1},\kappa_{m+1}}).
\end{aligned}
$$

From the induction hypothesis $\Psi_{1,\varepsilon}$ emerges from $\Psi^m_{\delta_J,\kappa_J}$, while by the above it also emerges from $\Psi_{2_{m+1},\delta_{m+1}} \circ \Psi_{3_{m+1},\kappa_{m+1}}$. The result then follows from Lemma 4.2. □

4.4 Proof of Theorem 3.1

We can finally proof our emergence theorem. For the convenience of the reader, we state it again.

Theorem 3.1 (Emergence Theorem) *Let M be a compact and oriented manifold and let $\mathrm{GB}(M)$ be a generalized background of (ℓ,k)-type, with $k>0$. Let $\Psi_{1,\varepsilon(\ell)}$ be a GPT of degree ℓ and let $\Psi_{2,\delta(\ell)}$ be a PPT of degree (l,ℓ') in r variables, where $\ell' = k'\ell$, with $0 < k' \leq k$, defined on $\mathrm{GB}(M)$. Suppose that:*

1. *$\Psi_{1,\varepsilon(\ell)}$ is homomorphic;*
2. *$\Psi_{2,\delta(\ell)}$ is right-invertible and the coefficient functions $f_\alpha : \mathrm{Par}(P)^{k'\ell} \to \mathbb{K}$ belongs to the functional calculus $C_{k'\ell;k}(P;\mathbb{K})$.*

Then $\Psi_{1,\varepsilon(\ell)}$ *emerges from* $\Psi_{2,\delta(\ell)}$.

Proof. The proof will be done by induction in r. The base of induction is Lemma 4.3. Suppose that the theorem holds for each $r = 1,...,q$ and let us show that it holds for $r = q+1$. Let $p^{l}_{\ell';q+1}[x_1,...,x_{r+1}] = \sum_{|\alpha|\leq l} f_\alpha \cdot x^\alpha$ be a multivariate polynomial with coefficients in $\mathrm{Map}(\mathrm{Par}(P)^{\ell'};\mathbb{K})$, which actually belong to $C_{k'\ell;k}(P;\mathbb{K})$. Since for every commutative ring R we have $R[x_1,...,x_{q+1}] \simeq R[x_1,...,x_q][x_{q+1}]$, given right-invertible generalized operators $\Psi_1,...,\Psi_{q+1} \in \mathrm{Op}(E)$ one can write

$$\Psi_{2,\delta(\ell)} = p^{l}_{\ell';q+1}[\Psi_1,...,\Psi_{r+1}] = \sum_j p^{l_j}_{\ell';q,j}[\Psi_1,...,\Psi_q] \cdot \Psi^{j}_{q+1},$$

where each $p^{l_j}_{\ell';q,j}[x_1,...,x_q] \in \mathrm{Map}_{l_j}(\mathrm{Par}(P)^{\ell'};\mathbb{K})[x_1,...,x_q]$ has coefficients which belongs to belongs to $C_{k'\ell;k}(P;\mathbb{K})$. Thus, by the induction hypothesis, $\Psi_{1,\varepsilon(\ell)}$ emerges from $p^{l_j}_{\ell';q,j}[x_1,...,x_q]$. Since Ψ_{q+1} is right-invertible and since the functional calculus is unital, from Lemma 4.4 we see that $\Psi_{1,\varepsilon(\ell)}$ emerges from Ψ^{j}_{q+1}. The result then follows from Lemma 4.5. □

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5.4 SECOND WORK

The following is a recompilation of the ArXiv preprint

- Y.X. Martins, L.F. A. Campos, R.J. Biezuner, *On Extensions of Yang-Mills-Type Theories, Their Spaces and Their Categories*, Jul 2020, arXiv:2007.01660

On Extensions of Yang-Mills-Type Theories, Their Spaces and Their Categories

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Abstract

In this paper we consider the classification problem of extensions of Yang-Mills-type (YMT) theories. For us, a YMT theory differs from the classical Yang-Mills theories by allowing an arbitrary pairing on the curvature. The space of YMT theories with a prescribed gauge group G and instanton sector P is classified, an upper bound to its rank is given and it is compared with the space of Yang-Mills theories. We present extensions of YMT theories as a simple and unified approach to many different notions of deformations and addition of correction terms previously discussed in the literature. A relation between these extensions and emergence phenomena in the sense of [34] is presented. We consider the space of all extensions of a fixed YMT theory S^G and we prove that for every additive group action of $\mathbb{G}$ in $\mathbb{R}$ and every commutative and unital ring R, this space has an induced structure of $R[\mathbb{G}]$-module bundle. We conjecture that this bundle can be continuously embedded into a trivial bundle. Morphisms between extensions of a fixed YMT theory are defined in such a way that they define a category of extensions. It is proved that this category is a reflective subcategory of a slice category, reflecting some properties of its limits and colimits.

1 Introduction

Although the Standard Model of Elementary Particle Physics (SM) is at present probably the most accurate and tested physical theory [54], there are indications that it must be viewed as an effective theory of a more fundamental theory [16, 4]. A lot of different approaches have been suggested along the last decades, e.g., string theory [7] and other approaches involving higher-dimensional spacetime [44, 43], loop quantum theory and other attempts to quantum gravity [28, 47], symmetry breaking extensions such as the Standard-Model Extension [10, 11, 30], symmetry enlargement models such as the Minimal Supersymmetric Standard Model and other supersymmetric extensions with higher supersymmetry [17, 41, 9], and so on. This *zoo* of extensions naturally leads one to consider the *classification problem* of all already existing and all the possible extensions that could

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eventually be discovered in the future, avoiding spending time studying informal or superficial theories (which is clearly not the case of the approaches cited above). In this paper we propose the beginning of a program attempting to formalize and give a strategy of attack for such a classification problem.

Notice that since the SM can be built by coupling spinorial field theories with Yang-Mills (YM) theories and with its classical Higgs fields, one can see that the basic strategy is to begin by classifying the possible extensions in each component piece. In this paper and in the next two [36, 35] we will focus on the YM part. Actually, in this work we prove that a classification scheme for the Higgs sector and for the YM part with a fixed spinorial background follows from a classification for the YM part (see Sections 3.2 and 3.3, respectively), so that the next step is to work on the classification of extensions of the spinorial sector for fixed YM background.

In order to motivate our definition of extension, let us begin by noticing that in the current literature we find many ways to extend Yang-Mills theories, such as those in [27, 25, 8, 49, 50, 51, 52, 19, 22, 53, 42, 15, 20]. We note that most of them can be organized into three classes: *deformations*, *addition of a correction term* and *extension of the gauge group*. In the first class there exists a theory $\hat{S}$ which is the limit $\lim_{i\to\infty}\hat{S}_i$ of a family of other theories $\hat{S}_i$ and such that the first term $\hat{S}_0$ on the sequence is precisely a YM theory S^G. Typically, the sequence is formed by partial sums, i.e, $\hat{S}_i = \sum_{j\leq i} f_j(\lambda_j)S_j$ depending on certain *fundamental* (or *deformation*) parameters λ_j and on functions of them such that $f_0(\lambda_0)=1$ and $\lim_{\lambda_j\to 0} f_j = 0$. In this case, $\hat{S}=\sum_{i\geq 0} f_i(\lambda_i)S_i$ and the YM theory S^G is recovered in the limit $(\lambda_I)\to 0$, where $I=\{0,1,...\}$. In the second class of examples, we begin with a YM theory S^G and we add some sort of correction term C, typically breaking some symmetry, so that the extended theory is the sum $\hat{S}=S^G+C$. In the third one there is a theory $\hat{S}^{\hat{G}}$ depending on a larger group $\hat{G}$, with $G\subseteq\hat{G}$, and such that there is some process of reduction from $\hat{G}$ to G such that, when applied to $\hat{S}^{\hat{G}}$ we recover S^G. Typically $\hat{G}=G\times K$ for some other group K and the reduction process is some dimensional reduction.

Now, let $\hat{S}$ be an extension by deformation (in the sense above) of a YM theory S^G and note that $\hat{S}=S_0+\sum_{i\geq 1} f_i(\lambda_i)S_i$. Let δ be the map which assigns a configuration $\varphi(\lambda_I)$ in the domain of $\hat{S}$ to its limit $\lim_{\lambda_I\to 0}\varphi(\lambda_I)$. Thus, the composition $S^G\circ\delta$ is precisely S_0, so that one can write

$$\hat{S}=S^G\circ\delta+\sum_{i\geq 1} f_i(\lambda_i)S_i, \tag{1}$$

which basically says that, except for the presence of the δ-map, an extension by deformation is an extension by adding the correction term $C=\sum_{i\geq 1} f_i(\lambda_i)S_i$. This suggests that there must be a wider notion of “extension” which unifies both classes, as we will propose in this paper.

Actually, we will work in a more general setting. Most of the discussion is about the geometry underlying the YM theories, so that it does not depends on the way used to attach this geometry in order to build the action functional. More precisely, our entire discussion applies for gauge theories whose Lagrangian is given by some quadratic form $q(F_D)$ of the curvature, not necessarily that given by $\mathrm{tr}(F_D\wedge\star F_D)$. This has the great advantage of avoiding the requirements on the existence of a semi-riemannian structure in the spacetime (used to build the Hodge star) and on a compactness and/or semi-simpleness hypothesis on the Lie algebra (needed to make the Killing form a nice pairing). Thus, throughout the paper we work with what we call *Yang-Mills-type* (YMT) theories. As we show the additional degrees of freedom on the choice of the pairing have real meaning, in the sense that there are really many more YMT theories than YM theories.

Looking at the two classes of extensions discussed in the literature described above, one can see that deformations are about *restriction of scales*, while addition of correction terms is about summing terms in the *entire domain*. Thus, if we are searching for an unifying notion of extension this should be about *adding correction terms in some scales allowing an enlarging of the gauge group*. This is the core of our definition. Indeed, given a YMT theory S^G with gauge group G, an *extension* for it is defined by the following:

1. a possibly larger group $\hat{G}$ such that $G \subseteq \hat{G}$, representing the enlargement of the gauge group;
2. a space $\widehat{\mathrm{Conn}}_{\hat{G}}$ containing all $\hat{G}$-connections and which is invariant by global gauge transformations of $\hat{G}$, called the *extended domain* and describing the domain in which the extended theory is defined;
3. a gauge invariant action functional $\hat{S}^{\hat{G}} : \widehat{\mathrm{Conn}}_{\hat{G}} \to \mathbb{R}$, called the *extended functional* and defining the extended theory;
4. a smaller space $C_{\hat{G}} \subset \widehat{\mathrm{Conn}}_{\hat{G}}$, called the *correction domain* and playing the role of a special regime, or scale;
5. another functional $C_{\hat{G}} \to \mathbb{R}$, called the *correction functional*, corresponding to the additional term;
6. a map $\delta : C_{\hat{G}} \to \mathrm{Conn}_G$ connecting the configurations at a special scale with the configurations of the starting YMT theory,

all of this subjected to the condition $\hat{S}^{\hat{G}}|_{C_{\hat{G}}} = S^G \circ \delta + C$, which is the analogue of (1).

Once a class of objects is introduced, the main classification problem is about finding a bijective correspondence between this class and some other set which is known a priori. Thus, the primary classification problem for extensions of a YMT theory S^G is about studying the space $\mathrm{Ext}(S^G; \hat{G})$ of its extensions relative to a fixed extended gauge group $\hat{G}$. We prove that if a group $\mathbb{G}$ acts in the set of real numbers $\mathbb{R}$ preserving the sum, then it induces a structure of a $R[\mathbb{G}]$-module bundle for any commutative unital ring R and we make the conjecture that for certain R and $\mathbb{G}$ this bundle is a continuous subbundle of a specific trivial $R[\mathbb{G}]$-module bundle. Thus, in order to classify the extensions of S^G one can analyze the algebraic-topological properties of the corresponding module bundles, which should be another natural next step in this program.

On the other hand, recall that there is another (more refined) way to work on the classification of a class of objects: by means of considering how each object interacts with each other. In other words, one can search for natural notions of "morphisms" between such objects in such a way that they constitute a category $\mathbb{C}$. In this case, one can try to classify the category itself or the starting class of objects, now up to isomorphisms. Following this philosophy, we prove that there really exists a notion of *morphisms between extensions* so that $\mathrm{Ext}(S^G; \hat{G})$ is the collection of objects of a category $\mathbf{Ext}(S^G; \hat{G})$ of extensions of S^G. We embed this category in a conservative way into a slice category, which produces some constraints on the possible categorical constructions that can be done between extensions. We also conjecture that this category can be regarded, for certain additive $\mathbb{G}$-actions on $\mathbb{R}$ and certain rings R, a category internal to the category $\mathbf{Bun}_{R[\mathbb{G}]}$.

There is also a third approach, more constructive, to the classification of $\mathrm{Ext}(S^G; \hat{G})$. It is about first classifying more easy subclasses $\mathcal{E}(S^G; \hat{G}) \subset \mathrm{Ext}(S^G; \hat{G})$. Notice that the categorical approach also applies to this case, since each subclass of the class of objects of a category **C** induces a full subcategory. In our context, notice that $\mathrm{Ext}(S^G;\hat{G})$ can be decomposed into two disjoint subclasses: those with vanishing correction term (i.e, such that $C \equiv 0$) and those with non-null correction term. The first one is the class of the so-called *complete extensions*, while the second one are the *incomplete extensions*. In [36] we propose a list of four additional problems which should be studied for a given subclass $\mathcal{E}(S^G;\hat{G}) \subset \mathrm{Ext}(S^G;\hat{G})$ in view to the classification problem: existence problem, universality problem, maximality problem and universality problem and we show that they have solutions internal to any "coherent" class of complete extensions. This proves, in particular, that well-behaved complete extensions exist with some generality. The study of incomplete extensions is made in [35], where it is shown that equivariant extensions are always complete, but there is a canonical class of incomplete equivariant extensions and even incomplete gauge-breaking extensions, suggesting that the class of incomplete extensions is more nasty.

Let us finish this introduction with a brief description of how the paper is organized. In Section 2 the notion of Yang-Mills-type theory is formally defined and we study the space of all of them. We prove that they define a nontrivial fibration over the set of all triples (M, G, P), where M is the manifold, G a Lie group and P a principal G-bundle. We show that if $\dim G \geq 2$ and if G is not discrete, then the corresponding fibers are infinite-dimensional as real vector spaces, but finitely-generated as $C^\infty(M)$-modules and in this case we provide an upper bound for their rank. In Section 3 the concept of extension is introduced in more precise terms and we explicitly show that the most notions of extensions arising in the literature are particular examples of ours. We also show that the emergence phenomena (in the sense of [34]) are a source of examples for nontrivial extensions. In Section 4 the properties of the space of extensions $\mathrm{Ext}(S^G;\hat{G})$ described above are proved. Finally, in Section 5 the category of extensions $\mathbf{Ext}(S^G;\hat{G})$ is introduced and proved to be a conservative subcategory of a product of slice categories.

2 Yang-Mills-Type Theories

We begin by recalling that a *Yang-Mills theory* (YM) is given by

1. a n-dimensional compact, orientable semi-Riemannian smooth manifold (M, g), regarded as the spacetime;
2. a real or complex finite-dimensional Lie group G, regarded as the gauge group of internal symmetries;
3. a principal G-bundle P over M, called the *instanton sector* of the theory.

The *action functional* is the map $S : \mathrm{Conn}(P;\mathfrak{g}) \to \mathbb{R}$, defined on the space of G-connections on P, given by

$$S[D] := \int_M \langle F_D, F_D\rangle_{\mathfrak{g}} dvol_g. \tag{2}$$

Here, $dvol_g$ is the volume form induced by g, $F_D = dD + \frac{1}{2}D[\wedge]_{\mathfrak{g}} D$ is the field strength of D and $[\wedge]_{\mathfrak{g}}$ is the wedge product induced by the Lie bracket of $\mathfrak{g}$ (we are using the notations and conventions of [33, 32]). Furthermore, $\langle \cdot,\cdot\rangle_{\mathfrak{g}} : \Omega^2_{heq}(P;\mathfrak{g}) \otimes \Omega^2_{heq}(P;\mathfrak{g}) \to C^\infty(M)$ is the standard $C^\infty(M)$-linear pairing on the space of horizontal G-equivariant $\mathfrak{g}$-valued 2-forms on P, whose definition we recall briefly (see [21, 14] for further details). The semi-riemannian metric g induces a pairing

$\langle \cdot, \cdot \rangle_g : \Omega^2(M) \otimes \Omega^2(M) \to C^\infty(M)$, given by $\langle \alpha, \beta \rangle_g = \alpha \wedge \star_g \beta$. On the other hand, the Killing form of $B_{\mathfrak{g}} : \mathfrak{g} \otimes \mathfrak{g} \to \mathbb{R}$ of $\mathfrak{g}$ extends to a pairing $\overline{B}_{\mathfrak{g}} : \Gamma(E_{\mathfrak{g}}) \otimes \Gamma(E_{\mathfrak{g}}) \to C^\infty(M)$ on the global sections of the adjoint bundle $E_{\mathfrak{g}} = P \times_M \mathfrak{g}$. Since $\Omega^2(M; E_{\mathfrak{g}}) \simeq \Omega^2(M) \otimes \Gamma(E_{\mathfrak{g}})$, together the pairings $\langle \cdot, \cdot \rangle_g$ and $\overline{B}_{\mathfrak{g}}$ induce a pairing $\langle \cdot, \cdot \rangle_g \otimes \overline{B}_{\mathfrak{g}} \equiv \langle \cdot, \cdot \rangle_{g,\mathfrak{g}}$ on $\Omega^2(M; E_{\mathfrak{g}})$. But we have a canonical isomorphism $\tau : \Omega^2_{heq}(P; \mathfrak{g}) \simeq \Omega^2(M; E_{\mathfrak{g}})$, leading us to take the pullback of $\langle \cdot, \cdot \rangle_{g,\mathfrak{g}}$ and define the desired pairing $\langle \cdot, \cdot \rangle_{\mathfrak{g}}$ by $\langle \omega, \omega' \rangle_{\mathfrak{g}} = \langle \tau(\omega), \tau(\omega') \rangle_{g,\mathfrak{g}}$. In the literature this is typically written as $\langle \omega, \omega' \rangle_{\mathfrak{g}} = \mathrm{tr}(\omega [\wedge]_{\mathfrak{g}} \star_g \omega')$. We also recall that the functional (2) is invariant by the group $\mathrm{Gau}_G(P)$ of global gauge transformations of P, i.e, the group of its G-principal M-bundle automorphisms (we are using notations and conventions of [31]).

Throughout this paper we will work with *Yang-Mills-type* (YMT) *theories* which differ from the YM theories by replacing the standard pairing $\langle \cdot, \cdot \rangle_{\mathfrak{g}}$ with an arbitrary (possibly degenerate and non-symmetric) $\mathbb{R}$-linear pairing $\langle \cdot, \cdot \rangle : \Omega^2_{heq}(P, \mathfrak{g}) \otimes \Omega^2_{heq}(P, \mathfrak{g}) \to C^\infty(M)$. The *action functional* of a YMT theory is defined analogously to (2): it is given the map $S : \mathrm{Conn}(P; \mathfrak{g}) \to \mathbb{R}$ such that

$$S[D] := \int_M \langle F_D, F_D \rangle dvol_g. \tag{3}$$

Remark 1. We emphasize that in a YMT theory we assume that the pairing is only $\mathbb{R}$-linear. If the pairing is actually $C^\infty(M)$-linear we will say that the corresponding YMT is *tensorial* or *linear*. This additional assumption is important when one does local computations, as in [35]. Since all results of this paper do not involve such local computations, we will work in the general $\mathbb{R}$-linear setting unless explicit stated otherwise.

Remark 2. Since the Killing form is an invariant polynomial, it follows that the classical YM theories are invariant by the action of the group $\mathrm{Gau}_G(P)$ of global gauge transformations.We also emphasize that we will *avoid* the assumption of invariance under global gauge transformations on the pairings $\langle \cdot, \cdot \rangle$, i.e, we will *not* require that $\langle f^*(F_D), f^*(F_D) \rangle = \langle F_D, F_D \rangle$ for every $f \in \mathrm{Gau}_G(P)$. This means that for us, *a YMT theory can be gauge-breaking*. This generality is important when studying gauge-breaking extensions of YMT theories [35]. In this work, when a YMT theory is such that his pairing is gauge invariant we will refer to it explicitly as a *gauge invariant YMT theory*.

Remark 3. After standard modifications (replacing arbitrary smooth functions by densities), all definitions above make sense for noncompact and non-orientable smooth manifolds. However, for simplicity we will work in the compact and oriented case.

2.1 Yang-Mills *vs* Yang-Mills-Type

In this subsection we will compare the concepts of YM and YMT theories. Let $\mathcal{X}$ be the space of all triples (M, G, P), where M is a (compact and oriented) semi-riemannian manifold, G is a finite-dimensional Lie group and P is an isomorphism class of principal G-bundle on M. Furthermore, let $\mathcal{YM}$, $\mathcal{YMT}$ and $\mathcal{YMT}_0$ be the spaces of all Yang-Mills, Yang-Mills-type and linear Yang-Mills theories respectively. We have obvious projections $\pi_{YM} : \mathcal{YM} \to \mathcal{X}$, $\pi_{YMT} : \mathcal{YMT} \to \mathcal{X}$ and $\pi_{YMT,0} : \mathcal{YMT}_0 \to \mathcal{X}$. Fixed $(M, G, P) \in \mathcal{X}$, let $\mathrm{YM}_G(P)$, $\mathrm{YMT}_G(P)$ and $\mathrm{YMT}_{G,0}(P)$ denote the corresponding fibers by π_{YM}, π_{YMT} and $\pi_{YMT,0}$, respectively. They are in bijection with the possible pairings on each context. Thus, $\mathrm{YM}_G(P)$ has a single object for every (M, G, P), characterized by the canonical pairing $\langle \omega, \omega' \rangle_{\mathfrak{g}} = \mathrm{tr}\, \omega \wedge \star_g \omega'$. In particular, $\pi_{YM} : \mathcal{YM} \simeq \mathcal{X}$ is a bijection. On the other hand, as consequence of the next lemma we will see that $\mathrm{YMT}_G(P)$ and

$\mathrm{YMT}_{G,0}(P)$ typically have a lot of elements, so that $\pi_{YMT}:\mathcal{YMT}\to\mathcal{X}$ and $\pi_{YMT,0}:\mathcal{YMT}_0\to\mathcal{X}$ are nontrivial fibrations.

Let R be a commutative ring, $\mathbb{K}\subset R$ be a subring which is also a field, V and Z two R-modules and $T:Z_{\mathbb{K}}\to\mathbb{K}$ a $\mathbb{K}$-linear map, where $Z_{\mathbb{K}}$ denotes the restriction of scalars. A $(T,\mathbb{K})$-*pairing* (resp. (T,R)-*pairing*) on V is a $\mathbb{K}$-linear map $B:V_{\mathbb{K}}\otimes_{\mathbb{K}}V_{\mathbb{K}}\to\mathbb{K}$ which factors through T, that is, such that $B=T\circ\overline{B}$ (resp. $B=T\circ\tilde{B}_{\mathbb{K}}$) for some $\mathbb{K}$-linear map (resp. R-linear map) $\overline{B}:V_{\mathbb{K}}\otimes_{\mathbb{K}}V_{\mathbb{K}}\to Z_{\mathbb{K}}$ (resp. $(\tilde{B}:V\otimes_R V\to Z)$), where $\tilde{B}_{\mathbb{K}}$ is the scalar restriction of $\tilde{B}$. Let $\mathrm{Pair}_{T,\mathbb{K}}(V)$ (resp. $\mathrm{Pair}_{T,R}(V)$) denote the space of all of them. Furthermore, let $\mathrm{Pair}_R(V)$ be the R-module of all R-linear maps $T:V\otimes_R V\to R$.

Lemma 2.1. *In the same notations above, let V, W and Z be R-modules and let $T:Z_{\mathbb{K}}\to\mathbb{K}$ be a $\mathbb{K}$-linear map. If T is injective, then we have isomorphisms of $\mathbb{K}$-vector spaces:*

$$\mathrm{Pair}_{(T,\mathbb{K})}(V\otimes_R W)\simeq\mathrm{Hom}_{\mathbb{K}}(V_{\mathbb{K}}^{\otimes^2_{\mathbb{K}}}\otimes_{\mathbb{K}}W_{\mathbb{K}}^{\otimes^2_{\mathbb{K}}};Z_{\mathbb{K}}) \tag{4}$$

$$\mathrm{Pair}_{(T,R)}(V\otimes_R W)\simeq[\mathrm{Hom}_R(V^{\otimes^2_R}\otimes_R W^{\otimes^2_R};Z)]_{\mathbb{K}} \tag{5}$$

where the right-hand side is the space of linear maps and $X^{\otimes^n}=X\otimes...\otimes X$, n-times. If in addition V, W and R are projectives as R-modules, then

$$\mathrm{Pair}_{(T,R)}(V\otimes_R W)\simeq[\mathrm{Pair}_R(V)\otimes_R\mathrm{Pair}_R(W)]_{\mathbb{K}}. \tag{6}$$

Proof. For the first case, notice that $\mathrm{Pair}_{T,\mathbb{K}}(V\otimes_R W)$ is precisely the image of the canonical map

$$T_*:\mathrm{Hom}_{\mathbb{K}}((V_{\mathbb{K}}\otimes_{\mathbb{K}}W)^{\otimes^2_{\mathbb{K}}};Z_{\mathbb{K}})\to\mathrm{Hom}_{\mathbb{K}}((V_{\mathbb{K}}\otimes_{\mathbb{K}}W)^{\otimes^2_{\mathbb{K}}};\mathbb{K}),$$

given by $T_*(\overline{B})=T\circ\overline{B}$. Since covariant hom-functors preserve monomorphisms, it follows that T_* is injective. The result then follows from the isomorphism theorem and from the commutativity up to isomorphisms of the tensor product. For the second case, notice that for every V,W and Z we have a canonical $\mathbb{K}$-linear injective map

$$\alpha:\mathrm{Hom}_{\mathbb{K}}([(V\otimes_R W)^{\otimes^2_R}]_{\mathbb{K}};Z_{\mathbb{K}})\to\mathrm{Hom}_{\mathbb{K}}((V_{\mathbb{K}}\otimes_{\mathbb{K}}W)^{\otimes^2_{\mathbb{K}}};Z_{\mathbb{K}}),$$

obtained as follows. Recall that restriction of scalars is right-adjoint to extension of scalars. Since tensor product is commutative up to isomorphisms, the extension of scalars functor is a strong monoidal functor relative to the monoidal structure given by tensor products (Chapter II of [6]), so that its left adjoint is lax monoidal (see Chapter 5 of [2]). Therefore, for every R-modules X,Y there is a $\mathbb{K}$-linear map $\mu:X_{\mathbb{K}}\otimes_{\mathbb{K}}Y_{\mathbb{K}}\to(X\otimes_R Y)_{\mathbb{K}}$ which in our case is a projection and then surjective. Thus, for $X=V\otimes_R W=Y$, a diagram chasing give us a surjective map

$$\mu:(V_{\mathbb{K}}\otimes_{\mathbb{K}}W)^{\otimes^2_{\mathbb{K}}}\to[(V\otimes_R W)^{\otimes^2_R}]_{\mathbb{K}},$$

so that $\alpha=\mu^*$, i.e, $\alpha(\varphi)=\varphi\circ\mu$. Finally, observe that $\mathrm{Pair}_{T,R}(V\otimes_R W)$ is the image of the following composition with $X=V\otimes_R W=Y$:

$$\begin{tikzcd}
\mathrm{Hom}_R(X\otimes_R Y;Z) \arrow[r, "(-)_{\mathbb{K}}"] \arrow[d, dashed] & \mathrm{Hom}_{\mathbb{K}}([X\otimes_R Y]_{\mathbb{K}};Z_{\mathbb{K}}) \arrow[r, "\alpha"] & \mathrm{Hom}_{\mathbb{K}}(X_{\mathbb{K}}\otimes_{\mathbb{K}}Y_{\mathbb{K}};Z_{\mathbb{K}}) \arrow[dll, bend left, "T_*"] \\
\mathrm{Hom}_{\mathbb{K}}(X_{\mathbb{K}}\otimes_{\mathbb{K}}Y_{\mathbb{K}};\mathbb{K}) & &
\end{tikzcd}$$

Since restriction of scalars is a faithful functor, the first of these maps is injective [45]. The second one is injective from the above disussion, while the third one is injective by the first case of this lemma, so that again by the isomorphism theorem and the commutativity up to isomorphisms of the tensor product we get the desired result. Now, for (6) recall that we always have a canonical morphism

$$\mathrm{Hom}_R(V;R)\otimes_R \mathrm{Hom}_R(W;R)\to \mathrm{Hom}_R(V\otimes_R W;R)$$

which is an isomorphism if the R-modules in question are projective [6]. Composing this isomorphism with (5) we get (6). ☐

Theorem 2.1. *If G is not discrete and $\dim M\geq 2$, then for every bundle P the fibers $\mathrm{YMT}_G(P)$ and $\mathrm{YMT}_{G,0}(P)$ of $\pi_{YMT}:\mathcal{YMT}\to\mathcal{X}$ and $\pi_{YMT,0}:\mathcal{YMT}_0\to\mathcal{X}$, respectively, are infinite-dimensional real vector spaces. Furthermore, as a $C^\infty(M)$-module $\mathrm{YMT}_{G,0}(P)$ is projective of finite rank. Otherwise, i.e, if G is discrete and/or $\dim M=0,1$, then the fibers are zero-dimensional as real vector spaces.*

Proof. First of all, notice that

$$\mathrm{YMT}_G(P)\simeq \mathrm{Pair}_{(\int_M,\mathbb{R})}(\Omega^2(M)\otimes_{C^\infty}\Gamma(E_{\mathfrak{g}})) \tag{7}$$

$$\mathrm{YMT}_{G,0}(P)\simeq \mathrm{Pair}_{(\int_M,C^\infty)}(\Omega^2(M)\otimes_{C^\infty}\Gamma(E_{\mathfrak{g}})), \tag{8}$$

where $\int_M : C^\infty(M)_{\mathbb{R}}\to\mathbb{R}$ is the integral and we wrote C^∞ instead of $C^\infty(M)$ in order to simplify the notation. Since the integral is injective up to sets with zero measure, a small change in the previous lemma allows us to conclude that

$$\mathrm{YMT}_G(P)\simeq \mathrm{Hom}_{\mathbb{R}}((\Omega^2(M)_{\mathbb{R}}^{\otimes^2_{\mathbb{R}}}\otimes_{\mathbb{R}}\Gamma(E_{\mathfrak{g}})_{\mathbb{R}}^{\otimes^2_{\mathbb{R}}};C^\infty(M)_{\mathbb{R}}) \tag{9}$$

$$\mathrm{YMT}_{G,0}(P)\simeq \mathrm{Hom}_{C^\infty}((\Omega^2(M)^{\otimes^2_{C^\infty}}\otimes_{C^\infty}\Gamma(E_{\mathfrak{g}})^{\otimes^2_{C^\infty}};C^\infty(M))_{\mathbb{R}} \tag{10}$$

Thus, if G is discrete, then $\dim G=0$, so that $\dim\mathfrak{g}=0$ and $E_{\mathfrak{g}}\simeq M\times 0$, implying $\Gamma(E_{\mathfrak{g}})_{\mathbb{R}}\simeq 0$, which means that the whole (9) is zero-dimensional. Since the rank of a $C^\infty(M)$-module is bounded from above by its rank as a $\mathbb{R}$-module, it follows that under the previous assumptions (10) is zero-dimensional too. Similarly, if $\dim M<2$, then $\Omega^2(M)_{\mathbb{R}}\simeq 0$, showing again that (9) and (10) are zero-dimensional. Thus, suppose that $\dim M\geq 2$ and that G is not discrete. In this case, the bundles $E_{\mathfrak{g}}$, Λ^2T^*M have positive rank. But if $E\to M$ is any vector bundle with positive rank, then $\dim_{\mathbb{R}}(\Gamma(E)_{\mathbb{R}})=\infty$, so that $\mathrm{YMT}_G(P)$ is infinite-dimensional. Notice that

$$\mathrm{Hom}_{C^\infty}((\Omega^2(M)^{\otimes^2_{C^\infty}}\otimes_{C^\infty}\Gamma(E_{\mathfrak{g}})^{\otimes^2_{C^\infty}};C^\infty(M))\simeq\Gamma(\mathrm{Hom}(\Lambda^2T^*M^{\otimes^2}\otimes E_{\mathfrak{g}}^{\otimes^2};M\times\mathbb{R})), \tag{11}$$

so that $\mathrm{YMT}_{G,0}(P)$ is also infinite-dimensional as a real vector space. Finally, recall that, regarded as a $C^\infty(M)$-module, the space of global sections $\Gamma(E)$ of any vector bundle is projective and of finite rank. Thus, from the isomorphism above, we see that under the hypotheses $\mathrm{YMT}_{G,0}(P)$ is a projective $C^\infty(M)$-module of finite rank. ☐

Corollary 1. *For discrete gauge groups (and therefore for instanton sectors given by covering spaces) Yang-Mills-type theories, linear Yang-Mills-type theories and Yang-Mill theories are all the same thing.*

Proof. By the last theorem, if G is discrete, then $\mathrm{YMT}_G(P)\simeq 0\simeq\mathrm{YMT}_{G,0}(P)$. But from the discussion at the beginning of Subsection 2.1 we know that $\mathrm{YM}_G(P)$ has always a single element. Thus, under the hypothesis all three sets are in bijection. ☐

2.2 Upper Bound

If $\dim M \geq 2$ and G is not discrete, we can actually give an upper bound to rank of $\mathrm{YMT}_{G,0}(P)$ when regarded as a $C^{\infty}(M)$-module. This goes as follows. By the Serre-Swan theorem (and its extensions to the non-compact case), the category of projective finitely generated $C^{\infty}(M)$-modules is equivalent to the category of vector bundles with finite rank over M, and the equivalence is given precisely by the functor of global sections [39]. Thus, if E is a bundle such that $\Gamma(E)$ is generated by N elements, then the module $\Gamma(E)$ has rank bounded from above by N, i.e, $\operatorname{rnk}\Gamma(E) \leq N$.

But in the construction of this generating set, we see that $N = m + k$, where k is the number of elements of a finite trivializing open covering for E. Let $\min_k(E)$ be the minimum of such k, i.e, the minimum number of elements in a trivilizing open covering of E. Thus, $1 \leq \min_k(E)$ and $\min_k(E) = 1$ if E is trivial. It can be proved that for $\min_k(E) \leq \mathrm{cat}(M)$ for every E, where $\mathrm{cat}(M)$ is the Lusternik-Schnirelmann category of M, which satisfies $\mathrm{cat}(M) \leq n+1$, where n is the dimension of M [12]. Thus, for every vector bundle E we have the upper bound $\operatorname{rnk}\Gamma(E) \leq m+n+1$. One can also prove that if M is q-connected with i.e, $\pi_i(M) = 0$ for $1 \leq i \leq q$, then $\mathrm{cat}(M) < (n+1/q+1)+1$, so that $\operatorname{rnk}\Gamma(E) < m + (n + 1/q + 1) + 1$.

Remark 4. We notice that if M is not contractible, then $q < n$. Indeed, every topological n-manifold is homotopic to a n-dimensional CW-complex [38, 57]. But if a n-dimensional CW-complex X is q-connected, with $q \geq n$, then X is contractible. On the other hand, if M is contractible, then M is not compact[1] and every bundle $E \to M$ is trivial. Thus, $\mathrm{rank}(E) = m + 1$.

Corollary 2. *For every l-dimensional Lie group G, with $l > 0$, every smooth n-dimensional manifold M, with $n \geq 2$ and every principal G-bundle P we have*

$$\operatorname{rnk}\mathrm{YMT}_{G,0}(P) \leq \frac{(n^2-n)^2 l^2}{4} + n + 1.$$

Furthermore,

1. *if the smooth manifold M is q-connected, we also have*
$$\operatorname{rnk}\mathrm{YMT}_{G,0}(P) < \frac{(n^2-n)^2 l^2}{4} + \frac{n+1}{q+1} + 1;$$

2. *if M is paralellizable and G is abelian, or if M is contractible, then*
$$\operatorname{rnk}\mathrm{YMT}_{G,0}(P) = \frac{(n^2-n)^2 l^2}{4} + 1. \tag{12}$$

Proof. For the main assertion and for (1), just apply the previous bounds to (11) noticing that the rank of the bundle
$$\mathrm{Hom}(\Lambda^2 T^* M^{\otimes^2} \otimes E_{\mathfrak{g}}^{\otimes^2}; M \times \mathbb{R}) \tag{13}$$
is given by $(n^2-n)^2 l^2/4$. For the case (2), notice that if M is paralellizable, then $\Lambda^2 T^* M$ is trivial. Furthermore, the adjoint bundle of a principal G-bundle with abelian G is also trivial. Thus, (13) is trivial and the result follows from the previous discussion. If, instead, M is contractible, then the bundle (13) is automatically trivial and the proof is done. □

[1] Recall that a compact manifold of positive dimension is never contractible.

Some interesting situations to keep in mind:

Example 1 (lower dimension examples). As we saw above, the lowest rank of $\mathrm{YMT}_{G,0}(M)$ is realized when M is contractible or when M is parallelizable and G is abelian. Assuming one of these conditions and fixing a upper bound z for (12) and find the integer solutions for the inequality

$$\frac{(n^2-n)^2 l^2}{4} + 1 \leq z,$$

where $n \geq 2$ and $l > 0$. For instance, if we fixes $z = 7$, we get the graphs below. In particular, we see that *there is no triple* $(M, G, P) \in \mathcal{X}$ *such that* $\mathrm{rnk}\,\mathrm{YMT}_{G,0}(M) = 1$.

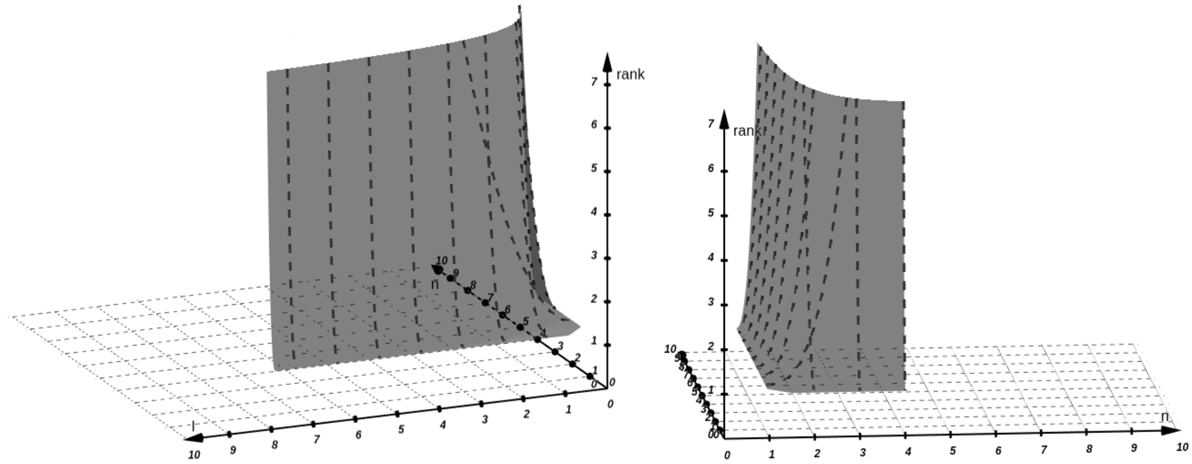

Figure 1: Two different views of the graphic for the values of n and l which produces a space of linear YMT theories with rank $z \leq 7$.

2.3 Gauge Invariant Case

This section is a remark on the classification of the subspace $\mathrm{YMT}_{G,0}(P)_{\mathrm{gau}} \subset \mathrm{YMT}_{G,0}(P)$ of gauge invariant linear YMT theories with prescribed Lie group G and instanton sector P. Since $\mathrm{YMT}_{G,0}(P)_{\mathrm{gau}} \subset \mathrm{YMT}_{G}(P)_{\mathrm{gau}} \subset \mathrm{YMT}_{G,0}(P)$, this can also be understood as lower bounds for the space $\mathrm{YMT}_{G}(P)_{\mathrm{gau}}$ of not necessaily linear gauge invariant YMT theories. We begin by noting that from (10) and (6) we get

$$\begin{aligned} \mathrm{YMT}_{G,0}(P) &= \mathrm{Pair}_{C^\infty}(\Omega^2_{heq}(P;\mathfrak{g})) && (14)\\ &\simeq \mathrm{Pair}_{C^\infty}((\Omega^2(M)) \otimes_{C^\infty} \mathrm{Pair}_{C^\infty}(\Gamma(E_{\mathfrak{g}})) && (15)\\ &\simeq \mathrm{Pair}_{C^\infty}((\Omega^2(M)) \otimes_{C^\infty} \Gamma(\mathrm{Pair}(E_{\mathfrak{g}})), && (16)\end{aligned}$$

where $\mathrm{Pair}(E)$ is the *bundle of pairs*, i.e, the bundle $\mathrm{Hom}(E \otimes E; M \times \mathbb{R})$. Since $\Omega^2(M)$ is independent of G, the action by pullbacks of $\mathrm{Bun}_G(P)$ on $\mathrm{Pair}_{C^\infty}(\Omega^2_{heq}(P;\mathfrak{g}))$ corresponds, via the decomposition above, to an adjoint action on $\Gamma(\mathrm{Pair}(E_{\mathfrak{g}}))$, i.e, $[f^*\langle\cdot,\cdot\rangle(s,s')]_p = \langle Ad_{f(p)}s(p), Ad_{f(p)}s'(p)\rangle$, where we are using the identification $\mathrm{Gau}_G(P) \simeq C^\infty_{eq}(M;G)$ of global gauge transformations with equivariant smooth maps. Let us define an *adjoint structure* on P as a C^∞-linear pairing $\langle\cdot,\cdot\rangle$ on $E_{\mathfrak{g}}$ which is

$\mathrm{Gau}_G(P)$-invariant, i.e, such that $\langle Ad_{f(p)}s(p), Ad_{f(p)}s'(p)\rangle = \langle s(p); s'(p)\rangle$ for every $p \in M$ and every sections $s, s' \in \Gamma(E_{\mathfrak{g}})$. Let $\mathrm{AdStr}(P;\mathfrak{g})$ be the set of all of them. It is actually a $C^\infty(M)$-submodule of $\Gamma(\mathrm{Pair}(E_{\mathfrak{g}}))$. Thus:

Lemma 2.2. *For every Lie group G and every principal G-bundle P, we have*

$$\mathrm{YMT}_{G,0}(P)_{\mathrm{gau}} \simeq \mathrm{Pair}_{C^\infty}((\Omega^2(M)) \otimes_{C^\infty} \mathrm{AdStr}(P;\mathfrak{g}).$$

Let $\mathrm{inv}(\mathfrak{g})$ be the semifree differential graded algebra of invariant polynomials of $\mathfrak{g}$. Every element of $\omega \in \mathrm{inv}^2(\mathfrak{g})$ degree 2 in $\mathrm{inv}(\mathfrak{g})$ induces an Ad-invariant pairing in $\mathfrak{g}$, which in turn induces an adjoint structure on P. Note that $\mathrm{inv}^2(\mathfrak{g})$ always contains the Killing form of $\mathfrak{g}$, but it is typically higher dimensional. For instance, even if we are looking for nondegenerate Ad-invariant pairings, one can find them not only on semi-simple Lie algebras (given in this case by the Killing form), but also on a large class of solvable Lie algebras [40].

3 Extensions

Let G be a Lie group. A *basic extension* of G is another Lie group $\hat{G}$ and an injective Lie group homomorphism $\imath : G \to \hat{G}$. A basic extension is called a *split* if there exists a Lie group isomorphism $\hat{G} \simeq \mathrm{Ker}(\imath) \ltimes G$. The relation between basic extensions and the classical notion of group extensions is as follows. Recall that a group $\hat{G}$ is called an *extension* of another group G if they belong to an short exact sequence of groups of the form

$$1 \longrightarrow N \xrightarrow{\imath} \hat{G} \xrightarrow{\pi} G \longrightarrow 1. \tag{17}$$

This implies that N is isomorphic to the normal subgroup $i(N) \subset G$ and that G is isomorphic to the group $\hat{G}/\imath(N)$. Hence G is not necessarily a subgroup of $\hat{G}$ *a priori*. It is precisely if the map $\pi : \hat{G} \to G$ has a global section, i.e. iff the extension is split. On the other hand an inclusion $i : G \to \hat{G}$ does not necessarily fulfill $\hat{G}$ as a classical extension of G, but it does iff G admits a semidirect complement in $\hat{G}$ and in this case the induced extension is necessarily split. More precisely, we have the dotted horizontal arrow below and its image is precisely the image of the vertical arrow. Thus, it factors producing the diagonal one, which is a bijection.

$$\begin{tikzcd} \{\text{split extensions}\} \arrow[r, hook, dashed] \arrow[dr, dotted, "\simeq"'] & \{\text{basic extensions}\} \\ & \{\text{split basic extension}\} \arrow[u, hook] \end{tikzcd} \tag{18}$$

In sum, we have the following conlusion:

Conclusion 1. *The typical examples of basic extensions of a Lie group are their split extensions. Actually, split extensions are in bijection with split basic extensions.*

Let $P \to M$ be a principal G-bundle over a smooth manifold M and let $\imath : G \hookrightarrow \hat{G}$ be a basic extension of G (for example, a split extension by the above). From the classification theorem of principal bundles, P is classified by a homotopy class of continuous maps $f : M \to BG$, where BG is the classifying space of G [31, 37]. Since B is functorial, we have a map $Bf : BG \to B\hat{G}$, which classifies a principal $\hat{G}$-bundle $\hat{P}$. In terms of cocycles this means that for every trivializing open covering U_i of M, the cocycles $\hat{g}_{ij} : U_{ij} \to \hat{G}$ are just the compositions $\imath \circ g_{ij}$. Since the classification of bundles is realized by pullbacks, from universality there exists a bundle map $\imath : P \hookrightarrow \hat{P}$, denoted by the same notation of the basic extension map, which is a monomorphism due to the stability of monomorphisms under pullbacks.

Furthermore, each basic extension also induces a group homomorphism $\xi : \mathrm{Gau}_G(P) \to \mathrm{Gau}_{\hat{G}}(\hat{P})$. Indeed, recall that $\mathrm{Gau}_G(P)$ can be regarded as the group of global sections of the group bundle $P_G = (P \times G)/G \to M$ [21]. Since $\hat{P}$ is the bundle induced by the basic extension, the actions of G on P and of $\hat{G}$ on $\hat{P}$ are compatible, so that there exists the dotted arrow below, which is a morphism of group bundles. Applying the functor of global sections we get ξ. A similar homotopical conclusion is the following. In [5] it is shown that we have a homotopy equivalence $\mathrm{Gau}_G(P) \simeq \Omega \operatorname{Map}(M; BG)_f$, where $f : M \to BR$ is the map whose homotopy class classifies P. Furthermore, Ω denotes the loop space, $\operatorname{Map}(X;Y)$ is the space of continuous maps in the compact-open topology and $\operatorname{Map}(X;Y)_f$ is path-component of f.

$$\begin{tikzcd}
P \times G \arrow[r, "\imath \times \imath"] \arrow[d, "\pi"'] & \hat{P} \times \hat{G} \arrow[d, "\hat{\pi}"] \\
(P \times G)/G \arrow[r, dashed] & (\hat{P} \times \hat{G})/\hat{G}
\end{tikzcd}$$

We can now define the main objects of this paper: *extensions of YMT theories*. Let G be a Lie group and let $S^G : \mathrm{Conn}(P;\mathfrak{g}) \to \mathbb{R}$ be the action functional of a YMT theory with gauge group G and instanton sector P. An *extension* of S^G consists of the following data:

1. a basic extension $\imath : G \hookrightarrow \hat{G}$ of G;
2. an equivariant extension of the space of connection 1-forms on $\hat{P}$. More precisely, a subset
$$\mathrm{Conn}(\hat{P};\hat{\mathfrak{g}}) \subseteq \widehat{\mathrm{Conn}}(\hat{P};\hat{\mathfrak{g}}) \subseteq \Omega^1_{eq}(\hat{P};\hat{\mathfrak{g}}), \tag{19}$$
called the *extended domain*, where $\hat{P}$ is the $\hat{G}$-bundle induced from P, which is invariant by the canonical action of $\mathrm{Gau}_{\hat{G}}(\hat{P})$ by pullbacks[2];
3. a $\mathrm{Gau}_{\hat{G}}(\hat{P})$-invariant functional $\hat{S}^{\hat{G}} : \widehat{\mathrm{Conn}}(\hat{P};\hat{\mathfrak{g}}) \to \mathbb{R}$, called the *extended action functional*;
4. a nonempty subset $0 \subseteq C^1(\hat{P};\hat{\mathfrak{g}}) \subseteq \widehat{\mathrm{Conn}}(\hat{P};\hat{\mathfrak{g}})$, called the *correction subspace* and whose inclusion map in $\widehat{\mathrm{Conn}}(\hat{P};\hat{\mathfrak{g}})$ we will denote by $\jmath$;
5. a *correction functional* $C : C^1(\hat{P};\hat{\mathfrak{g}}) \to \mathbb{R}$ and map $\delta : C^1(\hat{P};\hat{\mathfrak{g}}) \to \mathrm{Conn}(P;\mathfrak{g})$ such that
$$\hat{S}^{\hat{G}} \circ \jmath = S^G \circ \delta + C. \tag{20}$$

Thus, if $\hat{S}^{\hat{G}}$ is an extension of a YMT theory S^G we have the diagram below, representing the sequence of subspaces and maps defining it. If $\widehat{\mathrm{Conn}}(\hat{P};\hat{\mathfrak{g}}) \simeq \Omega^1_{eq}(\hat{P};\hat{\mathfrak{g}})$ we say that the extension is *full*. If the correction functional C is null, we say that the extension is *complete on the correction*

[2] Notice that although the YMT functionals are *not* necessarily gauge invariant, their extended functional $\hat{S}^{\hat{G}}$ *must be* gauge invariant

subspace $C^1(\hat{P};\hat{\mathfrak{g}})$ or simply that it is *complete*. If such subspace is such that $C^1(\hat{P};\hat{\mathfrak{g}}) \simeq \widehat{\mathrm{Conn}}(\hat{P};\hat{\mathfrak{g}})$ and $C \equiv 0$ we say that the extension is *fully complete*.

$$\begin{tikzcd}
0 \arrow[r] & \mathrm{Conn}(\hat{P};\hat{\mathfrak{g}}) \arrow[r, hook] & \widehat{\mathrm{Conn}}(\hat{P};\hat{\mathfrak{g}}) \arrow[r, hook] & \Omega^1_{eq}(\hat{P};\hat{\mathfrak{g}}) & \\
& \Omega^1_{eq}(P;\mathfrak{g}) & \mathrm{Conn}(P;\mathfrak{g}) \arrow[l, hook'] & C^1(\hat{P};\hat{\mathfrak{g}}) \arrow[l, "\delta"] \arrow[ul, hook', "\imath"'] & 0 \arrow[l]
\end{tikzcd}$$

Physically, a complete extension $\hat{S}^{\hat{G}}$ of a YMT theory S^G is one which reproduces S^G *exactly* (i.e, without any correction term) when restricted to the subspace $C^1(\hat{P};\hat{\mathfrak{g}})$. Furthermore, it is a genuine extension of S^G in the mathematical sense. If we think of this subspace as determining the configurations of $\hat{S}^{\hat{G}}$ in a certain scale, then we conclude that a complete extension of S^G is one which reproduces S^G exactly in *some scales*. In turn, a fully complete extension is one describing S^G exactly in *every scales*. This suggests a relation between emergence phenomena and complete extensions, which will be confirmed in Subsection 3.4.

Finally, a full extension of S^G is one whose vector potentials are not connection 1-forms on $\hat{P}$, but actually arbitrary equivariant 1-forms. Thus, in them, we have a physical meaning for 1-forms $\hat{D} : T\hat{P} \to \hat{g}$ which are not vertical.

Remark 5 (linear and equivariant extensions)**.** The definition of extension above has some variations. Indeed, as defined above, in an extension of a YMT theory the extended domain is required to be a $\mathrm{Gau}_{\hat{G}}(\hat{P})$-set such that the extended functional is $\mathrm{Gau}_{\hat{G}}(\hat{P})$-invariant. No structures are required on the correction subspace and the correction functional and the δ-map are just functions. The situation can be improved in two directions:

1. by requiring that the extended domain and the correction subspace are actually linear subspaces of $\Omega^1_{eq}(\hat{P};\hat{\mathfrak{g}})$. We can also require linearity of the correction function and/or of the δ-map;
2. by requiring that not only the extended domain, but also the correction subspace is a $\mathrm{Gau}_{\hat{G}}(\hat{P})$-set. In this situation it is typically useful to require equivariance of C and δ, so that the whole structure is equivariant.

In the first case, we say that the extension is *partially linear* (resp. *linear*) if the extended domain and the correction subspaces have linear structures (resp. if in addition C and δ are linear). Notice that since the action of $\mathrm{Gau}_{\hat{G}}(\hat{P})$ preserves the linear structure of $\Omega^1_{eq}(\hat{P};\hat{\mathfrak{g}})$, it follow that it also preserves the linear structure of the vector subspace $\mathrm{Conn}(\hat{P};\hat{\mathfrak{g}})$. In the second case, we say that the extension is *weakly equivariant* if the correction subspace is $\mathrm{Gau}_{\hat{G}}(\hat{P})$-invariant and *equivariant* if in addition C and δ are equivariant. Notice that if $\delta \neq id$, then we may have equivariant extensions of YMT which are not gauge invariant. The linear context is useful when studying complete extensions [36], while the equivariant context in useful in the study of incomplete extensions [35].

3.1 Trivial Extensions

Each YMT theory admits some trivial extensions, as described in the following examples.

Example 2 (*null-type extensions*). Every YMT S^G admits an extension relative to any basic extension $\imath : G \hookrightarrow \hat{G}$ and whose space is any given $\mathrm{Gau}_{\hat{G}}(\hat{P})$-invariant subset $\widehat{\mathrm{Conn}}(\hat{P};\hat{\mathfrak{g}})$ satisfying (19). It is obtained by taking the null extended functional $\hat{S}^{\hat{G}} = 0$, null correction subspace, i.e, $C^1(\hat{P};\hat{\mathfrak{g}}) = 0$, null correction functional $C = 0$ and $\delta = 0$. Condition 20 is immediately satisfied. We say that this is the *null extension of* S^G *with extended space* $\widehat{\mathrm{Conn}}(\hat{P};\hat{\mathfrak{g}})$. In the case when the extended space is the space of connections $\mathrm{Conn}(\hat{P};\hat{\mathfrak{g}})$, we say simply that it is the *null extension of* S^G. This is an example of an extension which is equivariant even if S^G is not gauge invariant.

Example 3 (identity extensions). For every gauge invariant[3] YMT S^G we can assign the *identity extension*, relative to the identity basic extension $id : G \hookrightarrow G$, defined as follows. The extended domain and the correction subspace are

$$\widehat{\mathrm{Conn}}(P;\mathfrak{g}) = \mathrm{Conn}(P;\mathfrak{g}) = C^1(P;\mathfrak{g}). \tag{21}$$

The δ-map is the identity of $\mathrm{Conn}(P;\mathfrak{g})$ and $C \equiv 0$. We clearly have $S^G \circ \delta + C = S^G$, so that this realizes S^G as a complete extension of itself.

Example 4 (equivariant choice extension). As in the last example, consider the identity basic extension $\mathrm{id} : G \hookrightarrow G$, with same correction subspace $C^1(P;\mathfrak{g}) = \mathrm{Conn}(P;\mathfrak{g})$, with same correction functional $C \equiv 0$ and with same δ-map, i.e, $\delta = \mathrm{id}$. But instead of taking the extended domain as in (21), take $\widehat{\mathrm{Conn}}(P;\mathfrak{g})$ as some $\mathrm{Gau}_G(P)$-invariant subset of $\Omega^1(P;\mathfrak{g})$ containing $\mathrm{Conn}(P;\mathfrak{g})$, i.e, which satisfies (19). Then, by the equivariant version of the Axiom of Choice[4], it follows that there are $\mathrm{Gau}_G(P)$-equivariant retracts, as below. If r is one such retract, define $\hat{S}^G = S^G \circ r$. Then $\hat{S}^G|_{C^1(P;\mathfrak{g})} = S^G = S^G \circ \delta + C$, which means that $\hat{S}^G$ is another extension of S^G. But notice that it is just the identity extension with a larger extended domain arising from the Axiom of Choice.

$$\begin{tikzcd} \mathrm{Conn}(P;\mathfrak{g}) \arrow[r, hook] \arrow[rr, bend right, "\mathrm{id}"'] & \widehat{\mathrm{Conn}}(P;\mathfrak{g}) \arrow[r, "r"] & \mathrm{Conn}(P;\mathfrak{g}) \end{tikzcd}$$

Remark 6. A look at the last example reveals that one can actually extend the domain of every equivariant extension such that $\delta = \mathrm{id}$. In particular, we can take $\widehat{\mathrm{Conn}}(P;\mathfrak{g}) = \Omega^1_{eq}(P;\mathfrak{g})$ in order to get a full extension. For a detailed discussion, see [35].

Example 5 (constant extensions). Notice that if $c \in \mathbb{R}$ is any real number and X is any G-set, then the constant function f with value c is G-invariant, because $f(g \cdot x) = c = f(x)$. With this in mind, fixed a basic extension $\imath : G \hookrightarrow \hat{G}$, let $\widehat{\mathrm{Conn}}(\hat{P};\hat{\mathfrak{g}})$ be any $\mathrm{Gau}_{\hat{G}}(\hat{P})$-invariant set satisfying (19) and let $C^1(\hat{P};\hat{\mathfrak{g}})$ be any subset of it. Given a real number $c \in \mathbb{R}$ and a connection D_0 in P we will build an extension for every YMT theory S^G whose extended domain and whose correction subspace are the above. Define $\delta : C^1(\hat{P};\hat{\mathfrak{g}}) \to \mathrm{Conn}(P;\mathfrak{g})$ as the constant map in D_0. Define $C : C^1(\hat{P};\hat{\mathfrak{g}}) \to \mathbb{R}$ as the constant map with value c. Define $\hat{S}^{\hat{G}} : \widehat{\mathrm{Conn}}(\hat{P};\hat{\mathfrak{g}}) \to \mathbb{R}$ as the constant map in $S^G[D_0] + c$, which is $\mathrm{Gau}_{\hat{G}}(\hat{P})$-invariant by the previous remark. Furthermore, we clearly have $\hat{S}^{\hat{G}} \circ \jmath = S^G \circ \delta + C$, so that the described data really defines an extension of S^G.

[3] Here is an example where the hypothesis of gauge invariance is needed.

[4] More precisely, if we assume the Axiom of Choice, then for every group G, in the corresponding category of G-sets, every mono and every epi are split [59].

An extension which is not of the types above is called *nontrivial*. Of course, we will be more interested in these ones. There are a lot of well-known examples of nontrivial extensions for specific kinds of YMT theories. In the following we will describe some of them. We begin with a detailed example.

3.2 A Detailed Example: Higgs Mechanism

It is well known that the choice of a specific scalar vacuum field with non zero expectation value coupling with the YM gauge field causes a spontaneous symmetry breaking (SSB) of the classical YM theory [23, 24, 18]. Here we will see that any equivariant correction of a YMT S^G theory involving the classical Higgs fields (in particular the corresponding Yang-Mills-Higgs (YMH) theory) can be viewed as an extension of S^G in the sense of the previous section.

Recall that for a principal $\hat{G}$-bundle $\hat{P}$ and for a closed Lie subgroup $G \subset \hat{G}$, the space of *classical Higgs fields*, denoted by $\mathrm{Higgs}(\hat{P}; G)$, is the space $\Gamma(\hat{P}/G)$ of global sections ϕ of the quotient bundle $\hat{P}/G$, which can be identified with the associated bundle $\hat{P} \times_{\hat{G}} (\hat{G}/G)$ [48, 55]. But sections of this associated bundle are in 1-1 correspodence with G-reductions of $\hat{P}$ [29], so that classical Higgs fields are in bijection with these G-reductions. In this sense, for a classical field theory $S[s]$ depending on $\hat{P}$, we say that *the $\hat{G}$-symmetry is spontanously broken to a G-symmetry* when such a classical Higgs field exists. In this case, the action functional S couples with the Higgs fields by means of adding a term $C[s, \phi]$ depending on both the fields, so that we have a total action $S_C = S + C$. Thus, in the case of a YM theory S^G on P, we have $S^G_C[D, \phi] = S^G[D] + C[D, \phi]$, suggesting that one can regard the functionals S^G_C as incomplete extensions of S^G.

Let P be a principal G-bundle and let $\imath : G \hookrightarrow \hat{G}$ be a basic extension of G. The space $\mathrm{Higgs}(\hat{P}; G)$ is clearly nonempty, since $\hat{P}$ arises from P by the basic extension. We say that $\imath : G \hookrightarrow \hat{G}$ is *coherent* if it becomes endowed with a map $\theta : \hat{G}/G \to \hat{\mathfrak{g}}$ satisfying the compatibility relation $\theta(aG) = Ad_a(\theta(G))$, where $a \in \hat{G}$. In this case, we have a bundle morphism $\Theta : \hat{P} \times_{\hat{G}} (\hat{G}/G) \to E_{\hat{\mathfrak{g}}}$, which is fiberwise given by θ. Consequently, we have also have an induced map $\Theta_* : \mathrm{Higgs}(\hat{P}; G) \to \Gamma(E_{\hat{\mathfrak{g}}})$. Now, notice that given a connection $\hat{D}$ in $\hat{P}$, its covariant derivative $d_{\hat{D}} : \Omega^0(\hat{P}; \hat{\mathfrak{g}}) \to \Omega^1(\hat{P}; \hat{\mathfrak{g}})$ induces a covariant derivative $\nabla_{\hat{D}} : \Gamma(E_{\hat{\mathfrak{g}}}) \to \Omega^1(M; E_{\hat{\mathfrak{g}}})$ on the adjoint bundle. We can then take the image of the following composition:

$$\begin{tikzcd} \mathrm{Higgs}(\hat{P}; G) \arrow[r, "\Theta_*"] \arrow[rrrr, bend right, "\Theta_{*,\hat{D}}"'] & \Gamma(E_{\hat{\mathfrak{g}}}) \arrow[r, "\nabla_{\hat{D}}"] & \Omega^1(M; E_{\hat{\mathfrak{g}}}) \arrow[r, "\simeq"] & \Omega^1_{heq}(\hat{P}; \hat{\mathfrak{g}}) \arrow[r, hook] & \Omega^1_{eq}(\hat{P}; \hat{\mathfrak{g}}). \end{tikzcd}$$

In particular, one can take the union of $\mathrm{img}(\Theta_{*,\hat{D}})$ over all $\hat{D} \in \mathrm{Conn}(\hat{P}; \hat{\mathfrak{g}})$, obtaining a subset $\Theta^1_{\mathrm{conn}}(\hat{P}; \hat{\mathfrak{g}}) \subset \Omega^1_{eq}(\hat{P}; \hat{\mathfrak{g}})$ independent of the choice of $\hat{D}$. As one can check, since Θ_* is equivariant, this subset is invariant under the canonical action by pullbacks of $\mathrm{Gau}_{\hat{G}}(\hat{P})$. Thus, the union $\Theta^1_{\mathrm{conn}}(\hat{P}; \hat{\mathfrak{g}}) \cup \mathrm{Conn}(\hat{P}; \hat{\mathfrak{g}})$ is also $\mathrm{Gau}_{\hat{G}}(\hat{P})$-invariant and clearly satisfies (19). Notice that this union is disjoint except maybe for the null 1-form. This means that the first inclusion in the diagram below is well defined, so that we can consider the corresponding composition.

$$\begin{tikzcd} \Theta^1_{\mathrm{conn}}(\hat{P}; \hat{\mathfrak{g}}) \arrow[r, hook] \arrow[rr, bend right, "\delta"'] & \Theta^1(\hat{P}; \hat{\mathfrak{g}}) \times \mathrm{Conn}(\hat{P}; \hat{\mathfrak{g}}) \arrow[r, "\mathrm{pr}_2"] & \mathrm{Conn}(\hat{P}; \hat{\mathfrak{g}}) \end{tikzcd} \tag{22}$$

Given any $\mathrm{Gau}_{\hat{G}}(\hat{P})$-invariant functional $C : \Theta^1_{\mathrm{conn}}(\hat{P};\hat{\mathfrak{g}}) \to \mathbb{R}$ and any gauge invariant YMT theory $S^{\hat{G}}$, we see that the construction above realizes the sum $\hat{S}^{\hat{G}}_C = S^{\hat{G}} \circ \delta + C$ as an extension of the YMT theory $S^{\hat{G}}$, relative to the trivial basic extension $id : \hat{G} \hookrightarrow \hat{G}$, with extended domain and correction subspaces given by

$$\widehat{\mathrm{Conn}}(\hat{P};\hat{\mathfrak{g}}) = \Theta^1_{\mathrm{conn}}(\hat{P};\hat{\mathfrak{g}}) = C^1(\hat{P};\hat{\mathfrak{g}}),$$

whose δ-map is that given in (22) and whose correction functional is the one given C. In particular, if $S^{\hat{G}}$ is a classical Yang-Mills theory with pairing $\langle\cdot,\cdot\rangle_{\hat{\mathfrak{g}}}$ and

$$C(\hat{D},\phi) = \langle \Theta_{*,\hat{D}}\phi, \Theta_{*,\hat{D}}\phi \rangle + V(\phi),$$

where $V(\phi)$ is some potential, then $\hat{S}^{\hat{G}}_C$ is precisely the action functional of the corresponding Yang-Mills-Higgs theory. Therefore, *the Yang-Mills-Higgs theory is an example of an extension of YM theories in the sense of previous section.* Notice that in this case, we actually obtain an example of equivariant extension.

Another practical example of this construction is the following.

Example 6 (t'Hooft-Polyakov monopole)**.** Recall the identification $SO(3)/SO(2) \simeq \mathbb{S}^2$, so that there is an injective map $SO(3)/SO(2) \to \mathbb{R}^3$. Under the Lie algebra identification $\mathbb{R}^3 \simeq \mathfrak{so}(3)$ one can check that $\theta(aSO(2)) = Ad_a(\theta(SO(2)))$, so that the basic extension $SO(2) \hookrightarrow SO(3)$ is coherent and we have the bundle morphism $\Theta : \hat{P} \times_{SO(3)} (SO(3)/SO(2)) \to E_{\mathfrak{so}(3)}$. Fixing a local trivialization for $\hat{P}$ and parameterizing $SO(3)$ by the Euler angles φ, α, ψ, we see that the map $\Theta_* : \Gamma(\mathrm{Higgs}(\hat{P}; SO(2)) \to \Gamma(E_{\mathfrak{so}(3)})$, via the fiberwise identification of θ, takes the following form, which is the classical expression of the normalized Higgs field in the t'Hooft-Polyakov monopole description [55]:

$$\Theta_*(\phi) = \begin{pmatrix} 0 & -\cos(\alpha) & -\cos(\varphi)\sin(\alpha) \\ \cos(\alpha) & 0 & -\sin(\varphi)\sin(\alpha) \\ \cos(\varphi)\sin(\alpha) & \sin(\varphi)\sin(\alpha) & 0 \end{pmatrix} \simeq \begin{pmatrix} \sin(\varphi)\sin(\alpha) \\ -\cos(\varphi)\sin(\alpha) \\ \cos(\alpha) \end{pmatrix}.$$

3.3 Further Examples

Some additional manifestations of extensions of YMT theories are the following. It is not our aim to exhaust the list of all known examples, but only to show the wide range of applicability of the definitions in Section 3.

Example 7 (BF Theory)**.** Let P be a principal G-bundle on a compact oriented 4-dimensional manifold (M,ω) and let $\langle\cdot,\cdot\rangle$ be a pairing in horizontal equivariant 2-forms. In analogy to the classical BF theories [3, 58], define the *BF-type theory* with the given pairing as the functional

$$S_{BF} : \mathrm{Conn}(P,\mathfrak{g}) \times \Omega^2_{he}(P,\mathfrak{g}) \to \mathbb{R} \quad \text{such that} \quad S_{BF}(D,B) = \int_M \langle F_D, B \rangle \omega. \tag{23}$$

Now, consider the identity basic extension $id : G \hookrightarrow G$ and for $n = 4$ look at the map $\mathrm{curv} : \mathrm{Conn}(P;\mathfrak{g}) \to \Omega^2_{heq}(P;\mathfrak{g})$ assigning to each connection D in P its curvature 2-form F_D. Let $\mathrm{grph}(\mathrm{curv})$ be its graph. The projection $\mathrm{pr} : \mathrm{grph}(\mathrm{curv}) \to \mathrm{Conn}(P;\mathfrak{g})$ is a bijection whose inverse is the diagonal map Δ_{curv}. Notice that $S_{BF} \circ \Delta_{\mathrm{curv}} = S^G$, where S^G is a gauge invariant

YMT theory with the given pairing. Thus, with the same extended domain, same correction subspace, same δ-map and same correction functional, we see that $S_{BF} \circ \Delta_{\text{curv}}$ is an extension of S^G which in some sense is *equivalent* to the identity extension of Example 3. In Section 5 we will see that they are actually isomorphic as objects of the category of extensions.

Example 8 (Higgs again)**.** Let us revisit our discussion of Subsection 3.2. For the present discussion, we will look at Higgs fields $\phi \in \text{Higgs}(\hat{P}; G)$ which are *$\hat{D}$-Higgs vacuum*, i.e., which are parallel with respect to some exterior covariant derivative $\nabla_{\hat{D}}$, i.e, $\nabla_{\hat{D}}\phi = 0$. In our context we note that the existence of a Higgs field is guaranteed by the definition of the extension, once that P is realized as a G-reduction of $\hat{P}$ by the monomorphism $\imath : P \to \hat{P}$, but, a priori, those *$\hat{D}$-Higgs vacuum* need not exist. Indeed, a $\hat{G}$-connection $\hat{D}$ in $\hat{P}$ reduces to a G-connection in P iff P is obtained from a $\hat{D}$-parallel Higgs field vacuum [29]. Thus, let $\text{Higgs}_0(\hat{P}; G) \subset \text{Higgs}(\hat{P}; G)$ be the set of Higgs vacuum. For a fixed $\phi_0 \in \text{Higgs}_0(\hat{P}; G)$, define $C^1(\hat{P}, \hat{\mathfrak{g}}) = \{\hat{D} \in \text{Conn}(\hat{P}, \hat{\mathfrak{g}}) | \nabla_{\hat{D}}\phi_0 = 0\}$. By the equivariance of the exterior covariant derivative it is clear that $C^1(\hat{P}, \hat{\mathfrak{g}})$ is $\text{Gau}_{\hat{G}}(\hat{P})$-invariant. By construction, there exists the map $\delta : C^1(\hat{P}, \hat{\mathfrak{g}}) \to \text{Conn}(P; \mathfrak{g})$ which takes a connection in which ϕ_0 is parallel and gives it reduction to P. Let S^G be a gauge invariant YMT theory whose instanton sector is P. Then, for every $\text{Gau}_{\hat{G}}(\hat{P})$-invariant set $\widehat{\text{Conn}}(\hat{P}; \hat{\mathfrak{g}})$ satisfying (19), and every $\text{Gau}_{\hat{G}}(\hat{P})$-invariant functional $C : C^1(\hat{P}, \hat{\mathfrak{g}}) \to \mathbb{R}$, the sum $\hat{S}^{\hat{G}} = S^G \circ \delta + C$ is an extension of S^G.

Remark 7. Note the difference: while in Subsection 3.2 the Yang-Mills-Higgs theory with group $\hat{G}$, relatively to the basic extension $\imath : G \hookrightarrow \hat{G}$, was realized as an extension of YMT theory with the *same* group $\hat{G}$, in example above the Yang-Mills-Higgs with group $\hat{G}$ was realized as an extension of the YMT with *smaller* group G. This naturally leads one to ask whether a YMT $S^{\hat{G}}$ can be itself regard as an extension of S^G. This was considered and studied in [35].

Remark 8. Higgs-like mechanisms were observed to emerge as extensions of YMT theories in some other contexts. For instance, by the method of dimension reduction if the extension is allowed to lie in a higher dimensional space-time [19, 22] and if the Lie algebra of the YM theory is replaced by a Leibniz algebra giving the 2-Higgs mechanism introduced in [53].

Example 9 (Full YM theories)**.** Let P be a G-principal, (M, g) a semi-Riemannian manifold and let S^G the classical YM theory, whose pairing is given by $\langle \alpha, \beta \rangle_{\mathfrak{g}} = \text{tr}(\alpha[\wedge] \star_g \beta)$. In dimension $n = 4$ we have another pairing given by $\langle \alpha, \beta \rangle'_{\mathfrak{g}} = \text{tr}(\alpha[\wedge]\beta)$. In the literature, the action functional

$$S^G_{\text{full}} : \text{Conn}(P; \mathfrak{g}) \to \mathbb{R} \quad \text{given by} \quad S^G_{\text{full}}[D] = \int_M \langle F_D, F_D \rangle_{\mathfrak{g}} dvol_g + \int_M \langle F_D, F_D \rangle'_{\mathfrak{g}} dvol_g \tag{24}$$

$$= S^G[D] + S^G_{\text{top}}[D] \tag{25}$$

is referred as the *full YM theory* with gauge group G. Let us consider the identity basic extension $id : G \hookrightarrow G$ with extended domain and correction subspace as in (21). Thus, taking $\delta = id$ we see that the full YM theory is an extension of the YM theory with $C = S_{\text{top}}$.

Example 10 (topological extensions)**.** The functional S^G_{top} of the last example is topological, since by Chern-Weil homomorphism the form $\text{tr}(F_D[\wedge]F_D)$ is closed and therefore is realized in de Rham cohomology. More generally, supposing $n = 2k$, let $\kappa \in \text{inv}(\mathfrak{g})$ be a homogeneous invariant polynomial of degree k in $\mathfrak{g}$ and define $S^G_\kappa[D] = \int_M \kappa(F_D[\wedge] \cdots [\wedge]F_D)$. Then there is an extension of the classical YM theory whose correction functional is $C = S^G_\kappa$.

Example 11 (background field)**.** Let S^G be a gauge invariant YMT theory and let $E \to M$ be some field bundle. Let $C : \mathrm{Conn}(P;\mathfrak{g}) \times \Gamma(E) \to \mathbb{R}$ be any $\mathrm{Gau}_G(P)$-invariant functional, where the gauge groups acts trivially on $\Gamma(E)$. Then, for every fixed $s \in \Gamma(E)$, the sum $S^G + C_s$, where $C_s[D] = C[D,s]$, is an extension of S^G relatively to the identity basic extension $id : G \hookrightarrow G$. Thus, in particular, *every background theory minimally coupled with a gauge invariant YMT theory induces an extension.*

In our definition of extension for a YMT, we constrained the extended functional $\hat{S}^{\hat{G}}$ to have domain satisfying (19). Since the correction subspace is required to be a subset of the extended domain, the upper bound in (19) also applies to $C^1(\hat{P};\hat{\mathfrak{g}})$. We notice that if we avoid this upper bound in both the extended domain and correction subspace, then a lot of new situations can be regarded as new examples. Let us call them "partial examples" of YMT extensions.

Example 12 (non-commutative YM via Seiberg-Witten maps)**.** In [52] it was argued the remarkable existence of a map assigning a noncommutative analogue S_{nc} to every classical gauge theory S. Furthermore, this noncommutative theory is supposed to be expanded in a noncommutative parameter θ, i.e, $S_{\mathrm{nc}} = \sum_{i\geq 0} \theta^i S_i$, such that $S_0 = S$. Let $\mathrm{NConn}(P;\mathfrak{g})$ be the space of those "noncommutative connections" in which S_{nc} is defined and consider the map $\delta : \mathrm{NConn}(P;\mathfrak{g}) \to \mathrm{Conn}(P;\mathfrak{g})$ given by the "commutative limit" $\theta \to 0$. Thus, in the case of a YMT theory one could write $S_{\mathrm{nc}} = S^G \circ \delta + C$, where $C = \sum_{i\geq 1} \theta^i S_i$, leading us to ask if S_{nc} can be considered as an extension of S^G. Although the space $\mathrm{Conn}(P;\mathfrak{g})$ was not rigorously defined, there is a more fundamental problem in regarding S_{nc} as an extension. Notice that a priori one can take each $S^G \circ \delta + \theta^i S_i$ as an extension of S^G. Since as in Subsection 4.1 we will prove that the sum of extensions remains an extension, we conclude that $\hat{S}^G_k = S^G \circ \delta + S_{\mathrm{nc},k}$, where $S_{\mathrm{nc},k} = \sum_{1\leq i\leq k} \theta^i S_i$, is an extension of S^G for every k. The problem may occur in the limiting process $k \to \infty$, since in this case we could analyze the convergence of a series in the space of all extensions of S^G. In Subsection 4.3 we prove that this space has a natural topology when P is compact, so that this convergence could be really considered. This, however, is outside of the scope of this work, so that we only present an speculation:

- **Speculation**. *Suppose P compact. Then $\hat{S}^G_k$ is an extension of S^G for every $k > 0$. Furthermore, the limit $\lim_{k\to\infty} \hat{S}^G_k$ exists in the topology of Subsection 4.3 and is $S^G \circ \delta + S_{\mathrm{nc}}$.*

Example 13 (deformations)**.** Analogous discussions and speculations of the last example apply to other ways to deform the action functional of YMT theory via auxiliary parameters, such as those in [27, 25, 8, 46].

Example 14 (tensorial YM theories)**.** In the sequence of papers [49, 50, 51] it was introduced and studied what the authors called *non-abelian tensor gauge theories*. These are some version of higher gauge theory such that vector potentials of arbitrary degree, such as $A_{i_1,i_2,\ldots}$, are allowed. In particular, its Lagrangian density is suppose to be the sum $\mathcal{L} = \sum_{i\geq 1} g_i \mathcal{L}_i$, where $g_1 = 1$ and $\mathcal{L}_1$ is the Lagrangian of the classical YM theory. Furthermore, $\mathcal{L}_i$, with $i > 1$ depends on a gauge field of higher degree. Let $\mathrm{Tens}(P;\mathfrak{g})$ be the space of all those "higher gauge fields" and consider the map $\delta : \mathrm{Tens}(P;\mathfrak{g}) \to \mathrm{Conn}(P;\mathfrak{g})$ which projects onto the space of gauge-fields of degree one. Then $\int \mathcal{L} = S^G \circ \delta + \sum_{i\geq 2} \int \mathcal{L}_i$ and one could ask, in analogy to the previous examples, whether this realizes $\mathcal{L}$ as an extension of S^G. Here we have an additional problem: $\delta : \mathrm{Tens}(P;\mathfrak{g})$ is too large to be contained in $\Omega^1_{eq}(P;\mathfrak{g})$. Thus, this cannot be an extension domain. But, even if we admit larger extended domains, the results of Subsection 4.1 and of Subsection 13 cannot be applied here.

In trying to avoiding misunderstandings, we close this section giving some non-examples in which the expression "Yang-Mills extensions" is used in a sense which is (as far as the authors know) completely unrelated with the notion introduced here: the well-known notion of *supersymmetric extensions*, the *stringy extensions* of [20], the *Yang-Mills families* of [15] and the extensions studied in [42].

3.4 The Role of Emergence Phenomena

Emergence phenomena arise in the most different sciences: biology, arts, philosophy, physics, and so on. In each of these contexts the term "emergence" has a different meaning [1, 56, 13], but all are about observing characteristics of a system in another system in an unexpected way. In [34] the authors suggested an axiomatization for the notion of emergence phenomena between field theories. Here we will show the surprising fact that in the theories emerging from a YMT theory the S^G itself can be regarded as a new nontrivial example of an extension of S^G.

Following [34] we begin by recalling that a *parameterized field theory* over a smooth manifold M consists of bundles $E \to M$ (the field bundle) and $P \to M$ (the *parameter bundle*), a set of *parameters* $\mathrm{Par}(F) \subset \Gamma(F)$, a set of *field configurations* $\mathrm{Conf}(E)$ and a family of functionals $S_\varepsilon : \mathrm{Conf}(E) \to \mathbb{R}$, with $\varepsilon \in \mathrm{Par}(F)$. The most interesting cases occur when dependence of S_ε on ε is only on certain differential operators. More precisely, we say that a parameterized theory is of *differential type* if there is a map $D_{-} : \mathrm{Par}(F) \to \mathrm{Diff}(E)$ assigning to each parameter ε a differential operator $D_\varepsilon : \Gamma(E) \to \Gamma(E)$ in E and a pairing $\langle \cdot, \cdot \rangle$ in $\Gamma(E)$ such that $S_\varepsilon(s) = \int \langle s, D_\varepsilon s \rangle$. In this case, $\mathrm{Conf}(E) = \Gamma(E)$.

Let S^G be a YMT. We will say that it has perfect pairing if the underlying pairing defining it is a perfect pairing. For instance, every Yang-Mills theory has perfect pairing.

Lemma 3.1. *Under the choice of a connection D_0 in the instanton sector P, every YMT theory S^G with perfect pairing can be naturally regarded as a parameterized field theory of differential type for any given set of parameters* $\mathrm{Par}(F)$.

Proof. Recalling that $\mathrm{Conn}(P; \mathfrak{g})$ is an affine space of $\Omega^1(M; E_{\mathfrak{g}})$, the choice of D_0 allows us to regard the action functional S^G as defined in $\Omega^1(M; E_{\mathfrak{g}}) \simeq \Gamma(TM^* \otimes E_{\mathfrak{g}})$. Furthermore, since the underlying pairing $\langle \cdot, \cdot \rangle$ is perfect, d_D has an adjoint which we denote by d_D^*, so that

$$S^G(D) = \int \langle d_D D, d_D D \rangle = \int \langle D, d_D^*(d_D D) \rangle.$$

For every $\mathrm{Par}(F)$ take the constant function $\mathrm{cst} : \mathrm{Par}(F) \to \mathrm{Diff}(TM^* \otimes E_{\mathfrak{g}})$ given by $\mathrm{cst}(\varepsilon) = d_D^* \circ d_D$. Thus $S_\varepsilon(D) = S^G(D)$ for every ε, which clearly regards S^G as a paremeterized theory of differential type with set of parameters $\mathrm{Par}(F)$. □

We say that a parameterized theory S_1, with field bundle E_1 and set of parameters $\mathrm{Par}(F_1)$, *strongly emerges* from another parameterized theory S_2, with field bundle E_2 and set of parameters $\mathrm{Par}(F_2)$, if there are maps $F : \mathrm{Par}(F_1) \to \mathrm{Par}(F_2)$ and $G : \mathrm{Conf}(E_1) \to \mathrm{Conf}(E_2)$ such that $S_{1,\varepsilon}(s) = S_{F(\varepsilon)}(G(s))$ for every ε and s. The pair (F, G) is called a *strong emergence phenomenon* between S_1 and S_2.

Theorem 3.1. *Let S^G be a YMT theory with perfect pairing, let F_1 be a bundle and let $\mathrm{Par}(F_1) \subset \Gamma(F_1)$ be a subset. Regard S^G as a parameterized theory with set of parameters $\mathrm{Par}(F_1)$, as in Lemma 3.1. Suppose that S^G emerges from another parameterized theory S_2, with*

$$\mathrm{Conn}(\hat{P};\hat{\mathfrak{g}}) \subseteq \mathrm{Conf}(E_2) \subseteq \Omega^1_{eq}(\hat{P};\hat{\mathfrak{g}})$$

being $\mathrm{Gau}_{\hat{G}}(\hat{P})$-equivariant. Then for every emergence phenomena (F,G) between S_2 and S^G, for every $\varepsilon \in \mathrm{Par}(F_1)$ the action functional $S_{2,\varepsilon}$ is a complete extension of S^G.

Proof. From Lemma 3.1 $S^G_\delta = S^G$ for every δ, while from the definition of emergence phenomena we have $S_{2,\varepsilon}(s) = S^G 2, F(\varepsilon)(G(s))$. Thus, for every ε we have $S_{2,\varepsilon} = S^G \circ G$. Therefore, $S_{2,\varepsilon}$ is an extension of S^G with $\widehat{\mathrm{Conn}}(\hat{P};\hat{\mathfrak{g}}) = \mathrm{Conf}(E_2) = C^1(\hat{P};\hat{\mathfrak{g}})$ with $\delta = G$ and $C = 0$. □

4 Space of Extensions

Let S^G be a YMT and, for a fixed basic extension $\imath : G \hookrightarrow \hat{G}$ of G, let $\mathrm{Ext}(S^G;\hat{G})$ denote the space of all extensions of S^G whose underlying basic extension is $\imath$. Here we study the properties of this space. In Subsection 4.1 we prove that for each fixed extended domain $\widehat{\mathrm{Conn}}(\hat{P};\hat{\mathfrak{g}})$ the corresponding subset $\mathrm{Ext}(S^G;\widehat{\mathrm{Conn}}(\hat{P};\hat{\mathfrak{g}}))$ of extensions which has this extended domain is a commutative monoid with pointwise sum. In Subsection 4.2 we prove that, given a group $\mathbb{G}$, then every $\mathbb{G}$-monoid structure on $\mathbb{R}$ induces for every commutative unital ring R a $R[\mathbb{G}]$-module structure on $\mathrm{Ext}(S^G;\widehat{\mathrm{Conn}}(\hat{P};\hat{\mathfrak{g}}))$. Finally, in Subsection 4.3 we show that by means of varying the extended spaces $\widehat{\mathrm{Conn}}(\hat{P};\hat{\mathfrak{g}})$ we get a $R[\mathbb{G}]$-module bundle with no typical fiber which is continuous in a natural topology if $\hat{P}$ is compact, and we conjecture that it can be regarded as a subbundle of a trivial $R[\mathbb{G}]$-module bundle.

4.1 Commutative Monoid Structure

Notice that, since an extension is a triple of maps $(\hat{S}^{\hat{G}}, C, \delta)$ valued in spaces where addition is well-defined[5], we could try to define the sum of two extensions as the pointwise sum of the underlying functions, which are considered in the intersection of the corresponding domains. We clearly have

$$(\hat{S}^{\hat{G}}_1 + \hat{S}^{\hat{G}}_2)|_{C^1_1(\hat{P};\hat{\mathfrak{g}}) \cap C^1_2(\hat{P};\hat{\mathfrak{g}})} = S^G \circ (\delta_1 + \delta_2) + (C_1 + C_2),$$

but the intersection of two extended domains satisfying (19) need not satisfy (19) nor need to be $\mathrm{Gau}_{\hat{G}}(\hat{P})$-invariant. This leads us to restrict our attention to extensions which are *compatible* meaning that

$$\mathrm{Conn}(\hat{P};\hat{\mathfrak{g}}) \subseteq \widehat{\mathrm{Conn}}_1(\hat{P};\hat{\mathfrak{g}}) \cap \widehat{\mathrm{Conn}}_2(\hat{P};\hat{\mathfrak{g}}) \subseteq \Omega^1_{eq}(\hat{P};\hat{\mathfrak{g}})$$

and that its intersection is $\mathrm{Gau}_{\hat{G}}(\hat{P})$-invariant. Furthermore, if $\hat{S}^{\hat{G}}_2$ is the some null extension of Example 2, then $\hat{S}^{\hat{G}}_1 + \hat{S}^{\hat{G}}_2$ is the restriction of $\hat{S}^{\hat{G}}_1$ to $C^1_1(\hat{P};\hat{\mathfrak{g}}) \cap C^1_2(\hat{P};\hat{\mathfrak{g}})$. Notice that if two extensions have the same extended space, then they are automatically compatible. Let us call a $\mathrm{Gau}_{\hat{G}}(\hat{P})$-invariant set satisfying (19) a *extended domain relative to* $\imath : G \hookrightarrow \hat{G}$. Thus:

[5] Here we are working with a fixed connection D_0 in $\hat{P}$, so that $\mathrm{Conn}(\hat{P};\hat{\mathfrak{g}})$ becomes a real vector space.

Lemma 4.1. *For every extended domain $\widehat{\mathrm{Conn}}(\hat{P};\hat{\mathfrak{g}})$ relative to $\imath : G \hookrightarrow \hat{G}$, the collection $\mathrm{Ext}(S^G;\widehat{\mathrm{Conn}}(\hat{P};\hat{\mathfrak{g}}))$ of extensions of S^G which has $\widehat{\mathrm{Conn}}(\hat{P};\hat{\mathfrak{g}})$ as extended space is an abelian monoid with the sum operation above.*

Let $\hat{S}^{\hat{G}}$ be an extension of a YMT S^G and let $c \in \mathbb{R}$ a real number. We could try to define a new extension by taking the pointwise multiplications $c \cdot \hat{S}^{\hat{G}}$, $c \cdot C$ and $c \cdot \delta$ and the same extended domain and correction subspace. But we do not have

$$(c \cdot \hat{S}^{\hat{G}})|_{C^1(\hat{P};\hat{\mathfrak{g}})} = c \cdot (S^G \circ \delta + C), \tag{26}$$

precisely because the action functional of a YMT is not $\mathbb{R}$-linear, but generally quartic. Indeed, assuming without of generality that the pairing is symmetric,

$$S^G[c \cdot D] = c^2 \ll dD, dD \gg +2c^3 \ll dD, \frac{1}{2}D[\wedge]D \gg +c^4 \ll \frac{1}{2}D[\wedge]D, \frac{1}{2}D[\wedge]D \gg,$$

while

$$cS^G[D] = c \ll dD, dD \gg +2c \ll dD, \frac{1}{2}D[\wedge]D \gg +c \lll \frac{1}{2}D[\wedge]D, \frac{1}{2}D[\wedge]D \gg,$$

so that the extensions of a YMT S^G are closed by scalar multiplication iff the action functional S^D is restricted to the subspace $\mathrm{ScConn}(P;\mathfrak{g}) \subset \mathrm{Conn}(P;\mathfrak{g})$ of those connections D which are *scalar invariant*, i.e, such that

$$(c^2 - c) \ll dD, dD \gg +(2c^3 - c) \ll dD, \frac{1}{2}D[\wedge]D \gg +(c^4 - c) \ll \frac{1}{2}D[\wedge]D, \frac{1}{2}D[\wedge]D \gg = 0. \tag{27}$$

for every $c \in \mathbb{R}$. But this set is typically empty. Indeed, equation (27) lead us to look at the roots of the following polynomial over some ring $C(X;\mathbb{R})$ of functions.

$$p(t) = a(x)(t^2 - t) + b(x)(2t^3 - t) + c(x)(t^4 - t). \tag{28}$$

Recall that polynomial rings over fields satisfies the unique polynomial factorization theorem and therefore a degree n univariate polynomial over fields have at most n roots. Therefore, if we look at (28) on the field $C_{>}(X;\mathbb{R}) \subset C(X;\mathbb{R})$ of functions which are null or nowhere vanishing we conclude that p has at most 4 roots. On the other hand, if $b(x) = 0 = c(x)$ or if $a(x) = 0 = b(x)$, then the real roots are $t = 0, 1$. Furthermore, if $a(x) = 0 = c(x)$, then the roots are $t = 0, \pm\sqrt{2}/2$. Let $I(S^G) \subset \mathbb{R}$ be the subset of those c such that (27) is satisfied for every connection in P. The discussion above reveals that under typical hypothesis on $\langle \cdot, \cdot \rangle$, we have $|I(S^G)| \leq 4$ and that -1 is typically not in $I(S^G)$. This implies that the commutative monoid structure of Lemma 4.1 is typically *not* an abelian group. Even so, we can consider its Grothendieck group.

Just to exemplify, let us say that a YMT theory S^G is *semi-nondegenerate* if its pairing $\langle \cdot, \cdot \rangle$ is such that for every connection $D \neq 0$ in P the smooth functions $\langle dD, dD \rangle$ and $\langle D[\wedge]D, D[\wedge]D \rangle$ are nowhere vanishing (and therefore by continuity strictly positive or strictly negative). This is the case, for instance, if S^G is actually a Yang-Mills theory. A connection D in P is *proper* relatively to the pairing $\langle \cdot, \cdot \rangle$ if $\langle dD, D[\wedge]D \rangle$ is null or nowhere vanishing. Let $\mathrm{PConn}(P; \langle \cdot, \cdot, \rangle) \subset \mathrm{Conn}(P;\mathfrak{g})$ be the space of them. Of course, for semi-nondegenerate YMT theories, flat connections are proper. We say that S^G is *proper* if it is semi-nondegenerate and if each connection is proper relatively to the underlying pairing. E.g, if G is abelian, then every YMT theory S^G which is semi-nondegenerate is also proper.

Lemma 4.2. *Let S^G be a proper YMT theory. Then $|I(S^G)| \leq 4$ and $I(S^G) = 0, 1$ if G is abelian.*

Proof. Just notice that under the hypothesis of non-degeneracy, since the integral is order-preserving, it follows that the functions $a(D) = \ll dD, dD \gg$ e $c(D) = \ll D[\wedge]D, D[\wedge]D \gg$ are nowhere vanishing. The same happens with $b(D) = \ll dD, D[\wedge]D \gg$ if D is proper. In this case, the polynomial (28) with $x = D$ has coefficients in a ring of nowhere vanishing functions, so that it has at most four roots, i.e, $|I(S^G)| \leq 4$. If G is abelian, then $D[\wedge]D = 0$, so that (28) becomes $p(t) = a(D)(t^2 - t)$, with roots $t = 0, 1$. $\square$

4.2 $R[\mathbb{G}]$-Module Structure

In the last section we saw that $\operatorname{Ext}(S^G; \widehat{\operatorname{Conn}}_1(\hat{P}; \hat{\mathfrak{g}}))$ is always a commutative monoid with pointwise sum, but almost never closed under the action of $\mathbb{R}$ by pointwise scalar multiplication. It is natural to ask if it can be invariant under other group action. The previous discussion on scalar multiplication give us the hint that the obstruction occurred precisely at the functionals S^G of the YMT theories, which do not commute with the action of $\mathbb{R}$, i.e, we do not have $S^G[c \cdot D] = c \cdot S^G[D]$. Thus, the idea is to consider groups acting in such a way that the property above is satisfied. The "trick" here is to remember that we can take pullbacks of action. More precisely, let $\mathbb{G}$ be a group acting on a set Y and let $f : X \to Y$ a map. We can then define a new map $*^f : \mathbb{G} \times X \to Y$ by $g *^f x = g * f(x)$, i.e, such that the first diagram below is commutative. In our context, given an action $*$ of $\mathbb{G}$ on $\mathbb{R}$ we get the action $*^{S^G}$ which satisfies the desired property.

$$
\begin{tikzcd}
\mathbb{G} \times X \arrow[d, "id \times f"'] \arrow[dr, dashed, "*^f"] & \\
\mathbb{G} \times \mathbb{R} \arrow[r, "*"'] & \mathbb{R}
\end{tikzcd}
\qquad
\begin{tikzcd}
\mathbb{G} \times C^1(\hat{P}; \hat{\mathfrak{g}}) \arrow[d, "id \times \delta"] \arrow[dd, bend right=80, "\hat{S}^{\hat{G}} \circ \jmath"'] \arrow[dd, bend right=60, "C"'] \arrow[dd, bend right=40, "S^G \circ \delta"'] \arrow[ddr, bend left=50, "*^{(\hat{S}^{\hat{G}} \circ \jmath)}"] \arrow[ddr, bend left=35, "*^C"] \arrow[ddr, bend left=20, "*^{(S^G \circ \delta)}"] \arrow[ddr, "(*^{S^G})^\delta"] & \\
\mathbb{G} \times \operatorname{Conn}(P; \mathfrak{g}) \arrow[d, "id \times S^G"] \arrow[dr, "*^{S^G}"] & \\
\mathbb{G} \times \mathbb{R} \arrow[r, "*"'] & \mathbb{R}
\end{tikzcd}
$$

Thus, for each extension $\hat{S}^{\hat{G}}$ of G we can form the second diagram above, where in the curved left-hand side arrows we omitted "$id\times$" in the labels and $\jmath : C^1(\hat{P}; \hat{\mathfrak{g}}) \hookrightarrow \widehat{\operatorname{Conn}}(\hat{P}; \hat{\mathfrak{g}})$ is the inclusion. Another important fact to note is that pullback of actions is functorial. More precisely, if $f : X \to Y$ is a map as above and $\delta : X' \to X$, then $*^{f \circ \delta} = (*^f)^\delta$. Thus, in particular, in the second diagram above we have $*^{(S^G \circ \delta)} = (*^{S^G})^\delta$. Now, notice that the rule $f \mapsto *^f$ preserves the same structures that are preserved by $*$. More precisely, for fixed X, let $\operatorname{Map}(X; \mathbb{R})$ be the set of functions $f : X \to \mathbb{R}$. With pointwise operations it becomes an associative unital real algebra. Let $*^- : \operatorname{Map}(X; \mathbb{R}) \to \operatorname{Map}(\mathbb{G} \times X; \mathbb{R})$ be the rule $*^-(f) = *^f$. As one can quickly check, if $* : \mathbb{G} \times \mathbb{R} \to \mathbb{R}$ is *additive*, i.e, if it preserves the sum of $\mathbb{R}$, then $*^-$ is too, i.e, $*^{f+f'} = *^f + *^{f'}$, and similarly for the multiplication. In our context, this means that $*^{(S^G \circ \delta + C)} = *^{S^G \circ \delta} + *^C$.

Finally, note that for every fixed $a \in \mathbb{G}$, we have an inclusion $\imath_a : a \times X \hookrightarrow \mathbb{R} \times \mathbb{R}$. Composing with the action $* : \mathbb{G} \times \mathbb{R} \to \mathbb{R}$ and with the identification $a \times X \simeq X$, for every map $f : X \to \mathbb{R}$ we have an induced map $a * f : X \to \mathbb{R}$ given by $(a * f)(x) = (*^f)(a, x) = a * f(x)$. Putting all this together we get:

Theorem 4.1. *Let S^G be a YMT theory, $\imath : G \hookrightarrow \hat{G}$ a basic extension of G and $\widehat{\mathrm{Conn}}(\hat{P};\hat{\mathfrak{g}})$ an extended domain. For every group action $* : \mathbb{G} \times \mathbb{R} \to \mathbb{R}$ we have an induced action*

$$\overline{*} : \mathbb{G} \times \mathrm{Ext}(S^G;\widehat{\mathrm{Conn}}(\hat{P};\hat{\mathfrak{g}})) \to \mathrm{Ext}(S^G;\widehat{\mathrm{Conn}}(\hat{P};\hat{\mathfrak{g}})). \tag{29}$$

Furthermore, if $*$ is additive, then $\overline{*}$ is too with respect to the commutative monoid structure of Lemma 4.1.

Proof. After the previous discussion the proof is basically done. Indeed, if $\mathcal{S} = (\hat{S}^{\hat{G}}, C, \delta)$ is an extension of S^G with extended domain $\widehat{\mathrm{Conn}}(\hat{P};\hat{\mathfrak{g}})$, for each $a \in \mathbb{G}$ define $a\overline{*}\mathcal{S} = (a\overline{*}\hat{S}^{\hat{G}}, a\overline{*}C, a\overline{*}\delta)$, where

$$(a\overline{*}\hat{S}^{\hat{G}})(\alpha) = \begin{cases} [a * (\hat{S}^{\hat{G}} \circ \jmath)](\beta) & \text{if } \alpha = \jmath(\beta), \text{ with } \beta \in C^1(\hat{P};\hat{\mathfrak{g}}) \\ \hat{S}^{\hat{G}}(\alpha) & \text{otherwise,} \end{cases}$$

while $a\overline{*}C = a * C$ and $[a\overline{*}\delta](\beta) = [a *^{S^G} \delta](\beta) = a * [S^G(\delta(\beta))]$. One can check that, if defined in this way, $a\overline{*}\mathcal{S}$ is another extension of S^G which by construction has the same extended domain. Furthermore, by the above expressions we see that $e\overline{*}\mathcal{S} = \mathcal{S}$, where e is the neutral element of $\mathbb{G}$, and that $a\overline{*}(b\overline{*}\mathcal{S}) = (a*b)\overline{*}\mathcal{S}$, so that $\overline{*}$ really defines the desired group action (29). Finally, since the commutative monoid structure of Lemma 4.1 is given by pointwise sum, the previous remark that $*$ is additive implies $*^{f+f'} = *^f + *^{f'}$ which shows that $\overline{*}$ is additive if $*$ is, finishing the proof. □

Now, let X be some additive commutative monoid endowed with an additive group action by $\mathbb{G}$, i.e, let X be a $\mathbb{G}$-monoid. Recall that for every $\mathbb{G}$ the category of $\mathbb{G}$-monoids can be identified with the category of $\mathbb{Z}[\mathbb{G}]$-monoids, where $\mathbb{Z}[\mathbb{G}]$ is the group ring of $\mathbb{G}$, which is unital. Furthermore, notice that every R-monoid, where R is a unital ring with unit e, can be turned a R-group, i.e, an R-module. Indeed, for every $x \in X$ define its additive inverse by $-x := (-e) * x$, where $-e$ is the additive inverse of e in R[6]. Thus,

Corollary 3. *For every ring R and every additive group action $* : \mathbb{G} \times \mathbb{R} \to \mathbb{R}$, i.e, for every $\mathbb{G}$-monoid structure in $\mathbb{R}$, we have an induced $R[\mathbb{G}]$-module structure in $\mathrm{Ext}(S^G;\widehat{\mathrm{Conn}}(\hat{P};\hat{\mathfrak{g}}))$.*

Proof. By the above discussion, for every $*$ we have a $\mathbb{Z}[\mathbb{G}]$-module structure. Then recall that $\mathbb{Z}[X] \otimes_{\mathbb{Z}} R \simeq R[X]$. □

Remark 9. If an action $* : \mathbb{G} \times \mathbb{R} \to \mathbb{R}$ is not only additive, but also multiplicative, then it corresponds to a 1-dimensional real irreductive representation of $\mathbb{G}$. Thus, in this setting, or the action is trivial (which is not desired) or $\mathbb{G}$ is not semi-simple.

4.3 Towards a Bundle Structure

Fixed a YMT theory S^G and a basic extension $\imath : G \hookrightarrow \hat{G}$ for G, let $\mathrm{ED}(S^G;\hat{G})$ be the set of all extended domains $\widehat{\mathrm{Conn}}(\hat{P};\hat{\mathfrak{g}})$. We have a projection $\pi : \mathrm{Ext}(S^G;\hat{G}) \to \mathrm{ED}(S^G;\hat{G})$ assigning to each extension of S^G its underlying extended domain. The fibers of this projection are precisely the sets $\mathrm{Ext}(S^G;\widehat{\mathrm{Conn}}(\hat{P};\hat{\mathfrak{g}}))$. But in the last subsection we proved that an additive $\mathbb{G}$-action in $\mathbb{R}$ induces a $R[\mathbb{G}]$-module structure in each of these fibers. Thus, it is very natural to ask whether $\pi : \mathrm{Ext}(S^G;\hat{G}) \to \mathrm{ED}(S^G;\hat{G})$ is actually a $R[\mathbb{G}]$-module bundle.

[6] We clearly have $0 = 0 * x = [e + (-e)] * x = (e * x) + [(-e) * x] = x + (-x) = (-x) + x$.

First of all, notice that by the definition of extended domain, one can regard $\mathrm{ED}(S^G;\hat{G})$ as the subset of the power set $\mathcal{P}(\Omega^1(\hat{P};\hat{\mathfrak{g}}))$ consisting of those subsets $X \subset \Omega^1(\hat{P};\hat{\mathfrak{g}})$ satisfying (19). Since $\Omega^1(\hat{P};\hat{\mathfrak{g}}) \simeq \Gamma(T^*\hat{P} \otimes (\hat{P} \times \hat{\mathfrak{g}}))$, if $\hat{P}$ is compact, then $\Omega^1(\hat{P};\hat{\mathfrak{g}})$ becomes a Fréchet space and therefore a complete metric space [31]. In particular, it can be regarded as a complete uniform space, which means that its power set $\mathcal{P}(\Omega^1(\hat{P};\hat{\mathfrak{g}}))$ has an induced uniform structure and, in particular, a natural (and typically non-Hausdorff) topology [26]. One can then take in $\mathrm{ED}(S^G;\hat{G})$ the subspace topology. Let $\Pi : \mathrm{Ext}(S^G;\hat{G}) \to \mathcal{P}(\Omega^1(\hat{P};\hat{\mathfrak{g}}))$ be the composition of π with the inclusion $\mathrm{ED}(S^G;\hat{G}) \hookrightarrow \mathcal{P}(\Omega^1(\hat{P};\hat{\mathfrak{g}}))$ and give $\mathrm{Ext}(S^G;\hat{G})$ the initial topology defined by Π. Thus, the closed sets of $\mathrm{Ext}(S^G;\hat{G})$ are preimages of closed sets on $\mathcal{P}(\Omega^1(\hat{P};\hat{\mathfrak{g}}))$. Furthermore, since π is surjective, it follows that it is continuous in the initial topology of Π. This topology is better than the initial topology of π because it is completely determined on the subspace of closed sets of $\mathcal{P}(\Omega^1(\hat{P};\hat{\mathfrak{g}}))$, which constitute a more well-behaved environment. E.g, the induced uniform structure arises from a metric, which is complete since $\Omega^1(\hat{P};\hat{\mathfrak{g}})$ a Fréchet space [31, 26].

Thus, fixed a $\mathbb{G}$-monoid structure in $\mathbb{R}$ we get, for every commutative unital ring R, a continuous projection $\pi_R : \mathrm{Ext}(S^G;\hat{G}) \to \mathrm{ED}(S^G;\hat{G})$ whose fibers are $R[\mathbb{G}]$-modules. We should not expect that this projection is locally trivial, since for different extended domains we have fibers with different sizes, so that a priori there is no typical fiber. On the other hand, we speculate that it can be regarded as a subbundle of a trivial bundle. More precisely, notice that $\mathrm{ED}(S^G;\hat{G})$ is a directed set with the inclusion relation. Furthermore, given two extended domains such that $\imath_{1,2} : \widehat{\mathrm{Conn}}_1(\hat{P};\hat{\mathfrak{g}}) \hookrightarrow \widehat{\mathrm{Conn}}_2(\hat{P};\hat{\mathfrak{g}})$ we have a corresponding map

$$\imath_{1,2}^* : \mathrm{Ext}(S^G;\widehat{\mathrm{Conn}}_2(\hat{P};\hat{\mathfrak{g}})) \to \mathrm{Ext}(S^G;\widehat{\mathrm{Conn}}_1(\hat{P};\hat{\mathfrak{g}})) \tag{30}$$

given by restriction, i.e, such that for every $\mathcal{S}_2 = (\hat{S}_2^{\hat{G}}, C_2, \delta_2)$ we have $\imath_{1,2}^*\mathcal{S}_2 = (\hat{S}_2^{\hat{G}} \circ \imath_{1,2}, C_2 \circ \imath_{1,2}, \delta_2 \circ \imath_{1,2})$, where the compositions are in the proper domains. One can check that this map actually preserves the $R[\mathbb{G}]$-module structures of last section. Thus, we have an inverse limit in the category of $R[\mathbb{G}]$-modules, whose limit is $\mathrm{Ext}(S^G;\Omega^1_{eq}(\hat{P};\hat{\mathfrak{g}}))$, since the direct set $\mathrm{ED}(S^G;\hat{G})$ has $\Omega^1_{eq}(\hat{P};\hat{\mathfrak{g}})$ as its greatest element.

Now, consider the projection

$$\mathrm{pr}_1 : \mathrm{ED}(S^G;\hat{G}) \times \mathrm{Ext}(S^G;\Omega^1_{eq}(\hat{P};\hat{\mathfrak{g}})) \to \mathrm{ED}(S^G;\hat{G}). \tag{31}$$

Let PR_1 be the composition of pr_1 with the inclusion $\mathrm{ED}(S^G;\hat{G}) \hookrightarrow \mathcal{P}(\Omega^1(\hat{P};\hat{\mathfrak{g}}))$ and consider its initial topology. Thus, (31) becomes continuous and defines a continuous trivial $R[\mathbb{G}]$-module bundle. We have a map $\imath^*$ such that $\imath^*(\widehat{\mathrm{Conn}}(\hat{P};\hat{\mathfrak{g}}), \mathcal{S}) = (\widehat{\mathrm{Conn}}(\hat{P};\hat{\mathfrak{g}}), \imath^*\mathcal{S})$, i.e, which is fiberwise of the form (30), clearly making commutative the diagram below.

$$\begin{tikzcd}
\mathrm{ED}(S^G;\hat{G}) \times \mathrm{Ext}(S^G;\Omega^1_{eq}(\hat{P};\hat{\mathfrak{g}})) \arrow[r, "\imath^*"] \arrow[dr, "\mathrm{pr}_1"'] & \mathrm{Ext}(S^G;\hat{G}) \arrow[d, "\pi_R"] \\
& \mathrm{ED}(S^G;\hat{G})
\end{tikzcd} \tag{32}$$

Notice that, if $\imath^*$ is continuous, it then defines a morphism of $R[\mathbb{G}]$-module bundles. Furthermore, suppose that we found R and $\mathbb{G}$ such that $\mathrm{Ext}(S^G;\Omega^1_{eq}(\hat{P};\hat{\mathfrak{g}}))$ is an injective module. Thus, $\imath^*$ becomes fiberwise an epimorphism of $R[\mathbb{G}]$-modules. If, in addition each $\mathrm{Ext}(S^G;\widehat{\mathrm{Conn}}(\hat{P};\hat{\mathfrak{g}}))$ is a projective $R[\mathbb{G}]$-module (except maybe for $\widehat{\mathrm{Conn}}(\hat{P};\hat{\mathfrak{g}}) = \Omega^1_{eq}(\hat{P};\hat{\mathfrak{g}})$), then $\imath^*$ is fiberwise an split epimorphism and we can define a section s for $\imath^*$ in such a way that diagram (32) extends to the following commutative diagram. Furthermore, if the fiberwise splits of $\imath^*$ are such that s is continuous, then s embeds the bundle π_R of extensions of S^G as a $R[\mathbb{G}]$-subbundle of the trivial $R[\mathbb{G}]$-bundle (31).

$$\begin{tikzcd} \mathrm{ED}(S^G;\hat{G}) \times \mathrm{Ext}(S^G;\Omega^1_{eq}(\hat{P};\hat{\mathfrak{g}})) \arrow[r, "\imath^*"] \arrow[dr, "\mathrm{pr}_1"'] & \mathrm{Ext}(S^G;\hat{G}) \arrow[r, "s"] \arrow[d, "\pi_R"] & \mathrm{ED}(S^G;\hat{G}) \times \mathrm{Ext}(S^G;\Omega^1_{eq}(\hat{P};\hat{\mathfrak{g}})) \arrow[dl, "\mathrm{pr}_1"] \\ & \mathrm{ED}(S^G;\hat{G}) & \end{tikzcd} \tag{33}$$

Thus, after all the previous discussion we are lead to speculate the following:

Conjecture 4.1. *For every YMT theory S^G, every basic extension $\imath : G \hookrightarrow \hat{G}$ of G such that $\hat{P}$ is compact, every ring R and every group $\mathbb{G}$, the corresponding map $\imath^*$ defined above is continuous in the described topologies.*

Conjecture 4.2. *Given a YMT theory S^G and a basic extension $\imath : G \hookrightarrow \hat{G}$ of G such that $\hat{P}$ is compact, there exists a ring R and a group $\mathbb{G}$ such that $\mathrm{Ext}(S^G;\Omega^1_{eq}(\hat{P};\hat{\mathfrak{g}}))$ is injective as $R[\mathbb{G}]$-module and such that $\mathrm{Ext}(S^G;\widehat{\mathrm{Conn}}(\hat{P};\hat{\mathfrak{g}}))$ (except maybe for $\widehat{\mathrm{Conn}}(\hat{P};\hat{\mathfrak{g}}) = \Omega^1_{eq}(\hat{P};\hat{\mathfrak{g}})$) is a projective as $R[\mathbb{G}]$-module. Furthermore, the induced splits can be choosen such that the corresponding map s of diagram (33) is continuous in the described topologies.*

5 Category of Extensions

In previous sections we studied the space of extensions. Here we will study how the elements of this space interact with each other. More precisely, we will build the category of extensions of a given YMT theory.

Let S^G be a YMT theory and let $\imath : G \hookrightarrow \hat{G}$ be a basic extension of G, which from now on will be fixed. Let $\hat{S}^{\hat{G}}_i \in \mathrm{Ext}(S^G;\hat{G})$, with $i = 1, 2$ be extensions of S^G. A *morphism* between them, denoted by $F : \hat{S}^{\hat{G}}_1 \to \hat{S}^{\hat{G}}_2$ is given by a pair $F = (f, g)$, where $f : \widehat{\mathrm{Conn}}_1(\hat{P};\hat{\mathfrak{g}}) \to \widehat{\mathrm{Conn}}_2(\hat{P};\hat{\mathfrak{g}})$ is a $\mathrm{Gau}_{\hat{G}}(\hat{P})$-equivariant map and $g : C^1_1(\hat{P};\hat{\mathfrak{g}}) \to C^1_2(\hat{P};\hat{\mathfrak{g}})$ is a function making commutative the first diagram below. Compositions and identities are defined componentwise, i.e, $(f, g) \circ (f', g') = (f' \circ f, g \circ g)$. It is straightforward to check that this data really defines a category $\mathbf{Ext}(S^G;\hat{G})$.

$$\begin{tikzcd} & & C^1_1(\hat{P};\hat{\mathfrak{g}}) \arrow[rr, hook] \arrow[dll, bend right, "\delta_1"'] \arrow[dl, "C_1"'] \arrow[dd, "g"] & & \widehat{\mathrm{Conn}}_1(\hat{P};\hat{\mathfrak{g}}) \arrow[dl, "\hat{S}^{\hat{G}}_1"'] \arrow[dd, "f"] \\ \mathrm{Conn}(P;\mathfrak{g}) & \mathbb{R} & & \mathbb{R} & \\ & & C^1_2(\hat{P};\hat{\mathfrak{g}}) \arrow[rr, hook] \arrow[ull, bend left, "\delta_2"] \arrow[ul, "C_2"] & & \widehat{\mathrm{Conn}}_2(\hat{P};\hat{\mathfrak{g}}) \arrow[ul, "\hat{S}^{\hat{G}}_2"] \end{tikzcd} \tag{34}$$

Given a group G, let $\mathbf{Set}_G$ denote the category of G-sets, i.e, whose objects are sets X with an action $G \times X \to X$ and whose morphisms are G-equivariant maps. Furthermore, if $\mathbf{C}$ is a category and $A \in \mathbf{C}$ is an object, let $\mathbf{C}/A$ denote the corresponding slice category, whose objets are morphisms $f : X \to A$ to A and whose morphisms $h : f \Rightarrow g$ are commutative triangles, as below, i.e, are morphisms $h : X \to Y$ in $\mathbf{C}$ such that $f = h \circ g$. We also recall that a subcategory $\mathbf{C}_0 \subset \mathbf{C}$ is *conservative* if the inclusion functor $\imath : \mathbf{C}_0 \to \mathbf{C}$ is conservative, i.e, if reflects isomorphisms, meaning that if a map $f : X \to Y$ is such that $\imath(f)$ is an isomorphism in $\mathbf{C}$, then f is an isomorphism in $\mathbf{C}_0$.

$$\begin{tikzcd}
 & X \arrow[dl, "f"'] \arrow[dd, "h"] \\
A & \\
 & Y \arrow[ul, "g"]
\end{tikzcd}$$

Theorem 5.1. *For every given YMT S^G and basic extension $\imath : G \hookrightarrow \hat{G}$, the corresponding category of extensions $\mathbf{Ext}(S^G; \hat{G})$ is a conservative subcategory of*

$$(\mathbf{Set}_{\mathrm{Gau}_{\hat{G}}(\hat{P})}/\mathbb{R}) \times (\mathbf{Set}/\mathbb{R}) \times (\mathbf{Set}/\operatorname{Conn}(P; \mathfrak{g})). \tag{35}$$

Proof. First of all, notice that an object of $\mathbf{Ext}(S^G; \hat{G})$ can be regarded as a triple $(\hat{S}^{\hat{G}}, C, \delta)$ in (35) such that $s(C) = s(\delta)$, $s(\delta) \subset s(\hat{S}^{\hat{G}})$ and $\hat{S}^{\hat{G}}|_{s(C)} = S^G \circ \delta + C$, where $s(f)$ denoted the domain of a map and

$$\operatorname{Conn}(\hat{P}; \hat{\mathfrak{g}}) \subset s(\hat{S}^{\hat{G}}) \subset \Omega^1_{eq}(\hat{P}; \hat{\mathfrak{g}}).$$

Furthermore, looking at diagram (34) a morphism (f, g) in $\mathbf{Ext}(S^G; \hat{G})$ can be regarded as a morphism (f, g, h) in (35) such that $h = g$ and $f|_{s(C)} = g$. Since compositions and identities in $\mathbf{Ext}(S^G; \hat{G})$ are componentwise, it follows that $\mathbf{Ext}(S^G; \hat{G})$ is a subcategory of (35). It is straighforward to check that the inclusion is faithful. Therefore, it reflects monomorphisms and epimorphisms. But epimorphisms (resp. monomorphisms) on the category of sets and G-sets are surjective (resp. monomorphism) maps [59]. Since in slice category limits and colimits are computed objectwise, we see that (f, g, h) is an epimorphism (resp. monomorphism) iff f, g and h are surjections (resp. injections). Thus, a morphism (f, g) in $\mathbf{Ext}(S^G; \hat{G})$ is an epimorphism (resp. monomorphism) iff f and g are surjections (resp. injections). In particular, (f, g) is a bimorphism iff both f and g are bijections. But it is straighforward to check that, in this case, (f, g) is an isomorphism. Thus, $\mathbf{Ext}(S^G; \hat{G})$ is balanced. The result then follows from the fact that very faithful functor defined on a balanced category is conservative [45]. $\square$

Corollary 4. *Suppose that a diagram J in $\mathbf{Ext}(S^G; \hat{G})$ admits limit (resp. colimit) and is such that for every limiting cone (resp. cocone) L the image $\imath(L)$ is a limiting cone (resp. cocone) for $\imath \circ J$, where $\imath$ is the inclusion functor to (35). In this case, the reciprocal holds, i.e, if L is such that $\imath(L)$ is a limiting cone (resp. cocone), then L must be a limiting cone (resp. cocone).*

Proof. It follows from the general fact that conservative functor $F : \mathbf{C} \to \mathbf{D}$ reflects every limit and colimit that exists in $\mathbf{C}$ and which are preserved by F. $\square$

There are, however, some limits and colimits which exists in $\mathbf{Ext}(S^G; \hat{G})$ but which are not preserved (and therefore not reflected) by the inclusion functor.

Example 15. As it is straightforward to check, the null extension of Example 2 is a terminal object in the category of extensions $\mathbf{Ext}(S^G; \hat{G})$. We assert that it is not preserved by $\imath$ if S^G is not null. Indeed, a terminal object in a slice category $\mathbf{C}/A$ is the identity id_A, so that a terminal object in (35) is the triple $id = (id_{\mathbb{R}}, id_{\mathbb{R}}, id_{\mathrm{Conn}(P;\mathfrak{g})})$. This means that if the inclusion functor preserves terminal objects, then id must belong to $\mathbf{Ext}(S^G; \hat{G})$, i.e, there exists an extension of S^G such that $\hat{S}^{\hat{G}} = id_{\mathbb{R}}$, $C = id_{\mathbb{R}}$ and $\delta = id_{\mathrm{Conn}(P;\mathfrak{g})}$. But the condition (20) would implies $S^G = 0$, which is a contradiction.

Remark 10. In the case of equivariant extensions (see Remark 5), Theorem 5.1 can be improved by replacing $\mathbf{Set}/\mathbb{R}$ with $\mathbf{Set}_{\mathrm{Gau}_{\hat{G}}(\hat{P})}/\mathbb{R}$ and $\mathbf{Set}/\mathrm{Conn}(P;\mathfrak{g})$ with $\mathbf{Set}_{\mathrm{Gau}_{\hat{G}}}/\mathrm{Conn}(P;\mathfrak{g})$, where the $\mathrm{Gau}_{\hat{G}}(\hat{P})$-action in $\mathrm{Conn}(P;\mathfrak{g})$ is that induced by the homorphism $\xi : \mathrm{Gau}_G(P) \to \mathrm{Gau}_{\hat{G}}(\hat{P})$ described in Section 3.

Remark 11 (speculation)**.** In Subsection 4.3 we proved that $\mathrm{Ext}(S^G; \hat{G})$ is an $R[\mathbb{G}$-module bundle for every ring R and every additive group action $\mathbb{G}$ in $\mathbb{R}$ and we conjectured that it can be regarded as continuous subundle of a trivial bundle. In the present section, on the other hand, we showed that $\mathrm{Ext}(S^G; \hat{G})$ is the collection of objects of a category $\mathbf{Ext}(S^G; \hat{G})$. Thus, it is natural to ask whether the collection of morphisms $\mathrm{Mor}(S^G; \hat{G})$ of $\mathbf{Ext}(S^G; \hat{G})$ can also be endowed with a $R[\mathbb{G}]$-module bundle structure such that the source, target and composition maps are bundle morphisms. In other words, it is natural to ask if $\mathbf{Ext}(S^G; \hat{G})$ cannot be internalized in the category $\mathbf{Bun}_{R[\mathbb{G}]}$ of $R[\mathbb{G}]$-module bundles. We do not have a complete answer and a careful analysis could motivate future research.

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5.5 THIRD WORK

The following is a recompilation of the ArXiv preprint

- Y.X. Martins, L.F. A. Campos, R.J. Biezuner, *On Maximal, Universal and Complete Extensions of Yang-Mills-Type Theories*, Jul 2020, arXiv:2007.08651

On Maximal, Universal and Complete Extensions of Yang-Mills-Type Theories

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Abstract

In this paper we continue the program on the classification of extensions of the Standard Model of Particle Physics started in [24]. We propose four complementary questions to be considered when trying to classify any class of extensions of a fixed Yang-Mills-type theory S^G: existence problem, obstruction problem, maximality problem and universality problem. We prove that all these problems admits a purely categorical characterization internal to the category of extensions of S^G. Using this we show that maximality and universality are dense properties, meaning that if they are not satisfied in a class $\mathcal{E}(S^G;\hat{G})$, then they are in their "one-point compactification" $\mathcal{E}(S^G;\hat{G}) \cup \hat{S}$ by a specific trivial extension $\hat{S}$. We prove that, by means of assuming the Axiom of Choice, one can get another maximality theorem, now independent of the trivial extension $\hat{S}$. We consider the class of almost coherent extensions, i.e, complete, injective and of pullback-type, and we show that for it the existence and obstruction problems have a complete solution. Using again the Axiom of Choice, we prove that this class of extensions satisfies the hypothesis of the second maximality theorem.

Keywords: Yang-Mills theories, classification, universality, complete extensions.

1 Introduction

While Standard Model of Elementary Particle Physics (SM) is probably the most accurate and tested physical theory [35], there are indications that it must be viewed as an effective theory of a more fundamental extended theory [10, 2]. In the last decades a cascade of models were proposed [5, 28, 27, 18, 29, 8, 9, 20, 11, 26, 7]. In [24] the authors started a program aiming to classify all the possible extensions of SM, starting with the Yang-Mills part. We showed that most of the existing proposals of extensions for YM theories, such as [17, 15, 6, 30, 31, 32, 33, 12, 14, 34], are particular examples of a simple and unifying notion of extension. We also proved that the Higgs mechanism and the YM theories in an spinorial background can also be regarded as examples of this unifying notion of extension, defined as follows.

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Let $S^G : \mathrm{Conn}(P;\mathfrak{g}) \to \mathbb{R}$ the action functional of a YM theory with gauge group G and instanton sector P. An *extension* of S^G consists of an extension $\hat{G}$ of G, which itself induces an extension $\hat{P}$ of $\hat{G}$, and another action functional $\hat{S}^{\hat{G}} : \widehat{\mathrm{Conn}}(\hat{P};\hat{\mathfrak{g}}) \to \mathbb{R}$, playing the role of the *extended action funcional*, such that, at least when restricted to a subspace $C^1(\hat{P};\hat{\mathfrak{g}})$, it can be written as

$$\hat{S}^{\hat{G}}[\alpha] = S^G[D(\alpha)] + C[\alpha], \tag{1}$$

i.e, as a sum of the original YM theory (evaluated in connections depending on α) and a correction term. In other words, we have a map $\delta : C^1(\hat{P};\hat{\mathfrak{g}}) \to \mathrm{Conn}(P;\mathfrak{g})$, which is part of the data defining an extension, such that $\hat{S}^{\hat{G}} \circ \jmath = S^G \circ \delta + C$, where $\jmath$ is the inclusion of the correction subspace in the extended domain. Actually, in [24] the authors considered the extensions above not for YM theories, but actually for what they called *Yang-Mills-type* (YMT) theories, which differ from the YM theories in allowing an arbitrary pairing of the curvature in the action funcional. This has the advantage of avoiding the usual requirements of compactness and semi-simplicity on G and the existence of a semi-Riemannian metric on M[1]. For a fixed extension $\hat{G}$ of the gauge group G we considered the space $\mathrm{Ext}(S^G;\hat{G})$ of extensions of a given YMT theory S^G in the above sense. It was shown that for every group $\mathbb{G}$ acting on $\mathbb{R}$ and every commutative unital ring R, the space $\mathrm{Ext}(S^G;\hat{G})$ has an induced structure of continuous $R[\mathbb{G}]$-module bundle. Therefore, in view of the classification problem described above, the main strategy is to try a direct topological classification of this bundle. For instance, in [24] the authors conjectured that for every S^G and every $\hat{G}$ the bundle of extensions can be regarded as a subbbundle of a trivial bundle. We notice, however, that another approach is to look at particular subclasses $\mathcal{E}(S^G;\hat{G}) \subset \mathrm{Ext}(S^G;\hat{G})$ and try to get some information about it. This is the approach adopted in this paper. More precisely, given a class $\mathcal{E}(S^G;\hat{G})$ of extensions, we propose to consider four questions for it:

1. *Existence Problem*: Does it contain nontrivial elements?
2. *Obstruction Problem*: For an arbitrarily given action functional $\hat{S}$, what are the obstructions to regard it as an element of $\mathcal{E}(S^G;\hat{G})$?
3. *Maximality Problem*: Is it internally maximal to some other class of extensions?
4. *Universality Problem*: Is it internally universal to some other class of extensions?

The first two problems are closely related to the classification problem restricted to the class $\mathcal{E}(S^G;\hat{G})$. Indeed, recalling that some trivial extensions exist for any S^G and $\hat{G}$ [24], the existence problem is about identifying if $\mathcal{E}(S^G;\hat{G})$ contains *at least one* nontrivial extension. Assuming that the existence problem has a positive solution, the obstruction problem is about the study of the space of *all* these nontrivial extensions.

The other two problems (the maximality problem and the universality problem) are more refined. They are about searching for extensions in the class $\mathcal{E}(S^G;\hat{G})$ which are not only nontrivial, but actually special in some sense. Indeed, we say that $\mathcal{E}(S^G;\hat{G})$ is *internally maximal* relative to other class $\mathcal{E}'(S^G;\hat{G})$ if there is a map $\mu : \mathcal{E}(S^G;\hat{G}) \to \mathcal{E}'(S^G;\hat{G})$, corresponding to a way to internalize one class into the other, meaning that for each other function $\nu : \mathcal{E}(S^G;\hat{G}) \to \mathcal{E}'(S^G;\hat{G})$, every extension $\hat{S}^{\hat{G}} \in \mathcal{E}(S^G;\hat{G})$ is such that $\mu(\hat{S}^{\hat{G}})$ can be regarded as an extension of $\nu(\hat{S}^{\hat{G}})$. Thus, this

[1] Which can suffer of topological obstructions [?] that become very strong if coupled with other geometric requirements [22, 21].

means that $\mu(\hat{S}^{\hat{G}})$ is the maximal way to regard $\hat{S}^{\hat{G}}$ as an element of $\mathcal{E}'(S^G;\hat{G})$. Notice that $\mu(\hat{S}^{\hat{G}})$ is required to be just an extension of $\nu(\hat{S}^{\hat{G}})$, which a priori can be realized in many different ways. If for every $\hat{S}^{\hat{G}}$ there is a unique way to view $\mu(\hat{S}^{\hat{G}})$ as an extension of $\nu(\hat{S}^{\hat{G}})$ we say that $\mathcal{E}(S^G;\hat{G})$ is not only internally maximal, but *internally universal*, relative to $\mathcal{E}'(S^G;\hat{G})$.

Notice that by the definition of maximality and universality, it is suggestive to believe that the fact of a class $\mathcal{E}(S^G;\hat{G})$ being maximal or universal depends crucially on the ambient class $\mathcal{E}'(S^G;\hat{G})$. We will prove, however, that maximality and universality are *dense* properties. If X is a set and $\mathcal{P}(X)$ is its power set, we say that a property ρ is *dense in* X if there exists an element $x\in X$ such that ρ holds in every $X_0\cup x$ for every $X_0\in\mathcal{P}(X)$. Equivalently, if there exists $x\in X$ such that if $x\in X_0$, then ρ holds in $X_0\in\mathcal{P}(X)$[2]. Thus, our first main result is the following:

Theorem A. *Given a YMT theory S^G and an extension $G\hookrightarrow\hat{G}$ of G, there exists an extension $\hat{S}^{\hat{G}}_0$ of S^G such that if it belongs to a class $\mathcal{E}'(S^G;\hat{G})$, then every other class $\mathcal{E}(S^G;\hat{G})$ is internally maximal and universal relative to $\mathcal{E}'(S^G;\hat{G})$.*

The theorem above depends crucially on the fact that the extensions of a YMT theory S^G constitute a category $\mathbf{Ext}(S^G;\hat{G})$, as described in [24]. In fact, $\hat{S}^{\hat{G}}_0$ is precisely the terminal object of this category, which is isomorphic to a certain null-type extension and, therefore, to a trivial extension. Although this does not contradict the existence problem, it would be interesting to have maximality and universality criteria independent of trivial extensions. We prove that if we assume the Axiom of Choice in the form of Zorn's Lemma, then such a criterion exists and is obtained by replacing the presence of null-type extensions with certain coherence conditions on the ambient class $\mathcal{E}'(S^G;\hat{G})$, which produces constraints on the size of the classes which are internally maximal/universal to it. More precisely, we prove the following result:

Theorem B. *Given a YMT theory S^G and an extension $G\hookrightarrow\hat{G}$ of G, if a class of extensions $\mathcal{E}'(S^G;\hat{G})$ is I-coherent for maximality (resp. for universality), then every other class $\mathcal{E}(S^G;\hat{G})$ satisfying $|\mathcal{E}(S^G;\hat{G})|\leq|I|$ is internally maximal (resp. universal) to $\mathcal{E}'(S^G;\hat{G})$.*

Notice that the collection $\mathrm{Ext}(S^G;\hat{G})$ of all extensions of a YMT theory S^G can be partitioned into two disjoint classes: those extensions whose correction term is null, i.e, such that $C\equiv 0$, and those extensions such that $C\neq 0$. Following the terminology of [24], we say that the first class is of the *complete extensions*, while the second one is of the *incomplete extensions*. In this article we give a special focus on classes of complete extensions, leaving the class of incomplete extensions to a parallel work [25]. We considered the class $\mathrm{Coh}(S^G;\hat{G})$ of *almost coherent extensions*, consisting of those extensions of S^G which are complete, whose extended action functional becomes injective on the quotient space of $\widehat{\mathrm{Conn}}(\hat{P};\hat{\mathfrak{g}})$ and which can be built from a categorical pullback. We give a complete description of the existence problem and of the obstruction problem for this class and we prove that if I is small enough, then $\mathrm{Coh}(S^G;\hat{G})$ is I-coherent for maximality. In fact, we prove:

Theorem C. *Let S^G be a YMT theory and let $\imath: G\hookrightarrow\hat{G}$ be an extension of G. Every extension of S^G induces an almost coherent extension of S^G which is nontrivial if the former is. Let $I\subset$ $\mathrm{ED}(S^G;\hat{G})$ be any subset of extended domains which are mutually disjoint. Then $\mathrm{Coh}(S^G;\hat{G})$ is I-coherent for maximality, so that every class $\mathcal{E}(S^G;\hat{G})$ with cardinality at least $|I|$ is internally maximal to $\mathrm{Coh}(S^G;\hat{G})$.*

The paper is organized as follows. In Section 2 we review the basic definitions and the basic

[2] The relation between this notion of density and the usual notion from topology will be discussed on Section 4.

3

facts of [24] concerning YMT theories, the space of extensions and the category of extensions. In Section 3 the four extension problems described above are presented in more detail and we discuss how to rewrite them in purely categorical language. This is a crucial step in proving Theorems A and B, which is the main objective of Section 4. Finally, in Section 5 we introduce the class of almost coherent extensions and we prove Theorem C by means of several steps.

2 Background

As defined in [24] and following the differential geometry conventions of [19, 22, 21], a Yang-Mills-type theory over a n-dimensional smooth manifold M is given by:

1. a real or complex Lie group G, regarded as the gauge group of internal symmetries;
2. a principal G-bundle $\pi : P \to M$, called *instanton sector* of the theory, and whose automorphism group, called the *group of global gauge transformations* is denoted $\mathrm{Gau}_G(P)$;
3. a $\mathbb{R}$-linear pairing $\langle \cdot, \cdot \rangle : \Omega^2_{he}(P, \mathfrak{g})_{\mathbb{R}} \otimes_{\mathbb{R}} \Omega^2_{he}(P, \mathfrak{g})_{\mathbb{R}} \to \mathrm{Dens}(M)_{\mathbb{R}}$.

Above, $\Omega^k_{he}(P, \mathfrak{g})$ denotes the vector space of horizontal and G-equivariant $\mathfrak{g}$-valued k-forms in P regarded as a $C^\infty(M)$-module via the restriction of scalars by $\pi^* : C^\infty(M) \to C^\infty(P)$, $\mathfrak{g}$ is the Lie algebra of G and $\mathrm{Den}(M)$ is the space of 1-densities on M, also regarded as a $C^\infty(M)$-module. Furthermore, $V_{\mathbb{R}}$ will denote scalar restriction by the inclusion $\mathbb{R} \hookrightarrow C^\infty(M)$. The *action functional* of a YMT over M is the functional

$$S^G : \mathrm{Conn}(P; \mathfrak{g}) \to \mathbb{R} \quad \text{given by} \quad S^G(D) = \int_M \langle F_D, F_D \rangle,$$

where $\mathrm{Conn}(P; \mathfrak{g}) \subset \Omega^1(P; \mathfrak{g})$ is the set of connection 1-forms on P and $F_D = d_D D$ is the field-strength of D, where $d_D : \Omega^k(P; \mathfrak{g}) \to \Omega^{k+1}_{he}(P; \mathfrak{g})$ is the *exterior covariant derivative* of D, such that $d_D\alpha = d\alpha + D[\wedge]_{\mathfrak{g}}\alpha$. Following [22, 21], $[\wedge]_{\mathfrak{g}}$ denotes the wedge product induced by the Lie bracket of $\mathfrak{g}$. Without risk of confusion, it will be denoted simply by $[\wedge]$.

Example 1. Every Yang-Mills theory over a semi-Riemannian manifold (M, g) is a YMT theory with the pairing $\langle \alpha, \beta \rangle = \mathrm{tr}(\alpha [\wedge] \star_g \beta)$. As shown in [24], for every triple (M, G, P) the corresponding collection $\mathrm{YM}_G(P)$ of YMT theories over M which has G as the gauge group and P as the instanton sector is a real vector space. If G is discrete or $\dim M < 2$, then this space is zero-dimensional, implying that under such conditions this YMT theory is a YM theory. Otherwise it is infinite-dimensional, meaning that when working with YMT theories instead of only with YM theories we will have an infinity amount of new degrees of freedom to work with.

Remark 1. In some cases, such as for those discussed in [24], it is better to work with a more restricted class of YMT theories. Indeed, we say that S^G is *tensorial* (resp. *gauge invariant*) if its pairing $\langle \cdot, \cdot \rangle$ is $C^\infty(M)$-linear (resp. invariant by the action of $\mathrm{Gau}_G(P)$, i.e, $\langle f^*(F_D), f^*(F_D) \rangle = \langle F_D, F_D \rangle$ for every $f \in \mathrm{Gau}_G(P)$). In the following, however, we will work in full generality. Notice that by avoiding the gauge invariance hypothesis on $\langle \cdot, \cdot \rangle$ we are allowing that S^G be gauge breaking in the classical sense. Such hypotheses of linearity and gauge invariance are important for doing local computations, as in [25].

2.1 Extensions

Let G be a Lie group. A *basic extension* of G is another Lie group $\hat{G}$ and an injective Lie group homomorphism $i : G \to \hat{G}$.

Example 2. Every split Lie group extension induces a basic extension. Actually, a Lie group extension is a basic extension iff it is split [24]. On the other hand, there are many basic extensions which are not Lie group extensions.

Let $P \to M$ be a principal G-bundle. Each basic extension $\imath : G \hookrightarrow \hat{G}$ induces a $\hat{G}$-bundle $\hat{P}$, which becomes equipped with a monomorphism $\imath : P \hookrightarrow \hat{P}$ and a group homomorphism $\xi : \mathrm{Gau}_G(P) \to \mathrm{Gau}_{\hat{G}}(\hat{P})$, as described in [13, 3, 24]. Let S^G be a YMT theory with gauge group G and instanton sector P. As defined in [24], an *extension* of S^G relative to a basic extension $\imath : G \hookrightarrow \hat{G}$ of G is given by:

1. a $\mathrm{Gau}_{\hat{G}}(\hat{P})$-invariant subspace of equivariant 1-forms in $\hat{P}$, called *extended domain*, such that:
$$\mathrm{Conn}(\hat{P};\hat{\mathfrak{g}}) \subseteq \widehat{\mathrm{Conn}}(\hat{P};\hat{\mathfrak{g}}) \subseteq \Omega^1_{eq}(\hat{P};\hat{\mathfrak{g}}), \tag{2}$$
2. a $\mathrm{Gau}_{\hat{G}}(\hat{P})$-invariant functional $\hat{S}^{\hat{G}} : \widehat{\mathrm{Conn}}(\hat{P};\hat{\mathfrak{g}}) \to \mathbb{R}$, called the *extended action functional*;
3. a subspace $0 \subseteq C^1(\hat{P};\hat{\mathfrak{g}}) \subseteq \widehat{\mathrm{Conn}}(\hat{P};\hat{\mathfrak{g}})$, called the *correction subspace*;
4. a *correction functional* $C : C^1(\hat{P};\hat{\mathfrak{g}}) \to \mathbb{R}$ and map $\delta : C^1(\hat{P};\hat{\mathfrak{g}}) \to \mathrm{Conn}(P;\mathfrak{g})$ such that
$$\hat{S}^{\hat{G}}\big|_{C^1(\hat{P};\hat{\mathfrak{g}})} = S^G \circ \delta + C. \tag{3}$$

Remark 2. As discussed in the Introduction and in Section 3 of [24], this definition was proposed as an unification for three kinds of "extensions of YM theories" arising in the literature: extensions by deformations, extensions by adding correction terms and extensions by augmenting the gauge group.

Remark 3. Let $\mathrm{Ext}(S^G;\mathfrak{g})$ be the collection of all extensions of a given YMT theory S^G relative to a fixed basic extension $\imath : G \hookrightarrow \hat{G}$ and let $\mathrm{ED}(S^G;\hat{G})$ be the space of all extended domains, i.e, of all $\mathrm{Gau}_{\hat{G}}(\hat{P})$-invariant sets of 1-forms in $\hat{P}$ satisfying (2). In [24] it was proven that the obvious projection $\pi : \mathrm{Ext}(S^G;\mathfrak{g}) \to \mathrm{ED}(S^G;\hat{G})$ is a bundle of commutative monoids. Furthermore, each additive action of a group $\mathbb{G}$ on $\mathbb{R}$ lifts the above structure of a bundle of commutative monoids to a structure of $R[\mathbb{G}]$-module bundle, where $R[\mathbb{G}]$ is the group R-algebra and R is any commutative unital ring.

Fixing extensions $\hat{S}^{\hat{G}}_i \in \mathrm{Ext}(S^G;\hat{G})$, with $i = 1, 2$, a *morphism* between them is a pair of functions (f, g) such that $f : \widehat{\mathrm{Conn}}_1(\hat{P};\hat{\mathfrak{g}}) \to \widehat{\mathrm{Conn}}_2(\hat{P};\hat{\mathfrak{g}})$ and $g : C^1_1(\hat{P};\hat{\mathfrak{g}}) \to C^1_2(\hat{P};\hat{\mathfrak{g}})$, and such that the underlying structures are preserved, i.e, such that f is $\mathrm{Gau}_{\hat{G}}(\hat{P})$-equivariant and the diagram below commutes. This defines a category $\mathbf{Ext}(S^G;\hat{G})$ which can be realized as a conservative subcategory of a product of slice categories [24].

$$
\begin{tikzcd}
 & & C_1^1(\hat{P};\hat{\mathfrak{g}}) \arrow[dll, bend right, "\delta_1"'] \arrow[dl, "C_1"'] \arrow[dd, "g"] \arrow[rr, hook] & & \widehat{\mathrm{Conn}}_1(\hat{P};\hat{\mathfrak{g}}) \arrow[dl, "\hat{S}_1^{\hat{G}}"'] \arrow[dd, "f"] \\
\mathrm{Conn}(P;\mathfrak{g}) & \mathbb{R} & & \mathbb{R} & \\
 & & C_2^1(\hat{P};\hat{\mathfrak{g}}) \arrow[ull, bend left, "\delta_2"] \arrow[ul, "C_2"] \arrow[rr, hook] & & \widehat{\mathrm{Conn}}_2(\hat{P};\hat{\mathfrak{g}}) \arrow[ul, "\hat{S}_2^{\hat{G}}"]
\end{tikzcd}
\tag{4}
$$

Remark 4. For every YMT S^G and every basic extension $\imath : G \hookrightarrow \hat{G}$ the category $\mathbf{Ext}(S^G;\hat{G})$ has a terminal object given by the null extension such that $\widehat{\mathrm{Conn}}(\hat{P};\hat{\mathfrak{g}}) = \Omega^1_{eq}(\hat{P};\hat{\mathfrak{g}})$, $C^1(\hat{P};\hat{\mathfrak{g}}) = 0$ and such that the maps $\hat{S}^{\hat{G}}$, C and δ are all null. See Example 2 and Example 15 of [24].

Remark 5. The null extension is an example of a *trivial extension* which exists for any YMT theory S^G. In addition to the null extensions (where the maps involved are null) the trivial extensions also include the identity extensions and the constant extensions, whose maps are all identities or constants, respectively. See Section 3.1 of [24]. Extensions which are not of null-type, constant-type or identity-type are called *nontrivial*. In Section 3.4 of [24] it was shown that emergence phenomena (in the sense of [23]) play an important role in building nontrivial extensions.

3 The Extension Problems

Let S^G be a YMT and let $\imath : G \hookrightarrow \hat{G}$ be a basic extension of G. A *class* of extensions of S^G, relative to $\imath : G \hookrightarrow \hat{G}$, is just a subset $\mathcal{E}(S^G;\hat{G})$ of $\mathrm{Ext}(S^G;\hat{G})$.

Remark 6. Throughout the paper we will always consider classes of extensions of the same YMT S^G and relative to the same basic extension $\hat{G}$ of G. Therefore, in order to simplify the notation, when S^G and $\hat{G}$ are implicitly understood we will write $\mathcal{E}$ instead of $\mathcal{E}(S^G;\hat{G})$ to denote a class of extensions.

The basic problems concerning classes of extensions of a given YMT theory S^G relative to a basic extension $\imath : G \hookrightarrow$ are the following.

Existence Problem. *Let $\mathcal{E}$ be a class of extensions. Under which conditions it has nontrivial elements in the sense of Remark 5?*

Obstruction Problem. *Let $\hat{S}$ be an action funcional. What are the obstructions to realize it as an element of $\mathcal{E}$?*

Besides these basic questions, there are others which are more refined. Let $\mathrm{Clss}(S^G;\hat{G})$ be the collection of all classes of extensions of S^G relative to $\imath : G \hookrightarrow \hat{G}$, i.e, the power set of $\mathrm{Ext}(S^G;\hat{G})$, and notice that we have a relation of *maximality* on it, denoted by "$\hookrightarrow_{\max}$". Indeed, we say that a class $\mathcal{E}_0$ is *internally maximal* to other class $\mathcal{E}_1$, and we write $\mathcal{E}_0 \hookrightarrow_{\max} \mathcal{E}_1$, if there exists a function $\mu : \mathcal{E}_0 \to \mathcal{E}_1$ satisfying the following condition: for every other function $\nu : \mathcal{E}_0 \to \mathcal{E}_1$ and every $\hat{S}^{\hat{G}} \in \mathcal{E}_0$, we can regard $\mu(\hat{S}^{\hat{G}})$ as an extension of $\nu(\hat{S}^{\hat{G}})$. We can also consider another relation, called *universality relation*, denoted by $\hookrightarrow_{\text{unv}}$, in which $\mathcal{E}_0 \hookrightarrow_{\text{unv}} \mathcal{E}_1$ iff $\mathcal{E}_0$ is internally maximal to $\mathcal{E}_1$ in an universal way, meaning that $\mu(\hat{S}^{\hat{G}})$ can be regarded as an extension of $\nu(\hat{S}^{\hat{G}})$ in a *unique* way.

Maximality Problem. *Given two classes of extensions of S^G, under which conditions one of them is internally maximal to the other? In other words, what are the classes of $\hookrightarrow_{\max}$?*

Universality Problem. *Given two classes of extensions of S^G, under which conditions one of them is internally universal to the other? In other words, what are the classes of $\hookrightarrow_{\text{unv}}$?*

Concretely, when the maximality problem has a solution, this means that every extension of S^G belonging to $\mathcal{E}_0$ can be regarded as an element of $\mathcal{E}_1$ in such a way that it itself admits extensions. Not only this, it admits a special extension which is maximal. The physical interpretation of the universality problem is analogous, but now the special extension can be regarded as universal among all other extensions.

Notice, however, that the relations $\hookrightarrow_{\max}$ and $\hookrightarrow_{\text{unv}}$ are transitive, but typically not reflexive, symmetric or antisymmetric. Therefore, the decomposition of $\mathrm{Ext}(S^G;\hat{G})$ into their classes is very complex, which makes the direct analysis of the maximality and universality problems very difficult. Recalling that $\mathrm{Ext}(S^G;\hat{G})$ is the collection of objects of the category $\mathbf{Ext}(S^G;\hat{G})$ and that the notion of "extension $\hat{S}_2^{\hat{G}}$ of an extension $\hat{S}_1^{\hat{G}}$" of S^G, used to describe the maximality and universality conditions above, can be interpreted by the existence of a morphism $\hat{S}_1^{\hat{G}} \to \hat{S}_2^{\hat{G}}$ in $\mathbf{Ext}(S^G;\hat{G})$, we conclude that the maximality and universality problems is the study of decompositions of the the collection of objects of a category which are induced by relations determined by morphisms between the objects.

Since for every category $\mathbf{C}$ there are natural and more well-behaved relations in $\mathrm{Ob}(\mathbf{C})$ (e.g, which are partial order relations and equivalence relations), it is natural to ask if the problems described above cannot be rewritten in purely categorical terms, such that the bad-behaved relations $\hookrightarrow_{\max}$ and $\hookrightarrow_{\text{unv}}$ become determined by these natural and well-behaved relations. Doing this is the main objective of the remaining of this section.

Remark 7 (The empty case)**.** Let $\mathcal{E} = \varnothing$ be the empty class. Then the existence and obstruction problems have obvious negative solutions. Furthermore, by vacuous truth this class is internally universal (and in particular maximal) to any other. Thus, from now on we will consider only nonempty classes of extensions.

3.1 Rewriting the Extension Problems

Recall that the full subcategories of a given category $\mathbf{C}$ are in bijection with the subclasses of the class of objects $\mathrm{Ob}(\mathbf{C})$ of $\mathbf{C}$. Thus, to give a class $\mathcal{E}$ of extensions whose underlying basic extension is $\imath : G \hookrightarrow \hat{G}$ is equivalent to giving a full subcategory of $\mathbf{Ext}(S^G;\hat{G})$, which in the following will be denoted by the same notation. We will say that a full subcategory of $\mathbf{Ext}(S^G;\hat{G})$ is *nontrivial* if its class of objects is nontrivial in the sense of Remark 5, i.e, if it does not contain the null-type, identity-type and constant-type extensions. With this, the existence problem and the obstruction problem can be rewritten in the following equivalent way:

Existence Problem (categorical form)**.** *Is a given full subcategory $\mathcal{E}$ of $\mathbf{Ext}(S^G;\hat{G})$ nontrivial?*

Obstruction Problem (categorical form)**.** *What are the subcategories of* $\mathbf{Ext}(S^G;\hat{G})$ *which are adjoint to a given full subcategory* $\mathcal{E}$*?*

The direct categorical analogues of the maximality and universality conditions described in the last section are the following. We say that a full subcategory $\mathbf{C}_0$ of $\mathbf{C}$ is *weakly internally maximal* to other full subcategory $\mathbf{C}_1$ if there exists a functor $\mu : \mathbf{C}_0 \to \mathbf{C}_1$ such that for every other functor $\nu : \mathbf{C}_0 \to \mathbf{C}_1$ there exists a natural transformation $u : \nu \Rightarrow \mu$. If this natural transformation is unique, we say that $\mathbf{C}_0$ is *weakly universal*. Thus:

Maximality Problem (weak categorical form)**.** *Given two full subcategories of* $\mathbf{Ext}(S^G;\hat{G})$*, are they weakly internally maximal one to the other?*

Universality Problem (weak categorical form)**.** *Are two given full subcategories of* $\mathbf{Ext}(S^G;\hat{G})$ *weakly internally universal one to the other?*

Although the direct categorical analogues of the existence and universality problems are *equivalent* to the original problems, we note that the above direct categorical analogues of the maximality and universality problems are a bit *different* of those in Subsection 3, which is the reason why we used the adjective "weakly". Indeed, notice that every functor $F : \mathbf{C}_0 \to \mathbf{C}_1$ induces a map between the underlying sets of objects and other map between the sets of morphisms. But in the original universality problem we only require information between objects. Therefore, if the map on objects lifts uniquely to a functor, then the weak maximality and weak universality imply maximality and universality in the original sense.

The problem is that a function $f : \mathrm{Ob}(\mathbf{C}_0)$ does not need extending to a functor $F : \mathbf{C}_0 \to \mathbf{C}_1$. Furthermore, even if such lifts exist, they are typically highly non-unique. Indeed, let $\mathrm{Aut}(\mathrm{img}(f))$ be the disjoint union of the automorphism groups $\mathrm{Aut}(f(X))$ of $f(X) \in \mathbf{D}$ for every $X \in \mathbf{C}$ and consider the obvious projection $\pi : \mathrm{Aut}(\mathrm{img}(f)) \to \mathrm{Ob}(\mathbf{C})$. Each section for this projection can be used to modify the functor F lifting f only on morphisms. Furthermore, the modified functor is not equal to F iff the section is nontrivial, which means that the automorphism groups $\mathrm{Aut}(f(X))$ are nontrivial. Only in very rare cases can we ensure that any two functors lifting f are naturally isomorphic [4].

Thus, if the basic obstruction in identifying maps on objects with functors (and therefore the weakly maximality/universality problem with the original maximality/universality problem) are the automorphisms, the idea is to work up to automorphisms.

3.2 Maximality and Universality Up to Isomorphisms

Let $\mathbf{C}$ be a category and let $\mathrm{Iso}(\mathbf{C})$ denote the quotient space $\mathrm{Ob}(\mathbf{C})/\simeq$ of objects by the isomorphism relation $\simeq$. Notice that the relation "*there exists a morphism between two objects* X *and* Y" is a partial order $\leq$ on $\mathrm{Ob}(\mathbf{C})$. It does not need to have a maximal element. Let $\mathbf{C}_0 \subset \mathbf{C}$ be a full subcategory, so that we have an induced partial order on $\mathrm{Ob}(\mathbf{C}_0) \subset \mathrm{Ob}(\mathbf{C})$. It is compatible with the isomorphism relation, meaning that if $X \simeq X'$ and $Y \simeq Y'$, then $X \leq Y$ iff $X' \leq Y'$. Thus, we have an induced partial order $\lesssim$ on $\mathrm{Iso}(\mathbf{C})$. We say that $\mathbf{C}_0$ is a *greatest subcategory* if $(\mathrm{Iso}(\mathbf{C}_0), \lesssim)$ has a greatest element, i.e, if there exists an object $X_0 \in \mathbf{C}_0$ and a morphism $Y \to X_0$ for each other Y.

Now, let $\mathbf{C}_0$ and $\mathbf{C}_1$ be full subcategories of $\mathbf{C}$. We say that $\mathbf{C}_0$ is *internally maximal up to isomorphism* to $\mathbf{C}_1$ if there exists a map on objects $f : \mathrm{Ob}(\mathbf{C}_0) \to \mathrm{Ob}(\mathbf{C}_1)$ such that the full subcategory defined by its image is a greatest subcategory of $\mathbf{C}_1$.

Maximality Problem (categorical form). *Given two full subcategories of* $\mathbf{Ext}(S^G;\hat{G})$*, is one internally maximal up to isomorphisms to the other?*

Given full categories $\mathbf{C}_0, \mathbf{C}_1$ of $\mathbf{C}$, notice that $\mathbf{C}_0$ is internally maximal up to isomorphisms to $\mathbf{C}_1$ iff there exists $Y_0 \in \mathbf{D}$ such that $Y_0 \simeq f(X_0)$ for some $X_0 \in \mathbf{C}$ and such that there exists a morphism $Y \to Y_0$ for each other $Y \in \mathbf{D}$ such that $Y \simeq f(X)$ for some $X \in \mathbf{C}$. Due to the compatibility between $\simeq$ and $\leq$, the previous condition is also equivalent to the folllowing: there exists $X_0 \in \mathbf{C}_0$ such that for each other $X \in \mathbf{C}_0$ there exists a morphism $f(X) \to f(X_0)$. But this is exactly the maximality condition of Subsection 3. Thus, the classical and the categorical forms of the maximality problem are equivalent. Explicitly, we have proved the following:

Proposition 1. *Let S^G be a YMT with gauge group G and let $\imath : G \hookrightarrow \hat{G}$ be a basic extension of G. A class $\mathcal{E}_0$ of extensions of S^G is internally maximal to other class $\mathcal{E}_1$ iff the full subcategory defined by $\mathcal{E}_0$ is internally maximal up to isomorphisms to the subcategory defined by $\mathcal{E}_1$.*

If we try to do a similar analysis with the universality condition we find a problem. Indeed, we should replace the relation *"there exists a monomorphism between X and Y"* with the relation *"there exists a* ***unique*** *morphism between X and Y"*. As we saw, the first one is a partial relation which behaves very well relative to isomorphisms. But the second one is typically not a partial relation precisely because isomorphisms are involved. Actually, it defines a partial relation in $\mathbf{C}$ iff $\mathbf{C}$ is a category without isomorphisms[3], i.e, if a morphism $f : X \to Y$ is not an isomorphism unless $Y = X$ and $f = id_X$. But this is a very strong assumption. For instance, it implies that the automorphism group $\mathrm{Aut}(X)$ is trivial for every $X \in \mathbf{C}$. Furthermore, $\mathbf{C}$ cannot be the codomain of a functor $F : \mathbf{D} \to \mathbf{C}$ where $\mathbf{D}$ has at least one isomorphism $f : X \to Y$ with $F(X) \neq F(Y)$. Even so, if we assume this hypothesis we can apply the discussion to the universality condition.

Thus, let $\mathbf{C}_0$ and $\mathbf{C}_1$ be full subcategories $\mathbf{C}$ and assume that $\mathbf{C}_1$ is without isomorphisms. Let $\leqq$ be the partial relation in $\mathrm{Ob}(\mathbf{C}_1)$ such that $X \leqq Y$ iff there exists a unique morphism $X \to Y$. Since the full subcategories are without isomorphisms, we have $\mathrm{Ob}(\mathbf{C}_1) \simeq \mathrm{Iso}(\mathbf{C}_1)$, with $i = 0, 1$ so that $\leqq$ trivially pass to quotient and define another partial relation $\lessapprox$. We say that $\mathbf{C}_0$ is *internally universal up to isomorphisms* to $\mathbf{C}_1$ if there is a map on objects $f : \mathrm{Ob}(\mathbf{C}_0) \to \mathrm{Ob}(\mathbf{C}_1)$ such that $\mathrm{img}(f) \subset \mathrm{Ob}(\mathbf{C}_1)$ has a greatest element when considered as a subset of $(\mathrm{Iso}(\mathbf{C}_1), \lessapprox)$. Since $\mathbf{C}_1$ is without isomorphisms, it happens that $\mathrm{img}(f)$ has a greatest element in $(\mathrm{Ob}(\mathbf{C}_1), \leqq)$.

Universality Problem (categorical form). *Given two full subcategories $\mathcal{E}_0$ and $\mathcal{E}_1$ of* $\mathbf{Ext}(S^G;\hat{G})$*, where the second one is without isomorphisms, under which conditions $\mathcal{E}_0$ is internally universal up to isomorphisms to $\mathcal{E}_1$?*

Under the hypothesis that $\mathcal{E}_1$ is without isomorphisms, one can check that Proposition 1 remains true if we replace "maximality" with "universality". More precisely, one can check that the solutions of the above problem correspond precisely to extensions $\hat{S}_0^{\hat{G}} \in \mathcal{E}_0$ such that there exists a map $\mu : \mathcal{E}_0 \to \mathcal{E}_1$ with the property that for each other extension $\hat{S}^{\hat{G}}$ there exists a unique morphism $\mu(\hat{S}^{\hat{G}}) \to \mu(\hat{S}_0^{\hat{G}})$.

Remark 8. Since by Proposition 1 maximality and maximality up to isomorphisms are equivalent conditions, in the following we will refer to both as "*maximality*". Since universality and universality up to isomorphisms are conditions which are not comparable when $\mathcal{E}_1$ is not without isomorphisms, we will mantain the distinction between them.

[3]A category without isomorphisms is also called a *gaunt category*. They appear in abstract homotopy theory and higher category theory as the class of categories whose simplicial nerve defines a complete Segal space [1].

4 Conditions for Maximality and Universality

After the prevous categorical characterizations of the fundamental problems involving extensions of YMT theories, we are ready to try to solve them. Here we will present some sufficient conditions to ensure maximality or universality between classes of extensions of YMT theories. We will first show that if we are analyzing universality internal to some class of extensions which contains the null-type extensions of Remark 4, then universality is immediate in both categorical senses, i.e, weak universality and universality up to isomorphisms.

Theorem 4.1 (Theorem A)**.** *Let $\mathcal{E}_1$ be a class of extensions of S^G relative to $\imath : G \hookrightarrow \hat{G}$.*

1. *If it contains the null-type extension of Remark 4 or any extension isomorphic to it, then each other class $\mathcal{E}_0$ is internally universal and weakly universal (in particular maximal and weakly maximal) to $\mathcal{E}_1$;*
2. *If in addition $\mathcal{E}_1$ is a category without isomorphisms, then every $\mathcal{E}_0$ is also universal up to isomorphisms.*

Proof. First of all, notice that the second assertion follows directly from the first one by recalling that, by the discussion of Section 3.2, under those hypotheses universality holds iff universality up to isomorphisms holds. Thus, it is enough to work on the first assertion. We affirm that the first hypothesis implies that the functor category $\mathrm{Func}(\mathcal{E}_0;\mathcal{E}_1)$ has a terminal object. Indeed, recall that the null extension of Remark 5 is a terminal object in the category $\mathbf{Ext}(S^G;\hat{G})$ [24]. Since by hypothesis it belongs to $\mathcal{E}_1$, the fact that limits are reflected by fully faithful functors implies that it is actually a terminal object in $\mathcal{E}_1$. Once limits of a functor category are determined by limits of the codomain category, it follows that the null extension defines a terminal object in $\mathrm{Func}(\mathcal{E}_0;\mathcal{E}_1)$. Since objects isomorphic to terminal objects are also terminal, it follows that the conclusion above remains valid if we replace the null extension by any other isomorphic to it. Now, notice that each terminal object in that functor category is a realization of $\mathcal{E}_0$ as weakly internally universal to $\mathcal{E}_1$, proving the first assertion in the weakly universal case. For the universal case, recall that from Remark 7 if $\mathcal{E}_0$ is empty, then there is nothing to do. Thus, assume $\mathcal{E}_0 \neq \varnothing$. Let $\hat{S}_0^{\hat{G}}$ be an extension isomorphic to the null extension (which by hypothesis belongs to $\mathcal{E}_1$) and let $\mu : \mathcal{E}_0 \to \mathcal{E}_1$ be any map whose image contains $\hat{S}_0^{\hat{G}}$. Let $\hat{S}_{0,*}^{\hat{G}} \in \mu^{-1}(\hat{S}_0^{\hat{G}})$ and notice that since $\hat{S}_0^{\hat{G}}$ is a terminal object, for every other $\hat{S}^{\hat{G}} \in \mathcal{E}_0$ there exists a unique morphism $\mu(\hat{S}^{\hat{G}}) \to \mu(\hat{S}_{0,*}^{\hat{G}})$, which proves universality. $\square$

Let X be a set, $\mathcal{P}(X)$ its power set and $\mathcal{P}^2(X)$ be the power set of its power set. Consider a property ρ in $\mathcal{P}(X)$. We say that ρ is dense at $\mathcal{P}_0(X) \in \mathcal{P}^2(X)$ if there exists an element $x_0 \in X$ such that for every $Y \subset X$ in $\mathcal{P}_0(X)$ the property ρ holds in $Y \cup x_0$. If $\mathcal{P}_0(X) = \mathcal{P}(X)$ we say simply that ρ is *dense*. The terminology is due to the following fact: let $\mathcal{P}_\rho(X)$ be the subset of those subsets of X in which property ρ is verified. For each $x_0 \in X$ consider the map $\cup_{x_0} : \mathcal{P}(X) \to \mathcal{P}(X)$ such that $\cup_{x_0} Y = Y \cup x_0$ obtained by taking the union with x_0. Take a topology in $\mathcal{P}_\rho(X)$ for which the map $\cup_{x_0}$ is continuous. If $\mathcal{P}_0(X)$ is dense[4], then its image by $\cup_{x_0}$ is also dense. But ρ being dense at $\mathcal{P}_0(X)$ means that this image is contained in $\mathcal{P}_\rho(X)$, so that $\mathcal{P}_\rho(X)$ is dense too.

[4]This hypothesis can be avoided if we consider a topology in $\mathcal{P}(X)$ in which evaluation of $\cup$ at points is always dense.

Given a relation $\mathcal{R}$ in $\mathcal{P}(X)$, we can consider the induced property $\rho_{\mathcal{C}}$ "*being $\mathcal{R}$-related with a fixed $X_0 \subset X$*". We say that $\mathcal{R}$ is *dense at* $\mathcal{P}_0(X)$ (resp. *dense*) if $\rho_{\mathcal{R}}$ is.

Corollary 1. *For every YMT S^G and every basic extension $\imath : G \hookrightarrow \hat{G}$, both the universality relation, the maximality relation, the weak universality relation and the weak maximality relations are all dense. The universality up to isomorphisms relation is dense in the classes of extensions whose induce full subcategories are without isomorphisms.*

Proof. Straightforward from the above discussion and from Theorem 4.1. □

4.1 Zorn's Lemma Approach

Let us now give a condition for maximality and universality even in the case when a class of extensions $\mathcal{E}$ does not contains an object isomorphic to the null extension. Recalling that maximality and universality up to isomorphism conditions are about existence of greatest elements, the main idea is to use Zorn's lemma. This is precisely what we will do.

Given a set of indexes I and a class of extensions $\mathcal{E}$, let $\mathcal{E}(I)$ denote the union of all subclasses $X \subset \mathcal{E}$ such that $|X| \leq |I|$. Then this is another subclass of extensions of $\mathcal{E}$. We say that $\mathcal{E}$ is *I-coherent for maximality* if it has J-coproducts for every $J \subset I$ and if the partial order $\lesssim$ of Subsection 3.2 is actually a total order when restricted to $\mathcal{E}(I)$. We say that it is *coherent for maximality* if it is I-coherent for I containing the class of objects of $\mathcal{E}$. Similarly, we say that it is *I-coherent for universality up to isomorphisms* if it has J-coproducts, $\mathcal{E}$ is without isomorphisms and the partial order $\lessapprox$ is a total order.

Example 3. If $\mathcal{E}$ is a coherent category in the sense of [16], then it is clearly coherent for maximality. The product and the slice of coherent categories are coherent categories. The category of sets is coherent. By Theorem 5.1 of [24], every $\mathcal{E}$ is a conservative subcategory of a product of slices of **Set**, so that it is a conservative subcategory of a coherent category. Thus, under additional requirements it can itself be made coherent [16].

Remark 9. Notice that if we allow $J = \varnothing$ in the definition of I-coherence above, then this implies that the category have initial objects, which is not covered by Theorem 4.1, since there we assumed existence of terminal objects.

Theorem 4.2 (Theorem B)**.** *Let $\mathcal{E}_1$ be a class of extensions as above. If it is I-coherent for maximality (resp. I-coherent for universality up to isomorphisms), then every $\mathcal{E}_0$ such that $|\mathcal{E}_0| \geq |\mathcal{E}_1(I)|$ is internally maximal (resp. universal up to isomorphisms) to $\mathcal{E}_1(I)$.*

Proof. Consider first the case $\mathcal{E}_0 = \mathrm{Ob}(\mathbf{Ext}(S^G; \hat{G}))$. By the closeness under J-coproducts, with $J \subset I$, it follows that each subset X of $\mathcal{E}_1$ such that $|X| \leq |I|$ is bounded from above. Indeed, let $b(X) = \coprod_{x \in X} x$, so that for every $x \in X$ we have the canonical morphism $x \to b(X)$ and therefore $x \lesssim b(X)$. Thus, each subset of $\mathcal{E}(I)$ is bounded from above. The total order hypothesis on $\lesssim$ implies that each subset is a chain. Therefore, by Zorn's lemma we see that $\mathcal{E}_1(I)$ has a maximal element which is actually a greatest element $\hat{S}_0^{\hat{G}}$. By the Axiom of Choice there exists a function $r : \mathrm{Ob}(\mathbf{Ext}(S^G; \hat{G})) \to \mathcal{E}_1(I)$ which is a retraction for the reciprocal inclusion $\mathcal{E}_1 \hookrightarrow \mathrm{Ob}(\mathbf{Ext}(S^G; \hat{G}))$ and therefore a surjective map. Since r is surjective, $r^{-1}(\hat{S}_0^{\hat{G}})$ is nonempty and each element on it satisfies the desired maximality or universality up to isomorphism condition. Finally, note that the same argument holds if $\mathcal{E}_0$ is any class in which $\mathcal{E}_1(I)$ can be embedded, i.e, such that $|\mathcal{E}_0| \leq |\mathcal{E}_1(I)|$. □

Remark 10. We can recover a version of Theorem 4.1 from the last theorem. Indeed, since I-coproducts with $I = \varnothing$ is the same as an initial object and since the null-extension is actually a null object in the category of extensions, it follows that $\mathcal{E}(\varnothing) \simeq *$ is the class containing the null extension. Suppose $\mathcal{E}_0$ nonempty, so that $|\mathcal{E}_0| \geq 1 = |\mathcal{E}(\varnothing)|$. Thus, by the last theorem, $\mathcal{E}_0$ is internally maximal to $\mathcal{E}_1(\varnothing)$. But note that if a class $\mathcal{E}_0$ is internally maximal to $\mathcal{E}_1$, then it is also internally maximal to any class containing $\mathcal{E}_1$. Therefore, we conclude that every $\mathcal{E}_0$ is internally maximal to any class containing the null extension, which is essentially Theorem 4.1.

5 A Concrete Situation

In this section we will analyze the four problems of Section 3 for a concrete class of extensions. More precisely, we prove the following, which basically constitutes Theorem C:

1. for every group G, every YMT theory S^G and every basic extension $\imath : G \hookrightarrow \hat{G}$, the class $\mathrm{Inj}(S^G;\hat{G})$ of $\mathrm{Gau}_{\hat{G}}(\hat{P})$-*injective* extensions of S^G (defined below) is such that the existence and the obstruction problem have a complete and positive solution;

2. there is a subclass $\mathrm{Coh}(S^G;\hat{G})$ of $\mathrm{Inj}(S^G;\hat{G})$ consisting of the so-called *coherent extensions* which are I-coherent for maximality in the sense of Section 4.1 for certain I.

We begin by defining what we mean by an *injective extension.* Let $* : G \times X \to X$ be the action of a group G on a set X, and let $f : X \to \mathbb{R}$ be a function. We say that f is *G-invariant in a subset* $X_0 \subset X$ (or that X_0 is a *G-invariant subset for f*) if X_0 is G-invariant and $f(g * x) = f(x)$ for every $x \in X_0$. This means that f factors as in the diagram below. We say that X_0 is an *injective G-invariant subset for f* if the induced map $\overline{f}_0$ is injective. Finally, we recall that if X_0/G is the orbit space of a group action, a *gauge fixing* for the underlying action is a global section for the projection map $\pi : X_0 \to X_0/G$, i.e, it is a function $\sigma : X_0/G \to X_0$ such that $\pi(\sigma(x)) = x$ for every $x \in X_0$. In the category of sets, gauge fixings always exists. This is a consequence of the Axiom of Choice, which states that every surjection of sets has a section. When the group action is implicit, we say simply that σ is a *gauge fixing in* X_0.

$$\begin{tikzcd}
X_0 \arrow[r, "\pi"] \arrow[d, hook] & X_0/G \arrow[d, hook] \arrow[dr, dashed, "\overline{f}_0"] & \\
X \arrow[r, "\pi"'] \arrow[rr, bend right, "f"'] & X/G & \mathbb{R}
\end{tikzcd}$$

Let S^G be a YMT theory and let $\hat{S}^{\hat{G}}$ be an extension of S^G. We say that $\hat{S}^{\hat{G}}$ is *injective* if its domain $\widehat{\mathrm{Conn}}(\hat{P};\hat{\mathfrak{g}})$ is an injective $\mathrm{Gau}_{\hat{G}}(\hat{P})$-invariant subset for $\hat{S}^{\hat{G}}$. The class of all of them will be denoted by $\mathrm{Inj}(S^G;\hat{G})$. The obstruction problem for this class is characterized by the following lemma, which will be proved in Subsection 5.1.

Lemma 5.1. *Let S^G be a YMT theory with instanton sector P, let $\imath : G \hookrightarrow \hat{G}$ be a basic extension of G and let $\mathrm{Conn}(\hat{P};\hat{\mathfrak{g}}) \subset X_0 \subset \Omega^1(\hat{P};\hat{\mathfrak{g}})$ be an injective $\mathrm{Gau}_{\hat{G}}(\hat{P})$-invariant subset for a map $S : \Omega^1(\hat{P};\hat{\mathfrak{g}}) \to \mathbb{R}$. Then*

1. *Each gauge fixing σ in* $\mathrm{Conn}(\hat{P};\hat{\mathfrak{g}})$ *induces a representation of the restriction* $S_0 : X_0 \to \mathbb{R}$ *of* S *to* X_0 *as a complete extension of* S^G *on a certain subspace* $X_0(\sigma)$*;*

2. *the subspace* $X_0(\sigma)$ *is independent of* σ*, i.e, for every two gauge fixings* σ, σ' *there exists a bijection* $X_0(\sigma) \simeq X_0(\sigma')$*.*

Now, let us consider the class of coherent extensions. We say $\hat{S}^{\hat{G}}$ is

1. of *pullback type* if it is of the form of the last theorem, i.e, if it is complete, $\widehat{\mathrm{Conn}}(\hat{P};\hat{\mathfrak{g}}) = X_0$ and there exists a gauge fixing σ such that

$$C^1(\hat{P};\hat{\mathfrak{g}}) \subset X_0(\sigma) \simeq \mathrm{Pb}(S^G, \overline{S}_{0,\jmath}).$$

The class of all of them will be denoted by $\mathrm{Pb}(S^G;\hat{G})$;

2. *small* (or that it has has a *small correction subspace*) if $C^1(\hat{P};\hat{\mathfrak{g}})$ is actually contained in $\mathrm{Conn}(\hat{P};\hat{\mathfrak{g}})$. The class of these extensions will be denoted by $\mathrm{Small}(S^G;\hat{G})$.

3. *coherent* if it is injective and of pullback-type. The class of all of them will be denoted by $\mathrm{Coh}(S^G;\hat{G})$, so that $\mathrm{Coh}(S^G;\hat{G}) = \mathrm{Inj}(S^G;\hat{G}) \cap \mathrm{Pb}(S^G;\hat{G})$;

4. *coherently small* if it is coherent and small. The class of them will be denoted by $\mathrm{SCoh}(S^G;\hat{G})$, so that $\mathrm{SCoh}(S^G;\hat{G}) = \mathrm{Coh}(S^G;\hat{G}) \cap \mathrm{Small}(S^G;\hat{G})$

With this nomenclature, Lemma 5.1 has the following meaning:

Corollary 2. *For every YMT* S^G *and every basic extension* $\imath : G \hookrightarrow \hat{G}$*, the only obstruction to belonging to* $\mathrm{Coh}(S^G;\hat{G})$ *is belonging to* $\mathrm{Inj}(S^G;\hat{G})$*.*

Notice that this corollary (and therefore Lemma 5.1) is only about the obstruction problem because a priori we do not know anything about the existence problem for the class $\mathrm{Inj}(S^G;\hat{G})$ of injective extensions. But in Subsection 5.2 we will prove the following lemma, showing that to every extension one can assign a canonical injective extension.

Lemma 5.2. *There exists a canonical map*

$$\mathcal{I} : \mathrm{Ob}(\mathbf{Ext}(S^G;\hat{G})) \to \mathrm{Inj}(S^G;\hat{G}) \cap \mathrm{Comp}(S^G;\hat{G})$$

assigning to each extension $\hat{S}^{\hat{G}}$ *of* S^G *the largest complete injective extension* $\mathcal{I}(\hat{S}^{\hat{G}})$ *for which there exists a monomorphism* $\mathcal{I}(\hat{S}^{\hat{G}}) \to \hat{S}^{\hat{G}}$*. Furthermore, this map preserves small extensions.*

Directly from the proof of Lemma 5.1 in Subsection 5.1 we will conclude that every pullback-type extension is small, so that $\mathrm{SCoh}(S^G;\hat{G}) = \mathrm{Coh}(S^G;\hat{G})$ and we have the inclusion

$$\mathrm{Pb}(S^G;\hat{G}) \hookrightarrow \mathrm{SCoh}(S^G;\hat{G}) = \mathrm{Coh}(S^G;\hat{G}). \tag{5}$$

In Subsection 5.3 we will then prove the following lemma, where we recall that a function $f : \mathrm{Ob}(\mathbf{C}) \to \mathrm{Ob}(\mathbf{D})$ between objects of categories is *essentially injective* if it factors through isomorphisms and the induced map $\overline{f} : \mathrm{Iso}(\mathbf{C}) \to \mathrm{Iso}(\mathbf{D})$ is a bijection.

Lemma 5.3. *The choice of a gauge fixing σ as in Lemma 5.1 induces a retraction r_σ for the inclusion (5). Furthermore, this retraction is an essentially injective map for the underlying full subcategories of* $\mathbf{Ext}(S^G;\hat{G})$.

In the sequence, in Subsection 5.4 we will give a proof for the following result:

Lemma 5.4. *The partial order $\lesssim$ is a total order when restricted to the class $\mathrm{Pb}(S^G;\hat{G})$ of pullback extensions.*

As a consequence of all these steps we have the following conclusion, where $\mathrm{ED}(S^G;\hat{G})$ denotes the collection of all extended domains of a YMT S^G relatively to a basic extension $\imath : G \hookrightarrow \hat{G}$.

Theorem 5.1 (Theorem C). *Let S^G be a YMT theory, $\imath : G \hookrightarrow \hat{G}$ a basic extension of G and $I \subset \mathrm{ED}(S^G;\hat{G})$ a subset of mutually disjoint extended domains. Then every class $\mathcal{E}_0$ of extensions such that $|\mathcal{E}_0| \geq |I|$ is internally maximal to the class $\mathrm{Coh}(S^G;\hat{G})(I)$ of coherent extensions.*

Proof. Since I is a subset of disjoint extended domains, its union coincides with the disjoint union and is also an extended domain. Thus, given extensions $\hat{S}_i^{\hat{G}}$, with $i \in I$, each of them with extended i, one can define its disjoint union $\coprod_i \hat{S}_i^{\hat{G}}$, which remains an extension of S^G. Furthermore, if each $\hat{S}_i^{\hat{G}}$ is injective (resp. of pullback-type), then the coproduct is injective (resp. of pullback-type too). Therefore, $\mathrm{Coh}(S^G;\hat{G}) = \mathrm{Inj}(S^G;\hat{G}) \cap \mathrm{Pb}(S^G;\hat{G})$ has I-coproducts. By Lemma 5.4 the class $\mathrm{Pb}(S^G;\hat{G})$ is totally ordered, while by Lemma 5.3 there exists an essentially injective map between pullback-type extensions and coherent extensions, showing that $\mathrm{Coh}(S^G;\hat{G})$ is totally ordered too. Thus, it is actually I-coherent for maximality and the result follows from Theorem 4.2. □

5.1 Proof of Lemma 5.1

The proof consists essentially in taking the pullback between S^G and a suitable monomorphism. Indeed, notice that we have the commutative diagram below, where $\overline{S}_0$ exists and is injective due to the injectivity hypothesis on X_0. Since the composition of injective maps is injective, we see that $\overline{S}_{0,\jmath} = \overline{S}_0 \circ \overline{\jmath}$ is itself injective.

$$\begin{tikzcd}
\mathrm{Conn}(\hat{P};\hat{\mathfrak{g}}) \arrow[r, "\pi"] \arrow[d, hook, "\jmath"'] & \mathrm{Conn}(\hat{P};\hat{\mathfrak{g}})/\,\mathrm{Gau}_{\hat{G}}(\hat{P}) \arrow[d, hook, "\overline{\jmath}"] \\
X_0 \arrow[r, "\pi"'] \arrow[d, "S_0"'] & X_0/\,\mathrm{Gau}_{\hat{G}}(\hat{P}) \arrow[dl, bend left, "\overline{S}_0"] \\
\mathbb{R} &
\end{tikzcd} \tag{6}$$

Consider now the pullback between S^G and $\overline{S}_{0,\jmath}$, as in the diagram below. Due to the stability of monomorphisms under pullbacks, it follows that $\overline{\jmath}_0$ is an inclusion. Let σ be a gauge fixing in $\mathrm{Conn}(\hat{P};\hat{\mathfrak{g}})$. Since sections are always monomorphisms, we see that σ is injective and then the composition $\overline{\sigma}_0 = \sigma \circ \overline{\jmath}_0$ below is an injection. Thus, the composition $\jmath \circ \overline{\sigma}_0$ is injective too, so that it embeds the pullback $\mathrm{Pb}(S^G, \overline{S}_{0,\jmath})$ as a subset of X_0, which we denote by $X_0(\sigma)$. Due to the commutativity of the left-hand side square, we have $S_0|_{X_0(\sigma)} = S^G \circ \delta$. Thus, if $X_0(\sigma)$ is invariant under $\mathrm{Gau}_G(P)$, then S_0 is a complete extension of S^G on the correction subspace $X_0(\sigma)$. This will follow from the universality of pullbacks.

$$
\begin{tikzcd}
\mathrm{Pb}(S^G, \overline{S}_{0,\jmath}) \arrow[r, dashed, "\overline{\sigma}_0"'] \arrow[rr, bend left, "\bar{\jmath}_0"] \arrow[dd, "\delta"'] & \mathrm{Conn}(\hat{P}; \hat{\mathfrak{g}}) \arrow[r, shift left, "\pi"] \arrow[d, hook, "\jmath"'] & \mathrm{Conn}(\hat{P}; \hat{\mathfrak{g}}) / \mathrm{Gau}_{\hat{G}}(\hat{P}) \arrow[l, shift left, "\sigma"] \arrow[d, hook, "\bar{\jmath}"] \\
& X_0 \arrow[r, "\pi"'] \arrow[d, "S_0"'] & X_0 / \mathrm{Gau}_{\hat{G}}(\hat{P}) \arrow[dl, "\overline{S}_0"] \\
\mathrm{Conn}(P; \mathfrak{g}) \arrow[r, "S^G"'] & \mathbb{R} &
\end{tikzcd}
\tag{7}
$$

Notice that what we have to prove is the existence of the dotted arrow below, where we replaced $X_0(\sigma)$ with $\mathrm{Pb}(S^G, \overline{S}_{0,\jmath})$ since there is bijective correspondence between them.

$$
\begin{tikzcd}
\mathrm{Gau}_G(P) \times \mathrm{Pb}(S^G, \overline{S}_{0,\jmath}) \arrow[r, dashed, "*"] \arrow[d, "id\times\delta"'] & \mathrm{Pb}(S^G, \overline{S}_{0,\jmath}) \arrow[d, "\delta"] \\
\mathrm{Gau}_G(P) \times \mathrm{Conn}(P; \mathfrak{g}) \arrow[r, "(-)^*"'] & \mathrm{Conn}(P; \mathfrak{g})
\end{tikzcd}
$$

Now, let $\xi : \mathrm{Gau}_G(P) \to \mathrm{Gau}_{\hat{G}}(\hat{P})$ be the map discussed in Section 2.1 and notice that the exterior diagram below commutes, which follows precisely from the realization of the pullback as the subset in which the functionals S^G and $\overline{S}_0$ coincide. Thus, by the universality of pullbacks, there exists the dotted arrow, which is the desired map. This concludes the first part of the proof.

$$
\begin{tikzcd}
\mathrm{Gau}_G(P) \times \mathrm{Pb}(S^G, \overline{S}_{0,\jmath}) \arrow[dr, dashed, "*"] \arrow[rrr, bend left, "\pi\circ\xi"] \arrow[ddd, bend right, "(-)^*\circ(id\times\delta)"'] & & & \\
& \mathrm{Pb}(S^G, \overline{S}_{0,\jmath}) \arrow[r, "\overline{\sigma}_0"'] \arrow[rr, bend left, "\bar{\jmath}_0"] \arrow[dd, "\delta"'] & \mathrm{Conn}(\hat{P}; \hat{\mathfrak{g}}) \arrow[r, shift left, "\pi"] \arrow[d, hook, "\jmath"'] & \mathrm{Conn}(\hat{P}; \hat{\mathfrak{g}}) / \mathrm{Gau}_{\hat{G}}(\hat{P}) \arrow[l, shift left, "\sigma"] \arrow[d, hook, "\bar{\jmath}"] \\
& & X_0 \arrow[r, "\pi"'] \arrow[d, "S_0"'] & X_0 / \mathrm{Gau}_{\hat{G}}(\hat{P}) \arrow[dl, "\overline{S}_0"] \\
& \mathrm{Conn}(P; \mathfrak{g}) \arrow[r, "S^G"'] & \mathbb{R} &
\end{tikzcd}
$$

For the second part, given gauge fixing σ and σ', let us prove that the induced extensions have the same correction subspaces, i.e, there exists a bijection $X_0(\sigma) \simeq X_0(\sigma')$. Looking at diagram (6) we see that $X_0(\sigma) \simeq \mathrm{img}(\jmath \circ \sigma \circ \bar{\jmath})$, while $X_0(\sigma') \simeq \mathrm{img}(\jmath \circ \sigma' \circ \bar{\jmath})$. Since those maps are injective, it follows that their image are in bijection with their domain. But both maps have the same domain: the pullback $\mathrm{Pb}(S^G, \overline{S}_{0,\jmath})$. Thus, $X_u(\sigma) \simeq X_0(\sigma')$ and the proof is done.

5.2 Proof of Lemma 5.2

Let $\hat{S}^{\hat{G}}$ be an extension of S^G with extended domain $X \in \mathrm{ED}(S^G; \hat{G})$, correction term $C : C^1(\hat{P}; \hat{\mathfrak{g}}) \to \mathbb{R}$ and δ-map $\delta : C^1(\hat{P}; \hat{\mathfrak{g}}) \to \mathrm{Conn}(P; \mathfrak{g})$. From it, define a new extension $\hat{S}_0^{\hat{G}}$ with same data, but now with a null correction term. This is obviously a complete extension of S^G. In turn, let $X' \subset X$ be the largest $\mathrm{Gau}_{\hat{G}}(\hat{P})$-invariant subset restricted to which $\hat{S}_0^{\hat{G}}$ remains $\mathrm{Gau}_{\hat{G}}$-invariant and such that it becomes injective on the quotient space $X'/\mathrm{Gau}_{\hat{G}}$. Since X' does not necessarily contains $\mathrm{Conn}(\hat{P}; \hat{\mathfrak{g}})$, it does not need to be an extended domain. On the other hand, the union $X'' = X' \cup \mathrm{Conn}(\hat{P}; \hat{\mathfrak{g}})$ it is. Thus, we can try to build an injective extension by considering the restriction $\hat{S}_0^{\hat{G}}|_{X''}$ as extended functional. Again, this may fail since a priori the correction space $C^1(\hat{P}; \hat{\mathfrak{g}}) \subset X$ needs not be contained in X''. This leads us to consider the intersection $C^1_{''} = C^1(\hat{P}; \hat{\mathfrak{g}}) \cap X''$ as correction subspace and the restrictions $C|_{C^1_{''}}$ and $\delta|_{C^1_{''}}$ as correction functional and δ-map, respectively. It is straightforward to check that this really defines a complete and injective extension of S^G, which we denote by $\mathcal{I}(\hat{S}^{\hat{G}})$. By construction, if $C^1(\hat{P}; \hat{\mathfrak{g}}) \subset \mathrm{Conn}(\hat{P}; \hat{\mathfrak{g}})$, i.e, if the starting extension is small, then the same happens to $C^1_{''}$, so that the rule $\hat{S}^{\hat{G}} \mapsto \mathcal{I}(\hat{S}^{\hat{G}})$ preserves the class of small extensions. Of course, we have a morphism $F : \mathcal{I}(\hat{S}^{\hat{G}}) \to \hat{S}^{\hat{G}}$ given by the inclusions $X'' \hookrightarrow X$ and $C^1_{''} \hookrightarrow C^1(\hat{P}; \hat{\mathfrak{g}})$, which is clearly a monomorphism, finishing the proof.

5.3 Proof of Lemma 5.3

Let $\hat{S}^{\hat{G}} : \widehat{\mathrm{Conn}}(\hat{P}; \hat{\mathfrak{g}}) \to \mathbb{R}$ be an injective extension of S^G with small correction subspace $C^1(\hat{P}; \hat{\mathfrak{g}})$ and map $\delta : C^1(\hat{P}; \hat{\mathfrak{g}}) \to \mathrm{Conn}(P; \mathfrak{g})$. We can then form the diagram below, which is commutative. For each gauge fixing σ we can compute the pullback and then there exists a unique map μ, as below. This map is injective. Indeed, for every global section $\overline{\sigma}$ fo $\overline{\pi}$ one can write $S^G \circ \pi_1 \circ \mu = \overline{\hat{S}^{\hat{G}}} \circ \overline{\sigma} \circ \jmath \circ \mathrm{inc}$. Since $\hat{S}^{\hat{G}}$ is injective, the right-hand side is an injection, which implies that the first map of the left-hand side, μ, is an injection too. By Lemma 5.1, the pullback defines a pullback-type extension. Thus, because μ is an injection, $\hat{S}^{\hat{G}}$ is of pullback-type. The construction above shows that the choice of a gauge fixing σ induces a map $r_\sigma : \mathrm{Ext}_0(S^G; \hat{G}) \to \mathrm{PbExt}(S^G; \hat{G})$. Now, let $\hat{S}^{\hat{G}}$ be a pullback-type extension, regard it as a injective extension and notice that the construction above applied to it does not do anything. This means that r_σ is a retraction, as desired.

$$
\begin{tikzcd}
C^1(\hat{P};\hat{\mathfrak{g}}) \arrow[r, dashed, "\mu"'] \arrow[rr, bend left, "\pi\circ\mathrm{inc}"] \arrow[rrr, bend left=30, hook, "\mathrm{inc}"] \arrow[dr, "\delta"'] & \mathrm{Pb}_\sigma \arrow[r, "\pi_2"] \arrow[d, "\pi_1"] & \mathrm{Conn}(\hat{P};\hat{\mathfrak{g}})/\mathrm{Gau}_{\hat{G}}(\hat{P}) \arrow[r, shift right, "\sigma"'] \arrow[d, hook, "\overline{\jmath}"'] & \mathrm{Conn}(\hat{P};\hat{\mathfrak{g}}) \arrow[l, shift right, "\pi"'] \arrow[d, hook, "\jmath"] \\
 & & \widehat{\mathrm{Conn}}(\hat{P};\hat{\mathfrak{g}})/\mathrm{Gau}_{\hat{G}}(\hat{P}) \arrow[d, "\overline{\hat{S}^{\hat{G}}}"'] & \widehat{\mathrm{Conn}}(\hat{P};\hat{\mathfrak{g}}) \arrow[l, "\overline{\pi}"'] \arrow[dl, "\hat{S}^{\hat{G}}"] \\
 & \mathrm{Conn}(P;\mathfrak{g}) \arrow[r, "S^G"'] & \mathbb{R} &
\end{tikzcd}
\tag{8}
$$

Let $\hat{S}_1^{\hat{G}}$ and $\hat{S}_2^{\hat{G}}$ be complete, injective and with small correction subspace extensions of S^G. Thus, for each of them we have the commutative diagram (8). If they are equivalent, both commutative

16

diagrams are linked via the bijections f and g, producing a bigger diagram[5]. It is commutative due to the commutativity of (4). Furthermore, it reduces to a diagram (4) for $\hat{S}_i^{\hat{G}}$ regarded as pullback-type extensions. This means that $\hat{S}_1^{\hat{G}} \simeq \hat{S}_2^{\hat{G}}$ implies $r_\sigma(\hat{S}_1^{\hat{G}}) \simeq r_\sigma(\hat{S}_2^{\hat{G}})$, so that r_σ passes to the quotient. Let $\overline{r}_\sigma$ be the induced map on the quotient spaces. Since r_σ is a retraction, $\overline{r}_\sigma$ is too. Thus, it is enough to prove that $\overline{r}_\sigma$ is injective. Assume $r_\sigma(\hat{S}_1^{\hat{G}}) \simeq r_\sigma(\hat{S}_2^{\hat{G}})$. This means that we have a diagram (4) for $r_\sigma(\hat{S}_i^{\hat{G}})$. But we also have the commutative diagram (8) corresponding to each $r_\sigma(\hat{S}_i^{\hat{G}})$. And these diagrams are linked due to the maps defining the equivalence $r_\sigma(\hat{S}_1^{\hat{G}}) \simeq r_\sigma(\hat{S}_2^{\hat{G}})$. Therefore, we have again a bigger commutative diagram, which again reduces to a diagram (4), now for $\hat{S}_1^{\hat{G}}$ and $\hat{S}_2^{\hat{G}}$. Thus, $\hat{S}_1^{\hat{G}} \simeq \hat{S}_2^{\hat{G}}$, finishing the proof.

5.4 Proof of Lemma 5.4

We have to prove that the relation $\lesssim$ is a total order in the class of pullback-type extensions. Thus, given two pullback-type extensions we have to prove that there exists a morphism between them. This is a direct consequence of the following lemma.

Lemma 5.5. *In the notations of Theorem 5.1, let* $\mathrm{Conn}(\hat{P};\hat{\mathfrak{g}}) \subset X_i \subset \Omega^1(\hat{P};\hat{\mathfrak{g}})$, *with* $i = 0, 1$, *be two injective* $\mathrm{Gau}_{\hat{G}}(\hat{P})$*-invariant subsets for a map* $S : \Omega^1(\hat{P};\hat{\mathfrak{g}}) \to \mathbb{R}$. *If* $X_0 \subset X_1$, *then for every gauge fixing* σ *we have* $X_0(\sigma) \subset X_1(\sigma)$, *where* $X_i(\sigma)$ *are the subsets of previous theorem. Furthermore, the diagram below commutes, meaning that* S_1 *is a complete extension of* S_0.

$$\begin{tikzcd}
X_0(\sigma) \arrow[r, hook, "\jmath_0 \circ \overline{\sigma}_0"] \arrow[d, hook] & X_0 \arrow[r, "S_0"] \arrow[d, hook] & \mathbb{R} \arrow[d, equal] \\
X_1(\sigma) \arrow[r, hook, "\jmath_1 \circ \overline{\sigma}_1"'] & X_1 \arrow[r, "S_1"'] & \mathbb{R}
\end{tikzcd} \tag{9}$$

Idea of proof. The proof is a bit straightforward. The idea is the following. First build diagrams (6) for X_0 and X_1 and notice that they are linked by the inclusion $X_0 \hookrightarrow X_1$, as below. Since S_0 and S_1 arose as restrictions of the same functional S, it follows that the entire diagram commutes.

$$\begin{tikzcd}
\mathrm{Conn}(\hat{P};\hat{\mathfrak{g}}) \arrow[r, "\pi"] \arrow[d, hook, "\jmath_0"] \arrow[dd, bend right, "\jmath_1"'] & \mathrm{Conn}(\hat{P};\hat{\mathfrak{g}})/\,\mathrm{Gau}_{\hat{G}}(\hat{P}) \arrow[d, hook, "\overline{\jmath}_0"] \arrow[dd, bend left, "\overline{\jmath}_1"] \\
X_0 \arrow[r, "\pi"] \arrow[d, hook] \arrow[dd, bend right, "S_0"'] & X_0/\,\mathrm{Gau}_{\hat{G}}(\hat{P}) \arrow[d, hook] \arrow[ddl, "\overline{S}_0"'] \\
X_1 \arrow[r, "\pi"'] \arrow[d, "S_1"] & X_1/\,\mathrm{Gau}_{\hat{G}}(\hat{P}) \arrow[dl, "\overline{S}_1"] \\
\mathbb{R} &
\end{tikzcd}$$

Taking the pullback of each diagram as in (7) and using the universality of pullbacks we get an universal map $\eta : \mathrm{Pb}(S^G, \overline{S}_{0,\jmath_0}) \to \mathrm{Pb}(S^G, \overline{S}_{1,\jmath_1})$ such that $\overline{\jmath}_0 = \overline{\jmath}_1 \circ \eta$. Since $\overline{\jmath}_0$ is a monomorphism,

[5] This bigger diagram is best visualized as a 3D diagram whose base and roof are the copies (8) corresponding to $\hat{S}_1^{\hat{G}}$ and $\hat{S}_2^{\hat{G}}$.

this implies that η is a monomorphism too and therefore an injection. The commutativity of the entire diagram involving both pullbacks implies the commutativity of (9), finishing the proof. $\square$

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6

GEOMETRIC REGULARITY

6.1 SYNTHESIS

Recall that Hilbert's Program was motivated by its axiomatization of geometry [76, 77]. In the same formalistic perspective, one could ask for an axiomatization of differential geometry. Two main approaches have been developed, both for the internal logic of certain topos admitting a notion of "infinitesimal neighborhood" around its objects:

1. *synthetic differential geometry*, for smooth toposes whose infinitesimal neighborhood is given by infinitesimal interval objects;
2. *differential cohesive geometry*, for arbitrary topos with differential cohesion and infinitesimal shape modality. In this case, the infinitesimal neighborhood is the canonical one.

Both approaches provides axiomatic definitions of tangent bundles, jet bundle, differential equations, differential forms, and so on, including even an axiomatic approach to Lie theory - see [102, 95, 126, 31] for the case of synthetic differential geometry and [145, 91, 113, 55] for differential cohesive geometry. In addition, differential cohesion also provides axiomatization of principal connections, Cartan connections, and many other geometric entities[1] [145, 161, 113].

We note, however, that both approaches do not seem to capture a general notion of "regular object". This follows from the fact that they focus on providing an axiomatization for *smooth* differential geometry. But, in practical situations, one needs to consider differential geometric entities which are not smooth, but regular in some other sense.

For instance, one could consider differential equations whose coefficients are of class C^k, with $k < \infty$, or which are uniformly bounded, or which have bounded derivatives up to a certain order, and so on. And then one could search for solutions satisfying some regularity, which may exist or not depending on the required regularity. This motivates the following questions:

- *Can synthetic differential geometry and/or differential cohesive geometry be extended in such a way that they provide "regular neighborhoods" to objects, leading us to talk about "regular differential equations", "regular solutions", and so on?*

[1] Recall form the discussion in Chapter 2 that ∞-topos with differential cohesion are the models for the Modal Homotopy Type Theory presuposed to be the natural formal language to work on Axiomatization Program of Physics.

The first step is to give a formal definition of what is "regular manifold" and what is "regular object" over a "regular manifold". In the sequence, one should give a characterization of these definitions purely in terms of certain toposes, allowing an axiomatization in the internal languages of these toposes, in such a way that the starting notions constitute an interpretation.

6.1.1 Hairer's Regularity Structures

Between 2013 and 2014 Martin Hairer developed a theory of what he called *regularity structures* aiming to study semilinear (stochastic) partial differential equations (PDEs) and to search for solutions which are regular in this new more ample sense [68, 67, 69]. For instance, normally the solutions of PDEs are supposed to be distributional, but the product between two distributions is not globally defined. Instead, Hairer's theory allows the search for solutions that are in correspondence with distributions (i.e., which can be used to "model" distributions) at the same time that their multiplication is well-defined - see Theorem 1.3 and Theorem 1.4 of [67].

The above suggests us to think of a family B_i of Banach spaces, describing different layers of "abstract functions", with i varying on a set $A \subset \mathbb{R}$, and continuous linear maps $\Pi : B \to \mathcal{S}'(\mathbb{R}^m)$ assigning to each abstract function $f \in B = \oplus_i B_i$ a genuine distribution $\Pi(f)$ on $\mathbb{R}^m$ for some $m > 0$. This idea is part of the essence of Hairer's notions of regularity structures and models over it. He also assumed that:

1. for each $x \in \mathbb{R}^m$ there is one of such continuous linear maps $\Pi_x : B \to \mathcal{S}'(\mathbb{R}^m)$;
2. for $x, y \in \mathbb{R}^m$ there is an automorphism $\Gamma_{xy} : B \to B$ satisfying the cocycle conditions $\Gamma_{xx} = id$ and $\Gamma_{xy} \circ \Gamma_{yz} = \Gamma_{xz}$;
3. for different $x, y \in \mathbb{R}^m$ the maps Π_x and Π_y are related by $\Pi_y = \Pi_x \circ \Gamma_{xy}$.

Some technical analytical conditions, needed for his purposes, are also supposed - see Definition 2.1 and Definition 2.17 of [67].

Hairer's theory on regularity structures was used to solve important problems, turning the method very used and studied since then - see, e.g., [58, 16, 24, 36] and the other more than 280 Crossref citations and more than 700 Google Scholar citations of his main work [67] which we found in a search in January 07 of 2021.

Actually, Hairer awarded 2014 Fields Medal [6]:

> *For his outstanding contributions to the theory of stochastic partial differential equations, and in particular for the creation of a theory of regularity structures for such equations.*

He was also awarded with 2021 Breakthrough Prize [5]:

> *For transformative contributions to the theory of stochastic analysis, particularly the theory of regularity structures in stochastic partial differential equations.*

Thus, it would be minimum to say that Hairer's theory was very succeeded in its main purpose of providing a tool to solve problems on PDEs. However, as far as we could observe, Hairer had not objected to building a general theory of regular object thinking in providing an extension of synthetic differential geometry and/or differential cohesive geometry. Indeed, although he introduced an abstract object, it seems that his focus was mainly on concrete applications instead of on the development of a general theory for such abstract objects.

For example, in a theory of "regular objects," one expects to have a notion of morphism between them in such a way that they fit into a category **Reg**. One could them study limits and colimits in **Reg** corresponding to the possibility of doing constructions between "regular objects". One could also ask if **Reg** is equivalent to some sheaf category, meaning that "regular objects" are local entities, and that **Reg** is a Grothendieck topos. And so on.

However, as far as we could understand, in his works, Hairer does not provide a notion of morphism between his regularity structures. In the very beginning of Section 2.4 of [67], p. 28, Hairer says (quotation marks by him):

> *There is a natural notion of "automorphism" of a given regularity structure.*

This notion of "automorphism" is presented in Definition 2.28, p. 29, by means of what he called *the automorphism group* $\mathrm{Aut}(\mathcal{T})$ of a regularity structure $\mathcal{T}$. Inevitably one could ask if there is a category of regularity structure whose automorphism group of $\mathcal{T}$ is $\mathrm{Aut}(\mathcal{T})$. However, in the sequence, in Remark 2.29 he noticed that $\mathrm{Aut}(\mathcal{I})$ may not be a group, but

> *[...] In most cases of interest however, the index* $\mathrm{ind}(A)$ *is finite, in which case* $\mathrm{Aut}(\mathcal{T})$ *is always an actual group.*

Thus, such a category could not exist if one thinks in considering all regularity structures and not only of those which are "in most cases of interest". On the other hand, this does not deny the existence of a **Reg** of all regularity structures and a subcategory $\mathbf{Reg}_0 \subset \mathbf{Reg}$ of those "cases of interest" whose categorical automorphism group of $\mathcal{T} \in \mathbf{Reg}_0$ is exactly $\mathrm{Aut}(\mathcal{T})$. In other words, this does not imply the impossibility to put Hairer's theory in the context of a "theory of regular objects", but it is indicative that it was not formulated for that purpose.

Alternatively, instead of taking a theory which is very well succeeded in terms of its applications and then make a lot of efforts to build from it an abstract theory for the underlying "regular objects" (which in this case were already defined thinking in the applications), one could try to extract the essence of these "regular objects" and propose a new notion of "regular objects", now focusing *a priori* in building a theory for them.

6.1.2 Γ-Spaces

In [120, 112] we started a program aiming to provide a general notion of *regular objects* on *regular manifolds* and to study existence and multiplicity of arbitrarily regular objects on sufficiently regular manifolds.

The first versions of these works are from 2019 and we were unknown on Hairer's work until January 2021 during the preparation of a talk for the I Encontro Brasileiro em Teoria das Categorias [114]. Curiously, as will be commented below, our notion of regular object can be viewed as a generalization of part of the data defining Hairer's regularity structures, while part of Hairer's model on a regularity structure is captured as a morphism from a regular object to an a priori fixed regular object.

As Hairer, our aim was not to develop a formal theory of regular objects to be axiomatized and then possibly provide an extension of synthetic differential geometry and/or differential cohesive geometry. Indeed, we were interested in finding normal form theorems for physical observables described by sufficiently regular geometric objects on differentiable manifolds. This naturally leads us to ask what is a regular geometric object and when it exists. Normally the regularity of an object over a manifold M is constrained by the regularity of M (e.g., C^k-maps can exist only between C^l-manifolds, with $l \leq k$), so that we would also ask "what is a regular manifold".

However, maybe for our personal preference for categorical language, instead of focusing on our main application, we spent some time trying the study the category of those regular objects and analyzing some properties of it. In this sense, our approach to "regular objects" is maybe closer to providing an axiomatic theory than Hairer's regularity structure theory, but at present moment we have not focused on this point.

As Hairer, we also consider a family of spaces B_i, with i varying on a set Γ of indexes, which will describe different layers of regularity. But, to us, Γ is an arbitrary set, while for Hairer it is a bounded from below subset of real numbers with no accumulation points. Furthermore, for Hairer each B_i is a Banach space, while to us they are just nuclear Fréchet space. We need Fréchet spaces (instead of Banach spaces) in order to include spaces of smooth functions as examples; the nuclear hypothesis is used to ensure a nice topological tensor product and, therefore, a nice way to talk about multiplication.

We call $(B_i)_{i\in\Gamma}$ a Γ-*space* in contrast to Hairer's *model space*. We can define a category of Γ-spaces by noticing that they are equivalent to functors $B : \mathbf{E}\Gamma \to \mathbf{NFre}$ from the discrete category of Γ to the category of nuclear Fréchet spaces and continuous linear maps. Thus, Γ-spaces defines a functor category which we denote by $\mathbf{NFre}_{\Gamma}$.

Remark 6.1. A morphism between B_i and B'_i is then just a family of continuous linear maps $f_i : B_i \to B'_i$. We can define constant Γ-spaces as $B_i = B_0$ for every $i \in \Gamma$. A morphism between two constant Γ-spaces B_0 and B_1 is a family of maps $f_i : B_0 \to B_1$, one for each $i \in \Gamma$. Let (B_i) be a Γ-space and consider a constant $\mathbb{R}^m$-space on $B = \oplus_i B_i$. The maps $\Pi_x : B \to \mathcal{S}'(\mathbb{R}^m)$, with $x \in \mathbb{R}^m$, appearing as part of Hairer's definition of a model to a regularity structure (see above) are equivalent to a morphism between constant $\mathbb{R}^m$-spaces B and $\mathcal{S}'(\mathbb{R}^m)$.

The topological tensor product on nuclear Fréchet spaces makes $\mathbf{NFre}$ a symmetric monoidal category and it induces a functor

$$\otimes_{\Gamma} : \mathbf{NFre}_{\Gamma} \times \mathbf{NFre}_{\Gamma} \to \mathbf{NFre}_{\Gamma\times\Gamma},$$

so that $\mathbf{NFre}_{\Gamma}$ has not an induced monoidal structure and,therefore, we cannot talk directly about monoid objects in the category $\mathbf{NFre}_{\Gamma}$. These would be Γ-spaces B endowed with morphims $* : B \otimes_{\Gamma} B \to B$. The problem is that the right-hand side need to be some $(\Gamma \times \Gamma)$-space and, therefore, it cannot be B. A replacement for B can be obtained noticing that each fixed $f : \Gamma \to \Gamma'$ induces a functor $f^* : \mathbf{NFre}_{\Gamma'} \to \mathbf{NFre}_{\Gamma}$. Thus, fixed $\epsilon : \Gamma \times \Gamma \to \Gamma$ we have a way to turn every Γ-space B into a $(\Gamma \times \Gamma)$-space $\epsilon^* B$, leading us to define a "monoid" in $\mathbf{NFre}_{\Gamma}$ as a Γ-space B endowed with a map $\epsilon : \Gamma \times \Gamma \to \Gamma$ and with a multiplication $*_{\epsilon} : B \otimes_{\Gamma} B \to \epsilon^* B$.

We say that $(\epsilon, *_{\epsilon})$ defines a *multiplicative structure* on B. A *multiplicative* Γ-*space* is one in which a multiplicative structure was chosen. Categories of (B, ϵ)-spaces and of multiplicative (B, ϵ)-spaces can be defined analogously. We can also study the compatibility between two multiplicative structures using distributivity conditions, and providing a notion similar to distributive monoids in a distributive monoidal category (or, more generally, in a rig category), leading to the notion of *distributive* Γ-*space*.

Example 6.2. As discussed in Example 2.2 - 2.6 of [120] and in Example 2.1 - 2.5 of [112], for any function $p : \Gamma \to [0, \infty]$ and any well-behaved open set $U \subset \mathbb{R}^n$ the sequences B_i given by $C^{p(i)}$, $L^{p(i)}$ and $W^{p(i),q}$ are Γ-spaces. For specific p, they admit multiplicative and distributive structures, which in the case of $L^{p(i)}$ and $W^{p(i),q}$ are obtained from Young's inequality, Hölder's inequality and Sobolev embeeding theorems.

Remark 6.3.

1. About the extension of a multiplicative structure to a distributive structure, see also Example 2.9 of [120].

2. Morphism of multiplicative and distributive Γ-spaces are defined analogously to morphisms between monoid objects and distributive monoid objects - see diagrams on pages 5-8 of [120] or, equivalently, equations in page 4 of [112]. Thus, we have categories $\mathbf{NFre}_{\Gamma,*}$ and $\mathbf{NFre}_{\Gamma,+,*}$ of multiplicative and distributive Γ-spaces, respectively.

6.1.3 Presheaves of Γ-Spaces

The absence of sheaf-theoretic methods in Hairer's works of regularity structures is probably because he was mainly interested in stochastic PDEs on a fixed open set $U \subset \mathbb{R}^n$. The extension of his methods to stochastic PDEs to manifolds would naturally lead to introducing sheaves. But, as remarked by Hairer and Steele in a research project at the Department of Mathematics of Imperial College London [70]:

> *Over the past few years, significant progress has been made in understanding the behavior and construction of solutions to very singular stochastic partial differential equations [...] Unfortunately, besides a few specific examples, current techniques are mostly restricted to the case in which the ambient space is flat. The goal of this project is to attempt to extend them to curved spaces. Such an extension would be a significant feat as it is not at all clear how to define even the most basic objects of the theory in this case.*

On the other hand, as commented above, we aim to study regular objects on regular manifolds. This leads us to consider presheaves of Γ-spaces defined on some model space. In the case of manifolds, this is $\mathbb{R}^n$.

Definition 6.4.

1. A *presheaf of* Γ*-spaces* on $\mathbb{R}^n$ is a functor $B : \mathrm{Op}(\mathbb{R}^n)^{op} \to \mathbf{NFre}_\Gamma$, where $\mathrm{Op}(\mathbb{R}^n)$ is the category of open sets and inclusions. Their category is denoted by $\mathrm{PShv}_\Gamma(\mathbb{R}^n)$;

2. A *presheaf of multiplicative (resp. distributive)* Γ*-spaces* is one which takes values into $\mathbf{NFre}_{\Gamma,*}$ (resp. $\mathbf{NFre}_{\Gamma,+,*}$). Their category is denoted by $\mathrm{PShv}_{\Gamma,*}(\mathbb{R}^n)$ (resp. Their category is denoted by $\mathrm{PShv}_{\Gamma,+,*}(\mathbb{R}^n)$).

The manifolds we are interested in are C^k-manifolds for some $0 \leq k \leq \infty$, so that being regular would mean being C^k and belonging to some Γ-space B. Thus, we need some way to talk about *intersection* between C^k and a Γ-structures B, so that about B^k*-structures*, as we pass to discuss.

6.1.4 Intersecting Γ-Spaces

Recall that if A and B are subjects of the same object X in a category with pullbacks, then the intersection $A \cap_X B$ is just the pullback of the inclusions. Thus, if $\mathbf{NFre}_\Gamma$ has pullbacks we can define the intersection $B \cap_X B'$ between two Γ-spaces B, B' which are embedded in a third Γ-space X.

Since $\mathbf{NFre}_\Gamma$ is a functor category, it follows that limits/colimits are given objectwise and, therefore, are determined by limits/colimits in $\mathbf{NFre}$. The category $\mathbf{Fre}$ of Fréchet spaces a quasi-abelian category and therefore it has all finite limits and colimits. In particular, it has pullbacks. The inclusion **NFre** $\hookrightarrow$ **Fre** is closed under binary products and taking subspaces (in particular, kernels). Thus, it is also closed by equalizers, which means that **NFre** has pullbacks, allowing us to define the intersection $B \cap_X B'$ between two Γ-spaces B, B' which are embedded in a third Γ-space X.

But, *what if B, B' does not belong to the same space? What if they belong to different spaces?* This motivates us to consider some way to transport this intersection to some other category:

Definition 6.5. A Γ-*ambient* is given by:

1. a category $\mathbf{X}$ with pullbacks;
2. a functor $\gamma_0 : \mathbf{NFre}_\Gamma \to \mathbf{X}$;

Definition 6.6. Given two Γ-spaces B, B', an *abstract intersection structure* between them, denoted by $\mathbb{X}$, is given by:

1. a Γ-ambient $(\mathbf{X}, \gamma_0)$;
2. an object $X \in \mathbf{X}$;
3. monomorphisms $\imath : \gamma_0(B) \to X$ and $\imath' : \gamma_0(B') \to X$.

Definition 6.7. The *abstract intersection* between two Γ-spaces B, B' relatively to a given abstract intersection structure is the pullback $\mathrm{pb}(\imath, \imath')$.

In the case of multiplicative/distributive Γ-spaces we want to intersect not only the Γ-spaces, but also their multiplication/distributive structures:

Definition 6.8. A *distributive* Γ-*ambient* is given by:

1. a category $\mathbf{X}$ with pullbacks;
2. a functor $\gamma_0 : \mathbf{NFre}_\Gamma \to \mathbf{X}$;
3. a functor $\gamma_1 : \mathbf{NFre}_{\Gamma\times\Gamma} \to \mathbf{X}$.

Definition 6.9. Given two distributive Γ-spaces B, B', an *abstract intersection structure* between them, denoted by $\mathbb{X}$, is given by:

1. a distributive Γ-ambient $(\mathbf{X}, \gamma_0, \gamma_1)$;
2. objects $X, X_*, X_+ \in \mathbf{X}$;
3. monomorphisms $\imath_0 : \gamma_0(B) \to X$ and $\imath'_0 : \gamma_0(B') \to X$;
4. monomorphisms $\imath_{1,*} : \gamma_1(B_\epsilon) \to X_*$ and $\imath'_{1,*} : \gamma_1(B'_{\epsilon'}) \to X_*$;
5. monomorphisms $\imath_{1,+} : \gamma_1(B_\delta) \to X_+$ and $\imath'_{1,+} : \gamma_1(B'_{\delta'}) \to X_+$.

Definition 6.10. The *abstract intersection* between two distributive Γ-spaces B, B' relatively to a given abstract intersection structure is the tuple of pullbacks given by $\mathrm{pb}(\imath_0, \imath'_0)$, $\mathrm{pb}(\imath_{1,*}, \imath'_{1,*})$ and $\mathrm{pb}(\imath_{1,+}, \imath'_{1,+})$.

By definition, the resulting abstract intersection exists only in the ambient category $\mathbf{X}$. We would then like to "concrectify" it, in the sense that there exists some (distributive) Γ-space realizing the pullbacks describing the abstract intersection. This is clearly the case if γ_0 and γ_1 reflect pullback, but this would be equivalent to saying that B, B' could, a priori, be embedded in some Γ-space, which is what we are trying to avoid.

Definition 6.11. An abstract intersection $\mathrm{pb}(\imath_0, \imath_0')$ between two Γ-spaces B, B' is said to be *concrete* if there is a Γ-space $B \cap_{X,\gamma} B'$ and morphisms $\theta_0 : \gamma_0(B \cap_{X,\gamma} B') \to B$ and $\theta_0' : \gamma_0(B \cap_{X,\gamma} B') \to B'$ such that the map u coming from universality is an isomorphism, as below.

$$\begin{tikzcd}
\gamma_0(B \cap_{X,\gamma_0} B') \arrow[dr, dashed, "u", "\simeq"'] \arrow[drr, bend left, "\gamma_0(\theta_0')"] \arrow[ddr, bend right, "\gamma_0(\theta_0)"'] & & \\
& \mathrm{pb}(\imath_0, \imath_0') \arrow[r] \arrow[d] & \gamma_0(B') \arrow[d, "\imath_0'"] \\
& \gamma_0(B) \arrow[r, "\imath_0"'] & X
\end{tikzcd}$$

Remark 6.12.

1. The discussion above on abstract intersections naturally extends from the category $\mathbf{NFre}_\Gamma$ of Γ-spaces to the category $\mathrm{PShv}_\Gamma(\mathbb{R}^n)$ of presheaves of Γ-spaces, as similarly for multiplicative/distributive Γ-spaces;
2. the definition of concrete instersection also extends to multiplicative/distributive Γ-spaces and to the corresponding presheaf setting. See pages 8-10 of [120] and Section 2.3 of [112].

6.1.5 B^k-Structure

To define regular objects on regular manifolds, let us first define what is a regular manifold. Recall that a C^k-manifold is a paracompact Hausdorff topological space M with a C^k-structure $\mathcal{A}$. This, in turn, is a collection $\varphi_a : U_a \to \mathbb{R}^n$ of continuous maps, which are homeomorphisms onto their images, and such that transition functions $\varphi_{ab} : \varphi_a(U_{ab}) \to \mathbb{R}^n$, where $U_{ab} = U_a \cap U_b$ and $\varphi_{ab} = \varphi_b \circ \varphi_a^{-1}$, are of class C^k. This *implies* that the derivatives $\partial^i \varphi_{ab}$ with $0 \leq i \leq k$ are of class $k - i$.

Let $C^{k-}_m : \mathrm{Ob}(\mathbb{R}^n) \to \mathbf{NFre}_{[0,k]}$ be the presheaf of $[0,k]$-spaces given by $(C^{k-}_m)_i(U;\mathbb{R}^n) = C^{k-i}(U;\mathbb{R}^m)$. By the above, a C^k-structure on M is equivalently:

1. an open covering (U_a) of M;
2. to each U_a a continuous map $\varphi_a : U_a \to \mathbb{R}^n$ which is a homeomorphism onto its image
3. whose induced transition functions satisfies $\partial^\mu \varphi_{ab} \in (C^{n^{k-})_i(U_{ab})}$ for every $i = 0, ..., k$ and every μ such that $|\mu| = i$.

This provides a direct extension to the notion of C^k-strucure by replacing the presheaf of $[0,k]$-spaces[2] C^{k-}_n with some intersection $C^{k-}_n \cap_X B$. A priori this would only makes sense if B was a presheaf of $[0,k]$-spaces. But, for any set Γ and any map $\beta : \Gamma \to [0,k]$ we can define a presheaf of Γ-spaces $C^{k-\beta}_n$ by $(C^{k-\beta}_n)_i(U) = C^{k-\beta(i)}(U;\mathbb{R}^n)$.

Definition 6.13. Let B be a presheaf of Γ-spaces and $\alpha : \Gamma \to \Gamma$ and $\beta : \Gamma \to [0,k]$ two maps. Suppose given a concrete intersection structure $\mathbb{X}$ between B_α and $C^{k-\beta}$. A function $f : U \to \mathbb{R}^m$, with $U \subset \mathbb{R}^n$, is a *(B,k,α,β)-function in* $\mathbb{X}$ if $\partial^\mu f$ belongs to

$$(C^{k-\beta}_m \cap_X B)_i(U) = (C^{k-\beta}_m)_i(U) \cap_{X(U)} B_i(U) = C^{k-\beta(i)}(U;\mathbb{R}^m) \cap_{X(U)} B_i(U)$$

for every $i = 1, .., k$ and every μ such that $|\mu| = i$.

[2]Here, $[s,t] := s, ..., t$ is the integer interval.

Definition 6.14. Let M be a paracompact Hausdorff topological space. A *n-dimensional $B^k_{\alpha,\beta}$-structure* on M, denoted by $\mathcal{B}^k_{\alpha,\beta}(\mathbb{X})$, is given by:

1. a presheaf B of Γ-spaces;
2. functions $\alpha : \Gamma \to \Gamma$ and $\beta : \Gamma \to [0,k]$;
3. a concrete intersection structure $\mathbb{X}$ between $C^{k-\beta}_n$ and B_α;
4. an open covering $(U_a)_a$ of M;
5. to each U_a a continuous map $\varphi_a : U_a \to \mathbb{R}^n$ such that
   (a) they are homeomorphism onto their images;
   (b) for every a, b such that $U_{ab} \neq \varnothing$ the transition functions φ_{ab} are (B,k,α,β)-functions.

Definition 6.15. A $(B^k_{\alpha,\beta},\mathbb{X})$*-manifold of dimension n* is a topological space endowed with a n-dimensional $B^k_{\alpha,\beta}$-structure $\mathcal{B}^k_{\alpha,\beta}(\mathbb{X})$.

Remark 6.16. The obvious next step would think in defining morphisms between $(B^k_{\alpha,\beta},\mathbb{X})$-manifolds and then ask whether they provide a category of $(B^k_{\alpha,\beta},\mathbb{X})$-manifolds. This is discussed in Section 5-6 of [120] and will be reviewed below.

6.1.6 B^k-Objects

Now we defined regular manifolds one could define regular geometric objects on them. To do so, notice that the typical geometric objects on a n-dimensional C^k-manifold M include bundles, connections, tensors, and differential operators. They have a common feature: are *local* objects and, therefore, are determined by the amount of information defined on an open covering $(U_a)_a$ of M.

This local data is actually a bunch of functions $f^I_J : \varphi_a(U_{abc..}) \to \mathbb{R}^N$, where I and J are collections of indexes and N is natural depending on n. Thus, the main idea to define regular objects is to require that these functions are actually (B,k,α,β)-functions. This makes perfect sense to define geometric objects on C^k*-manifolds*:

Definition 6.17 (roughly). Let $(M,\mathcal{A})$ be a C^k-manifold, B be a presheaf of Γ-space, α,β functions and $\mathbb{X}$ a concrete intersection structure between B_α and $C^{k-\beta}_N$. We say that a local geometry object Θ on M is a $(B^k_{\alpha,\beta},\mathbb{X})$-object if there exists an open covering of M by charts $\varphi_a \in \mathcal{A}$ such that the local expression $\Theta^I_J : \varphi_a(U_{abc..}) \to \mathbb{R}^N$ of Θ are (B,k,α,β)-functions in $\mathbb{X}$.

But our deal was to define regular objects on *regular manifolds*. Thus, we need to define regular objects when $(M,\mathcal{A})$ is not only a C^k-manifold, but actually a $B^k_{\alpha,\beta}$-manifold $(M,\mathcal{B}^k_{\alpha,\beta}(\mathbb{X}))$. The natural idea is to reproduce definition above but requiring a condition on the local expression $f^I_J : \varphi_a(U_{abc..}) \to \mathbb{R}^N$ defined for charts on $\mathcal{B}^k_{\alpha,\beta}(\mathbb{X})$. However, as discussed in Section 2.8 of [112], this would provide a rigid notion, leading us to keeping the presheaf of Γ-structures B fixed, but allowing a changing on the functions α,β and on the intersection structure $\mathbb{X}$.

Definition 6.18 (roughly). Let $(M,\mathcal{B}^k_{\alpha,\beta}(\mathbb{X})$ be a $(B^k_{\alpha,\beta},\mathbb{X})$-manifold. Given functions $\alpha' : \Gamma \to \Gamma$ and $\beta' : \Gamma \to [0,k]$ and a concrete instersection structure $\mathbb{Y}$ between $B_{\alpha'}$ and $C^{k'-\beta}_N$, a $(B^k_{\alpha',\beta'},\mathbb{Y})$-geometric object on M is a geometric object Θ on the underlying C^k-manifold whose local expression $\Theta^I_J : \varphi_a(U_{abc..}) \to \mathbb{R}^N$ on charts of $\mathcal{B}^k_{\alpha,\beta}(\mathbb{X})$ are (B,k,α',β')-functions in $\mathbb{Y}$.

Remark 6.19. This approach excludes nonlocal objects such as pseudo-differential operators and other more general nonlocal operators. These, however, are contemplated by Hairer's regularity structures. Thus, in certain sense, one could say that B^k-structures and regularity structures work on complementary domains: while currently there is no extension of Hairer's theory to manifolds, currently B^k-structures can be used applied only to local objects.

6.1.7 Main Results

The basic problems concerning any new mathematical concept are existence and uniqueness/multiplicity. The next one, provided existence and lack of uniqueness, is the classification problem. The latter is more interesting when the underlying concepts are part of a category. This leads us to ask:

- *there is some notion of morphism between arbitrary $(B^k_{\alpha,\beta},\mathbb{X})$-manifolds such that they fit into a category $\mathbf{Diff}^{B,k}_{\alpha,\beta}(\mathbb{X})$?*

Of course, since $(B^k_{\alpha,\beta},\mathbb{X})$-manifolds are C^k-manifolds with extra stuff, we also expect a forgetful functor $U : \mathbf{Diff}^{B,k}_{\alpha,\beta}(\mathbb{X}) \to \mathbf{Diff}^k$.

The obvious approach is the following:

Definition 6.20. A *morphism* between two $(B^k_{\alpha,\beta},\mathbb{X})$ manifolds $(M,\mathcal{B}^k_{\alpha,\beta}(\mathbb{X}))$ and $(N;\mathcal{B}'^k_{\alpha,\beta}(\mathbb{X}))$ is a C^k-function $f : M \to N$ between the underlying C^k-manifolds whose local expression $f : \varphi(U) \to \mathbb{R}^m$ by charts $\varphi \in \mathcal{B}^k_{\alpha,\beta}(\mathbb{X})$ and $\psi \in \mathcal{B}'^k_{\alpha,\beta}(\mathbb{X})$ are (B,k,α,β)-functions in $\mathbb{X}$.

This is a good definition in the sense that it generalizes the notion of C^k-maps in a natural way. It is enough to show that the composition of morphisms is not only a C^k-map, but also a morphism, and that the identity map is a morphism.In this case, it will not only define the category $\mathbf{Diff}^{B,k}_{\alpha,\beta}(\mathbb{X})$ but also provide the forgetful functor to C^k-manifolds.

However, Lemma 5.1 of [120] shows that, for arbitrary B, α, β and $\mathbb{X}$, the composition of morphisms is not a morphism. This happens if (B,k,α,β) is *order preserving* in $\mathbb{X}$, in a certain sense made explicit in p.20-21 of [112]. Also, the identity map id_M is not a morphism unless (B,k,α,β) is *unital* in $\mathbb{X}$, i.e., if for every $U \subset \mathbb{R}^n$ the identity map $id_U : U \to \mathbb{R}^n$ is a (B,k,α,β)-function in $\mathbb{X}$.

Proposition 6.21. *If (B,k,α,β) is unital and order preserving in $\mathbb{X}$, then the $(B^k_{\alpha,\beta},\mathbb{X})$-manifolds and their morphism defines a category $\mathbf{Diff}^{B,k}_{\alpha,\beta}(\mathbb{X})$ which comes with a forgetful functor to $\mathbf{Diff}^k$.*

Proof. This is Proposition 5.2 of [120]. □

Knowing that the category of $B^k_{\alpha,\beta},\mathbb{X})$-manifolds is well-defined, one could then ask about existence of a $B^k_{\alpha,\beta},\mathbb{X})$-structure on a C^k-manifold.

Lemma 6.22. *A C^k-manifold $(M,\mathcal{A})$ admits a $B^k_{\alpha,\beta},\mathbb{X})$-structure iff the identity map id_M is a (B,k,α,β)-function in $\mathbb{X}$.*

Proof. Straightforward. See also p.19 of [112]. □

One should not expect that this is satisfied for every C^k-manifold M. Actually, one should expect a lot of topological and geometric obstructions depending on B, since as particular examples of $B^k_{\alpha,\beta}, \mathbb{X})$-manifolds one has, e.g., manifolds with G-structures and therefore (almost) symplectic manifold, semi-Riemannian manifolds, etc., whose existence suffer from that kind of obstructions.

In particular, one should not expect the existence of a left or right adjoint to the forgetful functor $U : \mathbf{Diff}^{B,k}_{\alpha,\beta}(\mathbb{X}) \to \mathbf{Diff}^k$. In [112], however, we provide conditions on B, k, α, β and $\mathbb{X}$ for which such left/right adjoint exists and are isomorphic, meaning not only that there is no obstructions to the existence of a $(B^k_{\alpha,\beta}, \mathbb{X})$-structures, but also that each C^k-manifolds has a canonical of such structures.

Theorem 6.23. *If (B, k, α, β) is order preserving, fully absorbing and has retractible (B, k, α, β)-diffeomorphisms, all of this in $\mathbb{X}$, then the forgetful functor $U : \mathbf{Diff}^{B,k}_{\alpha,\beta}(\mathbb{X}) \to \mathbf{Diff}^k$ is ambidextrous adjoint.*

Proof. This is a version of Theorem B., p. 25 of [112]. □

Concerning the existence of regular local geometric objects Θ on $(B^k_{\alpha,\beta}, \mathbb{X})$, recall that, since they are local, they are determined in terms of data on an open covering. Thus, based on the local-to-global proof in the C^k case we can try to build a strategy for the general case.

Theorem 6.24. *On every C^k-manifold there exists C^l geometric objects Θ for certain l depending on k.*

Proof. It consists in the following steps:

1. local existence: showing that Θ exists in each open set U with a local expression f^I_J;
2. *global existence*: given an open covering $\varphi_a : U_a \to \mathbb{R}^n$, take Θ_a in each U_a. Let (ψ_a) a partition of unity of class C^k subordinate to $(U_a)_a$. Showing that $\Theta = \sum_a \psi_a \cdot \Theta_a$ is a globally defined geometric object of class C^k. Denote its local expression in each U_a by $f(a)^I_J$.
3. *local expression*: to determine the differentiabily class l of Θ, for each a let $(f_a)^I_J$ be the local expression of Θ_a. For $U_{ab} \neq \varnothing$, find the coordinate change expression $f_{a;b}(\varphi_{ab}, \partial^{\mu}_{ab})$ such that $(f_a)^I_J = \sum f_{a;b}(f_a)^I_J$. Notice that $f(a)^I_J = \sum_b \psi_b \cdot f_{a;b}$. Thus, the class of $f(a)^I_J$ (and therefore of Θ) is $l \leq |\mu|$. □

Adapting the steps above we find a sketch of how to prove the existence of regular objects on regular manifolds:

Theorem 6.25 (sketch). *On every $(B^k_{\alpha,\beta}, \mathbb{X})$-manifold there are $(B^k_{\alpha',\beta'}, \mathbb{Y})$-objects for α', β' depending on α, β and $\mathbb{Y}$ depending on $\mathbb{X}$.*

Proof. It consists, again, of some steps:

1. *local existence*: using local existence in the C^k case, notice that it extends to any (B, k, α', β') and *any* intersection structure $\mathbb{Y}$, with none restrictions: $\mathbb{Y}$ may be equal to $\mathbb{X}$ and α', β' are whatever;

2. *global existence with weak regularity*: similarly to the globalization step, take an open covering $\varphi_a : U_a \to$ (now on $\mathcal{B}^k_{\alpha,\beta}(\mathbb{X})$) and a subordinate partition unity (ψ_a) and use $\Theta = \sum \psi_a \cdot \Psi_a$ to get a globally defined object. The local expression $(f_a)^I_J = \sum f_{a;b}(f_a)^I_J$, with $f_{a;b}(\varphi_{ab}, \partial^\mu_{ab})$, and the behavior of (B, k, α', β')-functions relatively to addition, multiplication and multiplication by bump functions determines that
$$(f_a)^I_J \in C^{k-\beta'_0}(U_{ab}; \mathbb{R}^N) \cap_{X(U_{ab})B_{\alpha'_0}} (U_{ab}) \tag{6.1}$$
for some $\alpha'_0 \in \Gamma$ and some $\beta'_0 \in [0, k]$;
3. *weak regularity implies regularity*: given $\alpha'_0 \in \Gamma$, $\beta'_0 \in [0, k]$ and and a concrete intersection structure $\mathbb{X}$ between $C^{k-\beta}_N$ and B_α, there are:
   (a) a set $\Gamma' \subset \Gamma$ with $\alpha'_0 \in \Gamma'$;
   (b) functions $\alpha' : \Gamma' \to \Gamma'$ and $\beta' : \Gamma' \to [0, k]$;
   (c) an element $0 \in \Gamma'$ such that $\alpha'(0) = \alpha'_0$ and $\beta'(0) = \beta'_0$;
   (d) a concrete intersection structure $\mathbb{Y}$ between $C^{k-\beta'}_N$ and $B_{\alpha'}$,

   for which to each function $f : U \to \mathbb{R}^N$, with $U \subset M$, which belongs to
$$C^{k-\beta'_0}(U) \cap_{X(U)} B_{\alpha'_0}(U),$$
   corresponds a (B, k, α', β')-function in $\mathbb{Y}$ in such a way that sums, multiplications, and multiplication by bump functions are preserved.
4. *global existence*: applying last step in the functions $(f_a)^I_J$ in (6.1) to get (B, k, α', β')-functions $(g_a)^I_J$ in $\mathbb{Y}$, for some α', β' and $\mathbb{Y}$. Since in the correspondence of last step sums, multiplications and multiplication by bump functions, it follows that, as $(f_a)^I_J$, the maps $(g_a)^I_J$ are the local expression of a geometric object Θ on M, which therefore is a $(B^k_{\alpha',\beta'}, \mathbb{Y})$-object. □

For the case of affine connections, this is essentially the statement and proof of Theorem 3.2 of [112], with steps 1-3 corresponding, respectively, to Proposition 3.1, Theorem 3.1, and Lemma 2.4.

Remark 6.26.

1. The sketch above shows that locally regular objects may exist without restrictions on their regularity, meaning for every $\alpha', \beta', \mathbb{Y}$. However, they can be globalized only for certain $\alpha', \beta', \mathbb{Y}$.
2. This restriction appears precisely when proving that weak regularity implies regularity, which is so far the most technical part of [112] which we call Regularity Globalization Lemma.
3. However, notice that among steps 1-4, the third one is the only which is not about a specific type of geometric object Θ, but actually about conditions on the underlying structural presheaves and on relations between intersection structures. Thus, the Regularity Globalization Lemma proved in [112] could be used, with minor modifications, to a geometric object other than affine connections.

4. Theorem 6.25 can be read as follows: given $\alpha, \beta, \mathbb{X}$, there exists $\alpha'(\alpha)$, $\beta'(\beta)$ and $\mathbb{Y}(\mathbb{X})$ such that if M has a $(B^k_{\alpha,\beta}, \mathbb{X})$-structure, then it admits $(B^k_{\alpha',\beta'}, \mathbb{Y})$-objects. In other words, *arbitrarily regular manifolds admits sufficiently regular geometric objects*. The dependence between α and α', and β and β', can be inverted allowing us to conclude that *arbitrarily regular objects exist on sufficiently regular manifolds*.

Finally, concerning the multiplicity of regular objects, what the experience acquired working on affine connections in Section 4 of [112] reveals, is that many of the algebraic properties of the "space of geometric C^k-objects" extends to the case of arbitrary regularity. For instance, the space of affine connections on M is an affine space of $\mathrm{End}(M)$-valued 1-forms. Similarly, the space of $(B^k_{\alpha',\beta'}, \mathbb{Y})$ is an affine space of the vector space of $\mathrm{End}(M)$-valued (B, α', β') 1-forms in $\mathbb{Y}$ - see Proposition 4.1.

Section 4.1 of [120] closes with a discussion on what we should believe that the space of affine connections can be embedded as an open set in a larger class of regular geometric objects. We do not know anything about denseness results.

6.2 QUESTIONS AND SPECULATIONS

- In [120] we introduce what are regular manifolds and in [112] we formalized and studied what are regular affine connections on them. Of course, it would be interesting to extend the discussion to other kinds of geometric objects as sketched in Sections 6.1.6 and 6.1.7. The first step is to study regular bundles.

  Q6.1 *Define in formal terms $(B^l_{\alpha',\beta'}, \mathbb{X}')$-vector bundles over a $(B^k_{\alpha,\beta}, \mathbb{X})$-manifold, with $l \leq k$. Given a C^l-vector bundle $E \to M$, use of strategy in Theorem 6.25 and of the results in [120] to provide conditions under which E has a $(B^l_{\alpha',\beta'}, \mathbb{X}')$-structure.*

  Q6.2 *Define $(B, l, \alpha'', \beta'')$-morphisms between two C^l-budles $E_1, E_2 :\to M$. Assume that E_i, with $i = 1, 2$ is a $(B^l_{\alpha'_i}, \beta'_i)$-bundle over a $B^k_{\alpha,\beta}, \mathbb{X})$-manifold. Following a discussion similar to that of Section 5 of [120], determine conditions such that these morphims exist and define a category $\mathbf{Bund}^l_{\alpha',\beta'}(M)$ of regular bundles over regular manifolds.*

- The next step would be to study regular sections of regular bundles. Recall that to each C^l-bundle E the space $\Gamma^l(E)$ of C^l-global sections is a vector space and the rule $E \mapsto \Gamma(E)$ is a functor $\Gamma : \mathbf{Vec}(M) \to \mathbf{Vec}_{\mathbb{R}}$. In the *smooth* case, $k = \infty$, $C^\infty(M)$ is a commutative ring and each $\Gamma^l(E)$ is actually a $C^\infty(M)$-module[3].

  Q6.3 *Define $(B, l, \alpha'', \beta'', \mathbb{X}'')$-section of a C^l-vector bundle $E \to M$. Suppose that E is actually a $(B^l_{\alpha',\beta'}, \mathbb{X}')$-bundle over a $(B^k_{\alpha,\beta}, \mathbb{X})$-manifold. Using strategy of Theorem 6.25, prove existence of nontrivial these regular sections of regular bundles over regular manifolds. In the same lines of Proposition 4.1 of [112], show that they actually form vector subspace $\Gamma^l_{\alpha'',\beta''}(E)$ of $\Gamma^l(E)$. If $k = \infty$, show that $\Gamma^l_{\alpha'',\beta''}(E)$ is actually a $C^\infty(M)$-submodule.*

  Q6.4 *Find conditions such that the rule $E \mapsto \Gamma^l_{\alpha'',\beta''}(E)$ extends to a functor $\Gamma^l_{\alpha'',\beta''}$: $\mathbf{Bund}^l_{\alpha',\beta'}(M) \to \mathbf{Vec}_{\mathbb{R}}$.*

- The third step would be to ask whether the linear algebra of vector bundles can be extended to the regular case. More precisely, recall that direct sum $E_1 \oplus E_1$, tensor

[3] If $k < \infty$, then $C^k(M)$ is a ring nor a $\mathbb{R}$-algebra, so that we cannot define vector fields as derivations. An alternative is presented in [128]. As far as we know, there are few books on differential geometry taking the class of differentiabily into account. Lang's book [101] is an exception.

product $E_1 \otimes E_2$ and hom $(Hom)(E_1; E_2)$ are defined for C^l-bundles, being themselves C^l-bundles. Actually, $\oplus$ and $\otimes$ makes $\mathbf{Vec}^l_M$ a distributive monoidal category, while (Hom) provides an internal hom, turning $\mathbf{Vec}^l_M$ into a closed monoidal category.

Similarly, the categories $\mathbf{Vec}_{\mathbb{R}}$ and $\mathbf{Mod}_{C^\infty(M)}$ also are distributive closed monoidal categories with the corresponding direct sums, tensor products, and internal homs. Furthermore, the global section functors $\Gamma^l : \mathbf{Vec}^l(M) \to \mathbf{Vec}_{\mathbb{R}}$ preserves all those stuff.

Q6.5 *Let E_1 and E_2 be C^l-vector bundles and let $E_1 \times E_2$ their tensor bundle. Assume that E_i, with $i = 1, 2$ is a $(B^l_{\alpha'_i}, \beta'_i)$-bundle over a $B^k_{\alpha,\beta}, \mathbb{X})$-manifold. Using that $E_1 \otimes E_2$ is locally the tensor product between the cocycles of E_i, and following Theorem 6.25, find conditions such that $E_1 \times E_2$ is a $(B^l_{\alpha''}, \beta'')$-bundle.*

Q6.6 *Assuming Questions 6.2 and 6.2, analyze whether the tensor product $\otimes$ makes $\mathbf{Bund}^l_{\alpha',\beta'}(M)$ into a symmetric monoidal category. Assuming in addition Question 6.2, prove that the functor of global sections $\Gamma^l_{\alpha'',\beta''} : \mathbf{Bund}^l_{\alpha',\beta'}(M) \to \mathbf{Vec}_{\mathbb{R}}$ is symmetric monoidal.*

Q6.7 *Do the same for the direct sum $\oplus$. Provide conditions such that $\oplus$ and $\otimes$ makes $\mathbf{Bund}^l_{\alpha',\beta'}(M)$ into a distributive monoidal category.*

Q6.8 *Also do the same for the internal hom and give conditions such that $\mathbf{Bund}^l_{\alpha',\beta'}(M)$ becomes a closed symmetric monoidal category.*

- Now we can look at the existence problem of regular tensors on regular bundles. In the following, if $E \to M$ is a C^l-bundle, then For given $r, s \geq 0$, E^r_s is the tensor bundle $E^{\otimes_r} \otimes (E*)^{\otimes_s}$, where $E^{\otimes_n} = E \otimes ... \otimes E$ and $E* = \mathrm{Hom}(E; \underline{\mathbb{R}})$ is the dual bundle and $\underline{\mathbb{R}} = M \times \mathbb{R}$ is the monoidal unity of the tensor product $\otimes$.

Q6.9 *Let E be $(B^l_{\alpha',\beta'}, \mathbb{X}')$-bundle over a $(B^k_{\alpha,\beta}, \mathbb{X})$-manifold M. Assuming Questions 6.2 and 6.2, for every given $r, s \geq 0$, the E^r_s tensor bundle has an induced structure of $(B^l_{\alpha',\beta'}, \mathbb{X}')$-bundle. Use of Questions 6.2 abd 6.2 to show that E^r_s admit $B^l,'', \beta'', \mathbb{X}''$-sections for certain $(\alpha'', \beta'', \mathbb{X}'')$ independently of r, s.*

Q6.10 *On the other hand, for given $r, s \geq 0$, we could considered the C^l-bundle E^r_s and then used Question 6.2 to get directly a $(B^l_{\alpha'_{r,s}, \beta'_{r,s}}, \mathbb{X}'_{r,s})$-structure on E^r_s. From Question 6.2 we would then prove existence of $(B, l.\alpha''_{r,s}, \beta''_{r,s}, \mathbb{X}''_{r,s})$-sections. Compare this regularity with that $(\alpha'', \beta'', \mathbb{X}'')$ of Question 6.2. Determine the critical values (r, s) in which they agree.*

Recall that in the non-smooth case $k < \infty$, the space of global sections $\Gamma(E)$ is just a real vector space. In particular, the functor Γ is monoidal on $\mathbf{Vec}_{\mathbb{R}}$. This means that sections E^r_s are identified with $\mathbb{R}$-multilinear maps so that with *nonlinear tensors* on E. In the smooth setting, however, Γ take values into $\mathbf{Mod}_{C^\infty(M)}$.

Thus, Questions 6.2 and 6.2 argue the existence of nonlinear regular tensors on regular non-smooth manifolds and regular tensors on regular smooth manifolds. In the non-smooth setting one could try to make a change of coefficients on the real vector space $\Gamma(E)$, from $\mathbb{R}$ to $C^\infty(M)$, getting a $C^\infty(M)$-module $\Gamma(E)_{C^\infty(M)}$ which is isomorphic to the canonical $C^\infty(M)$-module $\Gamma(E)$ in the smooth setting. We could then define (linear) tensors on E as $C^\infty(M)$-multilinear maps on $\Gamma(E)_{C^\infty(M)}$, and then use questions above to prove the existence of regular (linear) tensors on regular bundles on non-smooth manifolds.

We have the obvious inclusion $\imath : \mathbb{R} \hookrightarrow C^{\infty}(M)$ regarding numbers as constant maps, which is a ring homomorphism. Thus, the obvious idea would be to take $\Gamma(E)_{C^{\infty}(M)}$ as the extension of scalars induced by $\imath$, i.e., $\imath_*(\Gamma(E)) = \Gamma(E) \otimes_{\mathbb{R}} C^{\infty}(M)$. However, the corresponding scalar restriction is not well-behaved to provide the existence of regular tensors.

Questions 6.2-6.2 and the discussion on how to provide the existence of regular (linear) tensor on the non-smooth setting are being considered in working in progress.

- Nonlinear differential operators of class C^r and degree d on a C^l-vector bundle E, with $r \leq l$ and $d \leq l - r$, can be viewed as nonlinear tensors $D : \Gamma^r(E) \to \Gamma^{r-d}(E)$ which are generated by linear connections $\nabla_1, ..., \nabla_d$ on E. Thus, we could use questions above to study existence problem of regular nonlinear differential operators on regular bundles over regular manifolds.

Q6.11 *Define $(B^l_{\alpha'',\beta''}, \mathbb{X}'')$-connection on a C^l-vector bundle E. Extend the existence results of [112] from regular affine connections on M to regular linear connections on E.*

Q6.12 *Define what is a nonlinear $(B^r_{\alpha'',\beta''}, \mathbb{X}'')$-differential operator of degree d on a C^l-vector bundle E over a C^k-manifold. Suppose that E is actually a $(B^l_{\alpha',\beta'}, \mathbb{X}')$-bundle and that M is a $(B^k_{\alpha,\beta}, \mathbb{X})$-manifold. Given d, r, using of Questions 6.2 and 6.2, find conditions such that nonlinear $(B^r_{\alpha''_d,\beta''_d}, \mathbb{X}''_d)$-differential operators of degree d exist on E. Show that they actually form a vector subspace $\mathrm{Diff}^d_{\alpha''_d,\beta''_d}(E)$ of the space of all $(B^r_{\alpha''_d,\beta''_d}, \mathbb{X}''_d)$-nonlinear tensors.*

Q6.13 *Assuming Question 6.2, find a triple $(\alpha'', \beta'', \mathbb{X}'')$, independent of d, such that there exist nonlinear $(B^r_{\alpha'',\beta''}, \mathbb{X}'')$-differential operators of every degree d. In the smooth case $k, l, r = \infty$, analyze whether the graded vector space $\mathrm{Diff}_{\alpha'',\beta''}(E) = \oplus_d \mathrm{Diff}^d_{\alpha'',\beta''}(E)$ is a graded-algebra with the composition operation.*

- C^k-manifolds are classified by Cech cocycles given by their transition functions. Similarly, vector bundles and vector bundles with connections also are classified by Cech cocycles. It would be interesting to find a "nonlinear Cech cohomology" whose cocycles classify nonlinear tensors and such that it can be extended to the regular setting. In this case, the globalization step in Theorem 6.25 could be studied cohomologically.

- We finish with very vague speculation concerning the possibility of using our notion of "regular object" to extend synthetic differential geometry/differential cohesive geometry to the regular setting.

  Indeed, notice that in [120, 112, 114], in Section 6.1.7 and in questions above we considered regular objects on regular manifolds with "regularity" being given by a structural presheaf $B : \mathrm{Open}(\mathbb{R}^n)^{op} \to \mathbf{NFre}_{\Gamma}$ of Γ-spaces on $\mathbb{R}^n$. A regular manifold was then certain kind of pair (M, B). The idea is try to give a more intrinsic definition, where M and B are part of the *same* entity.

  Recall that a n-dimensional manifold can also be viewed as a sheaf on $\mathbb{R}^n$, but with differential maps instead of continuous inclusions: it is a set. More precisely, let $\mathbf{CSp}^k$ be the category whose objects are $\mathbb{R}^n$, with $n \geq 0$, and whose morphisms are C^k-maps $f : \mathbb{R}^n \to \mathbb{R}^m$. A *smooth set* of class C^k is a presheaf $F : (\mathbf{CSp}^k)^{op} \to \mathbf{Set}$ which is a sheaf relative to the Grothendieck topology given by open covers. Let $\mathrm{Shv}(\mathbf{CSp}^k, J)$

be the Grothendieck topos of smooth sets of class C^k.

Let $\mathbf{Diff}^k$ be the category of C^k-manifolds and C^k-maps between then. Take on it the Grothendieck topology given by open covers. Then $\mathrm{Shv}(\mathbf{Diff}^k, J) \simeq \mathrm{Shv}(\mathbf{CSp}^k, J)$ (see [3]). Yoneda embedding then provide a full embedding $\mathbf{Diff}^k \hookrightarrow \mathrm{Shv}(\mathbf{Diff}^k, J)$ allowing us to regard C^k-manifolds as smooth sets of class C^k.

On the other hand, Let $\mathbf{Op}^k$ be the category of all open sets of $\mathbb{R}^n$, for every $n \geq 0$, with morphisms given by C^k-maps. Take again the Grothendieck topology given by open covers. Then $\mathrm{Shv}(\mathbf{Op}^k, J) \simeq \mathrm{Shv}(\mathbf{CSp}^k, J)$. Furthermore, if Γ is a set with a distinguished element $0 \in \Gamma$, we have a forgetful functor $U : \mathbf{NFre}_\Gamma \to \mathbf{Set}$ such that $U(B) = B(0)$. In this case, the "intrinsic" pairs (M, B) should be those in the external pullback below. and the definition could be immediately extended to arbitrary smooth sets of class C^k, as in the internal pullback below.

$$
\begin{tikzcd}
\mathbf{RegDiff}^k \arrow[rr, hook, dotted] \arrow[rrr, dashed, bend left] \arrow[dd, dashed] & & \mathbf{RegSmooth}^k \arrow[r, dotted] \arrow[dd, dotted] & \mathrm{PShv}(\mathbf{Op}^k; \mathbf{NFre}_\Gamma) \arrow[d, "U"] \\
 & & & \mathrm{PShv}(\mathbf{Op}^k; \mathbf{Set}) \arrow[d, "\simeq"] \\
\mathbf{Diff}^k \arrow[r, hook] & \mathrm{Shv}(\mathbf{Diff}^k, J) \arrow[r, "\simeq"'] & \mathrm{Shv}(\mathbf{CSp}^k, J) \arrow[r, hook] & \mathrm{PShv}(\mathbf{CSp}^k; \mathbf{Set})
\end{tikzcd}
$$

If we try to regard these intrinsic pairs (M, B) as regular objects, we see that they are related to B^k-manifolds, but do not are equivalent to them.

6.3 FIRST WORK

The following is a preliminary version of the published paper

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Existence of $B^k_{\alpha,\beta}$-Structures on C^k-Manifolds

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Abstract

In this paper we introduce $B^k_{\alpha,\beta}$-manifolds as generalizations of the notion of smooth manifolds with G-structure or with k-bounded geometry. These are C^k-manifolds whose transition functions $\varphi_{ji} = \varphi_j \circ \varphi_i^{-1}$ are such that $\partial^\mu \varphi_{ji} \in B_{\alpha(r)} \cap C^{k-\beta(r)}$ for every $|\mu| = r$, where $B = (B_r)_{r\in\Gamma}$ is some sequence of presheaves of Fréchet spaces endowed with further structures, $\Gamma \subset \mathbb{Z}_{\geq 0}$ is some parameter set and α, β are functions. We present embedding theorems for the presheaf category of those structural presheaves B. The existence problem of $B^k_{\alpha,\beta}$-structures on C^k-manifolds is studied and it is proved that under certain conditions on B, α and β, the forgetful functor from C^k-manifolds to $B^k_{\alpha,\beta}$-manifolds has adjoints.

1 Introduction

One can think of a "n-dimensional manifold" as a topological space M in which we assign to each neighborhood U_i in M a bunch of coordinate systems $\varphi_i : U_i \to \mathbb{R}^n$. The regularity of M is determined by the space in which the transition functions $\varphi_{ji} : \varphi_i(U_{ij}) \to \mathbb{R}^n$ live, where $U_{ij} = U_i \cap U_j$ and $\varphi_{ji} = \varphi_j \circ \varphi_i^{-1}$. Indeed, if $B : \mathrm{Open}(\mathbb{R}^n)^{op} \to \mathbf{Fre}$ is some presheaf of Fréchet spaces in $\mathbb{R}^n$, we can define a *n-dimensional B-manifold* as one whose regularity is governed by B, i.e, whose transition functions φ_{ji} belong to $B(\varphi_i(U_{ij}))$. For instance, given $0 \leq k \leq \infty$, a C^k-manifold (in the classical sense) is just a C^k-manifold in this new sense if we regard C^k as the presheaf of k-times continuouly differentiable functions.

We could study B-manifolds when $B \subset C^k$, calling them *B^k-manifolds.* For instance, if $k \geq 1$, then any C^k-manifold can always be regarded as a C^∞-manifold [1], showing that smooth manifolds belong to that class of B-manifolds, since $C^\infty \subset C^k$. Another example are the analytic manifolds, for which $B = C^\omega$. Recall that for $B = C^k$, with $k > 1$, we have regularity conditions not only on the transition functions φ_{ji}, but also on the derivatives $\partial^\mu \varphi_{ji}$, for each $|\mu| \leq k$. In fact, $\partial^\mu \varphi_{ji} \in C^{k-|\mu|}(\varphi_i(U_{ij}))$. We notice, however, that if $B \subset C^k$ is a subpresheaf, then $B \subset C^{k-|\mu|}$, but it is not necessarily true that $\partial^\mu \varphi_{ji} \in B(\varphi_i(U_{ij}))$. For instance, if $B = L^p \cap C^\infty$, then φ_{ji} are smooth L^p-integrable functions, but besides being smooth, the derivatives of a L^p-integrable smooth functions need not be L^p-integrable [13].

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The discussion above motivates us to consider n-dimensional manifolds which satisfy *a priori* regularity conditions on the transition functions and also on their derivatives. Indeed, let now $B = (B_r)_{r\in\Gamma}$ be a sequence of presheaves of Fréchet spaces with $r \in \mathbb{Z}_{\geq 0}$ and redefine a B^k-*manifold* as one in which $\partial^\mu \varphi_{ji} \in B_r(\varphi_i(U_{ij})) \cap C^{k-r}(\varphi_i(U_{ij}))$ for every $r \leq k$ and $|\mu| = r$. Many geometric objects can be put in this new framework. Just to exemplify we mention two of them.

Example 1.1. Given a linear group $G \subset GL(n;\mathbb{R})$, if we take $B_0 = C^k$, B_1 as the presheaf of C^1-functions whose jacobian matrix belongs to G and $B_r = C^{k-r}$, with $r \geq 2$, then a B^k-manifold describes a C^k-manifold endowed with a G-structure in the sense of [15, 20]. This example includes many interesting situations, such as semi-riemannian manifolds, almost-complex manifolds, regularly foliated manifolds, etc.

Example 1.2. For a fixed $p \in [1,\infty]$, consider $B_r = L^p$ for every r. Then a B^k-manifold is a C^k-manifold whose transition functions and its derivatives are L^p-integrable. This means that $\varphi_{ji} \in W^{k,p}(\varphi_i(U_{ij})) \cap C^k(\varphi_i(U_{ij}))$, where $W^{k,p}(\Omega)$ are the Sobolev spaces, so that a $(L^p)^k$-manifold is a "Sobolev manifold". The case $p = \infty$ corresponds to the notion of k-bounded structure on a n-manifold [9, 19, 12].

This is the first of a sequence of articles which aim to begin the development of a general theory of B^k-manifolds, including the unification of certain aspects of G-structures and Sobolev structures on C^k-manifolds. Our focus is on regularity results for geometric objects on B^k-manifolds. The motivation is as follows. When studying geometry on a smooth manifold M we usually know that geometric objects τ on M (e.g, affine connections, bundles, nonlinear differential operators of a specific type, solutions of differential equations, etc.) exist with certain regularity, meaning that their local coefficients $\tau^{i_1,\dots,i_r}_{j_1,\dots,j_s}$ belong to some space, but in many situations we would like to have existence of more regular objects, meaning to have τ such that $\tau^{i_1,\dots,i_r}_{j_1,\dots,j_s}$ belong to a smaller space.

For instance, in gauge theory and when we need certain uniform bounds on the curvature, it is desirable to work with connections whose coefficients are not only smooth, but actually L^p-integrable or even uniformly bounded [22, 9]. In the study of holomorphic geometry, we consider holomorphic or hermitian connections over complex manifolds, whose coefficients are holomorphic [14, 3]. It is also interesting to consider symplectic connections, which are important in formal deformation quantization of symplectic manifolds [6, 10]. These examples immediately extend to the context of differential operators, since they can be regarded as functions depending on connections on vector bundles over M [5]. The regularity problem for solutions of differential equations is also very standard and by means of Sobolev-type embedding results and bootstrapping methods it is strongly related to the regularity problem for the underlying differential operator defining the differential equation [13].

Since the geometric structures are defined on M, the existence of some τ such that $\tau^{i_1,\dots,i_r}_{j_1,\dots,j_s}$ satisfy a specific regularity condition should depend crucially on the existence of a (non-necessarily maximal) subatlas for M whose transition functions themselves satisfy additional regularity conditions, leading us to consider B^k-manifolds in the sense introduced above. Let S be a presheaf of Fréchet spaces and let us say that τ is S-*regular* if there exists an open covering of M by charts $\varphi_i : U_i \to \mathbb{R}^n$ in which $\tau^{i_1,\dots,i_r}_{j_1,\dots,j_s} \in S(\varphi_i(U_{ij}))$. The geometric motivation above can then be rephrased into the following problem:

- let M be a C^k-manifold. Given S, find some presheaf $B(S)$, depending on S, such that if M has a $B(S)^k$-structure, then there exist S-regular geometric structures on it.

In this sequence of articles we intend to work on this question for different kinds of geometric structures: affine connections, nonlinear tensors, differential operators and fiber bundles. Notice, on the other hand, that if S is a presheaf for which the previous problem has a solution for some kind of S-regular geometric structures, we have an immediate existence question:

- under which conditions does a C^k-manifold admit a $B(S)^k$-strucure?

Opening this sequence of works, in the present one we will discuss the general existence problem of B^k-structures on C^k-manifolds. Actually, we will work on more general entities, which we call $B^k_{\alpha,\beta}$-*manifolds*, where α and β are functions and the transition functions satisfy $\partial^\mu \varphi_{ji} \in B_{\alpha(r)}(\varphi_i(U_{ij})) \cap C^{k-\beta(r)}(\varphi_i(U_{ij}))$. Thus, for $\alpha(r) = r = \beta(r)$ a $B^k_{\alpha,\beta}$-manifold is the same as a B^k-manifold. We also study some aspects of the presheaf-category $\mathbf{C}^{k,\alpha}_{n,\beta}$ of the presheaves B. The present paper has the following two main results:

Theorem A. *There are full embeddings*

(1) $\imath : \mathbf{C}^{k,\alpha}_{n,\beta;l} \hookrightarrow \mathbf{C}^{l,\alpha}_{n,\beta}$, *if* $l \leq k$;

(2) $f_* : \mathbf{C}^{k,\alpha}_{n,\beta} \hookrightarrow \mathbf{C}^{k,\alpha}_{r,\beta}$, *for any continuous injective map* $f : \mathbb{R}^n \to \mathbb{R}^r$;

(3) $\jmath : \mathbf{C}^{k,\alpha}_{n,\beta;\beta'} \hookrightarrow \mathbf{C}^{k,\alpha}_{r,\beta'}$, *if* $\beta' \leq \beta$.

Theorem B. *If B is ordered, fully left-absorbing (resp. fully right-absorbing) and has retractible (B,k,α,β)-diffeomorphisms, all of this in the same intersection presheaf $\mathbb{X}$, then the choice of a retraction $\overline{r}$ induces a left-adjoint (resp. right-adjoint) for the forgetful functor F from $B^k_{\alpha,\beta}$-manifolds to C^k-manifolds, which actually independs of $\overline{r}$. In particular, if B is fully absorbing, then F is ambidextrous adjoint.*

This article is structured as follows. In Section 2 we introduce the classes of Fréchet spaces which will be used in the next sections. These are the (Γ,ϵ)-*spaces*, where Γ is a set of indexes (in general $\Gamma \subset \mathbb{Z}_{\geq 0}$) and $\epsilon : \Gamma \times \Gamma \to \Gamma$ is some function. The (Γ,ϵ)-spaces are itself sequences $B = (B_i)_{i \in \Gamma}$ of nuclear Fréchet spaces. The map ϵ is used in order to consider *multiplicative structures* on B, which are given by a family of continuous linear maps $*_{ij} : B_i \otimes B_j \to B_{\epsilon(i,j)}$, where the tensor product is the projective one. Many other structures on B are required, such as additive structures, distributive structures, intersection structures and closure structures.

In Section 3 we define the $C^k_{\alpha,\beta}$-presheaves, which are the structural presheaves for the $B^k_{\alpha,\beta}$-manifolds, and study the presheaf category of them. We begin by constructing a sheaf-theoretic version of the concepts and results of Section 2. Thus, in this section B is not a single (Γ,ϵ)-space, but a presheaf of them on $\mathbb{R}^n$. The $C^k_{\alpha,\beta}$-presheaves are those presheaves of (Γ,ϵ)-spaces which are well-related with the presheaf $C^{k-\beta}$ in a sense defined in that section. In Section 4 Theorem A is proved.

In Section 5 $B^k_{\alpha,\beta}$-manifolds are more precisely defined and some basic properties are proven. For instance, we establish conditions on B, α and β, under which the category $\mathbf{Diff}^{B,k}_{\alpha,\beta}$ of $B^k_{\alpha,\beta}$-manifolds and $B^k_{\alpha,\beta}$-morphisms between them becomes well-defined. Some examples are also given. Finally, in Section 6 we introduce the notions of ordered presheaf, fully left/right-absorbing presheaf and presheaf with retractible (B,k,α,β)-diffeomorphisms, which are needed for the first theorem.

A proof of the Theorem B is given and as a consequence we get the existence of some limits and colimits in the category of $B^k_{\alpha,\beta}$-manifolds.

2 (Γ, ϵ)-Spaces

Let **Set** be the category of all sets and let $\Gamma \in \mathbf{Set}$ be a set of indexes[1]. A *nuclear Fréchet* Γ*-space* (or Γ*-space*, for short) is a family $(B_i)_{i\in\Gamma}$ of nuclear Fréchet spaces. Equivalently, it is a Γ-graded vector space $B = \oplus_i B_i$ whose components are nuclear Fréchet Γ-spaces. In other words, it is a functor $B : \Gamma \to \mathbf{NFre}$ from the discrete category defined by Γ to the category of nuclear Fréchet spaces and continuous linear maps. Morphisms are pairs (ξ, μ), where $\mu : \Gamma \to \Gamma$ is a function and ξ is a family of continuous linear maps $\xi_i : B_i \to B'_{\mu(i)}$, with $i \in \Gamma$. Thus, it is an endofunctor μ together with a natural transformation $\xi : B \Rightarrow B' \circ \mu$. Composition is defined by horizontal composition of natural transformations.

Let $\mathbf{Fre}_\Gamma$ be the category of Γ-spaces and notice that we have an inclusion $\imath : [\Gamma; \mathbf{NFre}] \hookrightarrow \mathbf{NFre}_\Gamma$ given by $\imath(B) = B$ and $\imath(\xi) = (\xi, id_\Gamma)$, where $[\mathbf{C}; \mathbf{D}]$ denotes the functor category. Since $\imath$ is faithful it reflects monomorphisms and epimorphisms, which means that if a morphism (ξ, μ) in $\mathbf{NFre}_\Gamma$ is such that each ξ_i is a monomorphism (resp. epimorphism) in **NFre**, then it is a monomorphism (resp. epimorphism) in $\mathbf{NFre}_\Gamma$. We have a functor $\mathcal{F} : \mathbf{Set}^{op} \to \mathbf{Cat}$ such that $\mathcal{F}(\Gamma) = \mathbf{Fre}_\Gamma$.

More generally, let us define a (Γ, ϵ)*-space* as a Γ-space B endowed with a map $\epsilon : \Gamma \times \Gamma \to \Gamma$. A morphism between a (Γ, ϵ)-space B and a (Γ, ϵ')-space B' is a morphism (ξ, μ) of Γ-spaces such that the first diagram below commutes. Let $\mathbf{Fre}_{\Gamma,\epsilon}$ be the category of (Γ, ϵ)-spaces for a fixed ϵ and let $\mathbf{NFre}_{\Gamma,\Sigma}$ be the category of (Γ, ϵ)-spaces for all ϵ. Notice that $\mathbf{NFre}_{\Gamma,\Sigma} \simeq \coprod_\epsilon \mathbf{NFre}_{\Gamma,\epsilon}$ and that there exists a forgetful functor $U : \mathbf{NFre}_{\Gamma,\Sigma} \to \mathbf{NFre}_\Gamma$ that forgets ϵ.

$$
\begin{tikzcd}
\Gamma \times \Gamma \arrow[r, "\epsilon"] \arrow[d, "\mu\times\mu"'] & \Gamma \arrow[d, "\mu"] \\
\Gamma \times \Gamma \arrow[r, "\epsilon'"'] & \Gamma
\end{tikzcd}
\qquad
\begin{tikzcd}
\mathbf{NFre}_{\Gamma,\Sigma} \arrow[r, "\simeq"] \arrow[rrr, bend right, "F_\Sigma"'] & \coprod_\epsilon \mathbf{NFre}_{\Gamma,\epsilon} \arrow[r, "\coprod_\epsilon F_\epsilon"] & \coprod_\epsilon \mathbf{NFre}_{\Gamma\times\Gamma} \arrow[r, "\nabla"] & \mathbf{NFre}_{\Gamma\times\Gamma}
\end{tikzcd}
$$

For any $\epsilon : \Gamma \times \Gamma \to \Gamma$ we have a functor $F_\epsilon : \mathbf{NFre}_{\Gamma,\epsilon} \to \mathbf{NFre}_{\Gamma\times\Gamma}$ defined by $F_\epsilon(B, \epsilon) = B \circ \epsilon$ and $F_\epsilon(\xi, \mu) = (\xi \circ \epsilon, \mu \circ \epsilon)$. In the following we will write B_ϵ to denote $B \circ \epsilon$. Notice that F_ϵ is well-defined precisely because of the commutativity of the first diagram. Thus, we also have a functor $F_\Sigma : \mathbf{NFre}_{\Gamma,\Sigma} \to \mathbf{NFre}_{\Gamma\times\Gamma}$ given by the composition above, where ∇ is the codiagonal. Furthermore, recalling that the category of Fréchet spaces is symmetric monoidal with the projective tensor product $\otimes$ [21, 8], given Γ, Γ' we have a functor

$$\otimes_{\Gamma,\Gamma'} : \mathbf{NFre}_\Gamma \times \mathbf{NFre}_{\Gamma'} \to \mathbf{NFre}_{\Gamma\times\Gamma'}$$

playing the role of an *external product*, given by $(B \otimes B')_{(i,j)} = B_i \otimes B_j$ on objects and by $(\xi, \mu) \otimes (\eta, \nu) = (\xi \otimes \eta, \mu \times \nu)$ on morphisms. By composing with the diagonal and the forgetful functor U,

[1] In all the article and in the examples considered Γ will be $\mathbb{Z}_{\geq 0}$ or some finite product of copies of $\mathbb{Z}_{\geq 0}$. However, all the results of this section can be generalized to the case in which Γ is an arbitrary (i.e, not necessarily discrete) category.

for each Γ we get the functor $\otimes_\Gamma$ below.

$$\begin{tikzcd} \mathbf{NFre}_{\Gamma,\Sigma} \arrow[r, "U"] \arrow[rrr, bend right, "\otimes_\Gamma"'] & \mathbf{NFre}_\Gamma \arrow[r, "\Delta"] & \mathbf{NFre}_\Gamma \times \mathbf{NFre}_\Gamma \arrow[r, "\otimes_{\Gamma,\Gamma}"] & \mathbf{NFre}_{\Gamma\times\Gamma} \end{tikzcd}$$

A *multiplicative structure* in a (Γ,ϵ)-space (B,ϵ) is a morphism $(*,\epsilon) : \otimes_\Gamma(B,\epsilon) \to F_\Sigma(B,\epsilon)$, i.e, a family $*_{ij} : B_i \otimes B_j \to B_{\epsilon(i,j)}$ of continuous linear maps, with $i,j \in \Gamma$. A *multiplicative* Γ*-space* is a (Γ,ϵ) space in which a multiplicative structure has been fixed. A *weak morphism* between two multiplicative Γ-spaces $(B,\epsilon,*)$ and $(B',\epsilon',*')$ is an arbitrary morphism in the arrow category of $\mathbf{NFre}_{\Gamma\times\Gamma}$, i.e, it is given by morphisms $(\zeta,\psi) : \otimes_\Gamma(B,\epsilon) \to \otimes_\Gamma(B',\epsilon')$ and $(\eta,\nu) : F_\Sigma(B,\epsilon) \to F_\Sigma(B,\epsilon)$ in the category $\mathbf{NFre}_{\Gamma\times\Gamma}$ such that the diagrams below commute.

$$\begin{tikzcd} B_i \otimes B_j \arrow[r, "*_{ij}"] \arrow[d, "\zeta_{ij}"'] & B_{\epsilon(i,j)} \arrow[d, "\eta_{\epsilon(i,j)}"] \\ (B' \otimes B')_{\psi(i,j)} \arrow[r, "*'_{\psi(i,j)}"'] & B'_{\epsilon'(\psi(i,j))} \end{tikzcd} \qquad \begin{tikzcd} \Gamma \times \Gamma \arrow[r, "\epsilon"] \arrow[d, "\psi"'] & \Gamma \arrow[d, "\nu"] \\ \Gamma \times \Gamma \arrow[r, "\epsilon'"'] & \Gamma \end{tikzcd}$$

A *strong morphism* (or simply *morphism*) between multiplicative Γ-spaces is a weak morphism such that there exists $(\xi,\mu) : (B,\epsilon) \to (B',\epsilon')$ for which $(\zeta,\psi) = \otimes_\Gamma(\xi,\mu)$ and $(\eta,\nu) = F_\Sigma(\xi,\mu)$. In explicit terms this means that $\nu = \mu$, $\psi = \mu \times \mu$, $\zeta = \xi \otimes \xi$ and $\eta = \xi$, so that the diagrams above become the diagrams below. Notice that the second diagram only means that (ξ,μ) is a morphism of (Γ,ϵ)-spaces. We will denote by $\mathbf{NFre}_{\Gamma,*}$ the category of multiplicative Γ-spaces and strong morphisms.

$$\begin{tikzcd} B_i \otimes B_j \arrow[r, "*_{ij}"] \arrow[d, "\xi_i\otimes\xi_j"'] & B_{\epsilon(i,j)} \arrow[d, "\xi_{\epsilon(i,j)}"] \\ B'_{\mu(i)} \otimes B'_{\mu(j)} \arrow[r, "*'_{\mu(i)\mu(j)}"'] & B'_{\epsilon'(\mu(i)\mu(j))} \end{tikzcd} \qquad \begin{tikzcd} \Gamma \times \Gamma \arrow[r, "\epsilon"] \arrow[d, "\mu\times\mu"'] & \Gamma \arrow[d, "\mu"] \\ \Gamma \times \Gamma \arrow[r, "\epsilon'"'] & \Gamma \end{tikzcd}$$

An *additive* Γ*-space* is a multiplicative Γ-space $(B,+,\epsilon)$ such that $\epsilon_{ii} = i$ and such that $+_{ii} : B_i \otimes B_i \to B_i$ is the addition $+_i$ in B_i. Let $\mathbf{NFre}_{\Gamma,+} \subset \mathbf{NFre}_{\Gamma,*}$ denote the full subcategory of additive Γ-spaces. The inclusion has a left-adjoint $F(*,\epsilon) = (+,\delta)$ given by

$$\delta(i,j) = \begin{cases} \epsilon(i,j), & i \neq j \\ i & j = i \end{cases} \quad \text{and} \quad +_{ij} = \begin{cases} *_{ij}, & i \neq j \\ +_i & j = i \end{cases}.$$

Example 2.1. Any Γ-space $B = (B_i)_{i\in\Gamma}$ admits many trivial additive structures, defined as follows. Let $\theta : \Gamma \times \Gamma \to \Gamma$ be any function and define an additive structure by

$$\delta_\theta(i,j) = \begin{cases} \theta(i,j), & i \neq j \\ i & j = i \end{cases} \quad \text{and} \quad +_{ij} = \begin{cases} 0_{\theta(i,j)}, & i \neq j \\ +_i & j = i \end{cases},$$

where $0_{\theta(i,j)}$ is the function constant in the zero vector of $B_{\theta(i,j)}$. All these trivial structures are actually strongly isomorphic. Indeed, let B_θ denote B with the trivial additive structure defined by θ. As one can see, we have a natural bijection

$$\mathrm{IMor}_{\Gamma,+}(B_\theta, B'_{\theta'}) \simeq \mathrm{IMor}_\Gamma(B,B'),$$

where the right-hand side is the set of *μ-injective morphisms* between B and B', i.e, morphisms (ξ, μ) of Γ-spaces such that $\mu : \Gamma \to \Gamma$ is injective. Furthermore, this bijection preserve isomorphisms, so that we also have a bijection

$$\mathrm{IIso}_{\Gamma,+}(B_\theta, B'_{\theta'}) \simeq \mathrm{IISo}_\Gamma(B, B').$$

Since for $B' = B$ the right-hand side contains at least the identity, it follows that $B_\theta \simeq B_{\theta'}$ for every θ and θ'.

Example 2.2. Suppose that Γ has a partial order $\leq$. Any increasing or decreasing sequence of nuclear Fréchet spaces, i.e, such that $B_i \subset B_j$ if either $i \leq j$ or $j \leq i$, has a canonical additive structure, where $+_{ij}$ is given by the sum in $B_{\max(i,j)}$ or $B_{\min(i,j)}$, respectively, so that $\delta(i,j) = \max(i,j)$ or $\delta(i,j) = \min(i,j)$. In particular, for any open set $U \subset \mathbb{R}^n$ and any order-preserving integer function $p : \Gamma \to [0, \infty]$, the sequences $B_i = L^{p(i)}(U)$ and $B'_i = C^{p(i)}(U)$ are decreasing, where in $L^{p(i)}(U)$ we take the Banach structure given by the $L^{p(i)}$-norm and in $C^{p(i)}(U)$ we consider Fréchet family of seminorms

$$\|f\|_{r,l} = \sup_{|\mu|=r} \sup_{x \in K_l} \|\partial^\mu f(x)\|, \tag{1}$$

where $0 \leq r \leq p(i)$ and (K_l) is any countable sequence of compacts such that every other compact $K \subset U$ is contained in K_l for some l [21]. Thus, both sequences become an additive Γ-space in a natural way. They will be respectively denoted by $L^p(U)$ and $C^p(U)$. We will be specially interested in the function $p(i) = k - \alpha(i)$, for $i \leq k$, where α is some other function. In this case, we will write $C^{k-\alpha}(U)$ instead of $C^p(U)$. More precisely, we will consider the sequence given by $B_i = C^{k-\alpha(i)}(U)$, if $\alpha(i) \leq k$, and $B_i = 0$, if $\alpha(i) \geq k$. In this situation, $\delta(i,j) = k - \max(\alpha(i), \alpha(j))$.

Example 2.3. Similarly, if $U \subset \mathbb{R}^n$ is a nice open set such that the Sobolev embeddings are valid [13], then for any fixed integers $p, q, r > 0$ with $p < q$, we have a continuous embedding $W^{r,p}(U) \hookrightarrow W^{\ell(r),q}(U)$, where $\ell(r) = r + n\frac{(p-q)}{pq}$ and $W^{k,p}(U)$ is the Sobolev space. Thus, by defining $B_i = W^{l(i),q}(U)$, where $l(i) = \ell^i(r)$, i.e, $l(0) = r$, $l(1) = \ell(r)$ and $l(i) = \ell(\ell(...\ell(r)...))$, we get again a decreasing sequence of Banach spaces and therefore an additive $\mathbb{Z}_{\geq 0}$-space.

Example 2.4. As in Example 2.2, suppose Γ ordered and consider an integer function $p : \Gamma \to [0, \infty]$. In any open set $U \subset \mathbb{R}^n$ pointwise multiplication of real functions give us bilinear maps $\cdot_{ij} : C^{p(i)}(U) \times C^{p(j)}(U) \to C^{\min(p(i),p(j))}(U)$, which are continuous in the Fréchet structure (1), so that with $\delta(i,j) = \min(p(i), p(j))$, the $\mathbb{Z}_{\geq 0}$-space $C^p(U)$ is multiplicative. If $p(i) = k - \alpha(i)$, then $\delta(i,j) = k - \max(\alpha(i), \alpha(j))$.

Example 2.5. From Hölder's inequality, pointwise multiplication of real functions also defines continuous bilnear maps $\cdot_{ij} : L^i(U) \times L^j(U) \to L^{i \cdot j/(i+j)}(U)$ for $i, j \geq 2$ such that $i \star j = i \cdot j/(i+j)$ is an integer [13, 21]. Let $\mathbb{Z}_S$ be the set of all integers $m \geq 2$ such that for every two given $m, m' \in \mathbb{Z}_S$ the sum $m + m'$ divides the product $m \cdot m'$. Suppose Γ ordered and choose a function $p : \Gamma \to \mathbb{Z}_S$. Then $L^p(U)$ has a multiplicative structure with $\epsilon(i,j) = p(i) \star p(j)$. Even if $i \star j$ is not an integer we get a multiplicative structure. Indeed, let $\epsilon'(i,j) = \mathrm{int}(\epsilon(i,j))$, where $\mathrm{int}(k)$ denotes the integer part of a real number. Then $\epsilon'(i,j) \leq \epsilon(i,j)$, so that $L^{\epsilon(i,j)}(U) \subset L^{\epsilon'(i,j)}(U)$, and we can assume $\cdot_{ij}$ as taking values in $L^{\epsilon'(i,j)}(U)$.

Example 2.6. Given $1 \leq i, j \leq \infty$, notice that $i \star j \geq 1/2$, which is equivalent to saying the number $r(i,j) = i \cdot j/(i+j-i\cdot j)$, i.e, the solution of

$$\frac{1}{i} + \frac{1}{j} = \frac{1}{r} + 1,$$

also satisfies $1 \leq r \leq \infty$. Thus, from Young's inequality, convolution product defines a continuous bilinear map [13, 21]

$$*_{ij} : L^i(\mathbb{R}^n) \times L^j(\mathbb{R}^n) \to L^{r(i,j)}(\mathbb{R}^n),$$

so that for any function $p : \Gamma \to [1, \infty]$ the sequence $L^p(\mathbb{R}^n)$ has a multiplicative structure with $\epsilon(i,j) = r(p(i), p(j))$.

Example 2.7. A *Banach* Γ-*space* is a Γ-space $B = (B_i)_{i\in\Gamma}$ such that each B_i is a Banach space. Suppose that Γ is actually a monoid $(\Gamma, +, 0)$. Notice that a multiplicative structure in B with $\epsilon(i,j) = i+j$ is a family of continuous bilinear maps $*_{ij} : B_i \times B_j \to B_{i+j}$. Since the category of Banach spaces and continuous linear maps has small coproducts, the coproduct $\mathcal{B} = \oplus_{i\in\Gamma} B_i$ exists as a Banach space and $(\mathcal{B}, *)$ is actually a Γ-graded normed algebra.

We say that two multiplicative structures $(*, \epsilon)$ and $(+, \delta)$ on the same Γ-space B are *left-compatible* if the following diagrams are commutative. The first one makes sense precisely because the second one is commutative (σ is the map $\sigma(x,(y,z)) = ((x,y),(x,z))$). A *left-distributive* Γ-*space* is a Γ-space endowed with two left-compatible multiplicative structures. Morphisms are just morphisms of Γ-spaces which preserve both multiplicative structures. Let $\mathbf{Fre}^l_{\Gamma,*}$ be the category of all of them. Our main interest will be when $(+, \delta)$ is actually an additive structure, justfying the notation.

$$\begin{tikzcd}
B_i \otimes (B_j \otimes B_k) \arrow[r, "\sigma"] \arrow[d, "id\otimes +_{jk}"'] & (B_i \otimes B_j) \otimes (B_i \otimes B_k) \arrow[r, "*_{ij}\otimes *_{ik}"] & B_{\epsilon(i,j)} \otimes B_{\epsilon(i,k)} \arrow[d, "+_{\epsilon(i,j)\epsilon(i,k)}"] \\
B_i \otimes B_{\delta(j,k)} \arrow[rr, "*_{i\delta(j,k)}"'] & & B_{\epsilon(i,\delta(j,k))}
\end{tikzcd}$$

$$\begin{tikzcd}
\Gamma \times (\Gamma \times \Gamma) \arrow[r, "\sigma"] \arrow[d, "id\times\delta"'] & (\Gamma \times \Gamma) \times (\Gamma \times \Gamma) \arrow[r, "\epsilon\times\epsilon"] & \Gamma \times \Gamma \arrow[d, "\delta"] \\
\Gamma \times \Gamma \arrow[rr, "\epsilon"'] & & \Gamma
\end{tikzcd}$$

In a similar way, we say that $(*, \epsilon)$ and $(+, \delta)$ are *right-compatible* if the diagrams below commute, with $\sigma((x,y),z) = ((x,z),(y,z))$. A *right-distributive* Γ-*space* is a Γ-space together with a right-compatible structure. Morphisms are morphisms of Γ-spaces preserving those structures. Let $\mathbf{Fre}^r_{\Gamma,*}$ be the corresponding category. Finally, let $\mathbf{Fre}_{\geq,+,*} = \mathbf{Fre}^l_{\Gamma,+,*} \cap \mathbf{Fre}^r_{\Gamma,+,*}$ be the category of *distributive* Γ-*spaces*, i.e, the category of Γ-spaces endowed with two multiplicative structures which are both left-compatible and right-compatible.

$$\begin{tikzcd}
(B_i \otimes B_j) \otimes B_k \arrow[r, "\sigma"] \arrow[d, "+_{ij}\otimes id"'] & (B_i \otimes B_k) \otimes (B_j \otimes B_k) \arrow[r, "*_{ik}\otimes *_{jk}"] & B_{\epsilon(i,k)} \otimes B_{\epsilon(j,k)} \arrow[d, "+_{\epsilon(i,k)\epsilon(j,k)}"] \\
B_{\delta(i,j)} \otimes B_k \arrow[rr, "*_{\delta(i,j)k}"'] & & B_{\epsilon(\delta(i,j),k)}
\end{tikzcd}$$

$$\begin{tikzcd}
(\Gamma \times \Gamma) \times \Gamma \arrow[r, "\sigma"] \arrow[d, "\delta \times id"'] & (\Gamma \times \Gamma) \times (\Gamma \times \Gamma) \arrow[r, "\epsilon \times \epsilon"] & \Gamma \times \Gamma \arrow[d, "\delta"] \\
\Gamma \times \Gamma \arrow[rr, "\epsilon"'] & & \Gamma
\end{tikzcd}$$

Example 2.8. The additive and multiplicative structures for C^p described in Example 2.2 and Example 2.4 define a distributive structure. Any Γ-space B, when endowed with a trivial additive structure and the induced multiplicative structure, becomes a distributive Γ-space.

Example 2.9. The additive structure in L^p given in Example 2.2 is generally *not* left/right-compatible with the multiplicative structures of Example 2.5 and Example 2.6. Indeed, as one can check, the sum $+$ in L^p is compatible with the pointwise multiplication iff the function $p : \Gamma \to \mathbb{Z}_{\geq 0}$ is such that $p(i) \star p(j) \leq p(i) \star p(k)$ whenever $p(j) \leq p(k)$, where $i, j, k \in \Gamma$. For fixed p we can clearly restrict to the subset Γ_p in which the desired condition is satisfied, so that $(L^{p(i)})_{i \in \Gamma_p}$ is distributive for every p. Simlarly, the $+$ is compatible with the convolution product $*$ in $L^{p(\cdot)}$ iff $r(p(i), p(j)) \leq r(p(i), p(k))$ whenever $p(j) \leq p(k)$.

A Γ-*ambient* is a pair $(\mathbf{X}, \gamma)$, where $\mathbf{X}$ is a category with pullbacks and γ is an embedding $\gamma : \mathbf{NFre}_\Gamma \hookrightarrow \mathbf{X}$. Let $X \in \mathbf{X}$ and consider the corresponding slice category $\gamma(\mathbf{NFre}_\Gamma)/X$, i.e, the category of morphisms $\gamma(B) \to X$ in $\mathbf{X}$, with B a Γ-space, and commutative triangles with vertex X. Let $\mathrm{Sub}_\Gamma(X, \gamma)$ be the full subcategory whose objects are monomorphisms $\imath : \gamma(B) \hookrightarrow X$. If $(\gamma(B), \imath)$ belongs to $\mathrm{Sub}_\Gamma(X, \gamma)$ we say that it is a Γ-*subspace* of (X, γ). Finally, let $\mathrm{Span}_\Gamma(X, \gamma)$ be the category of spans $\gamma(B) \hookrightarrow X \hookleftarrow \gamma(B')$ of Γ-subspaces of (X, γ). If $(\gamma(B), \imath, \gamma(B'), \imath')$ is a span, we say that $(\mathbf{X}, \gamma, X, \imath, \imath')$ is an *intersection structure between* B *and* B' *in* X and we define the corresponding *intersection in* X as the object $B \cap_{X,\gamma} B' \in \mathbf{X}$ given by the pullback between $\imath$ and $\imath'$; the object X itself is called the *base object* for the intersection. We say that a span in $\mathrm{Span}_\Gamma(X, \gamma)$ is *proper* if the corresponding pullback actually belongs to $\mathbf{NFre}_\Gamma$, i.e, if there exists some $L \in \mathbf{NFre}_\Gamma$ such that $\gamma(L) \simeq B \cap_{X,\gamma} B'$. Notice that, since γ is an embedding, when L exists it is unique up to isomorphisms. Thus, we will write $B \cap_X B'$ in order to denote any object in the isomorphism class. We will also require that the universal maps $u_\gamma : B \cap_{X,\gamma} B' \to \gamma(B)$ and $u'_\gamma : B \cap_{X,\gamma} B' \to \gamma(B')$ also induce maps $u : B \cap_X B' \to B$ and $u' : B \cap_X B' \to u'$, which clearly exist if the embedding γ is full. Let $\mathrm{PSpan}_\Gamma(X, \gamma)$ be the full subcategory of proper spans.

$$\begin{tikzcd}
B \cap_{X,\gamma} B' \arrow[r] \arrow[d] & \gamma(B') \arrow[d, hook, "\imath'"] \\
\gamma(B) \arrow[r, hook, "\imath"'] & X
\end{tikzcd}
\qquad
\begin{tikzcd}
B_\epsilon \cap_{X,\gamma_\Sigma} B'_{\epsilon'} \arrow[r] \arrow[d] & \gamma_\Sigma(B'_{\epsilon'}) \arrow[d, hook, "\jmath'"] \\
\gamma_\Sigma(B_\epsilon) \arrow[r, hook, "\jmath"'] & X
\end{tikzcd} \tag{2}$$

Recall the functor $F_\Sigma : \mathbf{NFre}_{\Gamma,\Sigma} \to \mathbf{NFre}_{\Gamma \times \Gamma}$ assigning to each (Γ, ϵ)-space (B, ϵ) the corresponding $(\Gamma \times \Gamma)$-space $B \circ \epsilon \equiv B_\epsilon$. Let $(\mathbf{X}, \gamma_\Sigma)$ be a $(\Gamma \times \Gamma)$-ambient. Let $\mathrm{Sub}_{\Gamma,\Sigma}(X, \gamma_\Sigma)$ be the category of monomorphisms $\jmath : F_\Sigma(\gamma_\Sigma(B, \epsilon)) \hookrightarrow X$ for a fixed X, i.e, the category of (Γ, ϵ)-subspaces of X. The *intersection in* X between two of them is the second pullback above, where for simplicity we write $\gamma_\Sigma(B) \equiv F_\Sigma(\gamma_\Sigma(B, \epsilon))$. The intersection is *proper* if the resulting object belongs to $\mathbf{NFre}_{\Gamma,\Sigma}$. Let $\mathrm{PSpan}_{\Gamma,\Sigma}(X, \gamma_\Sigma)$ be the category of those spans. Now, let $(B, *, \epsilon)$ and $(B', *', \epsilon')$ be two multiplicative Γ-spaces. An *intersection structure* between them is a tuple $\mathbb{X} = (\mathbf{X}, \gamma_\Sigma, X, \jmath, \jmath')$ consisting of a $(\Gamma \times \Gamma)$-ambient $(\mathbf{X}, \gamma_\Sigma)$, and object X and a span $(\jmath, \jmath')$, as in (2). The *intersection* *object* between $*$ and $*'$ in $\mathbb{X}$ is the pullback $\mathrm{pb}(*,*';X,\gamma_\Sigma)$ below. By universality we get the dotted arrow $*\cap*'$. We say that an intersection structure is *proper* if not only the span $(\jmath,\jmath')$ is proper, but also the intersection object belongs to $\mathbf{NFre}_{\Gamma,\Sigma}$ (in the same previous sense). In this case, the object in $\mathbf{NFre}_{\Gamma,\Sigma}$ will be denoted by $\mathrm{pb}(*,*';X)$, its components by $\mathrm{pb}_{ij}(*,*';X)$ and we will define the *intersection number function* between $*$ and $*'$ in $\mathbb{X}$ as the function $\#_{*,*';\mathbb{X}}:\Gamma\times\Gamma\to[0,\infty]$ given by $\#_{*,*';\mathbb{X}}(i,j)=\dim_{\mathbb{R}}\mathrm{pb}_{ij}(*,*';X)$. We say that two multiplicative Γ-spaces $(B,*,\epsilon)$ and $(B',*',\epsilon')$ (or that $*$ and $*'$) have *nontrivial intersection in* $\mathbb{X}$ if $\#_{*,*';\mathbb{X}}\geq 1$. Finally, we say that $*$ and $*'$ are *nontrivially intersecting* if they have nontrivial intersection in some intersection structure.

$$
\begin{tikzcd}
\mathrm{pb}(*,*';X,\gamma_\Sigma) \arrow[rr] \arrow[dr, dashed, "*\cap*'"] \arrow[dd] & & \gamma_\Sigma(B'_{\epsilon'}\otimes B'_{\epsilon'}) \arrow[d, "\gamma_\Sigma(*')"] \\
 & B_\epsilon\cap_{X,\gamma_\Sigma}B'_{\epsilon'} \arrow[r] \arrow[d] & \gamma_\Sigma(B'_{\epsilon}) \arrow[d, hook, "\jmath'"] \\
\gamma_\Sigma(B_\epsilon\otimes B_\epsilon) \arrow[r, "\gamma_\Sigma(*)"'] & \gamma_\Sigma(B_\epsilon) \arrow[r, hook, "\jmath"'] & X
\end{tikzcd}
$$

Example 2.10. Let $(\mathbf{X},\gamma)$ be any Γ-ambient such that $\mathbf{X}$ has coproducts. Given B,B', let $X=\gamma(B)\oplus\gamma(B')$ and notice that we have monomorphisms $\imath:\gamma(B)\to X$ and $\imath':\gamma(B')\to X$, so that we can consider the intersection $B\cap_{X,\gamma}B'$ having the coproduct as a base object. However, this is generally trivial. For instance, if $\mathbf{X}=\mathbf{Set}_\Gamma$ is the category of Γ-sets, i.e, sequences $(X_i)_{i\in\Gamma}$ of sets, then the intersection above is the empty Γ-set, i.e, $X_i=\varnothing$ for each $i\in\Gamma$. Furthermore, if $\mathbf{X}$ is some category with null objects which is freely generated by Γ-sets, i.e, such that there exists a forgetful functor $U:\mathbf{X}\to\mathbf{Set}_\Gamma$ admitting a left-adjoint, then the intersection $B\cap_{X,\gamma}B'$ is a null object. In particular, if $\mathbf{X}=\mathbf{Vec}_{\mathbb{R},\Gamma}$ is the category of Γ-graded real vector spaces, with Span_Γ denoting the left-adjoint, then the intersection object $\mathrm{Span}_\Gamma(B\cap_{X,\gamma}B')$ is the trivial Γ-graded vector space. Let us say that an intersection structure $\mathbb{X}$ is *vectorial* if is proper and defined in a Γ-ambient $(\mathbf{Vec}_{\mathbb{R},\Gamma},\gamma)$ such that γ create null-objects (in other words, B is the trivial Γ-space iff $\gamma(B)$ is the trivial Γ-vector space, i.e, iff $\gamma(B)_i\simeq 0$ for each i). In this case, it then follows that for the base object $X=\gamma(B)\oplus\gamma(B')$ we have $B\cap_{X,\gamma}B'\simeq 0$ when regarded as a Γ-space.

Example 2.11. The intersection between two Γ-spaces in a vectorial intersection structure is not necessarily trivial; it actually depends on the base object X. Indeed, suppose that B and B' are such that there exists a Γ-set S for which $B\subset S$ and $B'\subset S$. Let $X=\mathrm{Span}_\Gamma(B\cup B')$, where the union is defined componentwise. We have obvious inclusions and the corresponding intersection object is given by $B\cap_{X,\gamma}B'\simeq\mathrm{Span}_\Gamma(B\cap B')$, where the right-hand side is the intersection as Γ-vector spaces, defined componentwise, which is nontrivial if $B\cap B'$ is nonempty as Γ-sets. For instance, $C^p(U)\cap_{X,\gamma}C^q(U)$ and $C^p(U)\cap_{X,\gamma}L^q(U)$ are nontrivial in them for every p,q. We will say that a vectorial intersection structure with base object $X=\mathrm{Span}_\Gamma(B\cup B')$ is a *standard intersection structure*. Notice that two standard intersection structures differ from the choice of γ and from the maps $\imath$ and $\imath'$ which define the span in X.

Vectorial intersection structures have the following good feature:

Proposition 2.1. *Let $(B,\epsilon,*)$ and $(B',\epsilon',*')$ be multiplicative structures and let $\mathbb{X}$ be a vectorial intersection structure between them. If at least B or B' is nontrivial, then the intersection space*

$\mathrm{pb}(*,*';X)$ *is nontrivial too, indenpendently of the base object* X. *In other words, nontrivial multiplicative structures have nontrivial intersection in any vectorial intersection structure.*

Proof. Since γ_Σ creates null objects, it is enough to work in the category of Γ-vector spaces. On the other hand, since a Γ-vector space $V = (V_i)_{i\in\gamma}$ is nontrivial iff at least one V_i is nontrivial, it is enough to work with vector spaces. Thus, just notice that if $T : V \otimes W \to Z$ and $T' : V' \otimes W' \to Z$ are arbitrary linear maps, then the pullback between them contains a copy of $V \oplus W \oplus V' \oplus V'$. $\square$

Corollary 2.1. *Nontrivial multiplicative structures are always nontrivially intersecting.*

Proof. Straightforward from the last proposition and from the fact that vectorial intersection structures exist. $\square$

Remark 2.1. The proposition explains that the intersection space $\mathrm{pb}(*,*';X)$ can be nontrivial even if the intersection $B_\epsilon \cap_X B'_{\epsilon'}$ is trivial.

Let $(B,*,+,\epsilon,\delta)$ and $(B',*',+',\epsilon',\delta')$ be two distributive Γ-spaces. A *distributive intersection structure* between them is an intersection structure $\mathbb{X}_{*,*'}$ between $*$ and $*'$ together with an intersection structure $X_{+,+'}$ between $+$ and $+'$ whose underlying Γ_Σ-ambient are the same. In other words, it is a tuple $\mathbb{X}_{*,+} = (\mathbf{X},\gamma_\Sigma,X_*,X_+,\jmath_*,\jmath'_*,\jmath_+,\jmath'_+)$ such that $(X_*,\jmath_*,\jmath'_*)$ and $(X_+,\jmath_+,\jmath'_+)$ are spans as in (2). A *full intersection structure* between distributive Γ-spaces is a pair $\mathbb{X} = (\mathbb{X}_0,\mathbb{X}_{*,+})$, where $\mathbb{X}_{*,+} = (\mathbb{X}_*,\mathbb{X}_+)$ is a distributive intersection structure and $\mathbb{X}_0 = (\mathbf{X}_0,\gamma,X,\imath,\imath')$ is an intersection structure between the Γ-spaces B and B', whose underlying ambient category $\mathbf{X}_0$ is equal to the ambient category $\mathbf{X}$ of $\mathbb{X}_*$ and $\mathbb{X}_+$. We say that the triple $(\mathbf{X},\gamma,\gamma_\Sigma)$ is a *full* Γ*-ambient* and that (X,X_*,X_+) is the *full base object*. We also say that $\mathbb{X}$ is *vectorial* if both $\mathbb{X}_0$, $\mathbb{X}_*$ and $\mathbb{X}_+$ are vectorial. The *full intersection space* of the distributive structures B and B' in the full intersection structure $\mathbb{X}$ is the triple consisting of the intersection spaces $B\cap_{X,\gamma}B'$ in $\mathbb{X}_0$, $\mathrm{pb}(*,*';X_*,\gamma_\Sigma)$ in $\mathbb{X}_*$ and $\mathrm{pb}(+,+';X_+,\gamma_\Sigma)$ in $\mathbb{X}_+$, denoted simply by $B\cap_{\mathbb{X}}B'$. It is *nontrivial* (and in this case we say that B and B' have *nontrivial intersection in* $\mathbb{X}$) if each of the three componentes are nontrivial. When $\mathbb{X}$ is vectorial, the object representing $B\cap_{\mathbb{X}}B'$ in $\mathbf{NFre}_\Gamma \times \mathbf{NFre}_{\Gamma,\Sigma} \times \mathbf{NFre}_{\Gamma,\Sigma}$ will also be denoted by $B\cap_{\mathbb{X}}B'$, i.e,

$$B\cap_{\mathbb{X}}B' = (B\cap_X B', \mathrm{pb}(*,*';X_*), \mathrm{pb}(+,+';X_+))$$

In some situations we will need to work with a special class of ambient categories $\mathbf{X}$ which becomes endowed with a monoidal structure that has a nice relation with some closure operator. We finish this section introducing them. Let $(\mathbf{X},\circledast,1)$ be a monoidal category, let $\mathrm{Ar}(\mathbf{X})$ be the arrow category and recall that we have two functors $s,t : \mathrm{Ar}(\mathbf{X}) \to \mathbf{X}$ which assign to each map f its source $s(f)$ and target $t(f)$. A *weak closure operator* in $\mathbf{X}$ is a functor $\mathrm{cl} : \mathrm{Ar}(\mathbf{X}) \to \mathrm{Ar}(\mathbf{X})$ such that $t(\mathrm{cl}(f)) = t(t)$. If $s(f) = B$ and $t(f) = X$, we write $\mathrm{cl}_X(B)$ to denote $s(\mathrm{cl}(f))$. A *weak closure structure* is a pair (cl,μ) given by a weak closure operator endowed with a natural transformation $\mu : id \Rightarrow \mathrm{cl}$, translating the idea of embedding a space onto its closure. The monoidal product $\circledast$ in $\mathbf{X}$ induces a functor $\overline{\circledast}$ in the arrow category given $f\circledast f'$ on objects and as below on morphisms.

$$\begin{tikzcd} X \arrow[r, "h_x"] \arrow[d, "f"'] & X' \arrow[d, "f'"] \\ Y \arrow[r, "h_y"'] & Y' \end{tikzcd} \ \overline{\circledast}\ \begin{tikzcd} Z \arrow[r, "l_z"] \arrow[d, "g"'] & Z' \arrow[d, "g'"] \\ W \arrow[r, "l_w"'] & W' \end{tikzcd} = \begin{tikzcd} X\circledast Z \arrow[r, "h_x\circledast l_z"] \arrow[d, "f\circledast g"'] & X'\circledast Z' \arrow[d, "f'\circledast g'"] \\ Y\circledast W \arrow[r, "h_y\circledast l_w"'] & Y'\circledast W' \end{tikzcd}$$

A *lax closure operator* in $\mathbf{X}$ is a weak closure operator which is lax monoidal relative to $\overline{\circledast}$. This means that for any pair of arrows (f, f') we have a corresponding arrow morphism $\phi_{f,f'} : \mathrm{cl}(f) \circledast \mathrm{cl}(f') \to \mathrm{cl}(f \circledast f')$ which is natural in f, f'. In particular, if $s(f) = B$, $s(f') = B'$, $t(f) = X$ and $t(f') = X'$, we have a morphism $\phi : \mathrm{cl}_X(B) \circledast \mathrm{cl}_{X'}(B') \to \mathrm{cl}_{X \circledast X'}(B \circledast B')$. A *lax closure structure* is a weak closure structure (cl, μ) whose weak closure operator is actually a lax closure operator (cl, ϕ) such that the first diagram below commutes, meaning that ϕ and μ are compatible. If f and f' have source/target as above, we have in particular the second diagram.

$$\begin{tikzcd} & \mathrm{cl}(f) \circledast \mathrm{cl}(f') \arrow[d, "\phi_{f,f'}"] \\ f \circledast f' \arrow[ur, "\mu_f \circledast \mu_{f'}"] \arrow[r, "\mu_{f \circledast f'}"'] & \mathrm{cl}(f \circledast f') \end{tikzcd} \qquad \begin{tikzcd} & \mathrm{cl}_X(B) \circledast \mathrm{cl}_{X'}(B') \arrow[d, "\phi_{B,B'}"] \\ B \circledast B' \arrow[ur, "\mu_B \circledast \mu_{B'}"] \arrow[r, "\mu_{B \circledast B'}"'] & \mathrm{cl}_{X \circledast X'}(B \circledast B') \end{tikzcd} \tag{3}$$

A *monoidal* Γ*-ambient* is a Γ-ambient $(\mathbf{X}, \gamma)$ whose ambient category $\mathbf{X}$ is a monoidal category $(\mathbf{X}, \circledast, 1)$ such that $\gamma : \mathbf{NFre}_\Gamma \to \mathbf{X}$ is a strong monoidal functor in the sense of [16, 2], i.e, lax and oplax monoidal in a compatible way. Thus, for any two Γ-spaces B, B' we have an isomorphism $\psi_{B,B'} : \gamma(B) \circledast \gamma(B') \simeq \gamma(B \otimes B')$. Let $\mathbf{X}$ be a monoidal Γ-ambient $(\mathbf{X}, \gamma, \circledast)$. A *closure structure* for a Γ-space B in $\mathbf{X}$ is a lax closure structure (cl, μ, ϕ) in $\mathbf{X}$ such that any morphism $f : \gamma(B) \to X$ (not necessarily a subobject) admits an extension $\hat{\mu}_f$ relative to $\mu_f : \gamma(B) \to \mathrm{cl}_X(\gamma(B))$ as in the diagram below. A Γ*-space with closure in* $\mathbf{X}$ is Γ-space B endowed with a closure structure $\mathbf{c} = (\mathrm{cl}, \mu, \phi)$ in $\mathbf{X}$. Notice that the diagrams above make perfect sense in the category $\mathbf{NFre}_{\Gamma,\Sigma}$ of (Γ, ϵ)-spaces, so that we can also define (Γ, ϵ)-spaces with closure in a monoidal Γ-ambient $(\mathbf{X}, \gamma, \circledast)$. Let $\mathbf{NFre}_{\Gamma,\Sigma,\mathbf{c}}(\mathbf{X}) \subset \mathbf{NFre}_{\Gamma,\Sigma}$ be the full subcategory of those (Γ, ϵ)-spaces.

$$\begin{tikzcd} & X \\ \gamma(B) \arrow[ur, "f"] \arrow[r, "\mu_f"'] & \mathrm{cl}_X(\gamma(B)) \arrow[u, dashed, "\hat{\mu}_f"'] \end{tikzcd} \tag{4}$$

3 $C^{k,\alpha}_{n,\beta}$-Presheaves

- Let Γ be a set such that $\Gamma \cap \mathbb{Z}_{\geq 0} \neq \varnothing$ and let $\Gamma_{\geq 0} = \Gamma \cap \mathbb{Z}_{\geq 0}$. In the following we will condiser only $\Gamma_{\geq 0}$-spaces.

We begin by introducting a presheaf version of the previous concepts. A *presheaf of* $\Gamma_{\geq 0}$*-spaces in* $\mathbb{R}^n$ is just a presheaf $B : \mathrm{Open}(\mathbb{R}^n)^{op} \to \mathbf{NFre}_{\Gamma_{\geq 0}}$ of $\Gamma_{\geq 0}$-spaces. Let $\mathbf{B}_n$ be the presheaf category of them. Given a $\Gamma_{\geq 0}$-ambient $(\mathbf{X}, \gamma)$, let $X : \mathrm{Open}(\mathbb{R}^n)^{op} \to \mathbf{X}$ be a presheaf, i.e, let $X \in \mathrm{Psh}(\mathbb{R}^n; \mathbf{X})$. We say that $B \in \mathbf{B}^k_n$ is a *subobject of* (X, γ) if it becomes endowed with a natural trasformation $\imath : \gamma \circ B \hookrightarrow X$ which is objectwise a monomorphism. As in the previous section, let $\mathrm{Span}_\Gamma(X, \gamma)$ be the category of spans of those subobjects. We have functors

$$\mathrm{Base} : \mathrm{Span}_\Gamma(X, \gamma) \to \mathrm{Psh}(\mathbb{R}^n; \mathbf{X}) \quad \text{and} \quad \mathrm{Pb} : \mathrm{Span}_\Gamma(X, \gamma) \to \mathrm{Psh}(\mathbb{R}^n; \mathbf{X})$$

which to each span $(\imath, \imath')$ assigns the *base presheaf* $\mathrm{Base}(\imath, \imath') = X$ and which evaluate the pullback of $(\imath, \imath')$, i.e, $\mathrm{Pb}(\imath, \imath')(U) = B(U) \cap_{X(U),\gamma} B'(U)$. In the following we will write $\mathrm{Pb}(\imath, \imath') \equiv B \cap_{X,\gamma} B'$

whenever there is no risck of confusion. We say that a span $(\imath, \imath')$ is *proper* if it is objectwise proper, i.e, if there exists $L \in \mathbf{B}_n$ such that $\gamma \circ L \simeq B \cap_{X,\gamma} B'$. If exists, then L is unique up to natural isomorphisms and will be denoted by $B \cap_X B'$. Furthermore, we also demand that there exists $u : B \cap_X B' \Rightarrow B$ and $u' : B \cap_X B' \Rightarrow B'$ such that $\gamma \circ u \simeq u_\gamma$ and $\gamma \circ u' \simeq u'_\gamma$. We call $\mathbb{X} = (\mathbf{X}, \gamma, X, \imath, \imath')$ and $B \cap_{X,\gamma} B'$ the *intersection structure presheaf* (ISP) and the *intersection presheaf* between B and B' in (X, γ), respectively. We say that B and B' have *nontrivial intersection in* $\mathbb{X}$ if objectwise they have nontrivial intersection, i.e, if $(B \cap_X B')(U)$ have positive real dimension for every U.

If (B, ϵ) and (B', ϵ') are now presheaves of $(\Gamma_{\geq 0}, \epsilon)$-spaces in $\mathbb{R}^n$, i.e, if they take values in $\mathbf{NFre}_{\Gamma_{\geq 0},\Sigma}$ instead of in $\mathbf{NFre}_{\Gamma_{\geq 0}}$, let $\mathbf{B}_{n,\Sigma}$ be the associated presheaf category. We can apply the same strategy in order to define the category $\mathrm{Span}_{\Gamma,\Sigma}(X, \gamma_\Sigma)$ of spans of subobjects of (X, γ_Σ). If $\jmath : \gamma_\Sigma \circ B_\epsilon \Rightarrow X$ and $\jmath' : \gamma_\Sigma \circ B'_{\epsilon'} \Rightarrow X$ are two of those spans, we define an *intersection structure presheaf* between (B, ϵ) and (B', ϵ') as the tuple $\mathbb{X} = (\mathbf{X}, \gamma_\Sigma, X, \jmath, \jmath')$. Similarly, pullback and projection onto the base presheaf define functors

$$\mathrm{Base}_\Sigma : \mathrm{Span}_{\Gamma,\Sigma}(X, \gamma_\Sigma) \to \mathrm{Psh}(\mathbb{R}^n; \mathbf{X}) \quad \text{and} \quad \mathrm{Pb}_\Sigma : \mathrm{Span}_{\Gamma,\Sigma}(X, \gamma_\Sigma) \to \mathrm{Psh}(\mathbb{R}^n; \mathbf{X}).$$

We will write $\mathrm{Pb}_\Sigma(\jmath, \jmath') \equiv B_\epsilon \cap_{X,\gamma_\Sigma} B'_{\epsilon'}$. If the span is proper, the representing object in $\mathbf{B}_{n,\Sigma}$ is denoted simply by $B_\epsilon \cap_X B'_{\epsilon'}$, so that $\gamma_\Sigma \circ (B_\epsilon \cap_X B'_{\epsilon'}) \simeq B_\epsilon \cap_{X,\gamma_\Sigma} B'_{\epsilon'}$. When $(B, \epsilon, *)$ and $(B', \epsilon', *')$ are presheaves of multiplicative $\Gamma_{\geq 0}$-spaces, meaning that they take values in $\mathbf{NFre}_{\Gamma_{\geq 0},*}$, we denote their presheaf category $\mathbf{B}_{n,*}$ and define the *intersection space presheaf* for an intersection structure presheaf $\mathbb{X}_{*,*'} = (\mathbf{X}, \gamma_\Sigma, X, \jmath, \jmath')$ between $*$ and $*'$ as the presheaf which objectwise is the pullback below in the category $\mathrm{Psh}(\mathbb{R}^n; \mathbf{X})$. In other words, it is $\mathrm{Pb}_\Sigma((\jmath \circ \gamma_\Sigma \circ *, \jmath' \circ \gamma_\Sigma \circ *'))$. If these spans are proper we denote the representing object in $\mathbf{B}_{n,\Sigma}$ by $\mathrm{pb}(*, *'; X)$ and we say that $*$ and $*'$ are *nontrivially intersecting* in $\mathbb{X}$ if that representing object has objectwise positive dimension.

$$\begin{tikzcd}
\mathrm{pb}(*, *'; X, \gamma_\Sigma) \arrow[rr, Rightarrow] \arrow[dd, Rightarrow] & & \gamma_\Sigma \circ (B'_{\epsilon'} \otimes B'_{\epsilon'}) \arrow[d, Rightarrow, "\gamma_\Sigma \circ *'"] \\
& B_\epsilon \cap_{X,\gamma_\Sigma} B'_{\epsilon'} \arrow[r, Rightarrow] \arrow[d, Rightarrow] & \gamma_\Sigma \circ B'_\epsilon \arrow[d, Rightarrow, "\jmath'"] \\
\gamma_\Sigma \circ (B_\epsilon \otimes B_\epsilon) \arrow[r, Rightarrow, "\gamma_\Sigma \circ *"'] & \gamma_\Sigma \circ B_\epsilon \arrow[r, Rightarrow, "\jmath"'] & X
\end{tikzcd}$$

Let us now consider presheaves of distributive $\Gamma_{\geq 0}$-spaces in $\mathbb{R}^n$, i.e, which assigns to each open subset $U \subset \mathbb{R}^n$ a corresponding distributive $\Gamma_{\geq 0}$-space. Let $\mathbf{B}_{n,*,+}$ be the presheaf category of them. Define a *full intersection structure presheaf* (full ISP) as a tuple

$$\mathbb{X} = (\mathbf{X}, \gamma, \gamma_\Sigma, X, X_*, X_+, \imath, \imath', \jmath_*, \jmath'_*, \jmath_+, \jmath'_+),$$

where $(\mathbf{X}, \gamma, \gamma_\Sigma)$ is a full Γ-ambient, $X, X_*, X_+ \in \mathrm{Psh}(\mathbb{R}^n; \mathbf{X})$ are presheaves and $(\imath, \imath')$, $(\jmath_*, \jmath'_*)$ and $(\jmath_+, \jmath'_+)$ are spans of subobjects of X, X_* and X_+, respectively. Given $B, B' \in \mathbf{B}_{n,*,+}$ we say that a full ISP $\mathbb{X}$ is *between* B *and* B' if B is on the domain of $\imath$, $\jmath_*$ and $\jmath_+$, while B' is on the domain of $\imath'$, $\jmath'_*$ and $\jmath'_+$. Thus, e.g, $\imath : \gamma \circ B \Rightarrow X$, $\jmath_* : \gamma_\Sigma \circ B_\epsilon \Rightarrow X_*$ and $\jmath'_+ : \gamma_\Sigma \circ B'_{\epsilon'} \Rightarrow X_+$. We can also write $\mathbb{X}$ as $\mathbb{X} = (\mathbb{X}_0, \mathbb{X}_*, \mathbb{X}_+)$, where

$$\mathbb{X}_0 = (\mathbf{X}, \gamma, X, \imath, \imath'), \quad \mathbb{X}_* = (\mathbf{X}, \gamma_\Sigma, X_*, \jmath_*, \jmath'_*) \quad \text{and} \quad \mathbb{X}_+ = (\mathbf{X}, \gamma_\Sigma, X_+, \jmath_+, \jmath'_+)$$

are ISP between B and B', between $*$ and $*'$, and between $+$ and $+'$, respectively. The *full intersection presheaf* between B and B' in a full ISP $\mathbb{X}$ is the triple consisting of the intersection presheaves between B and B' in $\mathbb{X}_0$, between $*$ and $*'$ in $\mathbb{X}_*$ and between $+$ and $+'$ in $\mathbb{X}_+$. When $\mathbb{X}$ is proper, the corresponding full intersection presheaf has a representing object $B \cap_{\mathbb{X}} B'$ in $\mathbf{B}_n \times \mathbf{B}_{n,\Sigma} \times \mathbf{B}_{n,\Sigma}$, given by

$$B \cap_{\mathbb{X}} B' = (B \cap_{\mathbb{X}_0} B', \mathrm{pb}(*, *'; X_*), \mathrm{pb}(+, +'; X_+)).$$

We say that B and B' have *nontrivial intersection* in a proper full ISP $\mathbb{X}$ if each of the three presheaves in $B \cap_{\mathbb{X}} B'$ are nontrivial, i.e, have objectwise positive real dimension.

Example 3.1. By means of varying U and restricting to $\Gamma_{\geq 0}$, for every fixed $n \geq 0$, each multiplicative Γ-space in Examples 2.2-2.6 defines a presheaf of multiplicative $\Gamma_{\geq 0}$-spaces in $\mathbb{R}^n$. Furthermore, by the same process, from Example 2.8 and Example 2.9 we get presheaves of distributive $\Gamma_{\geq 0}$-structures in $\mathbb{R}^n$. Vectorial ISP can be build by following Example 2.10 and Example 2.11.

Let us introduce $C^{k,\alpha}_{n,\beta}$-presheaves, which will be the structural presheaves appearing in the definition of $B^k_{\alpha,\beta}$-manifold. Le $K', B' \in \mathbf{B}_n$. An *action* of K' in B' is just a morphism $\kappa' : K' \otimes B' \Rightarrow B'$, where the tensor product is defined objectwise. A *morphism* between actions $\kappa : K \otimes B \Rightarrow B$ and $\kappa' : K' \otimes B' \Rightarrow B'$ is just a morphism (f, g) in the arrow category such that $f = \ell \otimes g$. More precisely, it is a pair (ℓ, g), where $\ell : K \Rightarrow K'$ and $g : B \Rightarrow B'$ are morphisms in $\mathbf{B}_n$ such that the diagram below commutes. Let $\mathbf{Act}_n$ be the category of actions and morphisms between them.

$$\begin{tikzcd} K' \otimes B' \arrow[r, Rightarrow, "\kappa'"] & B' \\ K \otimes B \arrow[u, Rightarrow, "\ell\otimes g"] \arrow[r, Rightarrow, "\kappa"'] & B \arrow[u, Rightarrow, "g"'] \end{tikzcd}$$

Given $B, B' \in \mathbf{B}_n$, an action $\kappa' : K' \otimes B' \Rightarrow B'$ and a proper ISP $\mathbb{X}_0$ between B and B', we say that B is *compatible* with κ' in $\mathbb{X}_0$ if there exists an action $\kappa : A \otimes B \cap_{\mathbb{X}_0} B' \Rightarrow B \cap_{\mathbb{X}_0} B'$ in the intersection presheaf $B \cap_{\mathbb{X}_0} B'$ and a morphism $\ell : A \Rightarrow K'$ such that (ℓ, u') is a morphism of actions, where $u' : B \cap_{\mathbb{X}_0} B' \Rightarrow B'$ is the morphism such that $\gamma \circ u' \simeq u'_\gamma$, existing due to the properness hypothesis. If it is possible to choose (A, ℓ) such that is objectwise a monomorphism, B is called *injectively compatible* with κ' in $\mathbb{X}_0$. Given $B, B' \in \mathbf{B}_{n,*,+}$ we say that B is *compatible with* B' in a proper full ISP $\mathbb{X} = (\mathbb{X}_0, \mathbb{X}_*, \mathbb{X}_+)$ if:

1. the proper full ISP $\mathbb{X}$ is between them;

2. they have nontrivial intersection in $\mathbb{X}$;

3. there exists an action $\kappa' : K' \otimes B' \Rightarrow B'$ in B' such that B is injectively compatible with κ' in $\mathbb{X}_0$.

Example 3.2. Suppose that C is a presheaf of increasing $\Gamma_{\geq 0}$-Fréchet algebras, i.e, such that $C(U)$ is a $\Gamma_{\geq 0}$-space for which each $C(U)_i$ is a Fréchet algebra such that there exists a continuous embedding of topological algebras $C(U)_i \subset C(U)_j$ if $i \leq j$. Let $p : \Gamma_{\geq 0} \to \Gamma_{\geq 0}$ be any function such that $i \leq p(i)$. For $K' = C$ and $B' = C_p$ we have an action $\kappa' : K' \otimes B' \Rightarrow B'$ such that $\kappa'_{U,i} : C(U)_i \otimes C(U)_{p(i)} \to C(U)_{p(i)}$ is obtainned by embedding $C(U)_i$ in $C(U)_{p(i)}$ and then using the Fréchet algebra multiplication of $C(U)_{p(i)}$. In other words, under the hypothesis C is actually a presheaf of multiplicative $\Gamma_{\geq 0}$-spaces, with $\epsilon(i,j) = \max(i,j)$, so that $\kappa'_{U,i} = *_{i,p(i)}$. An analogous discussion holds for decreasing presheaves.

As a particular case of the last example, we see that pointwise multiplication induces an action of the presheaf C^k, such that $C^k(U)_i = C^k(U)$ for every $U \subset \mathbb{R}^n$ and $i \in \Gamma_{\geq 0}$, on the presheaf $C^{k-\beta}$, such that $C^{k-\beta}(U)_i = C^{k-\beta(i)}(U)$, where $\beta : \Gamma_{\geq 0} \to [0,k]$ is some fixed integer function. Given $\alpha : \Gamma_{\geq 0} \to \Gamma_{\geq 0}$ and $\beta : \Gamma_{\geq 0} \to [0,k]$, we define a *$C^{k,\alpha}_{n,\beta}$-presheaf in a proper full ISP* $\mathbb{X}$ as a presheaf $B \in \mathbf{B}_{n,*,+}$ which such that B_α is compatible with $C^{k-\beta}$ in $\mathbb{X}$. Thus, B is a $C^{k,\alpha}_{n,\beta}$-presheaf in $\mathbb{X}$ if for every $U \subset \mathbb{R}^n$:

1. *the intersection spaces $B_\alpha(U) \cap_{X(U)} C^{k-\beta}(U)$, etc., have positive real dimension.* This means that at least in some sence (i.e, internal to X) the abstract spaces defined by B have a concrete interpretation in terms of differentiable functions satisfying some further properties/regularity. Furthermore, under this interpretation, the sum and the multiplication in B are the pointwise sum and multiplication of differentiable functions added of properties/regularity;

2. *there exists a subspace $A(U) \subset C^k(U)$ and for every $i \in \Gamma_{\geq 0}$ a morphism $\star_{U,i}$ making commutative the diagram below.* This means that if we regard the abstract multiplication of B as pointwise multiplication of functions with additional properties, then that multiplication becomes closed under a certain set of smooth functions.

$$\begin{tikzcd}
C^k(U) \otimes C^{k-\beta(i)}(U) \arrow[r, "\cdot_{U,i}"] & C^{k-\beta(i)}(U) \\
A(U) \otimes B_{\alpha(i)}(U) \cap_{X(U)} C^{k-\beta(i)}(U) \arrow[u, hook] \arrow[r, dashed, "\star_{U,i}"'] & B_{\alpha(i)}(U) \cap_{X(U)} C^{k-\beta(i)}(U) \arrow[u, hook]
\end{tikzcd} \tag{5}$$

If, in addition, the intersection structure $\mathbb{X}_0$ between B_α and $C^{k-\beta}(U)$ is such that the image of each $u' : B_{\alpha(i)} \cap_{X(U)} C^{k-\beta(i)}(U) \to C^{k-\beta(i)}(U)$ is closed, we say that B is a *strong $C^{k,\alpha}_{n,\beta}$-presheaf in* $\mathbb{X}$. Finally we say that B is a *nice $C^{k,\alpha}_{n,\beta}$-presheaf in* $\mathbb{X}$ if it is possible to choose A such that $\dim_{\mathbb{R}} A(U) \cap C^\infty_b(U) \geq 1$, where $C^\infty_b(U)$ is the space of bump functions. Notice that being nice does not depends on the ISP $\mathbb{X}$.

Example 3.3. From Example 3.2 and Proposition 2.1 we see that the presheaf of distributive structures C^{k-} is a $C^{k,\alpha}_{n,\beta}$-presheaf in the full ISP which is objectwise the standard vectorial intersection structure of Example 2.11, with $A = C^\infty$. Since $C^\infty_b(U) \subset C^\infty(U)$ for every U, we conclude that C^{k-} is actually a nice $C^{k,\alpha}_{n,\beta}$-presheaf.

Example 3.4. The same arguments of the previous example can be used to show that the presheaf L of distributive $\Gamma_{\geq 0}$-spaces $L(U)_i = L^i(U)$, endowed with the distributive structure given by pointwise addition and multiplication, is a nice $C^{k,\alpha}_{n,\beta}$-presheaf in the standard ISP, for $A(U) = C^\infty_b(U)$ or $A(U) = \mathcal{S}(U)$, where $\mathcal{S}(U)$ is the Schwarz space [13]. An analogous conclusion is valid if we replace pointwise multiplication with convolution product.

Let $\mathbf{C}^{k,\alpha}_{n,\beta}$ be the category whose objects are pairs $(B, \mathbb{X})$, where $\mathbb{X}$ is a proper full ISP and $B \in \mathbf{B}_{n,*,+}$ is a presheaf of distributive $\Gamma_{\geq 0}$-spaces which is a $C^{k,\alpha}_{n,\beta}$-presheaf in $\mathbb{X}$. Morphisms ξ :

$(B, \mathbb{X}) \Rightarrow (B', \mathbb{X}')$ are just morphisms $\xi : B \Rightarrow B'$ in $\mathbf{B}_{n,*,+}$. We have a projection $\pi_n : \mathbf{C}^{k,\alpha}_{n,\beta} \to \mathbf{Cat}$ assigning to each pair $(B, \mathbb{X})$ the ambient category $\mathbf{X}$ in the full Γ-ambient $(\mathbf{X}, \gamma, \gamma_\Sigma)$ of $\mathbb{X}$, where $\mathbf{Cat}$ is the category of all categories. Notice that such projection really depends only on n (and not on α, β and k). Let $\mathbf{Cat}_n$ be the image of π_n and for each $\mathbf{X} \in \mathbf{Cat}_n$ let $\mathbf{C}^{k,\alpha}_{n,\beta}(\mathbf{X})$ be the preimage $\pi_n^{-1}(\mathbf{X})$. Closing the section, let us study the dependence of the fiber $\pi_n^{-1}(\mathbf{X})$ on the variables k, n and β. We need some background.

Let $\mathbb{X}$ and $\mathbb{Y}$ be two (not necessarily proper) ISP in a Γ-ambient $(\mathbf{X}, \gamma)$. A *connection* between them is a transformation $\xi : X \Rightarrow Y$ between the underlying base presheaves such that for every two presheaves B, B' of Γ-subspaces for which both $\mathbb{X}$ and $\mathbb{Y}$ are between B and B', the diagram below commutes. Thus, by universality we get the dotted arrow.

$$\begin{tikzcd}
B \cap_{X,\gamma} \gamma \circ B' \arrow[rr, Rightarrow] \arrow[dr, Rightarrow, dashed, "\overline{\xi}"] \arrow[dd, Rightarrow] & & \gamma \circ B' \arrow[dd, Rightarrow, "\jmath'"] \arrow[ddd, Rightarrow, bend left, "\ell'"] \\
& B \cap_{Y,\gamma} B' \arrow[ur, Rightarrow] \arrow[dl, Rightarrow] & \\
\gamma \circ B \arrow[rr, Rightarrow, "\jmath"'] \arrow[drr, Rightarrow, bend right, "\ell"'] & & X \arrow[d, Rightarrow] \\
& & Y
\end{tikzcd} \tag{6}$$

Suppose now that $\mathbf{X}$ is a monoidal Γ-ambient and that B and B' are presheaves of $(\Gamma, \mathbf{c})$-spaces in $\mathbf{X}$, i.e, which objectwise belong to $\mathbf{NFre}_{\Gamma,\mathbf{c}}(\mathbf{X})$ (recall the notation in the end of Section 2). Notice that for any connection $\xi : X \Rightarrow Y$ we have the first commutative diagram below, where $\mathbf{c} = (\mathrm{cl}, \mu)$ is the closure structure. We say that a presheaf of Γ-spaces Z is *central* for $(B, B', \mathbb{X}, \mathbb{Y}, \mathbf{c})$ if it becomes endowed with an embedding $\imath : B \cap_{X,\gamma} B' \Rightarrow Z$ and a morphism η such that the second diagram below commutes.

$$\begin{tikzcd}
B \cap_{X,\gamma} B' \arrow[r, Rightarrow, "\mu \circ \imath_X"] \arrow[d, Rightarrow, "\overline{\xi}"'] & \mathrm{cl}_X(B \cap_{X,\gamma} B') \arrow[d, Rightarrow, "\mathrm{cl} \circ \overline{\xi}"] \\
B \cap_{Y,\gamma} B' \arrow[r, Rightarrow, "\mu \circ \imath_Y"'] & \mathrm{cl}_Y(B \cap_{Y,\gamma} B')
\end{tikzcd}
\qquad
\begin{tikzcd}
B \cap_{X,\gamma} B' \arrow[rr, Rightarrow, "\mu \circ \imath_X"] \arrow[dr, Rightarrow, "\imath"] \arrow[dd, Rightarrow, "\overline{\xi}"'] & & \mathrm{cl}_X(B \cap_{X,\gamma} B') \arrow[dd, Rightarrow, "\mathrm{cl} \circ \overline{\xi}"] \\
& Z \arrow[dr, Rightarrow, dashed, "\eta"] & \\
B \cap_{Y,\gamma} B' \arrow[rr, Rightarrow, "\mu \circ \imath_Y"'] & & \mathrm{cl}_Y(B \cap_{Y,\gamma} B'_{\epsilon'})
\end{tikzcd}$$

Remark 3.1. If $\mathbf{X}$ is complete/cocomplete then η always exists, at least up to universal natural transformations. Indeed, notice that η is actually the extension of $\mu_Y \circ \overline{\xi}$ by $\imath$ (equivalently, the extension of $\mathrm{cl} \circ \overline{\xi} \circ \mu_X$ by $\imath$), so that due to completeness/cocompleteness we can take the left/right Kan extension [16, 7].

Remark 3.2. In an analogous way we can define connections between presheaves of (Γ, ϵ)-spaces and central (Γ, ϵ)-presheaves relative to some closure structure. These can also be regarded as Kan extensions, so that if $\mathbf{X}$ is complete/cocomplete they will also exist up to universal natural transformations.

4 Theorem A

We say that a ISP $\mathbb{X} = (\mathbf{X}, \gamma, X, \imath, \imath')$ is *monoidal* if the underlying Γ-ambient $(\mathbf{X}, \gamma)$ is monoidal. A full ISP $\mathbb{X} = (\mathbb{X}_0, \mathbb{X}_*, \mathbb{X}_+)$ is *partially monoidal* if the ISP $\mathbb{X}_0$ is monoidal. Fixed monoidal Γ-ambient $(\mathbf{X}, \gamma, \circledast)$ and given an integer function $p : \Gamma_{\geq 0} \to \mathbb{Z}_{\geq 0}$ such that $p(i) \leq k - \beta(i)$ for each $i \in \Gamma_{\geq 0}$, define a *p-structure* on a $C^{k,\alpha}_{n,\beta}$-presheaf B in a partially monoidal full proper ISP $\mathbb{X} = (\mathbb{X}_0, \mathbb{X}_*, \mathbb{X}_+)$, as another non-necessarily proper monoidal ISP $\mathbb{Y}_0 = (\mathbf{X}, \gamma, Y)$, together with a closure structure $\mathbf{c} = (\mathrm{cl}, \mu, \phi)$ and a connection $\xi : Y_0 \Rightarrow X$ such that $Z = B_\alpha \cap_{X,\gamma} C^p$ is central for $(B_\alpha, C^{k-\beta}, \mathbb{X}_0, \mathbb{Y}_0, \mathbf{c})$ relative to the canonical embedding $\imath : B_\alpha \cap_{X,\gamma} C^{k-\beta} \hookrightarrow Z$ induced by universality of pullbacks applied to $C^{k-\beta} \hookrightarrow C^p$, which exists due to the condition $p \leq k-\beta$. Define a $C^{k,\alpha}_{n,\beta;p}$*-presheaf in* $\mathbb{X}$ as a $C^{k,\alpha}_{n,\beta}$-presheaf in B in $\mathbb{X}$ endowed with a p-structure and let $\mathbf{C}^{k,\alpha}_{n,\beta;p}$ be the full subcategory of them. Furthermore, let $\mathbf{C}^{k,\alpha}_{n,\beta;p}(\mathbf{X}_{\mathrm{full}}) \subset \mathbf{C}^{k,\alpha}_{n,\beta}(\mathbf{X})$ the corresponding full subcategories of pairs $(B, \mathbb{X})$ for which $\pi_n(B, \mathbb{X}) = \mathbf{X}$ and γ is a full functor.

We can now prove Theorem A. It will be a consequence of the following more general theorem:

Theorem 4.1. *For any category with pullbacks* $\mathbf{X}$*, there are full embeddings*

(1) $\imath : \mathbf{C}^{k,\alpha}_{n,\beta;p}(\mathbf{X}_{\mathrm{full}}) \hookrightarrow \mathbf{C}^{k,\alpha}_{n,p}(\mathbf{X})$*, if* $p - \beta \leq k$*;*

(2) $f_* : \mathbf{C}^{k,\alpha}_{n,\beta}(\mathbf{X}) \hookrightarrow \mathbf{C}^{k,\alpha}_{r,\beta}(\mathbf{X})$*, for any continuous injective map* $f : \mathbb{R}^n \to \mathbb{R}^r$*.*

Proof. We begin by proving (1). Notice that the fiber $\pi_n^{-1}(\mathbf{X})$ is a nonempty category only if $\mathbf{X}$ for every k, α, β is part of a partially monoidal proper full ISP, with γ full, in which at least one $C^{k,\alpha}_{n,\beta}$-presheaf is defined. Since for empty fibers the result is obvious, we assume the above condition. Thus, given $(B, \mathbb{X}) \in \mathbf{C}^{k,\alpha}_{n,\beta;p}(\mathbf{X}_{\mathrm{full}})$ (which exists by the assumption on $\mathbf{X}$), we will show that it actually belongs to $\mathbf{C}^{k,\alpha}_{n,p}(\mathbf{X})$. Since we are working with full subcategories this will be enough for (1). We assert that B_α and C^p have nontrivial intersection in $\mathbb{X}$. Since the functor γ creates null-objects, it is enough to prove that $B_\alpha \cap_{X,\gamma} C^p$, $(B_\alpha)_\epsilon \cap_{X_*,\gamma_\Sigma} (C^p)_{\epsilon'}$ and $(B_\alpha)_\delta \cap_{X_*,\gamma_\Sigma} (C^p)_{\delta'}$. But, since $\delta' = \epsilon' = \min$ and since $p \leq \beta - k$, from universality and stability of pullbacks under monomorphisms we get monomorphisms $B_\alpha \cap_{X,\gamma} C^{k-\beta} \hookrightarrow B_\alpha \cap_{X,\gamma} C^p$, etc. Now, being B a $C^{k,\alpha}_{n,\beta}$-presheaf in $\mathbb{X}$,the left-hand sides $B_\alpha \cap_{X,\gamma} C^{k-\beta}$, etc., are nontrivial, which implies B_α and C^p have nontrivial intersection in $\mathbb{X}$. By the same arguments we get the commutative diagram below, where A is the presheaf arising from the $C^{k,\alpha}_{n,\beta}$-structure of B. Our task is to extend $\star_\beta$ by replacing $k - \beta$ with p. If $p(i) - \beta(i) = k$ there is nothing to do for such i. Thus, suppose $p(i) - \beta(i) < k$ for all i. More precisely, our task is to get the dotted arrow in the second diagram below, where the long vertical arrows arise from the universality of pullbacks, as above. We use simple arrows intead of double arrows in order to simplify the notation.

$$\begin{tikzcd}
C^\infty \otimes C^p \arrow[r, "\cdot"] & C^p \\
C^\infty \otimes C^{k-\beta} \arrow[r, "\cdot"] \arrow[u, hook] & C^{k-\beta} \arrow[u, hook] \\
A \otimes (B_\alpha \cap_X C^{k-\beta}) \arrow[r, "\star_\beta"'] \arrow[u, hook] & B_\alpha \cap_X C^{k-\beta} \arrow[u, hook]
\end{tikzcd}
\qquad
\begin{tikzcd}
A \otimes (B_\alpha \cap_X C^p) \arrow[r, dashed, "\star_p"] \arrow[d, hook] & B_\alpha \cap_X C^p \arrow[d, hook] \\
C^\infty \otimes C^p \arrow[r, "\cdot"] & C^p \\
C^\infty \otimes C^{k-\beta} \arrow[r, "\cdot"] \arrow[u, hook] & C^{k-\beta} \arrow[u, hook] \\
A \otimes (B_\alpha \cap_X C^{k-\beta}) \arrow[r, "\star_\beta"'] \arrow[u, hook] \arrow[uuu, bend left] & B_\alpha \cap_X C^{k-\beta} \arrow[u, hook] \arrow[uuu, bend right]
\end{tikzcd}$$

Since $(B, \mathbb{X}) \in \mathbf{C}^{k,\alpha}_{n,\beta;p}(\mathbf{X}_{\text{full}})$, there exists a ISP $\mathbb{Y}_0$, a connection $\xi : Y \Rightarrow X$ and a closure structure $\mathbf{c} = (\text{cl}, \mu, \phi)$ such that $B_\alpha \cap_{X,\gamma} C^p$ is central for $(B_\alpha, C^{k-\beta}, \mathbb{X}_0, \mathbb{Y}_0, \mathbf{c})$. Since any morphism extends to the closure and recalling that γ is strong monoidal, we have the diagram below. It commutes due to the commutativity of (3) and (4) and due to the naturality of ψ. We are also using that $\gamma \circ B \cap_X B' \simeq B \cap_{X,\gamma} B'$. Furthermore, μ_A is $\mu_{\gamma(\imath_A)}$, where $\imath_A : A \hookrightarrow C^\infty$ is the embedding. Again we use simple arrows instead of double arrows.

$$\begin{tikzcd}
\text{cl}_{\gamma(C^\infty)}(\gamma(A)) \circledast \text{cl}_X(B_\alpha \cap_{X,\gamma} C^{k-\beta}) \arrow[d, "\phi"'] & \text{cl}_{\gamma(C^\infty)}(\gamma(A)) \circledast \text{cl}_Y(B_\alpha \cap_{Y,\gamma} C^{k-\beta}) \arrow[l, "id \circledast (\text{cl} \circ \bar{\xi})"'] \\
\text{cl}_{\gamma(C^\infty) \circledast X}(\gamma(A) \circledast (B_\alpha \cap_{X,\gamma} C^{k-\beta})) \arrow[r, "\hat{\mu}_{\gamma(\star_\beta) \circ \psi}"] & B_\alpha \cap_{X,\gamma} C^{k-\beta} \\
\gamma(A \otimes (B_\alpha \cap_X C^{k-\beta})) \arrow[r, "\gamma(id \otimes \imath)"'] \arrow[ur, "\gamma(\star_\beta)"] \arrow[d, shift left, "\psi^{-1}"] & \gamma(A \otimes (B_\alpha \cap_X C^p)) \arrow[d, shift left, "\psi^{-1}"] \\
\gamma(A) \circledast (B_\alpha \cap_{X,\gamma} C^{k-\beta}) \arrow[r, "\gamma(id) \circledast \gamma(\imath)"'] \arrow[u, shift left, "\psi"] \arrow[uu, bend left, "\mu_{A \circledast X}"] & \gamma(A) \circledast (B_\alpha \cap_{X,\gamma} C^p) \arrow[u, shift left, "\psi"] \arrow[uuu, bend right, "\mu_A \circledast \eta"']
\end{tikzcd}$$

Let us consider the composition morphism

$$f^p_\beta = \hat{\mu}_{\gamma(\star_\beta) \circ \psi} \circ \phi \circ (id \circledast (\text{cl} \circ \bar{\xi})) \circ (\mu_A \circledast \eta) \circ \psi^{-1} : \gamma(A \otimes B_\alpha \cap_X C^p) \to \gamma(B_\alpha \cap_X C^{k-\beta}).$$

On the other hand, we have $\gamma(\imath_{b,p}) : \gamma(B_\alpha \cap_X C^{k-\beta}) \hookrightarrow \gamma(B_\alpha \cap_X C^p)$ arising from the embedding $C^{k-\beta} \hookrightarrow C^p$. Composing them we get a morphism

$$\gamma(\imath_{b,p}) \circ f^p_\beta : \gamma(A \otimes B_\alpha \cap_X C^p) \to \gamma(B_\alpha \cap_X C^p). \tag{7}$$

Since γ is fully-faithful, there exists exactly one $\star_p : A \otimes B_\alpha \cap_X C^p \to B_\alpha \cap_X C^p$ such that $\gamma(\star_p) = \gamma(\imath_{b,p}) \circ f^p_\beta$, which is our desired map. That it really extends $\star_\beta$ follows from the commutativity of all diagrams involved in the definition of $\star_p$. For (2), recall that any continuous map $f : \mathbb{R}^n \to \mathbb{R}^r$ induces a pushforward functor $f^{-1} : \mathbf{Psh}(\mathbb{R}^n) \to \mathbf{Psh}(\mathbb{R}^r)$ between the corresponding presheaf categories of presheaves of sets, which becomes an embedding when f is injective [17]. Since $(f^{-1}F)(U) = F(f^{-1}(U))$, it is straighforward to verify that if $(B, \mathbb{X}, A)$ belongs to $\mathbf{C}^{k,\alpha}_{n,\beta}(\mathbf{X})$, then $(f^{-1}B, f^{-1}A, f^{-1}\mathbb{X}) \in \mathbf{C}^{k,\alpha}_{r,\beta}(\mathbf{X})$, where $f^{-1}\mathbb{X}$ is defined componentwise. Thus, we have an injective on objects functor $f^{-1} : \mathbf{C}^{k,\alpha}_{n,\beta}(\mathbf{X}) \to \mathbf{C}^{k,\alpha}_{r,\beta}(\mathbf{X})$. Because we are working with full subcategories, it follows that f^{-1} is full and therefore a full embedding. $\square$

Corollary 4.1. *Let $\mathbf{C}^{k,\alpha}_{n,\beta;l}(\mathbf{X}_{\text{full}})$ and $\mathbf{C}^{k,\alpha}_{n,\beta;\beta'}(\mathbf{X}_{\text{full}})$ be the category $\mathbf{C}^{k,\alpha}_{n,\beta;p}(\mathbf{X}_{\text{full}})$ for $p_l(i) = l - \beta(i)$ and $p_{\beta'}(i) = k - \beta'(i)$, respectively. Then there are full embeddings*

- $\jmath : \mathbf{C}^{k,\alpha}_{n,\beta;\beta'}(\mathbf{X}_{\text{full}}) \hookrightarrow \mathbf{C}^{k,\alpha}_{n,\beta}(\mathbf{X}_{\text{full}})$, if $\beta' \leq \beta$;
- $\imath : \mathbf{C}^{k,\alpha}_{n,\beta;l}(\mathbf{X}_{\text{full}}) \hookrightarrow \mathbf{C}^{l,\alpha}_{n,\beta}(\mathbf{X}_{\text{full}})$, if $l \leq k$.

Proof. Just notice that if $\beta' \leq \beta$ implies $p_{\beta'} - \beta \leq k$ and that $l \leq k$ implies $p_l - \beta \leq k$, and then uses (1) of Theorem 4.1. $\square$

Theorem A. *There are full embeddings*

(1) $\imath : \mathbf{C}^{k,\alpha}_{n,\beta;l} \hookrightarrow \mathbf{C}^{l,\alpha}_{n,\beta}$, *if $l \leq k$;*

(2) $f_* : \mathbf{C}^{k,\alpha}_{n,\beta} \hookrightarrow \mathbf{C}^{k,\alpha}_{r,\beta}$, *for any continuous injective map $f : \mathbb{R}^n \to \mathbb{R}^r$;*

(3) $\jmath : \mathbf{C}^{k,\alpha}_{n,\beta;\beta'} \hookrightarrow \mathbf{C}^{k,\alpha}_{r,\beta'}$, *if $\beta' \leq \beta$.*

Proof. Straighforward from Corollary 4.1 and condition (2) of Theorem 4.1. $\square$

Remark 4.1. The requirement of γ being full is a bit strong. Indeed, our main examples of Γ-ambients are the vectorial ones. But requiring a full embedding $\gamma : \mathbf{NFre}_\Gamma \to \mathbf{Vec}_{\mathbb{R},\Gamma}$ is a really strong condition. When looking at the proof of Theorem 4.1 we see that the only time when the full hypothesis on γ was needed is to ensure that the morphism (7) in $\mathrm{Psh}(\mathbb{R}^n; \mathbf{X})$ is induced by a morphism in $\mathbf{B}_n$. Thus, the hypothesis of being full can be clearly weakened. Actually, the hypothesis on γ being an embedding can also be weakened. In the end, the only hypothesis needed in order to develop the previous results is that γ creates null-objects and some class of monomorphisms. Since from now on we will not focus on the theory of $C^{k,\alpha}_{n,\beta}$-presheaves itself, we will not modify our hypothesis. On the other hand, in futures works concerning the study of the categories $\mathbf{C}^{k,\alpha}_{n,\beta}$ this refinement on the hypothesis will be very welcome.

5 $B^k_{\alpha,\beta}$-Manifolds

Let $\mathbb{X}$ be a proper full intersection structure and let $B \in \mathbf{C}^{k,\alpha}_{n,\beta}$ be a $C^{k,\alpha}_{n,\beta}$-presheaf in $\mathbb{X}$. A C^k-function $f : U \to \mathbb{R}^m$ is called a *(B, k, α, β)-function in $(\mathbb{X}, m)$* if $\partial^\mu f_j \in B_{\alpha(i)}(U) \cap_{X(U)} C^{k-\beta(i)}(U)$ for all $|\mu| = i$ and $i \in \Gamma_{\geq 0}$. Due to the compatibility between the operations of B_α and $C^{k-\beta}$ at the intersection, it follows that the collection $B^k_{\alpha,\beta}(U; \mathbb{X}, m)$ of all (B, k, α, β)-functions in $(\mathbb{X}, m)$ is a real vector space. This will become more clear in the next proposition. First, notice that by varying $U \subset \mathbb{R}^n$ we get a presheaf (at least of sets) $B^k_{\alpha,\beta}(-; \mathbb{X}, m)$. Recall that a *strong* $C^{k,\alpha}_{n,\beta}$-presheaf is one in which $B_\alpha \cap_X C^{k-\beta} \hookrightarrow C^{k-\beta}$ is objectwise closed.

Proposition 5.1. *For every $B \in \mathbf{C}^{k,\alpha}_{n,\beta}$, every $\mathbb{X}$ and every $m \geq 0$, the presheaf of (B, k, α, β)-functions in $(\mathbb{X}, m)$ is a presheaf of real vector spaces. If B is strong, then it is actually a presheaf of nuclear Fréchet spaces.*

Proof. Consider the following spaces:

$$\mathbb{C}^k_{\alpha,\beta}(U,m)=\prod_{\kappa(m)} C^{k-\beta(i)}(U) \quad \text{and} \quad \mathbb{BC}^k_{\alpha,\beta}(U;\mathbb{X},m)=\prod_{\kappa(m)} B_{\alpha(i)}(U)\cap_{X(U)} C^{k-\beta(i)}(U),$$

where $\prod_{\kappa(m)}=\prod_{j=1}^{m}\prod_i\prod_{|\alpha|=i}\prod_{m^i}$. Since they are countable products of nuclear Fréchet spaces, they have a natural nuclear Fréchet structure. Consider the map $j^k : C^k(U;\mathbb{R}^m)\to \mathbb{C}^{k,\alpha}(U,m)$, given by $j^k f=(\partial^\mu f_j)_{\mu,j}$, with $|\mu|=i$, and notice that $B^k_{\alpha,\beta}(U;\mathbb{X},m)$ is the preimage of j^k by $\mathbb{BC}^k_{\alpha,\beta}(U;\mathbb{X},m)$. The map j^k is linear, so that any preimage has a linear structure, implying that the space of (B,k,α,β)-functions is linear. But j^k is also continuous in those topologies, so that if $\mathbb{BC}^k_{\alpha,\beta}(U;\mathbb{X},m)$ is a closed subset in $\mathbb{C}^k_{\alpha,\beta}(U,m)$, then $B^k_{\alpha,\beta}(U;\mathbb{X},m)$ is a closed subset of a nuclear Fréchet spaces and therefore it is also nuclear Fréchet. This is ensured precisely by the strong hypothesis on B. □

Example 5.1. Even if B is not strong, the space of (B,k,α,β)-functions may have a good structure. Indeed, let $p\in[1,\infty]$ be fixed, let $\alpha(i)=p$, $\beta(i)=i$ and let L be the presheaf $L(U)_i=L^i(U)$, regarded as a $C^{k,\alpha}_{n,\beta}$-presheaf in the standard ISP. Thus, $L_\alpha(U)\cap_{X(U)} C^{k-\beta}(U)$ is the $\Gamma_{\geq 0}$-space of components $L^p(U)\cap C^{k-i}(U)$. A (B,k,α,β)-function is then a C^k-map such that $\partial^\mu f_j\in L^p(U)\cap C^{k-i}(U)$ for every $|\mu|=i$. Therefore, the space of all of them is the strong (in the sense of classical derivatives) Sobolev space $W^{k,p}(U;\mathbb{R}^m)$, which is a Banach space [13]. But $L^p(U)\cap C^{k-i}(U)$ is clearly not closed in $C^{k-i}(U)$, since sequences of L^p-integrable C^k-functions do not necessarily converge to L^p-integrable maps [13].

Remark 5.1. In order to simplify the notation, if $m=n$ the space of (B,k,α,β)-functions will be denoted by $B^k_{\alpha,\beta}(U;\mathbb{X})$ instead of $B^k_{\alpha,\beta}(U;\mathbb{X},n)$.

Let M be a Hausdorff paracompact topological space. A *n-dimensional C^k-structure* in M is a C^k-atlas in the classical sense, i.e, a family $\mathcal{A}$ of coordinate systems $\varphi_i : U_i\to\mathbb{R}^n$ whose domains cover M and whose transition functions $\varphi_j\circ\varphi_i^{-1}:\varphi(U_{ij})\to\mathbb{R}^n$ are C^k, where $U_{ij}=U_i\cap U_j$. A *n-dimensional C^k-manifold* is a pair $(M,\mathcal{A})$, where $\mathcal{A}$ is a maximal C^k-structure. Given a $C^{k,\alpha}_{n,\beta}$-presheaf B in a proper full ISP $\mathbb{X}$, define a *$(B^k_{\alpha,\beta},\mathbb{X})$-structure* on a C^k-manifold $(M,\mathcal{A})$ as a subatlas $\mathcal{B}^k_{\alpha,\beta}(\mathbb{X})\subset\mathcal{A}$ such that $\varphi_j\circ\varphi_i^{-1}\in B^{k,n}_{\alpha,\beta}(\varphi_i(U_{ij});\mathbb{X}(\varphi_i(U_{ij})))$. A *$(B^k_{\alpha,\beta},\mathbb{X})$-manifold* is one in which a $(B^k_{\alpha,\beta},\mathbb{X})$-structure has been fixed. A *$(B^k_{\alpha,\beta},\mathbb{X})$-morphism* between two C^k-manifolds is a C^k-function $f:M\to M'$ such that $\phi f\varphi^{-1}\in B^{k,n}_{\alpha,\beta}(\varphi(U);\mathbb{X}(\varphi(U)))$, for every $\varphi\in\mathcal{B}^k_{\alpha,\beta}(\mathbb{X})$ and $\phi\in\mathcal{B}'^k_{\alpha,\beta}(\mathbb{X})$. The following can be easily verified:

1. A C^k-manifold $(M,\mathcal{A})$ admits a $(B^k_{\alpha,\beta},\mathbb{X})$-structure iff there exists a subatlas $\mathcal{B}^k_{\alpha,\beta}(\mathbb{X})$ for which the identity $id:M\to M$ is a (B,k,α,β)-morphism in $\mathbb{X}$.

2. If M_j is a $(B^k_{\alpha_j,\beta_j},\mathbb{X})$-manifold, with $j=1,2$, then there exists a $(B^k_{\alpha,\beta},\mathbb{X})$-morphism between them only if $\beta(i)\geq\max_j\{b_j(i)\}$ for every i. In particular, if $\beta_j(i)=i$, then such a morphism exists only if $i\leq\beta(i)$.

Example 5.2. In the standard vectorial ISP $\mathbb{X}$ the Example 1.1 and Example 1.2 contains the basic examples of $(B^k_{\alpha,\beta},\mathbb{X})$-manifolds.

We would like to consider the category $\mathbf{Diff}^{B,k}_{\alpha,\beta}(\mathbb{X})$ of $(B^k_{\alpha,\beta},\mathbb{X})$-manifolds with $(B^k_{\alpha,\beta},\mathbb{X})$-morphisms between them. The next lemma reveals, however, that the composition of those morphisms is not well-defined.

Lemma 5.1. *Let B be a $C^{k,\alpha}_{n,\beta}$-presheaf in a proper full ISP $\mathbb{X}$, let $\Pi(i)$ be the set of partitions of $[i]=\{1,..,i\}$. For each $\mu[i]\in\Pi(i)$, write $\mu[i;r]$ to denote its blocks, i.e, $\mu[i]=\sqcup_r\mu[i;r]$. Let $\leq$ be a function assigning to each $i\in\Gamma_{\geq 0}$ orderings $\leq_i$ in $\Pi(i)$ and $\leq_{\mu[i]}$ in the set of blocks of $\mu[i]$, as follows:*

$$\begin{array}{rcl}
\mu[i]_{\min} & \leq_i & \mu[i]_{\min+1}\leq_i\cdots\leq_i\mu[i]_{\max-1}\leq_i\mu[i]_{\max}\\
\mu[i;\min] & \leq_{\mu[i]} & \mu[i;\min+1]\leq_{\mu[i]}\cdots\leq_{\mu[i]}\mu[i;\max-1]\leq_{\mu[i]}\mu[i;\max].
\end{array}$$

Then, for any given open sets $U,V,W\subset\mathbb{R}^n$, composition induces a map[2]

$$\bar{\circ}: B^{k,n}_{\alpha,\beta}(U,V;\mathbb{X})\times B^{k,n}_{\alpha,\beta}(V,W;\mathbb{X})\to B^{k,n}_{\alpha_{\leq},\beta_{\leq}}(U,W;\mathbb{X}),$$

where $B^{k,n}_{\alpha,\beta}(U,V;\mathbb{X})$ is the subspace of (B,k,α,β)-functions $f:U\to\mathbb{R}^n$ such that $f(U)\subset V$, and

$$\begin{array}{rcl}
\alpha_{\leq}(i) & = & \delta(\bar{\epsilon}_{\mu[i]_{\max}},\delta(\bar{\epsilon}_{\mu[i]_{\max-1}},\delta(\bar{\epsilon}_{\mu[i]_{\max-2}},\cdots,\delta(\bar{\epsilon}_{\mu[i]_{\min+1}},\bar{\epsilon}_{\mu[i]_{\min}})\cdots)))\\
\beta_{\leq}(i) & = & \displaystyle\max_{\mu[i]}\max_{\mu[i;r]}\{\beta(|\mu[i]|),\beta(|\mu[i;r]|)\}.
\end{array}$$

Here, if $\mu[i]$ is some partition of $[i]$, then

$$\bar{\epsilon}_{\mu[i]}=\epsilon(\alpha(|\mu[i]|),\epsilon(\mu_{\alpha}[i])),$$

where for any block $\mu[i;r]$ of $\mu[i]$ we denote $|\mu_{\alpha}[i;r]|\equiv\alpha(|\mu[i;r]|)$ and

$$\epsilon(\mu_{\alpha}[i])\equiv\epsilon(|\mu_{\alpha}[i;\max]|,\epsilon(|\mu_{\alpha}[i;\max-1]|,|\mu_{\alpha}[i;\max-2]|,\cdots\epsilon(|\mu_{\alpha}[i;\min+1]|,|\mu_{\alpha}[i;\min]|)\cdots))).$$

Proof. The proof follows from Faà di Bruno's formula giving a chain rule for higher order derivatives [13] and from the compatibility between the multiplicative/additive structures of B_{α} and $C^{k-\beta}$. First of all, notice that if $C^k(U;V)$ is the set of all C^k-functions $f:U\to\mathbb{R}^n$ such that $f(U)\subset V$, then for any $g\in C^k(V;W)$ the composition $g\circ f$ is well-defined. Thus, our task is to show that there exists the dotted arrow making commutative the diagram below.

$$\begin{tikzcd}
C^k(U;V)\times C^k(V;W) \arrow[r, "\circ"] & C^k(U;W)\\
B^k_{\alpha,\beta}(U,V;\mathbb{X})\times B^k_{\alpha,\beta}(V,W;\mathbb{X}) \arrow[u, hook] \arrow[r, dashed, "\bar{\circ}"'] & B^k_{\alpha,\beta}(U,W;\mathbb{X}) \arrow[u, hook]
\end{tikzcd}$$

Now, let $f\in C^k(U;V)$ and $g\in C^k(V;W)$, so that from Faà di Bruno's formula, for any $0\leq i\leq k$ and any multi-index μ such that $|\mu|=i$, we have

$$\partial^{\mu}(g\circ f)_j=\sum_j\sum_{\mu[i]}\partial^{\mu[i]}g_j\prod_{\mu[i;r]\in\mu[i]}\partial^{\mu[i;r]}f_j\equiv\sum_j\sum_{\mu[i]}\partial^{\mu[i]}g_jf_j^{\mu[i]}.$$

[2] In the subspace topology this is actually a continuous map, but we will not need that here.

Consequently, if f and g are actually (B,k,α,β)-functions in $\mathbb{X}$, then

$$\partial^{\mu[i]}g_j \in B_{\alpha(|\mu[i]|)}(V)\cap_{X(V)} C^{k-\beta(|\mu[i]|)}(V) \quad \text{and} \quad \partial^{\mu[i;r]}f_j \in B_{\alpha(|\mu[i;r]|)}(U)\cap_{X(U)} C^{k-\beta(|\mu[i;r]|)}(U).$$

Under the choice of ordering functions $\leq$, from the compatibility of multiplicative structures we see that

$$\begin{aligned} f_j^{\mu[i]} &\in B_{\epsilon(\mu_\alpha[i])}(U)\cap_{X(U)} C^{k-\max\limits_{\mu[i;r]}\{\beta(|\mu[i;r]|)\}}(U) \\ \partial^{\mu[i]}g_j f_j^{\mu[i]} &\in B_{\overline{\epsilon}_{\mu[i]}}(U)\cap_{X(U)} C^{k-\max\limits_{\mu[i;r]}\{\beta(|\mu[i;r]|),\beta(|\mu[i]|)\}}(U). \end{aligned}$$

Finally, compatibility of additive structures shows that $\partial^\mu(g\circ f)_j \in B_{\alpha_\leq(i)}(U)\cap_{X(U)} C^{k-\beta_\leq(i)}(U)$, so that by varying i we conclude that $g\circ f$ is a $(B,k,\alpha_\leq,\beta_\leq)$-function in $\mathbb{X}$. ☐

Remark 5.2. Unlike $B^k_{\alpha,\beta}(-;\mathbb{X})$, the rule $U\mapsto B^k_{\alpha,\beta}(U,U;\mathbb{X})$ is generally not a presehaf, since the restriction of a map $f:U\to U$ to an open set $V\subset U$ needs not take values in V.

The problem with the composition can be avoided by imposing conditions on B. Indeed, given a ordering function $\leq$ as above, let us say that a $C^{k,n}_{\alpha,\beta}$-presheaf B *preserves* $\leq$ (or that it is *ordered*) *in* $\mathbb{X}$ if there exists an embedding of presheaves $B_{\alpha_\leq}\cap_X C^{k-\beta_\leq}\hookrightarrow B_\alpha\cap_X C^{k-\beta}$.

Example 5.3. We say that B is *increasing* (resp. *decreasing*) if for any U we have embeddings $B(U)_i\hookrightarrow B(U)_j$ (resp. $B(U)_j\hookrightarrow B(U)_i$) whenever $i\leq j$, where the order is the canonical order in $\Gamma_{\geq 0}\subset\mathbb{Z}_{\geq 0}$. Suppose that $\beta_\leq(i)\leq\beta(i)$, that B is increasing (resp. decreasing) and that $\alpha_\leq(i)\leq\alpha(i)$ (resp. $\alpha_\leq(i)\geq\alpha(i)$). Thus, for any U and any i we have embeddings $B(U)_{\alpha_\leq(i)}\hookrightarrow B(U)_{\alpha(i)}$ and $B(U)_{\beta_\leq(i)}\hookrightarrow B(U)_{\beta(i)}$, so that by the universality of pullbacks and stability of monomorphisms we see that any intersection presheaf makes B ordered.

Corollary 5.1. *With the same notations and hypotheses of the previous lemma, if B is ordered relative to some intersection presheaf $\mathbb{X}$, then the composition induces a map*

$$\overline{\circ}: B^k_{\alpha,\beta}(U,V;\mathbb{X})\times B^k_{\alpha,\beta}(V,W;\mathbb{X})\to B^k_{\alpha,\beta}(U,W;\mathbb{X}).$$

Proof. Straightforward. ☐

On the other hand, we also have problems with the identities: the identity map $id:U\to U$ is not necessarily a (B,k,α,β)-function in an arbitrary intersection structure $\mathbb{X}$ for an arbitrary B. We say that B is *unital in* $\mathbb{X}$ if $id_U\in B^k_{\alpha,\beta}(U;\mathbb{X})$ for every open set $U\subset\mathbb{R}^n$. Thus, with this discussion we have proved:

Proposition 5.2. *If B is as $C^{k,\alpha}_{n,\beta}$-presheaf which is ordered and unital in some intersection presheaf $\mathbb{X}$, then the category $\mathbf{Diff}^{B,k}_{\alpha,\beta}(\mathbb{X})$ is well-defined.*

6 Theorem B

In this section we will finally prove an existence theorem of $(B^k_{\alpha,\beta},\mathbb{X})$-structures on C^k-manifolds under certain conditions on B, meaning absorption and retraction conditions, culminating in Theorem B. Given a $C^{k,\alpha}_{n,\beta}$-presheaf B in $\mathbb{X}$ and open sets $U,V\subset\mathbb{R}^n$, let $\mathrm{Diff}^k_{\alpha,\beta}(U,V;\mathbb{X})$ be the set of (B, k, α, β)-*diffeomorphisms* from U to V in $\mathbb{X}$, i.e, the largest subset for which there exists the dotted arrow below.

$$\begin{tikzcd}
B^k_{\alpha,\beta}(U,V;\mathbb{X}) \arrow[r, hook] & C^k(U;V) \\
\mathrm{Diff}^k_{\alpha,\beta}(U,V;\mathbb{X}) \arrow[u, hook] \arrow[r, dashed] & \mathrm{Diff}^k(U;V) \arrow[u, hook]
\end{tikzcd}$$

We say that B is *left-absorbing* (resp. *right-absorbing*) *in* $\mathbb{X}$ if for every U, V, W there also exists the dotted arrow in the lower (resp. upper) square below, i.e, $g \circ f$ remains a (B, k, α, β)-diffeomorphism whenever f (resp. g) is a (B, k, α, β)-diffeomorphism and g (resp. f) is a C^k-diffeomorphism. If B is both left-absorbing and right-absorbing, we say simply that it is *absorbing in* $\mathbb{X}$.

$$\begin{tikzcd}
\mathrm{Diff}^k(U;V) \times \mathrm{Diff}^k_{\alpha,\beta}(V,W;\mathbb{X}) \arrow[r, dashed, "\circ_r"] \arrow[d, hook, "id\times\imath"'] & \mathrm{Diff}^k_{\alpha,\beta}(U,W;\mathbb{X}) \arrow[d, hook] \\
\mathrm{Diff}^k(U;V) \times \mathrm{Diff}^k(V;W) \arrow[r, "\circ"] & \mathrm{Diff}^k(U;W) \\
\mathrm{Diff}^k_{\alpha,\beta}(U,V;\mathbb{X}) \times \mathrm{Diff}^k(V;W) \arrow[r, dashed, "\circ_l"'] \arrow[u, hook, "\imath\times id"] & \mathrm{Diff}^k_{\alpha,\beta}(U,W;\mathbb{X}) \arrow[u, hook]
\end{tikzcd}$$

A more abstract description of these absorbing properties is as follows. Let $\mathbf{C}$ be an arbitrary category and let $\mathrm{Iso}(\mathbf{C})$ the set of isomorphisms in $\mathbf{C}$, i.e,

$$\mathrm{Iso}(\mathbf{C}) = \coprod_{X,Y \in \mathrm{Ob}(\mathbf{C})} \mathrm{Iso}_{\mathbf{C}}(X;Y).$$

Let $\mathrm{Com}(\mathbf{C}) \subset \mathrm{Iso}(\mathbf{C}) \times \mathrm{Iso}(\mathbf{C})$ be the pullback between the source and target maps $s, t : \mathrm{Iso}(\mathbf{C}) \to \mathrm{Ob}(\mathbf{C})$. Composition gives a function $\circ : \mathrm{Com}(\mathbf{C}) \to \mathrm{Iso}(\mathbf{C})$. If $\mathbf{C}$ has a distinguished object $*$, we can extend $\circ$ to the whole $\mathrm{Iso}(\mathbf{C}) \times \mathrm{Iso}(\mathbf{C})$ by defining $\circ_*$, such that $g \circ_* f = g \circ f$ when $(f, g) \in \mathrm{Com}(\mathbf{C})$ and $g \circ_* f = id_*$, otherwise. Thus, $(\mathrm{Iso}(\mathbf{C}), \circ_*)$ is a magma. Define a *left $*$-ideal* (resp. *right $*$-ideal*) in $\mathbf{C}$ as a map I assigning to each pair of objects $X, Y \in \mathbf{C}$ a subset $I(X;Y) \subset \mathrm{Iso}_{\mathbf{C}}(X;Y)$ such that the corresponding subset

$$I(\mathbf{C}) = \coprod_{X,Y \in \mathrm{Ob}(\mathbf{C})} I(X;Y)$$

of $\mathrm{Iso}(\mathbf{C})$ is actually a left ideal (resp. right ideal) for the magma structure induced by $\circ_*$. A *bilateral $*$-ideal* (or *$*$-ideal*) in $\mathbf{C}$ is a map I which is both left and right $*$-ideal. When $*$ is an initial object, we say simply that I is a left-ideal, right-ideal or ideal in $\mathbf{C}$.

Proposition 6.1. *A $C^{k,\alpha}_{n,\beta}$-presheaf B is left-absorbing (resp. right-absorbing or absorbing) in a proper full ISP $\mathbb{X}$ iff the induced rule*

$$U, V \mapsto \mathrm{Diff}^{k,n}_{\alpha,\beta}(U,V;\mathbb{X})$$

is a left ideal (resp. right ideal or ideal) in the full subcategory of $\mathbf{Diff}^k$, consisting of open sets of $\mathbb{R}^n$ and C^k-maps between them.

Proof. Immediate from the definitions above. □

Notice that in the context of vector spaces, since these are free abelian objects, the short exact sequence below always split, so that from the splitting lemma we conclude the existence of a retraction r_U, such that $r_U \circ \imath = id$, for every U [11].

$$\begin{tikzcd} 0 \arrow[r] & B^k_{\alpha,\beta}(U;\mathbb{X}) \arrow[r, hook, "\imath"] & C^k(U;\mathbb{R}^n) \arrow[l, bend right, "r_U"'] \arrow[r, "\pi"] & C^k(U.\mathbb{R}^n)/B^k_{\alpha,\beta}(U;\mathbb{X}) \arrow[r] & 0 \end{tikzcd}$$

By restriction, for each V we have an induced retraction $r_{U,V}$, as in the first diagram below. On the other hand, the dotted arrow does not necessarily exists. In other words, $r_{U,V}$ need not preserve diffeomorphisms. We say that B has *retractible* (B,k,α,β)*-diffeomorphisms in* $\mathbb{X}$ if for every U,V there exist $\overline{r}_{U,V}$ in the second diagram, not necessarily making the first diagram commutative. A *retraction presheaf in* $\mathbb{X}$ for B is a rule $\overline{r}$, assigning to each U,V a retraction $\overline{r}_{U,V}$.

$$\begin{tikzcd} B^k_{\alpha,\beta}(U,V;\mathbb{X}) \arrow[r, hook, "\imath"] & C^k(U;V) \arrow[l, bend right, "r_{U,V}"'] \\ \mathrm{Diff}^k_{\alpha,\beta}(U,V;\mathbb{X}) \arrow[u, hook] \arrow[r, hook] & \mathrm{Diff}^k(U;V) \arrow[u, hook] \arrow[l, dashed, bend right, "\overline{r}_{U,V}"'] \end{tikzcd} \qquad \begin{tikzcd} \mathrm{Diff}^k_{\alpha,\beta}(U,V;\mathbb{X}) \arrow[r, hook, "\imath_{U,V}"'] & \mathrm{Diff}^k(U;V) \arrow[l, dashed, bend right, "\overline{r}_{U,V}"'] \end{tikzcd} \tag{8}$$

Given a C^k-manifold $(M,\mathcal{A})$ and a $C^{k,\alpha}_{n,\beta}$-presheaf B in $\mathbb{X}$, let $C^k(\mathcal{A})$ and $B^{k,n}_{\alpha,\beta}(\mathcal{A};\mathbb{X})$ denote the collection of not necessarily maximal C^k-structures $\mathcal{A}' \subset \mathcal{A}$ and $(B^k_{\alpha,\beta},\mathbb{X})$-structures $\mathcal{B}^k_{\alpha,\beta}(\mathbb{X}) \subset \mathcal{A}$, respectively. Observe that there is an inclusion $\imath_{\mathcal{A}} : B^k_{\alpha,\beta}(\mathcal{A};\mathbb{X}) \hookrightarrow C^k(\mathcal{A})$, which take a $(B^k_{\alpha,\beta},\mathbb{X})$-structure and regard it as a C^k-structure. We can now finally prove that for certain classes of B the set $B^{k,n}_{\alpha,\beta}(\mathcal{A};\mathbb{X})$ is non-empty.

Theorem 6.1. *Let B be a $C^{k,\alpha}_{n,\beta}$-presheaf which is ordered, left-absorbing or right-absorbing, and which has retractible (B,k,α,β)-diffeomorphisms, all of this in the same proper full ISP $\mathbb{X}$. In this case, for any C^k-manifold $(M,\mathcal{A})$, the choice of a retraction presheaf $\overline{r}$ induces a function $\kappa_{\overline{r}} : C^k(\mathcal{A}) \to B^{k,n}_{\alpha,\beta}(\mathcal{A};\mathbb{X})$ which is actually a retraction for $\imath_{\mathcal{A}}$. In particular, under this hypothesis every C^k-manifold has a $(B^k_{\alpha,\beta},\mathbb{X})$-structure.*

Proof. Let $\mathcal{A}' \subset \mathcal{A}$ be some not necessarily maximal C^k-structure and let $\varphi_i : U_i \to \mathbb{R}^n$ be its charts. The transition functions are given by $\varphi_{ji} = \varphi_j \circ \varphi_i^{-1} : \varphi_i(U_{ij}) \to \varphi_j(U_{ij})$. Notice that the restricted chart $\varphi_j|_{U_{ij}}$ can be recovered by $\varphi_j|_{U_{ij}} = \varphi_{ji} \circ \varphi_i|_{U_{ij}}$ for each i such that $U_{ij} \neq \varnothing$. This motivate us to define new functions $\overline{\varphi}_{j;i} : U_{ij} \to \mathbb{R}^n$ by $\overline{\varphi}_{j;i} = \overline{r}(\varphi_{ji}) \circ \varphi_i|_{U_{ij}}$, where $\overline{r}$ is a fixed restriction presheaf. They are homeomorphisms onto their images because they are composites of them. We assert that when varying i and j, the maps $\overline{\varphi}_{j;i}$ generate a $(B^k_{\alpha,\beta},\mathbb{X})$-structure, which we denote by $\overline{r}(\mathcal{A}')$. Indeed, for each i,j,k,l, then transition functions $\overline{\varphi}_{k;l} \circ \overline{\varphi}_{j;i}^{-1} : \overline{\varphi}_{j;i}(U_{ijkl}) \to \overline{\varphi}_{k;l}(U_{ijkl})$ are given by

$$\begin{aligned} \overline{\varphi}_{k;l} \circ \overline{\varphi}_{j;i}^{-1} &= [\overline{r}(\varphi_{kl}) \circ \varphi_l|_{U_{ijkl}}] \circ [\varphi_i^{-1}|_{U_{ijkl}} \circ \overline{r}(\varphi_{ji})^{-1}] \\ &= \overline{r}(\varphi_{kl}) \circ (\varphi_{li})|_{U_{ijkl}} \circ \overline{r}(\varphi_{ji})^{-1}, \end{aligned}$$

which are (B,k,α,β)-functions in $\mathbb{X}$, due to the absorbing properties of B in $\mathbb{X}$. More precisely, if B is left-absorbing, then $\overline{r}(\varphi_{kl}) \circ (\varphi_{li})|_{U_{ijkl}}$ is a (B,k,α,β)-function in $\mathbb{X}$. But $\overline{r}(\varphi_{ji})^{-1}$ is also a (B,k,α,β)-function in $\mathbb{X}$ and by the ordering hypothesis on B the composite remains a (B,k,α,β)-function in $\mathbb{X}$. In this case, define $\kappa_{\overline{r}}(\mathcal{A}') = \overline{r}(\mathcal{A}')$. If B is right-absorbing, similar argument holds. That $\kappa_{\overline{r}}$ is a retraction for $\imath_{\mathcal{A}}$, i.e, that $\overline{r}(\imath_{\mathcal{A}}(\mathcal{B}^k_{\alpha,\beta}(\mathbb{X}))) = \mathcal{B}^k_{\alpha,\beta}(\mathbb{X})$ follows from the fact that $\overline{r}$ is a retraction presheaf. □

Observe that if B is left-absorbing or right-absorbing, then it is automatically unital, so that from Proposition 5.2 under the hypotheses of the last theorem the category $\mathbf{Diff}^{B,k}_{\alpha,\beta}(\mathbb{X})$ is well-defined. We have an obvious forgetful functor $F^{B,k}_{\alpha,\beta} : \mathbf{Diff}^{B,k}_{\alpha,\beta}(\mathbb{X}) \to \mathbf{Diff}^k$ which takes a $(B^k_{\alpha,\beta}\mathbb{X})$-manifold $(M,\mathcal{A},\mathcal{B}^k_{\alpha,\beta}(\mathbb{X}))$ and forgets $\mathcal{B}^k_{\alpha,\beta}(\mathbb{X})$ (this is essentally an extension of the inclusion $\imath_{\mathcal{A}}$). Our task is to show that this functor has adjoints. We begin by proving existence of adjoints on the core.

We recall that if $\mathbf{C}$ is a category, then its *core* is the subcategory $\mathrm{C}(\mathbf{C}) \subset \mathbf{C}$ obtained by forgetting all morphisms which are not isomorphisms. Every functor $F : \mathbf{C} \to \mathbf{D}$ factors through the core, so that we have an induced functor $\mathrm{C}(F) : \mathrm{C}(\mathbf{C}) \to \mathrm{C}(\mathbf{D})$. Actually, the core construction provides a functor $\mathrm{C} : \mathbf{Cat} \to \mathbf{Cat}$, where $\mathbf{Cat}$ denotes the category of all categories [16, 7].

Theorem 6.2. *With the same notations and hypotheses of Theorem 6.1, for every restriction presheaf $\overline{r}$ the rule $\kappa_{\overline{r}}$ induces a functor $K_{\overline{r}} : \mathrm{C}(\mathbf{Diff}^k) \to \mathrm{C}(\mathbf{Diff}^{B,k}_{\alpha,\beta}(\mathbb{X}))$. If B is left-absorbing (resp. right-absorbing), then $K_{\overline{r}}$ is a left (resp. right) adjoint for the core of the forgetful functor. In particular, if B is absorbing, then $\mathrm{C}(F^{B,k}_{\alpha,\beta})$ is ambidextrous adjoint.*

Proof. Define $K_{\overline{r}}$ by $K_{\overline{r}}(M,\mathcal{A}) = (M,\mathcal{A},\overline{r}(\mathcal{A}))$ on objects and by $K_{\overline{r}}(f) = f$ on morphisms. On objects it is clearly well-defined. On morphisms it is too, because for any $\overline{\varphi}_{j;i}, \overline{\varphi}_{k;l} \in \overline{r}(\mathcal{A})$ we have

$$\begin{aligned} \overline{\varphi}_{k;l} \circ f \circ \overline{\varphi}^{-1}_{j;i} &= [\overline{r}(\varphi_{kl}) \circ \varphi_l|_{U_{ijkl}}] \circ f \circ [\varphi_i^{-1}|_{U_{ijkl}} \circ \overline{r}(\varphi_{ji})^{-1}] \\ &= \overline{r}(\varphi_{kl}) \circ (\varphi_l f \varphi_i)|_{U_{ijkl}} \circ \overline{r}(\varphi_{ji})^{-1}. \end{aligned} \tag{9}$$

Since we are in the core, $(\varphi_l f \varphi_i)$ is a C^k-diffeomorphism, so that we can use the same arguments of that used in Theorem 6.1 to conclude that (9) is a (B,k,α,β)-function in $\mathbb{X}$. Preservation of compositions and identities is clear, so that $K_{\overline{r}}$ really defines a functor. Suppose that B is left-absorbing. Given a C^k-manifold $(M,\mathcal{A})$ and a $({B'}^k_{\alpha,\beta},\mathbb{X})$-manifold $(M',\mathcal{A}',{\mathcal{B}'}^k_{\alpha,\beta}(\mathbb{X}))$ we assert that there is a bijection

$$\mathrm{Diff}^{B,k}_{\alpha,\beta}(K_{\overline{r}}(M,\mathcal{A});(M',\mathcal{A}',{\mathcal{B}'}^k_{\alpha,\beta}(\mathbb{X})) \underset{\imath_{M,M'}}{\overset{\xi_{M,M'}}{\rightleftarrows}} \mathrm{Diff}^k((M,\mathcal{A});(M',\mathcal{A}')) \tag{10}$$

which is natural in both manifolds. Define $\imath_{M,M'}(f) = f$ and notice that this is well-defined, since locally it is given by the inclusions $\imath_{U,V}$ in (8). Define $\xi_{M,M'}(f) = f$. In order to show that this is also well-defined, let $f : (M,\mathcal{A}) \to (M',\mathcal{A}')$ a C^k-diffeomorphism and let $\overline{\varphi}_{j;i} \in \overline{r}(\mathcal{A})$ and $\phi \in {\mathcal{B}'}^k_{\alpha,\beta}(\mathbb{X})$ charts. Thus,

$$\phi \circ f \circ \overline{\varphi}^{-1}_{j;i} = (\phi \circ f \circ \varphi_i^{-1}) \circ \overline{r}(\varphi_{ji})^{-1}. \tag{11}$$

Due to the inclusion $\imath_{U,V}$, the chart ϕ is a C^k-chart, so that $\phi \circ f \circ \varphi_i^{-1}$ is a C^k-diffeomorphism (since f is a C^k-diffeomorphism). The left-absorption property then implies that (11) is a (B,k,α,β)-function in $\mathbb{X}$, meaning that $\xi_{M,M'}$ is well-defined. That (10) holds is clear; naturality follows from the fact that the maps $\imath_{M,M'}$ and $\xi_{M,M'}$ do not depends on the manifolds. The case in which B is right-absorbing is completely analogous. □

Corollary 6.1. *With the same notations and hypotheses of Theorem 6.1, the function $\kappa_{\overline{r}}$ independs of $\overline{r}$. More precisely, if $\overline{r}$ and $\overline{r}'$ are two retraction presheaves, then there exists a natural isomorphism $K_{\overline{r}} \simeq K_{\overline{r}'}$, so that for every C^k-manifold $(M,\mathcal{A})$ we have a corresponding (B,k,α,β)-diffeomorphism $(M,\overline{r}(\mathcal{A})) \simeq (M,\overline{r}'(\mathcal{A}))$.*

Proof. Straighforward from the uniqueness of the left and right adjoints [16, 7]. □

We would like to extend Theorem 6.2 to the whole category $\mathbf{Diff}^{B,k}_{\alpha,\beta}(\mathbb{X})$. In order to do this, notice that when proving Theorem 6.2 the hypothesis that we are working on the core was used only to conclude that the local expressions $\phi f \varphi$ are C^k-diffeomorphisms, leading us to use the diffeomorphism-absorption properties. But, if instead absorbing only diffeomorphisms we can absorb every C^k-map, we will then be able to absorb $\phi f \varphi$ for every f, meaning that the same proof will still work in $\mathbf{Diff}^{B,k}_{\alpha,\beta}(\mathbb{X})$.

We say that a $C^{k,\alpha}_{n,\beta}$-presheaf B is *fullly left-absorbing* (resp. *fully right-absorbing*) *in* $\mathbb{X}$ if for every U,V,W there exists the dotted arrow in the lower (resp. upper) square below. If B is both fully left-aborving and fully right-absorbing, we say simply that it is *fully absorbing in* $\mathbb{X}$. There is also an abstract characterization in terms of left/right/bilateral ideals, but now considered in the magma $\mathrm{Mor}(\mathbf{C})$ of all morphisms instead of on the magma $\mathrm{Iso}(\mathbf{C})$ of isomorphisms.

$$\begin{tikzcd}
\mathrm{C}^k(U;V)\times \mathrm{Diff}^{k,n}_{\alpha,\beta}(V,W;\mathbb{X}) \arrow[r, dashed, "\circ_r"] \arrow[d, hook, "id\times\imath"'] & \mathrm{Diff}^{k,n}_{\alpha,\beta}(U,W;\mathbb{X}) \arrow[d, hook] \\
\mathrm{C}^k(U;V)\times \mathrm{C}^k(V;W) \arrow[r, "\circ"] & \mathrm{C}^k(U;W) \\
\mathrm{Diff}^{k,n}_{\alpha,\beta}(U,V;\mathbb{X})\times \mathrm{C}^k(V;W) \arrow[r, dashed, "\circ_l"'] \arrow[u, hook, "\imath\times id"] & \mathrm{Diff}^{k,n}_{\alpha,\beta}(U,W;\mathbb{X}) \arrow[u, hook]
\end{tikzcd}$$

Theorem B. *If B is ordered, fully left-absorbing (resp. fully right-absorbing) and has retractible (B,k,α,β)-diffeomorphisms, all of this in the same intersection presheaf $\mathbb{X}$, then the choice of a retraction $\overline{r}$ induces a left-adjoint (resp. right-adjoint) for the forgetful functor F from $B^k_{\alpha,\beta}$-manifolds to C^k-manifolds, which actually independs of $\overline{r}$. In particular, if B is fully absorbing, then F is ambidextrous adjoint.*

Proof. Immediate from the results and discussions above. □

Corollary 6.2. *With the same notations of the last theorem, if B is fully left-absorbing (resp. fully right-absorbing), then $\mathbf{Diff}^{B,k}_{\alpha,\beta}(\mathbb{X})$ has all small colimits (resp. small limits) that exist in $\mathbf{Diff}^k$. If B is fully absorbing, then the same applies for limits and colimits simultaneously. In particular, in this last case $\mathbf{Diff}^{B,k}_{\alpha,\beta}(\mathbb{X})$ has finite products and coproducts.*

Proof. Just apply Theorem B together with the preservation of small colimits/limits by left/right-adjoint functors [16, 7] and recall that the category of C^k-manifolds has finite products and coproducts [18, 4]. □

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6.4 SECOND WORK

The following is a preliminary version of the accepted paper

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Geometric Regularity Results on $B^k_{\alpha,\beta}$-Manifolds, I: Affine Connections

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Abstract

In this paper we consider existence and multiplicity results concerning affine connections on C^k-manifolds M whose coefficients are as regular as one needs, following the regularity theory introduced in [1]. We show that if M admits a $B^k_{\alpha,\beta}$-structure, then the existence of such regular connections can be established in terms of properties of the structural presheaf B. In other words, we propose a solution to the existence problem in this setting. With regard to the multiplicity problem, we show that the space of regular affine connections is an affine space of the space of regular $\mathrm{End}(TM)$-valued 1-forms, and that if two regular connections are locally additively different, then they are actually locally different. The existence of a topology in which the space of regular connections is a nonempty open dense subset of the space of all regular $\mathrm{End}(TM)$-valued 1-forms is suggested.

1 Introduction

In a naive sense, a "regular" mathematical object in one which is described by functions. The space where these structural functions live is the "regularity" of the defined mathematical object. Thus, n-manifolds M are "regular" because they are described by its charts $\varphi_i : U_i \to \mathbb{R}^n$. Furthermore, its regularity is given by the space where the transition functions $\varphi_{ji} = \varphi_j \circ \varphi_i^{-1}$ live. Similarly, tensors T, affine connections ∇ and partial differential operators D in M are also "regular", because they are described by their local coefficients.

In [1] the authors proposed a formalization to this general notion of "regularity" and begin its study in the context of C^k-manifolds. In particular, we considered the existence problem of subatlases satisfying prescribed regularity conditions. In the present work we will continue this study by considering existence and multiplicity problems for regular affine connections on regular C^k-manifolds.

Such problems are of wide interest in many areas. For instance, in gauge theory and when one needs certain uniform bounds on the curvature, it is desirable to work with connections ∇ whose coefficients are not only C^{k-2}, but actually L^p-integrable or even uniformly bounded [3, 4]. In

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other situations, as in the study of holomorphic geometry, one considers holomorphic or Hermitian connections over complex manifolds whose coefficients are holomorphic [5, 6]. If M has a G-structure, one can also consider G-principal connections ω on the frame bundle FM, which induce an affine connection ∇_ω in M, whose coefficients $\omega_b^c(\partial_a) = \Gamma_{ab}^c$ are such that for every a, $\Gamma_a \in C^{k-2}(U) \otimes \mathfrak{g}$, where $(\Gamma_a)_{cb} = \Gamma_{ab}^c$ [7]. Interesting examples of these are the symplectic connections, which are specially important in formal deformation quantization of symplectic manifolds [8, 9]. As a final example, a torsion-free affine connection defines a L_3-space structure in a smooth manifold M (in the sense of [10]) iff its coefficients satisfy a certain system of partial differential equations [11]. Thus, the theory of L_3-spaces is about certain regular connections.

In the formulation of "regular" object of [1], the spaces describing the regularity are modeled by nuclear Fréchet spaces, since their category **NFre** is a closed symmetric monoidal relatively to the projective tensor project $\otimes$, and since most of the function spaces arising from geometry belongs to **NFre**. Furthermore, since one such space for every open set $U \subset M$, this leads to consider presheaves $B : \mathrm{Open}(\mathbb{R}^n)^{op} \to \mathbf{NFre}$. On the other hand, notice that if a mathematical concept is described by C^k-functions f, then the regularity of this object is typically described not only by the space where the functions f live, but also by the space where their derivatives $\partial^i f$, $1 \leq i \leq k$, belong. Thus, we should consider sequences of presheaves $B_i : \mathrm{Open}(\mathbb{R}^n)^{op} \to \mathbf{NFre}$, with $i = 0, ..., k$. But, for reasons which will be more clear later (see Remark 2.9), it is better to allow sequences B_i indexed in more general sets Γ. These are called *presheaves of* Γ*-spaces*.

Example 1.1. The prototypical example of $[0, k]$-presheaf, where $[0, k] = 0, ..., k$, is the sequence C^{k-i}. If $\beta : \Gamma \to [0, k]$ is any function we get the Γ-presheaf $C^{k-\beta(i)}$. One can also consider the sequences $L^i(U)$, $W^{i,p}(U)$, and so on.

We would also like to sum and multiply in the regularity spaces in a distributive way. We will consider presheaves of *distributive* Γ*-spaces*, which are presehaves of Γ-spaces endowed with a distributive structure. On the other hand, recall that we will be interested in regularity structures on C^k-objets. Thus, we need some way to make sense of the intersections $B_{\alpha(i)}(U) \cap C^{k-\beta(i)}(U)$, where the index α is necessary to ensure that both are presheaves of Γ-spaces for the same Γ. This will be done with the help of an ambient category $\mathbb{X}$ with pullbacks (typically the category of Γ-graded real vector spaces) and of an ambient object X, in which both B_i and $C^{k-\beta(i)}$ can be included, being the intersection $B_i \cap_X C^{k-\beta(i)}$ given by the pullbacks of the inclusions. In other words, we will need an *intersection structure* between B_i and $C^{k-\beta(i)}$, denoted by $\mathbb{X}$.

With all this stuff, we define a "regular" function $f : U \to \mathbb{R}$, here called (B, k, α, β)*-function in* $\mathbb{X}$, as a C^k-function such that $\partial^i f \in B_{\alpha(i)}(U) \cap_{X(U)} C^{k-\beta(i)}(U)$ for each i. Similarly, a "regular" C^k-manifold, called $(B^k_{\alpha,\beta}, \mathbb{X})$*-manifold* is one whose transition functions φ_{ji} are regular functions, i.e., are (B, k, α, β)-functions in $\mathbb{X}$. Finally, a "regular" affine connection in a $(B^k_{\alpha,\beta}, \mathbb{X})$-manifold M is a C^k affine connection ∇ whose coefficients $(\Gamma_\varphi)^c_{ab}$ in each regular chart φ of M are (B, k, α', β')-functions in some other ambient $\mathbb{Y}$. We also say that ∇ is an affine $(B^k_{\alpha',\beta'}, \mathbb{Y})$-connection in M.

In this new language, the existence and multiplicity problems for regular affine connections on regular manifolds have the following description:

- **Existence.** Let M be a $(B^k_{\alpha,\beta}, \mathbb{X})$-manifolds. Given α' and β' and $\mathbb{Y}$, can we find conditions on B, k, α and β such that M admits affine $(B^k_{\alpha',\beta'}, \mathbb{Y})$-connections?
- **Multiplicity.** What can be said about the dimension of the space $\mathrm{Conn}^k_{\alpha',\beta'}(M; \mathbb{Y})$ of $(B^k_{\alpha',\beta'}, \mathbb{Y})$-connections in M?

We prove that arbitrarily regular connection exist in sufficiently regular manifolds. The will be based on four steps:

1. existence of arbitrarily regular affine connections in each $U \subset M$ (Proposition 3.1);
2. existence of global affine connections in M whose coefficients $(\Gamma_{\varphi})^c_{ab}$ are regular, but whose derivatives are not (Theorem 3.1);
3. proving a Regularity Globalization Lemma, ensuring that the derivatives $\partial^i(\Gamma_{\varphi})^c_{ab}$ can be made regular (Lemma 2.4);
4. putting together the previous steps to get a global regular connection (Theorem 3.2).

Concerning the multiplicity problem, in Proposition 4.1 we show that the space of regular affine connections is an affine space of the space of regular $\mathrm{End}(TM)$-valued 1-forms, and in Theorem 4.1 we prove that if two regular connections are locally additively different, then they are locally different.

The paper is organized as follows. In Section 2 we present the needed background for our main results. In parts, it contain concepts introduced in [1], but now described in a more concrete way. Subsection 2.9 and Subsection 2.9 are fully new in content if compared to [1]. In particular, in Subsection 2.9 the Regularity Globalization Lemma is stated and proved. Section 3 deals with the existence problem and the four steps described above. Section 4 is about the multiplicity problem. The paper ends with an informal discussion of why we should believe that there exists a topology in which $\mathrm{Conn}^k_{\alpha',\beta'}(M;\mathbb{Y})$ is a nonempty open dense subset of the space of tuples of (B, k, α', β')-functions in $\mathbb{Y}$.

- **Convention**: remarks concerning notations and assumptions will be presented using a bullet.

2 Background

In this section we will present all that is necessary to formally state and prove the existence and multiplicity results described in the introduction. We begin by recalling in detail the definitions and results from [1] that will be needed in the next sections, which are basically the notion of $B^k_{\alpha,\beta}$-manifold and the underlying concepts.

- **Notations**. In the following Γ denotes an arbitrary set, regarded as a set of indexes. Given $k, l \geq 0$ with $k \leq l$, $[k, l]$ will denote the integer closed interval from k to l. **NFre** will denote the category of nuclear Fréchet spaces with continuous linear maps, and $\otimes$ is the projective tensor product, which makes **NFre** a symmetric monoidal category [12].

2.1 Γ-Spaces

In this section we will recall what are distributive Γ-spaces. For a complete exposition, see Section 2 of [1].

Definition 2.1. A Γ-*space* is a family $B = (B_i)_{i\in\Gamma}$ of nuclear Fréchet spaces. A *morphism* $f : B \Rightarrow B'$ between two Γ-spaces B and B' is a family $f_i : B_i \to B'_i$ of morphisms in **NFre**, with $i \in \Gamma$. Composition and identities are defined componentwise, so that we have a category $\mathbf{NFre}_{\Gamma}$.

Example 2.1. The fundamental examples are the $[0,k]$-spaces given by $B_i = C^i(M;\mathbb{R}^m)$, where M is a C^k-manifold, with the standard family of semi-norms

$$\|f\|_{r,l} = \sum_j \sup_{|\mu|=r} \sup_{x\in K_l} |\partial^\mu f_j(x)| \tag{1}$$

with $0 \le r \le i$ and (K_l) some nice sequence of compact sets of M. More generally, given a set Γ and a function $\beta : \Gamma \to [0,k]$, we can define the Γ-space $B_i = C^{k-\beta(i)}(M;\mathbb{R}^m)$.

Example 2.2. Similarly, we can consider in $C^{k-\beta(i)}(U;\mathbb{R})$ other Banach norms, as L^p-norms or Sobolev norms.

Example 2.3. More generally, any sequence of Banach spaces defines a Γ-space, where Γ is the set in which the sequence is indexed.

Example 2.4. Given a Γ-space B and a function $\alpha : \Gamma' \to \Gamma$ we define a new Γ'-space B_α by reindexing B, i.e., $(B_\alpha)_i = B_{\alpha(i)}$.

Definition 2.2. A *distributive structure* in a Γ-space B consists of:

1. maps $\epsilon, \delta : \Gamma \times \Gamma \to \Gamma$;
2. continuous linear maps $*_{ij} : B_i \otimes B_j \to B_{\epsilon(i,j)}$ and $+_{ij} : B_i \otimes B_j \to B_{\delta(i,j)}$, for every $i,j \in \Gamma$

such that the following compatibility equations are satisfied for every $x \in B_i$, $y \in B_j$ and $z \in B_k$, and every $i,j,k \in \Gamma$ (notice that the first two equations, which describe left and right distributivity, respectively, makes sense only because of the last two).

$$\begin{aligned}
x *_{i\delta(j,k)} (y +_{jk} z) &= (x *_{ij} y) + (x *_{ik} z) \\
(x +_{ij} y) *_{\delta(i,j)k} z &= (x *_{ik} z) + (y *_{jk} z) \\
\epsilon(i, \delta(j,k)) &= \delta(\epsilon(i,j), \epsilon(i,k)) \\
\epsilon(\delta(i,j), k) &= \delta(\epsilon(i,k), \epsilon(j,k)).
\end{aligned}$$

We also require that $+_{ii}$ coincides with the sum $+_i$ in B_i.

Definition 2.3. A *distributive* Γ*-space* is a Γ-space B endowed with distributive structure $(\epsilon, \delta, +, *)$. A *morphism* between distributive Γ-spaces $(B, \epsilon, \delta, +, *)$ and $(B', \epsilon', \delta', +', *')$ is a pair (f, μ), where $\mu : \Gamma \to \Gamma$ is a function and $f = (f_i)_{i\in\Gamma}$ is a family of continuous linear maps $f_i : B_i \to B'_{\mu(i)}$ such that the following equations are satisfied (notice again that the first two equations, which describe left and right distributivity, respectively, makes sense only because of the last two):

$$\begin{aligned}
f(x *_{ij} y) &= f(x) *'_{\epsilon'(\mu(i),\mu(j))} f(y) \\
f(x +_{ij} y) &= f(x) +'_{\delta'(\mu(i),\mu(j))} f(y) \\
\mu(\epsilon(i,j)) &= \epsilon'(\mu(i), \mu(j)) \\
\mu(\delta(i,j)) &= \delta'(\mu(i), \mu(j)).
\end{aligned}$$

Example 2.5. With pointwise sum and multiplication, the Γ-spaces $C^{k-\beta(i)}(M;\mathbb{R})$ are distributive for $\delta(i,j) = k - \max(\beta(i),\beta(j)) = \epsilon(i,j)$. Hölder's inequality implies that the $\mathbb{Z}_{\geq 0}$-spaces $L^i(U)$ also are distributive, with $\delta(i,j) = \min(i,j)$ and $\epsilon(i,j) = i \star j = (i \cdot j)/(i+j)$, now viewed as a Γ-space for $\Gamma \subset \mathbb{Z}_{\geq 0}$ being the set of all $i,j \geq 0$ such that $i \star j$ is an integer. From Yong's inequality one can considers a different distributive structure in $L^i(\mathbb{R}^n)$ given by convolution product and such that $\epsilon(i,j) = r(i,j) = (i \cdot j)/(i + j - (i \cdot j))$, where now $\Gamma \subset \mathbb{Z}_{\geq 0}$ is such that $r(i,j)$ is an integer.

Definition 2.4. Let B and B' be a Γ-space and a Γ'-space, respectively. The *external tensor product* between them is the $\Gamma \times \Gamma'$-space $B \otimes B'$ such that $(B \otimes B')_{i,j} = B_i \otimes B'_j$.

Remark 2.1. A distributive Γ-space $(B,\epsilon,\delta,+,*)$ induces two $(\Gamma \times \Gamma)$-spaces B_ϵ and B_δ, defined by $(B_\epsilon)_{ij} = B_{\varepsilon(i,j)}$ and $(B_\delta)_{ij} = B_{\delta(i,j)}$. The family of maps $*_{ij}$ and $+_{ij}$ defining a distributive Γ-structure are actually morphisms of $\Gamma \times \Gamma$-spaces $* : B \otimes B \Rightarrow B_\epsilon$ and $+ : B \otimes B \Rightarrow B_\delta$.

2.2 Abstract Intersections

We now recall abstract intersection structures between presheaves of distributive Γ-spaces. For a complete exposition, see Section 2 of [1].

As a motivation, recall that the intersection $A \cap B$ of vector subspaces $A, B \subset X$ of a fixed vector space X can be viewed as the pullback between the inclusions $A \hookrightarrow X$ and $B \hookrightarrow X$. However, if A and B are arbitrary vector spaces, the "abstract intersection" $A \subset B$ is a priori not well-defined. In order to define it we first consider an ambient space X in which both can be embedded and one fixes embeddings $\alpha : A \hookrightarrow X$ and $\beta : B \hookrightarrow X$. The *abstract intersection* between A and B inside X, relative to the fixed embeddings α and β, is then defined as the pullback $\mathrm{pb}(\alpha,\beta)$ between α and β.

Notice that if the category $\mathbf{C}$ in which the objects A and B live does not have pullbacks, then the above notion of abstract intersection may not exist for certain triples (X,α,β). In these cases we first need to regard both A and B as objects of another category $\mathbf{X}$ with pullbacks. Thus, we take a faithful functor $\gamma : \mathbf{A} \to \mathbf{X}$, a fixed ambient object $X \in \mathbf{X}$ and fixed monomorphisms $\alpha : \gamma_{\mathbf{A}}(A) \hookrightarrow X$ and $\alpha : \gamma(A) \hookrightarrow X$ $\alpha : \gamma(B) \hookrightarrow X$. The abstract intersection can then be defined as the pullback $\mathrm{pb}(\alpha,\beta)$ between α and β, but now it depends additionally on the faithful functor γ. Furthermore, a priori it exists only as an object of $\mathbf{X}$, which is why we call it "abstract".

Definition 2.5. A Γ-*ambient* is a tuple $\mathcal{X} = (\mathbf{X},\gamma,\gamma_\Sigma)$, where $\mathbf{X}$ is a category with pullbacks and γ and γ_Σ are faithful functors $\gamma : \mathbf{NFre}_\Gamma \to \mathbf{X}$ and $\gamma_\Sigma : \mathbf{NFre}_{\Gamma\times\Gamma} \to \mathbf{X}$. We say that a Γ-ambient $\mathcal{X}$ *has null objects* if the ambient category $\mathbf{X}$ has a null object 0.

Example 2.6. As discussed in [1], the typical example of Γ-ambient is such that $\mathbb{X}$ is the category $\mathbf{Vec}_{\mathbb{R},\Gamma\times\Gamma}$ of $\Gamma \times \Gamma$-graded real vector spaces, $\gamma_\Sigma : \mathbf{NFre}_{\Gamma\times\Gamma} \to \mathbf{Vec}_{\mathbb{R},\Gamma\times\Gamma}$ is the forgetful functor and $\gamma : \mathbf{NFre}_\Gamma \to \mathbf{Vec}_{\mathbb{R},\Gamma\times\Gamma}$ is the composition of Γ_Σ with the inclusion $\mathbf{NFre}_\Gamma \hookrightarrow \mathbf{NFre}_{\Gamma\times\Gamma}$ given by $\imath(B)_{i,j} = B_i \oplus 0$.

Example 2.7. More generally, it is interesting to consider the class of *vector* Γ-*ambients* [1]. These are such that $\mathbf{X} = \mathbf{Vec}_{\mathbb{R},\Gamma\times\Gamma}$, γ_Σ creates null objects and $\gamma = \gamma_\Sigma \circ F$, where F is another functor that creates null objects.

Definition 2.6. Let B, B' be two Γ-spaces and let $\mathcal{X} = (\mathbf{X},\gamma,\gamma_\Sigma)$ be a Γ-ambient. An *intersection structure* between B and B' in $(\mathbf{X},\gamma,\gamma_\Sigma)$ is given by an object $X \in \mathbf{X}$ and monomorphisms

$\imath : \gamma(B) \hookrightarrow X$ and $\imath' : \gamma(B') \hookrightarrow X$. We write $\mathbb{X}$ in order to denote $(X, \imath, \imath')$ and we say that X is the *ambient object* of $\mathbb{X}$.

Notice that in the definition above the functor γ_Σ, which is part of the Γ-ambient, was not used yet. It will be used when taking distributive structures into account.

Definition 2.7. Let $(B, \epsilon, \delta, +, *)$ and $(B', \epsilon', \delta', +', *')$ be distributive Γ-spaces. An *intersection structure* between them, denoted by $\mathbb{X}$, consists of[1]

1. a Γ-ambient $\mathcal{X} = (\mathbf{X}, \gamma, \gamma_\Sigma)$;
2. an intersection structure $\mathbb{X}_0 = (X, \imath, \imath')$ between B and B';
3. an intersection structure $\mathbb{X}_* = (X_*, \jmath_*, \jmath'_*)$ between B_ϵ and $B'_{\epsilon'}$;
4. an intersection structure $\mathbb{X}_+ = (X_+, \jmath_+, \jmath'_+)$ between B_δ and $B'_{\delta'}$.

The *abstract intersection* between $(B, \epsilon, \delta, +, *)$ and $(B', \epsilon', \delta', +', *')$ in $\mathcal{X}$, relatively to $\mathbb{X}$, is:

1. the abstract intersection $\mathrm{pb}(X, \imath, \imath'; \mathcal{X}, \mathbb{X}_0)$ between B and B';
2. the abstract intersection $\mathrm{pb}(X_*, \jmath_*, \jmath'_*; \mathcal{X}, \mathbb{X}_*)$ between B_ϵ and $B'_{\epsilon'}$;
3. the abstract intersection $\mathrm{pb}(X_+, \jmath_+, \jmath'_+; \mathcal{X}, \mathbb{X}_+)$ between B_δ and $B'_{\delta'}$;
4. the additional pullbacks between $\gamma_\Sigma(*) \circ \jmath_*$ and $\gamma_\Sigma(*') \circ \jmath'_*$, and between $\gamma_\Sigma(+) \circ \jmath_+$ and $\gamma_\Sigma(+') \circ \jmath'_+$, as below[2], describing the abstract intersections between the multiplications $*$ and $*'$, and the sums $+$ and $+'$, respectively[3], in $\mathcal{X}$ and relatively to $\mathbb{X}$.

$$\begin{tikzcd}
\mathrm{pb}(\gamma_\Sigma(*) \circ \jmath_*, \gamma_\Sigma(*') \circ \jmath'_*) \arrow[rr] \arrow[ddd] & & \gamma_\Sigma(B' \otimes B') \arrow[d, "\gamma_\Sigma(*')"] \\
& \mathrm{pb}(\jmath_*, \jmath'_*) \arrow[r] \arrow[dd] & \gamma_\Sigma(B'_\epsilon) \arrow[dd, hook, "\jmath'_*"] \\
& & \\
\gamma_\Sigma(B \otimes B) \arrow[r, "\gamma_\Sigma(*)"'] & \gamma_\Sigma(B_\epsilon) \arrow[r, hook, "\jmath_*"'] & X_*
\end{tikzcd} \tag{2}$$

- The abstract intersections between $*$ and $*'$, and between $+$ and $+'$, as above, will be denoted simply by $\mathrm{pb}(*, *'; \mathcal{X}, \mathbb{X}_*)$ and $\mathrm{pb}(+, +'; \mathcal{X}, \mathbb{X}_+)$, respectively.

Example 2.8. The abstract intersection usually depends strongly on the ambient object X. E.g, in the Γ-ambient of Example 2.6, if $X = \gamma(B) \oplus \gamma(B')$, then $\mathrm{pb}(X, \imath, \imath'; \mathcal{X}, \mathbb{X}) \simeq 0$ independently of $\imath$ and $\imath'$. On the other hand, if $X = \mathrm{span}(\gamma(B) \cup \gamma(B'))$, then the abstract intersection is a nontrivial graded vector space if $B_i \cap B'_i$ is nontrivial for some $i \in \Gamma$. See Examples 12-13 of [1].

Remark 2.2. In typical examples, if there is some $i \in \Gamma$ such that B_i or B'_i is a nontrivial vector space, then the abstract intersection between B and B' is nontrivial too. This depends on how we "concretify" the abstract intersections: see Lemma 2.2.

[1] Recall the definitions of B_ϵ and B_δ in Remark 2.1.
[2] We presented only the first of these pullbacks, since the second one is fully analogous.
[3] Recall from Remark 2.1 that $*$ and $+$ can be viewed as morphisms defined on external tensor products.

2.3 Concretization

Here we will recall the concretization procedures for abstract intersection structures. See Section 2 of [1].

In the last section we described how we can intersect two objects $A, B \in \mathbf{C}$ in a category without pullbacks by means of considering spans in an ambient category $\mathbf{X}$ with pullbacks via a faithful functor $\gamma : \mathbf{C} \to \mathbf{X}$. The resulting abstract intersection, however, exists a priori only in $\mathbf{X}$, but in many situations we actually need to work with it in $\mathbf{C}$. Thus we need some "concretization" procedure. An obvious approach would be to require that $\gamma : \mathbf{C} \to \mathbf{X}$ reflects pullbacks, but this would assume that $\mathbf{C}$ has pullbacks, which is precisely what we are avoiding. One could also assume the presence of an opposite directed functor $F : \mathbf{X} \to \mathbf{C}$ which preserves pullbacks, but again this assumes that $\mathbf{C}$ has pullbacks. A final obvious attempt would be consider only flat functors $\gamma : \mathbf{C} \to \mathbf{X}$, but this is too restrictive for our purposes. The exact condition that we need is the following[4]:

Definition 2.8. Let $\gamma : \mathbf{C} \to \mathbf{X}$ be a faithful functor and $X \in \mathbf{X}$. A *concreteness structure* for γ in X is given by a set $W_\gamma(A; X)$ of morphisms $\gamma(A) \to X$ for each $A \in \mathbf{C}$. Let

$$W_\gamma(X) = \bigcup_{A \in \mathbf{C}} W_\gamma(A; X).$$

We say γ satisfies the *concreteness property* in the concreteness structure $W_\gamma(X)$ if for every pair $f : \gamma(A) \hookrightarrow X$ and $f' : \gamma(B) \hookrightarrow X$ in $W_\gamma(X)$ there exist:

1. an object $A \cap_{X,\gamma} B \in \mathbf{C}$ and an isomorphism $u : \gamma(A \cap_{X,\gamma} B) \simeq \mathrm{pb}(f, f')$;
2. morphisms $\theta_1 : A \cap_{X,\gamma} B \to A$ and $\theta_2 : A \cap_{X,\gamma} B \to B$,

such that the diagram below commutes.

$$\begin{tikzcd}
\gamma(A \cap_{X,\gamma} B) \arrow[dr, "u", "\simeq"'] \arrow[drr, bend left, "\gamma(\theta_2)"] \arrow[ddr, bend right, "\gamma(\theta_1)"'] & & \\
& \mathrm{pb}(f,f') \arrow[r, "\pi_2"] \arrow[d, "\pi_1"'] & \gamma(B) \arrow[d, "f'"] \\
& \gamma(A) \arrow[r, "f"'] & X
\end{tikzcd} \tag{3}$$

Remark 2.3. The cospan in $\mathbf{C}$ defined by (θ_1, θ_2) is called *concrete cospan* of the span (f, f'). Thus, the concreteness property allow us to replace a span in $\mathbf{X}$ by a not necessarily universal cospan (i.e., not necessarily a limit) in $\mathbf{C}$.

Lemma 2.1. *If γ satisfies the concreteness property in $W_\gamma(X)$ and if f and f' are monomorphisms, then the corresponding θ_1 and θ_2 are too.*

Proof. By the comutativity of the diagram above, we have $\gamma(\theta_i) = \pi_i \circ u$, with $i = 1, 2$. Since f and f' are monomorphisms, it follows that π_i are monomorphisms too. Thus, $\gamma(\theta_i)$ also are. Since γ is faithful, it reflect monomorphisms. $\square$

[4]One can actually work in a more general setting, as those discussed in [1]. However, the present conditions are sufficient for essentially everything, including our main results.

We can now put the previous discussion in our context.

Definition 2.9. Let $\mathcal{X} = (\mathbf{X}, \gamma, \gamma_\Sigma)$ be a Γ-ambient and let $X \in \mathbf{X}$. Define

1. $W_\gamma(B; X)$ as the collection of all monomorphisms $\imath : \gamma(B) \hookrightarrow X$, where $B \in \mathbf{NFre}_\Gamma$, and $W_\gamma(X)$ as the union of all of them;
2. $W_{\gamma_\Sigma}(B; X)$ as the collection of all morphisms $f : \gamma_\Sigma(B) \to X$, for $B \in \mathbf{NFre}_{\Gamma\times\Gamma}$ which are of the form $f = \imath \circ g$, where $\imath : \gamma_\Sigma(B') \hookrightarrow X$ is a monomorphism and $g : \gamma_\Sigma(B) \to \gamma_\Sigma(B')$ is an arbitrary morphism. Let $W_{\gamma_\Sigma}(X)$ be the union of all of them.

We say that $\mathcal{X}$ is *γ-concrete* (resp. *γ_Σ-concrete*) in X if the functor γ (resp. γ_Σ) satisfies the concreteness property in $W_\gamma(X)$ (resp. $W_{\gamma_\Sigma}(X)$). We say that $\mathcal{X}$ is *concrete* if it is both γ-concrete and γ_Σ-concrete.

Remark 2.4. The class $W_{\gamma_\Sigma}(B; X)$ contains morphisms of the form $f = \imath \circ g$ (instead of only monomorphism) precisely because we will need to concrectify diagrams like (2), where $g = \gamma_\Sigma(*)$. Notice that if g was assumed to be a monomorphism, then by the arguments of Lemma 2.1, the multiplication $* : B \otimes B \to B$ would be a monomorphism too, which is not true in full generality.

Definition 2.10. Let $\mathbb{X} = (\mathcal{X}, X, \imath, \imath')$ be an intersection structure between two fixed Γ-spaces B and B'. We say that it is *γ-concrete*[5] (resp. *γ_Σ-concrete* or concrete) if Γ-ambient $\mathcal{X}$ is γ-concrete (resp. γ_Σ-concrete or concrete) in X.

Consider, now, an intersection structure $\mathbb{X} = (\mathcal{X}, \mathbb{X}_0, \mathbb{X}_*, \mathbb{X}_+)$ between two distributive Γ-spaces $(B, \epsilon, \delta, *, +)$ and $(B', \epsilon', \delta', *', +')$. We say that $\mathbb{X}$ is *concrete* if $\mathbb{X}_0$ is γ-concrete and $\mathbb{X}_*$ and $\mathbb{X}_+$ are γ_Σ-concrete. The *concrete intersection* between B and B' in $\mathbb{X}$ is given by the concrete cospans of the spans[6] $(\imath, \imath')$, $(\jmath_*, \jmath'_*)$, $(\jmath_+, \jmath'_+)$, $(\jmath_* \circ \gamma_\Sigma(*), \jmath'_* \circ \gamma_\Sigma(*'))$ and $(\jmath_+ \circ \gamma_\Sigma(+), \jmath'_+ \circ \gamma_\Sigma(+'))$. The vertices of these cospans will be respectively denoted by $B \cap_{X,\gamma} B'$, $B_\epsilon \cap_{X_*,\gamma_\Sigma} B_{\epsilon'}$, $B_\delta \cap_{X_+,\gamma_\Sigma} B_{\delta'}$, $* \cap_{X_*,\gamma_\Sigma} *'$ and $+ \cap_{X_+,\gamma_\Sigma} +'$.

Remark 2.5. Notice that if an abstract intersection between Γ-spaces B and B' is concrete, then it can itself be regarded as a Γ-space. However, if an abstract intersection between *distributive* Γ-space is concrete, then it is not necessarily a *distributive* Γ-space; it is only a $\Gamma \times \Gamma$-space with further stuff. This is enough for our purposes.

We close with a useful lemma which shows that for concrete vector ambients, concrete intersections of nontrivial distributive Γ-spaces are nontrivial too.

Lemma 2.2. *Let $\mathbb{X}$ be a concrete intersection structure between distributive Γ-spaces $(B, \epsilon, \delta, *, +)$ and $(B', \epsilon', \delta', *', +')$, whose underlying Γ-ambient $\mathcal{X}$ is vectorial. If there exist $i, j \in \Gamma$ such that B_i or B'_j are nontrivial, then $\mathbb{X}$ is nontrivial too.*

Proof. See Proposition 1, page 10, of [1]. □

[5] In [1] this corresponds to the notion of *proper* intersection structure.
[6] Take a look at diagrams (2) and (3), and at Remark 2.4.

2.4 Extension to Sheaves

In the last sections we discussed that in order to formulate a notion of "regularity" on a C^k-manifold M we have to consider nontrivial intersections between the existing C^k regularity and the additional one. Notice, on the other hand, that by the very notion of regularity, they are *local* properties. This means that a manifold M is "regular" when the regularity is satisfied on the local pieces $U_s \subset M$ in such a way that coherence conditions hold at the intersections $U_{ss'} = U_s \cap U_{s'}$. Thus, we do not need a *global* Γ-space $B(M)$, but actually one $B(\varphi_s(U_s))$ for each local chart $\varphi_s : U_s \to \mathbb{R}^n$. Therefore, we have to work with presheaves $B : \mathrm{Open}(\mathbb{R}^n)^{op} \to \mathbf{NFre}_\Gamma$ of Γ-spaces.

Example 2.9. The examples of Γ-spaces in Example 2.1 and Example 2.4 extends, naturally, to presheaves of Γ-spaces. E.g, given m, k and $\beta : \Gamma \to [0, k]$, we have the presheaf $U \mapsto (C^{k-\beta(i)}(U))_i$. It will be denoted by $C^{k-\beta}_m$. or simply $C^{k-\beta}$ if $m = 1$. If, in addition, $\Gamma = [0, k]$ and $\beta = id$ it will be denoted by C^{k-}.

Example 2.10. Let B be a presheaf of Γ-spaces. Then every function $\alpha : \Gamma' \to \Gamma$ defines a new presheaf of Γ'-spaces B_α by $(B_\alpha)(U) = B(U)_\alpha$.

Notice that all the discussion above can be internalized in the presheaf category of presheaves in $\mathbb{R}^n$ by means of just taking a parametrization of definitions and results in terms of opens sets of $\mathbb{R}^n$. Thus, we can talk about presehaves of distributive Γ-spaces, presheaves of intersection structures between presheaves of distributive Γ-spaces, and so on. We refer the reader to Section 3 of [1] for more details to this extension to the presheaf setting.

Remark 2.6. To maintain compatibility with notations of [1], throughout this paper a presheaf of intersection structures between presheaves of distributive Γ-spaces B and B' will be called an *intersection structure presheaf* (ISP). Furthermore, it will be denoted simply by $\mathbb{X} = (\mathcal{X}, X)$, where $\mathcal{X} = (\mathbf{X}, \gamma, \gamma_\Sigma)$ is the Γ-ambient and $X : \mathrm{Op}(\mathbb{R}^n)^{op} \to \mathbf{NFre}_\Gamma$ is the presheaf of ambient Γ-spaces.

2.5 $B^k_{\alpha,\beta}$-Presheaves

In order to prove the existence of a geometric object on a smooth manifold one typically first proves local existence and then uses a partition of unity to globalize the construction. To extend the use of partition functions to the present regular setting we will work with presheaves of Γ-spaces which are $B^k_{\alpha,\beta}$*-presheaves*. See Section 3 of [1] for more details.

- In the following: $C^k_b(U) \subset C^k(M)$ denotes the space of C^k-bump functions with the Fréchet structure induced by the seminorms (1).

Definition 2.11. We say that a set of indexes Γ' has *degree* r for some $0 \leq r \leq k$ if it contains the interval $[0, r]$.

- In the following we will write Γ_r to denote a generic set with degree r, while Γ will remain an arbitrary set.

Definition 2.12. Let $B = (B_i)_{i \in \Gamma}$ be a presheaf of Γ-spaces and let $\alpha : \Gamma' \to \Gamma$ and $\beta : \Gamma' \to [0, k]$ be functions. We say that the tuple (B, α, β, k) has *degree* r if the domain Γ' of the functions α and β has degree r, i.e., if they are defined in some Γ_r.

Definition 2.13. Let (B, α, β, k) be a tuple as above. Let $\mathbb{X} = (\mathcal{X}, X)$ be a concrete ISP between B_α and $C^{k-\beta}$. We say that (B, k, α, β) is a $B^k_{\alpha,\beta}$*-presheaf* in $\mathbb{X}$ if it has degree k and for every open set $U \subset \mathbb{R}^n$ and every $i\Gamma_k$ there exists the dotted arrow $\star_{U,i}$ making commutative the diagram below, where $\theta_{2,U}$ is part of the concrete cospan[7].

$$\begin{tikzcd} C^k(U) \otimes C^{k-\beta(i)}(U) \arrow[rr, "\cdot_{U,i}"] & & C^{k-\beta(i)}(U) \\ C^k_b(U) \otimes B_{\alpha(i)}(U) \cap_{X(U)} C^{k-\beta(i)}(U) \arrow[u, hook, "\imath_U \otimes \theta_{2,U}"] \arrow[rr, dashed, "\star_{U,i}"'] & & B_{\alpha(i)}(U) \cap_{X(U)} C^{k-\beta(i)}(U) \arrow[u, hook, "\theta_{2,U}"'] \end{tikzcd} \tag{4}$$

Intuitively, a $B^k_{\alpha,\beta}$-presheaf is a presheaf of Γ-spaces $B(U)$, which, when regarded as a presheaf B_α of Γ_k-spaces via α, has an intersection space with $C^{k-\beta}(U)$ that is closed under multiplication of C^k-bump functions.

Example 2.11. For every k, β, α the presheaf of $C^{k-\alpha}$ is a $B^k_{\beta\beta}$-presheaf in the ISP which is objectwise the standard vectorial intersection structure of Example 2.7.

Example 2.12. Similarly, the presheaves $L_\alpha(U)_i = L^{\alpha(i)}(U)$ with the distributive structure given by pointwise sum and multiplication, is a nice $B^{k,\alpha}_\beta$-presheaf in the standard ISP, for $A(U) = C^\infty_b(U)$. An analogous conclusion is valid if we replace pointwise multiplication with convolution product.

We close this section with two remarks concerning Definition 2.13.

1. It has a generalization where one requires invariance under multiplication only by a subvector space $A(U) \subset C^k(U)$ with a nuclear Fréchet structure. In this case, for instance, in Example 2.12 we would get a different $B^k_{\alpha,\beta}$-structure in L_α taking $A(U) = \mathcal{S}(U)$ as the Schwartz space. Following [1], a *nice* $B^k_{\alpha,\beta}$-presheaf should be one such that $A(U)$ nontrivially intersects $C^k_b(U)$. In the following, however, we will work only with $A(U) = C^k_b(U)$, so that every $B^k_{\alpha,\beta}$-preshef will be nice.

2. Following the conventions of [1] a $B^k_{\alpha,\beta}$-presheaf in $\mathbb{X}$ whose presheaves of intersection spaces $B_\alpha \cap_X C^{k-\beta}$ are nontrivial is called a $C^k_{\alpha,\beta}$*-presheaf*. In this paper we will not work directly with them. However, from Lemma 2.2, for vectorial ISP $\mathbb{X}$, if B_α is nontrivial (i.e., if for every $U \subset \mathbb{R}^n$ there is $i_U \in \Gamma$ such that $B_{\alpha(i_U)}(U) \neq 0$), then every $B^k_{\alpha,\beta}$-presheaf in $\mathbb{X}$ is a $C^k_{\alpha,\beta}$-presheaf. In particular, the $B^k_{\alpha,\beta}$-presheaves of Examples 2.11-2 are $C^k_{\alpha,\beta}$-presheaves.

2.6 $B^k_{\alpha,\beta}$-Functions

As discussed in the introduction, in order to formalize the notion of regularity on manifolds we have to demand conditions not only on the transition functions $\varphi^a_{s's} : \varphi_s(U_{ss'}) \to \mathbb{R}$, with $a = 1, ..., n$, but also on their derivatives $\partial^\mu \varphi^a_{s's}$, with $\mu = 1, ..., k$, where $\varphi_{s's} = \varphi_{s'} \circ \varphi_s^{-1}$. In our context this is described as follows. See Section 5 of [1] for a complete exposition.

[7] which therefore is a monomophisms due to Lemma 2.1.

Definition 2.14. Let (B, k, α, β) be a $B^k_{\alpha,\beta}$-presheaf in a concrete ISP $\mathbb{X}$. Let $U \subset \mathbb{R}^n$ be an open set and let $f : U \to \mathbb{R}$ be a real C^k-function. Let $S \subset [0, k] \subset \Gamma_k$ be a subset[8]. We say that f is a $(B^k_{\alpha,\beta}|S)$-*function* (or $(B, k, \alpha, \beta|S)$-*function*) in $\mathbb{X}$ if for every $i \in S$ we have $\partial^i f \in B_{\alpha(i)}(U) \cap_{X(U)} C^{k-\beta(i)}(U)$. A vectorial function $f : U \to \mathbb{R}^m$, with $m \geq 1$ is a $(B^k_{\alpha,\beta}|S)$-*function* in $\mathbb{X}$ if its coordinate functions $f^a : U \to \mathbb{R}$, with $a = 1, ..., m$, are.

Thus, for instance, if $\Gamma = [0, k] = \Gamma_k$ and $\alpha(i) = i = \beta(i)$, then f is a $(B, k, \alpha, \beta|S)$-function in $\mathbb{X}$ precisely if $\partial^\mu f$ belongs to $B_i(U) \cap_X (U) C^{k-i}(U)$. This means that the set S and the functions α and β determine how the derivatives $\partial^\mu f$ of f lose regularity when μ increases.

2.7 $B^k_{\alpha,\beta}$-Manifolds

We are now ready to define what is a $B^k_{\alpha,\beta}$-manifold. See Section 5 of [1].

Definition 2.15. A C^k-*manifold* is a paracompact Hausdorff topological space M endowed with a maximal atlas $\mathcal{A}$ with charts $\varphi_s : U_s \to \mathbb{R}^n$, whose transition functions $\varphi_{s's} = \varphi_{s'} \circ \varphi_s^{-1} : \varphi_s(U_{ss'}) \to \mathbb{R}^n$ are C^k, where $U_{ss'} = U_s \cap U_{s'}$.

Definition 2.16. Let $(M, \mathcal{A})$ be a C^k-manifold. A subatlas $\mathcal{A}' \subset \mathcal{A}$ is *closed under restrictions* if restrictions of charts in $\mathcal{A}'$ to smaller open sets remains in $\mathcal{A}'$. More precisely, if $(\varphi, U) \in \mathcal{A}'$ and $V \subset U$, then $(\varphi|_V, V) \in \mathcal{A}'$.

Definition 2.17. Let (B, k, α, β) be a $B^k_{\alpha,\beta}$-presheaf in a concrete ISP $\mathbb{X}$. A $(B^k_{\alpha,\beta}, \mathbb{X})$-*structure* in a C^k-manifold $(M, \mathcal{A})$ is a subatlas $\mathcal{B}^k_{\alpha,\beta}(\mathbb{X})$ closed under restrictions whose transition functions $\varphi_{s's}$ are (B, k, α, β)-functions in $\mathbb{X}$. Explicitly, this means that if $\varphi_s \in \mathcal{B}^k_{\alpha,\beta}(\mathbb{X})$, then for each other $\varphi_{s'} \in \mathcal{B}^k_{\alpha,\beta}(\mathbb{X})$ such that $U_{ss'} \neq \varnothing$, we have

$$\partial^i \varphi^a_{ss'} \in B_{\alpha(i)}(\varphi_s(U_{ss'})) \cap_{X(\varphi_s(U_{ss'}))} C^{k-\beta(i)}(\varphi_s(U_{ss'}))$$

for every $i \in [0, k] \subset \Gamma_k$.

Definition 2.18. A $(B^k_{\alpha,\beta}, \mathbb{X})$-*manifold* is a C^k manifold endowed with a $(B^k_{\alpha,\beta}, \mathbb{X})$-structure.

In [1] the authors define *morphisms* between two $(B^k_{\alpha,\beta}, \mathbb{X})$-manifolds as C^k-maps whose local representation by charts in the corresponding $(B^k_{\alpha,\beta}, \mathbb{X})$-structures are (B, k, α, β)-functions in $\mathbb{X}$. Here we will need only the particular case between C^k-manifolds.

Definition 2.19. Let $(M, \mathcal{A})$ and $(M', \mathcal{A}')$ be C^k-manifolds, let $\mathbb{X}$ a concrete ISP, (B, k, α, β) a $B^k_{\alpha,\beta}$-presheaf in $\mathbb{X}$ and $S \subset [0, k] \subset \Gamma_k$ a subset. A $(B, k, \alpha, \beta|S)$-*morphism*, $(B, k, \alpha, \beta|S)$-*function* or $(B^k_{\alpha,\beta}|S)$-*morphism* between M and M' is a C^k-function $f : M \to M'$ such that for every $\varphi \in \mathcal{A}$ and $\varphi' \in \mathcal{A}'$ the corresponding C^k-function $\varphi' \circ f \circ \varphi^{-1}$ is actually a $(B^k_{\alpha,\beta}|S)$-function in $\mathbb{X}$. If $S = [0, k]$ we say simply that f is a $B^k_{\alpha,\beta}$-*morphism*.

Remark 2.7. From the examples above it is clear that a C^k-manifold does not necessarily admits any $(B^k_{\alpha,\beta}, \mathbb{X})$-structure. Thus, there are obstructions which typically can be characterized as topological

[8]Here is where we are using that a $B^k_{\alpha,\beta}$-presheaf has degree k.

or geometric obstructions on the underlying C^k-structure. A complete characterization of these obstructions was not found yet. On the other hand, in [1] the authors give sufficient conditions on B, α, β, k and on the underlying C^k-structure ensuring the existence of $B^k_{\alpha,\beta}$-structures, specially in the case where (B, k, α, β) is a $C^k_{\alpha,\beta}$-presheaf.

2.8 Affine $B^k_{\alpha,\beta}$-Connections

We now introduce the objects of study in this paper: affine ∇ connections on C^k-manifolds M whose local coefficients Γ^c_{ab} satisfies additional regularity conditions, possibly depending on some regularity on the underlying manifold M. In other words, we will consider "regular" affine connections on $(B^k_{\alpha,\beta}, \mathbb{X})$-manifolds. The obvious idea would then to be take affine connections whose coefficients $(\Gamma_\varphi)^c_{ab} : \varphi(U) \to \mathbb{R}$ in each $(\varphi, U) \in \mathcal{B}^k_{\alpha,\beta}(\mathbb{X})$ are $B^k_{\alpha,\beta}$-functions. This, however, is too restrictive to the global existence of such objects due to two reasons.

1. *The regularity rate decay of the derivatives $\partial^\mu \Gamma^c_{ab}$, when μ grows, is typically different of that $\partial^\mu \varphi^a_{ji}$ of the transition functions φ^a_{ji}.* This leads us instead to consider affine connections whose coefficients are $(B^k_{\alpha',\beta'}|S)$-functions, where α', β' are functions typically *different* from α, β in shape and/or in domain/codomain. For instance, φ^a_{ji} are C^k, while Γ^c_{ab} is C^{k-2}. Thus, if $\beta(i) = i$, then β' must be $\beta'(i) = i + 2$. Equivalently, we can regard β' as defined in $[2, k]$ instead of in $[0, k]$ and take $\beta'(i) = i$.

2. *The ISP appearing in the regularity of Γ^c_{ab} is usually different from that describing the regularity of φ_{ij}.* For instance, we could consider intersections in the same Γ-ambient $(\mathbf{X}, \gamma, \gamma_\Sigma)$, but in different presheaves of ambient Γ-spaces X and Y.

Definition 2.20. Let (B, k, α, β) be a $B^k_{\alpha,\beta}$-presheaf in a concrete ISP $\mathbb{X}$. Let $(M, \mathcal{B}^k_{\alpha,\beta}(\mathbb{X}))$ be a $B^k_{\alpha,\beta}$-manifold in $\mathbb{X}$. Let $S \subset [0, k] \subset \Gamma_k$ be a subset and consider functions $\alpha' : \Gamma_k \to \Gamma$ and $\beta' : \Gamma_k \to [2, k]$. Let $\mathbb{Y}$ be a concrete ISP between $B_{\alpha'}$ and $C^{k-\beta'}$. An *affine* $(B^k_{\alpha',\beta'}|S)$*-connection* in M, relative to $\mathbb{Y}$, is an affine connections ∇ in the underlying C^k-manifold such that for every $(\varphi, U) \in \mathcal{B}^k_{\alpha,\beta}(\mathbb{X})$ the corresponding local coefficients $(\Gamma_\varphi)^c_{ab} : \varphi(U) \to \mathbb{R}$ are $(B, k, \alpha', \beta'|S)$-functions in $\mathbb{Y}$.

2.9 Regularity Globalization Lemma

Condition 1 and the nice hypotheses on $B^k_{\alpha,\beta}$ suffice to ensure existence of connections ∇ which are *locally regular* in $\mathbb{X}$, following the classical construction of affine connections via partitions of unity. Condition 2 will be used to prove global regularity. More precisely, we show that if B_α admits some universal way to connect two ISP from it to $C^{k-\beta}$, then every affine connection ∇ *locally regular* in a given $\mathbb{X}$ induces another affine connection $\overline{\nabla}$ which is *globally regular* in an ISP $\mathbb{Y}$ "compatible" with $\mathbb{X}$. Here we introduce these notions of universality and compatibility between two ISP.

Definition 2.21.

1. Two ISP $\mathbb{X}$ and $\mathbb{Y}$ between presheaves of Γ-spaces B and B' are *compatible* if they have the same underlying Γ-ambient and *strongly compatible* if they also have the same presheaf of ambient spaces X;

2. A *strongly compatible sequence of* ISP (or scISP, for short) between sequences $B = (B_i)$ and $B = (B'_i)$ of presheaves of Γ-spaces is a sequence $\underline{\mathbb{X}} = (\mathbb{X}_i)$ such that $\mathbb{X}_i$ and $\mathbb{X}_j$ are strongly compatible for every i, j;

3. Two sequences $\underline{\mathbb{X}} = (\mathbb{X}_i)$ and $\underline{\mathbb{Y}} = (\mathbb{Y}_i)$ of ISP between $B = (B_i)$ and $B' = (B'_i)$ are *compatible* if $\mathbb{X}_i$ and $\mathbb{Y}_i$ are compatible for every i.

Definition 2.22. Let B and B' be two presheaves of Γ-spaces and of Γ'-spaces, respectively. Let Γ_k be a set of degree k. A Γ_k*-connective structure* between B and B' is given by:

1. sets of functions $\mathcal{O} \subset \mathrm{Mor}(\Gamma_k, \Gamma)$ and $\mathcal{O} \subset \mathrm{Mor}(\Gamma_k, \Gamma')$;

2. functions $D_{\mathcal{O}}$ and $D_{\mathcal{Q}}$ assigning to each pairs $\theta, \theta' \in \mathcal{O}$ and $\vartheta, \vartheta' \in \mathcal{Q}$ corresponding morphisms $D_{\mathcal{O}}(\theta, \theta') : B_\theta \Rightarrow B_{\theta'}$ and $D_{\mathcal{Q}}(\vartheta, \vartheta') : B'_\vartheta \Rightarrow B'_{\vartheta'}$ which satisfy the composite laws

$$D_{\mathcal{O}}(\theta, \theta'') = D_{\mathcal{O}}(\theta', \theta'') \circ D_{\mathcal{O}}(\theta, \theta'),$$

meaning that the following triangles are commutative:

$$\begin{tikzcd}
B_\theta \arrow[r, Rightarrow, "D_{\mathcal{O}}(\theta,\theta')"] \arrow[dr, Rightarrow, "D_{\mathcal{O}}(\theta,\theta'')"'] & B_{\theta'} \arrow[d, Rightarrow, "D_{\mathcal{O}}(\theta',\theta'')"] \\
& B_{\theta''}
\end{tikzcd}
\qquad
\begin{tikzcd}
B_{\vartheta'} \arrow[d, Rightarrow, "D_{\mathcal{Q}}(\vartheta',\vartheta'')"'] & B_\vartheta \arrow[l, Rightarrow, "D_{\mathcal{Q}}(\vartheta,\vartheta')"'] \arrow[dl, Rightarrow, "D_{\mathcal{Q}}(\vartheta,\vartheta'')"] \\
B_{\vartheta''} &
\end{tikzcd} \tag{5}$$

3. for every $(\theta, \vartheta) \in \mathcal{O} \times \mathcal{Q}$ a concrete ISP $\mathbb{X}_{\theta,\vartheta}$ between B_θ and B'_ϑ whose corresponding sequence $\underline{\mathbb{X}} = (\mathbb{X}_{\theta,\vartheta})$ is strongly compatible. Let X denote the underlying presheaf of ambient spaces;

4. for every other scISP $\underline{\mathbb{Y}} = (\mathbb{Y}_{\theta,\vartheta})$ between B and B', indexed in $\mathcal{O} \times \mathcal{Q}$ and compatible with $\underline{\mathbb{X}} = (\mathbb{X}_{\theta,\vartheta})$, a morphism $\mathcal{D}(X;Y) : X \Rightarrow Y$ of presheaves, where Y is the presheaf of ambient spaces of each $\mathbb{Y}_i$, such that the diagram below commutes for every $\theta, \theta', \vartheta, \vartheta'$.

$$\begin{tikzcd}
(B_\theta \cap_{X,\gamma} B'_\vartheta) \arrow[rr, Rightarrow] \arrow[dd, Rightarrow] \arrow[dr, Rightarrow, dashed, "\xi^{\theta,\vartheta}_{\theta'\vartheta'}"] & & B'_\vartheta \arrow[d, Rightarrow, "D_{\mathcal{Q}}(\vartheta,\vartheta')"'] \arrow[ddd, Rightarrow, bend left, "\jmath'_\vartheta"] \\
& (B_{\theta'} \cap_Y B'_{\vartheta'}) \arrow[r, Rightarrow] \arrow[d, Rightarrow] & B'_{\vartheta'} \arrow[d, Rightarrow, "\imath'_{\vartheta'}"] \\
B_\theta \arrow[r, Rightarrow, "D_{\mathcal{O}}(\theta,\theta')"] \arrow[drr, Rightarrow, bend right, "\jmath_\theta"'] & B_{\theta'} \arrow[r, Rightarrow, "\imath_{\theta'}"'] & Y \\
& & X \arrow[u, Rightarrow, "\mathcal{D}(X;Y)"']
\end{tikzcd} \tag{6}$$

Remark 2.8. By the universality of pullbacks, there exists the dotted arrow $\xi^{\theta,\vartheta}_{\theta',\vartheta'}$ in diagram (6). Furthermore, the composition laws (5) and the uniqueness of pullbacks imply a composition law for those dotted arrows:

$$\xi^{\theta,\vartheta}_{\theta'',\vartheta''} = \xi^{\theta',\vartheta'}_{\theta'',\vartheta''} \circ \xi^{\theta,\vartheta}_{\theta',\vartheta'}.$$

Definition 2.23. We say that the sequence $\underline{X} = (\mathbb{X}_{\theta,\vartheta})$ is the *base* scISP of the Γ_k-connective structure.

- In the following we will say "consider a Γ_k-connective structure between B and B' in the scISP $\underline{\mathbb{X}}$", meaning that $\underline{\mathbb{X}}$ is the base scISP of the refereed connective structure.

In the case where B and B' are presheaves of distributive Γ-spaces and Γ'-spaces, we need to require that the connection between them preserves sum and multiplication.

Definition 2.24. Let B and B' be presheaves of distributive Γ-spaces and Γ'-spaces, respectively. A Γ_k-connective structure between B and B' in $\underline{\mathbb{X}} = (\mathbb{X}_{\theta,\vartheta})$ is *distributive* if for every $\theta, \theta' \in \mathcal{O}$ and every $\vartheta, \vartheta' \in \mathcal{Q}$ there are objective monomorphisms $(\gamma_*)_{\theta,\theta'} : B_{\epsilon(\theta,\theta)} \Rightarrow B_{\epsilon(\theta',\theta')}$ and $(\gamma'_{*'})_{\vartheta,\vartheta'} : B'_{\epsilon'(\vartheta,\vartheta)} \Rightarrow B'_{\epsilon'(\vartheta',\vartheta')}$ making commutative the diagram below, and also objectwise monomorphisms $(\gamma_+)_{\theta,\theta'} : B_{\delta(\theta,\theta)} \Rightarrow B_{\delta(\theta',\theta')}$ and $(\gamma'_{+'})_{\vartheta,\vartheta'} : B'_{\delta'(\vartheta,\vartheta)} \Rightarrow B'_{\delta'(\vartheta',\vartheta')}$ making commutative the analogous diagram for the additive structures.

$$\begin{tikzcd}
B'_{\epsilon'(\vartheta,\vartheta)} \arrow[d, Rightarrow, "(\gamma'_{*'})_{\vartheta,\vartheta'}"'] & B'_\vartheta \otimes B'_\vartheta \arrow[l, Rightarrow, "*'"'] & (B_\theta \cap_X B'_\vartheta) \otimes (B_\theta \cap_X B'_\vartheta) \arrow[l, Rightarrow] \arrow[r, Rightarrow] \arrow[d, Rightarrow, "\xi^{\theta,\vartheta}_{\theta',\vartheta'} \otimes \xi^{\theta,\vartheta}_{\theta',\vartheta'}"'] & B_\theta \otimes B_\theta \arrow[r, Rightarrow, "*"] & B_{\epsilon(\theta,\theta)} \arrow[d, Rightarrow, "(\gamma_*)_{\theta,\theta'}"] \\
B'_{\epsilon'(\vartheta',\vartheta')} & B'_{\vartheta'} \otimes B'_{\vartheta'} \arrow[l, Rightarrow, "*'"] & (B_{\theta'} \cap_Y B'_{\vartheta'}) \otimes (B_{\theta'} \cap_Y B'_{\vartheta'}) \arrow[l, Rightarrow] \arrow[r, Rightarrow] & B_{\theta'} \otimes B_{\theta'} \arrow[r, Rightarrow, "*"'] & B_{\epsilon(\theta',\theta')}
\end{tikzcd}$$

Let us now return to our context of $B^k_{\alpha,\beta}$-presheaves. First, a notation:

- given a set Γ_k of degree k, a function $\beta_0 : \Gamma_k \to [2,k]$ and $j \in \Gamma_k$, let $[\beta_0; j]_k$ denote the subset of all $i \in [0,k] \subset \Gamma_k$ such that $0 \le \beta_0(j) - i \le k$, i.e., the interval $[0, k - \beta_0(j)]$.
- Thus, by restriction each presheaf B of Γ'-spaces and the presheaf[9] C^{k-} of $[0,k]$ can both be regarded as presheaves of $[\beta_0; j]_k$-spaces.

Definition 2.25. Let $\mathbb{X}$ be a concrete ISP and let (B,k,α,β) be a $B^k_{\alpha,\beta}$-presheaf in $\mathbb{X}$. Given functions $\alpha_0 : \Gamma_k \to \Gamma$ and $\beta_0 : \Gamma_k \to [2,k]$, for each $j \in \Gamma_k$ regard B and $(C^{k-i})_i$ as presheaves of $[\beta_0; j]_k$-spaces. A $(\alpha_0, \beta_0; j)$-*connection* in B is a connection between B and C^k- in some scISP $\underline{\mathbb{X}}$, such that:

1. $\alpha, \alpha_0 \in \mathcal{O}$, $\beta, \beta_0 \in \mathcal{Q}$ and $\mathbb{X}_{\alpha,\beta} = \mathbb{X}$;
2. $\alpha_{0,j} \in \mathcal{O}$ and $\beta_{\#,j} \in \mathcal{Q}$, where $\alpha_{0,j}(i) = \alpha_0(j)$ is the constant function and $\beta_{,j}(i) = \beta_0(j) - i$ is the shifting function;
3. if $\vartheta\mathcal{Q}$ is any function bounded from above by β_0, i.e., if $\vartheta(i) \le \beta_0(i)$, then the morphism $D_{\mathcal{Q}}(\beta_0, \vartheta) : C^{h-\beta_0} \Rightarrow C^{k-\vartheta}$ is the canonical inclusion.

Definition 2.26. A (α_0, β_0, j)-connection in B is *distributive* if it is a distributive connection between B and C^{k-} in the sense of Definition 2.24.

[9]Recall the notation in Example 2.9.

The globalization of the local regularity of an affine connection in a $(B^k_{\alpha,\beta},\mathbb{X})$-manifold will depends on the existence of a distributive $(\alpha_0,\beta_0;j)$-connection in B. Furthermore, the extended regular will have local coefficients which are $(B,\theta,\vartheta,\mathbb{Y}|S)$-functions for functions $(\theta,\vartheta)\in\mathcal{O}\times\mathcal{Q}$ which are "ordinary" and for a set S which is of a very special shape, in the the following sense.

Definition 2.27. An *additive structure* in a set Γ_k of degree k consists of

1. a set Γ_{2k} of degree $2k$ containing Γ_k;
2. a map $\mp:\Gamma_k\times\Gamma_k\to\Gamma_{2k}$ extending sum of nonnegative integers, i.e., such that if $z,l\in[0,k]$, then $z\mp l=z+l$.

An *additive set of degree* k is a set of degree k where an additive structure where fixed.

- To make notations simpler, in the following we will write $z+l$ instead of $z\mp l$ even if z,l are arbitrary elements of Γ_k.
- Let $(\Gamma_k,+)$ be an additive set of degree k and $\beta_0:\Gamma_k\to[2,k]$ be a map. Given $z\in[\beta_0;j]_k$, let $\Gamma_k[z]\subset\Gamma_k$ be the set of every $l\in\Gamma_k$ such that $z+l\in[\beta_0;j]_k$.

Lemma 2.3. *In the same notations above, for every $z\in[\beta_0;j]_k$ we have*

$$\Gamma_k[z]\subset(\Gamma_k-\Gamma_k\cap[\beta_0(j)-z+1,\infty)). \tag{7}$$

Proof. Straightforward: just a simple inequalities study. □

Example 2.13. In particular, if $\Gamma_k=[0,k]$, then $\Gamma_k[z]\subset[0,\beta_0(j)-z]$. More specifically, if $z=0$, then $\Gamma_k[z]=[\beta_0;j]_k$, while if $z=k-\beta_0(j)$, then $\Gamma_k[z]=0$.

Remark 2.9 (Important Remark). Sets like $\Gamma_k[z]$ will be the sets S in which the regular affine connections that we will build will have coefficients as $(B,k,\theta,\vartheta|S)$-functions. Thus, for larger $\Gamma_k[z]$ we will have more control on the regularities of higher order derivatives of the coefficient functions. On the other hand, from Lemma 2.3 we see that when z increases, the set $\Gamma_k[z]$ becomes smaller. Furthermore, in Example 2.13 we saw that if $\Gamma_k=[0,k]$ is the obvious set of degree k, then $\Gamma_k[z]$ strongly depends on z. In particular, for $z=k-\beta_0(j)$ we have $\Gamma_k[z]=0$ which would produce a regular affine connection with no control on the regularity of derivatives! This is one of the main reasons for working in the setup of general Γ_k-spaces: *to have more control on the regularity of derivatives of the new affine connections, specially independing on z.* Indeed, notice that if Γ_k increases, then the difference (2.13) becomes closer to Γ_k and, therefore, is independent of z.

Example 2.14. If $k=\infty$, then $[\beta_0;j]_\infty=[0,\infty)$ for every β_0. Thus $[0,\infty)\subset\Gamma_\infty[z]$ for every $z\in[\beta_0;j]_\infty=[0,\infty)$, which follows from the fact that the additive structure of Γ_∞ is compatible with sum of integers.

Definition 2.28. Let Γ_k be a set of degree k, Γ a set and let $\mathcal{O}\subset\mathrm{Mor}(\Gamma_k;\Gamma)$ and $\mathcal{Q}\subset\mathrm{Mor}(\Gamma_k;[0,k])$. Let $X\subset\Gamma_r$ a subset. We say that a sequence of pairs $(\theta_l,\vartheta_l)_l$, with $\theta_l\in\mathcal{O}$ and $\vartheta_l\in\mathcal{Q}$, is *ordinary in* X if each $l\in X$ and the corresponding pair (θ_*,ϑ_*), given by $\theta_*(l)=\theta_l(z)$ and $\vartheta_*(l)=\vartheta_l(z)$ also belongs to $\mathcal{O}\times\mathcal{Q}$. Finally, we say that $(\theta,\vartheta)\in\mathcal{O}\times\mathcal{Q}$ is *ordinary in* X if $\theta=\theta'_*$ and $\vartheta=\vartheta'_*$ for some θ'_* and ϑ'_*, i.e, if it is the induced pair of a sequence of ordinary pairs in X.

We can now state and prove the fundamental step in the globalization of the regularity.

Lemma 2.4 (Regularity Globalization Lemma). *Let M be a $(B^k_{\alpha,\beta}, \mathbb{X})$-manifold and suppose that B admits a $(\alpha_0, \beta_0; j)$-connection in some* scISP $\underline{\mathbb{X}}$*. Then, for every $U \subset \mathbb{R}^n$, every $z \in [\beta_0; j]_k$ and every* scISP $\underline{\mathbb{Y}} = (\mathbb{Y}_{\theta,\vartheta})$ *between B and C^{k-}, compatible with $\underline{\mathbb{X}}$, the elements of*

$$B_{\alpha_0(j)}(U) \cap_{X(U)} C^{k-\beta_0(j)-z}(U)$$

can be regarded as $(B, k, \theta, \vartheta|\Gamma_k[z])$-functions in $\mathbb{Y}_{\theta,\vartheta}$, for every $(\theta, \vartheta) \in \mathcal{O} \times \mathcal{Q}$ ordinary in $\Gamma_k[z]$.

Proof. Since the B and C^{k-} admits a Γ_k-connective structure in some $\underline{\mathbb{X}}$, by definition it follows that for every scISP $\underline{\mathbb{Y}}$ compatible with $\underline{\mathbb{X}}$ and every $\theta, \theta' \in \mathcal{O}$ and $\vartheta, \vartheta' \in \mathcal{Q}$ we have morphisms $\xi_{U,i} : B_{\theta(i)}(U) \cap_{X(U)} C^{k-\vartheta(i)} \to B_{\theta'(i)}(U) \cap_{Y(U)} C^{k-\vartheta'(i)}(U)$ making commutative the diagram (6). Since the Γ_k-connective structure is actually a $(\alpha_0, \beta_0; j)$-connection we have $\alpha_{0,j} \in \mathcal{O}$ and $\beta_{\#,j} \in \mathcal{Q}$. Thus, for $\theta = \alpha_{0,j}$ and $\vartheta = \beta_{\#,j}$ we see that for every $\theta' \in \mathcal{O}$, $\vartheta' \in \mathcal{Q}$, and $i \in [\beta_0; j]_k$ we have a morphism from $B_{\alpha_0(j)}(U) \cap_{X(U)} C^{k-\beta_0(j)-i}(U)$ to $B_{\theta'(i)}(U) \cap_{Y(U)} C^{k-\vartheta'(i)}(U)$. Notice that if $f \in B_{\alpha_0(j)}(U) \cap_{X(U)} C^{k-\beta_0(j)-i}(U)$, then $\partial^l f \in C^{k-\beta_0(j)-(i+l)}(U)$ for every l that $i + l \in [\beta_0; j]_k$. Thus, by universality, for every such l we can regard $\partial^l f \in B_{\theta'_l(i)}(U) \cap_{Y(U)} C^{k-\vartheta'_l(i)}(U)$, where $\theta'_l \in \mathcal{O}$ and $\vartheta'_l \in \mathcal{Q}$. If $z \in [\beta_0; j]_k$, then by definition we have $z + l \in [\beta_0; j]_k$ for every $l \in \Gamma_k[z]$, so that the functions $\theta(l) = \theta'_l(z)$ and $\vartheta(l) = \vartheta'_l(z)$ are defined on $\Gamma_k[z]$. Suppose that this pair (θ, ϑ) belongs to $\mathcal{O} \times \mathcal{Q}$, i.e, suppose that it is ordinary in $\Gamma_k[z]$. Thus, $\partial^l f \in B_{\theta(l)}(U) \cap_{Y(U)} C^{k-\vartheta(l)}(U)$, meaning that f can be regarded as a $(B, k, \theta, \vartheta|\Gamma_k[z])$-function in $\mathbb{Y}$. $\square$

The Regularity Globalization Lemma will ensure that in each open set $U \subset \mathbb{R}^n$ the local regularity can be globalized. The collage of these conditions when U varies will be made gain, as expected, via partitions of unity. Since we are introducing a new structure (connections between presheaves of Γ-spaces in an scISP), we need to require some compatibility between them, the partitions of unity and the regular charts of the underlying regular manifold.

Definition 2.29. Let B be a $B^k_{\alpha,\beta}$-presheaf in a concrete ISP $\mathbb{X}$. We say that a $(\alpha_0, \beta_0; j)$-connection in some scISP $\underline{\mathbb{X}}$ is:

1. *Support preserving* if for every pair $(\theta, \vartheta) \in \mathcal{O} \times \mathcal{Q}$, every scISP $\underline{\mathbb{Y}}$ compatible with $\underline{\mathbb{X}}$ and every $U \subset \mathbb{R}^n$, if $f \in B_{\alpha_0(i)}(U) \cap_{X(U)} C^k_b(U)$, then $\operatorname{supp}(\xi^{\alpha_0,\beta_0}_{\theta,\vartheta}(f)) \subset \operatorname{supp}(f)$[10].

2. *Bump preserving* if for every pair $(\theta, \vartheta) \in \mathcal{O} \times \mathcal{Q}$, every scISP $\underline{\mathbb{Y}}$ compatible with $\underline{\mathbb{X}}$ and every $U \subset \mathbb{R}^n$, the map $\xi^{\alpha_0,\beta_0}_{\theta,\vartheta}$ factors as below[11].

$$\begin{tikzcd}
B_{\alpha_0(i)}(U) \cap_{X(U)} C^{k-\beta_0(i)}(U) \arrow[d, "\xi^{\alpha_0,\beta_0}_{\theta,\vartheta}"'] & B_{\alpha_0(i)}(U) \cap_{X(U)} C^k_b(U) \arrow[l, hook'] \arrow[d, dashed] \\
B_{\theta(i)}(U) \cap_{Y(U)} C^{k-\vartheta(i)}(U) & B_{\theta(i)}(U) \cap_{X(U)} C^k(U) \arrow[l, hook']
\end{tikzcd}$$

3. *Unital* if given $f \in B_{\alpha_0(i)}(U) \cap_{X(U)} C^{k-\beta_0(i)}(U)$ and $p \in U$ such that $f(p) = 1$, then $(\xi^{\alpha_0,\beta_0}_{\theta,\vartheta}(f))(p) = 1$ for every $(\theta, \vartheta) \in \mathcal{O} \times \mathcal{Q}$ and every scISP $\underline{\mathbb{Y}}$ compatible with $\underline{\mathbb{X}}$.

[10] Recall that $\alpha_0 \in \mathcal{O}$ and $\beta_0 \in \mathcal{Q}$, so that the map ξ^{α_0,β_0} above exists.
[11] Compare with Definition 2.13. See Remark 1.

4. *Nice* if it is support preserving, bump preserving and unital.

Next lemma shows that distributive nice $(\alpha_0,\beta_0;j)$-connections preserve partitions of unity.

Lemma 2.5. *Let B be $B^k_{\alpha,\beta}$ in $\mathbb{X}$ and suppose that it admits a distributive and nice $(\alpha_0,\beta_0;j)$-connection in some* scISP $\underline{\mathbb{X}}$. *In this case, if $(U_s)_s$ is a covering of $\mathbb{R}^n$ and $(\psi_s)_s$ is a C^k-partition of unity in $\mathbb{R}^n$ subordinate to it, then for every $(\theta,\vartheta)\in\mathcal{O}\times\mathcal{Q}$ the corresponding sequence $(\xi^{\alpha_0,\beta_0}_{\theta,\vartheta}(\psi_s))_s$ is a partition of unity subordinate to the same covering.*

Proof. Support preserving ensures that each $\xi^{\alpha_0,\beta_0}_{\theta,\vartheta}(\psi_s)$ is a bump function with support in U_s, which is C^k due to the bump preserving condition. Support preserving and locally finiteness of $(\psi_s)_s$ imply that $(\xi^{\alpha_0,\beta_0}_{\theta,\vartheta}(\psi_s))_s$ is locally finite too. Furthermore, since the $(\alpha_0,\beta_0;j)$-connection is distributive and unital, we have

$$\sum_s \xi^{\alpha_0,\beta_0}_{\theta,\vartheta}(\psi_s)(p)=\xi^{\alpha_0,\beta_0}_{\theta,\vartheta}(\sum_s \psi_s(p))=\xi^{\alpha_0,\beta_0}(1)=1,$$

so that $(\xi^{\alpha_0,\beta_0}_{\theta,\vartheta}(\psi_s))_s$ is a partition of unity, as desired. □

- Let $(M,\mathcal{A})$ be a C^k-manifold. For every $0\leq r\leq k$, let $\partial^r\mathcal{A}(U)\subset C^{k-r}(U)$ be the set of rth derivatives $\partial^r\varphi^a_{ji}$ of transitions functions $\varphi_{ji}=\varphi_j\circ\varphi_i^{-1}$ by charts $\varphi_i,\varphi_j\in\mathcal{A}$ such that $\varphi_i(U_{ij})=U$. Note that if $\mathcal{B}\subset\mathcal{A}$ is a subatlas, then $\partial^r\mathcal{B}(U)\subset\partial^r\mathcal{A}(U)$ for each r.

Definition 2.30. Let $(M,\mathcal{B}^k_{\alpha,\beta}(\mathbb{X}))$ be a $(B^k_{\alpha,\beta},\mathbb{X})$-manifold and take $0\leq r\leq k$. We say that a $(\alpha_0,\beta_0;j)$-connection in B in some scISP $\underline{\mathbb{X}}$ has *degree* r if for every par $(\theta,\vartheta)\in\mathcal{O}\times\mathcal{Q}$ such that $i\leq\vartheta(i)$, with $0\leq i\leq r$, every scISP $\underline{\mathbb{Y}}$ compatible with $\mathbb{X}$ and every $U\subset\mathbb{R}^n$, there exists the dotted arrow below (we omitted U to simplify the notation). In other words, $\xi^{\alpha,\beta}_{\theta,\vartheta}(\partial^l\varphi_{ji})=\partial^l\phi_{ji}$, for every transition function φ_{ji} by charts $\varphi_i,\varphi_j\in\mathcal{B}^k_{\alpha,\beta}(\mathbb{X})$, where ϕ_{ji} are transition functions of charts $\phi_i,\phi_j\in\mathcal{A}$.

$$\begin{tikzcd}
(B_{\alpha(i)}\cap_X C^{k-\beta(i)})\cap_{C^{k-\beta(i)}}\partial^i\mathcal{B}^k_{\alpha,\beta}(\mathbb{X}) \arrow[r, hook] \arrow[d, dashed] & B_{\alpha(i)}\cap_X C^{k-\beta(i)} \arrow[d, "\xi^{\alpha,\beta}_{\theta,\vartheta}"] \\
(B_{\theta(i)}\cap_X C^{k-\vartheta(i)})\cap_{C^{k-\vartheta(i)}}\partial^i\mathcal{A} \arrow[r, hook] & B_{\theta(i)}\cap_X C^{k-\vartheta(i)}
\end{tikzcd} \tag{8}$$

3 Existence

We are now ready to prove the existence theorem of $(B^k_{\alpha_0,\beta_0}|S)$-connections in $B^k_{\alpha,\beta}$-manifolds in the same lines described in the introduction, now presented in a much more precise form:

1. *Existence of regular locally defined connections.* In Proposition 3.1 we will show that every coordinate open set $U\subset M$ of a C^k-manifold admits a $B^k_{\alpha_0,\beta_0}$-connection for every α_0,β_0.

2. *Existence of weakly locally regular globally defined connections.* In Theorem 3.1 we will prove, using the first step and a partition of unity argument, that every $(B^k_{\alpha,\beta},\mathbb{X})$-manifold admits, for every given functions α_0,β_0, affine connections ∇ whose coefficients in each $(\varphi,U)\in\mathcal{B}^k_{\alpha,\beta}(\mathbb{X})$ belongs to 9, where α_0' and β_0' are numbers depending on α_0 and β_0.

3. *Existence of locally regular globally defined connections.* The connections obtained in the last step are *almost* locally regular because their local coefficients are *not* (B, k, α', β')-functions for certain α', β', but only belong to (9). In order to be (B, k, α', β')-functions we also need regularity on the derivative of the coefficients. It is at this moment that Lemma 2.4 plays its role and, therefore, the moment when we need to add their hypotheses and additional objects ($(\alpha_0, \beta_0; j)$-connections). This is done in Proposition 3.2.

4. *Existence of globally regular globally defined connections.* Finally, in Theorem 3.2 we glue the locally regular connections in each $(\varphi, U) \in \mathcal{B}^k_{\alpha,\beta}(\mathbb{X})$ getting the desired $B^k_{\alpha',\beta'}$-connections. It is in this step that we will need to assume that the $(\alpha_0, \beta_0; j)$-connections are compatible with partitions of unity, i.e., that they are nice in the sense of Definition 2.29.

Proposition 3.1 (Regular Local Existence)**.** *Let $(M, \mathcal{A})$ be a C^k-manifold, with $k \geq 2$, and $(\varphi, U) \in \mathcal{A}$ a chart, regarded as a C^k-manifold. Let Γ be a set, $S \subset [0, k] \subset \Gamma_k$ a subset, and consider functions $\alpha_0 : S \to \Gamma$ and $\beta_0 : S \to [2, k]$. Let B be a presheaf of distributive Γ-spaces. Then U admits a $(B^k_{\alpha_0,\beta_0}|S)$-connection ∇_φ relatively to every concrete ISP between B_{α_0} and $C^{k-\beta_0}$.*

Proof. Recall that any family of C^2-functions $f^c_{ab} : \mathbb{R}^n \to \mathbb{R}$ defines a connection in $\mathbb{R}^n$ [2]. Thus, any family of C^2-functions $f^c_{ab} : V \to \mathbb{R}$ in an open set $V \subset \mathbb{R}^n$ defines a connection ∇_0 in V. If $\varphi : U \to \mathbb{R}^n$ is a chart in M with $\varphi(U) \subset V$, pulling back ∇_0 we get a connection ∇_φ in U whose coefficients are $f^c_{ab} \circ \varphi$. Consequently, if f^c_{ab} are chosen $(B, k, \alpha_0, \beta_0|S)$-functions in $\mathbb{X}$, then ∇_φ is a $(B^k_{\alpha_0,\beta_0}, \mathbb{X}|S)$-connection in U. □

- If $(*, \epsilon)$ is the multiplicative structure of B, let

$$\epsilon^r(l, m) = \epsilon(l, \epsilon(l, \epsilon(l, \epsilon(...\epsilon(l, m)...))).$$

Theorem 3.1 (Locally Weakly Regular Existence)**.** *Let $\mathbb{X}$ be a concrete ISP and let M be a $(B^k_{\alpha,\beta}, \mathbb{X})$-manifold, with $k \geq 2$. Let Γ_k be a set of degree k. Given functions $\alpha_0 : \Gamma_k \to \Gamma$ and $\beta_0 : \Gamma_k \to [2, k]$, there exists an affine connection ∇ in M whose coefficients in each $(\varphi, U) \in \mathcal{B}^k_{\alpha,\beta}(\mathbb{X})$ belong to*

$$B_{\alpha'_0}((\varphi(U')) \cap_{X(\varphi(U'))} C^{k-\beta'_0}(\varphi(U')) \tag{9}$$

for some nonempty open set $U' \subset U$,

$$\alpha'_0 = \delta(\epsilon^3(\alpha(1), \alpha_0(0)), \epsilon(\alpha(2), \alpha(1))), \quad \beta'_0 = \max(\beta(1), \beta(2), \beta_0(0)) \tag{10}$$

and (ϵ, δ) are part of the distributive structure of B.

Proof. Let $(\varphi_s, U_s)_s$ be an open covering of M by coordinate systems in $\mathcal{B}^k_{\alpha,\beta}(\mathbb{X})$ and let ∇_s be an affine connection in U_s with coefficients $(f_s)^c_{ab}$. Let (ψ_s) be a partition of unity of class C^k, subordinate to (φ_s, U_s), and recall that $\nabla = \sum_s \psi_s \cdot \nabla_s$ is a globally defined connection in M. We assert that if φ_s are in $\mathcal{B}^k_{\alpha,\beta}(\mathbb{X})$, then ∇ is a well-defined connection whose cofficients belong to (9) for certain U'. Notice that the coefficients $(\Gamma_s)^c_{ab}$ of ∇ in a fixed φ_s are obtained in the following way: let $N(s)$ be the finite set of every s' such that $U_{ss'} \equiv U_s \cap U_{s'} \neq \varnothing$. For each $s' \in N(s)$ we can do a change of coordinates and rewrite $(f_{s'})^c_{ab}$ in the coordinates φ_s as given by the functions

$$(f_{s';s})^c_{ab} = \sum_{l,m,o} \frac{\partial \varphi^c_s}{\partial \varphi^l_{s'}} \frac{\partial \varphi^m_{s'}}{\partial \varphi^a_s} \frac{\partial \varphi^o_{s'}}{\partial \varphi^b_s} (f_{s'})^l_{mo} + \sum_l \frac{\partial^2 \varphi^l_{s'}}{\partial \varphi^a_s \partial \varphi^b_s} \frac{\partial \varphi^c_s}{\partial \varphi^l_{s'}}, \tag{11}$$

where by abuse of notation $\varphi_s^c = (\varphi_s \circ \varphi_{s'}^{-1})^c$. We then have

$$(\Gamma_s)^c_{ab} = \sum_{s' \in N(s)} \psi_{s'}(f_{s';s})^c_{ab}. \tag{12}$$

From Proposition 3.1 we can take each ∇_s $(B^k_{\alpha_0,\beta_0}, \mathbb{X})$-connection, so that $(f_{s'})^l_{mn}$ can be choosen $(B, k, \alpha_0, \beta_0)$-functions in $\mathbb{X}$. Furthermore, since $\varphi_s \in \mathcal{B}^k_{\alpha,\beta}(\mathbb{X})$ we have

$$\frac{\partial \varphi_s^c}{\partial \varphi_{s'}^l} \in (B_{\alpha(1)} \cap_X C^{k-\beta(1)})(U^{s'}_{ss'}) \quad \text{and} \quad \frac{\partial^2 \varphi_{s'}^l}{\varphi_s^a \partial \varphi_s^b} \in (B_{\alpha(2)} \cap_X C^{k-\beta(2)})(U^{s'}_{ss'})$$

where $U^{s'}_{ss'} = \varphi_{s'}(U_{ss'})$. This is well-defined since $\beta(2) \leq k$. Due to the compatibility between the additive/multiplicative structures of B_α and $C^{k-\beta}$, the first and the second terms of the right-hand sice of (11) belong to

$$(B_{\epsilon^3(\alpha(1),\alpha_0(0))} \cap_X C^{k-\max(\beta(1),\beta_0(0))})(U^{s'}_{ss'})$$

and

$$(B_{\epsilon(\alpha(2),\alpha(1))} \cap_X C^{k-\max(\beta(2),\beta(1))})(U^{s'}_{ss'}),$$

respectively, so that the left-hand side of (11) belongs to

$$(B_{\tau(\epsilon^3(\alpha(1),\alpha_0(0)),\epsilon(\alpha(2),\alpha(1)))} \cap_X C^{k-\max(\beta(1),\beta(2),\beta_0(0))})(U_{N(s)}) = (B_{\alpha'_0} \cap_X C^{k-\beta'_0})(U_{N(s)}),$$

where $U_{N(s)} = \cap_{s' \in N(s)} U_{ss'}$. Since B is a $B^k_{\alpha,\beta}$-presheaf in $\mathbb{X}$, diagram (4) implies that each product $\psi_{s'}(f_{s';s})^c_{ab}$ in the sum (12) belongs to the same space, so that the entire sum also belongs to that space. Repeating the process for all coverings in $\mathcal{B}^k_{\alpha,\beta}(\mathbb{X})$ and noticing that α' and β' do not depend on s, s', we get the desired result. □

Proposition 3.2 (Locally Regular Existence)**.** *In the same notations and hypotheses of Theorem 3.1, if B admits a $(\alpha', \beta'; j)$-connection in some* scISP $\underline{\mathbb{X}}$*, where $\alpha' : \Gamma_k \to \Gamma$ and $\beta' : \Gamma_k \to [2, k]$ are functions such that $\alpha'(j) = \alpha'_0$ and $\beta'(j) = \beta'_0$, then for every* scISP $\underline{\mathbb{Y}}$ *compatible* $\underline{\mathbb{X}}$ *and every $z \in [\beta'; j]_k$, the elements of (9) can be regarded as $(B, k, \theta, \vartheta|\Gamma_k[z])$-functions in $\mathbb{Y}_{\theta,\vartheta}$, for every pair $(\theta, \vartheta) \in \mathcal{O} \times \mathcal{Q}$ which is ordinary in $\Gamma_k[z]$.*

Proof. Direct application of Lemma 2.4. □

Theorem 3.2 (Global Regular Existence)**.** *Let M be a $(B^k_{\alpha,\beta}, \mathbb{X})$-manifold, with $k \geq 2$, and, as above, suppose that the structural presheaf B admits a $(\alpha', \beta'; j)$-connection in some* scISP $\underline{\mathbb{X}}$*, where $\alpha'(j) = \alpha'_0$ and $\beta'(j) = \beta'_0$. Suppose, in addition, that this $(\alpha', \beta'; j)$-connection is distributive, nice and has degree $r \geq 2$. Then, M admits a $(B^k_{\theta,\vartheta}, \mathbb{Y}_{\theta,\vartheta}|\Gamma_k[z])$-connection for every* scISP $\underline{\mathbb{Y}}$ *compatible with $\underline{\mathbb{X}}$, every $z \in [\beta'; j]_k$ and every pair (θ, ϑ) ordinary in $\Gamma_k[z]$ such that $i \leq \vartheta(i)$.*

Proof. From Theorem 3.1 there exists an affine connection ∇ whose coefficients (12) belong to (9). By the first part of the hypotheses we can apply Proposition 3.2, so that for any pair (θ, ϑ) ordinary in $\Gamma_k[z]$, the coefficients $(\Gamma_s)^c_{ab}$ in (12) can be regarded as $(B, k, \theta, \vartheta, \mathbb{Y}_{\theta,\vartheta}|\Gamma_k[z])$-functions. But, from the proof of Lemma 2.4 we see that this way to regard elements of an intersection space as regular functions is made by taking their image under $\xi^{\alpha',\beta'}_{\theta,\vartheta}$. Thus, Proposition 3.2 is telling us precisely that, for every φ_s, the corresponding $\xi^{\alpha',\beta'}_{\theta,\vartheta}((\Gamma_s)^c_{ab})$ are $(B, k, \theta, \vartheta, \mathbb{Y}_{\theta,\vartheta}|\Gamma_k[z])$-functions. We will show that, under the additional hypotheses, the collection of these when varying s also defines an affine connection in M, which will be a $(B^k_{\theta,\vartheta},\mathbb{Y}_{\theta,\vartheta}|\Gamma_k[z])$-connection by construction. Notice that the expression (11) which describes how $(\Gamma_s)^c_{ab}$ changes when s varies involves multiplications, sums, bump functions and derivatives of the transition functions. But the additional hypotheses are precisely about the preservation of these data by ξ! More precisely, for $i\leq\vartheta(i)$ we have:

$$\begin{aligned}
\xi^{\alpha',\beta'}_{\theta\vartheta}((f_{s';s})^c_{ab}) &= \xi^{\alpha',\beta'}_{\theta\vartheta}\Big(\sum_{l,m,o}\frac{\partial\varphi^c_s}{\partial\varphi^l_{s'}}\frac{\partial\varphi^m_{s'}}{\partial\varphi^a_s}\frac{\partial\varphi^o_{s'}}{\partial\varphi^b_s}(f_{s'})^l_{mo}+\sum_l\frac{\partial^2\varphi^l_{s'}}{\partial\varphi^a_s\partial\varphi^b_s}\frac{\partial\varphi^c_s}{\partial\varphi^l_{s'}}\Big)\\
&= \sum_{l,m,o}\xi_{\theta\vartheta}\Big(\frac{\partial\varphi^c_s}{\partial\varphi^l_{s'}}\Big)\xi_{\theta\vartheta}\Big(\frac{\partial\varphi^m_{s'}}{\partial\varphi^a_s}\Big)\xi_{\theta\vartheta}\Big(\frac{\partial\varphi^o_{s'}}{\partial\varphi^b_s}\Big)\xi_{\theta\vartheta}((f_{s'})^l_{mo})+\\
&\quad+\sum_l\xi_{\theta\vartheta}\Big(\frac{\partial^2\varphi^l_{s'}}{\partial\varphi^a_s\partial\varphi^b_s}\Big)\xi_{\theta\vartheta}\Big(\frac{\partial\varphi^c_s}{\partial\varphi^l_{s'}}\Big)\\
&= \sum_{l,m,o}\frac{\partial\phi^c_s}{\partial\phi^l_{s'}}\frac{\partial\phi^m_{s'}}{\partial\phi^a_s}\frac{\partial\phi^o_{s'}}{\partial\phi^b_s}\xi^{\alpha',\beta'}_{\theta\vartheta}((f_{s'})^l_{mo})+\sum_l\frac{\partial^2\phi^l_{s'}}{\partial\phi^a_s\partial\phi^b_s}\frac{\partial\phi^c_s}{\partial\phi^l_{s'}},
\end{aligned}$$

where we omitted the superior indexes in the maps ξ to simplify the notation and ϕ_s are charts in the maximal C^k-atlas $\mathcal{A}$ of M. Furthermore, in the first step we used (11) and in the second one we used compatibility between the distributive structures of B and C^{k-} and also the distributive properties of ξ (which comes from the fact that the $(\alpha',\beta';j)$-connection is supposed distributive) and the composition laws (5). In the third step we used that the $(\alpha',\beta';j)$-connection has degree $r\geq 2$ (here is where we need $i\leq\vartheta(i)$). By the same kind of arguments,

$$\begin{aligned}
\xi^{\alpha',\beta'}_{\theta,\vartheta}((\Gamma_s)^c_{ab}) &= \xi^{\alpha',\beta'}_{\theta\vartheta}\Big(\sum_{s'\in N(s)}\psi_{s'}\cdot(f_{s';s})^c_{ab}\Big) && (13)\\
&= \sum_{s'\in N(s)}\xi^{\alpha',\beta'}_{\theta,\vartheta}(\psi_{s'})\Big(\sum_{l,m,o}\frac{\partial\phi^c_s}{\partial\phi^l_{s'}}\frac{\partial\phi^m_{s'}}{\partial\phi^a_s}\frac{\partial\phi^o_{s'}}{\partial\phi^b_s}\xi^{\alpha',\beta'}_{\theta\vartheta}((f_{s'})^l_{mo})+ && (14)\\
&\qquad+\sum_l\frac{\partial^2\phi^l_{s'}}{\partial\phi^a_s\partial\phi^b_s}\frac{\partial\phi^c_s}{\partial\phi^l_{s'}}\Big) && (15)\\
&= \sum_{s'\in N(s)}\xi^{\alpha',\beta'}_{\theta,\vartheta}(\psi_{s'})((\Gamma_{s'})_{\theta,\vartheta})^c_{ab}, && (16)
\end{aligned}$$

where $((\Gamma_{s'})_{\theta,\vartheta})^c_{ab}$ are the coefficients of the affine connection in $U_{s'}$ defined by the family of functions $\xi^{\alpha',\beta'}_{\theta\vartheta}((f_{s'})^l_{mo})$. Since the $(\alpha',\beta';j)$-connection in B is nice and distributive it follows from Lemma 2.5 that $(\xi^{\alpha',\beta'}_{\theta,\vartheta}(\psi_s))_s$ is a C^k-partition of unity for M. Thus, (16) is a globally defined affine connection in M, which by construction is a $(B^k_{\theta,\vartheta},\mathbb{Y}_{\theta,\vartheta}|\Gamma_k[z])$-connection. □

4 Multiplicity

The last result was about the existence of regular connections. In the following we will consider the multiplicity problem. More precisely, given a $B^k_{\alpha,\beta}$-manifold M in an ISP $\mathbb{X}$, $S\subset\Gamma_k$, functions $\theta:S\to\Gamma$ and $\vartheta:S\to[2,k]$, and a concrete ISP $\mathbb{Y}$ compatible with $\mathbb{X}$, we will study the set $\mathrm{Conn}^k_{\theta,\vartheta}(M;\mathbb{Y}|S)$ of $(B^k_{\theta,\vartheta},\mathbb{Y}_{\theta,\vartheta}|S)$-connections in M.

- Let $(\mathcal{B}_{\text{end}})^k_{\alpha,\beta}(\mathbb{X}) \subset \mathcal{B}^k_{\alpha,\beta}(\mathbb{X})$ be the collection of all charts $(\varphi, U) \in \mathcal{B}^k_{\alpha,\beta}(\mathbb{X})$ which trivializes the endomorphism bundle $\mathrm{End}(TM)$.

Definition 4.1. Let $S \subset \Gamma_k$, $\theta : S \to \Gamma$, $\vartheta : S \to [2, k]$ and $\mathbb{Y}$ as above. An $\mathrm{End}(TM)$-valued $(B^k_{\theta,\vartheta}, \mathbb{Y})$-form of degree one is an $\mathrm{End}(TM)$-valued 1-form ω of class C^k whose coefficients $(\omega_\varphi)^c_{ab}$ in every $(\varphi, U) \in (\mathcal{B}_{\text{end}})^k_{\alpha,\beta}(\mathbb{X})$ are $(B, k, \theta, \vartheta|S)$-functions in $\mathbb{Y}$. Let

$$\Omega^1_{\theta,\vartheta}(M; \mathrm{End}(TM), \mathbb{Y}|S) \tag{17}$$

denote the set of all of them.

We begin with an easy fact: in the same way as the set of affine connections in a C^k-manifold is an affine space of the vector space of $\mathrm{End}(TM)$-valued 1-forms, the set of regular affine connections on a regular manifold is an affine space of the vector space of regular $\mathrm{End}(TM)$-valued 1-forms. More precisely,

Proposition 4.1. *Let M be a $(B^k_{\alpha,\beta}, \mathbb{X})$-manifold. For every S, θ, ϑ and $\mathbb{Y}$ as above:*

1. *(17) is a vector subspace of $\Omega^1(M; \mathrm{End}(TM))$;*
2. *$\mathrm{Conn}^k_{\theta,\vartheta}(M; \mathbb{Y}|S)$ is an affine space of (17).*

Proof. For the first one, recall that in a distributive Γ-space $B = (B_i)$ the additive structure $+_{ij} : B_i \otimes B_j \to B_{\delta(i,j)}$ is such that $\delta(i, i) = i$ and $+_{ii}$ is the sum of B_i. For the second one, if ∇ and $\overline{\nabla}$ are two affine C^k-connections in M, then $\omega = \nabla - \overline{\nabla}$ has coefficients in $(\varphi, U) \in (\mathcal{B}_{\text{end}})^k_{\alpha,\beta}(\mathbb{X})$ given by $(\omega_\varphi)^c_{ab} = (\Gamma_\varphi)^c_{ab} - (\overline{\Gamma}_\varphi)^c_{ab}$. Thus, again by $+_{ii} = +_i$, if $(\Gamma_\varphi)^c_{ab}$ and $(\overline{\Gamma}_\varphi)^c_{ab}$ are $(B, k, \theta, \vartheta, \mathbb{Y}|S)$-functions, then $(\omega_\varphi)^c_{ab}$ are so. □

We will now prove that $(B^k_{\theta\vartheta}, \mathbb{Y}|S)$-connections ∇ and $\overline{\nabla}$ which are "locally additively different" are actually "locally different".

Definition 4.2. Let M be a $(B^k_{\alpha,\beta}, \mathbb{X})$-manifold. Given $S \subset \Gamma_k$, θ, δ and $\mathbb{Y}$ as above, a *3-parameter $(\theta, \vartheta, \mathbb{Y}|S)$-family* in M is just a collection $(\Omega_\varphi)^c_{ab}$ of $(B, k, \theta, \vartheta|S)$-functions in $\mathbb{Y}$, with $a, b, c = 1, ..., n$, for each $(\varphi, U) \in \mathcal{B}^k_{\alpha,\beta}(\mathbb{X})$.

- If $(\Omega_\varphi)^c_{ab}$ is a 3-parameter family, let $(\Omega_\varphi)^c = \sum_{a,b} (\overline{\Omega}_\varphi)^c_{ab}(p)$.

Definition 4.3. Two 3-parameter $(\theta, \vartheta, \mathbb{Y}|S)$-families $(\Omega_\varphi)^c_{ab}$ and $(\overline{\Omega}_\varphi)^c_{ab}$ are:

1. *locally additively different* if for every chart $(\varphi, U) \in \mathcal{B}^k_{\alpha,\beta}(\mathbb{X})$ and every $c = 1, ..., n$ we have $(\Omega_\varphi)^c \neq (\overline{\Omega}_\varphi)^c$;
2. *locally different* if for every chart $(\varphi, U) \in \mathcal{B}^k_{\alpha,\beta}(\mathbb{X})$ and every $a, b, c = 1, ..., n$ there exists a nonempty open set $U_{a,b,c} \subset U$ such that $(\Omega_\varphi)^c_{ab}(p) \neq (\overline{\Omega}_\varphi)^c_{ab}(p)$ for every $p \in U_{a,b,c}$.

Definition 4.4. Two $(B^k_{\theta,\vartheta}, \mathbb{Y}|S)$-connections ∇ and $\overline{\nabla}$ in M are *locally additively different* (resp. *locally different*) if the corresponding 3-parameter $(\theta, \vartheta, \mathbb{Y}|S)$- families of their coefficients are locally additively different (resp. locally different).

Theorem 4.1. *If two 3-parameter $(\theta, \vartheta, \mathbb{Y}|S)$- families $(\Omega_{\varphi})^c_{ab}$ and $(\overline{\Omega}_{\varphi})^c_{ab}$ in a smooth $(B^{\infty}_{\alpha,\beta}, \mathbb{X})$-manifold are locally additively different, then they are locally different.*

Proof. For every chart (φ, U) and every $c = 1, .., n$, let $U_c \subset U$ be the subset in which $(\Omega_{\varphi})^c(p) \neq (\overline{\Omega}_{\varphi})^c(p)$. Since the 3-parameters are locally additively different, the subsets U_c are nonempty. On the other hand, they are the complement of the closed sets $[(\Omega_{\varphi})^c - (\overline{\Omega}_{\varphi})^c]^{-1}(0)$. Thus, they are nonempty of sets. For each $p \in U_c$ we then have

$$0 < |\Omega^c_{\varphi}(p) - (\overline{\Omega}_{\varphi})^c(p)| \leq \sum_{a,b} |(\Omega_{\varphi})^c_{ab}(p) - (\overline{\Omega}_{\varphi})^c_{ab}(p)|,$$

implying the existence of functions $a_c, b_c : U_c \to [n]$ such that

$$(\Omega_{\varphi})^c_{a_c(p)b_c(p)}(p) \neq (\overline{\Omega}_{\varphi})^c_{a_c(p)b_c(p)}(p).$$

Regard $[n]$ as a 0-manifold and notice that if two smooth functions $f, g : V \to [n]$ are transversal, then they coincide in an open set $V' \subset V$. Since $[n]$ is a 0-manifold, any smooth map $f : V \to [n]$ is a submersion and therefore transversal to each other. In particular, a_c (resp. b_c) is transversal to each constant map $a' \in [n]$ (resp. b'), so that for each a' (resp. b') there exists $U_{a',c} \subset U_c$ (resp. $U_{b',c} \subset U_c$) in which

$$(\Omega_{\varphi})^c_{a'b_c(p)}(p) \neq (\overline{\Omega}_{\varphi})^c_{a'b_c(p)}(p) \quad \text{(resp.} \quad (\Omega_{\varphi})^c_{a_c(p)b'}(p) \neq (\overline{\Omega}_{\varphi})^c_{a_c(p)b'}(p))$$

Taking $U_{a',b',c} = U_{a',c} \cap U_{b',c}$ and noticing that this intersection is nonempty, the proof is done. □

4.1 Discussion

We close with a broad discussion which we would like to see formalized and enlarged in some future work.

- Let $\mathcal{C}_k(\theta, \vartheta; \mathbb{Y})$ denote the set of 3-parameter $(\theta, \vartheta, \mathbb{Y})$-families in some $(B^k_{\alpha,\beta}, \mathbb{X})$-manifold M. Let ξ be a nice and distributive $(\alpha', \beta'; j)$-connection in M with degree $r \geq 0$.

The construction of Section 3 shows that every 3-parameter $(\alpha_0, \beta_0, \mathbb{X})$-family $f = (f_{\varphi})^c_{ab}$ in M induces, for each ordinary pair (θ, ϑ), a corresponding $(B^k_{\theta,\vartheta}, \mathbb{Y}_{\theta,\vartheta})$-connection ∇^f_{ξ} in M, whose 3-parameter family of coefficients $(\Gamma^f_{\xi,\varphi})^c_{ab}$ is given by (16). Suppose, for a moment, that we found a class $\mathcal{S} \subset \mathcal{C}^k_3(\alpha_0, \beta_0; \mathbb{X})$ of 3-parameter $(\alpha_0, \beta_0, \mathbb{X})$-families such that for every $f \in \mathcal{S}$ the corresponding regular connection ∇^f_{ξ} has coefficients $(\Gamma^f_{\xi,\varphi})^c_{ab} \in \mathcal{C}_k(\theta, \vartheta; \mathbb{Y})$ which are locally additively different to each other $\Omega_{\varphi} \in \mathcal{C}_k(\theta, \vartheta; \mathbb{Y})$. Thus, by Theorem 4.1 it follows that they are also locally different. In particular, *every 3-parameter $(\theta, \vartheta, \mathbb{Y})$-family is locally additively different to some $(B^k_{\theta,\vartheta}, \mathbb{Y})$-connection and this "difference" is measured by elements in $\mathcal{S}$.* This is the typical shape of denseness theorems. More precisely, one could expect the existence of a topology in $\mathcal{C}_k(\theta, \vartheta; \mathbb{Y})$ whose basic open sets are parameterized by elements of $\mathcal{S}$ and such that $\mathrm{Conn}^k_{\theta,\vartheta}(M; \mathbb{Y}) \subset \mathcal{C}_k(\theta, \vartheta; \mathbb{Y})$ is a dense subset.

Actually, the existence of $\mathcal{S}$ as above also implies that *if $f \in \mathcal{S}$, then ∇^f_{ξ} is locally additively different to each other $\nabla \in \mathrm{Conn}^k_{\theta,\vartheta}(M; \mathbb{Y})$*, which is the shape of an openness theorem. Thus, we could expect that, if the topology described above exists, then $\mathrm{Conn}^k_{\theta,\vartheta}(M; \mathbb{Y}) \subset \mathcal{C}_k(\theta, \vartheta; \mathbb{Y})$ should be open and nonempty due Theorem 3.2.

Finally, looking again at the expressions (11) and (16) defining ∇^f_ξ, one concludes that the problem of finding $\mathcal{S}$ consists in studying the complement in $\mathcal{C}_k(\theta,\vartheta;\mathbb{Y})$ of the solution spaces $\mathcal{X}_\xi(\varphi,\Omega)$, for each $\varphi\in\mathcal{B}^k_{\alpha,\beta}(\mathbb{X})$ and each $\Omega\in\mathcal{C}_k(\theta,\vartheta;\mathbb{Y})$, of the system of nonhomogeneous equations

$$F^c_\xi(f;\varphi,\Omega)=(\Omega_\varphi)^c,\quad \text{where}\quad F^c_\xi(f;\varphi,\Omega)=\sum_{a,b}\xi^{\alpha',\beta'}_{\theta,\vartheta}((\Gamma^f_{\xi,\varphi})^c_{ab}), \tag{18}$$

whose coefficients belong to the space of $(B,k,\theta,\vartheta|\Gamma_k[z])$-functions. Since equations (18) strongly depend on the $(\alpha',\beta';j)$-connection ξ, the basic strategy to find $\mathcal{S}$ (and then to get the openness and denseness results described above) would be to introduce additional conditions in ξ which simplify (18). In other words, would be to look at (18) as parameterized systems of equations in the space of all $(\alpha',\beta';j)$-connections.

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7

FEYNMAN FUNCTORS

7.1 SYNTHESIS

Recall that, as discussed in Chapter 2, in an analogy of how Hilbert's Program extends Hilbert's second problem, we can consider an extension of Hilbert's sixth problem to an Axiomatization Program of Physics (APP). There we introduced an approach which emphasizes the role of the abstraction processes (see Corollary 2.21). We also commented that this approach ignores the existence of disjoint classes of physical systems. Here we will discuss another approach that takes this into account.

The idea goes as follows. Let $\mathcal{F}_{real}$ be the class of real physical systems, which by the empiricist ontology is the largest subclass of $\mathcal{F}$ in which diagram below can be completed with the dotted arrow.

$$\begin{tikzcd}
\mathcal{F}_{real} \arrow[r, dashed] \arrow[d, hook] & \mathrm{Exp}(M) \arrow[d, hook] & \\
\mathcal{F} \arrow[r] & \mathrm{Belief}(\mathcal{F}) & \mathrm{Know}(\mathcal{F}) \arrow[l, hook'] \arrow[ul, "e"']
\end{tikzcd}$$

Definition 7.1. A subclass $\mathcal{F}_0 \subset \mathcal{F}_{real}$ (resp. $\mathcal{E}_0 \subset \mathrm{Ext}(M)$) is said to be *maximal* (or *irreductible*) if it cannot be written as a union of two disjoint proper subclasses of $\mathcal{F}_{real}$ (resp. $\mathrm{Ext}(M)$). An empiricist class[1] $(\mathcal{F}_0, \mathcal{E}_0)$ is *maximal* (or *irreductible*) if both $\mathcal{F}_0$ and $\mathcal{E}_0$ are.

Let $\pi_0(\mathcal{F})$ denote the class of all maximal empiricist classes.

Definition 7.2. A *morphism* $\alpha : \mathcal{F}\mathcal{E}_0 \to \mathcal{F}\mathcal{E}_1$ between two empiricist classes $\mathcal{F}\mathcal{E}_0 = (\mathcal{F}_0, \mathcal{E}_0)$ and $\mathcal{F}\mathcal{E}_1 = (\mathcal{F}_1, \mathcal{E}_1)$ is given by:

1. a map $\alpha(\mathcal{F}) : \mathcal{F}_0 \to \mathcal{F}_1$;
2. a map $\alpha(\mathcal{E}) : \mathcal{E}_0 \to \mathcal{E}_1$,

[1] For the definition of empiricist class see Definition 2.15.

such that the following diagram commutes:

$$\begin{tikzcd}
 & & & \mathcal{E}_0 \arrow[d, "\alpha(\mathcal{E})"] \arrow[ddrr, bend left, "e_0"] & & \\
 & & & \mathcal{E}_1 \arrow[d, hook] \arrow[ddrr, bend left, "e_1"] & & \\
\mathcal{F}_0 \arrow[r, "\alpha(\mathcal{F})"'] \arrow[uurrr, bend left, "\iota_0"] & \mathcal{F}_1 \arrow[r, hook] \arrow[urr, bend left, "\iota_1"] & \mathcal{F}_{real} \arrow[r] \arrow[d, hook] & \mathrm{Exp}(M) \arrow[d, hook] \arrow[drr, "e"] & & \\
 & & \mathcal{F} \arrow[r] & \mathrm{Belief}(\mathcal{F}) & & \mathrm{Know}(\mathcal{F}) \arrow[ll, hook']
\end{tikzcd}$$

Remark 7.3. Thus, $\mathcal{FE}_1$ is an extension of $\mathcal{FE}_0$ (see Definition 2.18) precisely if there is an injective morphism $\alpha : \mathcal{FE}_0 \to \mathcal{FE}_1$.

Definition 7.4. Let $\mathcal{FE}_0$ and $\mathcal{FE}_1$ be two empiricist classes. A *morphism* $\beta : (\mathcal{E}'_0, L_0, f_0) \to (\mathcal{E}'_1, L_1, f_1)$ between a proposal of APP in $\mathcal{FE}_0$ and a proposal of APP in $\mathcal{FE}_1$ (see Definition 2.17) is given by:

1. a morphism $\beta(FE) : \mathcal{FE}_0 \to \mathcal{FE}_1$ factoring trough the classes of fundamental experiments $\mathcal{E}'_0$ and $\mathcal{E}'_1$;
2. a morphism $\beta(L) : L_0 \to L_1$ of axiomatic languages (see here),

which makes commutative the following diagram:

$$\begin{tikzcd}
\mathcal{F}_0 \arrow[rrr, "e_0"] \arrow[dr, "\iota_0"'] & & & \mathrm{Know}(\mathcal{F}) & & & \mathcal{F}_1 \arrow[lll, "e_1"'] \arrow[dl, "\iota_1"] \\
 & \mathcal{E}_0 \arrow[rrrr, "\beta(\mathcal{E})"] \arrow[dr, "f_0"] & & & & \mathcal{E}_1 \arrow[dl, "f_1"'] & \\
 & & \mathrm{Word}(L_0) \arrow[rr, "\beta(L)"] & & \mathrm{Word}(L_1) & & \\
 & & \mathrm{Axiom}(\mathrm{L}_0) \arrow[u, hook] \arrow[rr, "\bar{\beta}(L)"'] & & \mathrm{Axiom}(\mathrm{L}_1) \arrow[u, hook] & & \\
 & \mathcal{E}'_0 \arrow[uuu, hook] \arrow[ur, "\bar{f}_0"] \arrow[rrrr, dashed, "\beta(\bar{\mathcal{E}})"'] & & & & \mathcal{E}'_1 \arrow[uuu, hook] \arrow[ul, "\bar{f}_1"'] &
\end{tikzcd}$$

Now, suppose that:

1. there is a poset $(I, \leq)$ and a map bijection $I \simeq \pi_0(\mathcal{F})$;
2. for every $i, j \in I$ such that $i \leq j$ there is a morphism $\alpha_{ji} : \mathcal{FE}_i \to \mathcal{FE}_j$;
3. for every $i, j, k \in I$ such that $i \leq j \leq k$ we have $\alpha_{kj} \circ \alpha_{ji} = \alpha_{ki}$;
4. to each $i \in I$ corresponds a proposal $(\mathcal{E}'_i, L_i, f_i)$ to APP in $\mathcal{FE}_i$ (see Definition 2.17);
5. the morphisms $\alpha_{ji} : \mathcal{FE}_i \to \mathcal{FE}_j$ extends to morphisms $\beta_{ji} : (\mathcal{E}'_i, L_i, f_i) \to (\mathcal{E}'_j, L_j, f_j)$ between the corresponding proposals to APP.

From hypotheses 1, 2 and 3, the class $\pi_0(\mathcal{F})$ of maximal empiricist classes forms a direct system whose limit is $\mathcal{F}_{real}$. Hypotheses 4 and 5 tells us that this direct system extends to the class of all proposals to APP. Since morphisms of proposals are defined by commutative diagrams, one could expect that the corresponding limit exists as a proposal, and that if each proposal is actually a realization of APP, then it also exists as a realization of APP in the entire Physics. In this case:

- *to realize APP in the entire Physics it is enough to provide realizations in each maximal empiricist class in such a way that hypotheses above are satisfied.*

7.1.1 Quantization

It is a typical assumption (motivated by the knowledge obtained trough experiments) that there are at least two maximal empiricist classes $\mathcal{FE}_{cls} = (\mathcal{F}_{cls}, \mathcal{E}_{cls})$ and $\mathcal{FE}_{qnt} = (\mathcal{F}_{qnt}, \mathcal{E}_{qnt})$ describing *classical Physics* and *quantum Physics*, respectively.

As part of the previous approach to APP, we have to find a poset $(I, \leq)$ and a bijection $I \simeq \pi_0(\mathcal{F})$, so that a way to compare maximal empiricist classes (hypothesis 1). It is also a common assumption that, if this poset exists, then $\mathcal{FE}_{cls}$ and $\mathcal{FE}_{qnt}$ are comparable and $\mathcal{FE}_{cls} \leq \mathcal{FE}_{qnt}$. In this case, also as part of the previous approach to APP, there should exist:

1. realizations $(\mathcal{E}'_{cls}, L_{cls}, f_{cls})$ and $(\mathcal{E}'_{qnt}, L_{qnt}, f_{qnt})$ of APP in $\mathcal{FE}_{cls}$ and $\mathcal{FE}_{qnt}$, respectively (hypothesis 4);
2. a morphism $Q : \mathcal{FE}_{cls} \to \mathcal{FE}_{qnt}$ of empiricist classes (hypothesis 2) which extends to a morphism $\bar{Q} : (\mathcal{E}'_{cls}, L_{cls}, f_{cls}) \to (\mathcal{E}'_{qnt}, L_{qnt}, f_{qnt})$ between the corresponding realizations (hypothesis 5).

Remark 7.5.

1. In the literature, the morphism $Q : \mathcal{FE}_{cls} \to \mathcal{FE}_{qnt}$ above and its extension $\bar{Q}$ are known as *quantization procedures*. In the following we will say that Q is a *philosophical quantization* and that $\bar{Q}$ is its *axiomatic extension*, so that an *axiomatic quantization* relatively to given realizations of APP.
2. A philosophical quantization may exist even if an axiomatic quantization does not exists, exactly as a map $f : X \to X$ may exists without having an extension $\bar{f} : \bar{X} \to \bar{Y}$ relatively to given maps $\bar{X} \to X$ and $\bar{Y} \to Y$. For instance, if there is no way to realize APP in $\mathcal{FE}_{cls}$ or $\mathcal{FE}_{qnt}$, then the axiomatic quantization does not exist.
3. However, the nonexistence of an axiomatic quantization for specific realizations of APP, does not imply the inexistence of a philosophical quantization, in the same way as the nonexistence of an extension $\bar{f} : \bar{X} \to \bar{Y}$ for specific $\bar{X} \to X$ and $\bar{Y} \to Y$ does not implies the nonexistence of the map $f : X \to Y$.

Explicitly, a philosophical quantization is given by maps $Q(\mathcal{F}) : \mathcal{F}_{cls} \to \mathcal{F}_{qnt}$ and $Q(\mathcal{E}) : \mathcal{E}_{cls} \to \mathcal{E}_{qnt}$ such that the following diagram commutes. We say that $Q(\mathcal{F})$ maps a classical system in its *quantized system*, while $Q(\mathcal{E})$ maps a classical experiment into its *quantized experiment*. Thus, the existence of a philosophical quantization procedure means that:

1. *quantization commutes with taking experiments*. More precisely, the quantum experiments of a quantized classical system coincides with the quantum experiments of the quantized system (blue part of the diagram);

2. *experimental truth factors trough quantization*. In more precise words, the knowledge arising from a classical experiment can be recovered as the knowledge of the quantized experiment (red part of the diagram).

$$
\begin{tikzcd}
 & & & \mathcal{E}_{cls} \arrow[d, "Q(\mathcal{E})"'] \arrow[ddrr, bend left, "e_{cls}"] & & \\
 & & & \mathcal{E}_{qnt} \arrow[d, hook] \arrow[ddrr, bend left, "e_{qnt}"] & & \\
\mathcal{F}_{cls} \arrow[r, "Q(\mathcal{F})"'] \arrow[uurrr, bend left, "\imath_{cls}"] & \mathcal{F}_{qnt} \arrow[r, hook] \arrow[urr, bend left, "\imath_{qnt}"] & \mathcal{F}_{real} \arrow[r, "f"] \arrow[d, hook] & \mathrm{Exp}(M) \arrow[d, hook] \arrow[drr, "e"] & & \\
 & & \mathcal{F} \arrow[r] & \mathrm{Belief}(\mathcal{F}) \arrow[rr, leftrightarrow] & & \mathrm{Know}(\mathcal{F})
\end{tikzcd}
$$

Instead of finding a global quantization procedure which applies to all classical systems, one could look for *local quantizations*, which are defined between empiricist subclass $\mathcal{F}\mathcal{E}_{cls,0} \subset \mathcal{F}\mathcal{E}_{cls}$ and $\mathcal{F}\mathcal{E}_{qnt,0} \subset \mathcal{F}\mathcal{E}_{qnt}$.

Definition 7.6. Let $\mathcal{F}\mathcal{E}_{cls,0} = (\mathcal{F}_{cls,0}, \mathcal{E}_{cls,0})$ and $\mathcal{F}\mathcal{E}_{qnt,0} = (\mathcal{F}_{qnt,0}, \mathcal{E}_{qnt,0})$ be a classical and a quantum empiricist class, respectively.

1. A *local philosophical quantization* between them is is a morphism $Q_0 : \mathcal{F}\mathcal{E}_{cls,0} \to \mathcal{F}\mathcal{E}_{qnt,0}$ such that diagram below commutes, where $\imath_{cls,0}$ and $\imath_{qnt,0}$ are just restrictions of $\imath_{cls}$ and $\imath_{qnt}$.

2. An *axiomatic extension* of a local philosophical quantization Q_0 relatively to realizations $(\mathcal{E}'_{cls,0}, L_{cls,0}, f_{cls,0})$ and $(\mathcal{E}'_{qnt,0}, L_{qnt,0}, f_{qnt,0})$ of APP in $\mathcal{F}\mathcal{E}_{cls,0}$ and $\mathcal{F}\mathcal{E}_{qnt,0}$, respectively, is an extension of Q_0 to a morphism of $\bar{Q}_0$ of realizations.

3. an *local axiomatic quantization* between two realizations as above is the axiomatic extension of some local philosophical quantization.

$$
\begin{tikzcd}
 & & & \mathcal{E}_{cls,0} \arrow[d, "Q_0(\mathcal{E})"'] \arrow[dr] \arrow[ddddrr, bend left, "e_{cls,0}"] & & \\
 & & & \mathcal{E}_{qnt,0} \arrow[dr] \arrow[dddrr, "e_{qnt,0}"] & \mathcal{E}_{cls} & \\
\mathcal{F}_{cls,0} \arrow[r, "Q_0(\mathcal{F})"] \arrow[uurrr, bend left, "\imath_{cls,0}"] \arrow[dr, hook] & \mathcal{F}_{qnt,0} \arrow[urr, bend left, "\imath_{qnt,0}"] \arrow[dr, hook] & & & \mathcal{E}_{qnt} & \\
 & \mathcal{F}_{cls} & \mathcal{F}_{qnt} \arrow[r, hook] & \mathcal{F}_{real} \arrow[r] \arrow[d, hook] & \mathrm{Exp}(M) \arrow[d, hook] \arrow[dr, "e"] & \\
 & & & \mathcal{F} \arrow[r] & \mathrm{Belief}(\mathcal{F}) \arrow[r, leftrightarrow] & \mathrm{Know}(\mathcal{F})
\end{tikzcd}
$$

Example 7.7. As it is well-known, part $\mathcal{F}\mathcal{E}_{cls,0}$ of classical physics has a realization of APP in terms of symplectic manifolds and Hamiltonian functions, while part $\mathcal{F}\mathcal{E}_{qnt,0}$ of quantum physics has a realization in terms of separable Hilbert spaces and unbounded self-adjoint operators. Groenewold's theorem establishes the nonexistence of a local axiomatic quantization between these realizations. However this *does not means* that there is no philosophical quantization $Q_0 : \mathcal{FE}_{cls,0} \rightarrow \mathcal{FE}_{cls,1}$ nor that it cannot has some other axiomatic extension relatively to other realizations.

The existence problem of a global axiomatic quantization procedure whose quantized systems satisfy additional requirements is one of the most important open problems in Mathematical Physics. Actually, even the quantization of a single class of classical systems (a local quantization procedure) in such a way that the quantized system satisfies additional properties is of summary importance. For instance, in the language above, something closer to one of the unsolved Millennium Problems could be written as follows [86]:

Quantum Yang-Mills and Mass Gap *Show that:*

1. *there is a classical empiricist class $\mathcal{FE}_{cls,0}$ in which there exist a realization $(\mathcal{E}'_{cls,0}, L_{cls,0}, f_{cls,0})$ to APP, which contains Yang-Mills theories on the spacetime $\mathbb{R}^4$ for every compact simple Lie group G;*
2. *there is a quantum empiricist class $\mathcal{FE}_{qnt,0}$ in which there exist a realization $(\mathcal{E}'_{qnt,0}, L_{qnt,0}, f_{qnt,0})$ to APP;*
3. *there is a local philosophical quantization $Q_0 : \mathcal{FE}_{cls,0} \rightarrow \mathcal{FE}_{qnt,0}$ such that:*
   - *(a) it has an axiomatic extension $\bar{Q}$ relatively to the given realizations to APP;*
   - *(b) for every compact simple Lie group G, the image of the Yang-Mills theories on $\mathbb{R}^4$ with gauge group G by $\bar{Q}_0$, after projected on $\mathcal{FE}_{qnt,0}$, contains quantum experiments asserting that the least massive particle has positive mass.*

One idea to build a philosophical/axiomatic quantization procedure is to follows a local-to-global inductive strategy similar to that presented in section An Inductive Approach of Chapter 2, but now parameterized by pairs composed by a classical empiricist class and a quantum empiricist class, instead of by arbitrary empiricist classes.

Firstly one should prove the existence of philosophical quantization. For them, the inductive strategy is rough as follows:

1. *base of induction*: showing that there is a classical empiricist class $\mathcal{FE}_{cls,0}$ and a quantum empiricist class $\mathcal{FE}_{qnt,0}$ in which and a local philosophical quantization $Q_0 : \mathcal{FE}_{cls,0} \rightarrow \mathcal{FE}_{cls}$ between them;
2. *induction step*: showing that if $(\mathcal{FE}_{cls,1}, \mathcal{FE}_{qnt,1})$ is some pair extending $(\mathcal{FE}_{cls,0}, \mathcal{FE}_{qnt,0})$, then the local philosophical quantization $Q_0 : \mathcal{FE}_{cls,0} \rightarrow \mathcal{FE}_{cls}$ extends to it in such a way that the following diagram commutes:

$$
\begin{tikzcd}
 & & & \mathcal{E}_{cls,0} \arrow[dd, "Q_0(\mathcal{E})"'] \arrow[dr, hook] \arrow[ddddrr, bend left, "e_{cls,0}"] & & \\
 & & & & \mathcal{E}_{cls,1} \arrow[dd, dashed, "Q_1(\mathcal{E})"] \arrow[dddr, "e_{cls,1}"] & \\
\mathcal{F}_{cls,0} \arrow[uurrr, bend left, "\iota_{cls,0}"] \arrow[r, "Q_0(\mathcal{F})"] \arrow[dr, hook] & \mathcal{F}_{qnt,0} \arrow[rr, "\iota_{qnt,0}"] \arrow[dr, hook] & & \mathcal{E}_{qnt,0} \arrow[dr, hook] \arrow[ddrr, bend right, "e_{qnt,0}"'] & & \\
 & \mathcal{F}_{cls,1} \arrow[uuurr, bend left] \arrow[r, dashed, "Q_1(\mathcal{F})"'] & \mathcal{F}_{qnt,1} \arrow[rr] & & \mathcal{E}_{qnt,1} \arrow[dr, "e_{qnt,1}"] & \\
 & & & & & \mathrm{Know}(\mathcal{F})
\end{tikzcd}
$$

7.1.2 Perturbative Quantization

As discussed in the last section, one of the basic assumptions concerning the philosophical foundations of Physics is that it has at least two maximal empiricist classes given by *classical systems* and *quantum systems*. In the direct limit approach to APP, one should then provide realizations of APP in classical and quantum systems and a morphism (global philosophical quantization) between them which extends to the realizations (axiomatic extension to a global axiomatic quantization).

Notice that, a priori, this assumption does not imply that every physical system is either classical or quantum. In other words, it does not deny the existence of other maximal empiricist classes. Actually, another typical assumption when studying the foundations of Physics is that a third maximal empiricist class $\mathcal{F}\mathcal{E}_{prt} = (\mathcal{F}_{prt}, \mathcal{E}_{prt})$ exists: the class of the so-called *perturbative systems*. These are such that there is a span as below, where P is the *perturbative expansion*, while PQ is the *perturbative quantization*.

$$
\begin{tikzcd}
 & \mathcal{F}_{qnt} \arrow[d, "P"] \\
\mathcal{F}_{cls} \arrow[r, "PQ"'] & \mathcal{F}_{prt}
\end{tikzcd}
\tag{7.1}
$$

More explicitly, this spam means the commutativity of the following diagram:

$$
\begin{tikzcd}
\mathcal{E}_{cls} \arrow[r, "PQ(\mathcal{E})"'] \arrow[rrrr, bend left, "e_{cls}"] & \mathcal{E}_{prt} \arrow[rrr, bend left, "e_{prt}"] & \mathcal{E}_{qnt} \arrow[l, "P(\mathcal{E})"] \arrow[dr] & & \mathrm{Know}(\mathcal{F}) \arrow[dd, hook] \\
\mathcal{F}_{cls} \arrow[u, "\imath_{cls}"] \arrow[r, "PQ(\mathcal{F})"] \arrow[drr, bend right] & \mathcal{F}_{prt} \arrow[u, "\imath_{prt}"'] \arrow[dr] & \mathcal{F}_{qnt} \arrow[u, "\imath_{qnt}"'] \arrow[l, "P(\mathcal{F})"'] \arrow[d, hook] & \mathrm{Exp}(M) \arrow[ur, "e"] \arrow[dr] & \\
 & & \mathcal{F} \arrow[ur, "\imath"] \arrow[rr] & & \mathrm{Belief}(\mathcal{F})
\end{tikzcd}
$$

Observe that if there is an arrow $C : \mathcal{F}\mathcal{E}_{prt} \to \mathcal{F}\mathcal{E}_{qnt}$ is the opposite direction, then one can composite it with perturbative quantization to get a global quantization, as below. This arrow is should be called *completion*, since should take a perturbative quantization of a classical system and complete it to provide a genuine quantum system. To build this global completion constitute another approach to get a global (philosophical) quantization.

$$
\begin{tikzcd}
 & \mathcal{F}\mathcal{E}_{qnt} \arrow[d, dashed, "P"'] \\
\mathcal{F}\mathcal{E}_{cls} \arrow[ur, dashed, "Q"] \arrow[r, "PQ"'] & \mathcal{F}\mathcal{E}_{prt} \arrow[u, bend right, "C"']
\end{tikzcd}
$$

Suppose for a moment that our discussion is internal to some context which Axiom of Choice is satisfied. For instance, suppose that classes of physical systems are sets or classes in the context of ZFC, NBG or MK, that the arrows in the diagrams are functions, etc. In this case, if the perturbative extension PQ was an epimorphism, then Axiom of Choice would imply that it has a section that could be viewed as a completion.

Reciprocally, note that a completion C does not need to depend on the perturbative expansion P. But physically it is expected that the perturbative expansion of a completed perturbative quantization is the perturbative quantization itself, meaning that expanding perturbatively something which is already perturbative has no effect. However, this is precisely the assumption that C is a section for P and, therefore, that P is an epimorphism.

The conclusion is:

- *one should expect existence of a global completion iff perturbative expansion P is an epimorphism.*

Let us see what can be said without additional assumptions on the perturbative expansion. To do so, suppose that we are discussing internally to a context that not only exists a version of Axiom of Choice, but also that has pullbacks[2], as ZFC, NBG or MK.

Thus, take the pullback of the spam (7.1). Let $\mathcal{FE}'_{cls}$ be the projection of this pullback onto $\mathcal{FE}_{cls}$. From the Axiom of Choice there is a section s as below, so that we can compose with the projection onto quantum systems to get a morphism $Q' : \mathcal{FE}'_{cls} \to \mathcal{FE}_{qnt}$. This means that, under this hypothesis, there is a local quantization in empiricist class $\mathcal{FE}'_{cls}$, which is the base of induction for the strategy to build a global quantization discussed in the last section.

$$
\begin{tikzcd}
\mathcal{F}'_{cls} \arrow[r, "s"] \arrow[rr, dashed, bend left, "Q'(\mathcal{F})"] \arrow[dr] & pb \arrow[r] \arrow[d] & \mathcal{F}_{qnt} \arrow[d, "P(\mathcal{F})"] \\
& \mathcal{F}_{cls} \arrow[r, "PQ(\mathcal{F})"'] & \mathcal{F}_{prt}
\end{tikzcd}
\qquad
\begin{tikzcd}
\mathcal{E}'_{cls} \arrow[r, "s"] \arrow[rr, dashed, bend left, "Q'(\mathcal{E})"] & pb \arrow[r] \arrow[d] & \mathcal{E}_{qnt} \arrow[d, "P(\mathcal{E})"] \\
& \mathcal{E}_{cls} \arrow[r, "PQ(\mathcal{E})"'] & \mathcal{E}_{prt}
\end{tikzcd}
$$

Now, let $\mathcal{FE}'_{prt} \subset \mathcal{FE}_{prt}$ be the the empiricist subclass given by the the image $P(\mathcal{FE}_{prt})$ of perturbative expansion. Thus, the quantization Q_0 of $\mathcal{FE}'_{cls}$ induces a factorization of the perturbative quantization PQ' as below, so that $PQ' = P' \circ Q'$.

$$
\begin{tikzcd}
& pb \arrow[r] \arrow[d] \arrow[dl] & \mathcal{FE}_{qnt} \arrow[d, "P"] \arrow[dr, "P'"] & \\
\mathcal{FE}'_{cls} \arrow[ur, bend left, "s"] \arrow[urr, bend left=50, "Q'"] \arrow[r, hook] \arrow[rrr, dashed, bend right, "PQ'"'] & \mathcal{FE}_{cls} \arrow[r, "PQ"] & \mathcal{FE}_{prt} & \mathcal{FE}_{prt,0} \arrow[l, hook']
\end{tikzcd}
$$

On the other hand, again by the Axiom of Choice, we have a section C' to P', so that a "partial completion", as below.

$$
\begin{tikzcd}
pb \arrow[r] \arrow[d] & \mathcal{FE}_{qnt} \arrow[d, "P"] \arrow[dr, "P'"] & \\
\mathcal{FE}_{cls} \arrow[r, "PQ"'] & \mathcal{FE}_{prt} & \mathcal{FE}_{prt,0} \arrow[l, hook'] \arrow[ul, dashed, bend right, "C'"']
\end{tikzcd}
$$

[2] Actually, we will only need that the pullback between perturbative quantization and perturbative expansion exists.

Furthermore, C' can be chosen in such a way that the entire diagram becomes commutative, meaning that $Q' = C' \circ PQ'$, i.e., that C' is completion for PQ'.

$$
\begin{tikzcd}
 & \mathrm{pb} \arrow[r] \arrow[d] \arrow[dl, bend left] & \mathcal{FE}_{qnt} \arrow[d, "P"'] \arrow[dr, "P'"] & \\
\mathcal{FE}'_{cls} \arrow[r, hook] \arrow[ur, "s"] \arrow[urr, bend left, "Q'"] \arrow[rrr, dashed, bend right, "PQ'"'] & \mathcal{FE}_{cls} \arrow[r, "PQ"] & \mathcal{FE}_{prt} & \mathcal{FE}'_{prt} \arrow[l, hook'] \arrow[ul, dotted, bend right, "C'"']
\end{tikzcd}
$$

Remark 7.8. In terms of physicist literature, one could say that $\mathcal{FE}'_{prt}$ is the class of those perturbative empiricist classes which are *renormalizable*. Thus, the completion C' says that every renormalizable perturbative system can be completed to a quantum system. Furthermore, $\mathcal{FE}'_{cls}$ is the class of classical systems whose perturbative quantization is renormalizable and, therefore, for which a quantum system is well-defined.

7.1.3 Feynman Rules

All the discussion above was on the philosophical level. To be used in the APP it should be extended to the axiomatic level. This implies to find proposals of APP in $\mathcal{FE}_{cls}$, $\mathcal{FE}_{prt}$ and $\mathcal{FE}_{qnt}$ such that perturbative quantization and perturbative extension extends to morphisms of APP proposals, and so on. Furthermore instead of starting the discussion globally (i.e, in the entire $\mathcal{FE}_{cls}$), one could work on a subclass $\mathcal{FE}_{cls,0} \subset \mathcal{FE}_{cls}$ and then try to follow some inductive approach as discussed previously.

If we take in $\mathcal{FE}_{prt}$ the APP proposal where perturbative systems are modeled by "analytic expressions" on locally convex spaces with some topological tensorial product and $\mathcal{FE}_{cls,0}$ is the class of classical systems which can be modeled by Lagrangian field theories, then its perturbative quantization $PQ : \mathcal{FE}_{cls,0} \to \mathcal{FE}_{prt}$ has an axiomatic extension which assigns analytical expressions to each field theory through the so-called *Feynman rules*.

Definition 7.9. Let M be a compact Riemannian manifold, regarded as the spacetime. A *free Lagrangian field theory* on M, denoted by $\mathcal{L}_0$ is given by:

1. a $\mathbb{K}$-vector bundle $E \to M$, with $\mathbb{K} = \mathbb{R}, \mathbb{C}$, called *field bundle*;
2. a symmetric bundle map $\langle \cdot, \cdot \rangle : E \otimes E \to \Omega^n(M)$;
3. a generalized Laplacian $D : \Gamma(E) \to \Gamma(E)$ in E;
4. a differential operator $\mathfrak{D} : E \to E^!$, where $E^! = \mathrm{Hom}(E; \Omega^n(M))$ is the strong dual of E, such that:
   (a) is formally symmetric, i.e., $\mathfrak{D}^! = \mathfrak{D}$;
   (b) commutes with D, i.e., $\mathfrak{D} \circ D = D \circ \mathfrak{D}^! = D \circ \mathfrak{D}$.

Definition 7.10. A *term of interaction* on a vector bundle E is a symmetric function evaluated in the algebra of differential operators of $\Gamma(E)$, i.e., it is a formal series $I[l] = \sum_l p_l$, where each p_l is a symmetric polynomial of degree l evaluated in differential operators in E.

Definition 7.11. An *interacting Lagrangian field theory* is a free field theory $\mathcal{L}_0$ endowed with a term of interaction I to its field bundle, being denoted by $\mathcal{L} = \mathcal{L}_0 + I$.

Definition 7.12. Let X be a nuclear Fréchet space, $TX = \oplus T^k X$ its tensor algebra relatively to a topological tensor product $\otimes$ (which in this case coincides is the same for projective, injective or inductive topologies) and let $PX = \oplus_k P^k X$ be the corresponding symmetric tensor algebra. Let $\mathcal{O}(X) = P(X^\vee)$ where $X^\vee$ is the topological dual.

1. A *k-propagator* in X is an element of $P^k X$;
2. An *interacting term* in X relatively to a formal parameter h is an element of $\mathcal{O}(X)[[h]]$;
3. An *analytical expression of order* k in X is an element of $A^k X = \oplus_k P^k X \otimes \mathcal{O}^k X$.

The Feynman rules, as commented above, assign analytical expressions to Lagrangian field theories. Recall that if E is a vector bundle over a compact manifold M, then its space $\Gamma(E)$ of smooth global sections is a nuclear Fréchet space. Thus, of course, Feynman rules take a Lagrangian field theory with field bundle E and return a bunch of analytical expressions in the nuclear Fréchet space $\Gamma(E)$. These analytical expressions are organized into a class of graphs known as *Feynman graphs*.

Definition 7.13. A *Feynman graph* is a finite graph G endowed with the following structures:

1. a vertex-decomposition $VG = V^0 G \sqcup V^1 G$ into *external vertices* and *internal vertices*, respectively;
2. a function $g : VG \to \mathbb{Z}_{\geq 0}$ assigning to each vertex v its *genus* $g(v) \in \mathbb{Z}_{\geq 0}$,

such that the following conditions are satisfied:

1. if $v \in V^0 G$ is an external vertex, then $d(v) = 1$ and $g(v) = 0$, where $d(v)$ is the total number of adjacent edges;
2. if $v \in V^1 G$ is an internal vertex, then $d(v) \geq 3$;
3. there are no edges between two external vertices.

Remark 7.14. The condition that $d(v) = 1$ for every $v \in V^0 G$ establishes the existence of a subclass $E^0 G \subset EG$ of *external edges* such that $|E^0 G| = |V^0 G|$. We then also has an edge-decomposition $EG = E^0 G \sqcup E^1 G$ by external edges and internal edges, where $E^1 G = EG - E^0 G$.

Let $\mathcal{L} = \mathcal{L}_0 + I$ be a Lagrangian field theory with field bundle $E \to M$ and whose interacting term I is on a formal parameter h. To it corresponds a collection $G(\mathcal{L})$ of Feynman graphs, organized into powers of h, as follows.

Write the interaction term I as $I = \sum_k I_k$ with $I_k = \sum_n I_{k,n} h^n$. For each k, n let $G_{k,n}$ the collection of all Feynman graphs G such that there is an internal vertex $v \in V^1 G$ such that $d(v) = k$ and $g(v) = n$. Then take $G(\mathcal{L})$ as the union $\cup_{K(I)} G_{k,n}$ where $K(I)$ is the collection of all pairs (k, n) such that $I_{k,n} \neq 0$.

The *Feynman rules* of a Lagrangian field theory $\mathcal{L}$ in a field bundle E are given by the function $FR : G(\mathcal{L}) \to A(\Gamma(E)))$ assigning an analytical expression $FR(G)$ to each Feynman graph G of $\mathcal{L}$, obtained as follows:

1. since M is compact, the generalized Laplacian D of Definition 7.9 has a smooth kernel $K_t \in \Gamma(E^!) \otimes \Gamma(E)$ and the function $t \mapsto K_t$ is continuous. Applying $\mathfrak{D} \otimes id_{\Gamma(E)}$ we get an element $\mathfrak{K}_t \in \Gamma(E) \otimes \Gamma(E)$ and integrating over t in we find $\mathfrak{K} \in \Gamma(E) \otimes \Gamma(E)$. Because $\mathfrak{D}$ is symmetric and commutes with D, it follows that $\mathfrak{K}$ is symmetric too, so that $\mathfrak{K} \in P^2(\Gamma(E))$;

2. given $G \in G(\mathcal{L})$ consider the function $FR(G)_E : EG(\mathcal{L}) \to P^2(\Gamma(E))$ that to each edge e assigns a copy $\mathfrak{K}_{x,y}$ of the heat kernel of $\mathfrak{K}$ labeled by the vertices of $\{x, y\}$ of e;
3. consider the map $FG^1_V : V^1G(\mathcal{L}) \to \mathcal{O}(\Gamma(E))[[h]]$ that to each $v \in VG^1$ with $G \in G(\mathcal{L})$ assigns $I_{d(v),g(v)}$;
4. take any injective map $FG^0_V : V^0G(\mathcal{L}) \to \Gamma(E)$ assigning to each external vertice $v \in V^0G$ a section $FR_0(v)$;
5. finally, define $FR : G(\mathcal{L}) \to A(\Gamma(E))$ by tensoring the image of all the three maps above, i.e:
$$FR(G) = \bigotimes_{e\in EG} \bigotimes_{v\in V^1G} \bigotimes_{w\in V^0G} FG_E(e) \otimes FG^1_V(v) \otimes FG^0_V(w).$$

Remark 7.15. That the map FR define above really take values into $A(\Gamma(E))$ follows from the graph identity $\sum_{v\in VG} d(v) = 2|EG|$ and from the fact that $|VG| = |VG^0| + |VG^1|$.

7.1.4 Feynman Functors

Above we saw that there is a well-known way to provide an axiomatic extension of local perturbative quantization defined on the class of classical physical systems which can be modeled as field theories. Thus, Feynman rules are actually a correspondence that assigns an analytical expression to each Feynman graph of a field theory. The problem is then on how to extend this to a more ample class of classical physical systems.

Recall that the axiomatic perturbative quantization above was obtained building a map into two steps:

1. assigning to each Lagrangian field theory $\mathcal{L} = \mathcal{L}_0 + I$ in a field bundle E a collection $G(\mathcal{L})$ of Feynman graphs;
2. constructing, for every $\mathcal{L}$, a map $FR(\mathcal{L}) : G(\mathcal{L}) \to A(\Gamma(E))$ which assigns analytical expressions in E to Feynman graphs of $\mathcal{L}$: the *Feynman rules* of $\mathcal{L}$.

Graphs with graph morphisms define a category **Graph**. Define a *morphism* $f : G \to H$ between two Feynman graphs as a graph morphism which preserves the vertex and the genus, i.e., such that $f : VG \to VH$ maps internal/external vertices into internal/external vertices in such a way that $g_H(f(v)) = g_G(v)$. This gives a category **Feyn** of Feynman graphs.

On the other hand, for X a nuclear Fréchet space, notice that an element $x \in X$ is equivalently a morphism $\bar{x} : \mathbb{R} \to X$. Thus, we can define a *morphism* $f : \bar{x} \to \bar{y}$ between two elements $x, y \in X$ as a morphism in the slice category, i.e., as a morphism $f : X \to X$ such that $\bar{y} = f \circ \bar{x}$, which in this only means that $y = f(x)$. This produces a category $\mathbb{R}/X$.

Thus, for a given nuclear Fréchet space X, let $\mathbf{Analy}_X$ the category of analytical expressions in X with morphisms as above, i.e., $\mathbf{Analy}_X = 1/A(X)$. The Feynman map $FR : \mathrm{Feyn} \to A(X)$ is, therefore, a map between classes of objects of two categories, which by its defining expression (7.2) extends to a functor $FR : \mathbf{Feyn} \to \mathbf{Analy}_X$ that we could call a *Feynman functor*.

The category **Graph** of graphs has a symmetric monoidal structure $(\sqcup,)$ given by disjoint union and the empty graph, which induces a corresponding symmetric monoidal structure on Feynman graphs. On the other hand, the category **NFre** of nuclear Fréchet spaces also becomes symmetric monoidal $(\otimes, \mathbb{R})$. Again, the defining properties (7.2) reveal that the induced functor $\mathbf{Feyn} \to \mathbf{Analy}_X$ is strong symmetric monoidal.

Remark 7.16. That Feynman rules defines monoidal functors between Feynman graphs and topological vector spaces were also discussed from a perspective of groupoidification by Baez and Dolan in [13] and then in [14, **?**].

Remark 7.17.

The above reveals that the axiomatic perturbative quantization discussed in the last section was obtained assigning to each Langrangian field theory $\mathcal{L}$ with field bundle E:

1. a subcategory $\mathbf{Feyn}(\mathcal{L}) \subset \mathbf{Feyn}$ of Feynman graphs[3];
2. a symmetric monoidal functor $FR : \mathbf{Feyn}(\mathcal{L}) \to \mathbf{Analy}_{\Gamma(E)}$.

Thus, the natural idea is to extend that axiomatic approach is to look for extensions of the functor $FR : \mathbf{Feyn}(\mathcal{L}) \to \mathbf{Analy}_{\Gamma(E)}$ from Feynman graphs to other classes of structured graphs and then search for an ampler class of classical theories to which corresponds these more general graphs in the same ways as Feynman graphs are assigned to Lagrangian field theories. It is in the first of these steps (extending Feynman functors) that our work [116] finds itself.

7.1.5 Abstract Feynman Functors

As Feynman functors were built from Feynman rules, the idea is:

1. to extend Feynman functors abstractly;
2. to extend Feynman rules;
3. to prove the existence of extended Feynman functors showing that extended Feynman rules induce extended Feynman functors.

Notice that the notion of "analytical expression" in a Frćhet space X depends only on the existence of a notion of topological tensor product $X \otimes Y$ on **NFre** an on a notion of topological dual $X^{\vee}$. Thus, we could extend not only the class of graphs in which Feynman functors are defined but also the ambient in which they take values.

Definition 7.18. A *functor of duals* in a monoidal category $(\mathbf{C}, \otimes, 1)$ is a functor $(-)^{\vee} : \mathbf{C}^{op} \to \mathbf{C}$.

1. A *pseudocontext for doing functional analysis* (or just *pseudocontext*) is a monoidal category with a functor of duals.
2. A *context for doing functional analysis* (or just *context*) is a pseudocontext endowed with a compatibility condition between $\otimes$ and $(-)^{\vee}$ given by a natural transformation $\xi : (-)^{\vee} \otimes (-)^{\vee} \Rightarrow (-) \otimes (-)$. Thus, a context is a pseudocontext whose functor of duals is lax monoidal;
3. A *strong context for doing functional analysis* (or simply *strong context*) is a context whose compatibility transformation ξ is a natural isomorphism, i.e., whose functor of duals is strong monoidal.

Remark 7.19. A pseudocontext/context/strong context is required to have countable limits which are presented by the tensor product and the functor of duals preserve.

Definition 7.20. Let $\mathbf{C}$ be a pseudocontext. Given $k \geq 0$ and $X \in \mathbf{C}$ writes $X^{\otimes_k} = ((X \otimes X) \otimes ...)$. Then:

[3]Notice that from the definition of the class $G(\mathcal{L})$ of Feynman graphs of $\mathcal{L}$, it is clealry closed by disjoint unions and contains φ, so that it is actually symmetric monoidal subcategory

1. the *object of tensors* of X is $TX = \prod_k X^{\otimes_k}$;
2. we say that $P^k X$ is an *object of k-progators* in X if there is functor $K : \mathbf{C} \to \mathbf{Set}$ and a group action $S_k \times K(X^{\otimes_k}) \to K(X^{\otimes_k})$ such that $K(P^k X) \simeq K(X^{\otimes_k})/S_k$. Writes $PX = \prod_k P^k X$ and $\mathcal{O}(X) = P(X^\vee)$;
3. the *object of analytical expressions* in X is $AX = PX \otimes P(X^\vee)$.

We define:

1. a *k-tensor* in X as a morphism $t : 1 \to T^k X$;
2. a*k-propagator* in X as a morphism $p : 1 \to P^k X$;
3. an *analytical expression* in X as a morphism $\psi : 1 \to AX$.

For every k, k-tensors, k-propagators and analytical expression defines slice categories $\mathbf{T}^k X$, $\mathbf{P}^k X$ and $\mathbf{Analy}_X$.

Remark 7.21. For every X the object TX is a monoid. In the following, we will suppose that the functor K is such that the monoid structure in TX descends to a monoid structure in PX. Consequently, $AX = PX \otimes P(X^\vee)$ is a monoid too. On the other hand, recall that if Y is a monoid, then the slice category $1/Y$ has an induced monoidal structure. It then follows that $\mathbf{T}X$, $\mathbf{P}X$, and $\mathbf{Analy}_X$ are monoidal categories.

Feynman graphs are graphs with extra stuff. Thus, we could think of defining Feynman functor for any class of structured graphs. If we think of higher-dimensional edges as a structure, this leads us to think about extending Feynman functors to structured hypergraphs.

Definition 7.22. Let $\mathbf{C}$ be a pseudocontext and let $\mathbf{Hyp}$ be a category of hypergraphs which is closed under disjoint unions. A *Feynman functor* in $\mathbf{Hyp}$ with coefficients in a monoid $X \in \mathbf{C}$ is a monoidal functor $F : \mathbf{Hyp} \to \mathbf{Analy}_X$.

Before discussing how to extend Feynman rules, notice that we are working in a more ample class of coefficients that goes beyond functional analysis and include categories of representations of algebraic objects:

Example 7.23.
1. The symmetric monoidal category $\mathbf{NFre}$ of nuclear Fréchet spaces is a strong context with the functor of duals given by the topological dual.
2. Similarly, the symmetric monoidal category $(\mathbf{FVec}_{\mathbb{K}}, \otimes_{\mathbb{K}}, \mathbb{K})$ of finite-dimensional $\mathbb{K}$-vector spaces is a strong context with the functor of duals given by linear duals.
3. The category $\mathbf{LCS}$ of locally convex spaces can be endowed with different symmetric monoidal structures corresponding to the different topological tensor products that it has. Each of these categories become a pseudocontext for the different topological duals that it also has;
4. If $(\mathbf{C}, \otimes, 1)$ is a monoidal category with left/right duals (i.e., a left/right autonomous category), then assigning an object to its left/right dual is an anafunctor. Assuming Axiom of Choice this is a functor and, therefore, it makes $(\mathbf{C}, \otimes, 1)$ a pseudocontext
5. If $(\mathbf{C}, \otimes, 1)$ is autonomous, then taking duals is a strong monoidal functor and therefore we actually have a strong context. In particular, symmetric autonomous categories (i.e., compact closed categories) define strong contexts.
6. If $(\mathbf{C}, \otimes, 1)$ is a pivotal category, then there is a natural isomorphism $(X*)* \simeq X$. Thus $(-*) * \otimes (-*)* \simeq (-) \otimes (-)$, so that if we take $(-*)*$ as a functor of duals we

get a strong context. Since fusion categories admit a pivotal structure [??], it follows that they define strong contexts;

7. Via Tannaka duality:

   (a) the category of representations of any Hopf algebra is a rigid monoidal category and therefore a strong context;

   (b) the category of representations of any weak Hopf algebra is a fusion category and therefore a strong context.

8. The category $\mathrm{Frob}(\mathbf{C},\otimes)$ of Frobenius monoids in a monoidal $(\mathbf{C},\otimes,1)$ is a monoidal subcategory of the monoidal category of object monoids $\mathrm{Mon}(\mathbf{C},\otimes)$. Furthermore, every Frobenius monoid X is self-dualizable, so that the category $\mathrm{Frob}(\mathbf{C},\otimes)$ is a rigid monoidal category and, therefore, defines a strong context.

To extend the Feynman rules, notice that from the previous definition of Feynman rules, we see that they can be generically defined for every class "Feyn" of Feynman graphs and taking values in any nuclear Fréchet space X:

Definition 7.24. Let X be a nuclear Fréchet space and let Feyn be the class of Feynman graphs. A *Feynman rule in* Feyn *with coefficients in* X is given by assigning to each each $G \in$ Feyn functions:

1. $FR(G)_E : EG \to P^2(X)$;
2. $FR(G)^1_V : V^1G \to \mathcal{O}(X)[[h]]$, such that $FR(G)^1_V(v) \in X_{d(v),g(v)}$;
3. $FR(G)^0_V : V^0G \to X$.

The induced *Feynman map* of a Feynman rule is the map $FR : \mathrm{Feyn} \to AX$ given by

$$FR(G) = \bigotimes_{e\in EG} \bigotimes_{v\in V^1G} \bigotimes_{w\in V^0G} [FR(G)_E](e) \otimes [FR(G)^1_V](v) \otimes [FR(G)^0_V](w). \tag{7.2}$$

As for the case of Feynman rules of Lagrangian field theories, the Feynman map $FR : \mathrm{Feyn} \to AX$ extends to a functor $F : \mathbf{Feyn} \to \mathbf{Analy}_X$, which is obtained from the defining expression (7.2) and the graph identity in Remark ??.

Notice that:

1. Expression (7.2) involves tensor products on the different parts in which VG decomposes. In the case of arbitrary hypergraphs, we do not need to have such an intrinsic decomposition. Thus, we have to fix one and the Feynman rule will depend on that.

2. In the definition of the map $FR^1_V(G)$ it is required that it take values into $X_{d(v),g(v)}$. The function $g : VG \to \mathbb{Z}_{\geq 0}$ is part of the data defining a Feynman graph, so that it also does not need to exist in an arbitrary hypergraph. Furthermore, the function g becomes determined by its restrictions $g^i : V^iG \to \mathbb{Z}_{\geq 0}$. Thus, in the case of hypergraph the additional functions will depend on the decomposition.

With this in mind we can extend Feynman rules:

Definition 7.25. Let **Hyp** be a category of hypergraphs and let **C** be a pseudocontext. A *Feynman rule* in **Hyp** with coefficients in $X \in \mathbf{C}$ is a rule assigning to each $G \in \mathbf{Hyp}$ the following data:

1. a decomposition $VG = V^0G \sqcup ... V^mG$ of its vertex set and, for each $j \geq 1$, a decomposition $E_jG = E^0_jG \sqcup ... E^m_jG$ of its j-edges set into finite parts of same size;

2. for $i = 0, ..., m$ and $j \geq 1$ functions $g^i : V^i G \to \mathbb{Z}_{\geq 0}$ and $g^i_j : E^i_j G \to \mathbb{Z}_{\geq 0}$;
3. functions $FR(G)^i : V^i G \to \mathbf{P}X^{\vee}$, with $i = 0, ..., m$, such that $FR(G)^i_V(v) \in \mathbf{P}^{r^i(v)} X^{\vee}$;
4. functions $FR(G)^i_j : E^i_j G \to \mathbf{P}X$, for $i = 0, ...m$ and $j \geq 1$, such that $FR(G)^i_j(e) \in \mathbf{P}^{r^i_j(e)} X$.

The idea is then to mimic the previous construction of Feynman functors from Feynman maps to get extended Feynman functors from extended Feynman rules. This is precisely the deal of [116].

Remark 7.26. Another approach to axiomatize the Feynman rule and then abstract then was developed by Aluffi and Marcolli in [?] (see also [111, 110]) based on works of Connes and Kreimer [41, 46]. There is also an approach by Ionescu [83, 84, 85] based on works of Kontsevich [96].

7.1.6 Main Results

In the construction of the Feynman map, the fact that it takes values into analytic expressions follows from the graph identity $\sum_v d(v) = 2|EG|$ and from $|VG| = \sum_i vertV^i G|$, with $i = 0, 1$ (see Remark 7.15). To do the same with an extended Feynman rule we will need that the functions g^i and g^i_j also satisfies an identity, that we call *coherency identity* (see equation (10), p. 355 of [116]):

$$\sum_{i=0}^{m} \sum_{j>2} \sum_{e \in E^i_j G} g^i_j(e) = \sum_{i=0}^{m} \sum_{v \in V^i G} g^i(v).$$

Remark 7.27. The left-hand side makes sense as a sum only if j is bounded from above and if each $E^i_j G$ is finite, while the right-hand side makes sense only if each $V^i G$ is finite. This forces us to consider categories $\mathbf{Hyp}_b$ of finite hypergraphs which are bounded, i.e., for which $E_j G = \varnothing$ for $j > b$.

On the other hand, notice that the expression (7.2) defining the Feynman map of a Feynman rule involves a lot of tensor products. The ordering that they are considered does not matter since the monoidal category **NFre** of nuclear Fréchet spaces is symmetric. If **C** is a pseudocontext that is not symmetric we would have a problem dealing with these bunch of tensor products. An alternative is to assume that **C** is strict as monoidal category.

Alternatively, notice that the tensor products in (7.2) are indexed by vertex and edges of the Feynman graphs. In the extended case it would then be indexed by vertex and hyper-edges of the hypergraphs. If the hypergraphs in question are labeled, meaning that their vertex and hyper-edges are labeled, we could use the labeling of the vertex and hyper-edges to order the tensor products that they index.

After building the Feynman map from expression (7.2) it was immediately extended to a monoidal functor. In the general case, this need not be the case. Indeed, a morphism $f : G \to H$ induces a morphism between the corresponding expressions (7.2) only if it factors through the maps defining the Feynman rule. If the Feynman rule is such that this is satisfied for all morphisms we say that it is *functorial*.

Even for functorial Feynman rules, for which we have an analogous expression of (7.2) and it extends to a functor $FR : \mathbf{Hyp}_b \to \mathbf{Analy}_X$, it need not be lax/strong monoidal. Again, this can be obtained by adding some hypotheses on the functions defining the Feynman rule. This condition basically says that the functions $FR^i(G \sqcup G')$ and $FR^i_j(G \sqcup G')$

are related with $FR^i(G) \otimes FR^i(G')$ and $FR^i_j(G) \otimes F^i_j(G')$, respectively. A Feynman rule satisfying these conditions is called *monoidal*.

Theorem 7.28. *Let $\mathbf{Hyp}_b$ be a category of bounded hypergraphs and let $\mathbf{C}$ be a pseudocontext. If the hypergraphs in $\mathbf{Hyp}_b$ are labeled or if $\mathbf{C}$ is strict as a monoidal category, then every functorial Feynman rule in $\mathbf{Hyp}_b$ with coefficients in $X \in \mathbf{C}$ that satisfies the coherency identity induces a functor $FR : \mathbf{Hyp}_b \to \mathbf{Analy}_X$. If the Feynman rule is monoidal, then the induced functor is a Feynman functor.*

Proof. This is a particular case of Theorem 5.1 and Proposition 5.3 of [116]. □

Remark 7.29. For the Feynman functor induced by a classical Feynman rule of a Lagrangian field theory, the Feynman graphs determine its analytical expressions, but the analytical expressions are generally not enough to recover the Feynman graph. In other words, the obtained functor $FR : \mathbf{Feyn} \to \mathbf{Analy}_{\Gamma(E)}$ is generally not essentially injective.

One could ask whether the extended Feynman rules produce an extended Feynman functor which is essentially injective. These conditions were obtained in Proposition 5.2 of [116]. A Feynman functor which is essentially injective is called *complete*. A *complete* Feynman rule is one that induces a complete Feynman functor. Thus, around Proposition 5.2 of [116] these complete Feynman rules are described.

Having established existence, the next step is to consider the uniqueness of Feynman rules. Of course, since there is freedom in the choice of the functions FR^i and F^i_j defining a Feynman rule, one should expect to have a lot of Feynman functors. Also, a priori one may have Feynman functors which are not induced by Feynman rules. However, in [116] we proved that every two complete Feynman functors are "conjugated" in some nice sense. In particular, every complete Feynman functor is "conjugated" to some which is induced by a complete Feynman rule.

Definition 7.30. Two functors $F : \mathbf{C} \to \mathbf{D}$ and $F' : \mathbf{C}' \to \mathbf{D}'$ are *conjugated* if there are functors α, β and a natural isomorphism as below. We say that they are *quasi essentially injectively conjugated* (qeic) if there are subcategories $\mathbf{C}_0 \subset \mathbf{C}$ and $\mathbf{D}_0 \subset \mathbf{D}$ such that:

1. α and β are constant in $\mathbf{C}_0$ and $\mathbf{D}_0$, respectively;
2. α and β are essentially injective outside $\mathbf{C}_0$ and $\mathbf{D}_0$, respectively.

$$\begin{tikzcd} \mathbf{C} \arrow[r, "F"] \arrow[d, "\alpha"'] \arrow[dr, phantom, "\overset{\simeq}{\Longrightarrow}"] & \mathbf{D} \arrow[d, "\beta"] \\ \mathbf{C}' \arrow[r, "F'"'] & \mathbf{D}' \end{tikzcd}$$

Theorem 7.31. *Let $\mathbf{Hyp}_b$ be a class of bounded hypergraphs and let $\mathbf{C}$ be a pseudocontext. Assuming the Axiom of Choice, any two complete Feynman functors $FR, FR' : \mathbf{Hyp}_b \to \mathbf{Analy}_X$ are conjugated.*

Proof. This follows from Lemma 5.1 of [116]. □

A note on reconstruction conjectures

From Theorem 7.28 we know that Feynman rules defined on hypergraphs and taking values on any object of a strict pseudocontext induce functors $FR : \mathbf{Hyp}_b \to \mathbf{Analy}_X$. Furthermore, under certain conditions, these functors are monoidal and essentially injective.

The monoidalness property says that every finite decomposition $G \simeq G_1 \sqcup ... \sqcup G_m$ of an hypergraph $G \in \mathbf{Hyp}_b$ into finite parts G_i, each of them belonging to $\mathbf{Hyp}_b$, induces a decomposition of the analytical expression $FR(G) \simeq (...(FR(G_1) \otimes FR(G_2)) \otimes ...) \otimes FR(G_m)$ in terms of the analytical expression of the parts.

Suppose that the category of hypergraphs $\mathbf{Hyp}_b$ is closed under vertex deletion, i.e., suppose that if $G \in \mathbf{Hyp}_b$, then $G - v \in \mathbf{Hyp}_b$ for every $v \in VG$, where $G - v$ is obtained deleting the vertex v and all adjacent hyper-edges. Since we are working with finite hypergraphs, it follows that $\bar{G} = \sqcup_{v \in VG} G - v$ belongs to $\mathbf{Hyp}_b$ for every G. If $G \simeq H$, then $\bar{G} \simeq \bar{H}$. The question whether the reciprocal is true, i.e., whether $\bar{G} \simeq \bar{H}$ is known in Graph Theory as *reconstruction conjecture* [73, 22, 21].

Notice that if $FR : \mathbf{Hyp}_b \to \mathbf{Analy}_X$ is a complete Feynman functor, then $F(\bar{G}) \simeq F(\bar{H})$ implies $\bar{G} \simeq \bar{H}$. Thus:

- *the reconstruction conjecture in $\mathbf{Hyp}_b$ is true only if for every complete Feynman functor $FR : \mathbf{Hyp}_b \to \mathbf{Analy}_X$ and every $G, H \in \mathbf{Hyp}_b$, the condition $FR(\bar{G}) \simeq FR(\bar{H})$, i.e.,*

$$\bigotimes_{v \in VG} FR(G - v) \simeq \bigotimes_{v \in VH} FR(H - v), \tag{7.3}$$

  implies $G \simeq H$.

This then leads to the following:

Theorem 7.32. *Let $\mathbf{Hyp}_b$ be a category of bounded finite hypergraphs which is closed under vertex deletion. If there is a pseudocontext $\mathbf{C}$ and a complete Feynman functor $FR : \mathbf{Hyp}_b \to \mathbf{Analy}_X$, where $X \in \mathbf{C}$, and two hypergraphs G, H such that there is no isomorphism (7.3), then the reconstruction conjecture is false in $\mathbf{Hyp}_b$.*

On the other hand, recall that the initial aim to extend Feynman rules was to extend the axiomatization of perturbative quantization $PQ : \mathcal{FE}_{cls} \to \mathcal{FE}_{prt}$ from the class $\mathcal{FE}_{cls,0}$ of classical systems which are modeled by Lagrangian field theories to some other ampler class $\mathcal{FE}_{cls,1}$.

Suppose that we found this class $\mathcal{FE}_{cls,1}$ such that to each system $\mathcal{S} \in \mathcal{FE}_{cls,1}$ corresponds:

1. a category $\mathbf{Hyp}_b(\mathcal{S})$ of bounded finite hypergraphs in the same way as a category $G(\mathcal{L})$ of Feyman graphs corresponds to a Lagrangian field theory $\mathcal{L} \in \mathcal{FE}_{cls,0}$;
2. a monoidal functorial Feynman rule $\mathbf{Hyp}_b(\mathcal{S})$ with coefficients in some object X of some strict pseudocontext $\mathbf{C}$.

Thus, from Theorem 7.28 to each $\mathcal{S}$ there is also a Feynman functor $FR(\mathcal{S}) : \mathbf{Hyp}_b(\mathcal{S}) \to \mathbf{Analy}_X$ which defines an axiomatic extension of perturbative quantization $PQ : \mathcal{FE}_{cls,1} \to \mathbf{F}_{prt}$. This means that the analytical expressions $[FR(\mathcal{S})](G)$ are related with fundamental experiments of $\mathbf{F}_{prt}$.

In particular (assuming that $\mathcal{L}$ is such that $\mathbf{Hyp}_b(\mathcal{S})$ is closed under vertex deleting), given G, H, the condition (7.3) could be interpreted as a relation between fundamental experiments, meaning that the fundamental experiment of G is determined by the experiments of the parts $G - v$.

The conclusion is the following:

- *if the $\mathcal{FE}_{cls,1}$ with the corresponding axiomatic extension exists, then for each $\mathcal{S} \in$ such that the corresponding Feynman rule is complete, the validity of the reconstruction conjecture on $\mathbf{Hyp}_b(\mathcal{S})$ could be studied analyzing the fundamental experiments of $PQ(\mathcal{S})$.*

Remark 7.33. A more precise account of this discussion, which involves extending Feynman rules and Feynman functors from categories of hypergraphs to prestacks of hypergraphs, as well as a generalization on that to other types of reconstruction conjectures, is in Section 3 and Section 6 of [116].

7.2 QUESTIONS AND SPECULATIONS

(To be added)

7.3 THE WORK

The following is a recompilation of the ArXiv preprint

- Y.X. Martins, R.J. Biezuner, *Graph Reconstruction, Functorial Feynman Rules and Superposition Principles*, Mar 2019, arXiv:1903.06284

Graph Reconstruction, Functorial Feynman Rules and Superposition Principles

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Abstract

In this article functorial Feynman rules are introduced as large generalizations of physicists Feynman rules, in the sense that they can be applied to arbitrary classes of hypergraphs, possibly endowed with any kind of structure on their vertices and hyperedges. We show that the reconstruction conjecture for classes of (possibly structured) hypergraphs admit a sheaf-theoretic characterization, allowing us to consider analogous conjectures. We propose an axiomatization for the notion of superposition principle and prove that the functorial Feynman rules work as a bridge between reconstruction conjectures and superposition principles, meaning that a conjecture for a class of hypergraphs is satisfied only if each functorial Feynman rule defined on it induces a superposition principle. Applications in perturbative euclidean quantum field theory and graph theory are given.

1 Introduction

Graphs (and their generalizations such as hypergraphs) appear in the most different areas of mathematics and physics, generally parametrizing definitions and constructions. For instance: in algebraic geometry the Deligne-Mumford moduli stack of stable curves is stratified by certain graphs [1, 2]; in symplectic topology the stable maps, which play an important role in the study of J-holomorphic curves and Gromov-Witten theory, are defined by making use of graphs [3, 4]; Kontsevich's formula for deformation quantization of Poisson manifolds is parametrized by graphs [5, 6]; the Ribbon graphs (fat graphs) are used to compute the weak homotopy type of the geometric realization of mapping class group of surfaces with marked points [7, 8, 9]; in perturbative quantum field theory the Feynman graphs parametrize the possible worldlines of relativistic quantum particles [10, 11]; the moduli space of marked surfaces (and, therefore, stable graphs) is also used to parametrize the worldsheet of closed strings [12, 13]; the recent amplituhedrons, which parametrize the scattering amplitudes, are certain hypergraphs [14].

In the abstract study of graph theory there are important conjectures, known as *reconstruction conjectures*, stating that in order to describe a graph (belonging to a certain fixed class) it is necessary and sufficient to describe subgraphs obtained by some deleting process. It is known that these conjectures are true for some classes (such as trees, regular graphs, maximal planar

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graphs) and false for others (such as digraphs and infinite graphs), but the general classification remains broadly open (see [15, 16] for a review and an exposition).

On the other hand, in a completely different perspective, there are the so-called *superposition principles* which state that physical properties of a physical system are totally determined by the corresponding properties of certain subsystems. They typically occur when the physical property in question is described by a linear partial differential equation. For example, Maxwell's and Schrödinger's equations are linear, so that we have wave superposition and wave function superposition (quantum superposition). Thinking in this way, it is natural to regard a superposition principle as some kind of reconstruction phenomenom.

When considering systems of perturbative quantum field theory we have both graphs (or even hypergraphs) and physical properties: Feynman graphs and scattering amplitudes. They are related via certain rules, known as Feynman rules [10, 11]. It then makes sense to consider the reconstruction conjecture for Feynman graphs and to ask about superposition principles for scattering amplitudes. Furthermore, it is natural to ask if Feynman rules play some role between these two types of reconstruction processes. In this article, our objective is to give a positive answer, but in a much more general setup.

More precisely, we show that the physicists Feynman rules can be axiomatized under a very general frame, being regarded as functors defined in some category of structured hypergraphs and taking values into the category of analytic expressions (which will define the scattering amplitudes) of some monoidal category (which plays the role of a context where functional analysis can be done). We call these functors *Feynman functors*. We prove that the classic Feynman rules can be extended for structured hypergraphs and for any suitable context for functional analysis, establishing the general existence of Feynman functors. We also prove that any other Feynman functor is conjugated (in a very nice way, which we call quasi essentially injective conjugation) to that obtained extending the classic Feynman rules, establishing uniqueness.

There are two fundamental steps in showing that Feynman functors behave as a bridge between reconstruction conjectures and superposition principles:

s1) *showing that the reconstruction conjectures admit a sheaf-theoretic characterization.* For each finite set V, let $\mathbf{S}_{\mathfrak{s},V}$ denote the category of hypergraphs which have structure of type $\mathfrak{s}$ and vertex set V. Varying V we get a prestack $S_{\mathfrak{s}}$. By making use of a deleting process, say $\mathcal{D}$, we get a new prestack $\mathcal{D}S_{\mathfrak{s}}$ and a morphism $\mathcal{D}_{\mathfrak{s}} : S_{\mathfrak{s}} \Rightarrow \mathcal{D}S_{\mathfrak{s}}$, i.e, a family of functors $\mathcal{D}_{\mathfrak{s},V} : \mathbf{S}_{\mathfrak{s},V} \to \mathcal{D}\mathbf{S}_{\mathfrak{s},V}$. We show that different reconstruction conjectures consist in different choices of $\mathcal{D}$ and that the corresponding $\mathcal{D}_{\mathfrak{s}}$ is objectwise essentially injective;

s2) *proving that we can always consider Feynman functors which are not only arbitrary functors, but actually monoidal and essentially injective.* Monoidal property means that the analytic expression of two disjoint structured hypergraphs is the product of the corresponding analytic expressions. In turn, the essentially injectivity property means that the hypergraphs are totally described by their analytic expression.

Let $\mathbf{A}$ be a monoidal category endowed with a structure of context for doing functional analysis, and let $Z_V : \mathbf{S}_{\mathfrak{s},V} \to \mathbf{A}_{\tau V}$ be a Feynman functor assigning to each hypergraph with $\mathfrak{s}$-structure an analytic expression in $\mathbf{A}$. Let $\mathcal{D}$ be a deleting process and suppose that we have a canonical morphism from $\mathcal{D}_{\mathfrak{s},V}G$ to the disjoint union of the parts of G obtained via deleting (which generally happens). Since Z is monoidal, for each G we have a corresponding morphism

$$Z_V(G) \to \bigotimes_{\text{pieces}} Z(\text{pieces of } G).$$

Suppose now that the reconstruction conjecture induced by $\mathcal{D}$ is satisfied. Then, because Z_V is essentially injective, it follows that two hypergraphs G and G' are isomorphic iff their analytic expressions have the same decomposition in terms of the analytic expressions of the pieces. This

2

conclusion is precisely one example of a superposition principle (in the sense axiomatized here). Let us call it the *$\mathcal{D}$-superposition principle.* Thus, from s1) and s2) there follows our main result:

Theorem 1.1. *The $\mathcal{D}$-reconstruction conjecture for a prestack $S_{\mathfrak{s}}$ of $\mathfrak{s}$-structured hypergraphs is true only if for any Feynman functor the $\mathcal{D}$-superposition principle holds.*

This theorem can be regarded both as an obstruction to the validity of reconstruction conjectures and as a source of new superposition principles. This relation becomes more involved when we think of the role of quantum field theory. Indeed, consider $\mathbf{S}_{\mathfrak{s}}$ as the category of hypergraphs which parametrize the worldvolume of particles, strings or branes (for instance, of Feynman graphs of QED or some other gauge theory). Suppose we find $\mathcal{D}$ such that the $\mathcal{D}$-reconstruction conjecture is true. Then each Feynman functor (in particular that obtained by the Feynman rules of QED, etc.) will produce a new superposition principle for the scattering amplitudes.

On the other hand, since $\mathbf{S}_{\mathfrak{s}}$ are the Feynman graphs of a physical theory, we can analyze whether these superposition principles for the scattering amplitutes exist or not, looking at concrete experiments of LHC, trying to find a counterexample. If found, it will be a strong indicative that the $\mathcal{D}$-reconstruction conjecture is false. Another approach is to notice that the existence of new superposition principles in quantum theories produce many logic implications [17, 18], so that assuming the validity of the $\mathcal{D}$-conjecture we could verify if the induced logic implications contradict those that are experimentally realized.

This paper is organized as follows. In Section 2 we define what is meant by a $\mathfrak{s}$-structured hypergraph and give many examples of objects that can be regarded as such. We also show that the category of all categories that can be embedded into $\mathbf{S}_{\mathfrak{s}}$ for some $\mathfrak{s}$ is complete. In Section 3 the sheaf-theoretic characterization of the classical reconstruction conjecture is given and analogous conjectures are defined, as needed for step s1). We also prove that if one work with hypergraphs which contain labelings as part of their structures, then many reconstruction conjectures are true. In Section 4 the notions of context for functional analysis and analytic expressions are axiomatized and many examples are given. In Section 5 Feynman functors and functorial Feynman rules are defined and the existence and uniqueness up to quasi essentially injetive conjugation is established, as required for step s2). We also show how to recover the classic Feynman rules from this general approach. In Section 6 the notion of superposition principle is formalized and a formal proof of Theorem 1.1 is given. Finally, in Section 7 some applications of our results on hypergraph theory, manifold topology and perturbative quantum field theory are presented.

Remark 1.1. Along this article, by an oplax monoidal functor we mean one where only the morphisms of the products are reverted. So, $F : \mathbf{C} \to \mathbf{C}'$ is oplax if it becomes endowed with natural transformations $F(X \otimes Y) \to F(X) \otimes' F(Y)$ and a morphism $1' \to F(1)$ making the appropriated diagrams commutative.

2 Structured Hypergraphs

There are several ways to define a hypergraph. For us, a *hypergraph* G consists of a finite (possibly empty) set V of vertices and for each $j > 1$ a finite (possibly empty) set E_j of j-edges and a j-adjacency function $\psi_j : E_j \to \mathrm{bin}(V, j)$ such that $\psi_1 = id_V$, where $\mathrm{bin}(V, j)$ denotes the set of j-subsets of V. We usually write $E_1 = V$, so that a 1-edge is just a vertice. If $v \in \psi_j(e)$ we say that e is adjacent to e. For each $v \in V$ and each $j > 1$, let $d_j(v)$ denote the number of j-edges that are adjacent to v. Using these notations, we have [19]:

$$\sum_{j>1} \sum_{v} d_j(v) = \sum_{j>1} j|E_j|. \tag{1}$$

Let $\mathbb{N}$ be the set of natural number regarded as a discrete category and notice that for each set X the rule $j \mapsto \mathrm{bin}(X, j)$ extends to a functor $\mathrm{bin}(X, -) : \mathbb{N} \to \mathbf{Set}$. Thus, a hypergraph is equivalently a pair (E, ψ), where $E : \mathbb{N} \to \mathbf{FinSet}$ and $\psi : E \Rightarrow \mathrm{bin}(E(1), -)$ is a natural transformation such that $\psi_1 = id$. Here, $\mathbf{FinSet}$ denotes the category of finite sets. The equivalence between both definitions is obtained via the identifications $E(j) = E_j$.

The rule assigning to each set X its set $\mathrm{bin}(X, j)$ of j-subsets also extends to a functor $\mathrm{bin}(-, j) : \mathbf{Set} \to \mathbf{Set}$. We can then define a *morphism* $f : G \to G'$ of between hypergraphs (E, ψ) and (E', ψ') as a natural transformation $f : E \Rightarrow E'$ which commutes with adjacencies, i.e, such that $\psi' \circ f = \mathrm{bin}(f_1, -) \circ \psi$. We have the category $\mathbf{Hyp}$ of hypergraphs. Under the operation of taking disjoint unions of hypergraphs it becomes a symmetric monoidal category whose neutral object is the empty hypergraph.

Sometimes we will work with *bounded hypergraphs*. We say that G in *bounded* if there is some b such that $E_j = \varnothing$ for each $j \geq b$. The smallest of these b's is the *bounding degree* of G. For each $b > 0$ we have monoidal subcategories $\mathbf{Hyp}^b \subset \mathbf{Hyp}$ of *b-bounded hypergraphs* with fixed bounding degree b. For instance, if $b = 2$ this is the category of what is known as finite pseudographs or finite graphs, depending on the author and we will write $\mathbf{Grph}$ instead of $\mathbf{Hyp}^2$.

Remark 2.1. Let $\mathbf{C} \subset \mathbf{Hyp}$ be some category of hypergraphs. For fixed V we can consider the full subcategory $\mathbf{C}_V \subset \mathbf{C}$ of hypergraphs in $\mathbf{C}$ whose vertex set is V or empty. Even if $\mathbf{C}$ is a monoidal subcategory, if $V \neq \varnothing$ then $\mathbf{C}_V$ is not monoidal. Indeed, if G, G' have the same vertex set V, then $G \sqcup G'$ has vertex set $V \sqcup V$. This is one of the motivations for considering reconstruction conjectures in a sheaf-theoretic perspective, as will be discussed in the next section.

In the following we will work with *structured hypergraphs*, in that for any $j \geq 1$ we have functions $\varepsilon_j : E_j \to s_j$. We think of s_j as a set of j-structures in the set of j-edges. Thus, ε_j assigns to each j-edge a corresponding j-structure. We form the category $\mathbf{S} \subset \mathbf{Hyp}$ whose morphisms are hypergraph morphisms $f : G \to G'$ preserving j-structures. In more precisely terms, for each functor $s : \mathbb{N} \to \mathbf{Set}$, called a *functor of structures*, we define a category $\mathbf{S}_s$ as follows. Objects are *s-structured hygraphs*, i.e, pairs (G, ε), where $G = (E, \psi)$ is a hypergraph and $\varepsilon : E \Rightarrow s$ is a natural transformation. Morphisms $f : (G, \varepsilon) \to (G', \varepsilon')$ are hypergraph morphisms between the underlying hypergraphs such that $\varepsilon' \circ \mathrm{bin}(f, -) = \varepsilon$. If we are working with bounded hypergraphs we can consider $\mathbf{S}^b \subset \mathbf{Hyp}^b$, defined analogously.

Given s-structured hypergraphs (G, ε) and (G', ε'), the disjoint union $G \sqcup G'$ can be naturally regarded as a s-structured hypergraph with the transformation $(\varepsilon \sqcup \varepsilon')_j : E_j \sqcup E'_j \to s_j$ given by the composition below, where the second map is the codiagonal. Also, the empty hypergraph has a unique s-structure, so that $\mathbf{S} \subset \mathbf{Hyp}$ is actually a monoidal subcategory.

$$\begin{tikzcd} E_j \sqcup E'_j \arrow[r, "\varepsilon_j \sqcup \varepsilon'_j"] \arrow[rr, bend right, "(\varepsilon \sqcup \varepsilon')_j"'] & s_j \sqcup s_j \arrow[r] & s_j \end{tikzcd}$$

Remark 2.2. When $s_j : \mathbb{N} \to \mathbf{Ab}$, i.e, when the structural functor takes values in the category of abelian groups, the map $(\varepsilon \sqcup \varepsilon')_j$ can be identified with $\varepsilon_j + \varepsilon'_j$. In these situations we say that s is an *additive structure*. In this paper, essentially all structures will take values in $\mathbf{FinSet}$, so that we can always think of them as additive structures by replacing a finite set $[n]$ with the abelian group $\mathbb{Z}_n$ that it generates.

The last construction can be easily generalized by considering not only one functor of structure s, but a family of them. In fact, let $\mathfrak{s} : \mathbb{N} \to [\mathbb{N}; \mathbf{Set}]$ be a functor that to each k assigns a structure functor $\mathfrak{s}_k : \mathbb{N} \to \mathbf{Set}$, so that $\mathfrak{s}_{k,j} \in \mathbf{Set}$ (equivalently, $\mathfrak{s}$ can be regarded as a bifunctor $\mathbb{N} \times \mathbb{N} \to \mathbf{Set}$)[1]. Let $\mathrm{cst}_E : \mathbb{N} \to [\mathbb{N}; \mathbf{Set}]$ be the constant functor in E. We define a *$\mathfrak{s}$-structured hypergraph* as previously: it is a pair (G, ϵ), but now ϵ is a natural transformation $\epsilon : \mathrm{cst}_E \Rightarrow \mathfrak{s}$.

[1] Here $[\mathbf{C}; \mathbf{D}]$ denotes the functor category.

Thus, it is a rule that to each k assigns a natural transformation $\epsilon_k : E \Rightarrow s_k$, which means that for fixed $j \geq 1$ we have a family $\epsilon_{k,j} : E_j \to \mathfrak{s}_{k,j}$. Morphisms of $\mathfrak{s}$-structured hypergraphs also are defined analogously, so that we have a monoidal category $\mathbf{S}_{\mathfrak{s}}$. If we are working with bounded hypergraphs we have the corresponding monoidal category $\mathbf{S}^b_{\mathfrak{s}}$.

2.1 Embedded Subcategories

In this subsection we will discuss some category of hypergraphs that can be embedded into the category of structured hypergraphs in an essentially injective way. More precisely, we will give examples of subcategories $\mathbf{C} \subset \mathbf{Hyp}$ such that for each fixed V the corresponding $\mathbf{C}_V$ can be realized, up to equivalence, as a full subcategory of $\mathbf{S}_{\mathfrak{s},V}$ for some $\mathfrak{s}$, meaning that there exists a fully faithful (and, therefore, essentially injective) inclusion functor $\imath : \mathbf{C}_V \hookrightarrow \mathbf{S}_{\mathfrak{s},V}$.

Example 2.1 (colouring)**.** Recall that a (vertex) *colouring* for a hypergraph G consists of a finite set $A \subset \mathbb{N}$ of colors and a function $c : V \to S$ (assigning to each vertice its color) such that each hyperedge contains at least two vertices of distinct colors. In other words, for each j the composition $c_j = \mathrm{bin}(V, c) \circ \psi_j$ is non-constant. We will work with $A = [n] = \{1, 2, ..., n\}$. Notice that for graphs we recover the usual notion of graph coloring. A *n-colored* hypergraph is one in which a coloring with a set of n colors was fixed, i.e, it is a pair (G, c) with $c : V \to [n]$. A morphism $f : (G, c) \to (G', c')$ of n-colored hypergraphs is a hypergraph morphism which preserves the coloring, i.e, $c' \circ f = c$. Denote by $n\mathbf{Col}$ the category of n-colored hypergraphs. Define $\mathfrak{s}$ by $\mathfrak{s}_{1,1} = [n]$ and $\mathfrak{s}_{j,k} = 0$ if $j \neq 1$ or $k \neq 1$. Let $n\mathbf{S}_{\mathfrak{s},V}$ be the full subcategory of $\mathbf{S}_{\mathfrak{s},V}$ whose objects are $\mathfrak{s}$-structured graphs (G, ϵ) with $\epsilon_{1,1} = c$ and $\epsilon_{j,k} = 0$ if $j \neq 1$ or $k \neq 1$. The condition $\epsilon' \circ f = \epsilon$ means precisely that $c' \circ f = c$, so that $n\mathbf{Col}_V \simeq \mathbf{S}_{\mathfrak{s},V}$.

Example 2.2 (labeled hypergraphs)**.** By a *labeling* of a hypergraph G we mean a bijection $\varphi_1 : V \to [|V|]$, where $[n] = \{1, ..., n\}$. Notice that this map induces bijections $\mathrm{bin}(\varphi_1, j)$ for every $j > 1$ and, by restriction, corresponding bijections $\varphi_j : E_j \to [|E_j|]$. A *labeled hypergraph* is one in which a labeling has been chosen, i.e, it is a pair (G, φ_1). A morphism of labeled hypergraphs (G, φ_1) and (G', φ'_1) is a triple (f, α, α'), where $f : G \to G'$ is a morphism of hypergraphs and α and α' are families of maps such that the diagram below commutes. Notice that, since φ_j and φ_1 are bijective, α and α' are actually determined by f, φ and φ'.

$$\begin{tikzcd}
{[|E_j|]} \arrow[r, "\varphi_j"] \arrow[d, "\alpha_j"'] & E_j \arrow[r, "f_j"] \arrow[d, "\psi_j"'] & E'_j \arrow[d, "\psi'_j"] & {[|E'_j|]} \arrow[l, "\varphi'_j"'] \arrow[d, "\alpha'_j"] \\
\mathrm{bin}([|V|], j)) \arrow[r, "\mathrm{bin}(\varphi_1, j)"'] & \mathrm{bin}(V, j)) \arrow[r, "\mathrm{bin}(f_1, j)"'] & \mathrm{bin}(V', j)) & \mathrm{bin}([|V'|], j)) \arrow[l, "\mathrm{bin}(\varphi'_1, j)"]
\end{tikzcd}$$

We then have a category $\mathbf{Hyp}^\ell$ of labeled hypergraphs. Let us show that for every fixed finite set V we have a canonical functor $\imath : \mathbf{Hyp}^\ell_V \hookrightarrow \mathbf{S}_{\mathfrak{s},V}$ for some $\mathfrak{s}$. From the above will then follows that $\imath$ is essentially injective, as desired. We define $\mathfrak{s}_{k,j}$ as the group $\mathbb{Z}_{|V|}$, if $j = k = 1$, and the trivial group otherwise, i.e, if $j \neq 1$ or $k \neq 1$. Now, for every labeled hypergraph (G, φ) with vertex set $E(1) = V$, take $\epsilon_{k,j} : E_j \to \mathfrak{s}_{k,j}$ as the trivial map if $k \neq 1$ or $j \neq 1$, and $\epsilon_{1,1} = \varphi_1$. For a morphism f of labeled graphs, it is clear that $\epsilon'_{k,j} \circ \mathrm{bin}(f, j) = \epsilon_{k,j}$, so that it becomes a morphism of structured graphs, and thus $\mathbf{Hyp}^\ell_V \hookrightarrow \mathbf{S}_{\mathfrak{s},V}$.

Remark 2.3. If (G, φ) and (G', φ') are labeled hypergraphs such that $|E_j| = |E'_j|$ for some j, then it follows from the commutativity of the last diagram that any morphism $f : G \to G'$ has the jth component f_j completely determined by the labelings, i.e, $f_j = \varphi_j \circ \varphi_j^{-1}$. In particular, it is a bijection. Consequently, if there exists an isomorphism between two labeled graphs, then it is unique. Furthermore, if we are working in $\mathbf{Hyp}^\ell_V$, then we can always assume $f_1 = id_V$.

Example 2.3 (Feynman graphs)**.** A *Feynman graph* is a graph G that has further structure satisfying additional properties. Precisely, it has a decomposition $V = V^0 \sqcup V^1$ into *external* (or *false*) *vertices* and *internal* (or *fundamental*) *vertices*, respectively, and a function $g : V \to \mathbb{Z}_{\geq 0}$, called *genus map*. This data is required to satisfy:

c1) if $v \in V^0$, then $d(v) = 1$ and $g(v) = 0$;

c2) there are no edges between two external vertices.

From the conditions above we see that there exists $E^0 \subset E$ with $|E^0| = |V^0|$, so that we also have a decomposition $E = E^0 \sqcup E^1$ into *external edges* (or *tails*) and *internal edges*, respectively, where $E^1 = E - E^0$. A *morphism* of Feynman graphs is a hypergraph morphism preserving the additonal structure, i.e, such that f_1 maps V^i into V'^i and $g' \circ f_1 = g$. Consequently, f_2 preserves E^0 and, therefore, E^1. Let **Fyn** denote the category of Feynman graphs. Notice that to give a decomposition of V is equivalent to giving a surjective function $\pi : V \to \mathbb{Z}_2$ by $\pi_V^{-1}(i) = V^i$. Take

$$\mathfrak{s}_{k,1} = \begin{cases} \mathbb{Z}_2, & k = 1 \\ \mathbb{Z}_{\geq 0}, & k = 2 \\ 0, & k > 2 \end{cases}$$

with $\mathfrak{s}_{k,j} = 0$ if $j > 1$. If (G, π, g) is a Feynman graph on V, define $\epsilon_{1,1} = \pi$, $\epsilon_{2,1} = g$ and $\epsilon_{k,j} = 0$, otherwise. For a given graph morphism f the condition $\epsilon'_{k,j} \circ \mathrm{bin}(f, j) = \epsilon_{k,j}$ means precisely that f preserves π and g. Thus, f is a Feynman graph morphism iff it is a morphism of $\mathfrak{s}$-structured graphs. This gives the desired embedding $\mathbf{Fyn}_V \hookrightarrow \mathbf{S}_{\mathfrak{s},V}$.

Remark 2.4. If to conditions c1) and c2) above we add

c3) if $g(v) = 0$, then $d(v) \geq 3$;

c4) if $g(v) = 1$, then $d(v) \geq 1$,

then we have what is known as *stable graphs*, since this class of graphs contains those arising in the study of stable curves [1, 20]. If $\mathbf{Stb} \subset \mathbf{Fyn}$ is the full subcategory of stable graphs, then $\mathbf{Stb}_V$ can also be embedded in $\mathbf{S}_{\mathfrak{s},V}$ for the same $\mathfrak{s}$ as $\mathbf{Fyn}_V$.

Example 2.4 (structured Feynman hypergraphs)**.** Feynman graphs can be generalized in two directions: allowing hyperedges and allowing additional structures. In the first case we say that we have a *Feynman hypergraph*, while in the second one we say that we have a *structured Feynman graph*. It is straightforward to verify (following the same kind of construction used in the previous examples) that these classes of graphs produce categories $\mathbf{HypFyn}_V$ and $\mathbf{Fyn}_{\mathfrak{s},V}$ which can be embedded in $\mathbf{S}_{\mathfrak{s},V}$ for some $\mathfrak{s}$. Special examples are the so-called *generalized Feynman graphs* [21, 22] and *sectored Feynman graphs* [23, 24, 25][2]. For instance, in generalized Feynman graphs we have an additional decomposition of V^1 into *positive* and *negative* vertices, i.e, $V^{1,+} \sqcup V^{1,-}$ (which means that now we need to use a projection $\pi : V \to \mathbb{Z}_3$ onto $\mathbb{Z}_3$ instead of onto $\mathbb{Z}_2$) and the edge set E is constrained by the condition that there is no edges between vertices of the same signal.

Example 2.5 (ribbon graphs)**.** An alternative way of defining a graph G is as being given by a set V of vertices, a set E of edges, an incidence function $s : V \to E$ and an involution $i : E \to E$ without fixed points. Morphisms $f : G \to G'$ are defined as functions $f_V : V \to V'$ and $f_E : E \to E'$ such that $f_E \circ s = s' \circ f_V$ and $f_E \circ i = i' \circ f_E$. Denote by $\mathbf{Grph}'$ this category. So, $\mathbf{Grph}' \simeq \mathbf{Hyp}^2$. Of special interest is the subcategory **Ribb** of *ribbon* (or *fat*) graphs. They become endowed with a permutation $\sigma : E \to E$ satisfying the following property:

[2]The terminology is not standard: some authors use generalized Feynman graphs to refer to sectored Feynman graphs. There is also a notion of *generalized Feynman amplitudes*, introduced in [26], which (to the best of the author's knowledge) it is not directly related with the other ones.

r) let $\langle\sigma\rangle$ be the cyclic group generated by σ. It acts on E giving a decomposition, which must coincide with that induced by the fibers $s^{-1}(x)$ of s.

The morphisms between ribbon graphs are graph morphisms which preserve the permutations. Let $\mathbf{Ribb}_0$ be the category of graphs with a permutation $\sigma : E \to E$ but which does not necessarily satisfies r). It is clear that $\mathbf{Ribb}_0 \hookrightarrow \mathbf{S}^1_{\mathfrak{s}}$ for some $\mathfrak{s}$. But $\mathbf{Ribb}$ is a full subcategory of $\mathbf{Ribb}_0$, so that it can also be embedded into a category of structured graphs.

The intersection of two embedded subcategories of structured hypergraphs remains a category of structured hypergraphs. More precisely, if $\mathbf{C} \hookrightarrow \mathbf{S}_{\mathfrak{s}}$ and $\mathbf{C}' \hookrightarrow \mathbf{S}_{\mathfrak{s}'}$, then $\mathbf{C} \cap \mathbf{C}'$ can be embedded in both $\mathbf{S}_{\mathfrak{s}}$ and $\mathbf{S}_{\mathfrak{s}'}$. For instance, in the last example $\mathbf{Fyn}_{\mathfrak{s}} = \mathbf{Fyn} \cap \mathbf{S}_{\mathfrak{s}}$. More generally, arbitrary limits of categories of structured hypergraphs remain a category of structured hypergraphs. In fact, let $\mathfrak{S}$ be the category defined as follows. Its objects are categories $\mathbf{C}$ such that there exists a functor of structures $\mathfrak{s}$ and an essentially injective embedding $\imath : \mathbf{C} \hookrightarrow \mathbf{S}_{\mathfrak{s}}$, while the morphisms are functors. So $\mathfrak{S}$ it is actually a full subcategory of $\mathbf{Cat}$. Since full embeddings are monadic, it follows that they reflect limits [27]. Thus:

Proposition 2.1. *The category $\mathfrak{S}$ is complete.*

We end this section with a convention which will be specially important in the construction of Feynman functors in Section 5.

Remark 2.5. Let (G, φ) and (G', φ') two labeled hypergraphs with vertex set V. Regarding them as structured hypergraphs as done in Example 2.2 we find ambiguities when considering the induced labeling in $G \sqcup G'$. In order to fix this we will use the following convention: let $[|V|]' = \{1', ..., |V|'\}$ be a copy of $[|V|]$ and in the disjoint union

$$[|V|] \sqcup [|V|]' = \{1, ..., |V|, 1', ..., |V|'\}$$

consider the ordering

$$1 \leq 2 \leq ... \leq |V| \leq 1' \leq 2' \leq ... \leq |V|'.$$

Define in $V \sqcup V$ a similar ordering, i.e, $\varphi(1) \leq ... \leq \varphi(|V|) \leq \varphi(1') \leq ... \leq \varphi(|V|')$. Then define the labeling in $G \sqcup G'$ as the unique bijection such that this ordering is preserved.

3 Reconstruction Conjectures

Given two sets V, V', with $V' \subset V$, let $V'^c = V - V'$ denote the complement of V' in V. We have a pair of adjoint functors

$$D_{V',V} : \mathbf{Hyp}_V \rightleftharpoons \mathbf{Hyp}_{V'} : I_{V',V},$$

defined as follows. The right adjoint $I_{V',V}$ is just the inclusion functor. More precisely, if G is a hypergraph with vertex set V', then $I_{V',V}(G)$ is the graph obtained by adding the elements of V'^c as isolated vertices. On the other hand, $D_{V',V}$ is the functor that takes a hypergraph G, with vertex set V, and delete the vertices V'^c together with their adjacent hyperedges.

So, if we fix a set X and define $\mathbb{X}$ as the category whose objects are subsets $V \subset X$ and whose morphisms are inclusions, varying V, V' we get functors $I : \mathbb{X} \to \mathbf{Cat}$ and $D : \mathbb{X}^{op} \to \mathbf{Cat}$. Notice that the process of adding (resp. deleting) a finite number of vertices is equivalent to iterating the process of adding (resp. deleting) a single vertice. This means that we have a distinguished class $J \subset \mathrm{Mor}(\mathbb{X})$ of morphisms, given by inclusions $\jmath_x : V' \hookrightarrow V$, with $V' = V - x$ for some $x \in V$. We can think of J as a rule assigning to each $V \in \mathbb{X}$ the collection of morphisms $J(V) = (\jmath_x)_{x \in V}$, which we call the *covering family of* V. It is clear that if $V' \subset V$, then $V' - x \subset V - x$. Since the only morphisms in $\mathbb{X}$ are inclusions and the only covering families are

$(\jmath_x)$, the previous condition implies that given a covering family $J(V)$ and a morphism $V' \to V$, then for any covering family $J(V')$ and each $\jmath'_{x'} \in J(V')$ there exists some $\jmath_x \in J(V)$ (actually $\jmath_{x'}$) such that the diagram below commutes.

$$\begin{tikzcd} V' - x' \arrow[r, "f-x'"] \arrow[d, "\jmath'_{x'}"'] & V - x' \arrow[d, "\jmath'_x"] \\ V' \arrow[r, "f"'] & V \end{tikzcd}$$

In other words, J is a coverage for $\mathbb{X}$. Therefore, we can talk about stacks on the site $(\mathbb{X}, J)$. More precisely, we have a reflexive subcategory of presheaves

$$\imath : \mathbf{Stack}(\mathbb{X}, J) \rightleftharpoons [\mathbb{X}^{op}; \mathbf{Cat}] : L$$

whose reflection L preserve finite limits. This reflexive subcategory is just the localization of the presheaf category at the local isomorphism system associated with J. If $\jmath \in J$, then $Y(\jmath)$ belongs to this system, where Y denotes the Yoneda embedding. Therefore, $LY(\jmath)$ is an isomorphism, so that from Yoneda lemma $LF(\jmath)$ is an isomorphism for every $F : \mathbb{X}^{op} \to \mathbf{Cat}$. Particularly, it is for the deleting functor D above and for every subfunctor $C \subset D$. This means that *after localization, the process of deleting vertices becomes an equivalence.* Furthermore: the same holds for every subfunctor $C \subset D$. Such a subfunctor assigns to each $V \subset X$ a subcategory $\mathbf{C}_V \subset \mathbf{Hyp}_V$ which is invariant by vertex deleting, i.e, we get an induced functor $C_{V',V} : \mathbf{C}_V \to \mathbf{C}_{V-V'}$. So, after localizing, deleting vertices is an equivalence independently of the class of hypergraphs considered.

We should not expect the same result before localizing, since in general there are much more graphs with $|V|$ vertices than graphs with $|V| - 1$ vertices. But, we can ask if in a given class of graphs, i.e, for a given subfunctor $C \subset D$, for every $\jmath_x \in J$ the corresponding functor $C(\jmath_x) : \mathbf{C}_V \to \mathbf{C}_{V-x}$ is at least essentially injective. This remains a very strong requirement, since we are asking if any information of a hypergraph $G \in \mathbf{C}_V$ can be recovered from the information after deleting a single *fixed* vertice $x \in V$. Thus, we can think of taking all $x \in V$ into account simultaneously. More precisely, we can ask if the induced composition below is essentially injective.

$$\begin{tikzcd} \mathbf{C}_V \arrow[r, "\Delta"'] \arrow[rr, bend left, "\Delta C_V"] & \prod_{x \in V} \mathbf{C}_V \arrow[r, "\prod_{x\in V} C(\jmath_x)"'] & \prod_{x \in V} \mathbf{C}_{V-x} \end{tikzcd}$$

This is just a sheaf theoretically formulation of what is usually known in hypergraph theory as the Reconstruction Conjecture for the class of graphs defined by C. In fact, calling an isomorphism in the image of ΔC_V a *hypomorphism*, the assertion that ΔC_V is essentially injective is equivalent to:

Conjecture 3.1 (RC-C)**.** Two hypergraphs in $\mathbf{C}_V$ are isomorphic iff they are hypomorphic.

Remark 3.1. Of course, if the categories $\mathbf{C}_V$ actually belong to $\mathbf{Grph}_V$, i.e, if we are in the context of graphs instead of general hypergraphs, then the above discussion reproduces the same conjecture, but now in graph theory. Some consequences of this categorical description (not directly related with the sheaf structure and specially concerning obstructions to the existence of non-nilpotent graph invariants) are in a work in preparation.

3.1 Disjoint Reconstruction

In the last section we gave a sheaf-theoretically description of the classical graph reconstruction conjecture. Our approach, on the other hand, has a problem:

- the prestack D is morphismwise adjoint to I, so that it is natural to believe that I should appear in any fundamental construction involving D. However, it was not used in the construction of ΔD_V.

In order to fix these pathologies, notice that despite of $I : \mathbb{X} \to \mathbf{Cat}$ not being strong monoidal (since for arbitrary V, V' we do not have an equivalence between $\mathbf{Hyp}_{V \sqcup V'}$ and $\mathbf{Hyp}_V \times \mathbf{Hyp}_{V'}$), it is lax comonoidal (because for generic V, V' there are more hypergraphs over $V \sqcup V'$ than pairs of hypergraphs over V and V'). Consequently, we have a natural transformation

$$\xi^I_{V',V} : \mathbf{Hyp}_V \times \mathbf{Hyp}_{V'} \to \mathbf{Hyp}_{V \sqcup V'}$$

sending pairs (G, G') into its disjoint union $G \sqcup G'$. We introduced the upper index to emphasize that ξ^I depends on I. So, instead of ΔD_V we can consider the composition dD_V below.

$$\begin{tikzcd}
\mathbf{Hyp}_V \arrow[r, "\Delta"'] \arrow[rr, bend left, "\Delta D_V"] \arrow[drr, bend right, "dD_V"'] & \prod_{x \in V} \mathbf{Hyp}_V \arrow[r, "\prod_{x \in V} D(\jmath_x)"'] & \prod_{x \in V} \mathbf{Hyp}_{V-x} \arrow[d, "\xi^I"] \\
& & \mathbf{Hyp}_{\coprod_x (V-x)}
\end{tikzcd} \tag{2}$$

This clearly fixes the initial problem for D, but this does not makes sense for arbitrary $C \subset D$, since $C_{V,V'}$ may not commute with $\xi^I_{V',V}$. That is, given $G \in \mathbf{C}_V$ and $G' \in \mathbf{C}_{V'}$ there is no guarantee that $G \sqcup G' \in \mathbf{C}_{V \sqcup V'}$. If $C \subset D$ is a subfunctor satisfying this condition we will say that it is *proper*. In this case, for every V the map dC_V is well defined.

Example 3.1 (structured presheaves)**.** A prestack of $\mathfrak{s}$-structured hypergraphs is proper. More precisely, given any functor $\mathfrak{s} : \mathbb{N} \times \mathbb{N} \to \mathbf{Set}$, the preshaf $S_{\mathfrak{s}}$ that to any V assigns $\mathbf{S}_{\mathfrak{s},V}$ is proper. Indeed, given (G, ϵ) and (G', ϵ') in $\mathbf{S}_{\mathfrak{s},V}$ and $\mathbf{S}_{\mathfrak{s},V'}$, respectively, we can always introduce structure $\epsilon \sqcup \epsilon'$ in $G \sqcup G'$. In particular, the presheaf S^ℓ such that $S^\ell(X) = \mathbf{Hyp}^\ell_X$ is proper.

Example 3.2. The trivial prestack $\varnothing$ such that $\varnothing_V = \varnothing$ for every V is trivally proper.

Remark 3.2. Beware that subfunctors of proper subfunctors need not be proper. That is, if $C \subset D$ is proper, then for arbitrary $C' \subset C$ it is not true that $C' \subset D$ is proper. On the other hand, the intersection $S^\ell_{\mathfrak{s}} = S^\ell \cap S_{\mathfrak{s}}$ is proper, because $S^\ell_{\mathfrak{s}} = S_{\mathfrak{s}'}$ for some $\mathfrak{s}'$.

Knowing that the problem is fixed by replacing ΔC with $\mathbf{d}C$, we can think of a conjecture somewhat analogous to C-RC. In order to do this, let us say that the isomorphisms in the image of each $\mathbf{d}C_V$ are *weak hypomorphisms.* So, we can then conjecture the Disjoint Reconstruction Conjecture:

Conjecture 3.2 (dRC-C)**.** Two hypergraphs in a proper presheaf $C \subset D$ are isomorphic iff they are weakly hypomorphic. In other words, $\mathbf{d}C_V$ is essentially injective.

Remark 3.3. From now on, if C is any prestack of hypergraphs, we will use the following simplified notations:

1. $\sqcap \mathbf{C}_V$ instead of $\prod_{x \in V} \mathbf{C}_{V-x}$;
2. $\sqcup \mathbf{C}_V$ instead of $\coprod_{x \in V} \mathbf{C}_{V-x}$;
3. $\mathbf{C}_{dV}$ instead of $\mathbf{C}_{\coprod_{x \in V} V-x}$.

3.2 Category of Reconstruction Conjectures

Let us now see that both conjectures RC and dRC can be considered in an axiomatic background. For doing this we define a *hypergraph reconstruction context* as given by the following data:

1. for each set X we have a subcategory $\mathfrak{C} \subset [\mathbb{X}^{op}; \mathbf{Cat}]$ of *applicable prestacks*, such that if $C \in \mathfrak{C}$, then $C_V \subset \mathbf{Hyp}$ for each $V \subset X$;
2. a functor $\mathcal{D} : \mathfrak{C} \to [\mathbb{X}^{op}; \mathbf{Cat}]$, playing the role of a "deleting process" and that to any applicable prestack C it assigns the *prestack of pieces* $\mathcal{D}C$, such that for each $V \subset X$ we have the *category of pieces* $\mathcal{D}C_V$*;*
3. a natural transformation $\gamma : \imath_{\mathfrak{C}} \Rightarrow \mathcal{D}$, where $\imath_{\mathfrak{C}}$ is the inclusion of applicable prestacks into the category of prestacks.

We will represent the reconstruction context simply by its deleting process $\mathcal{D}$, except when we need more details. We say that two hypergraphs are *$\mathcal{D}$-hypomorphic* if their image by $\mathcal{D}C_V$ are isomorphic. So, given a reconstruction context $\mathcal{D}$ and an applicable prestack $C \in \mathfrak{C}$ we can consider the following $\mathcal{D}$-reconstruction conjecture for C.

Conjecture 3.3 ($\mathcal{D}$-RC-C)**.** For each V, two hypergraphs in C_V are isomorphic iff they are $\mathcal{D}$-hypomorphic, i.e, each functor $\gamma_V : C_V \to DC_V$ is essentially injective.

Before giving examples, two important remarks:

Remark 3.4. In the previous sections, the prestacks of hypergraphs $C : \mathbb{X}^{op} \to \mathbf{Cat}$ were such that for each $V \subset X$ the corresponding C_V is not only an arbitrary subcategory of $\mathbf{Hyp}$, but actually a subcategory of $\mathbf{Hyp}_V$, i.e, we worked with prestacks that assign to each V a category of hypergraphs with vertex set V. From now on, we will work with prestacks which a priori take values only in $\mathbf{Hyp}$. This will be specially important in proving existence and uniqueness of Feynman rules. In order to distinguish between these situations, we will say that C is a *concrete prestack* if $C_V \subset \mathbf{Hyp}_V$, using the bold notation $\mathbf{C}_V$ instead of C_V (as we have used in previous sections).

Remark 3.5. We say that an applicable prestack $C \in \mathfrak{C}$ is *proper* if it becomes endowed with a natural transformation $\xi : C_V \times C_{V'} \to C_{V \sqcup V'}$. Recall that in the last sections we defined a proper (concrete) prestack as such that $\xi : (G, G') \mapsto G \sqcup G'$ is well defined. This means that any proper concrete prestack (in the older context) is proper (in the newer sense). However, the reciprocal is not true, due to the last remark. In order to emphasize that this new concept is more general, we will call them *concretely proper* prestacks.

Example 3.3 (RC and dRC)**.** In order to recover the classical reconstruction conjecture, take $\mathfrak{C}$ as the whole category of prestacks, $\mathcal{D}C_V = \sqcap \mathbf{C}_V$ and $\gamma_V = \Delta \mathbf{C}_V$. For recovering the disjoint reconstruction conjecture, take $\mathfrak{C}$ as the subcategory of proper prestacks, $DC_V = \mathbf{C}_{dV}$ and $\gamma_V = d\mathbf{C}_V$.

Example 3.4 (trivial context)**.** We also have a trivial reconstruction context $\mathcal{I}$, in which $\mathfrak{C}$ is the whole category of prestacks and $\mathcal{D}$ and γ are the identity functors. Notice that in it any recontruction conjecture is satisfied.

Example 3.5 (restrictions)**.** Let $\mathcal{D}$ be a reconstruction context with category of applicable prestacks $\mathfrak{C}$ and natural transformation $\gamma : \imath_{\mathfrak{C}} \Rightarrow \mathcal{D}$. For any subcategory $\mathfrak{D} \subset \mathfrak{C}$ we get a context $\mathcal{D}|_{\mathfrak{D}}$ by restricting $\mathcal{D}$ and γ to $\mathfrak{D}$.

We will build some categories of reconstruction conjectures, allowing us to compare two of different conjectures. A *left morphism* between two reconstruction contexts $\mathcal{D}$ and $\mathcal{D}'$ is given by a functor $F : \mathfrak{C} \to \mathfrak{C}'$ between the categories of applicable prestacks, together with a natural transformation $\xi : \mathcal{D} \Rightarrow \mathcal{D}' \circ F$ between the deleting process, such that the first diagram below commutes, i.e, if for every $C \in \mathfrak{C}$ we have $\xi_C \circ \gamma_C = \gamma'_{F(C)}$. This gives us a category $L\mathbf{RC}$. By inverting the direction of ξ we define *right morphisms*, which produce a dual category $R\mathbf{RC}$, characterized by the second diagram below.

$$\begin{tikzcd} \imath_{\mathfrak{C}} \arrow[r, Rightarrow, "\gamma"] \arrow[d, Rightarrow, "\imath_F"'] & \mathcal{D} \arrow[d, Rightarrow, "\xi"] \\ \imath_{\mathfrak{C}'\circ F} \arrow[r, Rightarrow, "\gamma'\circ F"'] & \mathcal{D}'\circ F \end{tikzcd} \qquad \begin{tikzcd} \imath_{\mathfrak{C}} \arrow[r, Rightarrow, "\gamma"] \arrow[d, Rightarrow, "\imath_F"'] & \mathcal{D} \\ \imath_{\mathfrak{C}'\circ F} \arrow[r, Rightarrow, "\gamma'\circ F"'] & \mathcal{D}'\circ F \arrow[u, Rightarrow, "\xi"'] \end{tikzcd} \tag{3}$$

Example 3.6 (canonical morphisms)**.** For any reconstruction context $\mathcal{D}$ there is a canonical right morphism $I_R : \mathcal{D} \to \mathcal{I}$, defined as follows. In applicable prestacks it is the inclusion functor, i.e, $F = \imath_{\mathfrak{C}}$. Among deleting processes it is just the transformation $\xi_C = \gamma_C$ of $\mathcal{D}$. The existence of left morphisms $I_L : \mathcal{D} \to I$ is more restrictive: if we keep the canonical choice $F = \imath_{\mathfrak{C}}$, the condition $\xi_C \circ \gamma_C = id_C$ implies that ξ is a retraction for γ. In the general case, the commutativity condition is $\xi \circ \gamma = id \circ F$, which is also some kind of retraction requirement (let us say that γ has ξ as a F-*retraction*). So, we have the following proposition:

Proposition 3.1. *Given a prestack $F : \mathfrak{C} \to [\mathbb{X}^{op}; \mathbf{Cat}]$, there is a left morphism $D \to I$ of reconstruction contexts coinciding with F in applicable prestacks iff γ has a F-retraction.*

We could think of getting morphisms in the opposite direction, i.e, from the trivial context I to a given context D. This is an even more strong requirement. This is essentially because a priori we have no canonical functor $F : [\mathbb{X}^{op}; \mathbf{Cat}] \to \mathfrak{C}$. The commutativity conditions implies that if it exists, then it must be a retraction for the inclusion of $\mathfrak{C}$ in $[\mathbb{X}^{op}; \mathbf{Cat}]$. In the case of left morphisms this is enough. For right morphisms $I \to D$ we also need γ to have a F-retraction.

Example 3.7 (from RC to dRC)**.** Let $\mathfrak{C}$ be the category of proper stacks. Let Δ and d be the contexts of the RC and dRC, as defined in Example 3.3. We have a right morphism $F : d \to \Delta$, defined as follows. In applicable prestacks we define $F : \mathfrak{C} \to [\mathbb{X}^{op}; \mathbf{Cat}]$ as the inclusion of proper prestacks and $\xi : \Delta \Rightarrow d$ as the composition $\xi^I \circ u$ in diagram (2). Notice that the commutativity of (3) follows directly from the commutativity of (2).

Proposition 3.2. *Let $F : \mathcal{D} \to \mathcal{D}'$ be a morphism. For a given prestack $C \in \mathfrak{C}$:*

1. *if F is left and $\mathcal{D}'$-RC-$F(C)$ holds, then $\mathcal{D}$-RC-C holds;*
2. *if F is right and $\mathcal{D}$-RC-C holds, then $\mathcal{D}'$-RC-$F(C)$ holds.*

Proof. For the first case, notice that the commutativity of (3) gives us $\xi_C \circ \gamma_C = \gamma'_{F(C)}$ and recall that essentially injective functors behave as monomorphisms, so that if $\gamma'_{F(C)}$ is essentially injective, then γ_C is too. But the validity of $\mathcal{D}'$-RC-$F(C)$ is, by definition, the garantee that $\gamma'_{F(C)}$ is essentially injective. For the second case, (3) gives $\gamma_C = \xi_C \circ \gamma'_{F(C)}$. Now use the same argument of the first case. □

In some cases we have a left morphism, we know that $\mathcal{D}$-RC-C holds and we would like to conclude that $\mathcal{D}'$-RC-$F(C)$ holds. In other words, we would like to have conditions under which the hypothesis of the first part of the last proposition implis the conclusion of the second part, and vice-versa. This can be easily ensured if we work with a special class of morphisms. We say that a left morphism $F : \mathcal{D} \to \mathcal{D}'$ is a left C-*implication* if the transformation $\xi : D' \circ F \Rightarrow D$ is objectwise essentially injective. Right C-implications are defined analogously.

Proposition 3.3. *Let $F : \mathcal{D} \to \mathcal{D}'$ be a morphism. For a given prestack $C \in \mathfrak{C}$:*

1. *if F is left C-implication and $\mathcal{D}$-RC-C holds, then $\mathcal{D}'$-RC-$F(C)$ holds;*
2. *if F is right C-implication and $\mathcal{D}'$-RC-$F(C)$ holds, then $\mathcal{D}$-RC-C holds.*

Proof. As in the last proposition, for the first case (3) gives $\xi_C \circ \gamma_C = \gamma'_{F(C)}$. Since composition of essentially injective functors remains essentially injective, it is done. The second case is analogous. □

For any $\mathcal{D}$ and any $C \in \mathfrak{C}$, the conjecture $\mathcal{D}$-RC-C holds iff the canonical morphism $I_R : \mathcal{D} \to \mathcal{I}$ is a right C-implication.

Proof. Straightforward. □

As a final result, let us show that reconstruction conjectures are invariant by a certain base-change.

Proposition 3.4. *Let $\mathcal{D}$ be a reconstruction context and suppose that $\mathcal{D}$-RC-C holds for some applicable prestack $C \in \mathfrak{C}$. In this case, if $f : A \to C$ is some objectwise essentially injective morphism in $\mathfrak{C}$, then $\mathcal{D}$-RC-A holds.*

Proof. Since $\gamma : \imath_{\mathfrak{C}} \Rightarrow D$ is a natural transformation, we have $Df \circ \gamma_A = \gamma_C \circ f$. By hypothesis f and γ_C are essentially injective, so that $Df \circ \gamma_A$, and therefore γ_A, is also. □

3.3 Reconstruction of Labeled Structured Hypergraphs

In this subsection we will show that the RC is true for any prestack $S^\ell_{\mathfrak{s}}$ of labeled $\mathfrak{s}$-structured hypergraphs. As a consequence, since from Example 2.2 and Example 3.1 this prestack is proper, it will follow from Proposition 3.2 and Example 3.7 that dRC-$S^\ell_{\mathfrak{s}}$ holds.

Theorem 3.1. *For any functor of structures $\mathfrak{s}$, the $S^\ell_{\mathfrak{s}}$-RC holds. In other words, $\Delta S^\ell_{\mathfrak{s},V}$ is essentially injective for every V.*

Proof. Notice that $\mathbf{S}^\ell_{\mathfrak{s},V} = \mathbf{S}_{\mathfrak{s},V} \cap \mathbf{Hyp}^\ell_V$. Therefore, $\mathbf{S}^\ell_{\mathfrak{s},V}$ can be embbeded in an essentially injective way in $\mathbf{Hyp}^\ell_V$. This can also be verified explicitly by a counting process:

Addendum. Regard $\mathfrak{s}$ as a bifunctor $\mathfrak{s} : \mathbb{N} \times \mathbb{N} \to \mathbf{Set}$ and let $N : \mathbb{X}^{op} \times \mathbb{N} \times \mathbb{N} \to \mathbf{Cat} \times \mathbf{Ab}$ be the functor given by $N(V, i, j) = \varnothing \times \mathbb{N} \simeq \mathbb{N}$ for every V, i, j. We have a natural transformation $k : S^\ell_{\mathfrak{s}} \times \mathfrak{s} \Rightarrow N$ such that for every V, i, j the map $k^V_{i,j} : \mathbf{S}^\ell_{\mathfrak{s},V} \times s_{i,j} \to \mathbb{N}$ is the rule that to any labeled $\mathfrak{s}$-structured hypergraph (G, Φ, ϵ) and to any element $a_{ij} \in s_{i,j}$ it assigns the cardinality of $\epsilon^{-1}_{i,j}(a_{ij})$. We have

$$(n-1)k^V_{ij}(G, a_{ij}) = \sum_{x \in V} k^{V-x}_{ij}(G - x, a_{ij}), \tag{4}$$

where $n = |V|$. Suppose now that $G, G' \in \mathbf{S}^\ell_{\mathfrak{s},V}$ are isomorphic only as labeled hypergraphs. From Remark 2.3 there is a unique isomorphism, determined by the labelings. This implies that G and G' will have the same counting (4). In particular, $k^V_{ij}(G', a_{ij}) = k^V_{ij}(G, a_{ij})$, allowing us to conclude that $\epsilon'_{i,j} = \epsilon_{i,j}$ for every i, j, and therefore $\epsilon' = \epsilon$. Consequently, G' and G are also isomorphic as $\mathfrak{s}$-structured hypergraphs.

Returning to the proof, from Proposition 3.4 we only need to prove RC for labeled hypergraphs, i.e, that $\Delta D^\ell_V : \mathbf{Hyp}^\ell_V \to \sqcap \mathbf{Hyp}^\ell_{V-x}$ is essentially injective for every V. So, let (G, φ) and (G', φ') be two labeled hypergraphs and suppose that there exists $f_x : (G - x, \varphi_x) \simeq (G' - x, \varphi'_x)$ for every x. Since we are working with isomorphic labeled hypergraphs defined over the same vertex set, Remark 2.3 allows us to assume that the isomorphism on the vertices is given by the identity map , i.e, $(f_x)_1 = id$ for every x. Furthermore, the bijection $(f_x)_j : (E_j)_x \to (E'_j)_x$on the j-edges must be given by $(\varphi_j)^{-1}_x \circ (\varphi'_j)^{-1}_x$, so that it can be clearly extended to a bijection between E_j and E'_j preserving the labelings, which gives $G \simeq G'$. □

If $C \subset S^{\ell}_{\mathfrak{s}}$ is concretely proper and objectwise essentially injetive, then both conjectures RC-C and dRC-C holds.

Proof. It follows directly from Proposition 3.4. □

4 Contexts for Functional Analysis

When working with functional analysis we are dealing with certain classes of spaces, each one with an associated "dual space", and for which we know how to take tensor products. This leads us to define a *context for functional analysis* (or simply a *context*) as a monoidal category $(\mathbf{A}, \otimes, 1)$ endowed with a functor $\cdot^{\vee} : \mathbf{A}^{op} \to \mathbf{A}$, assigning to each object $U \in \mathbf{A}$ its *dual* $U^{\vee}$, which are compatible in the sense that we have a *compatibility transformation* $\cdot^{\vee} \otimes \cdot^{\vee} \Rightarrow (\cdot \otimes \cdot)^{\vee}$ and a distinguished morphism $i : 1 \to 1^{\vee}$.

We define a *morphism* between two contexts $(\mathbf{A}, \otimes, \cdot^{\vee})$ and $(\mathbf{B}, \circledast, \cdot^{*})$ as a functor $F : \mathbf{A} \to \mathbf{B}$ which weakly preserves tensor products and duals. In other words, it is an oplax monoidal functor[3] together with a transformation $F(\cdot^{\vee}) \Rightarrow F(\cdot)^{*}$ such that the diagram below commutes. We then have the category **Cnxt** of contexts and morphisms between them.

$$\begin{tikzcd}
F(U^{\vee} \otimes V^{\vee}) \arrow[r] \arrow[d] & F((U \otimes V)^{\vee}) \arrow[d] & & \\
F(U^{\vee}) \circledast F(V^{\vee}) \arrow[d] & F(U \otimes V)^{*} \arrow[d] & F(1_{\otimes}) \arrow[r] & F(1_{\otimes}^{\vee}) \\
F(U)^{*} \circledast F(V)^{*} \arrow[r] & (F(U) \circledast F(V))^{*} & 1_{\circledast} \arrow[r] \arrow[u] & 1_{\circledast}^{*} \arrow[u]
\end{tikzcd}$$

Remark 4.1. In some cases, the compatibility transformation $\cdot^{\vee} \otimes \cdot^{\vee} \Rightarrow (\cdot \otimes \cdot)^{\vee}$ and the distinguished map $1 \to 1^{\vee}$ are isomorphisms. In such cases we say that we have *strong contexts*. With the same notion of morphisms they define a full subcategory $\mathbf{SCnxt} \subset \mathbf{Cnxt}$.

Example 4.1 (linear algebra)**.** Given a field $\mathbb{K}$, the category $\mathbf{Vec}_{\mathbb{K}}$ of $\mathbb{K}$-vector spaces defines a context for functional analysis when endowed with the tensor product monoidal structure $(\otimes_{\mathbb{K}}, \mathbb{K})$, with $U^{\vee}$ being the linear dual U^{*}. This is *not* a strong context, since for arbitrary vector spaces the canonical transformation $U^{*} \otimes_{\mathbb{K}} V^{*} \to (U \otimes_{\mathbb{K}} V)^{*}$ is not an isomorphism. On the other hand, the full subcategory $\mathbf{FinVec}_{\mathbb{K}}$ of finite-dimensional $\mathbb{K}$-vector spaces is a strong context. More generally, the transformation $U^{*} \otimes_{\mathbb{K}} V^{*} \to (U \otimes_{\mathbb{K}} V)^{*}$ exists but may not be an isomorphism for arbitrary R-modules; but they are if we consider finitely generated projective R-modules [28]. So, with analogous structure, $\mathbf{Mod}_{R}$ is a context and the category of finitely generated projective R-modules is a strong context.

Example 4.2 (locally convex)**.** Now we can take $\mathbf{A}$ as some category of topological real vector spaces and continuous linear maps and think of defining $U \otimes V$ and $U^{\vee}$ as $U \otimes_{\mathbb{R}} V$ and $B(U;\mathbb{R}) \subset U^{*}$, respectively, endowed with some topology[4]. There are many possible choices of topology, of course. For locally convex spaces (lcs), there are at least three canonical ways to topologize $U \otimes V$: the projective, the injective and the inductive topologies. The choice of each of them produce a symmetric monoidal category $(\mathbf{LCS}, \otimes, \mathbb{R})$ [29]. Also, there are many topologies in $B(U;\mathbb{R})$, such as the topology of pointwise convergence and uniform convergence in bounded sets. With each of them, we get a pseudocontext structure in $\mathbf{LCS}$.

[3] Recall our convention that in an oplax monoidal functor the arrow between the neutral objects remains in the correct direction.

[4] Here, $B(U;V)$ denotes the space of bounded linear maps.

Example 4.3 (nuclear Fréchet)**.** When the spaces are nuclear, the injective and the projective topologies coincide [29, 30, 20], so that we have a more canonical pseudocontext structure. In general, none of the topologies in $U^\vee = B(U;\mathbb{R})$ will induce isomorphisms $\cdot^\vee \otimes \cdot^\vee \simeq (\cdot \otimes \cdot)^\vee$, but they exist if we restrict to the full subcategory **NucFrec** of nuclear Fréchet spaces, showing that $(\mathbf{Frec}, \otimes_p, \cdot^\vee)$ is a strong context [20, 30]. Also, for Fréchet spaces the projective and the inductive topologies coincide [31], meaning that in **NucFrec** we have a canonical symmetric monoidal structure.

Example 4.4 (subcontext)**.** Let $(\mathbf{A}, \otimes, \cdot^\vee)$ be a context. We define a *subcontext* as a full subcategory $\mathbf{C} \subset \mathbf{A}$ such that $1 \in \mathbf{C}$ and which is closed under $\otimes$ and $\cdot^\vee$, meaning that $U \otimes V$ and $U^\vee$ belongs to $\mathbf{C}$ when $U, V \in \mathbf{C}$. It then follows that $(\mathbf{C}, \otimes, \cdot^\vee)$ is a context and that the inclusion functor $\imath : \mathbf{C} \hookrightarrow \mathbf{A}$ is a morphism of contexts. Furthermore, if $\mathbf{A}$ is strong, then $\mathbf{C}$ is also (the reciprocal is false as the last example shows). In particular, the full subcategories **Ban** of Banach spaces and **Hilb** of Hilbert spaces are subcontexts of **Frec** and therefore define themselves strong contexts for functional analysis.

Example 4.5 (categories with duals)**.** There are many flavors of monoidal categories whose objects or morphisms have duals (in the monoidal sense), e.g, autonomous categories, pivotal categories, spherical categories, spacial categories, compact closed categories and dagger monoidal categories (see [32] for a survey). All of them define a version of strong contexts whose $\cdot^\vee$ is actually an ana-functor. Via Tannaka duality, they can be characterized as the representation category of certain monoid objects [33, 34].

We can consider monoidal categories $\mathbf{A}$ endowed with a functor $\cdot^\vee : \mathbf{A}^{op} \to \mathbf{A}$ without any compatibility condition. In this case we will say that we have a *pseudocontext*. A *morphism* between two of them is just a lax monoidal functor $F : \mathbf{A} \to \mathbf{B}$ together with a transformation $F(\cdot^\vee) \Rightarrow F(\cdot)^*$, giving us a category **PCntx** which contains **Cntx**.

4.1 Analytic Expressions

From now on we will assume that the pseudocontexts $(\mathbf{A}, \otimes, \cdot^\vee)$ are such that:

1. the category $\mathbf{A}$ have countable limits, coproducts and cokernels;
2. the tensor product $\otimes$ and the functor of duals $\cdot^\vee$ preserve limits.

For physical interpretation we will also require that they become endowed with an additional functor $K : \mathbf{A} \to \mathbf{Set}$ and for every $k \in \mathbb{Z}_{\geq 0}$ a transformation $S_k \times K(U^{\otimes_k}) \to K(U^{\otimes_k})$ which is objectwise a group action[5]. We require that the quotient $K(U^{\otimes_k})/S_k$ belongs to the image of K and have a single pre-image, so that it uniquelly defines an object in $\mathbf{A}$. The obvious notation for this object should be $\mathrm{Sym}^k(U)$, but here we will use $P^k U$ to denote it. For $k > 1$ we will say that this is the *object of k-propagators* in U (2-propagators are called propagators for short).

Given $X \in \mathbf{A}$ we define an *object of formal power series* as the countable product of copies of X, i.e, as the product $\prod_i X_i$, with $X_i \simeq X$. We usually use the powers of a formal parameter h to indicate the order of the products, writing $\prod_i X_i h^i$ or $X[[h]]$ for short. We can consider power series not only with a single parameter h, but with a family $\mathfrak{t} = (h_n)_{n \in \mathbb{Z}_{\geq 0}}$ of them. These will be given by $X[[\mathfrak{h}]] = \prod_n \prod_i X_i^n h_n^i$ with $X_i^n \simeq X$. In physical contexts $\mathfrak{h}$ will represent the family of fundamental parameters over which we will do perturbation theory.

Write $PU = \prod_k P^k U$ and $\mathcal{O}U = PU^\vee$. Given a family of parameters $\mathfrak{h}$ we define an *interacting term* in U relative to $\mathfrak{h}$ as a morphism $I : 1 \to \mathcal{O}U[[\mathfrak{h}]]$. A *morphism* of interacting terms is a morphism in the under category $1/\mathbf{A}$, giving us a full subcategory $\mathbf{Int}_{U,\mathfrak{h}}$ of $1/\mathbf{A}$.

Feynman rules will take structured hypergraphs and assigns to each of them interacting terms and propagators which together will fit into analytic expressions. Let $A^k U$ denote $(U^\vee)^{\otimes_k} \otimes U^{\otimes_k}$

[5] Here S_k is the permutation group and $U^{\otimes_k} = ((U \otimes U) \otimes U)...$, k times.

and let $AU = \prod_k A^k U$ (since the tensor product $\otimes$ preserves countable products, if $\mathbf{A}$ is a *strong context*, we can also write $AU = PU \otimes PU^{\vee}$). An *analytic expression* in U is a morphism $a : 1 \to AU$. A *morphism* is a morphism in $1/\mathbf{A}$, so that we have a full subcategory $\mathbf{A}_U$. We say that an analytic expression has *order k* if it takes values in $A^k U$ instead of AU, defining a category $\mathbf{A}_U^k$. We have $\mathbf{A}_U \simeq \prod_k \mathbf{A}_U^k$.

Proposition 4.1. *For every $U \in \mathbf{A}$, the category $\mathbf{A}_U$ acquires a canonical monoidal structure, induced from the monoidal structure in $\mathbf{A}$.*

Proof. Recall that if $(\mathbf{C}, \otimes, 1)$ is any monoidal category and then for any comonoid object $X \in \mathbf{C}$, say with coproduct $\mu : X \to X \otimes X$ and counit $\eta : X \to 1$, then the under category $X/\mathbf{C}$ can be endowed with a monoidal product $X/\otimes$, defined by tensoring with $\otimes$ and precomposing with μ, whose neutral object is η. Particularly, for $X = 1$ we get that $1/\mathbf{C}$ is monoidal. Let $(\mathbf{A}, \otimes, \cdot^{\vee})$ be a context. Let $\mathbf{A}U \subset \mathbf{A}$ be the full subcategory whose single object is AU. Since $\otimes$ and $\cdot^{\vee}$ preserve countable products, we see that AU is actually a monoid object, so that $\mathbf{A}U$ is a monoidal subcategory. On the other hand, $\mathbf{A}_U \simeq 1/\mathbf{A}U$, from where we get the desired monoidal structure on analytic expressions. $\square$

Proposition 4.2. *Every pseudocontext morphism $F : \mathbf{A} \to \mathbf{B}$ induces, for each $U \in \mathbf{A}$, a functor $F_U : \mathbf{A}_U \to \mathbf{B}_{F(U)}$ between the corresponding categories of analytic expressions.*

Proof. Given an analytic expression $a : 1 \to AU$ in $\mathbf{A}$, define $F_U(a)$ as the following composition.

$$1_{\circledast} \longrightarrow F(1) \xrightarrow{F(a)} F(AU) \xrightarrow{u} \prod_k F(A^k U) \longrightarrow \prod_k B^k F(U) = BF(U)$$

The first and the last arrows appear because F is oplax monoidal and because a pseudocontext morphism becomes endowed with a transformation $F(\cdot^{\vee}) \to F(\cdot)^*$. Furthermore, u is from the universality of products in $\mathbf{B}$. From the definition of $F_U(a)$ it immediately follows that F_U is functorial. $\square$

Remark 4.2. Unless the pseudocontext morphism $F : \mathbf{A} \to \mathbf{B}$ is a bilax monoidal functor (instead of only oplax monoidal), the induced functor $F_U : \mathbf{A}_U \to \mathbf{B}_{F(U)}$ generally will not be monoidal. Indeed, recall that the monoidal structure of $\mathbf{A}_U$ is essentially the monoid object structure of AU in $\mathbf{A}$. Therefore, saying that F_U is monoidal we are saying that F maps the monoid AU into the monoid $BF(U)$, which is not a typical property of oplax functors, but which is clearly satisfied when F is bilax or strong monoidal.

Remark 4.3. We could think of replacing the products $\prod_k$ by coproducts $\coprod_k$ in the above definitions and constructions. This would make life easier. But, in order to do this, we should assume that $\mathbf{A}$ has countable colimits (instead of countable limits) and that $\otimes$ and $\cdot^{\vee}$ preserve colimits (instead of limits). For our purposes, these assumptions are restrictive: if $\otimes$ preserves colimits, then it has right adjoint, meaning that $(\mathbf{A}, \otimes, 1)$ is a closed monoidal category and automatically excluding $(\mathbf{Frec}, \otimes_p, \cdot^{\vee})$ as a possible context.

5 Feynman Functors

We can finally introduce and prove existence and uniqueness of functorial Feynman functors. These assign to each presheaf of hypergraphs an analytic expression in some pseudocontext. If the pseudocontext is sufficiently well behaved, then it will be able to evaluate these analytic expressions, giving us some kind of "amplitude of probability" assigned to each hypergraph.

Given a prestack $C : \mathbb{X}^{op} \to \mathbf{Cat}$ of hypergraphs (generally regarded as an applicable prestack), a pseudocontext $(\mathbf{A}, \otimes, \cdot^{\vee})$ and a functor $\tau : \mathbf{FinSet} \to \mathbf{A}$, we define a *functorial*

Feynman functor (or *Feynman functor*) for C with values in $(\mathbf{A},\tau)$, denoted by $Z : C \to (\mathbf{A},\tau)$, as a function that to each $V \in \mathbf{FinSet}$ assigns a functor $Z_V : C_V \to 1/\mathbf{A}$ such that for every $G \in C_V$ we have $Z_V(G) \in \mathbf{A}_{\tau(VG)}$, where VG is the vertex set of G. We say that a Feynman functor is *complete* when each Z_V is essentially injective, meaning that the structured hypergraphs can be totally described by their associated analytic expressions.

We are also interested in *monoidal Feynman functors*. In order to define them we need to work with C proper, which means that it becomes endowed with a transformation $\xi_{V,V'} : C_V \times C_{V'} \to C_{V\sqcup V'}$, as discussed in Remark 3.5. From it we obtain the following transformation, which takes into account three finite sets instead of only two. Notice that $\xi_{(V,V'),V''} \simeq \xi_{V,(V',V'')}$, where the isomorphism means that any three hypergraphs have isomorphic image under these maps.

$$\begin{tikzcd}
(C_V \times C_{V'}) \times C_{V''} \arrow[r, "\xi_{V,V'}\times id_{V''}"] \arrow[dr, dashed, "\xi_{(V,V'),V''}"'] & C_{V\sqcup V'} \times C_{V''} \arrow[d, "\xi_{(V\sqcup V'),V''}"] \\
& C_{(V\sqcup V')\sqcup V''}
\end{tikzcd} \tag{5}$$

We say that Z is *oplax monoidal* if for every $V, V' \in \mathbf{FinSet}$, every $G \in C_V$ and every $G' \in C_{V'}$ we have morphisms

$$\mu_{V,V'} : Z_{V\sqcup V'}(\xi_{V,V'}(G,G')) \to Z_V(G) \otimes Z_{V'}(G') \quad \text{and} \quad \nu : Z_\varnothing(\varnothing) \to id_1 \tag{6}$$

in $1/\mathbf{A}$ satisfying the usual comonoid-like diagrams (e.g, the associativity diagram is that presented below[6]). Similarly, we define the situations when Z is *lax monoidal* and *strong monoidal* (or simply *monoidal*) by reverting the arrow or by requiring that they are isomorphisms, respectively.

$$\begin{tikzcd}
Z_{V,(V',V'')}(\xi_{V,(V',V'')}(G,G',G'')) \arrow[r, "\simeq"] \arrow[d, "\mu_{V,(V',V'')}"'] & Z_{(V,V'),V''}(\xi_{(V,V'),V''}(G,G',G'')) \arrow[d, "\mu_{(V,V'),V''}"] \\
Z_V(G) \otimes Z_{V',V''}(\xi_{V',V''}(G',G'')) \arrow[d, "id\times\mu_{V',V''}"'] & Z_{V,V'}(\xi_{V,V'}(G,G')) \otimes Z_{V''}(G'') \arrow[d, "\mu_{V,V'}\times id"] \\
Z_V(G) \otimes (Z_{V'}(G') \otimes Z_{V''}(G'')) \arrow[r, "\simeq"'] & (Z_V(G) \otimes Z_{V'}(G')) \otimes Z_{V''}(G'')
\end{tikzcd} \tag{7}$$

Remark 5.1. When C is concretely proper, $C_V \subset \mathbf{Hyp}_V$ for every V, so that

$$Z_{V\sqcup V'}(\xi(G,G')) \in \mathbf{A}_{\tau(V\sqcup V')} \quad \text{and} \quad Z_V(G) \otimes Z_{V'}(G') \in \mathbf{A}_{\tau V} \otimes \mathbf{A}_{\tau V'}.$$

Therefore, in order to formalize the notion of lax (resp. oplax) monoidal Feynman functors, instead of requiring the morphisms (6) to belong to $1/\mathbf{A}$, in these cases we could require that $\tau : (\mathbf{FinSet}, \sqcup, \varnothing) \to (\mathbf{A}, \otimes, 1)$ is lax (resp. oplax) monoidal. However, this would produce a much more rigid concept, e.g, τ could not be chosen as the constant functor in some $U \in \mathbf{A}$, with $U \neq 1$. And, as we will see, it is precisely when $\tau = \text{cst}$ that the explict connection with perturbative quantum field theory is made.

Example 5.1 (trivial Feynman functor)**.** For any C and any $(\mathbf{A},\tau)$ there always exists a trivial monoidal Feynman functor $cst_1 : C \to (\mathbf{A},\tau)$ assigning to each V the functor $cst_1 : C_V \to 1/\mathbf{A}$ constant in the neutral object id_1 of the monoidal category $1/\mathbf{A}$. In the next section we will show that nontrivial Feynman functors also exist.

[6] In order to simplify the notation we wrote $Z_{V,V'}$ instead of $Z_{V\sqcup V'}$. Furthermore, notice that it makes sense due to the commutativity of (5).

Example 5.2 (Feynman subfunctor)**.** Suppose given a Feynman functor $Z : C \to (\mathbf{A}, \tau)$. Precomposition defines a Feynman functor $Z' : C' \to (\mathbf{A}, \tau)$ for any subfunctor $C' \subset C$, and:

1. if Z is lax monoidal and $C' \subset C$ is proper, then Z' is lax monoidal. The same holds for oplax or strong monoidal;
2. if Z is complete and $C' \subset C$ is essentially injetive, then Z' is complete.

Our aim in this section is to show that when fixed a decomposition system in a prestack C, we can obtain Feynman functors $F : C \to (\mathbf{A}, \tau)$ constructively by means of following certain rules, called *Feynman rules*. So, let us begin by defining what we mean by a decomposition system.

Let $\mathbf{C} \subset \mathbf{Hyp}$ be a category of hypergraphs. A *decomposition system of order* n in $\mathbf{C}$ is a rule $\mathfrak{d}$ assigning to each $G \in \mathbf{C}$ decompositions $EG_j = EG_j^0 \sqcup EG_j^1 \sqcup ... \sqcup EG_j^n$, for each $j \geq 1$. In other words, we have a decomposition of the vertex set $VG = EG_1$ and of each set EG_j of j-edges into n parts. We do not require conditions on the adjacency functions ψ_j. We say that $v^0 \in EG_1^0$ and $e_j^0 \in EG_{j>1}^0$ are the *external vertices* and *external j-edges*, respectively. If $i > 0$, the elements v^i of EG_1^i and e_j^i of EG_j^i are called *internal vertices* and *internal j-edges of type i*.

We say that $\mathfrak{d}$ is *functorial* if each hypergraph morphism $f : G \to G'$ preserve the decomposition. It is straightforward to check that the choice of a functorial decomposition system induces an embedding $C_V \hookrightarrow \mathbf{S}_{\mathfrak{s}}$ for certain $\mathfrak{s}$. We define a functorial decomposition system $\mathfrak{d}$ for a prestack C as a rule that to any $V \subset X$ it assigns a functorial decomposition system $\mathfrak{d}_V$ for C_V.

Example 5.3 (trivial)**.** Each $\mathbf{C} \subset \mathbf{Hyp}$ possesses a trivial functorial decomposition system of arbitrary order: just define $VG^0 = VG$, $EG_{j>1}^0 = EG_{j>1}$ and the remaining pieces given by the empty set, i.e, $EG_j^i = \varnothing$ if $i > 0$.

Example 5.4 (incidence)**.** On the other hand, each $\mathbf{C}$ also possesses a nontrivial functorial decomposition of order 2. In fact, recall that to each hypergraph G we assign a bipartite graph IG: its *incidence graph*, whose set of vertices is $V_{IG} = VG \sqcup EG$, where $EG = \sqcup_j EG_j$. There exists a 2-edge between $x, x' \in V_{IG}$ iff $x \in VG$ and if $x' \in EG$ is some hyperedge adjacent to x. The rule $G \mapsto IG$ is functorial and actually an equivalence $\mathbf{Hyp} \simeq \mathbf{BipGrph}$ between hypergraphs and bipartite graphs [19]. In particular, each $\mathbf{C} \subset \mathbf{Hyp}$ is equivalent to some category of bipartite graphs. But bipartite graphs have, by definition, a decomposition of order 2. So, any $\mathbf{C}$ has an induced decomposition, which we denote by $\mathfrak{d}_{\mathfrak{i}}$. Furthermore, each prestack C can be endowed with this decomposition. Particicularly, each C can be embedded into a structured prestack $S_{\mathfrak{s}}$ for certain $\mathfrak{s}$.

Example 5.5 (Feynman)**.** For any V, the category $\mathbf{Fyn}_V$ (and, therefore the prestack) of Feynman graphs have a canonical functorial decomposition system of order 2, as explained in Example 2.3. Analogously for Feynman hypergraphs and generalized Feynman hypergraphs introduced in Example 2.4.

Example 5.6 (structured)**.** Any prestack $S_{\mathfrak{s}}$ of additively structured hypergraphs (e.g, those which are finitely structured - see Remark 2.2) can be endowed with a nontrivial functorial decomposition system of order 2. Indeed, define VG^0 as the subset of VG consisting of all vertices v such that

s1) $d_2(v) = 1$ and $d_{j>2}(v) = 0$;

s2) for every k, $\mathfrak{s}_{k,1}(v) = 0$;

s3) if exists j-edge between v and v', then v' is not in VG^0.

In other words, VG^0 is the collection of vertices that have trivial structure and that belong to a single 2-edge. Then take $VG^1 = VG - VG^0$. Conditions s1) and s2) establish a subset $EG_2^0 \subset EG_2$ and conditions s2) and s3) give us $|EG_2^0| = |VG^0|$. We then have a decomposition $EG_2 = EG_2^0 \sqcup EG_2^1$. Finally, for $j > 2$ define $EG_j^1 = EG_j$ and $EG_j^{i\neq 1} = \varnothing$. We will denote this decomposition system by $\mathfrak{d}_{\mathfrak{s}}$.

Remark 5.2. For future reference, we define a *standard decomposition* $\mathfrak{d}$ for a prestack as being one codifying the fundamental property of Examples 5.5 and 5.6

st1) the number of external vertices is equal to the number of external edges. In other words, $|VG^0| = \sum_j |EG_j^0|$;

st2) each j-edge in EG_j^0 is adjacent to at least one vertex in VG^0.

Remark 5.3. In our construction we will need to work with decompositions such that $|EG_j^0| \leq 1$, which cannot be satisfied by standard decompositions of order 2 with $|VG^0| > 1$. This leads us to consider decompositions such that $|EG_j^0| \leq 1$ and $|VG^0| \leq 1$. They will be called *normal.* If $\mathfrak{d}$ is a functorial normal decomposition of a prestack C we say that the pair $(C, \mathfrak{d})$ is *normal* and that $\mathfrak{d}$ is a *normal structure for* C. Starting with any pair $(C, \mathfrak{d})$, where $\mathfrak{d}$ is of order n, we can always build a normal pair $(\mathcal{N}C, \mathcal{N}\mathfrak{d})$, with $\mathcal{N}\mathfrak{d}$ also of order n, called the *normalization* of $(C, \mathfrak{d})$ and defined as follows: $\mathcal{N}C_V$ is the category of hypergraphs $\mathcal{N}_{\mathfrak{d}}G$ obtained from hypergrahs $G \in C_V$ by deleting VG^0 and EG_j^0, and consider the decompositon $\mathcal{N}\mathfrak{d}_V$ given by

$$V\mathcal{N}_{\mathfrak{d}}G^0 = \varnothing = E\mathcal{N}_{\mathfrak{d}}G_j^0, \quad V\mathcal{N}_{\mathfrak{d}}G^{i>0} = VG^{i>0}, \quad \text{and} \quad E\mathcal{N}_{\mathfrak{d}}G_j^{i>0} = EG_j^{i>0}.$$

Furthermore, any prestack C admits a normal structure $\mathfrak{d}_0$ of order 2: just take $VG^0 = \varphi = EG_j^0$ and $VG^1 = VG$ and $EG_j^1 = EG_j$.

Remark 5.4. From Example 5.4 any prestack C admits a decomposition $\mathfrak{d}_{\mathfrak{i}}$, implying $C \subset S_{\mathfrak{s}}$. By Example 5.6 C can then be endowed with a new decomposition $\mathfrak{d}_{\mathfrak{s}}$, which is standard by the last remark. In other words, each C admits a nontrivial standard decomposition.

Let $(C, \mathfrak{d})$ be a prestack of hypergraphs endowed with a (non necessarily functorial) decomposition system $\mathfrak{d}$, say of order n, let $(\mathbf{A}, \otimes, \cdot^{\vee})$ be a pseudocontext and let $\tau : \mathbf{FinSet} \to \mathbf{A}$ be a functor. A *Feynman rule* in $(C, \mathfrak{d})$ with coefficients in $(\mathbf{A}, \tau)$ is a rule FR that to each $V \in \mathbf{FinSet}$ and each hypergraph $G \in C_V$ it assigns:

1. functions $r^i : VG^i \to \mathbb{Z}_{\geq 0}$ and $r_j^i : EG_j^i \to \mathbb{Z}_{\geq 0}$, with $0 \leq i \leq n$ and $j \geq 1$, called the *degrees* or *weights* of the decomposition $\mathfrak{d}$;
2. functors assigning to each external vertice, etc., a tensor in $\tau VG \in \mathbf{A}$ or $(\tau VG)^{\vee} \in \mathbf{A}$ of corresponding degree. Precisely, functors
$$FR^i : VG^i \to 1/(\tau(VG)^{\vee})^{\otimes_{r^i}} \quad \text{and} \quad FR_j^i : EG_j^i \to 1/(\tau VG)^{\otimes_{r_j^i}},$$
regarding VG^i and EG_j^i as discrete categories, while
$$1/(\tau(VG)^{\vee})^{\otimes_{r^i}} \quad \text{and} \quad 1/(\tau VG)^{\otimes_{r_j^i}}$$
denotes the full subcategories of $1/\mathbf{A}$ consisting of all morphisms
$$1 \to 1/(\tau(VG)^{\vee})^{\otimes_{r^i(v^i)}} \quad \text{and} \quad 1/(\tau VG)^{\otimes_{r_j^i(e_j^i)}},$$
for every $v^i \in VG^i$ and $e_j^i \in EG_j^i$, respectively. Recall that $1/U^0$ is the category with a single object id_1.

Example 5.7 (classic Feynman rules)**.** Let C be any prestack endowed with a functorial decomposition $\mathfrak{d}$. Let $(\mathbf{A},\otimes,\cdot^{\vee})$ be any pseudocontext endowed with any functor $\tau:\mathbf{FinSet}\to\mathbf{A}$. An usual setup for Feynman rules $FR:(C,\mathfrak{d})\to(\mathbf{A},\tau)$ is to take for each $G\in C_V$ the degree functions as $r^0\equiv 0$, $r^0_j\equiv 1$, $r^{i\geq 0}_j\equiv j$ and $r^{i\geq 0}(v)=d(v)=\sum_j d_j(v)$, so that

$$FR^0:VG^0\to 1/(\tau(VG)^{\vee})^0$$

must be the constant functor in id_1 (i.e, it must assign the trivial tensor in $\tau(VG)$ to each external vertice). Furthermore, to each hypergraph G consider two formal parameters t,o such that if G is b-bounded and $\mathfrak{d}$ has order n, then $t^j=0$ for $j>b$ and $u^i=0$ for $i>n$. Take the object $\tau(VG)[[t,o]]$ of formal power series in t, which decomposes as a product $\prod_{j,i}X_{i,j}o^i t^j$, with $X_{i,j}\simeq\tau(VG)$. Then define FR^0_j as any rule assigning a generalized element $FR^0_j(e)$ of degree $(0,j)$ in $\tau(VG)$, i.e, as a morphism $1\to\tau(VG)[[t,o]]$ which actually take values in $X_{0,j}$. Moreover, define FR^i_j as any map assigning j-propagators in $\tau(VG)$ to internal j-edges. More precisely, recall that $P\tau(VG)[[t,o]]=\prod_{k,i,j}P^k_{i,j}o^i t^j$, with $P^k_{i,j}\simeq\mathrm{Sym}^k(\tau(VG))$, and take $FR^i_j(e)\in 1/P^j_{i,j}$. Finally, take $FR^{i\geq 0}$ as any rule assigning to each internal vertex v^i a corresponding interacting term in $\tau(VG)^{\vee}$, relative to t, of degree $d(v)$. In other words, for each v we have a morphism $FR^i(v):1\to\mathcal{O}\tau(VG)[[t]]$, where $\mathcal{O}\tau(VG)[[t,o]]=\prod_{k,i,j}I^k_{i,j}o^i t^j$, with $I^k_{i,j}\simeq\mathrm{Sym}^k(\tau(VG)^{\vee})$, which actually take values in $I^{d(v)}_{i,1}$.

Example 5.8 (structured Feynman rules)**.** We can particularize and consider C as a prestack $S_{\mathfrak{s}}$ of $\mathfrak{s}$-structured hypergraphs. Furthermore, we can take $\tau:\mathbf{FinSet}\to\mathbf{A}$ as constant in some object U, called *background object*, so that all data assigned by FR are tensors on U. We can also use $\mathfrak{s}$ to require more properties on $FR^{i\geq 0}$, as follows. To each $s_{k,j}$ we consider a new formal parameter $h_{k,j}$, so that

$$\mathcal{O}\tau(VG)[[\mathfrak{h}]]=\prod_{k,i,j,l_{1,1},l_{1,2},\dots}I^k_{i,j,l_{1,1},l_{2,1},\dots}t^j o^i h_{1,1}^{l_{1,1}}h_{1,2}^{l_{1,2}}\dots$$

Then, for a given (G,ϵ) a $\mathfrak{s}$-structure graph, take $FR^i(v)$ as a generalized element in $I^{d(v_i)}_{i,1,\epsilon_{1,1}(v),\epsilon_{2,1}(v),\dots}$.

Example 5.9 (physicists Feynman rules)**.** Let us now particularize even more and consider C as the concrete prestack $C_V=\mathbf{Fyn}_V$ of Feynman graphs of Example 2.3 endowed with the functorial decomposition of Example 5.5, so that $b=2$ and $n=1$. Also, we have two nontrivial structures $s_{1,1}=\mathbb{Z}_2$ and $s_{2,1}=\mathbb{Z}_{\geq 0}$, corresponding to the decomposition and the genus map, so that we have two parameters $h_{1,1}$ and $h_{2,1}$. In this particular setting, let us denote $h_{2,1}\equiv\hbar$. Furthermore, let us consider the pseudocontext of Example 4.2, allowing us to work with **LCS** instead of with $\mathbb{R}/$**LCS**, as explained in Example 5.10. Thus, according to the previous example, fixed a background object U, we get Feynman rules by giving functions $FR^0_2:EG^0\to U$, $FR^1_2:EG^1\to P^2U$ and $FR^1:VG^1\to\mathcal{O}U[\hbar]$ such that $FR^1(v)\in I^{d(v)}_{g(v)}$. In turn, these functions are exactly the data defining the Feynman rule of a perturbative QFT as will be more detailed explored in Section 7.

Suppose now that the decomposition system $\mathfrak{d}$ in C is functorial. In this case, for any Feynman rule FR in $(C,\mathfrak{d})$, given a morphism $f:G\to G'$ we get the induced diagrams below (without the segmented arrow). We say FR is *functorial* when for each f there exists the dotted arrows $r^i f$ and $r^i_j f$ completing these diagrams (of functors between discrete categories) in a commutative way. As we will see, if we assume Axiom of Choice, any Feynman rule becomes functorial.

$$\begin{tikzcd} VG^i \arrow[r, "FR^i"] \arrow[d, "f^i"'] & 1/(\tau VG^\vee)^{\otimes_{r^i}} \arrow[d, dashed, "r^i f"] & EG^i_j \arrow[r, "FR^i_j"] \arrow[d, "f^i_j"'] & 1/(\tau VG)^{\otimes_{r^i_j}} \arrow[d, dashed, "r^i_j f"] \\ VG'^i \arrow[r, "FR'^i"'] & 1/(\tau VG'^\vee)^{\otimes_{r'^i}} & EG'^i_j \arrow[r, "FR^i_j"'] & 1/(\tau VG')^{\otimes_{r'^i_j}} \end{tikzcd} \tag{8}$$

Let $\mathbf{A}$ be a category with a distinguished object $1 \in \mathbf{A}$. Let $\mathbb{D}$ be a collection of categories. We say that $(\mathbf{A}, 1)$ has the *Hahn-Banach property* (or simply that it is Hahn-Banach) *relative to* $\mathbb{D}$ when for any $\mathbf{D}, \mathbf{D}' \in \mathbb{D}$, any full subcategoriesb$\mathbf{C}, \mathbf{C}' \subset 1/\mathbf{A}$ and any functors $F : \mathbf{D} \to \mathbf{D}'$, $G : \mathbf{D}' \to \mathbf{C}'$ and $H : \mathbf{D} \to \mathbf{C}$, the corresponding e admit an extension in **Cat** relatively to any functor H, as in diagram (9). There, $e \circ m$ denotes the mono-epi decomposition of $G \circ F$ obtained by taking its image in **Cat**. We say that $(\mathbf{A}, 1)$ is *Hahn-Banach relative to a functor* $\sigma : \mathbf{FinSet} \to \mathbf{Cat}$ if it is Hahn-Banach relative to $\mathbb{D} = \mathrm{img}(\sigma)$.

$$\begin{tikzcd} \mathbf{D} \arrow[r, "H"] \arrow[dr, "e"] \arrow[dd, "F"'] & \mathbf{C} \arrow[d, dashed, "\hat{e}"] \\ & \mathrm{img}(G \circ F) \arrow[d, "m"] \\ \mathbf{D}' \arrow[r, "G"'] & \mathbf{C}' \end{tikzcd} \tag{9}$$

Example 5.10 (locally convex spaces)**.** Let **LCS** be the category of lcs endowed with the real line $\mathbb{R}$ as a distinguished object. We assert that it is Hahn-Banach relative to the subcategory $\mathbf{FinVec}_{\mathbb{R}}$ of finite-dimensional vector spaces. With the inductive topology in $\otimes$, $(\mathbf{LCS}, \otimes, \mathbb{R})$ is closed monoidal, whose internal hom $[U; U']$ between U and U' is given by $B(U; U')$ endowed with the topology of pointwise convergence. Therefore, $[\mathbb{R}; U] \simeq U$ and we can work directly with **LCS** instead of $\mathbb{R}/\mathbf{LCS}$. Since a continuous linear map taking values in a finite-dimensional vector space is the same thing as a finite combination of linear functionals, the Hahn-Banach condition above is a direct consequence of Hahn-Banach theorem for lcs [30, 31]. The inclusion $\mathbf{Vec}_{\mathbb{R}} \hookrightarrow \mathbf{Set}$ is reflective. Let $\mathbb{R}[\cdot] : \mathbf{FinSet} \to \mathbf{FinVec}_{\mathbb{R}}$ be the restriction of the the left adjoint to the category of finite sets and let $\imath : \mathbf{FinVec}_{\mathbb{R}} \hookrightarrow \mathbf{LCS}$ be the inclusion. We can verify that **LCS** is Hahn-Banach relative to $\imath \circ \mathbb{R}[\cdot]$.

Example 5.11 (full subcategories)**.** If $(\mathbf{A}, 1)$ is Hahn-Banach, then $(\mathbf{C}, 1)$ is also for every full subcategory $\mathbf{C} \subset \mathbf{A}$. So, in particular, the pseudocontexts from Example 4.3 and Example 4.4 define Hahn-Banach pairs.

Now, the fundamental fact is that, as a consequence of the axiom of choice, any pair $(\mathbf{A}, 1)$ is Hahn-Banach relative to the functor regarding finite sets as discrete categories. Indeed, since the domain category is discrete we can only work with objects, so that the extension problem (9) is equivalent to a problem is **Set** which has a solution, since e is an epimorphism and we are assuming Axiom of Choice.

Thus, if $FR : (C, \mathfrak{d}) \to (\mathbf{A}, \tau)$ is any Feynman rule, the following dotted arrows exist, so that

FR is a functorial Feynman rule for $r^i f = m^i \circ \hat{e}^i$ and $r^i_j f = m^i_j \circ \hat{e}^i_j$.

$$
\begin{tikzcd}
VG^i \arrow[r, "FR^i"] \arrow[dr, "e^i"'] \arrow[dd, "f^i"'] & 1/(\tau(VG)^\vee)^{\otimes_{r^i}} \arrow[d, dashed, "\hat{e}^i"] \\
 & \mathrm{img}(FR'^i \circ f^i) \arrow[d, "m^i"] \\
VG'^i \arrow[r, "FR'^i"'] & 1/(\tau(VG')^\vee)^{\otimes_{r^i}}
\end{tikzcd}
\qquad
\begin{tikzcd}
EG^i_j \arrow[r, "FR^i_j"] \arrow[dr, "e^i_j"'] \arrow[dd, "f^i_j"'] & 1/(\tau(VG)^\vee)^{\otimes_{r^i_j}} \arrow[d, dashed, "\hat{e}"] \\
 & \mathrm{img}(FR'^i_j \circ f^i_j) \arrow[d, "m^i_j"] \\
EG'^i_j \arrow[r, "FR'^i_j"'] & 1/(\tau(VG')^\vee)^{\otimes_{r^i_j}}
\end{tikzcd}
$$

5.1 Existence

We can now prove that functorial Feynman rules induce monoidal Feynman functors. Actually, we will need to work with Feynman rules $FR : (C, \mathfrak{d}) \to (\mathbf{A}, \tau)$ which are *coherent* in the sense that the degrees r^i and r^i_j of $\mathfrak{d}$ satisfy the following coherence equation:

$$\sum_{i \geq 0} \sum_{j > 2} \sum_{e^i_j} r^i_j(e^i_j) = \sum_{i \geq 0} \sum_{v_i} r^i(v_i). \tag{10}$$

Example 5.12 (standard Feynman rules)**.** We say that a Feynman rule is *standard* if the decomposition system $\mathfrak{d}$ is standard (in the sense of Remark 5.2) and if the degrees r^i and r^i_j are given by $r^0 \equiv 1$, $r^{i>0}(v^i) = d_i(v^i)$, $r^0_j \equiv 1$ and $r^{i>0}_j(e^i_j) = j$. For instance, the classic Feynman rules are standard. We assert if FR is standard then it is coherent. Due to the structure of the degrees, for any hypergraph $G \in C_V$, the left-hand side and the right-hand side (10) writes, respectively

$$\sum_{j>2} |EG^0_j| + \sum_{i>0} \sum_{j>2} j |EG^i_j| \quad \text{and} \quad |VG^0| + \sum_{i>0} \sum_{v_i} r^i(v_i).$$

Since $\mathfrak{d}$ is standard, the first term of both sides coincide. Let $G_{i>0}$ be the hypergraph obtained by deleting VG^0. The formula (1) applied to it shows that the second terms also coincide.

Theorem 5.1. *Let $(\mathbf{A}, \otimes, \cdot^\vee)$ be a pseudocontext endowed with a functor $\tau : \mathbf{FinSet} \to \mathbf{A}$ and let C^b be a prestack of b-bounded hypergraphs. Suppose one of the following conditions:*

c1) *$C^b \subset S^{\ell,b}$, i.e, our hypergraphs are labeled;*

c2) *the monoidal category $(\mathbf{A}, \otimes, 1)$ is strict.*

Then any coherent Feynman rule $FR : (C^b, \mathfrak{d}) \to (\mathbf{A}, \tau)$ defines a Feynman functor $Z : C^b \to (\mathbf{A}, \tau)$.

Proof. Assuming condition c1), let (G, φ) be a labeled b-bounded hypergraph in $C^b_V \subset S^{\ell,b}$. Let n be the order of $\mathfrak{d}$. The decompositions $VG = \sqcup_i V^i G$ and $E_{j>1}G = \sqcup_i E^i_j G$ with $0 \leq i \leq n$ defined by $\mathfrak{d}$, induce decompositions for the bijections φ_j. Without loss of generality we can assume that now we have $\varphi^i_1 : [|V^i G|] \to V^i G$ and $\varphi^i_{j>1}$. Since morphisms of labeled hypergraphs preserve the labelings, they will preserve these decomposed labelings. With this in mind, if $FR : (C^b, \mathfrak{d}) \to (\mathbf{A}, \tau)$ is a Feynman rule, for each set $V \subset X$ and each labeled hypergraph $(G, \varphi) \in C_V$, define $Z^i_V(G)$ as the following tensor product (in $1/\mathbf{A}$)

$$Z^i_V(G) = (...((FR^i(\varphi^i(1)) \otimes FR^i(\varphi^i(2))) \otimes ....) \otimes FR^i(\varphi^i(|V^i G|)). \tag{11}$$

In analogous way, define $Z^i_{j,V}(G)$ for every $1 < j \leq b$, i.e, by taking the tensor product in $1/\mathbf{A}$ of the images of FR^i_j, following the order given by the labelings φ^i_j. Furthermore, define

$$Z^I_V(G) = (...((Z^0_V(G) \otimes Z^1_V(G)) \otimes Z^2_V(G))...) \otimes Z^n_V(G) \tag{12}$$

and similarly $Z^i_{J,V}(G)$ for each $0 < i \leq n$. Then take

$$Z_V(G) = (...((Z^I_V(G) \otimes Z^1_{J,V}(G)) \otimes Z^2_{J,V}(G))...) \otimes Z^n_{J,V}(G). \tag{13}$$

From the coherence equation (10) we see that $Z_V(G) \in \mathbf{A}_{\tau VG}$. Therefore, we have a rule $G \mapsto Z_V(G)$ assigning to each hypergraph with vertex set VG an analytic expression in $\tau VG \in \mathbf{A}$. Its functoriality follows from the functoriality of the Feynman rules FR. Indeed, if $f : (G,\varphi) \to (G',\varphi')$ is a morphism in C^b_V, we have the morphisms $r^i f$ anf r^i_j in (8). Since f preserves the decomposed labelings, $r^i f$ induce morphisms $Z^i_V(f) : Z^i_V(G) \to Z^i_V(G')$ and analogously for r^i_j. Define $Z^I_V(f)$ by replacing the symbol "G" with the symbol "f" in (12) and define $Z^i_{J,V}(f)$ similarly. Finally, replacing "G" with "f" in (13) we get the desired morphism $Z_V(f) : Z_V(G) \to Z_V(G')$. It is straightforward to check that composition and identity morphisms are preserved, so that we have a Feynman functor $Z : C \to (\mathbf{A},\tau)$. Finally, notice that condition c1) was used only to fix an ordering in the tensor products (11). Suppose now that c2) is satisfied. Then $\mathbf{A}$ is strict, so that we can remove the parenthesis of tensor products, meaning that we do not need to care about ordering. So, the same construction works even without labelings. □

[of the proof] A Feynman functor induced by a Feynman rule is never trivial unless the prestack C^b is trivial.

Proof. Just look at the defining expressions (11), (12) and (13) and use the coherece equation. □

We say that two Feynman rules $Z : C \to (\mathbf{A},\tau)$ and $Z' : C' \to (\mathbf{A}',\tau')$ *differ by a change of coefficients* if they are defined in the same prestack, i.e, if $C' = C$, and if there exists an isomorphism $F : (\mathbf{A},\otimes,\cdot^\vee) \to (\mathbf{A}',\otimes',\cdot^{\vee'})$ of pseudocontexts and a natural isomorphism $\zeta : \tau \simeq \tau'$ such that for every hypergraph $G \in C_V$ we have $F_\zeta \circ Z = Z'$, where $F_\zeta : \mathbf{A}_{\tau VG} \simeq \mathbf{A}'_{\tau' VG}$ is the equivalence induced by F and ζ (this equivalence exists, since by Proposition 4.2 morphisms of pseudocontexts induce functors between the categories of analytic expressions).

Proposition 5.1. *Up to change of coefficients, there exists a nontrivial Feynman functor $Z : C \to (\mathbf{A},\tau)$ defined in any nontrivial proper prestack C (not necessarily bounded) and taking values in any pseudocontext $(\mathbf{A},\otimes,\cdot^\vee)$ endowed with any functor $\tau : \mathbf{FinSet} \to \mathbf{A}$.*

Proof. From Example 5.4 any C is subprestack of $S^2_\mathfrak{s}$ for some $\mathfrak{s}$. By Example 5.6 and by Remark 5.2, $S^2_\mathfrak{s}$ (and therefore C) can be endowed with a standard decomposition system $\mathfrak{d}$, and by Example 5.7 we have a Feynman rule (the classical one) $FR : (S^2_\mathfrak{s},\mathfrak{d}) \to (\mathbf{A},\tau)$ with coefficients in any $(\mathbf{A},\tau)$, which induces a Feynman rule in $(C,\mathfrak{d})$. From Example 5.12 this Feynman rule is coherent, so that if $\mathbf{A}$ is strict, then by Theorem 5.1 FR induces a Feynman functor $Z : C \to (\mathbf{A},\tau)$. But, since we are working up to change of coefficients, Mac Lane's strictification theorem for monoidal categories allows us to forget the strictness hypothesis on $(\mathbf{A},\otimes,1)$. Nontrivialily of Z is due Corollary 5.1. □

It would be interesting to find conditions on Feynman rules under which the induced Feynman functors are monoidal and/or complete. We start by analyzing completeness. Let $(\mathbf{C},\otimes,1)$ be a monoidal category and let $F : \mathbf{I} \to \mathbf{C}$ be some functor. We say that F is *decomposable* if for any isomorphism $f : I \to I'$ in $\mathbf{I}$ and any decompositions $F(I) \simeq X \otimes Y$ and $F(I') \simeq X' \otimes Y'$ there are isomorphisms $\alpha : X \simeq X'$ and $\beta : Y \simeq Y'$ in $\mathbf{C}$ such that $F(f) = \alpha \otimes \beta$, i.e, such that the diagram below commutes. If $(\mathbf{C}, \otimes, \vee)$ is actually a pseudocontext, we require decompositions not only for $F(I)$, but also for each $F(I)^{\otimes_r}$ and $(F(I)^{\vee})^{\otimes_r}$. We say that F is 1/decomposable if the induced functor $1/F : \mathbf{I} \to 1/\mathbf{C}$ is decomposable.

$$\begin{tikzcd}
F(I) \arrow[r, "\simeq"] \arrow[d, "F(f)"'] & X \otimes Y \arrow[d, "\alpha\otimes\beta"] \\
F(I') \arrow[r, "\simeq"'] & X' \otimes Y'
\end{tikzcd}$$

We say that a Feynman rule $Z : (C, \mathfrak{d}) \to (\mathbf{A}, \tau)$ is *complete* if for each hypergraph G the maps FR^i_j are equivalences over their images. This means that different internal vertices and different external and internal hyperedges have different tensorial representations. If, in addition, FR^i are also equivalences over their images, we say that Z is *strongly complete*.

Proposition 5.2. *In the same notations of Theorem 5.1, let $FR : (C^b, \mathfrak{d}) \to (\mathbf{A}, \tau)$ be a coherent Feynman rule such that τ is 1/decomposable and suppose one of the following conditions:*

- c1) *$C^b \subset S^{b,\ell}$ is concrete and FR is complete;*
- c2) *$C^b \subset S^{b,\ell}$ is FR is strongly complete;*
- c3) *$(\mathbf{A}, \otimes, 1)$ is strict and FR is strongly complete.*

Then FR induces a Feynman functor which is also complete.

Proof. It is clear that the induced monoidal Feynman functor exists, since conditions c1), c2) and c3) above contains the conditions c1) and c2) of Theorem 5.1. Let us show that the induced Feynman functor is complete. We will work first with c1). Fixed V, given a morphism $f : G \to G'$ in C^b_V, suppose that $Z_V(f) : Z_V(G) \to Z_V(G')$ is an isomorphism. From the definition of $Z_V(f)$ and the fact that τ is 1/decomposable we find that $r^i f$ and r^i_j, with $i = 0, 1..., n$ and $1 < j \leq b$, are all isomorphisms. Because FR is complete, each $FR^i_{j>1}$ is an equivalence over its image. From the commutativity of diagrams (8) we then see that f^i_j are bijections. Since the prestack is concrete, we are working with hypergraphs over the same vertex set. Furthermore, since the hypergraphs are labeled, from Remark 2.3 we can assume $f_V : V \to V$ equal to the identity id_V. Then each f^i is also a bijection, so that f is a hypergraph isomorphism. Assume c2) or c3). Then now FR is strongly complete, so that not only $FR^i_{j>1}$, but also FR^i, are equivalences over their images, implying directly that f^i_j and f^i are bijections. □

Let $(\mathbf{A}, \otimes, \cdot^{\vee})$ be a pseudocontext and let τ 1/decomposable. In this case:

- *c1)* if $C \subset S^{\ell,b}_{\mathfrak{s}}$ is bounded, labeled, structured, then for any normal structure $\mathfrak{d}$ in C there exists a nontrivial complete Feynman functor $Z : C \to (\mathbf{A}, \tau)$;
- *c2)* if C is arbitrary, then for any normal structure $\mathfrak{d}$ in C such a Feynman functor also exists, but only up to a change of coefficients.

Proof. From Example 5.7, if $C \subset S^b_{\mathfrak{s}}$, then there exist classic Feynman rules in C. Notice that whenever classic rules of Feynman exist, we can modify them in order to assume that $FR^{i>0}_j$ and $FR^{i>0}$ are equivalences over their images. If $(C, \mathfrak{d})$ is normal, the domains of FR^0 and FR^0_j are the empty category or the point category, so that both functors are automatically equivalences over their images. Therefore, FR becomes strongly complete. With this in mind, c1) and c2) follow, respectivly, from conditions c2) and c3) of Proposition 5.2. □

Recall from Example 5.3 that we have a rule $\mathcal{N}$ assigning to each pair $(C, \mathfrak{d})$ its normalization $(\mathcal{N}_{\mathfrak{d}} C, \mathcal{N}\mathfrak{d})$. We say that some assertion about prestacks with decompositions holds *up to* *normalization* if it holds for every $\mathcal{N}_{\mathfrak{d}}C$. Up to normalization and change of coefficients, there exists a nontrivial complete Feynman functor defined in any pair $(C,\mathfrak{d})$ and taking values in any pair $(\mathbf{A},\tau)$ with τ 1/decomposable.

Proof. Straightforward from condition c2) of Corollary 5.1. $\square$

Remark 5.5. The existence result for complete Feynman functors is much less general than that for Feynman functors: for the complete case we need to work with decomposable functors $\tau:\mathbf{FinSet}\to\mathbf{A}$. This hypothesis on τ cannot be avoided because we have no analogue of Mac Lane's strictification theorem establishing that any $(\mathbf{A},\tau)$ is equivalent to other $(\mathbf{A}',\tau')$ with a τ' 1/decomposable.

We will now discuss the monoidal property. Let (C,ξ) be a proper prestack endowed with a functorial decomposition $\mathfrak{d}$. We say that a Feynman rule $FR:(C,\mathfrak{d})\to(\mathbf{A},\tau)$ is *oplax monoidal* if for any every $V,V'\subset X$ and for every $G\in C_V$ and $G'\in C_{V'}$ we have functors

$$FR^i(G,G'):V\xi(G,G')^i \;\to\; FR^i(VG^i)\otimes FR^i(VG'^i) \tag{14}$$
$$FR^i_j(G,G'):E\xi(G,G')^i_j \;\to\; FR^i_j(EG^i_j)\otimes FR^i_j(EG'^i_j), \tag{15}$$

and also $FR^i(\varnothing)\to id_1$ and $FR^i_j(\varnothing)\to id_1$, fulfilling comonoid-like diagrams analogous to (7). Lax monoidal and strong monoidal Feynman rules are defined in a similar way.

Example 5.13 (concrete proper)**.** Let (C,ξ) be a concrete proper prestack. We say that a functorial decomposition $\mathfrak{d}$ in C is *compatible* with ξ if

$$V\xi(G,G')^i=V\xi(G,\varnothing)^i\sqcup V\xi(\varnothing,G')^i \;\text{ and }\; E\xi(G,G')^i_j=E\xi(G,\varnothing)^i_j\sqcup E\xi(\varnothing,G')^i_j \tag{16}$$

for $0\leq i\leq n$ and $j>1$, where n is the order of $\mathfrak{d}$. For instance, if C is concretely proper, which means that $\xi(G,G')=G\sqcup G'$, then the condition above becomes

$$V(G\sqcup G')^i=VG^i\sqcup VG'^i \;\text{ and }\; E(G\sqcup G')^i_j=EG^i_j\sqcup EG'^i_j.$$

We notice that any Feynman rule $FR:(C,\mathfrak{d})\to(\mathbf{A},\tau)$, defined in a concrete proper prestack endowed with a compatible decomposition, is oplax monoidal in a unique way. Indeed, notice that since we have (16), the maps (14) and (15) we are looking for will be defined in a coproduct. But maps defined in a coproduct are uniquely determined by its components. So, $FR^i(G,G')$ exists and it is totally determined by $FR^i(G)$ and $FR^i(G')$.

Proposition 5.3. *Under the same notations and hypotheses of Theorem 5.1, if C^b is proper and the coherent Feynman rule is oplax (resp. lax or strong) monoidal, then the induced Feynman functor is oplax (resp. lax or strong) monoidal.*

Proof. We will prove only the oplax case assuming condition c2) in Theorem 5.1. The other cases (lax, strong and condition c1) are analogous. So, let $FR:(C^b,\mathfrak{d})\to(\mathbf{A},\tau)$ be a coherent Feynman rule with $\mathbf{A}$ strict and FR oplax, and let Z be the induced Feynman rule. Notice that, by definition $Z_{V,V'}(\xi(G,G'))$ is the tensor product between $Z^i(\xi(G,G'))$ and $Z^i_j(\xi(G,G'))$, for $0\leq i\leq n$, where here we omit the subindices V,V' in order to simplify the notation (see expressions (13) and (12)). In turn, $Z^i(\xi(G,G'))$ is the tensor product between $FR^i(v)$, with $v\in V\xi(G,G')^i$, while $Z^i_j(\xi(G,G'))$ is the tensor product of $FR^i_j(e)$, for $e\in E\xi(G,G')^i_j$. Since FR is oplax, we have the maps (14), (15) and also $\nu:FR(\varnothing)\to id_1$. Therefore, tensoring $FR^i(v)$ for every v and $FR^i_j(e)$ for every e we get maps $Z^i(\xi(G,G'))\to Z^i(G)\otimes Z^i(G')$ and $Z^i_j(G,G')\to Z^i_j(G)\otimes Z^i_j(G')$. Tensoring them and varying i and j, we obtain a map

$$\mu_{V,V'}:Z_{V,V'}(\xi(G,G'))\to Z_V(G)\otimes Z_{V'}(G').$$

Furthermore, by definition of Z we see that ν induces another $\nu : Z_{\varnothing}(\varnothing) \to id_1$. These maps will satisfy the comonoid-like diagrams precisely because they are finite tensor products of maps that satisfy the diagrams. $\square$

Up to change of coordinates, there exists a nontrivial oplax monoidal Feynman functor defined in any concrete proper prestack C and taking values in any pseudocontex $\mathbf{A}$ endowed with any functor $\tau : \mathbf{FinSet} \to \mathbf{A}$.

Proof. Direct consequence of Example 5.8, Example 5.13 and Proposition 5.3. $\square$

Remark 5.6. We could think of getting a general existence result for Feynman functors which are simultaneously complete and oplax monoidal. Corollary 5.1 tell us that complete Feynman functors exist with the hypothesis that C is endowed with a normal structure. This condition was avoided in Corollary 5.1 by making use of a normalization process. Similarly, Corollary 5.1 states that oplax monoidal Feynman functors exist if C is concrete proper. So, we could try to build some kind of "concretization" and "propertification" processes, allowing us to say that up to them oplax Feynman functors always exist. The fundamental fact is that, even if these processes are built, we **cannot** use them to say that *up to normalization, "propertification" and "concretification" complete oplax Feynman functor always exist.* This happens because the normalization of an arbitrary nontrivial prestack cannot be concrete. Indeed, recall that being concrete means that $C_V \in \mathbf{Hyp}_V$ for any V. However, the normalization $\mathcal{N}C$ is such that $V\mathcal{N}_{\mathfrak{d}}G^0 = \varnothing$, so that $\mathcal{N}C_V \notin \mathbf{Hyp}_V$ unless the initial decomposition $\mathfrak{d}$ coincide with $\mathcal{N}\mathfrak{d}$, i.e, unless $(C, \mathfrak{d})$ is normal.

5.2 Uniqueness

Closing our discussion on Feynman functors, let us focus on the uniqueness problem. We will show that two complete Feynman functors are always conjugated in a suitable way.

We say that a functor $\alpha : \mathbf{C} \to \mathbf{C}'$ is *quasi essentially injective* (qei) if it is constant in a subcategory $c \subset \mathbf{C}$ and essentially injective in the remaining $\mathbf{C} - c$. We say that $F : \mathbf{C} \to \mathbf{D}$ is *quasi essentially injectively conjugated* (qeic) to another functor $F' : \mathbf{C}' \to \mathbf{D}'$ if there are qei functors α and β making the following square commutative up to natural isomorphisms:

$$
\begin{tikzcd}
\mathbf{C} \arrow[r, "F"] \arrow[d, "\alpha"'] & \mathbf{D} \arrow[d, "\beta"] \\
\mathbf{C}' \arrow[r, "G"'] \arrow[ur, Rightarrow, "\simeq"] & \mathbf{D}'
\end{tikzcd}
\tag{17}
$$

Lemma 5.1. *Let $\mathbf{D}$ a category with null objects. If a functor $F : \mathbf{C} \to \mathbf{D}$ is essentially injective and faithful, then it is qeic to any essentially injective functor $G : \mathbf{C} \to \mathbf{D}$.*

Proof. Set $\alpha = id_{\mathbf{C}}$ and for every $Y \in \mathbf{D}$ define

$$
\beta(Y) = \begin{cases} G(F^{-1}(Y)), & Y \in \operatorname{img} F \\ 0, & \text{otherwise.} \end{cases}
$$

Since F is essentially injective, this is well defined up to natural isomorphisms using the axiom of choice. Let $f : Y \to Z$ be a morphism in $\mathbf{D}$. If Y do not belong to the image of F, define $\beta(f) : \beta(Y) \to \beta(Z)$ as the unique map $0 \to \beta(Z)$. In a similar way define $\beta(f)$ when Z (or both Y and Z) do not belong to $F(\mathbf{C})$. Finally, notice that since F is essentially injective and faithful, it is an equivalence over its image, so that if both Y, Z belong to $F(\mathbf{C})$, then to any $f : Y \to Z$ corresponds a unique $F^{-1}(f)$. In this case, define $\beta(f) = G(F^{-1}(f))$. It is straightforward to check the functorial properties of G. By definition, β is essentially injective when restricted to $F(\mathbf{C})$ and constant (equal to the null object) in the remaining part. Thus, both α and β are qei. Furthermore, by construction the diagram (17) commutes up to isomorphisms, giving the desired conjugation and completing the proof. $\square$

We say that a monoidal category $(\mathbf{A}, \otimes, 1)$ is *τ-faithful,* where $\tau : \mathbf{FinSet} \to \mathbf{A}$ is a given functor, if the induced monoidal product in $1/\mathbf{A}$ is faithful in both variables when restricted to the image of $1/\tau$, i.e, if $f \otimes g = f' \otimes g'$ implies $f = f'$ and $g = g'$ for every $f, g, f', g' \in 1/\tau(\mathbf{FinSet})$.

Proposition 5.4. *Under the same notations of Proposition 5.2, suppose that $(\mathbf{A}, \otimes, 1)$ is τ-faithful and τ is $1/$decomposable. In this case, if $Z : C^b \to (\mathbf{A}, \tau)$ is the Feynman rule induced by a strongly complete Feynman rule FR fulfilling condition* c2) *or condition* c3)*, then it is qeic to any other complete Feynman functor $Z' : C^b \to (\mathbf{A}, \tau)$.*

Proof. Since Z is a complete Feynman functor, Z_V is essentially injective for every V. Therefore, by the previous lemma it is enough to prove that Feynman functors induced by strongly complete Feynman rules taking values into a τ-faithful are also faithful, i.e, are such that if $Z_V(f) = Z_V(g)$, then $f = g$. In order to do this we need to work case by case of Proposition 5.2. We will give the proof only for case c3). Case c2) is analogous, needing only some care with the parentheses. By definition, $Z_V(f)$ is a tensor product between $Z^i_V(f)$ and $Z^i_{V,j}(f)$, with $1 < j \leq b$ and $0 \leq i \leq n$. In turn, $Z^i_V(f)$ is the map between $\otimes_v FR^i(v)$ and $\otimes_w FR'^i(w)$, with $v \in VG^i$ and $w \in VG'^i$, given by the restriction of $\otimes_v r^i f$. Furthermore, $Z^i_{V,j}(f)$ is obtained in a similar way as a restriction of $\otimes_e r^i_j f$ with $e \in EG^i_j$. Therefore, $Z_V(f)$ is the restriction of $\otimes_{i,j} \otimes_{v,e} r^i f \otimes r^i_j f$ to an object depending only of G (and not of f). So, if $Z_V(f) = Z_V(g)$, then

$$\otimes_{i,j} \otimes_{v,e} r^i f \otimes r^i_j f = \otimes_{i,j} \otimes_{v,e} r^i g \otimes r^i_j g.$$

Since both sides are morphisms between $1/\tau VG$ and $1/\tau VG'$, the fact that $\mathbf{A}$ is τ-faithful implies $r^i f = r^i g$ and $r^i_j f = r^i_j g$. Finally, because FR is strongly proper, each FR^i and each FR^i_j is an equivalence over their images. Thus, commutativity of diagrams (8) gives us $f^i = g^i$ and $f^i_j = g^i_j$, i.e, $f = g$. $\square$

Putting together all parts of our construction we have the following general existence and uniqueness theorem.

Theorem 5.2 (existence and uniqueness)**.** *Let C be a proper, $(\mathbf{A}, \otimes, \cdot^\vee)$ be a τ-faithful pseudo-context with null objects, where $\tau : \mathbf{FinSet} \to \mathbf{A}$ is a $1/$decomposable functor. In this case:*

1. *up to normalization, change of coefficients and qeic, there exists a unique complete Feynman functor $Z : C \to (\mathbf{A}, \tau)$;*
2. *if C is concrete proper and becomes endowed with a normal structure $\mathfrak{d}$, then up to change of coefficients and qeic, there exists a unique monoidal and complete Feynman functor $Z : C \to (\mathbf{A}, \tau)$.*

6 Superposition

Let $(\mathbf{C}, \otimes, 1)$ be a monoidal category. We say that an object $X \in \mathbf{C}$ is the *superposition* of a family of objects $A_i \in \mathbf{C}$, with $1 \leq i \leq n$, if there exists an isomorphism

$$(...((A_1 \otimes A_2) \otimes A_3)...) \otimes A_n \simeq X. \tag{18}$$

The number n is called the *order* of the superposition.

Example 6.1 (trivial superposition)**.** Each object X admits a superposition of arbitrary order. Indeed, for given n, just take $A_1 = X$ and $A_{i>1} = 1$.

Let $\mathcal{C} \subset \mathrm{Ob}(\mathbf{C})$ be a collection of objects in $\mathbf{C}$. A *superposition principle* in $\mathcal{C}$ is given by

1. a bounded from above function $n : \mathcal{C} \to \mathbb{N}$, which is equivalent to saying that $n(\mathcal{C}) \subset \mathbb{N}$ is finite[7]. The maximum of $n(\mathcal{C})$ will be denoted by N;
2. a function $\mathcal{S} : \mathcal{C} \to \mathrm{Ob}(\mathbf{C})$ assigning to each object X in $\mathcal{C}$ a family of objects $\mathcal{S}X_i$ in $\mathbf{C}$, with $1 \leq i \leq N$, such that
   (a) for $i > n(X)$ we have $SX_i \simeq 1$;
   (b) after taking the tensor product between the SX_i, in the same order that in (18) we get a superposition for X.

We say that a superposition principle is *functorial* if $\mathcal{C}$ is actually a (non necessarily monoidal) subcategory of $\mathbf{C}$ such that the function $\mathcal{S} : \mathcal{C} \to \mathrm{Ob}(\mathbf{C})$ extends to a functor $\mathcal{S} : \mathcal{C} \to \mathbf{C}$.

Example 6.2 (trivial superposition principle)**.** In any set $\mathcal{C}$ we have a trivial superpostion principle, which assigns to each $X \in \mathcal{C}$ its corresponding trivial superposition.

In the following we will show that nontrivial Feynman functors behave as a bridge between reconstruction conjectures and nontrivial superposition principles. In order to do this, let $\mathcal{D}$ be a deleting process, which assigns to each applicable prestack $C \in \mathfrak{C}$ its prestack $\mathcal{D}C$ of pieces, and let us define a *structure of disjoint pieces* in some $C \in \mathfrak{C}$ as:

1. a rule that for each V associates a decomposition $V = \coprod_i \overline{V}_i$;
2. a subprestack $K \subset C$ (not necessarily applicable);
3. for each V a functor $\kappa_V : \mathcal{D}C_V \to \prod_i K_{\overline{V}_i}$.

Example 6.3 (representable reconstruction)**.** If each $\mathcal{D}C_V$ is representable, then it maps colimits into limits, so that for any decomposition $V = \coprod_i \overline{V}_i$ we have $\mathcal{D}C_{\coprod_i \overline{V}_i} \simeq \prod_i \mathcal{D}C_{\overline{V}_i}$. So, if $DC_V \subset C_V$, we can take $K_{\overline{V}_i} = \mathcal{D}C_{\overline{V}_i}$ and κ_V as the previous isomorphism.

Example 6.4 (disjoint reconstruction)**.** The disjoint context $\mathcal{D}C_V = \sqcap C_V$ has a canonical structure of disjoint pieces with $\overline{V}_i = V - v_i$, $K = DC$ and $\kappa = id$.

Theorem 6.1. *Let (C, ξ) be a proper prestack of hypergraphs which is an applicable prestack of certain reconstruction context $\mathcal{D}$. Suppose given a structure of disjoint pieces in C. Then each nontrivial strong monoidal Feynman functor $Z : C \to (\mathbf{A}, \tau)$ induces a nontrivial superposition principle in a set $\mathcal{A}_\tau$ of analytic expressions in $\mathbf{A}$, which becomes endowed with canonical map $\lambda_Z : \mathcal{G} \to \mathcal{A}_\tau$, where $\mathcal{G}$ is the set of all isomorphism classes of hypergraphs $G \in C_V$ for every V.*

Proof. We will give the proof for the case in which $(\mathbf{A}, \otimes, 1)$ is strict. The general case is analogous, only needing some care with the parentheses. For each V, we have the functor $\kappa_V : \mathcal{D}C_V \to \prod_i K_{\overline{V}_i}$. On the other hand, since C is proper for any W, W' we have $\xi : C_W \times C_{W'} \to C_{W \sqcup W'}$, which clearly extends to a functor $\xi : \prod_i C_{W_i} \to C_{\coprod_i W_i}$, where W_i is any finite family. Because $K \subset C$, such functors can be restricted to K, giving $\xi : \prod_i K_{W_i} \to K_{\coprod_i W_i}$. Taking $W_i = \overline{V}_i$, let us consider ξ restricted to the image of κ_V. Since Z is strong monoidal, for every $\mathcal{D}G \in \mathcal{D}C_V$ we have an isomorphism

$$Z_{\coprod_i \overline{V}_i = V}(\xi(\mathcal{D}G)) \simeq \otimes_i Z_{\overline{V}_i}(\mathcal{D}G_i), \tag{19}$$

[7] Indeed, due to the well-ordering principle a subset $S \subset \mathbb{N}$ is always bounded from below, hence it is bounded from above iff it is bounded. Furthermore, any infinite subset of $\mathbb{N}$ is in bijection with $\mathbb{N}$, so that it is unbounded. Finally, every finite subset of $\mathbb{N}$ is bounded.

where $\mathcal{D}G_i \in K_{\overline{V}_i}$ are the components of $\kappa_V(\mathcal{D}G)$. We can regard the isomorphism above as superposition principle for $Z(\xi(\mathcal{D}G))$. Recall that for every V and every $\mathcal{D}G \in \mathcal{D}C_V$, $Z_V(\mathcal{D}G) \in \mathbf{A}_{\tau V \mathcal{D}G}$. Therefore, if $\mathcal{A}_\tau \subset \mathrm{Ob}(1/\mathbf{A})$ is the set of all analytic expressions in τV, for every V, then $Z_V(\mathcal{D}G) \in \mathcal{A}_\tau$ for each V and each $\mathcal{D}G$. This means that when varying V and $\mathcal{D}G$ in (19) we get a superposition principle in $\mathcal{A}_\tau$. Let $\mathcal{D}\mathcal{G}$ be the set of all isomorphism classes of $\mathcal{D}G \in \mathcal{D}C_V$ for every V. So we have an obvious function $\Lambda_Z : \mathcal{D}\mathcal{G} \to \mathcal{A}_\tau$ assigning $Z_V(\mathcal{D}G)$ to each $\mathcal{D}G$. Finally, recall that in a reconstruction context there exists a transformation $\gamma_V : C_V \to DC_V$, which produces $\gamma : \mathcal{G} \to \mathcal{D}\mathcal{G}$. By composing with Λ_Z we get the desired $\lambda_Z : \mathcal{G} \to \mathcal{A}_\tau$. □

Let $C \in \mathfrak{C}$ be an applicable prestack of a reconstruction context $\mathcal{D}$. Suppose that C is proper and endowed with a structure of disjoint pieces. Then $\mathcal{D}$-RC-C holds only if for every nontrivial complete monoidal Feynman rule $Z : C \to (\mathbf{A}, \tau)$ the corresponding map λ_Z is injective.

Proof. Assume $\mathcal{D}$-RC-C holds, which means that $\gamma_V : C_V \to DG_V$ is essentially injective for every V, and let $Z : C \to (\mathbf{A}, \tau)$ be a complete monoidal Feynman rule. Recall that $\lambda_Z : \mathcal{G} \to \mathcal{A}_\tau$ is the composition between $\gamma : \mathcal{G} \to \mathcal{D}\mathcal{G}$ and $\Lambda_Z : \mathcal{D}\mathcal{G} \to \mathcal{A}_\tau$. Completeness of Z implies Λ_Z injective and essential injectivity of γ_V makes γ injective, so that λ_Z is also injective. Reciprocally, suppose that λ_Z is injective for some complete monoidal Feynman rule. By definition, $\lambda_Z = \Lambda_Z \circ \gamma$. Since injective functions are monomorphisms, γ_Z injective implies γ injective, which is equivalent to saying that γ_V is essentially injective for every V, so that $\mathcal{D}$-RC-C holds. □

Remark 6.1. As a consequence of Proposition 3.2 and Proposition 3.3, the previous corollary can be improve by replacing the hypothesis of $\mathcal{D}$-RC-C holding with the existence of left/right morphisms $F : \mathcal{D} \to \mathcal{D}'$ or left/right C-implications to some other reconstruction context.

Let $C \subset S^{b,\ell}_{\mathfrak{s}}$ be some concretely proper prestack of labeled structured bounded hypergraphs, regarded as an applicable prestack for the standard reconstruction context $\mathcal{D} = d$ and endowed with the canonical structure of disjoint pieces. Then for every complete monoidal Feynman functor $Z : C \to (\mathbf{A}, \tau)$ the induced map λ_Z in injective. Explicitly,

$$Z_V(G) \simeq Z_V(G') \quad \textit{iff} \quad \bigotimes_{v \in V} Z(G - v) \simeq \bigotimes_{v \in V} Z(G' - v).$$

Proof. Immediate from Corollary 3.3, Corollary 6 and from the definition of λ_Z. □

7 Applications

As the first application we present an existence result for representations of hypergraph categories in monoidal categories.

Proposition 7.1. *Let C^b be b-bounded prestack of hypergraphs. Let $(\mathbf{A}, \otimes, \cdot^\vee)$ be a pseudocontext which is τ-faithful relative to a 1/decomposable functor $\tau : \mathbf{FinSet} \to \mathbf{A}$. Suppose one of the following conditions:*

- *c1) $C^b \subset S^{b,\ell}_{\mathfrak{s}}$ for some functor of structures $\mathfrak{s}$;*
- *c2) $(\mathbf{A}, \otimes, 1)$ is a strict monoidal category.*

In this case, the choice of a normal structure $\mathfrak{d}$ for C^b induces an equivalence between the category C^b_V, for each V, and some subcategory of $1/\mathbf{A}$.

Proof. Notice that we are in the hypothesis of Proposition 5.4 and from its proof we see that if there exists some strongly complete Feynman rule FR in $(C^b, \mathfrak{d})$, then it induces a Feynman functor $Z : C^b \to (\mathbf{A}, \tau)$ such that each $Z_V : C^b_V \to 1/\mathbf{A}$ is essentially injective and faithful, so that they are equivalences over their images. Since $\mathfrak{d}$ is chosen normal, it follows from Corollary 5.1 that these Feynman rules really exist. $\square$

7.1 Mapping Class Group and Ribbon Graphs

One of the main problems of manifold topology is to determine the mapping class group $\mathrm{MCG}(M)$ of a given manifold M, i.e, the quotient of the diffeomorphism group $\mathrm{Diff}(M)$ by the path-component at the identity. It is a remarkable fact that for marked surfaces that object can be described by the category of ribbon graphs. More precisely, if $\mathbf{Ribb}^{con}_3 \subset \mathbf{Ribb}$ denotes the category of connected ribbon graphs whose vertices are at least trivalent and with morphisms given by ribbon graphs isomorphisms, then there exists a homotopy equivalence

$$|\mathbf{Ribb}^{con}_3| \simeq \coprod_{[\Sigma]\in \mathrm{Iso}(\mathbf{Diff}^*_2)} B\,\mathrm{MCG}(\Sigma),$$

where $|\mathbf{C}|$ and BG denotes, respectively, the geometric realization of a category and the classifying space of a topological group. Furthermore, the coproduct is taken over the diffeomorphic classes of all marked surfaces, except for two exceptional cases: the sphere $\mathbb{S}^2$ with one and with two marked points.

This result was proven using different methods in [7, 8, 9, 35, 36]. As a second application of the existence of complete Feynman rules we give an independent proof of a related result. Indeed, we will show that for any fixed vertex set V, the category $\mathbf{Ribb}_V$ can be regarded as a subcategory $\mathbf{A}_V$ of

$$\coprod_{[\Sigma]\in \mathrm{Iso}(\sqcup\mathbf{Diff}^*_2)} \mathbf{B}\,\mathrm{MCG}(\Sigma),$$

where $\mathbf{B}G$ is the delooping groupoid of a group and $\sqcup\mathbf{Diff}^*_2$ means that the coproduct is take over arbtirary finite coproducts of marked surfaces.

Let $\mathbf{Diff}^*$ be the category of marked manifolds, i.e, pairs (M, S), where $S \subset M$ is some finite subset of mutually distinct points, and morphisms given by smooth maps $f : M \to M'$ such that $f(S) \subset S'$. The category has coproducts given by $\coprod_i (M_i, S_i) = (\coprod_i M_i, \coprod_i S_i)$. Two marked manifolds are isomorphic only if $S \simeq S'$, which means that in the isomorphism classes only the number of marked points matter. We will write M^s instead of (M, S), where $s = |S|$. Let $\mathbf{Sing}$ be the proper class of singletons and fix an injective class function $\omega : \mathrm{Iso}(\mathbf{Diff}^*) \to \mathbf{Sing}$. For instance, we could take $\omega([M]) = \{[M]\}$. Define a category $\mathrm{MCG}^{\omega}_{\mathbf{Diff}^*}$ as follows. Objects are given by the image of ω, i.e, we have an object for each isomorphism class of marked manifolds and this object which is a singleton $\omega([M^s]) = *_{[M^s]}$. Furthermore, there are morphisms between $*_{[M^s]}$ and $*_{[N^r]}$ iff $[M^s] = [N^r]$ and in this case $\mathrm{Mor}(*_{[M^s]}, *_{[M^s]}) = \mathrm{MCG}(M^s)$. In particular, $(\mathrm{MCG}^{\omega}_{\mathbf{Diff}^*})^{op} = \mathrm{MCG}^{\omega}_{\mathbf{Diff}^*}$. Furthermore, since

$$\mathrm{MCG}^{\omega}_{\mathbf{Diff}^*} \simeq \coprod_{[M^s]\in \mathrm{Iso}(\mathbf{Diff}^*)} \mathbf{B}\,\mathrm{MCG}(M^s),$$

the left-hand side does not depends of ω. Coproducts pass to $\mathrm{MCG}^{\omega}_{\mathbf{Diff}^*}$ by taking $*_{[M^s]} \sqcup *_{[N^r]} = *_{[M^s \sqcup N^r]}$. Furthermore, if $f \in \mathrm{MCG}(M^s)$ and $g \in \mathrm{MCG}(N^r)$, define $f \sqcup g : *_{[M\sqcup N]} \to *_{[M\sqcup N]}$ as the coproduct in $\mathbf{Diff}^*$.

Consider a full subcategory $\mathbf{A} \subset \mathrm{MCG}^{\omega}_{\mathbf{Diff}^*}$ closed by finite coproducts, so that we can take the cocartesian monoidal structure and since $\mathbf{A}^{op} = \mathbf{A}$, we get a pseudocontext structure $(\mathbf{A}, \sqcup, \cdot^{\vee})$ with $\cdot^{\vee} : \mathbf{A} \to \mathbf{A}$ some endofunctor of $\mathbf{A}$. These functors are in 1-1 correspondence with rules assigning manifolds M^s such that $\omega([M^s]) \in \mathbf{A}$ to manifolds N such that $\omega([N^r]) \in \mathbf{A}$, together with a group homomorphism $\mathrm{MCG}(M^s) \to \mathrm{MCG}(N^r)$. For instance, if in $\mathbf{A}$ there exists some manifold X^o whose mapping class group is completely understood, we can take $\cdot^{\vee}$ as some functor constant in such a manifold, which is equivalent to giving a representation of every $\mathrm{MCG}(M^s)$ in $\mathrm{MCG}(X^o)$. But we can also simply take $\cdot^{\vee} = id_{\mathbf{A}}$. Let $S \neq \varnothing$ be a connected manifold such that $\omega([S^s]) \in \mathbf{A}$ for every s and define $\tau_S : \mathbf{FinSet} \to \mathbf{A}$ as follows. For each finite set X we take $\tau_S(X) = \omega([S^{|X|}])$. Furthermore, if $f : X \to Y$ is a map between finite sets, then $\tau_S(f)$ is nontrivial iff $X \simeq Y$ and in this case $\tau_S(f) = id_{S^{|X|}}$. objects and

Now, recall that each construction in this article was made in $1/\mathbf{A}$. The only reason for doing this was to have a frame closer to physics interpretation. Indeed, in this way we can talk about analytic expressions itself, while when working in $\mathbf{A}$ we can only talk about the object of all analytic expressions. Even so, all definitions, statements and demonstrations work *ipsi literis* in $\mathbf{A}$. With this in mind we can search for Feynman functors taking values in $(\mathbf{A}, \tau_S)$ for the pseudocontext $(\mathbf{A}, \sqcup, \cdot^{\vee})$ defined above.

Proposition 7.2. *For each connected non-empty manifold S such that $\omega(S^s) \in \mathbf{A}$ for every s, the pseudocontext $\mathbf{A} \subset \mathrm{MCG}^{\omega}_{\mathbf{Diff}^*}$ is τ_S-faithful and τ_S is decomposable.*

Proof. Notice that $\tau_S(\mathbf{FinSet})$ is the category whose objects are $\omega([S^s])$, with $s \geq 1$, and whose only morphisms are the identities id_{S^s}, for $s \geq 1$. Therefore, $(\mathbf{A}, \sqcup, \varnothing)$ is clearly τ_S-faithful. In order to see that τ_S is decomposable, given finite sets X and X', suppose we have decompositions

$$\tau_S(X) \simeq \omega([M^s]) \sqcup \omega([N^r]) \quad \text{and} \quad \tau_S(X') \simeq \omega([M'^{s'}]) \sqcup \omega([N'^{r'}]).$$

This implies $S^{|X|} \simeq M^s \sqcup N^r$ and $S^{|X'|} \simeq M'^{s'} \sqcup N'^{r'}$. Since S is connected, either we have the following configuration or we have one of the other seven permutations:

- $M^s \simeq S^{|X|}$ and $N^r = \varnothing$, together with $M'^{s'} \simeq S^{|X'|}$ and $N'^{r'} = \varnothing$.

We should proceed case by case, but since everything is analogous we will work only with the above configuration. Let $f : X \to X'$ be an isomorphism. Then $|X| = |X'|$, implying $M^s = M'^{s'} \simeq S^{|X|}$. Since $\tau_S(f) = id_{S^{|X|}}$, we just take $\alpha = id_{M^s}$. Furthermore, since $N^r = \varnothing = N'^{r'}$, there exists a unique $\beta : N^r \simeq N'^{r'}$ and we clearly have $\tau_S(f) = \alpha \sqcup \beta$. $\square$

Let C^b be a b-bounded prestack of hypergraphs. For each normal structure $\mathfrak{d}$ and each non-empty connected manifold S there exists an equivalence between C^b_V, for every V, and a subcategory $\mathbf{A}_V$ of any pseudocontext $\mathbf{A} \subset \mathrm{MCG}^{\omega}_{\mathbf{Diff}^*}$ such that $\omega(S^s) \in \mathbf{A}$ for every s.

Proof. Straighforward from Proposition 7.2 and Proposition 7.1. $\square$

Let $\mathrm{MCG}^{\omega}{}_{\sqcup\mathbf{Diff}^*_2} \subset \mathrm{MCG}^{\omega}_{\mathbf{Diff}^*}$ denote the subcategory generated by compact orientable marked surfaces and finite disjoint unions of them. The choice of a normal decomposition $\mathfrak{d}$ induces an equivalence between the category $\mathbf{Ribb}_V$ and some subcategory of $\mathrm{MCG}^{\omega}{}_{\sqcup\mathbf{Diff}^*_2}$.

Proof. Just recall that any prestack of graphs is 2-bounded and then apply the last corollary. $\square$

Remark 7.1. Our construction, however, does not give much information about the homotopy type of the classifying spaces $B\,\mathrm{MCG}(M)$. Indeed, since we included the empty manifold in $\mathrm{MCG}^{\omega}_{\mathbf{Diff}^*}$ and the empty graph in each C^b_V, both admit initial objects, so that their geometric realizations are automatically contractible.

7.2 Perturbative QFT

As a final application, let us show that the validity of a reconstruction conjecture induces a new superposition principle in perturbative QFT. A *classical field theory* $\mathcal{S}$ (following [20]) is given by the following data:

1. a vector bundle $\pi : E \to M$ over a compact riemannian manifold M;
2. a positive generalized laplacian $\mathcal{D}$ in $E^! = E^\vee \otimes \mathrm{Dens}_M^{s,+} \simeq \mathrm{Hom}(E; \mathrm{Dens}_M^{s,+})$, from which we build the free functional $S_0[s] = \int_M \mathcal{D}s$. Typical examples occur when $\mathcal{D}s = \langle s, Ds\rangle$, where D is a generalized laplacian in E and $\langle \cdot, \cdot \rangle : E \otimes E \to \mathrm{Dens}_M^{s,+}$ is a symmetric bundle map;
3. a differential operator $\mathfrak{D}$ between $E^!$ and E, which is formally symmetric, i.e, $\mathfrak{D}^! = \mathfrak{D}$ and such that $\mathcal{D} \circ \mathfrak{D}^! = \mathfrak{D} \circ \mathcal{D}^!$;
4. an element $I \in \Gamma(E)[[\hbar]]$, giving the full action functional $S = S_0 + I$.

Since $\Gamma(E)$ is a nuclear Fréchet space, it belongs to context $(\mathbf{NucFrec}, \otimes_p, \cdot^\vee)$ of Example 4.3. Let us take $\tau : \mathbf{FinSet} \to \mathbf{NucFrec}$ as constant in $\Gamma(E)$. From any classical field theory (in the above sense) we can extract a full subcategory $\mathbf{Feyn}(\mathcal{S}) \subset \mathbf{Feyn}$ simply by doing the standard Feynman graph expansion of an action funcional [20]. Varying V on $\mathbf{Feyn}(\mathcal{S})_V$ we get a prestack $C_{\mathcal{S}}$. Let us call the pair $(\mathcal{S}, C_{\mathcal{S}})$ the *pertubative QFT* of $\mathcal{S}$. Notice, on the other hand, that from data 1-3 above we can extract a tensor $P \in \mathrm{Sym}^2\,\Gamma(E)$, obtained as follows. Since M is compact, $\mathcal{D}$ has a smooth heat kernel $K_t \in \Gamma(E^!) \otimes \Gamma(E) \otimes C^\infty(\mathbb{R}_{\geq 0})$. Composing with $\mathfrak{D}$ we get an element of $\mathcal{K}_t \in \Gamma(E) \otimes \Gamma(E) \otimes C^\infty(\mathbb{R}_{\geq 0})$. Because $\mathfrak{D}$ is symmetric, $\mathcal{K}_t$ is symmetric too. By means of integrating we get $P \in \mathrm{Sym}^2(\Gamma(E))$. We can then get a Feynman rule $FR : (C_{\mathcal{S}}, \mathfrak{d}) \to (\mathbf{NucFrec}, \tau)$ from Example 5.9 by fixing FR_2^1 as constant in P and FR^1 as determined by I. We also take FR^0 constant due to the indistinguishability of quantum particles. Since $C_{\mathcal{S}}$ is concretely proper, it follows from Example 5.13 and Proposition 5.3 that the induced Feynman functor $Z_{\mathcal{S}} : C_{\mathcal{S}} \to (\mathbf{NucFrec}, \tau)$ is oplax monoidal. Let us define a *superposition principle* for $(\mathcal{S}, C_{\mathcal{S}})$ as a superposition principle in the image of $Z_{\mathcal{S}}$.

Proposition 7.3. *Let $(\mathcal{S}, C_{\mathcal{S}})$ be a perturbative QFT whose Feynman functor $Z_{\mathcal{S}}$ is strong monoidal. If $C_{\mathcal{S}}$ is an applicable prestack for a reconstruction context $\mathcal{D}$ we have a nontrivial superposition principle for $(\mathcal{S}, C_{\mathcal{S}})$.*

Proof. Just apply Theorem 6.1. □

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A

LIST OF QUESTIONS AND SPECULATIONS

The following is just a compilation of the questions presented in the section "Questions and Speculations" of each chapter.

A.1 AXIOMATIZATION OF PHYSICS

Q1.1 Show that every category $\mathbf{C}$ defines a system $\mathcal{S}(\mathbf{C})$ in the sense of Bunge. What kind of systems corresponds to some category? For instance, show if $\mathbf{C}$ and $\mathbf{C}'$ are two categories, then the systems $\mathcal{S}(\mathbf{C})$ and $\mathcal{S}(\mathbf{C}')$ are actually Σ_{cat}-systems for certain Σ_{cat}. Build a functor $F : \mathbf{Cat} \to \mathbf{Sys}_{\Sigma_{cat}}$. Has this functor left/right adjoints?

Q1.2 Given Σ, is the category $\mathbf{Sys}_{\Sigma}$ complete/cocomplete?

Q1.3 Do extend the rules u_* and u^* defined above to functors $u_* : \mathbf{Sys}_{\Sigma} \to \mathbf{Sys}_{\Sigma'}$ and $u^* : \mathbf{Sys}_{\Sigma'} \to \mathbf{Sys}_{\Sigma}$? Are them adjoint? Let $\mathbf{Set}$ be category of classes and sets. Varying u we get functors $(-)_* : \mathbf{Set} \to \mathbf{Cat}$ and $(-)^* : \mathbf{Set}^{op} \to \mathbf{Cat}$? This means that the study of systems can be viewed as a presheaf of categories?

Q1.4 Define morphisms between concrete systems of a fixed type Σ. Show that they fit into a category $\mathbf{CSys}_{\Sigma}$. Prove that there is an inclusion functor $\imath : \mathbf{CSys}_{\Sigma}\mathbf{Sys}_{\Sigma}$. This functor has a "concretification" left/right adjoint $L : \mathbf{Sys}_{\Sigma} \to \mathbf{CSys}_{\Sigma}$? Is the subcategory $\mathbf{CSys}_{\Sigma}$ of concrete systems a reflective/co-reflective subcategory?

Q1.5 Prove that every category $\mathbf{C}$ defines a concrete system $\mathcal{C}(\mathbf{C})$ of a specific type $\Sigma_{cat'}$. Show that the construction extends to a functor $\mathcal{C} : \mathbf{Cat} \to \mathbf{CSys}_{\Sigma_{cat'}}$. Find a map $u : \Sigma_{cat} \to \Sigma_{cat'}$ such that diagram below commutes. Supposing that the concrectification functor L exists (i.e., assuming Question 2.2), prove that the entire diagram remains commutative.

Q1.6 Reformulate what are ontological/epistemological/logical theories for an area of study now regarding them internally to categories instead of to classes. Show that they are characterized by a version of diagram (2.1) in $\mathbf{Cat}$.

Q1.7 Assume Question 2.2. Let $\mathbf{Sys}_{\Sigma_{cat}}$ category of systems which are defined by categories. Show that $\mathbf{Sys}_{\Sigma_{cat}}$ is actually a 2-category such that the functor $F : \mathbf{Cat} \to \mathbf{Sys}_{\Sigma_{cat}}$ is a 2-functor.

Q1.8 Assume Questions 2.2 and 2.2. Reformulate ontological/epistemological/logical theories for categorical areas of study internally to 2-categories instead of to categories.

Prove that they can be characterized by diagrams which are analogous of (2.1), but in $2\mathbf{Cat}$.

Q1.9 Define higher (or graded) system as a tuple $\mathcal{S} = (T, C, E, S)$ where $E = E_1, ...$ is a list of mutually disjoint classes, $C, E_i \subset T$ and S is a collection of relations on $\sqcup_i C \sqcup E_i$. Show that every ∞-category defines a higher system and that these higher systems are always of the same type $\Sigma_{\infty cat}$.

Q1.10 Prove that higher Σ-systems constitute a ∞-category $\infty\mathbf{Sys}_{\Sigma}$. Show that construction of Question 2.2 extends to a ∞-functor $F : \infty\mathbf{Cat} \to \infty\mathbf{Sys}_{\Sigma_{\infty cat}}$.

Q1.11 In analogy to Question 2.2, show that each $u : \Sigma \to \Sigma'$ induces adjoint ∞-functors u_* and u^*. Varying u build a ∞-functor $(-)^* : \mathbf{Set}^{op} \to \infty\mathbf{Cat}$ allowing us to view higher systemics as a certain ∞-presheaf of ∞-categories.

Q1.12 Define what is a higher ontological/epistemological/logical theory for an area of study defined by a sub-∞-category $\mathcal{O}$ of higher systems. Prove that they can be characterized by diagram (2.1), now on $\infty\mathbf{Cat}$.

Q1.13 Let $\mathcal{O}$ be an area of study consisting of higher systems endowed with a higher epistemological theory. Assuming Question 2.2 show that the diagram (5.3) characterizing an axiomatic approach to $\mathcal{O}$ is defined on $\infty\mathbf{Cat}$.

Q1.14 What kind of languages need to be used to build axiomatic approaches to areas of study defined by of higher systems?

Q1.15 Show that the construction of Section 2.1.6 can be reproduced internally to any category $\mathbf{H}$.

Q1.16 Define the class $\mathrm{Inj}(\mathbf{H})$ of "injective objects" of $\mathbf{H}$ by the property that if $f : X \to Y$ is a monomorphism and $X \in \mathrm{Inj}(\mathbf{H})$, then f is a split monomorphism. Assume Question 2.2. Extending Corollary 2.22 prove that if

1. the APP can be realized internally to $\mathbf{H}$ for an empiricist class $\mathcal{FE}_0 = (\mathcal{F}_0, \mathcal{E}_0)$ such that $\mathcal{E}_0 \in \mathrm{Inj}(\mathbf{H})$;
2. there exists a systematic way of extending any axiomatic language,

then APP can be realized internally to $\mathbf{H}$ in the entire Physics.

Q1.17 In order to contemplate higher systems, show that Question 2.2 remains true if $\mathbb{H}$ is $\infty\mathbf{Cat}$.

Q1.18 What are "higher injective objects" in $\infty\mathbf{Cat}$? Are they ∞-categories $\mathbf{C}$ such that every ∞-functor $F : \mathbf{C} \to \mathbf{D}$ which is a ∞-monomorphism it is actually a split ∞-monomorphism? But, what are ∞-monomorphisms and what are split ∞-monomorphisms? Show that for the right notion of higher injective objects we have a higher analog Corollary 2.22.

A.2 MATHEMATICAL ASTROPHYSICS

Q3.1 To each model M to gravity (General Relativity, EHP, $f(R)$-gravity, and so on) assigns a corresponding model $\mathcal{C}(M)$ to the axiomatics of Section 3.1.3.

Q3.2 Show that each model to gravity (General Relativity, EHP, $f(R)$-gravity, and so on) corresponds a model to Astrophysics such that this minimal class exists. Furthermore, show that the known concrete models of stellar systems belongs to a corresponding model $\mathcal{C}_{\alpha_{min}}(M)$ to the axiomatics of Section 3.1.3.

Q3.3 what is a morphism between realizations of Astrophysics in the sense of Section 3.1.3? Do they constitute a category? What can be said about the classification problem of this category? Under which conditions do two models to gravity induce isomorphic models to Astrophysics? And what about the induced models restricted to the subcategories of stellar systems?

Q3.4 What is "stellar evolution" in the astrophysical context of Section 3.1.3? How to characterize the stellar systems in $\mathcal{C}_{star} \subset \mathcal{C}_{min}$ which are in evolutional equilibrium? Show that this characterization is such that in the interpretations corresponding to gravity models we recover the corresponding equations of stellar evolution.

Q3.5 In terms of the axiomatics of Section 3.1.3, what does it means to say that a star is in "hydrodynamic equilibrium"? Prove that semantically the given notion contains the hydrodynamic equilibrium equations of models as General Relativity, etc. Give conditions under which this abstract notion of hydrodynamic equilibrium intersects evolutional equilibrium.

Q3.6 How to extend the classical models to stellar systems under hydrodynamic equilibrium from spherically symmetric objects to more general objects?

Q3.7 Verify if elliptic stellar systems can provide an attempt to Question Q3.6.

Q3.8 Show that Astrophysics in the sense of Section 3.1.3 admits a model on given spacetime M as a classical theory, i.e., as the critical points of functional.

A.3 GEOMETRY OF GRAVITY

Q4.1 Define in formal terms what (reductive) algebra-valued gauge theories are. Show that the Geometric Obstruction Theory method can be extended to them.

Q4.2 Show that Teleparallel Gravity and Einstein-Cartan extend to reductive algebra-valued gauge theories and apply Geometric Obstruction Theory on them. Compare the corresponding obstructions with those obtained for EHP theories in [117, 115].

Q4.3 Yang-Mills theories, Chern-Simons theories, and other non-reductive gauge theories can be extended to algebra-valued gauge theories? What are the corresponding obstructions?

Q4.4 Define what is a graded algebra-valued gauge theory on a graded manifold M. Prove that the Geometric Obstruction Theory extends to this case.

Q4.5 Show that the typical super gauge theories admitting a superspace formalism extend to a super algebra-valued gauge theory on a supermanifold. Find the corresponding obstructions.

Q4.6 Define what is a super-extension of an algebra-valued theory. Establish correspondence between obstructions on a theory and obstructions on its super-extensions.

Q4.7 Assuming Questions Q4.3 and Q4.6, find obstructions on super Yang-Mills theories. If a super Yang-Mills theory admits a superspace formalism, compare these obstructions with those obtained from Questions Q4.4 and Q4.5.

Q4.8 Define what is a higher-algebra-valued gauge theory. Show that the Geometric Obstruction Theory can be extended to this setting. Define what is a higher extension of an algebra-valued gauge theory. Prove that obstructions on an algebra-valued gauge theory induce obstructions on its higher extensions.

Q4.9 Define what is an "algebraic context" in which stacks taking values into their objects exist. Define the algebra-valued stack classifying an algebra-valued bundle P. Define what is a "refinement" of an algebra-valued stack. Show that the refinements of the algebra-valued stack classifying an algebra-valued bundle P correspond to geometric objects on P which locally can be written as higher-algebra-valued forms. Define *algebra-valued higher gauge theory* as the study of these stacks. Prove that the Geometric Obstruction Theory extends to this context.

Q4.10 Prove that the supergravity theories extend to algebra-valued higher gauge theories. Compute the obstructions on them.

Q4.11 Let S be a reductive algebra-valued gauge theory. Show that the geometric/algebraic duals S^* and ${}^*S$ are well-defined. For which classes of algebras do these action functionals induce equivalent perturbative series? Similar questions for super-algebra-valued gauge theories and higher-algebra-valued gauge theories.

Q4.12 Is EHP with e not necessarily a pointwise isomorphism physically equivalent to Singular Semi-Riemannian Geometry?

A.4 EMERGENCE AND EXTENSION

Q5.1 Show that an area of study $\mathcal{O}$ is ontological reduionist (resp. epistemological reduionist or logical redutionist) iff the rule $(-)_{real}$ (resp. Know or Logic) assigning to each system $X \in \mathcal{O}$ its class $(\mathcal{O}_x)_{real}$ (resp. Know$(\mathcal{O}_x)$ or Logic$(\mathcal{O}_x)$) of real subsystems (resp. knowledge or logically true propositions) is a ∞-sheaf. Dually, show that $\mathcal{O}$ is ontological/epistemological/logical emergentist iff the corresponding rules are not ∞-sheaves.

Q5.2 Prove that an effective structure in an area of study $\mathcal{O}$ induces a Grothendieck topology on it (if this is not the case, give conditions ensuring it). Furthermore, $\mathcal{O}$ is an effective ontological, epistemological, or logical reductionist in that effective structure iff the corresponding rules $(-)_{real}$, Know or Logic are ∞-sheaves in the induced Grothendieck topology. Dually, show that effective ontological/epistemological/logical emergentism is on the failure of those rules being ∞-sheaves.

Q5.3 Show that each morphism between effective structures in Definition 5.8 induces a morphism between sites relative to the corresponding Grothendieck topologies of Question (Q5.2). Use this to give a sheaf-theoretic proof for the fact that reductionist approaches can be pulled back and to the strategy to build emergentist approaches.

Q5.4 Use of global analysis and homotopy theory to provide general, but local, emergence theorems between arbitrary parameterized field theories.

Q5.5 Use of the discussion above and conditions on the homotopy of $\mathcal{S}(E)$ to find emergence relations between parameterized field theories whose fundamental parameters are cocycles of twisted generalized cohomologies of $\mathcal{S}(E)$.

Q5.6 Formalize discussion above and use of Questions (Q5.4) (Q5.5) to build local emergence theories between data investigation methods. What does it mean a Lagrangian field theory in this context? How could the emergence theorems of [121, 118] be interpreted? How does this relates to topological data analysis?

Q5.7 What can be said about the $R[\mathbb{G}]$-module bundle structures that can be put on $\mathrm{Ext}(\hat{S};\hat{G})$? How are they related? Are there bundle morphisms between them?

Q5.8 Use some kind of K-theory and characteristic classes for group-algebra bundles to study the $R[\mathbb{G}]$-module bundle structures on $\mathrm{Ext}(\hat{S};\hat{G})$.

Q5.9 Let $\mathrm{Inc}(\hat{S},\hat{G})$ the class of all incomplete extensions $\hat{S}$ of S with extended gauge group $\hat{G}$. What can be said about $\mathrm{Inc}(\hat{S},\hat{G})$? It inherits a sub-$R[\mathbb{G}]$-module bundle structure from $\mathrm{Ext}(\hat{S};\hat{G})$?

Q5.10 Construct equivariant extensions from equivariant cocycles of $\mathrm{Gaug}(\hat{G})$-equivariant K-theory of $\mathrm{Ext}(\hat{S};\hat{G})$.

Q5.11 Use the cohomological description of the anomaly-free condition to characterize the $\mathrm{Free}(\hat{S},\hat{G})$. In particular, characterize whether equivariant extensions $\hat{S}$ of anomaly-free gauge theories are anomaly-free.

Q5.12 Assuming Question (Q4.1), let S be an algebra-valued gauge theory with an algebra of infinitesimal symmetries A. Let $\hat{A}$ an algebra which has A as a subalgebra. Define what is an extension $\hat{S}$ of S with the algebra of infinitesimal symmetries given by $\hat{A}$.

Q5.13 What is the analog of the group of gauge transformations for algebra-valued gauge theories? Let $\mathrm{Gaug}_A(P)$ this group. Let $\mathrm{Equiv}(\hat{S};\hat{A})$ be the class of extensions of a given algebra-valued gauge theory S which are invariant under the action of $\mathrm{Gaug}_{\hat{A}}(\hat{P})$. Show that as above these equivariant extensions can be characterized as solutions of a system of partial differential equations, now depending on $\hat{A}$. Use Geometric Obstruction Theory to find classes of algebras in which these equations simplify to equations whose solution space is well-known, allowing us to classify $\mathrm{Equiv}(\hat{S};\hat{A})$.

Q5.14 Is it possible to extend the BV-BRST formalism to the case of algebra-valued gauge theories? For which class of algebras is this possible? For them, find an analogous cohomological characterization of anomaly-free theories. Use Geometric Obstruction Theory to find conditions on A for which the gauge anomaly vanishes identically, so that algebra-valued gauge theories with coefficients in A are anomaly-free. Similarly, find conditions on $\hat{A}$ such that extensions of anomaly-free algebra-valued gauge theories remain anomaly-free.

A.5 GEOMETRIC REGULARITY

Q6.1 Define in formal terms $(B^l_{\alpha',\beta'},\mathbb{X}')$-*vector bundles* over a $(B^k_{\alpha,\beta},\mathbb{X})$-manifold, with $l\leq k$. Given a C^l-vector bundle $E\to M$, use of strategy in Theorem 6.25 and of the results in [120] to provide conditions under which E has a $(B^l_{\alpha',\beta'},\mathbb{X}')$-structure.

Q6.2 Define (B,l,α'',β'')-*morphisms* between two C^l-budles $E_1,E_2:\to M$. Assume that E_i, with $i=1,2$ is a $(B^l_{\alpha'_i},\beta'_i)$-bundle over a $B^k_{\alpha,\beta},\mathbb{X})$-manifold. Following a discussion similar to that of Section 5 of [120], determine conditions such that these morphims exist and define a category $\mathbf{Bund}^l_{\alpha',\beta'}(M)$ of regular bundles over regular manifolds.

Q6.3 Define $(B,l,\alpha'',\beta'',\mathbb{X}'')$-*section* of a C^l-vector bundle $E\to M$. Suppose that E is actually a $(B^l_{\alpha',\beta'},\mathbb{X}')$-bundle over a $(B^k_{\alpha,\beta},\mathbb{X})$-manifold. Using strategy of Theorem 6.25, prove existence of nontrivial these regular sections of regular bundles over regular manifolds. In the same lines of Proposition 4.1 of [112], show that they actually form vector subspace $\Gamma^l_{\alpha'',\beta''}(E)$ of $\Gamma^l(E)$. If $k=\infty$, show that $\Gamma^l_{\alpha'',\beta''}(E)$ is actually a $C^\infty(M)$-submodule.

Q6.4 Find conditions such that the rule $E \mapsto \Gamma^l_{\alpha'',\beta''}(E)$ extends to a functor $\Gamma^l_{\alpha'',\beta''}$: $\mathbf{Bund}^l_{\alpha',\beta'}(M) \to \mathbf{Vec}_{\mathbb{R}}$.

Q6.5 Let E_1 and E_2 be C^l-vector bundles and let $E_1 \times E_2$ their tensor bundle. Assume that E_i, with $i=1,2$ is a $(B^l_{\alpha'_i},\beta'_i)$-bundle over a $B^k_{\alpha,\beta},\mathbb{X})$-manifold. Using that $E_1 \otimes E_2$ is locally the tensor product between the cocycles of E_i, and following Theorem 6.25, find conditions such that $E_1 \times E_2$ is a $(B^l_{\alpha''},\beta'')$-bundle.

Q6.6 Assuming Questions 6.2 and 6.2, analyze whether the tensor product $\otimes$ makes $\mathbf{Bund}^l_{\alpha',\beta'}(M)$ into a symmetric monoidal category. Assuming in addition Question 6.2, prove that the functor of global sections $\Gamma^l_{\alpha'',\beta''}:\mathbf{Bund}^l_{\alpha',\beta'}(M) \to \mathbf{Vec}_{\mathbb{R}}$ is symmetric monoidal.

Q6.7 Do the same for the direct sum $\oplus$. Provide conditions such that $\oplus$ and $\otimes$ makes $\mathbf{Bund}^l_{\alpha',\beta'}(M)$ into a distributive monoidal category.

Q6.8 Also do the same for the internal hom and give conditions such that $\mathbf{Bund}^l_{\alpha',\beta'}(M)$ becomes a closed symmetric monoidal category.

Q6.9 Let E be $(B^l_{\alpha',\beta'},\mathbb{X}')$-bundle over a $(B^k_{\alpha,\beta},\mathbb{X})$-manifold M. Assuming Questions 6.2 and 6.2, for every given $r,s \geq 0$, the E^r_s tensor bundle has an induced structure of $(B^l_{\alpha',\beta'},\mathbb{X}')$-bundle. Use of Questions 6.2 abd 6.2 to show that E^r_s admit $B^l,'',\beta'',\mathbb{X}''$-sections for certain $(\alpha'',\beta'',\mathbb{X}'')$ independently of r,s.

Q6.10 On the other hand, for given $r,s \geq 0$, we could considered the C^l-bundle E^r_s and then used Question 6.2 to get directly a $(B^l_{\alpha'_{r,s},\beta'_{r,s}},\mathbb{X}'_{r,s})$-structure on E^r_s. From Question 6.2 we would then prove existence of $(B,l.\alpha''_{r,s},\beta''_{r,s},\mathbb{X}''_{r,s})$-sections. Compare this regularity with that $(\alpha'',\beta'',\mathbb{X}'')$ of Question 6.2. Determine the critical values (r,s) in which they agree.

Q6.11 Define $(B^l_{\alpha'',\beta''},\mathbb{X}'')$-connection on a C^l-vector bundle E. Extend the existence results of [112] from regular affine connections on M to regular linear connections on E.

Q6.12 Define what is a nonlinear $(B^r_{\alpha'',\beta''},\mathbb{X}'')$-differential operator of degree d on a C^l-vector bundle E over a C^k-manifold. Suppose that E is actually a $(B^l_{\alpha',\beta'},\mathbb{X}')$-bundle and that M is a $(B^k_{\alpha,\beta},\mathbb{X})$-manifold. Given d,r, using of Questions 6.2 and 6.2, find conditions such that nonlinear $(B^r_{\alpha''_d,\beta''_d},\mathbb{X}''_d)$-differential operators of degree d exist on E. Show that they actually form a vector subspace $\mathrm{Diff}^d_{\alpha''_d,\beta''_d}(E)$ of the space of all $(B^r_{\alpha''_d,\beta''_d},\mathbb{X}''_d)$-nonlinear tensors.

Q6.13 Assuming Question 6.2, find a triple $(\alpha'',\beta'',\mathbb{X}'')$, independent of d, such that there exist nonlinear $(B^r_{\alpha'',\beta''},\mathbb{X}'')$-differential operators of every degree d. In the smooth case $k,l,r=\infty$, analyze whether the graded vector space $\mathrm{Diff}_{\alpha'',\beta''}(E)=\oplus_d \mathrm{Diff}^d_{\alpha'',\beta''}(E)$ is a graded-algebra with the composition operation.

A.6 FEYNMAN FUNCTORS

(to be added)

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